



US012403336B2

(12) **United States Patent**
Ciaramella et al.

(10) **Patent No.:** **US 12,403,336 B2**

(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **BETACORONA VIRUS MRNA VACCINES**

(71) Applicant: **ModernaTX, Inc.**, Cambridge, MA
(US)

(72) Inventors: **Giuseppe Ciaramella**, Sudbury, MA
(US); **Sunny Himansu**, Winchester,
MA (US)

(73) Assignee: **ModernaTX, Inc.**, Cambridge, MA
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **19/000,472**

(22) Filed: **Dec. 23, 2024**

(65) **Prior Publication Data**

US 2025/0121221 A1 Apr. 17, 2025

Related U.S. Application Data

(63) Continuation of application No. 18/762,372, filed on
Jul. 2, 2024, now Pat. No. 12,208,288, and a
continuation of application No. 18/762,384, filed on
Jul. 2, 2024, and a continuation of application No.
18/762,296, filed on Jul. 2, 2024, said application No.
18/762,384 is a continuation of application No.
18/528,323, filed on Dec. 4, 2023, said application
No. 18/762,372 is a continuation of application No.
18/528,323, filed on Dec. 4, 2023, said application
No. 18/762,296 is a continuation of application No.
18/528,323, filed on Dec. 4, 2023, which is a
continuation of application No. 16/897,734, filed on
Jun. 10, 2020, now Pat. No. 11,872,278, which is a
continuation of application No. 16/368,270, filed on
Mar. 28, 2019, now Pat. No. 10,702,599, which is a
continuation of application No. 16/040,981, filed on
Jul. 20, 2018, now Pat. No. 10,272,150, which is a
continuation of application No. 15/674,599, filed on
Aug. 11, 2017, now Pat. No. 10,064,934, which is a
continuation of application No. PCT/US2016/058327,
filed on Oct. 21, 2016.

(60) Provisional application No. 62/247,362, filed on Oct.
28, 2015, provisional application No. 62/247,483,
filed on Oct. 28, 2015, provisional application No.
62/247,297, filed on Oct. 28, 2015, provisional
application No. 62/247,394, filed on Oct. 28, 2015,
provisional application No. 62/245,031, filed on Oct.
22, 2015, provisional application No. 62/244,837,
filed on Oct. 22, 2015, provisional application No.
62/244,813, filed on Oct. 22, 2015, provisional
application No. 62/244,802, filed on Oct. 22, 2015,
provisional application No. 62/244,946, filed on Oct.
22, 2015.

(51) **Int. Cl.**

A61P 31/14 (2006.01)

A61K 39/12 (2006.01)

A61K 39/155 (2006.01)

A61K 39/215 (2006.01)

A61P 11/00 (2006.01)

C07K 16/10 (2006.01)

A61K 39/00 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 39/12** (2013.01); **A61K 39/155**
(2013.01); **A61K 39/215** (2013.01); **A61P**
11/00 (2018.01); **A61P 31/14** (2018.01); **C07K**
16/10 (2013.01); **C07K 16/1002** (2023.08);
C07K 16/1027 (2013.01); **A61K 2039/53**
(2013.01); **A61K 2039/55511** (2013.01); **A61K**
2039/55555 (2013.01); **A61K 2039/6018**
(2013.01); **A61K 2039/70** (2013.01); **C07K**
2317/76 (2013.01); **C12N 2760/18034**
(2013.01); **C12N 2760/18334** (2013.01); **C12N**
2760/18434 (2013.01); **C12N 2760/18534**
(2013.01); **C12N 2760/18634** (2013.01); **C12N**
2770/20034 (2013.01); **Y02A 50/30** (2018.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

3,906,092 A	9/1975	Hilleman et al.
4,790,987 A	12/1988	Compans et al.
5,169,628 A	12/1992	Wathen
5,427,782 A	6/1995	Compans et al.
6,225,091 B1	5/2001	Klein et al.
6,500,419 B1	12/2002	Hone et al.
6,514,948 B1	2/2003	Raz et al.
7,001,890 B1	2/2006	Wagner et al.

(Continued)

FOREIGN PATENT DOCUMENTS

AU	2015210364 A1	3/2017
CA	2473135	6/2003

(Continued)

OTHER PUBLICATIONS

[No Author Listed] fusion glycoprotein F0 [Human respirovirus 3].
GenBank Accession No. AHX22069. First seen on NCBI on May
14, 2014. 2 pages.

(Continued)

Primary Examiner — Nicole Kinsey White

(74) *Attorney, Agent, or Firm* — Foley & Lardner LLP

(57)

ABSTRACT

The disclosure relates to ribonucleic acid (RNA) vaccines
and combination vaccines, as well as methods of using the
vaccines and compositions comprising the vaccines. Spe-
cific embodiments relate to RNA betacoronavirus vaccines
formulated in a lipid nanoparticle.

8 Claims, 24 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

U.S. PATENT DOCUMENTS

7,208,161	B1	4/2007	Murphy et al.	11,497,807	B2	11/2022	Ciaramella et al.
7,449,324	B2	11/2008	Fouchier et al.	11,564,893	B2	1/2023	Smith
7,531,342	B2	5/2009	Fouchier et al.	11,576,961	B2	2/2023	Ciaramella et al.
7,671,186	B2	3/2010	Klein et al.	11,638,693	B2	5/2023	Geall
7,704,720	B2	4/2010	Tang et al.	11,638,694	B2	5/2023	Geall
8,217,016	B2	7/2012	Hoerr et al.	11,643,441	B1	5/2023	Ciaramella et al.
8,252,289	B2	8/2012	Eleouët et al.	11,666,534	B2	6/2023	Geall
8,691,966	B2	4/2014	Kariko et al.	11,696,946	B2	7/2023	Ciaramella
8,710,200	B2	4/2014	Schrum et al.	11,752,206	B2	9/2023	Ciaramella et al.
8,722,341	B2	5/2014	Fouchier et al.	11,766,401	B2	9/2023	Geall
8,734,853	B2	5/2014	Sood et al.	11,786,467	B2	10/2023	Geall
8,754,062	B2	6/2014	De Fougerolles et al.	11,786,607	B2	10/2023	Hoge et al.
8,822,663	B2	9/2014	Schrum et al.	11,851,694	B1	12/2023	Mauger et al.
8,841,433	B2	9/2014	Fouchier et al.	11,866,696	B2	1/2024	Issa et al.
8,889,146	B2	11/2014	Blais et al.	11,872,278	B2	1/2024	Ciaramella et al.
8,927,206	B2	1/2015	De Jong et al.	11,905,525	B2	2/2024	Brito et al.
8,980,864	B2	3/2015	Hoge et al.	11,911,453	B2	2/2024	Ciaramella et al.
8,999,380	B2	4/2015	Bancel et al.	11,912,982	B2	2/2024	Issa et al.
9,192,651	B2	11/2015	Chakraborty et al.	2003/0092653	A1	5/2003	Kisich et al.
9,192,661	B2	11/2015	Jain et al.	2003/0232061	A1	12/2003	Fouchier et al.
9,221,891	B2	12/2015	Bancel et al.	2004/0005545	A1	1/2004	Fouchier et al.
9,271,996	B2	3/2016	De Fougerolles et al.	2004/0096451	A1	5/2004	Young et al.
9,283,287	B2	3/2016	Bancel et al.	2005/0032730	A1	2/2005	Von Der Mulbe et al.
9,295,689	B2	3/2016	De Fougerolles et al.	2005/0059624	A1	3/2005	Hoerr et al.
9,303,079	B2	4/2016	Bancel et al.	2005/0250723	A1	11/2005	Hoerr et al.
9,376,726	B2	6/2016	Fouchier et al.	2006/0002958	A1	1/2006	Naylor et al.
9,464,124	B2	10/2016	Bancel et al.	2006/0172003	A1	8/2006	Meers et al.
9,512,456	B2	12/2016	Wang et al.	2006/0228367	A1	10/2006	Ulbrandt et al.
9,533,047	B2	1/2017	de Fougerolles et al.	2007/0280929	A1	12/2007	Hoerr et al.
9,567,653	B2	2/2017	Fouchier et al.	2008/0025944	A1	1/2008	Hoerr et al.
9,572,896	B2	2/2017	Bancel et al.	2008/0171079	A1	7/2008	Hanon et al.
9,597,380	B2	3/2017	Chakraborty et al.	2008/0171711	A1	7/2008	Hoerr et al.
9,623,095	B2	4/2017	Kallen et al.	2009/0123529	A1	5/2009	Xiaomao
9,669,089	B2	6/2017	Thess et al.	2009/0162395	A1	6/2009	Crowe et al.
9,675,668	B2	6/2017	Bancel et al.	2010/0203076	A1	8/2010	Fotin-Mleczeck et al.
9,790,531	B2	10/2017	Wang et al.	2010/0239608	A1	9/2010	Von Der Mulbe et al.
9,803,199	B2	10/2017	Koizumi et al.	2010/0272747	A1	10/2010	Chow et al.
9,868,691	B2	1/2018	Benenato et al.	2010/0291156	A1	11/2010	Barner et al.
9,872,900	B2	1/2018	Ciaramella et al.	2010/0305196	A1	12/2010	Probst et al.
9,937,196	B2	4/2018	Jain et al.	2011/0135645	A1	6/2011	Williamson et al.
10,023,626	B2	7/2018	Bolen et al.	2011/0250225	A1	10/2011	Fotin-Mleczeck et al.
10,064,934	B2	9/2018	Ciaramella et al.	2011/0269950	A1	11/2011	Von Der Mulbe et al.
10,064,935	B2	9/2018	Ciaramella et al.	2012/0009221	A1	1/2012	Hoerr et al.
10,124,055	B2	11/2018	Ciaramella et al.	2012/0045471	A1	2/2012	Haller et al.
10,207,010	B2	2/2019	Besin et al.	2012/0101148	A1	4/2012	Aking et al.
10,232,055	B2	3/2019	Kariko et al.	2012/0177701	A1	7/2012	Ilyinskii et al.
10,272,150	B2	4/2019	Ciaramella et al.	2012/0213812	A1	8/2012	Lipford et al.
10,273,269	B2	4/2019	Ciaramella	2012/0219573	A1	8/2012	Baumhof et al.
10,286,086	B2	5/2019	Roy et al.	2012/0276209	A1	11/2012	Cullis et al.
10,323,076	B2	6/2019	Ellsworth et al.	2013/0022538	A1	1/2013	Rossi
10,385,088	B2	8/2019	Fraley et al.	2013/0078281	A1	3/2013	He et al.
10,449,244	B2	10/2019	Ciaramella et al.	2013/0102034	A1	4/2013	Schrum et al.
10,465,190	B1	11/2019	Chen et al.	2013/0121988	A1	5/2013	Hoerr et al.
10,493,143	B2	12/2019	Ciaramella et al.	2013/0142818	A1	6/2013	Baumhof et al.
10,526,629	B2	1/2020	Rabideau et al.	2013/0183355	A1	7/2013	Jain et al.
10,543,269	B2	1/2020	Ciaramella et al.	2013/0195867	A1	8/2013	Hoerr et al.
10,653,712	B2	5/2020	Hoge	2013/0195967	A1	8/2013	Guild et al.
10,653,767	B2	5/2020	Ciaramella et al.	2013/0195969	A1	8/2013	Geall et al.
10,695,419	B2	6/2020	Ciaramella et al.	2013/0202684	A1	8/2013	Geall et al.
10,702,599	B2	7/2020	Ciaramella et al.	2013/0236974	A1	9/2013	De Fougerolles
10,702,600	B1	7/2020	Ciaramella et al.	2013/0243848	A1	9/2013	Lobovkina et al.
10,857,105	B2	12/2020	Benenato et al.	2013/0245103	A1	9/2013	de Fougerolles et al.
10,906,944	B2	2/2021	He et al.	2013/0259923	A1	10/2013	Bancel et al.
10,912,826	B2	2/2021	Thess et al.	2013/0266640	A1	10/2013	De Fougerolles et al.
10,925,958	B2	2/2021	Ciaramella	2013/0295043	A1	11/2013	Kallen et al.
10,960,070	B2	3/2021	Graham et al.	2013/0336998	A1	12/2013	Kallen et al.
11,027,025	B2	6/2021	Hoge et al.	2014/0024076	A1	1/2014	Tang et al.
11,045,540	B2	6/2021	Ciaramella	2014/0037660	A1	2/2014	Folin-Mleczeck et al.
11,103,578	B2	8/2021	Ciaramella et al.	2014/0147432	A1	5/2014	Bancel et al.
11,266,735	B2	3/2022	Kallen et al.	2014/0148502	A1	5/2014	Bancel et al.
11,351,242	B1	6/2022	Lori et al.	2014/0193482	A1	7/2014	Bancel et al.
11,384,352	B2	7/2022	Miracco	2014/0206752	A1	7/2014	Afeyan et al.
11,406,703	B2	8/2022	Kramarczyk et al.	2014/0255472	A1	9/2014	Geall et al.
11,464,848	B2	10/2022	Ciaramella et al.	2014/0271829	A1	9/2014	Lilja et al.
11,485,960	B2	11/2022	Dousis et al.	2014/0308304	A1	10/2014	Manoharan et al.
				2014/0370497	A1	12/2014	Fouchier et al.
				2014/0378538	A1	12/2014	Bancel
				2015/0051268	A1	2/2015	Bancel et al.
				2015/0056253	A1	2/2015	Bancel et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2015/0093413	A1	4/2015	Thess et al.	2020/0038499	A1	2/2020	Narayanan et al.
2015/0126589	A1	5/2015	Geiger et al.	2020/0054737	A1	2/2020	Ciaramella et al.
2015/0141499	A1	5/2015	Bancel et al.	2020/0061185	A1	2/2020	Graham et al.
2015/0307542	A1	10/2015	Roy et al.	2020/0069599	A1	3/2020	Smith et al.
2015/0315541	A1	11/2015	Bancel et al.	2020/0069793	A1	3/2020	Ciaramella
2015/0335728	A1	11/2015	Wong et al.	2020/0069794	A1	3/2020	Ciaramella et al.
2016/0022580	A1	1/2016	Ramsay et al.	2020/0071689	A1	3/2020	Miracco
2016/0024140	A1	1/2016	Issa et al.	2020/0085916	A1	3/2020	Martini et al.
2016/0024141	A1	1/2016	Issa et al.	2020/0109420	A1	4/2020	Brito et al.
2016/0032273	A1	2/2016	Shahrokh et al.	2020/0129445	A1	4/2020	Patel et al.
2016/0038612	A1	2/2016	Hoge et al.	2020/0129608	A1	4/2020	Ciaramella et al.
2016/0039884	A1	2/2016	Li et al.	2020/0129615	A1	4/2020	Ciaramella et al.
2016/0046675	A1	2/2016	Kwong et al.	2020/0163878	A1	5/2020	Baumhof et al.
2016/0151474	A1	6/2016	Kallen et al.	2020/0239869	A1	7/2020	Issa et al.
2016/0194368	A1	7/2016	Hoge et al.	2020/0254086	A1	8/2020	Hoge et al.
2016/0194625	A1	7/2016	Hoge et al.	2020/0282046	A1	9/2020	Ciaramella et al.
2016/0243221	A1	8/2016	Hoge et al.	2020/0282047	A1	9/2020	Ciaramella et al.
2016/0271272	A1	9/2016	Bancel et al.	2020/0297634	A1	9/2020	Karmali et al.
2016/0317647	A1	11/2016	Ciaramella et al.	2020/0306191	A1	10/2020	Schariter et al.
2016/0331828	A1	11/2016	Ciaramella et al.	2020/0338004	A1	10/2020	Hansson et al.
2016/0361411	A1	12/2016	Gindy et al.	2020/0368162	A1	11/2020	Martini
2017/0043037	A1	2/2017	Kariko et al.	2020/0399322	A1	12/2020	Baumhof et al.
2017/0065675	A1	3/2017	Bancel et al.	2020/0407402	A1	12/2020	He et al.
2017/0130255	A1	5/2017	Wang et al.	2021/0023199	A1	1/2021	Kallen et al.
2017/0202979	A1	7/2017	Chakraborty et al.	2021/0030866	A1	2/2021	Kallen et al.
2017/0204152	A1	7/2017	Nelson et al.	2021/0046173	A1	2/2021	Ciaramella et al.
2017/0340724	A1	11/2017	Ciaramella et al.	2021/0060175	A1	3/2021	Baumhof et al.
2018/0000953	A1	1/2018	Almarsson et al.	2021/0087135	A1	3/2021	Benenato et al.
2018/0002393	A1	1/2018	Bancel et al.	2021/0128716	A1	5/2021	Probst et al.
2018/0008694	A1	1/2018	Ciaramella et al.	2021/0139543	A1	5/2021	He et al.
2018/0028645	A1	2/2018	Ciaramella et al.	2021/0145982	A1	5/2021	Hoge et al.
2018/0028664	A1	2/2018	Besin et al.	2021/0162037	A1	6/2021	Jasny et al.
2018/0237849	A1	8/2018	Thompson	2021/0163919	A1	6/2021	Issa et al.
2018/0243225	A1	8/2018	Ciaramella	2021/0187097	A1	6/2021	Ciaramella et al.
2018/0243230	A1	8/2018	Smith	2021/0206818	A1	7/2021	Huang et al.
2018/0256628	A1	9/2018	Hoge et al.	2021/0217484	A1	7/2021	Giessel et al.
2018/0271795	A1	9/2018	Martini et al.	2021/0228707	A1	7/2021	Mektar et al.
2018/0271970	A1	9/2018	Ciaramella et al.	2021/0228708	A1	7/2021	Smith et al.
2018/0273977	A1	9/2018	Mousavi et al.	2021/0268086	A1	9/2021	Zhong et al.
2018/0274009	A1	9/2018	Marquardt et al.	2021/0309976	A1	10/2021	Dousis et al.
2018/0280496	A1	10/2018	Ciaramella et al.	2021/0378980	A1	12/2021	Horhota et al.
2018/0289792	A1	10/2018	Ciaramella et al.	2022/0031631	A1	2/2022	Almarsson et al.
2018/0303929	A1	10/2018	Ciaramella et al.	2022/0047518	A1	2/2022	Hennessy et al.
2018/0311336	A1	11/2018	Ciaramella et al.	2022/0054653	A1	2/2022	Martini et al.
2018/0311343	A1	11/2018	Huang et al.	2022/0062175	A1	3/2022	Smith et al.
2018/0318409	A1	11/2018	Valiante et al.	2022/0125899	A1	4/2022	Ashburn et al.
2018/0326039	A1	11/2018	Haruta	2022/0145381	A1	5/2022	Elich et al.
2018/0363019	A1	12/2018	Hoge	2022/0236253	A1	7/2022	Hopson
2018/0369374	A1	12/2018	Frederick et al.	2022/0241399	A1	8/2022	Lusso et al.
2018/0371047	A1	12/2018	Ticho et al.	2022/0347292	A1	11/2022	Panther et al.
2019/0002890	A1	1/2019	Martini et al.	2022/0348900	A1	11/2022	Shamashkin et al.
2019/0008938	A1	1/2019	Ciaramella et al.	2022/0349006	A1	11/2022	Amato et al.
2019/0085368	A1	3/2019	Bancel et al.	2023/0000970	A1	1/2023	Nachbagauer et al.
2019/0099481	A1	4/2019	Ciaramella et al.	2023/0142529	A1	5/2023	White et al.
2019/0125839	A1	5/2019	Frederick et al.	2023/0173060	A1	6/2023	Benmohamed
2019/0175517	A1	6/2019	Martini et al.	2023/0181481	A1	6/2023	White et al.
2019/0175727	A1	6/2019	Huang et al.	2023/0190761	A1	6/2023	Brader et al.
2019/0192646	A1	6/2019	Cohen et al.	2023/0212645	A1	7/2023	Marquardt et al.
2019/0192653	A1	6/2019	Hoge et al.	2023/0287437	A1	9/2023	Smith et al.
2019/0240317	A1	8/2019	Ciaramella et al.	2023/0338506	A1	10/2023	Shaw et al.
2019/0241633	A1	8/2019	Fotin-Mleczeck et al.	2023/0338512	A1	10/2023	Muik et al.
2019/0275170	A1	9/2019	Benenato et al.	2023/0346914	A1	11/2023	Stewart-Jones et al.
2019/0298657	A1	10/2019	Martini et al.	2023/0355743	A1	11/2023	Stewart-Jones et al.
2019/0298658	A1	10/2019	Benenato	2024/0100145	A1	3/2024	Bollman et al.
2019/0300906	A1	10/2019	Martini et al.	2024/0100151	A1	3/2024	Carfi et al.
2019/0314292	A1	10/2019	Benenato et al.	2024/0139309	A1	5/2024	Carfi et al.
2019/0314493	A1	10/2019	Ciaramella et al.	2024/0173400	A1	5/2024	Ciaramella et al.
2019/0336452	A1	11/2019	Brader	2024/0181030	A1	6/2024	Himansu et al.
2019/0336595	A1	11/2019	Ciaramella	2025/0121220	A1	4/2025	Ciaramella et al.
2019/0351040	A1	11/2019	Valiante et al.				
2019/0351047	A1	11/2019	Jasny et al.				
2019/0382774	A1	12/2019	Hoge et al.				
2019/0390181	A1	12/2019	Benenato et al.				
2020/0030432	A1	1/2020	Ciaramella et al.				
2020/0032274	A1	1/2020	Mauger et al.				

FOREIGN PATENT DOCUMENTS

CA	3194325	A1	5/2022
CN	110951756	A	4/2020
CN	110974950	A	4/2020
CN	110974954	A	4/2020
CN	111088283	A	5/2020
CN	111218458	A	6/2020
CN	111218459	A	6/2020

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	111303254	A	6/2020	WO	WO 2012/019780	A1	2/2012
CN	111518175	A	8/2020	WO	WO 2012/030901	A1	3/2012
CN	111592602	A	8/2020	WO	WO 2012/031043	A1	3/2012
CN	111848753	A	10/2020	WO	WO 2012/031046	A2	3/2012
CN	111939250	A	11/2020	WO	WO 2012/045075	A1	4/2012
CN	111978377	A	11/2020	WO	WO 2012/045082	A2	4/2012
CN	112023035	A	12/2020	WO	WO-2012/068295	A1	5/2012
CN	112028976	A	12/2020	WO	WO 2012/116714	A1	9/2012
CN	112043825	A	12/2020	WO	WO 2012/116715	A1	9/2012
CN	112048005	A	12/2020	WO	WO 2012/116810	A1	9/2012
CN	112094327	A	12/2020	WO	WO 2012/116811	A1	9/2012
CN	112220920	A	1/2021	WO	WO 2013/052523	A1	4/2013
CN	112226445	A	1/2021	WO	WO 2013/055905		4/2013
CN	112266411	A	1/2021		A1.		
CN	112321688	A	2/2021	WO	WO 2013/090186	A1	6/2013
CN	112358533	A	2/2021	WO	WO 2013/090648	A1	6/2013
CN	112480217	A	3/2021	WO	WO 2013/102203	A1	7/2013
CN	112546211	A	3/2021	WO	WO 2013/120628	A1	8/2013
CN	112626089	A	4/2021	WO	WO 2013/120629	A1	8/2013
CN	112794884	A	5/2021	WO	WO 2013/185069	A1	12/2013
EP	1026253		8/2000	WO	WO 2014/089239	A1	6/2014
EP	1083232		2/2005	WO	WO 2014/089486	A1	6/2014
EP	1905844	A2	2/2008	WO	WO 2014/144196	A1	9/2014
EP	2548960	A1	1/2013	WO	WO 2014/144711	A1	9/2014
EP	3608308	A	2/2020	WO	WO 2014/152027	A1	9/2014
JP	2015-518472	A	7/2015	WO	WO 2014/152030	A1	9/2014
JP	2016-515815	A	6/2016	WO	WO 2014/152200	A1	9/2014
WO	WO 1987/005326	A1	9/1987	WO	WO 2014/152774	A1	9/2014
WO	WO 1990/011092		10/1990	WO	WO 2014/152940	A1	9/2014
WO	WO 1993/014778		8/1993	WO	WO 2014/159813	A1	10/2014
WO	WO 1995/024485		9/1995	WO	WO 2014/160243	A1	10/2014
WO	WO 1995/026204		10/1995	WO	WO 2015/005253	A1	1/2015
WO	WO 1995/033835		12/1995	WO	WO 2015/024668	A2	2/2015
WO	WO 1996/040945	A2	12/1996	WO	WO-2015/081155	A1	6/2015
WO	WO 1998/058956		12/1998	WO	WO 2015/085318	A2	6/2015
WO	WO 1999/033982		7/1999	WO	WO 2015/095340	A1	6/2015
WO	WO 1999/33982		7/1999	WO	WO 2015/095346	A1	6/2015
WO	WO 2003/072720	A2	9/2003	WO	WO 2015/101414	A2	7/2015
WO	WO 2004/076645	A1	9/2004	WO	WO 2015/101415	A1	7/2015
WO	WO 2005/009346		2/2005	WO	WO 2015/130584	A2	9/2015
WO	WO 2005/027963	A2	3/2005	WO	WO 2015/143335	A1	9/2015
WO	WO 2005/118813	A2	12/2005	WO	WO 2016/103238		6/2016
WO	WO 2006/056027	A1	6/2006	WO	WO 2016/123864	A1	8/2016
WO	WO 2006/071903		7/2006	WO	WO 2016/138160	A1	9/2016
WO	WO 2006/095259		9/2006	WO	WO 2016/164762	A1	10/2016
WO	WO 2007/024708	A2	3/2007	WO	WO 2016/201377	A1	12/2016
WO	WO 2007/038862	A1	4/2007	WO	WO 2017/011773	A2	1/2017
WO	WO 2007/094854	A2	8/2007	WO	WO 2017/015457	A1	1/2017
WO	WO 2007/095976	A2	8/2007	WO	WO 2017/015463	A1	1/2017
WO	WO 2008/052770	A2	5/2008	WO	WO 2017/019935	A1	2/2017
WO	WO 2009/030254	A1	3/2009	WO	WO 2017/020026	A1	2/2017
WO	WO 2009/030481	A1	3/2009	WO	WO 2017/062513	A1	4/2017
WO	WO 2009/095226		8/2009	WO	WO 2017/066789	A1	4/2017
WO	WO 2009/127230	A1	10/2009	WO	WO 2017/070601	A1	4/2017
WO	WO 2010/037408	A1	4/2010	WO	WO 2017/070616	A1	4/2017
WO	WO 2010/037539	A1	4/2010	WO	WO 2017/070618	A1	4/2017
WO	WO 2010/042877	A1	4/2010	WO	WO 2017/070620	A1	4/2017
WO	WO 2010/054401	A1	5/2010	WO	WO 2017/070622	A1	4/2017
WO	WO 2010/054406	A1	5/2010	WO	WO 2017/070623	A1	4/2017
WO	WO 2010/088927	A1	8/2010	WO	WO 2017/070626	A1	4/2017
WO	WO 2010/129709	A1	11/2010	WO	WO 2017/075038	A1	5/2017
WO	WO 2010/144740	A1	12/2010	WO	WO 2017/075531	A1	5/2017
WO	WO 2010/149743	A2	12/2010	WO	WO 2017/081082	A1	5/2017
WO	WO 2011/005799	A2	1/2011	WO	WO 2017/112865	A1	6/2017
WO	WO 2011/026641	A9	3/2011	WO	WO 2017/127750	A1	7/2017
WO	WO 2011/068810	A1	6/2011	WO	WO 2017/174564	A1	10/2017
WO	WO 2011/069529	A1	6/2011	WO	WO 2017/201333	A1	11/2017
WO	WO 2011/069586	A1	6/2011	WO	WO 2017/201340	A1	11/2017
WO	WO 2011/140627	A1	11/2011	WO	WO 2017/201342	A1	11/2017
WO	WO 2011/144358	A1	11/2011	WO	WO 2017/201347	A1	11/2017
WO	WO 2012/006369	A2	1/2012	WO	WO 2017/201349	A1	11/2017
WO	WO 2012/006372	A1	1/2012	WO	WO 2018/006052	A1	1/2018
WO	WO 2012/006377	A2	1/2012	WO	WO 2018/053209	A1	3/2018
WO	WO-2012/006378	A1	1/2012	WO	WO 2018/075980	A1	4/2018
WO	WO 2012/019630	A1	2/2012	WO	WO 2018/081459	A1	5/2018
				WO	WO 2018/081462	A1	5/2018
				WO	WO 2018/089851	A1	5/2018
				WO	WO 2018/107088	A1	6/2018
				WO	WO 2018/111967	A1	6/2018

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO WO 2018/144082 A1 8/2018
 WO WO 2018/144778 A1 8/2018
 WO WO 2018/151816 A1 8/2018
 WO WO 2018/157009 A1 8/2018
 WO WO 2018/170245 A1 9/2018
 WO WO 2018/170256 A1 9/2018
 WO WO 2018/170260 A1 9/2018
 WO WO 2018/170270 A1 9/2018
 WO WO 2018/170347 A1 9/2018
 WO WO 2018/175783 A1 9/2018
 WO WO 2018/187590 A2 10/2018
 WO WO 2018/200737 A1 11/2018
 WO WO 2018/232355 A1 12/2018
 WO WO 2018/232357 A1 12/2018
 WO WO 2019/036670 A1 2/2019
 WO WO 2019/036682 A1 2/2019
 WO WO 2019/036683 A1 2/2019
 WO WO 2019/036685 A1 2/2019
 WO WO 2019/103993 A1 5/2019
 WO WO 2019/148101 A1 8/2019
 WO WO 2019/202035 A1 10/2019
 WO WO 2020/006242 A1 1/2020
 WO WO 2020/056370 A1 3/2020
 WO WO 2020/061284 A1 3/2020
 WO WO 2020/061295 A1 3/2020
 WO WO 2020/061367 A1 3/2020
 WO WO 2020/097291 A1 5/2020
 WO WO 2020/172239 A1 8/2020
 WO WO 2020/185811 A1 9/2020
 WO WO 2020/190750 A1 9/2020
 WO WO 2020/243561 A1 12/2020
 WO WO 2021/030533 A1 2/2021
 WO WO 2021/050864 A1 3/2021
 WO WO 2021/055811 A1 3/2021
 WO WO 2021/155243 A1 8/2021
 WO WO 2021/155274 A1 8/2021
 WO WO 2021/159040 A2 8/2021
 WO WO 2021/159130 A2 8/2021
 WO WO 2021/160346 A1 8/2021
 WO WO 2021/163365 A1 8/2021
 WO WO 2021/204175 A1 10/2021
 WO WO 2021/211343 A1 10/2021
 WO WO 2021/213945 A1 10/2021
 WO WO 2021/222304 A1 11/2021
 WO WO 2021/226436 A1 11/2021
 WO WO 2021/231901 A1 11/2021
 WO WO 2021/231929 A1 11/2021
 WO WO 2021/231963 A1 11/2021
 WO WO 2021/237084 A1 11/2021
 WO WO 2021/247817 A1 12/2021
 WO WO 2022/032154 A2 2/2022
 WO WO 2022/043551 A2 3/2022
 WO WO 2022/067010 A1 3/2022
 WO WO 2022/076562 A1 4/2022
 WO WO 2022/099003 A1 5/2022
 WO WO 2022/101469 A1 5/2022
 WO WO 2022/104265 A1 5/2022
 WO WO 2022/137133 A1 6/2022
 WO WO 2022/150717 A1 7/2022
 WO WO 2022/155524 A1 7/2022
 WO WO 2022/155530 A1 7/2022
 WO WO 2022/187698 A1 9/2022
 WO WO 2022/197624 A1 9/2022
 WO WO 2022/204491 A1 9/2022
 WO WO 2022/212191 A1 10/2022
 WO WO 2022/212442 A1 10/2022
 WO WO 2022/212711 A2 10/2022
 WO WO 2022/221335 A1 10/2022
 WO WO 2022/221336 A1 10/2022
 WO WO 2022/221359 A1 10/2022
 WO WO 2022/221440 A1 10/2022
 WO WO 2022/226277 A1 10/2022
 WO WO 2022/226318 A1 10/2022
 WO WO 2022/232585 A1 11/2022
 WO WO 2022/241103 A1 11/2022

WO WO 2022/245888 A1 11/2022
 WO WO 2022/266010 A1 12/2022
 WO WO 2022/266012 A1 12/2022
 WO WO 2022/266389 A1 12/2022
 WO WO 2023/283642 A2 1/2023
 WO WO 2023/283645 A1 1/2023
 WO WO 2023/283651 A1 1/2023
 WO WO 2023/014649 A1 2/2023
 WO WO 2023/018773 A1 2/2023
 WO WO 2023/018923 A1 2/2023
 WO WO 2023/019181 A1 2/2023
 WO WO 2023/056401 A1 4/2023
 WO WO 2023/069625 A1 4/2023
 WO WO 2023/069895 A1 4/2023
 WO WO 2023/069900 A1 4/2023
 WO WO 2023/076658 A1 5/2023
 WO WO 2023/081311 A1 5/2023
 WO WO 2023/092069 A1 5/2023
 WO WO 2023/094713 A2 6/2023
 WO WO 2023/107999 A2 6/2023
 WO WO 2023/114307 A1 6/2023
 WO WO 2023/132885 A1 7/2023
 WO WO 2023/135333 A1 7/2023
 WO WO 2023/137149 A1 7/2023
 WO WO 2023/147091 A1 8/2023
 WO WO 2023/150256 A1 8/2023
 WO WO 2023/154818 A1 8/2023
 WO WO 2023/196914 A1 10/2023
 WO WO 2023/201204 A1 10/2023
 WO WO 2023/201294 A1 10/2023
 WO WO 2023/201296 A1 10/2023
 WO WO 2023/212696 A1 11/2023
 WO WO 2023/225524 A1 11/2023
 WO WO 2023/250119 A1 12/2023
 WO WO 2024/010993 A1 1/2024
 WO WO 2024/015890 A1 1/2024
 WO WO 2024/026005 A1 2/2024
 WO WO 2024/030369 A1 2/2024
 WO WO 2024/050483 A1 3/2024
 WO WO 2024/097874 A1 5/2024

OTHER PUBLICATIONS

[No Author Listed] Lineage list. Last accessed Oct. 17, 2023 at https://cov-lineages.org/lineage_list.html. 77 pages.
 [No Author Listed], "Messenger RNA" definition. Collins Dictionary of Science. 2005:514.
 [No Author Listed], "Messenger RNA" definition. The Penguin Dictionary of Science. 2009: 3 pages.
 [No Author Listed], "Messenger RNA" definition. Webster's New World Medical Dictionary. 2008;3:271.
 [No Author Listed], 1st International mRNA Health Conference Programme. Universitat Tübingen. Oct. 23-24, 2013: 32 pages.
 [No Author Listed], 2022 Breakthrough Prize in Life Sciences Awarded to Penn Medicine mRNA Pioneers Drew Weissman and Katalin Kariko. Press Release. Sep. 9, 2021: 3 pages.
 [No Author Listed], 3rd International mRNA Health Conference Program. mRNA Conference. Nov. 11, 2015: 28 pages.
 [No Author Listed], A Phase 3, Randomized, Stratified, Observer-Blind, Placebo-Controlled Study to Evaluate the Efficacy, Safety, and Immunogenicity of mRNA-1273 SARS-CoV-2 Vaccine in Adults Aged 18 Years and Older. ModernaTX, Inc. Protocol mRNA-1273-P301, Amendment 3. Aug. 20, 2020. 135 pages.
 [No Author Listed], An Audience with Moncef Slaoui. Nat Rev Drug Discov Jul. 1, 2015;14:452-453. <https://doi.org/10.1038/nrd4669>.
 [No Author Listed], Background document on the mRNA-1273 vaccine (Moderna) against COVID-19. World Health Organization. Feb. 3, 2021. Accessed from <[https://www.who.int/publications/i/item/background-document-on-the-mrna-1273-vaccine-\(moderna\)-against-covid-19](https://www.who.int/publications/i/item/background-document-on-the-mrna-1273-vaccine-(moderna)-against-covid-19)>. 41 pages.
 [No Author Listed], BioNTech Expands Clinical Oncology Portfolio with First Patient Dosed in Phase 2 Trial of mRNA-based Individualized Immunotherapy BNT122 in Colorectal Cancer Patients. Press Release. BioNTech. Oct. 1, 2021: 2 pages.

(56)

References Cited**OTHER PUBLICATIONS**

[No Author Listed], BioNTech Fourth Quarter and Full-Year 2019 Results. BioNTech. Mar. 31, 2020: 15 pages.

[No Author Listed], BioNTech SE Annual Report (Form 20-F) 179, F-12. United States Securities and Exchange Commission. Mar. 30, 2021: 739 pages.

[No Author Listed], BioNTech SE Annual Report (Form 20-F) 81. United States Securities and Exchange Commission. Mar. 30, 2022: 700 pages.

[No Author Listed], BioNTech Signs Collaboration Agreement with Pfizer to Develop mRNA-based Vaccines for Prevention of Influenza. Press Release. Aug. 16, 2018: 3 pages.

[No Author Listed], BNT162 COVID-19 Vaccine Program Update. BioNTech. Apr. 23, 2020: 15 pages.

[No Author Listed], Canada Exercises Increased Option for 20 Million Doses of mRNA Vaccine Against COVID-19 (mRNA-1273). Moderna Press Release. Sep. 22, 2020. 2 pages.

[No Author Listed], CDC. SARS-CoV-2 Variant Classifications and Definitions. <https://www.cdc.gov/coronavirus/2019-ncov/variants/variant-classifications.html>. Updated Sep. 1, 2023. 5 pages.

[No Author Listed], Center for Drug Evaluation and Research Approval Package for New Drug Application No. 50-718/S-50 for Doxil®. Approved Dec. 28, 2015. 97 pages.

[No Author Listed], Clinical trial NCT04528719: A Dose Escalation Study to Evaluate Safety, Reactogenicity, and Immunogenicity of mRNA-1345 in Healthy Adults and in Children Who Are Respiratory Syncytial Virus Seropositive (ModernaTX, Inc.) First Posted Aug. 27, 2020. Retrieved online on Mar. 15, 2021 at <https://www.clinicaltrials.gov/ct2/show/NCT04528719?term=NCT04528719&draw=2&rank=1>.

[No Author Listed], Comirnaty (mRNA-1273) EMA SmPC (summary of product characteristics). Published Dec. 1, 2021. 276 pages.

[No Author Listed], COMIRNATY Dosing Schedule for People 12 Years of Age & Older (2023-2024 Formula). Oct. 2023: 2 pages.

[No Author Listed], COMIRNATY Prescribing Information. Jul. 2022: 27 pages.

[No Author Listed], COMIRNATY Prescribing Information. COMIRNATY. Oct. 2023: 36 pages.

[No Author Listed], Coronavirus (COVID-19) Update: FDA Authorizes Changes to Simplify Use of Bivalent mRNA COVID-19 Vaccines. FDA News Release. Apr. 18, 2023: 3 pages.

[No Author Listed], Coronavirus (COVID-19) Update: FDA Issues Policies to Guide Medical Product Developers Addressing Virus Variants. FDA News Release. Feb. 22, 2021. 5 pages.

[No Author Listed], Could COVID-19 vaccines be tweaked to cover new coronavirus variants? I Gavi, the Vaccine Alliance. Published Jan. 14, 2021. URL: <https://www.gavi.org/vaccineswork/could-covid-19-vaccines-be-tweaked-cover-new-coronavirus-variants>: Last accessed Apr. 26, 2022. 5 Pages.

[No Author Listed], COVID-19 Vaccine Development Program. Pfizer. Jul. 1, 2020: 18 pages.

[No Author Listed], DARPA Awards Moderna Therapeutics a Grant for up to \$25 Million to Develop Messenger RNA Therapeutics. News Release. Moderna. Oct. 2, 2013: 1 page.

[No Author Listed], Delivery means to avoid degradation of RNA preparations: Lipid nanoparticles (LNPs) for RNA delivery. CosmoBio. English-Language Machine Translation. Retrieved from the Internet: https://www.cosmobio.co.jp/product/detail/lipid-nanoparticles-for-rna-delivery-ecl.asp?entry_id=43081. Last Accessed: Oct. 23, 2023. 10 pages.

[No Author Listed], Dosing Schedule for Ages 12 Years of Age and Older with Pfizer-BioNTech COVID-19 Vaccine, Bivalent (Original and Omicron BA.4/BA.5). Apr. 2023: 7-9.

[No Author Listed], European Commission Authorizes COVID 19 Vaccine Moderna in Europe. Jan. 6, 2021. Accessed from <https://www.drugs.com/clinical_trials/european-commission-authorizes-covid-19-vaccine-moderna-europe-19152.html>. 4 pages.

[No Author Listed], Experimental COVID-19 Vaccine Protects Upper and Lower Airways in Nonhuman Primates. NIH News Release. Jul. 28, 2020. 4 pages.

[No Author Listed], Experimental COVID-19 Vaccine Safe, Generates Immune Response. NIH News Release. Jul. 14, 2020. 4 pages.

[No Author Listed], Fact Sheet for Healthcare Providers Administering Vaccine: Emergency Use Authorization of Pfizer-BioNTech COVID-19 Vaccine (2023-2024 Formula) for 6 Months through 11 Years of Age. Pfizer-BioNTech. Sep. 2023: 59 pages.

[No Author Listed], Goldman Sachs Virtual 41st Annual Global Healthcare Conference. Pfizer Inc. Jun. 9, 2020: 13 pages.

[No Author Listed], History of Changes for Study: NCT04283461, Safety and Immunogenicity Study of 2019-nCoV Vaccine (mRNA-1273) for Prophylaxis of SARS-CoV-2 Infection (COVID-19). Version submitted Dec. 8, 2022. 8 pages.

[No Author Listed], History of Changes for Study: NCT04283461, Safety and Immunogenicity Study of 2019-nCoV Vaccine (mRNA-1273) for Prophylaxis of SARS-CoV-2 Infection (COVID-19). Version submitted Mar. 11, 2021. 10 pages.

[No Author Listed], History of Changes for Study: NCT04405076, Dose-Confirmation Study to Evaluate the Safety, Reactogenicity, and Immunogenicity of mRNA-1273 COVID-19 Vaccine in Adults Aged 18 Years and Older. Version submitted Dec. 6, 2022. 6 pages.

[No Author Listed], Increase in Manufacturing Capacity for COVID-19 Vaccines from Janssen, Moderna, and BioNTech/Pfizer. European Medicines Agency. Dec. 16, 2021: 4 pages.

[No Author Listed], Influenza A virus A/Jiangxi/IPB13/2013(H10N8) hemagglutinin protein. UniProtKB Accession: A0A059T4A1. Sep. 3, 2014. 1 page.

[No Author Listed], Katalin Kariko and Drew Weissman Awarded Horwitz Prize for Pioneering Research on COVID-19 Vaccines. Columbia University. Aug. 16, 2021: 12 pages.

[No Author Listed], Katalin Kariko Laureate de la Grande Medaille 2021 de L'Academie des Sciences. Press Release. De L'Academie des Sciences. Sep. 28, 2021: 2 pages.

[No Author Listed], Messenger RNA Pioneer Katalin Kariko Receives European Inventor Award 2022 for Lifetime Achievement. Press Release. European Patent Office. Jun. 21, 2022: 4 pages.

[No Author Listed], Moderna Advances Late-Stage Development of its Vaccine (mRNA-1273) Against COVID-19. Moderna Press Release. Jun. 11, 2020. 3 pages.

[No Author Listed], Moderna and Lonza Announce Worldwide Strategic Collaboration to Manufacture Moderna's Vaccine (mRNA-1273) Against Novel Coronavirus. Moderna Press Release. May 1, 2020. 3 pages.

[No Author Listed], Moderna Announces Award from U.S. Government Agency BARDA for up to \$483 Million to Accelerate Development of mRNA Vaccine (mRNA-1273) Against Novel Coronavirus. Moderna Press Release. Apr. 16, 2020. 3 pages.

[No Author Listed], Moderna Announces Expansion of BARDA Agreement to Support Larger Phase 3 Program for Vaccine (mRNA-1273) Against COVID-19. Moderna Press Release. Jul. 26, 2020. 3 pages.

[No Author Listed], Moderna Announces FDA Authorization of Moderna COVID 19 Vaccine in U.S. Press Release. Dec. 19, 2020. 3 pages.

[No Author Listed], Moderna Announces First Participant Dosed in NIH-led Phase 1 Study of mRNA Vaccine (mRNA-1273) Against Novel Coronavirus. Moderna Press Release. Mar. 16, 2020. 2 pages.

[No Author Listed], Moderna Announces First Participant Dosed in Phase 2 Study of Omicron-Specific Booster Candidate and Publication of Data on Booster Durability Against Omicron Variant. News Release. Moderna. Jan. 26, 2022: 4 pages.

[No Author Listed], Moderna Announces First Participant Dosed in Phase 2 Study of Omicron-Specific Booster Candidate. News Release. Mar. 10, 2022: 3 pages.

[No Author Listed], Moderna Announces First Participants in Each Age Cohort Dosed in Phase 2 Study of mRNA Vaccine (mRNA-1273) Against Novel Coronavirus. Moderna Press Release. May 29, 2020. 2 pages.

(56)

References Cited**OTHER PUBLICATIONS**

[No Author Listed], Moderna Announces Funding Award from CEPI to Accelerate Development of Messenger RNA (mRNA) Vaccine Against Novel Coronavirus. News Release, Moderna. Jan. 23, 2020: 2 pages.

[No Author Listed], Moderna Announces IND Submitted to U.S. FDA for Phase 2 Study of mRNA Vaccine (mRNA-1273) Against Novel Coronavirus. Moderna Press Release. Apr. 27, 2020. 3 pages.

[No Author Listed], Moderna Announces it has Shipped Variant-Specific Vaccine Candidate, mRNA-1273.351, to NIH for Clinical Study. Moderna Press Release. Feb. 24, 2021. 3 pages.

[No Author Listed], Moderna Announces Phase 3 COVE Study of mRNA Vaccine Against COVID-19 (mRNA-1273) Begins. Moderna Press Release. Jul. 27, 2020. 3 pages.

[No Author Listed], Moderna Announces Positive Interim Phase 1 Data for its mRNA Vaccine (mRNA-1273) Against Novel Coronavirus. Moderna Press Release. May 18, 2020. 3 pages.

[No Author Listed], Moderna Announces Progress Across Broad Portfolio and all Three Clinical Stage Therapeutic Areas at 2020 R&D Day. Moderna Press Release. Sep. 17, 2020. 4 pages.

[No Author Listed], Moderna Announces Publication in The New England Journal of Medicine of Interim Results From Phase 1 Study of Its mRNA Vaccine Against COVID-19 (mRNA-1273). Moderna Press Release. Jul. 14, 2020. 5 pages.

[No Author Listed], Moderna Announces Publication in The New England Journal of Medicine of Non-Human Primate Preclinical Viral Challenge Study of its mRNA Vaccine Against COVID-19 (mRNA-1273). Moderna Press Release. Jul. 28, 2020. 3 pages.

[No Author Listed], Moderna Announces Strategy to Address Omicron (B.1.1.529) SARS-CoV-2 Variant. Press Release. Moderna. Nov. 26, 2021: 2 pages.

[No Author Listed], Moderna Announces Supply Agreement with U.S. Government for Initial 100 Million Doses of mRNA Vaccine Against COVID-19 (mRNA-1273). Moderna Press Release. Aug. 11, 2020. 3 pages.

[No Author Listed], Moderna Completes Enrollment of Phase 2 Study of its mRNA Vaccine Against COVID-19 (mRNA-1273). Moderna Press Release. Jul. 8, 2020. 2 pages.

[No Author Listed], Moderna Confirms Advanced Discussions with European Commission to Supply Europe with 80 Million Doses of mRNA Vaccine Against COVID-19 (mRNA-1273). Moderna Press Release. Aug. 24, 2020. 3 pages.

[No Author Listed], Moderna Confirms Discussions with the Ministry of Health, Labour and Welfare to Supply Japan with 40 Million Doses of mRNA Vaccine Against COVID-19 (mRNA-1273). Moderna Press Release. Aug. 28, 2020. 2 pages.

[No Author Listed], Moderna has no plans to share its COVID-19 vaccine recipe. Associated Press. Oct. 11, 2021: 8 pages.

[No Author Listed], Moderna Inc to Discuss Updates on Respiratory Syncytial Virus (RSV) Vaccine Program Call. Edited Transcript. Refinitiv. Oct. 8, 2020: 14 pages.

[No Author Listed], Moderna Patents. <https://www.modernatx.com/en-US/patents> [last accessed Jul. 1, 2024].

[No Author Listed], Moderna Provides Business Update and Announces Three New Development Programs in Infectious Disease Vaccines. Press Release. Jan. 11, 2021. 5 pages.

[No Author Listed], Moderna Receives FDA Authorization for Emergency Use of Omicron-Targeting Bivalent COVID-19 Booster Vaccine for Adults 18 Years and Older. News Release. Moderna. Aug. 31, 2022: 5 pages.

[No Author Listed], Moderna Receives FDA Fast Track Designation for mRNA Vaccine (mRNA-1273) Against Novel Coronavirus. Moderna Press Release. May 12, 2020. 2 pages.

[No Author Listed], Moderna Reports Fourth Quarter and Fiscal Year 2020 Financial Results and Provides Business Updates. Press Release. Feb. 25, 2021. 9 pages.

[No Author Listed], Moderna Ships mRNA Vaccine Against Novel Coronavirus (mRNA-1273) for Phase 1 Study. Moderna Press Release. Feb. 24, 2020. 2 pages.

[No Author Listed], Moderna's COVID-19 Vaccine Candidate Meets its Primary Efficacy Endpoint in the First Interim Analysis of the Phase 3 COVE Study. Press Release. Nov. 16, 2020. 3 pages.

[No Author Listed], Moderna's Update Patent Pledge. Press Release. Moderna. Mar. 7, 2022: 2 pages.

[No Author Listed], Modified mRNA Vaccines. 2021 Lasker-DeBakey Clinical Medical Research Award. 17 pages.

[No Author Listed], Nih Clinical Trial of Investigational Vaccine for COVID-19 begins. NIH News Release. Mar. 16, 2020. 3 pages.

[No Author Listed], NIH-Moderna Investigational COVID-19 Vaccine Shows Promise in Mouse Studies. NIH News Release. Aug. 5, 2020. 4 pages.

[No Author Listed], Penn mRNA Scientists Drew Weissman and Katalin Kariko Receive 2021 Lasker Award, America's Top Biomedical Research Prize. Penn News Release. Sep. 24, 2021: 6 pages.

[No Author Listed], Pfizer and BioNTech Announce Early Positive Data from an Ongoing Phase 1/2 Study of mRNA-based Vaccine Candidate against SARS-CoV-2. Press Release. Jul. 1, 2020: 8 pages.

[No Author Listed], Pfizer and BioNTech Announce Early Positive Update from German Phase 1/2 Study of COVID-19 Vaccine Study, Including First T Cell Response Data. Press Release. Jul. 20, 2020: 6 pages.

[No Author Listed], Pfizer and BioNTech Announce Phase 3 Trial Data Showing High Efficacy of a Booster Dose of Their COVID-19 Vaccine. Press Release. Pfizer. Oct. 21, 2021: 10 pages.

[No Author Listed], Pfizer and BioNTech Choose Lead mRNA Vaccine Candidate Against COVID-19 and Commence Pivotal Phase 2/3 Global Study. Press Release. Jul. 27, 2020: 7 pages.

[No Author Listed], Pfizer and BioNTech Granted FDA Emergency Use Authorization of Omicron BA.4/BA.5-Adapted Bivalent COVID-19 Vaccine Booster for Ages 12 Years and Older. Pfizer. Aug. 31, 2022: 12 pages.

[No Author Listed], Pfizer and BioNTech Receive Expanded U.S. Emergency Use Authorization for an Additional COVID-19 Vaccine Booster in Individuals Aged 50 Years and Older. Press Release. Pfizer. Mar. 29, 2022: 12 pages.

[No Author Listed], Pfizer and BioNTech Submit Applications to U.S. FDA for Omicron XBB.1.5-Adapted Monovalent COVID-19 Vaccine. Pfizer. Jun. 23, 2023: 8 pages.

[No Author Listed], Pfizer and BioNTech Submit Supplemental Biologics License Application for U.S. FDA Approval of Omicron BA.4/BA.5-Adapted Bivalent COVID-19 Vaccine for Ages 12 Years and Older as Primary Series or Booster. Pfizer. Feb. 24, 2023: 12 pages.

[No Author Listed], Pfizer Inc to discuss data from an ongoing phase 1/2 study of mRNA-based vaccine candidate against SARS-CoV-2 Call. Edited Transcript. Thomson Reuters. Jul. 1, 2020: 15 pages.

[No Author Listed], Pfizer reports fourth-quarter and full-year 2021 results. Pfizer. Feb. 8, 2022: 41 pages.

[No Author Listed], Pfizer-BioNTech COVID-19 Vaccine Information Sheet. 3 pages.

[No Author Listed], Phase 3 Clinical Trial of Investigational Vaccine for COVID-19 Begins. NIH News Release. Jul. 27, 2020. 3 pages.

[No Author Listed], Pioneering Scientists Awarded the Albany Prize, America's Most Distinguished Prize in Medicine, for Research Leading to Covid-19 Vaccines. The Albany Prize. Aug. 26, 2021: 4 pages.

[No Author Listed], Press release, Moderna, "Moderna Announces Publication in The New England Journal of Medicine of Interim Results From Older Adult Age Cohorts in Phase 1 Study of its mRNA Vaccine Against COVID-19 (mRNA-1273)", Sep. 29, 2020. 7 pages.

[No Author Listed], Programme of the 1st International mRNA Health Conference held in Germany, Oct. 2013, 32 pages.

[No Author Listed], Q2 2020 Earnings Call. BioNTech SE. Aug. 11, 2020: 26 pages.

[No Author Listed], RBC capital Markets Global Healthcare Conference. Pfizer Inc. May 19, 2020: 9 pages.

(56)

References Cited

OTHER PUBLICATIONS

[No Author Listed], Remarks by President Trump and Members of the Coronavirus Task Force in Meeting with Pharmaceutical Companies. The White House. Mar. 2, 2022: 31 pages.

[No Author Listed], SARS-CoV-2 variants of concern as of May 4, 2023. European Centre for Disease Prevention and Control. Last accessed May 16, 2023 at <https://www.ecdc.europa.eu/en/covid-19/variants-concern>. 7 pages.

[No Author Listed], Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1, complete genome. GenBank Accession No. 045512, published on Jul. 18, 2020. 15 pages.

[No Author Listed], Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1, complete genome. Genbank Accession No. MN908947.3. Jan. 13, 2020. 15 pages.

[No Author Listed], spike glycoprotein [Middle East respiratory syndrome coronavirus]. GenBank Accession No. AGN52936.1. Published Jun. 10, 2013. 2 pages.

[No Author Listed], SPIKE_CVHSA. UniProt KB Accession No. P59594. Published Oct. 14, 2015. 1 page.

[No Author Listed], Spikevax (mRNA-1273) EMA SmPC (summary of product characteristics). Published Jan. 20, 2021. 137 pages.

[No Author Listed], Statement by Moderna on Intellectual Property Matters during the COVID-19 Pandemic. Press Release. Moderna. Oct. 8, 2020: 1 page.

[No Author Listed], Statement from NIH and BARDA on the FDA Emergency Use Authorization of the Moderna COVID-19 Vaccines. COVID.gov. Dec. 18, 2020: 3 pages.

[No Author Listed], Two Pioneering RNA Scientists Win the 2021 Dr. Paul Janssen Award for Biomedical Research. Press Release. Johnson & Johnson. Sep. 28, 2021: 2 pages.

[No Author Listed], Vaccine information fact sheet for recipients and caregivers about COMIRNATY (COVID-19 Vaccine, mRNA) and the Pfizer-BioNTech COVID-19 vaccine to prevent coronavirus disease 2019 (COVID-19) for use in individuals 12 years of age and older. Jul. 8, 2022: 9 pages.

[No Author Listed], Who Drug Information, vol. 35, No. 2, 2021, 341 pages.

[No Author Listed], Who Drug Information. 2021;35(2):578-9.

[No Author Listed], World Health Organization. 2021. Classification of Omicron(B.1.1.529): SARS-CoV-2 Variant of Concern. Nov. 26, 2021. 3 pages.

[No Author Listed], Reflection paper on the regulatory requirements for vaccines intended to provide protection against variant strain(s) of SARS-CoV-2. European Medicines Agency. EMA/117973/2021. Feb. 23, 2021. 7 pages.

Abbasi, COVID-19 and mRNA Vaccines-First Large Test for a New Approach. JAMA. Sep. 22, 2020;324(12):1125-1127. doi: 10.1001/jama.2020.16866.

Adney et al., Efficacy of an Adjuvanted Middle East Respiratory Syndrome Coronavirus Spike Protein Vaccine in Dromedary Camels and Alpacas. Viruses. Mar. 2, 2019;11(3). pii: E212. doi: 10.3390/v11030212.

Ahmad et al., New multivalent cationic lipids reveal bell curve for transfection efficiency versus membrane charge density: lipid-DNA complexes for gene delivery. J Gene Med. Jun. 2005;7(6):739-48. doi: 10.1002/jgm.717.

Ahmed et al., Biochemistry, Lipids, STATPEARLS Publishing (2023) (<https://www.ncbi.nlm.nih.gov/books/NBK525952/> [last accessed Aug. 24, 2023]: 5 pages.

Akinc et al., A combinatorial library of lipid-like materials for delivery of RNAi therapeutics. Nat Biotechnol. May 2008;26(5):561-9. doi: 10.1038/nbt1402. Epub Apr. 27, 2008.

Al Kahlout et al., Comparative Serological Study for the Prevalence of Anti-MERS Coronavirus Antibodies in High- and Low-Risk Groups in Qatar. J Immunol Res. Feb. 18, 2019;2019:1386740. doi: 10.1155/2019/1386740. eCollection 2019.

Altieri et al., The influence of 4-thiouridine labeling on pre-mRNA splicing outcomes. PLoS One. Dec. 13, 2021;16(12):e0257503. doi: 10.1371/journal.pone.0257503. eCollection 2021.

Anderson et al., Safety and Immunogenicity of SARS-CoV-2 mRNA-1273 Vaccine in Older Adults. N Engl J Med. Sep. 29, 2020;NEJMoa2028436: 1-12. doi: 10.1056/NEJMoa2028436.

Andries et al., N(1)-methylpseudouridine-incorporated mRNA outperforms pseudouridine-incorporated mRNA by providing enhanced protein expression and reduced immunogenicity in mammalian cell lines and mice. J Control Release. Nov. 10, 2015;217:337-44. doi: 10.1016/j.jconrel.2015.08.051. Epub Sep. 3, 2015.

Arthur, Pfizer predicts \$54bn in 2022 revenue from Comirnaty and Paxlovid. BioPharmaReporter. Feb. 8, 2022: 2 pages.

Atsmon et al., Safety and immunogenicity of multimeric-001—a novel universal influenza vaccine. J Clin Immunol. Jun. 2012;32(3):595-603. doi: 10.1007/s10875-011-9632-5. Epub Feb. 9, 2012.

Baden et al., Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine. N Engl J Med. Feb. 4, 2021;384(5):403-416. doi: 10.1056/NEJMoa2035389. Epub Dec. 30, 2020.

Bancel, Our Global Commitment to Vaccine Access. Moderna. Oct. 8, 2021: 1 page.

Bangham et al., Diffusion of univalent ions across the lamellae of swollen phospholipids. J Mol Biol. Aug. 1965;13(1):238-52. doi: 10.1016/s0022-2836(65)80093-6.

Bear, Gene expression and regulation. In: Lewin's Cells. 2011. Cassimeris et al., Eds. Chapter 4:105-68.

Bennett et al., Cholesterol enhances cationic liposome-mediated DNA transfection of human respiratory epithelial cells. Biosci Rep. Feb. 1995;15(1):47-53. doi: 10.1007/BF01200214.

Bermudez, We decided not to enforce our patent during the pandemic: Nubur Afeyan, co-founder of Moderna. BBC News World. May 6, 2021: 11 pages.

Bharali et al., Nanoparticles and cancer therapy: a concise review with emphasis on dendrimers. Int J Nanomedicine. 2009;4:1-7. Epub Apr. 1, 2009.

Bhavana et al., COVID-19: Pathophysiology, treatment options, nanotechnology approaches, and research agenda to combating the SARS-CoV2 pandemic. Life Sci. Nov. 15, 2020;261:118336. doi: 10.1016/j.lfs.2020.118336. Epub Aug. 23, 2020.

Bourla, An Open Letter from Pfizer Chairman and CEO to Colleagues. Pfizer. May 7, 2021: 3 pages.

Bruxvoort et al., Real-world effectiveness of the mRNA-1273 vaccine against COVID-19: Interim results from a prospective observational cohort study. Lancet Reg Health Am. Feb. 2022;6:100134. doi: 10.1016/j.lana.2021.100134. Epub Nov. 25, 2021.

Buschmann et al., Nanomaterial Delivery Systems for mRNA Vaccines. Vaccines (Basel). Jan. 19, 2021;9(1):65. doi: 10.3390/vaccines9010065.

Callaway, Omicron likely to weaken COVID vaccine protection. Nature. Dec. 2021;600(7889):367-368. doi: 10.1038/d41586-021-03672-3.

Carvalho et al., The first 12 months of COVID-19: a timeline of immunological insights. Nat Rev Immunol. Apr. 2021;21(4):245-256. doi:10.1038/s41577-021-00522-1. Epub Mar. 15, 2021.

Cele et al., SARS-CoV-2 Omicron has extensive but incomplete escape of Pfizer BNT162b2 elicited neutralization and requires ACE2 for infection. medRxiv. 2021 doi: 10.1101/2021.12.08.21267417. 2021.12.08.21267417. 15 pages.

Cerutti et al., Potent SARS-CoV-2 neutralizing antibodies directed against spike N-terminal domain target a single supersite. Cell Host Microbe. May 12, 2021;29(5):819-833.e7. doi: 10.1016/j.chom.2021.03.005. Epub Mar. 12, 2021.

Chan et al., Genomic characterization of the 2019 novel human-pathogenic coronavirus isolated from a patient with atypical pneumonia after visiting Wuhan. Emerg Microbes Infect. Jan. 28, 2020;9(1):221-236. doi: 10.1080/22221751.2020.1719902. eCollection 2020.

Chen et al., A novel neutralizing monoclonal antibody targeting the N-terminal domain of the MERS-CoV spike protein. Emerg Microbes Infect. Jun. 28, 2017;6(6):e60. doi: 10.1038/emi.2017.50. Erratum for: Emerg Microbes Infect. May 24, 2017;6(5):e37.

Chen et al., Emerging coronaviruses: Genome structure, replication, and pathogenesis. J Med Virol. Apr. 2020;92(4):418-423. doi: 10.1002/jmv.25681. Epub Feb. 7, 2020. Erratum in: J Med Virol. Oct. 2020;92(10):2249.

(56)

References Cited

OTHER PUBLICATIONS

- Chen et al., Vaccine design of hemagglutinin glycoprotein against influenza. *Trends Biotechnol.* Sep. 2011;29(9):426-34. doi: 10.1016/j.tibtech.2011.04.007.
- Cheng et al., Selective ORgan Targeting (SORT) nanoparticles for tissue specific mRNA delivery and CRISPR/Cas gene editing. *Nat Nanotechnol.* Apr. 2020; 15(4): 313-320.
- Chi et al., A neutralizing human antibody binds to the N-terminal domain of the Spike protein of SARS-CoV-2. *Science.* Aug. 7, 2020;369(6504):650-655. doi: 10.1126/science.abc6952. Epub Jun. 22, 2020.
- Choi et al., 2'-O-methylation in mRNA disrupts tRNA decoding during translation elongation. *Nat Struct Mol Biol.* Mar. 2018;25(3):208-216. doi: 10.1038/s41594-018-0030-z. Epub Feb. 19, 2018.
- Choi et al., Safety and immunogenicity of SARS-CoV-2 variant mRNA vaccine boosters in healthy adults: an interim analysis. *Nat Med.* Nov. 2021;27(11):2025-2031. doi: 10.1038/s41591-021-01527-y. Epub Sep. 15, 2021.
- Choi et al., Serum Neutralizing Activity of mRNA-1273 against SARS-CoV-2 Variants. *J Virol.* Nov. 9, 2021;95(23):e0131321. doi: 10.1128/JVI.01313-21. Epub Sep. 22, 2021.
- Chu et al., Immune Memory Response After a Booster Injection of mRNA-1273 for Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2). 2021 MedRxiv doi: <https://doi.org/10.1101/2021.09.29.21264089>. 52 pages.
- Cohen, New crop of mRNA vaccines aim for accessibility. *Science.* Apr. 8, 2022;376(6589):120-121. doi: 10.1126/science.abq3935. Epub Apr. 7, 2022.
- Cohen, Scientists are moving at record speed to create new coronavirus vaccines—but they may come too late. *Science Insider.* Jan. 27, 2020. 6 pages.
- Cohen, Scientists question Moderna invention claim in COVID-19 vaccine dispute. *Science Insider.* Aug. 29, 2022: 3 pages.
- Cooper et al., Chapters 1, 7, and 8 in *The Cell: A Molecular Approach*, 5th Ed. (2009), 163 pages.
- Corbett et al., Evaluation of the mRNA-1273 Vaccine against SARS-CoV-2 in Nonhuman Primates. *N Engl J Med* 2020; 383:1544-55. doi: 10.1056/NEJMoa2024671.
- Corbett et al., SARS-CoV-2 mRNA vaccine design enabled by prototype pathogen preparedness. *Nature.* Oct. 2020;586(7830):567-571. doi: 10.1038/s41586-020-2622-0. Epub Aug. 5, 2020.
- Cox et al., FluBlok, a recombinant hemagglutinin influenza vaccine. *Influenza Other Respir Viruses.* Nov. 2008; 2(6): 211-219.
- Dai et al., Viral targets for vaccines against COVID-19. *Nat Rev Immunol.* Feb. 2021;21(2):73-82. doi: 10.1038/s41577-020-00480-0. Epub Dec. 18, 2020.
- Defrancesco, The 'anti-hype' vaccine. *Nat Biotechnol.* Mar. 2017;35(3):193-197. doi: 10.1038/nbt.3812. Epub Feb. 27, 2017.
- Degroot et al., Part II—The Positive Sense Single Stranded RNA Viruses, Family Coronaviridae, in *Virus Taxonomy: Ninth Report of the International Committee on Taxonomy of Viruses* 806 (2012) 25 pages.
- Dejnirattisai et al., Reduced neutralisation of SARS-CoV-2 omicron B.1.1.529 variant by post-immunisation serum. *Lancet.* Jan. 15, 2022;399(10321):234-236. doi: 10.1016/S0140-6736(21)02844-0. Epub Dec. 20, 2021.
- Dimitriadis et al., Translation of rabbit globin mRNA introduced by liposomes into mouse lymphocytes. *Nature.* Aug. 31, 1978;274(5674):923-4. doi: 10.1038/274923a0.
- Dolgin, Business: The billion-dollar biotech. *Nature.* Jun. 4, 2015;522(7554):26-8. doi: 10.1038/522026a.
- Dormitzer, Synthetic Vaccinology at Novartis, Presentation at The National Academy of Sciences Forum on Synthetic Biology. Oct. 21, 2013, The Keck Center, Washington, DC. 12 pages.
- Du et al., The spike protein of SARS-CoV—a target for vaccine and therapeutic development. *Nat Rev Microbiol.* Mar. 2009;7(3):226-36. doi: 10.1038/nrmicro2090. Epub Feb. 9, 2009.
- Dufourc, Sterols and membrane dynamics. *J Chem Biol.* Nov. 2008;1(1-4):63-77. doi: 10.1007/s12154-008-0010-6. Epub Sep. 23, 2008.
- El Sahly et al., Efficacy of the mRNA-1273 SARS-CoV-2 Vaccine at Completion of Blinded Phase. *N Engl J Med.* Nov. 4, 2021; 385:1774-1785.
- Ellebedy et al., Influenza Vaccines. *Vaccine.* Nov. 5, 2009; 27(Suppl 4): D65-D68.
- Engler et al., Half- vs full-dose trivalent inactivated influenza vaccine (2004-2005): age, dose, and sex effects on immune responses. *Arch Intern Med.* Dec. 8, 2008;168(22):2405-14. doi: 10.1001/archinternmed.2008.513.
- Erasmus et al., A Nanostructured Lipid Carrier for Delivery of a Replicating Viral RNA Provides Single, Low-Dose Protection against Zika. *Mol Ther.* Oct. 3, 2018;26(10):2507-2522. doi: 10.1016/j.yimthe.2018.07.010. Epub Aug. 2, 2018.
- Escriou et al., Protection from SARS coronavirus conferred by live measles vaccine expressing the spike glycoprotein. *Virology.* Mar. 2014;452-453:32-41. doi: 10.1016/j.virol.2014.01.002. Epub Jan. 28, 2014.
- Fauquet, Taxonomy, Classification and Nomenclature of Viruses. *Encyclopedia of Virology.* 2008:9-23. doi: 10.1016/B978-012374410-4.00509-4. Epub Jul. 30, 2008.
- Fujimori, How this Once Little-Known Molecule is Disrupting Medicine. *Pfizer.* Jun. 2022: 7 pages.
- Furuichi et al., Viral and cellular mRNA capping: past and prospects. *Adv Virus Res.* 2000;55:135-84. doi: 10.1016/s0065-3527(00)55003-9.
- Gallie, D.R., The cap and poly(A) tail function synergistically to regulate mRNA translational efficiency. *Genes Dev.* Nov. 1991;5(11):2108-16.
- Garde et al., The story of mRNA: How a once-dismissed idea became a leading technology in the Covid vaccine race. *STAT News.* Nov. 10, 2020: 23 pages.
- Geall et al., Using self-amplifying mRNA vaccines to facilitate a rapid response to pandemic influenza. *Eur Pharm Rev.* Jul. 3, 2014; 3: 5 pages.
- Gilboa, E. et al., Cancer immunotherapy with mRNA-transfected dendritic cells. *Immunol Rev.* Jun. 2004;199:251-63.
- Guevara et al., Advances in Lipid Nanoparticles for mRNA-Based Cancer Immunotherapy. *Front Chem.* 2020; 8: 589959.
- Guo et al., An Engineered Receptor-Binding Domain Improves the Immunogenicity of Multivalent SARS-CoV-2 Vaccines. *mBio.* May-Jun. 2021; 12(3): e00930-21.
- Guo et al., Nanoparticles escaping RES and endosome: Challenges for siRNA delivery for cancer therapy. *J Nanomat.* Jan. 2011;742895:1-12.
- Hashiba et al., The use of design of experiments with multiple responses to determine optimal formulations for in vivo hepatic mRNA delivery. *J Control Release.* Nov. 10, 2020;327:467-476. doi: 10.1016/j.jconrel.2020.08.031. Epub Aug. 25, 2020.
- Hatziantoniou et al., Anaphylactic reactions to mRNA COVID-19 vaccines: A call for further study. *Vaccine.* May 6, 2021;39(19):2605-2607. doi: 10.1016/j.vaccine.2021.03.073. Epub Apr. 6, 2021. PMID: 33846043; PMCID: PMC8023205.
- He et al., Receptor-binding domain of SARS-CoV spike protein induces highly potent neutralizing antibodies: implication for developing subunit vaccine. *Biochem Biophys Res Commun.* Nov. 12, 2004;324(2):773-81. doi: 10.1016/j.bbrc.2004.09.106.
- He et al., Single-component, self-assembling, protein nanoparticles presenting the receptor binding domain and stabilized spike as SARS-CoV-2 vaccine candidates. *Sci Adv.* Mar. 19, 2021;7(12):eabf1591. doi: 10.1126/sciadv.abf1591. Print Mar. 2021.
- Heinz et al., Distinguishing features of current COVID-19 vaccines: knowns and unknowns of antigen presentation and modes of action. *NPJ Vaccines.* Aug. 16, 2021;6(1):104. doi: 10.1038/s41541-021-00369-6.
- Hemmer et al., Minimal peptide length requirements for CD4(+) T cell clones—implications for molecular mimicry and T cell survival. *Int Immunol.* Mar. 2000;12(3):375-83. doi: 10.1093/intimm/12.3.375.
- Hoernes et al., Nucleotide modifications within bacterial messenger RNAs regulate their translation and are able to rewire the genetic code. *Nucleic Acids Res.* Jan. 29, 2016; 44(2): 852-862.

(56)

References Cited

OTHER PUBLICATIONS

- Hope et al., Reduction of Liposome Size and Preparation of Unilamellar Vesicles by Extrusion Techniques. In: *Liposome Technology*. 1993. Gregoriadis et al., Ed. vol. 1:123.
- Houser et al., Influenza Vaccines: Challenges and Solutions. *Cell Host Microbe*. Mar. 11, 2015; 17(3): 295-300. doi: 10.1016/j.chom.2015.02.012.
- Hsieh et al., Structure-based design of prefusion-stabilized SARS-CoV-2 spikes. *Science*. Sep. 18, 2020;369(6510):1501-1505. doi: 10.1126/science.abd0826. Epub Jul. 23, 2020.
- Huang et al., Asymmetric 1-alkyl-2-acyl phosphatidylcholine: a helper lipid for enhanced non-viral gene delivery. *Int J Pharm*. May 1, 2012;427(1):64-70. doi: 10.1016/j.ijpharm.2011.06.022. Epub Jun. 21, 2011.
- Huang et al., Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. *Lancet*. Jan. 24, 2020;395(10223):497-506.
- Huang et al., Construction and biological characterisation of recombinant porcine circovirus type 2 expressing the V5 epitope tag. *Virus Res*. Nov. 2011;161(2):115-23. doi: 10.1016/j.virusres.2011.05.015. Epub May 27, 2011.
- Jackson et al., An mRNA Vaccine against SARS-CoV-2—Preliminary Report. *N Engl J Med*. Jul. 14, 2020; NEJMoa2022483. doi: 10.1056/NEJMoa2022483. Online ahead of print.
- Javier et al., Phylogenetic Analysis and Structural Modeling of SARS-CoV-2 Spike Protein Reveals an Evolutionary Distinct and Proteolytically Sensitive Activation Loop. *J Mol Biol*. May 1, 2020;432(10):3309-3325. doi: 10.1016/j.jmb.2020.04.009. Epub Apr. 19, 2020.
- Jayaraman et al., Maximizing the potency of siRNA lipid nanoparticles for hepatic gene silencing in vivo. *Angew Chem Int Ed Engl*. Aug. 20, 2012;51(34):8529-33. doi: 10.1002/anie.201203263.
- Jiaming et al., The recombinant N-terminal domain of spike proteins is a potential vaccine against Middle East respiratory syndrome coronavirus (MERS-CoV) infection. *Vaccine*. Jan. 3, 2017;35(1):10-18. doi: 10.1016/j.vaccine.2016.11.064. Epub Nov. 26, 2016.
- Jiang et al., The Variation of SARS-CoV-2 and Advanced Research on Current Vaccines. *Front Med (Lausanne)*. Jan. 18, 2022;8:806641. doi: 10.3389/fmed.2021.806641. eCollection 2021.
- Johanning et al., A Sindbis virus mRNA polynucleotide vector achieves prolonged and high level heterologous gene expression in vivo. *Nucleic Acids Res*. May 11, 1995; 23(9): 1495-1501.
- Kanasty et al., Delivery materials for siRNA therapeutics. *Nat Mater*. Nov. 2013;12(11):967-77. doi: 10.1038/nmat3765.
- Kariko et al., Generating the optimal mRNA for therapy: HPLC purification eliminates immune activation and improves translation of nucleoside-modified, protein-encoding mRNA. *Nucleic Acids Res*. Nov. 2011;39(21):e142. doi: 10.1093/nar/gkr695. Epub Sep. 2, 2011.
- Kariko et al., Suppression of RNA recognition by Toll-like receptors: the impact of nucleoside modification and the evolutionary origin of RNA. *Immunity*. Aug. 2005;23(2):165-75. doi: 10.1016/j.immuni.2005.06.008.
- Kauffman et al., Materials for non-viral intracellular delivery of messenger RNA therapeutics. *J Control Release*. Oct. 28, 2016;240:227-234. doi: 10.1016/j.jconrel.2015.12.032. Epub Dec. 21, 2015.
- Keegan, The Lasker Awards: A History of "America's Nobels". *Cell Mentor*. Sep. 11, 2018: 7 pages.
- Khoury et al., Neutralizing antibody levels are highly predictive of immune protection from symptomatic SARS-CoV-2 infection. *Nat Med*. Jul. 2021;27(7):1205-1211. doi: 10.1038/s41591-021-01377-8. Epub May 17, 2021.
- King et al., Virus Taxonomy: Classification and Nomenclature of Viruses. Ninth Report of the International Committee on Taxonomy of Viruses. *Int Union of Microbiol Soc*. 2012: 50 pages.
- Kirchdoerfer et al., Stabilized coronavirus spikes are resistant to conformational changes induced by receptor recognition or proteolysis. *Sci Rep*. Oct. 24, 2018;8(1):15701. doi: 10.1038/s41598-018-34171-7.
- Kisich et al., Antimycobacterial agent based on mRNA encoding human beta-defensin 2 enables primary macrophages to restrict growth of *Mycobacterium tuberculosis*. *Infect Immun*. Apr. 2001;69(4):2692-9.
- Kolata, Kati Kariko Helped Shield the World from the Coronavirus. *New York Times*. Sep. 24, 2021: 5 pages.
- Koyama et al., Analysis on GENIE reveals novel recurrent variants that affect molecular diagnosis of sizable number of cancer patients. *BMC Cancer*. Feb. 1, 2019;19(1):114. doi: 10.1186/s12885-019-5313-1.
- Koyama et al., Emergence of Drift Variants That May Affect COVID-19 Vaccine Development and Antibody Treatment. *Pathogens*. Apr. 26, 2020;9(5):324. doi: 10.3390/pathogens9050324.
- Krammer et al., Chimeric hemagglutinin influenza virus vaccine constructs elicit broadly protective stalk-specific antibodies. *J Virol*. Jun. 2013;87(12):6542-50. doi: 10.1128/JVI.00641-13. Epub Apr. 10, 2013.
- Kresge, The Messenger RNA Pioneers Everyone Ignored. *Bloomberg*. Nov. 23, 2021: 4 pages.
- Kurimoto et al., PEG-OligoRNA Hybridization of mRNA for Developing Sterically Stable Lipid Nanoparticles toward In Vivo Administration. *Molecules*. Apr. 3, 2019;24(7): 1303.
- Leung et al., Lipid nanoparticles for short interfering RNA delivery. *Adv Genet*. 2014;88:71-110. doi: 10.1016/B978-0-12-800148-6.00004-3.
- Leuschner et al., Therapeutic siRNA silencing in inflammatory monocytes in mice. *Nat Biotechnol*. Oct. 9, 2011;29(11):1005-10. doi: 10.1038/nbt.1989.
- Li et al., Application of ultra-high performance liquid chromatography for chemical characterization of liposome-based therapeutic small-interfering RNA, 13 American Pharmaceutical Review 102 (2010). 13 pages.
- Li et al., Cell Attachment Domains of the Porcine Epidemic Diarrhea Virus Spike Protein Are Key Targets of Neutralizing Antibodies. *J Virol*. May 26, 2017;91(12):e00273-17. doi: 10.1128/JVI.00273-17. Print Jun. 15, 2017.
- Li, Receptor recognition mechanisms of coronaviruses: a decade of structural studies. *J Virol*. Feb. 2015;89(4):1954-64. doi: 10.1128/JVI.02615-14. Epub Nov. 26, 2014.
- Li, Structure, Function, and Evolution of Coronavirus Spike Proteins. *Annu Rev Virol*. Sep. 29, 2016;3(1):237-261. doi: 10.1146/annurev-virology-110615-042301. Epub Aug. 25, 2016.
- Lin et al., Glycan Masking of Epitopes in the NTD and RBD of the Spike Protein Elicits Broadly Neutralizing Antibodies Against SARS-CoV-2 Variants. *Front Immunol*. Dec. 2, 2021;12:795741. doi: 10.3389/fimmu.2021.795741. eCollection 2021.
- Lin et al., Three-dimensional imaging of lipid gene-carriers: membrane charge density controls universal transfection behavior in lamellar cationic liposome-DNA complexes. *Biophys J*. May 2003;84(5):3307-16. doi: 10.1016/S0006-3495(03)70055-1.
- Liu et al., Chimeric severe acute respiratory syndrome coronavirus (SARS-CoV) S glycoprotein and influenza matrix 1 efficiently form virus-like particles (VLPs) that protect mice against challenge with SARS-CoV. *Vaccine*. Sep. 2, 2011;29(38):6606-13. doi: 10.1016/j.vaccine.2011.06.111. Epub Jul. 14, 2011.
- Liu et al., Influence of polyethylene glycol density and surface lipid on pharmacokinetics and biodistribution of lipid-calcium-phosphate nanoparticles. *Biomaterials*. Mar. 2014;35(9):3027-34. doi: 10.1016/j.biomaterials.2013.12.022. Epub Jan. 2, 2014.
- Liu et al., Potent neutralizing antibodies against multiple epitopes on SARS-CoV-2 spike. *Nature*. Aug. 2020;584(7821):450-456. doi: 10.1038/s41586-020-2571-7. Epub Jul. 22, 2020.
- Loftus, Moderna Signals It May Enforce Covid-19 Vaccine Patents in Wealthy Nations. *Wall Street Journal*. Mar. 7, 2022. <https://www.wsj.com/articles/moderna-signals-it-may-enforce-covid-19-vaccine-patents-in-wealthy-nations-11646699609> [last accessed Jul. 1, 2024].
- Lu et al., Genomic characterisation and epidemiology of 2019 novel coronavirus: implications for virus origins and receptor binding. *Lancet*. Feb. 22, 2020;395(10224):565-574. doi: 10.1016/S0140-6736(20)30251-8. Epub Jan. 30, 2020.
- Lutz et al., Unmodified mRNA in LNPs constitutes a competitive technology for prophylactic vaccines. *NPJ Vaccines*. Oct. 19, 2017;2:29. doi: 10.1038/s41541-017-0032-6. eCollection 2017.

(56)

References Cited

OTHER PUBLICATIONS

- Ma et al., Intranasal vaccination with recombinant receptor-binding domain of MERS-CoV spike protein induces much stronger local mucosal immune responses than subcutaneous immunization: Implication for designing novel mucosal MERS vaccines. *Vaccine*. Apr. 11, 2014;32(18):2100-8. doi: 10.1016/j.vaccine.2014.02.004. Epub Feb. 18, 2014.
- Mandal et al., Reprogramming human fibroblasts to pluripotency using modified mRNA. *Nat Protoc*. Mar. 2013;8(3):568-82. doi: 10.1038/nprot.2013.019. Epub Feb. 21, 2013.
- Martinez-Flores et al., SARS-CoV-2 Vaccines Based on the Spike Glycoprotein and Implications of New Viral Variants. *Front Immunol*. Jul. 12, 2021;12:701501. doi: 10.3389/fimmu.2021.701501. eCollection 2021.
- McCallum et al., N-terminal domain antigenic mapping reveals a site of vulnerability for SARS-CoV-2. *Cell*. Apr. 29, 2021; 184(9): 2332-2347.e16.
- McKay et al., Self-amplifying RNA SARS-CoV-2 lipid nanoparticle vaccine candidate induces high neutralizing antibody titers in mice. *Nat Commun*. Jul. 9, 2020;11(1):3523. doi: 10.1038/s41467-020-17409-9.
- McLellan et al., Structure and function of respiratory syncytial virus surface glycoproteins. *Curr Top Microbiol Immunol*. 2013;372:83-104. doi: 10.1007/978-3-642-38919-1_4.
- McLellan et al., Structure of respiratory syncytial virus fusion glycoprotein in the postfusion conformation reveals preservation of neutralizing epitopes. *J Virol*. Aug. 2011;85(15):7788-96. doi: 10.1128/JVI.00555-11. Epub May 25, 2011.
- Mendonca et al., Design of lipid-based nanoparticles for delivery of therapeutic nucleic acids. *Drug Discov Today*. Mar. 2023;28(3):103505. doi: 10.1016/j.drudis.2023.103505. Epub Jan. 25, 2023.
- Miao et al., Delivery of mRNA vaccines with heterocyclic lipids increases anti-tumor efficacy by STING-mediated immune cell activation. *Nat Biotechnol* 37, 1174-1185 (2019). <https://doi.org/10.1038/s41587-019-0247-3>.
- ModernaTX, Inc. et al v. Pfizer Inc. et al.* (D. Mass. 22-11378-RGS) D.I. 105—District Court Memorandum and Order on Claim Construction, Aug. 1, 2023. 15 pages.
- ModernaTX, Inc. et al. v. Pfizer Inc. et al.* (D. Mass. 22-11378-RGS) D.I. 1—Complaint for Patent Infringement, Aug. 26, 2022. 39 pages.
- ModernaTX, Inc. et al. v. Pfizer Inc. et al.* (D. Mass. 22-11378-RGS) D.I. 45—Answer to the Complaint, Dec. 5, 2022. 81 pages.
- Modjarrad., MERS-CoV vaccine candidates in development: The current landscape. *Vaccine*. Jun. 3, 2016;34(26):2982-2987. doi: 10.1016/j.vaccine.2016.03.104. Epub Apr. 12, 2016.
- Mullard, COVID-19 vaccine success enables a bolder vision for mRNA cancer vaccines, says BioNTech CEO. *Nat Rev Drug Discov*. Jul. 2021;20(7):500-501. doi: 10.1038/d41573-021-00110-x.
- Mulligan et al., Phase I/II study of COVID-19 RNA vaccine BNT162b1 in adults. *Nature*. Oct. 2020;586(7830):589-593. doi: 10.1038/s41586-020-2639-4. Epub Aug. 12, 2020.
- Murphy, Excerpts from Janeway's Immunobiology, 8th Ed. (2012). 250 pages.
- Muthumani et al., A synthetic consensus anti-spike protein DNA vaccine induces protective immunity against Middle East respiratory syndrome coronavirus in nonhuman primates. *Sci Transl Med*. Aug. 19, 2015;7(301):301ra132. doi: 10.1126/scitranslmed.aac7462.
- Nance et al., Modifications in an Emergency: The Role of N1-Methylpseudouridine in COVID-19 Vaccines. *ACS Cent Sci*. May 26, 2021;7(5):748-756. doi: 10.1021/acscentsci.1c00197. Epub Apr. 6, 2021.
- Oostvogels et al., Phase 1 Assessment of the Safety and Immunogenicity of an mRNA-Lipid Nanoparticle Vaccine Candidate Against SARS-CoV-2 in Human Volunteers. *medRxiv* Nov. 9, 2020. 20228551;1-38. doi: <https://doi.org/10.1101/2020.11.09.20228551>.
- Opposition against European Patent No. 3324979 B1, filed by BioNTech SE (Opponent) against ModernaTX, Inc. (Patentee) on Jul. 10, 2023. 86 pages.
- Ostro et al., Evidence for translation of rabbit globin mRNA after liposome-mediated insertion into a human cell line. *Nature*. Aug. 31, 1978;274(5674):921-3. doi: 10.1038/274921a0.
- Pallesen et al., Immunogenicity and structures of a rationally designed prefusion MERS-CoV spike antigen. *Proc Natl Acad Sci U S A*. Aug. 29, 2017;114(35):E7348-E7357. doi: 10.1073/pnas.1707304114. Epub Aug. 14, 2017.
- Pardi et al., Nucleoside-modified mRNA vaccines induce potent T follicular helper and germinal center B cell responses. *J Exp Med*. Jun. 4, 2018;215(6):1571-1588. doi: 10.1084/jem.20171450. Epub May 8, 2018.
- Pardi et al., Zika virus protection by a single low-dose nucleoside-modified mRNA vaccination. *Nature*. Mar. 9, 2017;543(7644):248-251. doi: 10.1038/nature21428. Epub Feb. 2, 2017.
- Park et al., The Miracle Workers. Heroes of the Year. *TIME Magazine*. 2021: 17 pages.
- Pascolo., Messenger RNA-based vaccines. *Expert Opin Biol Ther*. Aug. 2004;4(8):1285-94. doi: 10.1517/14712598.4.8.1285.
- Patel et al., Naturally-occurring cholesterol analogues in lipid nanoparticles induce polymorphic shape and enhance intracellular delivery of mRNA. *Nat Commun*. Feb. 20, 2020;11(1):983. doi: 10.1038/s41467-020-14527-2.
- Pathak et al., Intra-host variability in global SARS-CoV-2 genomes as signatures of RNA editing: implications in viral and host response outcomes. *bioRxiv* Dec. 9, 2020; 2020.12.09.417519; doi: <https://doi.org/10.1101/2020.12.09.417519>. 32 pages.
- Peabody et al., Termination-reinitiation occurs in the translation of mammalian cell mRNAs. *Mol Cell Biol*. Jul. 1986;6(7):2695-703. doi: 10.1128/mcb.6.7.2695-2703.1986.
- Peek et al., Nanotechnology in vaccine delivery. *Adv Drug Deliv Rev*. May 22, 2008;60(8):915-28. doi: 10.1016/j.addr.2007.05.017. Epub Feb. 7, 2008.
- Pegu et al., Durability of mRNA-1273 vaccine-induced antibodies against SARS-CoV-2 variants. *Science*. Sep. 17, 2021;373(6561):1372-1377. doi: 10.1126/science.abj4176. Epub Aug. 13, 2021.
- Probst et al., Characterization of the ribonuclease activity on the skin surface. *Genet Vaccines Ther*. May 29, 2006;4:4. doi: 10.1186/1479-0556-4-4.
- Pulliam et al., Increased risk of SARS-CoV-2 reinfection associated with emergence of the Omicron variant in South Africa. *medRxiv* 2021. doi:10.1101/2021.11.11.21266068:2021.11.11.21266068. 42 pages.
- Raj et al., Dipeptidyl peptidase 4 is a functional receptor for the emerging human coronavirus—EMC. *Nature*. Mar. 14, 2013;495(7440):251-4. doi: 10.1038/nature12005.
- Regalado, Don't panic about the latest coronavirus mutations say drug companies. *MIT Technology Review*. Dec. 23, 2020. 5 pages.
- Regalado, Moderna believes it could update its coronavirus vaccine without a big new trial. *MIT Technology Review*. Jan. 13, 2021. 4 pages.
- Robbins et al., Moderna Backs down in its Vaccine Patent Fight with the NIH. *New York Times*. Dec. 17, 2021: 1 page.
- Rodriguez-Gascon et al., Development of nucleic acid vaccines: use of self-amplifying RNA in lipid nanoparticles. *Int J Nanomedicine*. 2014; 9: 1833-1843.
- Rossi et al., IRES-based Venezuelan equine encephalitis vaccine candidate elicits protective immunity in mice. *Virology*. Mar. 15, 2013;437(2):81-8. doi: 10.1016/j.virol.2012.11.013. Epub Jan. 22, 2013.
- Sahin et al., Personalized RNA mutanome vaccines mobilize poly-specific therapeutic immunity against cancer. *Nature*. Jul. 13, 2017;547(7662):222-226. doi: 10.1038/nature23003. Epub Jul. 5, 2017.
- Salomon et al., A liposomal RNA vaccine inducing neoantigen-specific CD4+ T cells augments the antitumor activity of local radiotherapy in mice. *Oncoimmunology*. Jun. 22, 2020;9(1):1771925. doi: 10.1080/2162402X.2020.1771925.
- Santoro et al., An efficient system for the evolution of aminoacyl-tRNA synthetase specificity. *Nat Biotechnol*. Oct. 2002;20(10):1044-8. doi: 10.1038/nbt742. Epub Sep. 16, 2002.

(56)

References Cited

OTHER PUBLICATIONS

- Sarkar et al., S glycoprotein diversity of the Omicron variant. medRxiv 2021. doi:10.1101/2021.12.04.21267284;2021.12.04.21267284. 30 pages.
- Schoenmaker et al., mRNA-lipid nanoparticle COVID-19 vaccines: Structure and stability. *Int J Pharm.* May 15, 2021;601:120586. doi: 10.1016/j.ijpharm.2021.120586. Epub Apr. 9, 2021.
- Schumann et al., Multiple links between 5-methylcytosine content of mRNA and translation. *BMC Biol.* Apr. 15, 2020;18(1):40. doi: 10.1186/s12915-020-00769-5.
- Schuurhuis et al., Ins and outs of dendritic cells. *Int Arch Allergy Immunol.* 2006;140(1):53-72. doi: 10.1159/000092002. Epub Mar. 13, 2006.
- Shidhadye et al., Solid lipid nanoparticles and nanostructured lipid carriers—innovative generations of solid lipid carriers. *Curr Drug Deliv.* Oct. 2008;5(4):324-31. doi: 10.2174/156720108785915087.
- Shim et al., Intranasal immunization with plasmid DNA encoding spike protein of SARS-coronavirus/polyethylenimine nanoparticles elicits antigen-specific humoral and cellular immune responses. *BMC Immunol.* Dec. 31, 2010;11:65. doi: 10.1186/1471-2172-11-65.
- Skidmore et al., Genomic Sequencing of SARS-CoV-2 E484K Variant B.1.243.1, Arizona, USA. *Emerg Infect Dis.* Oct. 2021;27(10):2718-2720. doi: 10.3201/eid2710.211189.
- Stephenson et al., Cross-reactivity to highly pathogenic avian influenza H5N1 viruses after vaccination with nonadjuvanted and MF59-adjuvanted influenza A/Duck/Singapore/97 (H5N3) vaccine: a potential priming strategy. *J Infect Dis.* Apr. 15, 2005;191(8):1210-5. doi: 10.1086/428948. Epub Mar. 14, 2005.
- Tegally et al., Emergence and rapid spread of a new severe acute respiratory syndrome-related coronavirus 2 (SARS-CoV-2) lineage with multiple spike mutations in South Africa. *medRxiv.* Dec. 22, 2020; Preprint:19 pages. doi: 10.1101/2020.12.21.20248640. Epub Dec. 22, 2020.
- Tews et al., Chapter 2: Self-Replicating RNA. *RNA Vaccines.* 2017; 1499: 15-35. Published online Jun. 11, 2016. doi: 10.1007/978-1-4939-6481-9_2.
- Thery et al., The cell biology of antigen presentation in dendritic cells. *Curr Opin Immunol.* Feb. 2001;13(1):45-51. doi: 10.1016/s0952-7915(00)00180-1.
- Van Boheemen et al., Genomic characterization of a newly discovered coronavirus associated with acute respiratory distress syndrome in humans. *mBio.* Nov. 20, 2012;3(6):e00473-12. doi: 10.1128/mBio.00473-12.
- Vardi et al., Moderna's Mysterious Medicines. *Forbes.* Dec. 14, 2016: 5 pages.
- Vardi, Moderna's Mysterious Coronavirus Vaccine Delivery System. *Forbes.* Jul. 29, 2020: 9 pages.
- Verma et al., Double-headed nucleosides: Synthesis and applications. *Beilstein J Org Chem.* Jun. 8, 2021;17:1392-1439. doi: 10.3762/bjoc.17.98. eCollection 2021.
- Vogel et al., BNT162b vaccines protect rhesus macaques from SARS-CoV-2. *Nature.* Apr. 2021;592(7853):283-289. doi: 10.1038/s41586-021-03275-y. Epub Feb. 1, 2021.
- Von Niessen et al., Improving mRNA-Based Therapeutic Gene Delivery by Expression-Augmenting 3' UTRs Identified by Cellular Library Screening. *Mol Ther.* Apr. 10, 2019;27(4):824-836. doi: 10.1016/j.ymthe.2018.12.011. Epub Dec. 18, 2018.
- Walls et al., Crucial steps in the structure determination of a coronavirus spike glycoprotein using cryo-electron microscopy. *Protein Sci.* Jan. 2017;26(1):113-121. doi: 10.1002/pro.3048. Epub Oct. 18, 2016.
- Walls et al., Tectonic conformational changes of a coronavirus spike glycoprotein promote membrane fusion. *Proc Natl Acad Sci U S A.* Oct. 17, 2017;114(42):11157-11162. doi: 10.1073/pnas.1708727114. Epub Oct. 3, 2017.
- Walsh et al., Safety and Immunogenicity of Two RNA-Based Covid-19 Vaccine Candidates. *N Engl J Med.* Oct. 14, 2020;NEJMoa2027906: 1-13. doi: 10.1056/NEJMoa2027906. Online ahead of print.
- Wang et al., An Evidence Based Perspective on mRNA-SARS-CoV-2 Vaccine Development. *Med Sci Monit.* May 5, 2020;26:e924700. doi: 10.12659/MSM.924700.
- Wang et al., Antibody Resistance of SARS-CoV-2 Variants B.1.351 and B.1.1.7. *bioRxiv [Preprint].* Feb. 12, 2021:2021.01.25.428137. doi: 10.1101/2021.01.25.428137. 24 pages. Update in: *Nature.* May 2021;593(7857):130-135.
- Wang et al., Chapter 3: Lipid Nanoparticles for the Delivery of Nucleic Acids. *Book: Nanoparticulate Drug Delivery Systems: Strategies, Technologies, and Applications.* 2013. 29 pages.
- Wang et al., Increased Resistance of SARS-CoV-2 Variants B.1.351 and B.1.1.7 to Antibody Neutralization. *bioRxiv.* Jan. 26, 2021; 23 pages. doi: 10.1101/2021.01.25.428137. Preprint.
- Wang et al., mRNA vaccine-elicited antibodies to SARS-CoV-2 and circulating variants. *Nature.* Apr. 2021;592(7855):616-622. doi: 10.1038/s41586-021-03324-6. Epub Feb. 10, 2021. PMID: 33567448; PMCID: PMC8503938.
- Wang et al., Structural Definition of a Neutralization-sensitive Epitope on the MERS-CoV S1-NTD. *Cell Rep.* Sep. 24, 2019;28(13):3395-3405.e6. doi: 10.1016/j.celrep.2019.08.052.
- Weissman, mRNA transcript therapy. *Expert Rev Vaccines.* Feb. 2015;14(2):265-81. doi: 10.1586/14760584.2015.973859. Epub Oct. 31, 2014.
- Whitehead et al., Knocking down barriers: advances in siRNA delivery. *Nat Rev Drug Discov.* Feb. 2009;8(2):129-38. doi: 10.1038/nrd2742.
- Whitt, Generation of VSV pseudotypes using recombinant AG-VSV for studies on virus entry, identification of entry inhibitors, and immune responses to vaccines. *J Virol Methods.* Nov. 2010;169(2):365-74. doi: 10.1016/j.jviromet.2010.08.006. Epub Aug. 13, 2010.
- Wilhelm et al., Reduced Neutralization of SARS-CoV-2 Omicron Variant by Vaccine Sera and Monoclonal Antibodies. *medRxiv.* 2021. doi:10.1101/2021.12.07.21267432;2021.12.07.21267432. 9 pages.
- Wrapp et al., Cryo-EM structure of the 2019-nCoV spike in the prefusion conformation. *Science.* Feb. 19, 2020. pii: eabb2507. doi: 10.1126/science.abb2507.
- Wu et al., A new coronavirus associated with human respiratory disease in China. *Nature.* Mar. 2020;579(7798):265-269. doi: 10.1038/s41586-020-2008-3. Epub Feb. 3, 2020.
- Wu et al., mRNA-1273 vaccine induces neutralizing antibodies against spike mutants from global SARS-CoV-2 variants. *bioRxiv preprint.* Jan. 25, 2021. doi: 10.1101/2021.01.25.427948. 20 pages.
- Wu et al., Serum Neutralizing Activity Elicited by mRNA-1273 Vaccine. *N Engl J Med.* Apr. 15, 2021;384(15):1468-1470. doi: 10.1056/NEJMc2102179. Epub Mar. 17, 2021.
- Wu et al., Variant SARS-CoV-2 mRNA vaccines confer broad neutralization as primary or booster series in mice. *Vaccine.* Dec. 17, 2021;39(51):7394-7400.
- Xia, Detailed Dissection and Critical Evaluation of the Pfizer/BioNTech and Moderna mRNA Vaccines. *Vaccines (Basel).* Jul. 3, 2021;9(7):734. doi: 10.3390/vaccines9070734.
- Yang et al., A DNA vaccine induces SARS coronavirus neutralization and protective immunity in mice. *Nature.* Apr. 1, 2004;428(6982):561-4. doi: 10.1038/nature02463.
- Yang et al., The structure and functions of coronavirus genomic 3' and 5' ends. *Virus Res.* Aug. 3, 2015;206:120-33. doi: 10.1016/j.virusres.2015.02.025. Epub Feb. 28, 2015.
- Ye et al., Rational development of a combined mRNA vaccine against COVID-19 and influenza. *NPJ Vaccines.* Jul. 26, 2022;7(1):84. doi: 10.1038/s41541-022-00478-w.
- Yin et al., Immunogenicity and protective efficacy of recombinant fusion proteins containing spike protein of infectious bronchitis virus and hemagglutinin of H3N2 influenza virus in chickens. *Virus Res.* Sep. 2, 2016;223:206-12. doi: 10.1016/j.virusres.2016.07.010. Epub Aug. 3, 2016.
- Yin et al., Non-viral vectors for gene-based therapy. *Nat Rev Genet.* Aug. 2014;15(8):541-55. doi: 10.1038/nrg3763. Epub Jul. 15, 2014.
- Zangi et al., Modified mRNA directs the fate of heart progenitor cells and induces vascular regeneration after myocardial infarction. *Nat Biotechnol.* Oct. 2013;31(10):898-907. doi: 10.1038/nbt.2682. Epub Sep. 8, 2013.

(56)

References Cited

OTHER PUBLICATIONS

- Zeng et al., Formulation and Delivery Technologies for mRNA Vaccines. *Curr Top Microbiol Immunol.* Jun. 2, 2020;10.1007/82_2020_217. doi: 10.1007/82_2020_217.
- Zhang et al., A Thermostable mRNA Vaccine against COVID-19. *Cell. Sep. 3, 2020;182(5):1271-1283.e16.* doi: 10.1016/j.cell.2020.07.024. Epub Jul. 23, 2020.
- Zhang et al., Delivery of mRNA vaccine with a lipid-like material potentiates antitumor efficacy through Toll-like receptor 4 signaling. *Proc Natl Acad Sci U S A.* Feb. 9, 2021;118(6):e2005191118. doi: 10.1073/pnas.2005191118.
- Zhang et al., Prefusion spike protein stabilization through computational mutagenesis. *Proteins.* Apr. 2021;89(4):399-408. doi: 10.1002/prot.26025. Epub Dec. 4, 2020.
- Zhao et al., Key Aspects of Coronavirus Avian Infectious Bronchitis Virus. *Pathogens.* May 11, 2023;12(5):698. doi: 10.3390/pathogens12050698.
- Zhou et al., Evidence of escape of SARS-CoV-2 variant B.1.351 from natural and vaccine-induced sera. *Cell.* Apr. 29, 2021; 184(9): 2348-2361.e6.
- Zhukovsky et al., Heterogeneity of early intermediates in cell-liposome fusion mediated by influenza hemagglutinin. *Biophys J.* Nov. 1, 2006;91(9):3349-58. doi: 10.1529/biophysj.106.088005. Epub Aug. 11, 2006.
- Ziganshina et al., Antibody-Dependent Enhancement with a Focus on SARS-CoV-2 and Anti-Glycan Antibodies. *Viruses.* Jul. 20, 2023;15(7):1584. doi: 10.3390/v15071584.
- International Search Report and Written Opinion for Application No. PCT/US2016/058327 mailed Jun. 29, 2017.
- [No Author Listed], "Messenger RNA", Internet: Wikipedia. Jun. 19, 2013, XP002699196, Retrieved from the Internet: URL: http://en.wikipedia.org/wiki/Messenger_RNA.
- [No Author Listed], Geneseq Accession No. BC153556. First entry: Jan. 14, 2016. 3 pages.
- [No Author Listed], Geneseq Accession No. BGP67180. First entry: Sep. 19, 2019. 3 pages.
- [No Author Listed], Katalin Karikó and Drew Weissman awarded Horwitz Prize for pioneering research on COVID-19 vaccines, Columbia University Irving Medical Center, Aug. 16, 2021, <https://www.cuimc.columbia.edu/news/horwitz-prize-2021> (last accessed Aug. 24, 2023). 12 pages.
- [No Author Listed], List of past and pending litigations. Prepared Oct. 24, 2023. 1 page.
- Agrawal et al., Immunization with inactivated Middle East Respiratory Syndrome coronavirus vaccine leads to lung immunopathology on challenge with live virus. *Hum Vaccin Immunother.* Sep. 2016;12(9):2351-6. doi:10.1080/21645515.2016.1177688. Epub Jun. 7, 2016.
- Aliprantis et al., A phase 1, randomized, placebo-controlled study to evaluate the safety and immunogenicity of an mRNA-based RSV prefusion F protein vaccine in healthy younger and older adults. *Hum Vaccin Immunother.* May 4, 2021;17(5):1248-1261. doi: 10.1080/21645515.2020.1829899. Epub Oct. 29, 2020.
- Alnylam Pharmaceuticals, Inc. v. Moderna, Inc.* (Fed. Circ. 2023-2357) D.I. 8—Docketing Statement, Sep. 21, 2023. 2 pages.
- Alnylam Pharmaceuticals, Inc. v. Moderna, Inc. et al.* (D. Del. 23-cv-580-CFC) D.I. 1—Complaint for Patent Infringement, May 26, 2023. 28 pages.
- Alnylam Pharmaceuticals, Inc. v. Moderna, Inc. et al.* (D. Del. 23-cv-580-CFC) D.I. 12—Answer to the Complaint, Jul. 17, 2023. 37 pages.
- Alnylam Pharmaceuticals, Inc. v. Moderna, Inc. et al.* (D. Del. 22-cv-335-CFC) D.I. 1—Complaint for Patent Infringement, Mar. 17, 2022. 14 pages.
- Alnylam Pharmaceuticals, Inc. v. Moderna, Inc. et al.* (D. Del. 22-cv-335-CFC) D.I. 128—Final Judgement, Aug. 30, 2023. 2 pages.
- Alnylam Pharmaceuticals, Inc. v. Moderna, Inc. et al.* (D. Del. 22-cv-335-CFC) D.I. 130—Notice of Appeal to the U.S. Court of Appeals for the Federal Circuit, Aug. 31, 2023. 3 pages.
- Alnylam Pharmaceuticals, Inc. v. Moderna, Inc. et al.* (D. Del. 22-cv-335-CFC) D.I. 87—Answer to the Complaint, May 10, 2023. 44 pages.
- Arbutus BioPharma Corp. et al. v. Moderna, Inc. et al.* (D. Del. 22-cv-252-MSG) D.I. 1—Complaint for Patent Infringement, Feb. 28, 2022. 51 pages.
- Arbutus BioPharma Corp. et al. v. Moderna, Inc. et al.* (D. Del. 22-cv-252-MSG) D.I. 31—District Court Memorandum, Nov. 2, 2022. 16 pages.
- Arbutus BioPharma Corp. et al. v. Moderna, Inc. et al.* (D. Del. 22-cv-252-MSG) D.I. 35—Answer to the Complaint, Nov. 30, 2022. 83 pages.
- Arbutus BioPharma Corp. et al. v. Moderna, Inc. et al.* (D. Del. 22-cv-252-MSG) D.I. 64—District Court Memorandum, Mar. 10, 2023. 4 pages.
- Archer, S.J., Induction of a T-cell specific antigen on bone marrow lymphocytes with thymus RNA. *Immunology.* Jan. 1978;34(1):123-9.
- Ashley, D.M. et al., Bone marrow-generated dendritic cells pulsed with tumor extracts or tumor RNA induce antitumor immunity against central nervous system tumors. *J Exp Med.* Oct. 6, 1997; 186(7): 1177-82.
- Austin et al., Split-Dose Administration Enhances Immune Responses Elicited by a mRNA/Lipid Nanoparticle Vaccine Expressing Respiratory Syncytial Virus F Protein. *Mol Pharm.* Jan. 2, 2023;20(1):279-289. doi: 10.1021/acs.molpharmaceut.2c00635. Epub Oct. 17, 2022.
- Barbier et al., The clinical progress of mRNA vaccines and immunotherapies. *Nat Biotechnol.* Jun. 2022;40(6):840-854. doi: 10.1038/s41587-022-01294-2. Epub May 9, 2022.
- Bettinger, T. et al., Peptide-mediated RNA delivery: a novel approach for enhanced transfection of primary and post-mitotic cells. *Nucleic Acids Res.* Sep. 15, 2001;29(18):3882-91.
- Bogers et al., Potent immune responses in rhesus macaques induced by nonviral delivery of a self-amplifying RNA vaccine expressing HIV type 1 envelope with a cationic nanoemulsion. *J Infect Dis.* Mar. 15, 2015;211(6):947-55. doi: 10.1093/infdis/jiu522. Epub Sep. 18, 2014.
- Bonehill, A., et al., Single-step antigen loading and activation of dendritic cells by mRNA electroporation for the purpose of therapeutic vaccination in melanoma patients. *Clin Cancer Res.* May 2009; 15(10): 3366-3375.
- Bose, S. et al., Role of nucleolin in human parainfluenza virus type 3 infection of human lung epithelial cells. *J Viral.* Aug. 2004;78(15):8146-58.
- Byoung-Shik et al., Intranasal immunization with plasmid DNA encoding spike protein of SARS-coronavirus/polyethylenimine nanoparticles elicits antigen-specific humoral and cellular immune responses. *BMC Immunol.* Dec. 31, 2010;11:65. doi: 10.1186/1471-2172-11-65.
- Chaudhary et al., mRNA vaccines for infectious diseases: principles, delivery and clinical translation. *Nature Reviews Drug Discovery* vol. 20, pp. 817-838 (2021).
- Clancy et al., Translation: DNA to mRNA to Protein. *Nature Education* 2008;1(1):101.
- Cobb et al., Who discovered messenger RNA? *Curr Biol.* Jun. 29, 2015;25(13):R526-32. doi:10.1016/j.cub.2015.05.032.
- Conry, R.M. et al., Characterization of a messenger RNA polynucleotide vaccine vector. *Cancer Res.* Apr. 1, 1995 ;55 (7):1397-1400.
- Cross, Without these lipid shells, there would be No. mRNA vaccines for COVID-19. *Chem Eng News.* Mar. 6, 2021; 99(8). 8 pages.
- Cullis et al., Lipid Nanoparticle Systems for Enabling Gene Therapies. *Mol Ther.* Jul. 5, 2017;25(7):1467-1475. doi: 10.1016/j.ymthe.2017.03.013. Epub Apr. 13, 2017.
- Dahlman, James E. et al., In vivo endothelial siRNA delivery using polymeric nanoparticles with low molecular weight, *Nature Nanotechnology*, 2014, No. Vol.#, pp. 1-8.
- Deering et al., Nucleic acid vaccines: prospects for non-viral delivery of mRNA vaccines. *Expert Opin Drug Deliv.* Jun. 2014;11(6):885-99. doi: 10.1517/17425247.2014.901308. Epub Mar. 26, 2014.

(56)

References Cited

OTHER PUBLICATIONS

- Del Pozo-Rodriguez et al., Lipid nanoparticles as vehicles for macromolecules: nucleic acids and peptides. *Recent Pat Drug Deliv Formul.* Sep. 2011;5(3):214-26. doi: 10.2174/18722111797200515.
- Diken et al., Current Developments in Actively Personalized Cancer Vaccination with a Focus on RNA as the Drug Format. *Prog Tumor Res.* 2015;42:44-54. doi: 10.1159/000437184. Epub Sep. 4, 2015. Review.
- Du et al., Recombinant adeno-associated virus expressing the receptor-binding domain of severe acute respiratory syndrome coronavirus S protein elicits neutralizing antibodies: Implication for developing SARS vaccines. *Virology.* Sep. 15, 2006;353(1):6-16. doi: 10.1016/j.virol.2006.03.049. Epub Jun. 21, 2006.
- Fleeton et al., Self-replicative RNA vaccines elicit protection against influenza A virus, respiratory syncytial virus, and a tickborne encephalitis virus. *J Infect Dis.* May 1, 2001;183(9):1395-8. Epub Mar. 30, 2001.
- Furuichi, Caps on Eukaryotic mRNAs. eLS. John Wiley & Sons. Jul. 1-12, 2014.
- Geall et al., Nonviral delivery of self-amplifying RNA vaccines. *Proc Natl Acad Sci U S A.* Sep. 4, 2012;109(36):14604-9. doi: 10.1073/pnas.1209367109. Epub Aug. 20, 2012.
- Geall et al., RNA: the new revolution in nucleic acid vaccines. *Semin Immunol.* Apr. 2013;25(2):152-9. doi: 10.1016/j.smim.2013.05.001. Epub Jun. 2, 2013.
- GenBank Accession No. AHX22069. First seen on NCBI on May 14, 2014.
- Glenn et al., Safety and immunogenicity of a Sf9 insect cell-derived respiratory syncytial virus fusion protein nanoparticle vaccine. *Vaccine.* Jan. 7, 2013;31(3):524-32. doi: 10.1016/j.vaccine.2012.11.009.
- Graham et al., Novel antigens for RSV vaccines. *Curr Opin Immunol.* Aug. 2015;35:30-8. doi: 10.1016/j.coi.2015.04.005. Epub Jun. 10, 2015.
- Graham et al., Novel Vaccine Technologies: Essential Components of an Adequate Response to Emerging Viral Diseases. *JAMA.* Apr. 10, 2018;319(14):1431-1432. doi: 10.1001/jama.2018.0345.
- Greer et al., Long-term protection in hamsters against human parainfluenza virus type 3 following mucosal or combinations of mucosal and systemic immunizations with chimeric alphavirus-based replicon particles. *Scand J Immunol.* Dec. 2007;66(6):645-53. Epub Oct. 17, 2007.
- Hartmann et al., *Handbook of RNA Biochemistry*, Second Edition, "Part I RNA Synthesis and Detection." 2014. p. 1-27.
- Hecker, J.G. et al., Non-Viral DNA and mRNA Gene Delivery to the CNS Pre-Operatively for Neuroprotection and Following Neurotrauma. *Molecular Therapy.* 2004; 9, S258-S258.
- Heiser, A. et al., Induction of polyclonal prostate cancer-specific CTL using dendritic cells transfected with amplified tumor RNA. *J Immunol.* Mar. 1, 2001; 166(5):2953-60.
- Heyes et al., Cationic lipid saturation influences intracellular delivery of encapsulated nucleic acids. *J Control Release.* Oct. 3, 2005;107(2):276-87.
- Hoerr, I. et al., In vivo application of RNA leads to induction of specific cytotoxic T lymphocytes and antibodies. *Eur J Immunol.* Jan. 2000;30(1):1-7.
- Hoerr, I. et al., Stabilized Messenger RNA (RNAActive™) as a Tool for Innovative Gene Delivery. *Tissue Engineering.* Apr. 2007; 13(4): 865-925.
- Hoerr, More than a messenger: A new class of drugs-mRNA-based therapeutics. *Genetic Engineering & Biotechnology News.* Jun. 18, 2013. <http://www.genengnews.com/gen-articles/more-than-a-messenger-a-new-class-of-drugs-mrna-based-therapeutics/4916/> [last accessed Mar. 25, 2016].
- Holtkamp, S. et al., Modification of antigen-encoding RNA increases stability, translational efficacy, and T-cell stimulatory capacity of dendritic cells. *Blood.* Dec. 15, 2006;108(13):4009-17.
- Hörr, RNA vaccine for the induction of specific cytotoxic T lymphocytes (CTL) and antibodies. Eberhard Karls University of Tübingen. Dissertation. English-Language Translation. 1999. 138 Pages.
- Hou et al., Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials* vol. 6, pp. 1078-1094 (2021).
- Human Metapneumovirus (HM PV) Clinical Features, CDC Website, <https://www.cdc.gov/surveillance/nrevss/hmpv/clinical.html>. Sep. 5, 2019. (Year: 2019).
- Jirikowski, G.F., et al., Reversal of diabetes insipidus in Brattleboro Rats: Intrahypothalamic injection of vasopressin mRNA. *Science.* Feb. 1992; 255(5047): 996-998.
- Jorquera et al., Nanoparticle vaccines encompassing the respiratory syncytial virus (RSV) G protein CX3C chemokine motif induce robust immunity protecting from challenge and disease. *PLoS One.* Sep. 10, 2013;8(9):e74905. doi: 10.1371/journal.pone.0074905. eCollection 2013.
- Kallen et al., A development that may evolve into a revolution in medicine: mRNA as the basis for novel, nucleotide-based vaccines and drugs. *Ther Adv Vaccines.* Jan. 2014;2(1):10-31. doi: 10.1177/2051013613508729.
- Kallen et al., A novel, disruptive vaccination technology: self-adjuvanted RNAActive® vaccines. *Hum Vaccin Immunother.* Oct. 2013;9(10):2263-76. doi: 10.4161/hv.25181. Epub Jun. 4, 2013. Review.
- Kalra et al., Virosomes: As a Drug Delivery Carrier. *American Journal of Advanced Drug Delivery.* 2013;1:29-35.
- Kanapathipillai, et al., Nanoparticle targeting of anti-cancer drugs that alter intracellular signaling or influence the tumor microenvironment. *Adv. Drug Deliv. Rev.* (2014), , pp. 1-12.
- Kariko et al., Incorporation of pseudouridine into mRNA yields superior nonimmunogenic vector with increased translational capacity and biological stability. *Mol Ther.* Nov. 2008; 16(11):1833-40. Epub Sep. 16, 2008.
- Kariko, K., et al., Generating the optimal mRNA for therapy: HPLC purification eliminates immune activation and improves translation of nucleoside-modified, protein-encoding mRNA. *Nucleic Acids Research*, Oxford University Press, GB, vol. 39, No. 21, Sep. 2, 2011 (Sep. 2, 2011), e142. doi: 10.1093/nar/gkr695. Epub Sep. 2, 2011.
- Kauffman et al., Optimization of Lipid Nanoparticle Formulations for mRNA Delivery in Vivo with Fractional Factorial and Definitive Screening Designs. *Nano Lett.* Nov. 11, 2015;15(11):7300-6. doi: 10.1021/acs.nanolett.5b02497. Epub Oct. 20, 2015.
- Keshwara et al., Rabies-based vaccine induces potent immune responses against Nipah virus. *NPJ Vaccines.* Apr. 15, 2019;4:15. Erratum in: *NPJ Vaccines.* May 13, 2019;4:18.
- Kozielski et al., Bioreducible cationic polymer-based nanoparticles for efficient and environmentally triggered cytoplasmic siRNA delivery to primary human brain cancer cells. *ACS Nano.* Apr. 22, 2014;8(4):3232-41. doi: 10.1021/nn500704t. Epub Apr. 3, 2014.
- Kramps et al., Messenger RNA-based vaccines: progress, challenges, applications. *Wiley Interdiscip Rev RNA.* Nov.-Dec. 2013;4(6):737-49. doi: 10.1002/wrna.1189. Epub Jul. 25, 2013.
- Kreiter, S., et al., Intranodal vaccination with naked antigen-encoding RNA elicits potent prophylactic and therapeutic antitumoral immunity. *Cancer Res.* 2010; 70: 9031-9040.
- Kreiter, S., et al., Tumor vaccination using messenger RNA: prospects of a future therapy. *Curr Opinion in Immun.* Jun. 2011; 23(3): 399-406.
- Kresge, The messenger RNA pioneers everyone ignored. *Bloomberg*, Nov. 23, 2021, <https://www.bloomberg.com/news/newsletters/2021-11-23/themessenger-rna-pioneers-everyone-ignored> (last accessed Aug. 24, 2023). 8 pages.
- Kuhn, A.N., et al., mRNA as a versatile tool for exogenous protein expression. *Current Gene Therapy.* Oct. 2012; 12 (5): 347-361.
- Leitner, W.W. et al., DNA and RNA-based vaccines: principles, progress and prospects. *Vaccine.* Dec. 10, 1999;18 (9-10):765-77.
- Li et al., Effects of local structural transformation of lipid-like compounds on delivery of messenger RNA. *Sci Rep.* Feb. 26, 2016;6:22137. doi: 10.1038/srep22137.
- Li et al., Lipid-based nanoparticles for nucleic acid delivery. *Pharm Res.* Mar. 2007;24(3):438-49. doi: 10.1007/s11095-006-9180-5.
- Li, L. et al., Overcoming obstacles to develop effective and safe siRNA therapeutics. *Expert Opin Biol Ther.* May 2009; 9(5): 609-19.

(56)

References Cited

OTHER PUBLICATIONS

- Liang et al., Enhanced Neutralizing Antibody Response Induced by Respiratory Syncytial Virus Prefusion F Protein Expressed by a Vaccine Candidate. *J Virol.* Sep. 2015;89(18):9499-510. doi: 10.1128/JVI.01373-15. Epub Jul. 8, 2015.
- Lin et al., Progress and challenges of mRNA vaccines. *Interdisc Med.* Dec. 22, 2022;1(1):e20220008. doi: 10.1002/INMD.20220008.
- Lorenzi, J.C., et al., Intranasal vaccination with messenger RNA as a new approach in gene therapy: Use against tuberculosis. *BMC Biotechnol.* Oct. 2010; 10(77): 1-11.
- Lu et al., Bat-to-human: spike features determining 'host jump' of coronaviruses SARS-CoV, MERS-CoV, and beyond. *Trends Microbiol.* Aug. 2015;23(8):468-78. doi: 10.1016/j.tim.2015.06.003. Epub Jul. 21, 2015.
- MacLachlan, Lipid Nanoparticle-mediated delivery of messenger RNA. Presentation. 1st International mRNA Health Conference. Tubingen, Germany. Oct. 24, 2013. http://files.shareholder.com/downloads/ABEA-50QJTB/2628241206x0x699789/47543d12-db34-4e6e-88a9-f3ae5d97b1d2/MacLachlan_mRNA_Conf_2013.pdf. Last accessed Dec. 22, 2016.
- Magini et al., Self-Amplifying mRNA Vaccines Expressing Multiple Conserved Influenza Antigens Confer Protection against Homologous and Heterosubtypic Viral Challenge. *PLoS One.* Aug. 15, 2016;11(8):e0161193. doi: 10.1371/journal.pone.0161193. eCollection 2016.
- Martinon, F. et al., Induction of virus-specific cytotoxic T lymphocytes in vivo by liposome-entrapped mRNA. *Eur J Immunol.* Jul. 1993;23(7):1719-22.
- Mas et al., Engineering, Structure and Immunogenicity of the Human Metapneumovirus F Protein in the Postfusion Conformation. *PLoS Pathog.* Sep. 9, 2016;12(9):e1005859. doi:10.1371/journal.ppat.1005859. eCollection Sep. 2016.
- McDonnell et al. DNA vaccines. *N Engl J Med.* Jan. 4, 1996;334(1):42-5. doi: 10.1056/NEJM199601043340110.
- McLellan et al., Structure-based design of a fusion glycoprotein vaccine for respiratory syncytial virus. *Science.* Nov. 1, 2013;342(6158):592-8. doi: 10.1126/science.1243283.
- Midoux et al., Lipid-based mRNA vaccine delivery systems. *Expert Rev Vaccines.* Feb. 2015;14(2):221-34. doi: 10.1586/14760584.2015.986104. Epub Dec. 26, 2014. Review.
- Mitchell, Da et al., RNA transfected dendritic cells as cancer vaccines. *Curr Opin Mal Ther.* Apr. 2000;2(2):176-81.
- Mitchell, Da et al., RNA-transfected dendritic cells in cancer immunotherapy. *J Clin Invest.* Nov. 2000;106 (9):1065-9.
- Mockey et al., mRNA-based cancer vaccine: prevention of B16 melanoma progression and metastasis by systemic injection of MART1 mRNA histidylated lipopolyplexes, *Cancer Gene Therapy*, 2007, 14, pp. 802-814.
- Mok et al., An Alphavirus Replicon-Based Human Metapneumovirus Vaccine is Immunogenic and Protective in Mice and Cotton Rats. *J Virol.* Nov. 2008; 82(22): 11410-11418.
- Muller, M.R. et al., Transfection of dendritic cells with RNA induces CD4- and COB-mediated T cell immunity against breast carcinomas and reveals the immunodominance of presented T cell epitopes. *J Immunol.* Jun. 15, 2003;170(12):5892-6.
- Narayanan et al., Interplay between viruses and host mRNA degradation. *Biochim Biophys Acta.* Jun.-Jul. 2013;1829(6-7):732-41. doi: 10.1016/j.bbagr.2012.12.003. Epub Dec. 26, 2012.
- Pardi et al., mRNA vaccines—a new era in vaccinology. *Nat Rev Drug Discov.* Apr. 2018;17(4):261-279. doi: 10.1038/nrd.2017.243. Epub Jan. 12, 2018.
- Pascolo, S. Vaccination with messenger RNA (mRNA). *Handb Exp Pharmacol.* 2008; 183:221-235.
- Petition For Inter Partes Review of U.S. Pat. No. 10,702,600. *BioNTech SE and Pfizer Inc. (Petitioners) v. ModernaTX, Inc. (Patent Owner)*. Filed Aug. 28, 2023. 87 pages.
- Petition For Inter Partes Review of U.S. Pat. No. 10,933,127. *BioNTech SE and Pfizer Inc. (Petitioners) v. ModernaTX, Inc. (Patent Owner)*. Filed Aug. 28, 2023. 89 pages.
- Petition For Inter Partes Review of U.S. Pat. No. 8,058,069 (IPR2019-00554). *Moderna Therapeutics, Inc. (Petitioner) v. Arbutus Biopharma Corporation (Patent Owner)*. Filed Nov. 15, 2011. 81 pages.
- Petsch et al., Protective efficacy of in vitro synthesized, specific mRNA vaccines against influenza A virus infection. *Nat Biotechnol.* Dec. 2012;30(12):1210-6. doi: 10.1038/nbt.2436. Epub Nov. 25, 2012.
- Phua et al., Messenger RNA (mRNA) nanoparticle tumour vaccination. *Nanoscale.* Jul. 21, 2014;6(14):7715-29. doi: 10.1039/c4nr01346h. Review.
- Promosoma, LLC v. Moderna, Inc.* (S.D. Cal. 23-cv-1047-JES-DDL) D.I. 1—Complaint for Patent Infringement, Jun. 6, 2023. 51 pages.
- Promosome, LLC v. Moderna, Inc.* (S.D. Cal. 23-cv-1047-JES-DDL) D.I. 30—Notice of Dismissal, Sep. 19, 2023. 4 pages.
- Pulford, B., et al., Liposome-siRNA-peptide complexes cross the blood-brain barrier and significantly decrease PrP^C on neuronal cells and PrP^{RES} in infected cell cultures. *PLoS ONE.* 2010; 5(6): e11085.
- Rabinovich, P.M., et al., Synthetic messenger RNA as a tool for gene therapy. *Hum. Gene Ther.* Oct. 2006; 17: 1027-1035.
- Ramamoorth et al., Nonviral vectors in gene therapy—an overview. *J Clin Diagn Res.* Jan. 2015; 9(1): GE01-GE06.
- Reichmuth et al., mRNA Vaccine Delivery Using Lipid Nanoparticles. *Ther Deliv.* 2016;7(5):319-34. doi: 10.4155/tde-2016-0006.
- Ren et al., Recent vaccine development for human metapneumovirus. *J Gen Virol.* Jul. 2015;96(Pt 7):1515-20. doi: 10.1099/vir.0.000083. Epub Feb. 9, 2015.
- Rittig et al., Intradermal vaccinations with RNA coding for TAA generate CD8+ and CD4+ immune responses and induce clinical benefit in vaccinated patients. *Mol Ther.* May 2011;19(5):990-9. doi: 10.1038/mt.2010.289. Epub Dec. 28, 2010.
- Sahin et al., mRNA-based therapeutics—developing a new class of drugs. *Nat Rev Drug Discov.* Oct. 2014;13(10):759-80. doi: 10.1038/nrd4278. Epub Sep. 19, 2014.
- Schlake et al., Developing mRNA-vaccine technologies. *RNA Biol.* Nov. 2012;9(11):1319-30. doi: 10.4161/rna.22269. Epub Oct. 12, 2012.
- Schmitt, W.E. et al., In vitro induction of a bladder cancer-specific T-cell response by mRNA-transfected dendritic cells. *J Cancer Res Clin Oncol.* 2001; 127(3):203-6.
- Schott, J.W., et al., Viral and non-viral approaches for transient delivery of mRNA and proteins. *Current Gene Ther.* 2011; 11 (5): 382-398.
- Segura, J., et al., Monitoring gene therapy by external imaging of mRNA: Pilot study on murine erythropoietin. *Ther Drug Monit.* Oct. 2007; 29(5): 612-8.
- Shin et al., Recent Advances in RNA Therapeutics and RNA Delivery Systems Based on Nanoparticles. *Adv. Therap., Nov.* 2018;1(7):1800065. Review.
- Smits, E., et al., RNA-based gene transfer for adult stem cells and T cells. *Leukemia.* 2004; 18: 1898-1902.
- Sohn, R.L., et al., In-vivo particle mediated delivery of mRNA to mammalian tissues: ballistic and biological effects. *Wound Rep and Regen.* Jul.-Aug. 2001; 287-296.
- Strong, V.T. et al., Incorporation of beta-globin untranslated regions into a Sindbis virus vector for augmentation of heterologous mRNA expression. *Gene Ther.* Jun. 1997;4(6):624-7.
- Sullenger, Ba et al., Emerging clinical applications of RNA. *Nature.* Jul. 11, 2002;418(6894):252-8.
- Tam et al., Advances in Lipid Nanoparticles for siRNA Delivery. *Pharmaceutics.* Sep. 18, 2013;5(3):498-507. doi: 10.3390/pharmaceutics5030498.
- Tavernier, G., et al., mRNA as gene therapeutic: How to control protein expression. *J. of Controlled Release.* Mar. 2011; 150(3): 238-247.
- Tenchov et al., Lipid Nanoparticles—From Liposomes to mRNA Vaccine Delivery, a Landscape of Research Diversity and Advancement. *ACS Nano.* Nov. 23, 2021;15(11):16982-17015. doi: 10.1021/acsnano.1c04996. Epub Jun. 28, 2021.

(56)

References Cited**OTHER PUBLICATIONS**

- Terada et al., Characterization of Lipid Nanoparticles Containing Ionizable Cationic Lipids Using Design-of-Experiments Approach. *Langmuir*. Jan. 26, 2021;37(3):1120-1128. doi: 10.1021/acs.langmuir.0c03039. Epub Jan. 13, 2021.
- Teufel, R. et al., Human peripheral blood mononuclear cells transfected with messenger RNA stimulate antigen-specific cytotoxic T-lymphocytes in vitro. *Cell Mol Life Sci*. Aug. 2005;62(15):1755-62.
- Thess et al., Sequence-engineered mRNA Without Chemical Nucleoside Modifications Enables an Effective Protein Therapy in Large Animals. *Mol Ther. Sep.* 2015;23(9):1456-64. doi: 10.1038/mt.2015.103. Epub Jun. 8, 2015.
- To et al., An overview of rational design of mRNA-based therapeutics and vaccines. *Expert Opin Drug Discov*. Nov. 2021;16(11):1307-1317. doi: 10.1080/17460441.2021.1935859. Epub Jul. 19, 2021.
- Ulmer et al., RNA-based vaccines. *Vaccine*. Jun. 22, 2012;30(30):4414-8. doi: 10.1016/j.vaccine.2012.04.060. Epub Apr. 28, 2012.
- Wang et al., Delivery of oligonucleotides with lipid nanoparticles. *Adv Drug Deliv Rev*. Jun. 29, 2015;87:68-80. doi: 10.1016/j.addr.2015.02.007. Epub Feb. 27, 2015.
- Wang et al., Systemic delivery of modified mRNA encoding herpes simplex virus 1 thymidine kinase for targeted cancer gene therapy. *Mol Ther*. Feb. 2013;21(2):358-67. doi: 10.1038/mt.2012.250. Epub Dec. 11, 2012.
- Wen et al., New Approaches for Immunization and Therapy against Human Metapneumovirus. *Clin Vaccine Immunol*. Aug. 2015;22(8):858-66. doi: 10.1128/CVI.00230-15. Epub Jun. 10, 2015.
- Williams et al., Human metapneumovirus isolate TN/92-4 fusion protein gene, complete eds. GenBank: EF051124.1. Nov. 29, 2007.
- Wong et al., An mRNA vaccine for influenza. *Nat Biotechnol*. Dec. 2012;30(12):1202-4. doi: 10.1038/nbt.2439.
- Yamamoto et al., Current prospects for mRNA gene delivery, *European Journal of Pharmaceutics and Biopharmaceutics* 71 (2009) 484-489.
- Youn et al., Modified mRNA as an alternative to plasmid DNA (pDNA) for transcript replacement and vaccination therapy. *Expert Opin Biol Ther*. 2015;15(9):1337-48. doi: 10.1517/14712598.2015.1057563. Epub Jun. 30, 2015.
- Yuan et al., Cryo-EM structures of MERS-CoV and SARS-CoV spike glycoproteins reveal the dynamic receptor binding domains. *Nat Commun*. Apr. 10, 2017;8:15092. doi: 10.1038/ncomms15092.
- Zhao et al., Chapter Two: Lipid Nanoparticles for Gene Delivery. Book: *Advances in Genetics*. Elsevier, 2014. 24 pages.
- Zhao et al., Nanoparticle vaccines. *Vaccine*. Jan. 9, 2014;32(3):327-37. doi: 10.1016/j.vaccine.2013.11.069. Epub Dec. 2, 2013.
- Zhou, W.Z. et al., RNA melanoma vaccine: induction of antitumor immunity by human glycoprotein 100 mRNA immunization. *Hum Gene Ther*. Nov. 1, 1999;10(16):2719-24.
- Zumla et al., Middle East respiratory syndrome. *Lancet*. Sep. 5, 2015;386(9997):995-1007. doi: 10.1016/S0140-6736(15)60454-8. Epub Jun. 3, 2015.
- [IPR-2023-01358] Institution Decision in IPR2023-01358 dated Mar. 19, 2024.
- [IPR-2023-01358] Patent Owner's Preliminary Response in IPR2023-01358 dated Dec. 8, 2023.
- [IPR-2023-01358] Patent Owner's Preliminary Statement Regarding Petitioner's Inconsistent Positions in IPR2023-01358 dated Jan. 16, 2024.
- [IPR-2023-01358] Patent Owner's Response in IPR2023-01358 dated May 31, 2024.
- [IPR-2023-01358] Patent Owner's Sur-Reply in IPR2023-01358 dated Oct. 16, 2024.
- [IPR-2023-01358] Patent Owner's Sur-Reply in Support of Preliminary Response in IPR2023-01358 dated Jan. 25, 2024.
- [IPR-2023-01358] Petitioner's Preliminary Statement Regarding Patent Owner's Inconsistent Positions in IPR-2023-01358 dated Jan. 23, 2024.
- [IPR-2023-01358] Petitioner's Reply to Patent Owner's Preliminary Response in IPR2023-01358 dated Jan. 12, 2024.
- [IPR-2023-01358] Petitioner's Reply to Patent Owner's Response in IPR2023-01358 dated Aug. 30, 2024.
- [IPR2023-01359] Institution Decision dated Mar. 19, 2024 in IPR2023-01359.
- [IPR2023-01359] Patent Owner's Preliminary Response in IPR2023-01359 dated Dec. 7, 2023.
- [IPR2023-01359] Patent Owner's Preliminary Statement Regarding Alleged Inconsistent Positions in IPR2023-01359 dated Jan. 16, 2024.
- [IPR2023-01359] Patent Owner's Response in IPR2023-01359 dated May 31, 2024.
- [IPR2023-01359] Patent Owner's Sur-Reply in IPR2023-01359 dated Oct. 16, 2024.
- [IPR2023-01359] Patent Owner's Sur-Reply in Support of Preliminary Response in IPR2023-01359 dated Jan. 25, 2024.
- [IPR2023-01359] Petitioner's Reply to Patent Owner's Preliminary Response in IPR2024-01359 dated Jan. 12, 2024.
- [IPR2023-01359] Petitioner's Reply to Patent Owner's Response in IPR2023-01359 dated Aug. 30, 2024.
- Agnihothram, S., Review Committee Chair, Summary Basis for Regulatory Action, 2022, STN 125752/0 (30 Pages).
- Ahmed et al., *Biochemistry, Lipids, Statpearls Publishing (2023)* (available at <https://www.ncbi.nlm.nih.gov/books/NBK525952/>).
- Akinc et al., "Targeted Delivery of RNAi Therapeutics With Endogenous and Exogenous Ligand-Based Mechanisms," 2010, vol. 18, Issue 7, *Molecular Therapy* (8 pages).
- Allen et al., "Liposomal drug delivery systems: From concept to clinical applications," 2013, *Adv. Drug Deliv. Rev.*, pp. 36-48.
- Atkins et al., "The molecular pathogenesis of Semliki Forest virus: a model virus made useful?," *Journal of General Virology* 2287, 1999 (11 Pages).
- Bahl et al., "Preclinical and Clinical Demonstration of Immunogenicity by mRNA Vaccines against H10N8 and H7N9 Influenza Viruses," 2017, vol. 25, Issue 6, *Mol Ther*.
- Bisht et al., "Severe acute respiratory syndrome coronavirus spike protein expressed by attenuated vaccinia virus protectively immunizes mice," 2004, *Biological Sciences* (6 Pages).
- Brito et al., "A Cationic Nanoemulsion for the Delivery of Next-generation RNA Vaccines," 2014, *The American Society of Gene & Cell Therapy* (12 Pages).
- Buchholz et al., "Contributions of the structural proteins of severe acute respiratory syndrome coronavirus to protective immunity," 2004, vol. 101, Issue 26, *Biological Sciences*.
- Buschmann et al., "Chitosans for Delivery of Nucleic Acids," 2013, *Advanced Drug Delivery Reviews*, pp. 1234-1270.
- Cassimeris et al., "Lewin & 2019s Cells," 2011, Jones and Bartlett Publishers, vol. 2 (109 Pages).
- Center for Drug Evaluation and Research Approval Package for New Drug Application No. 50-718/S-50 for Doxil (97 Pages).
- Chalkias et al., "Safety, immunogenicity and antibody persistence of a bivalent Beta-containing booster vaccine against COVID-19: a phase 2/3 trial," 2022, *Nature Medicine*, pp. 2388-2397.
- Chang et al., Recent Insights into the Developments of Therapeutics Against Coronavirus Disease by Targeting N Protein, 2016, *Drug Discovery Today*, No. 4, pp. 562-572.
- Cherry et al., "SARS: The First Pandemic of the 21st Century," 2004, *Pediatric Research*, vol. 56, Issue 1, pp. 1-5.
- Coelho et al., "Safety and Efficacy of RNAi Therapy for Transthyretin Amyloidosis," 2013, vol. 369, Issue 9, *New England J. Med.* (11 Pages).
- Curriculum Vitae of Daniel O. Griffin, M.D., Ph.D.
- Curriculum Vitae of James J. Moon, Ph.D.
- Curriculum Vitae of Deborah Fuller (Deposition Exhibit).
- Declaration of Amy K. Wigmore in IPR2023-01358 dated Apr. 16, 2024.
- Declaration of Amy K. Wigmore in IPR2023-01359 dated Apr. 16, 2024.
- Declaration of Andrew J. Danford in IPR2023-01358 dated Jul. 1, 2024.
- Declaration of Daniel O. Griffin, M.D., Ph.D. in IPR2023-01358 dated Aug. 27, 2023.
- Declaration of Daniel O. Griffin, M.D., Ph.D. in IPR2023-01359 dated Aug. 27, 2023.

(56)

References Cited**OTHER PUBLICATIONS**

Declaration of Deborah H. Fuller, Ph.D., in IPR2023-01358 dated May 30, 2024 (public redacted version of Ex. 2199).

Declaration of Deborah H. Fuller, Ph.D., in IPR2023-01359 dated May 30, 2024. (public redacted version of Ex. 2199).

Declaration of Douglas L. Sawyer in IPR2023-01358 dated Jul. 17, 2024.

Declaration of Douglas L. Sawyer in IPR2023-01359 dated Jul. 17, 2024.

Declaration of Gabrielle Landes in IPR2023-01358 dated Apr. 25, 2024.

Declaration of Gabrielle Landes in IPR2023-01359 dated Apr. 25, 2024.

Declaration of James E. Malackowski in IPR2023-01358 dated May 29, 2024.

Declaration of James E. Malackowski in IPR2023-01359 dated May 29, 2024.

Declaration of James J. Moon, Ph.D. in IPR2023-01358 dated Apr. 25, 2023.

Declaration of James J. Moon, Ph.D. in IPR2023-01359 dated Apr. 25, 2023.

Declaration of Joshua L. Stern in IPR2023-01358 dated Apr. 17, 2024.

Declaration of Joshua L. Stern in IPR2023-01359 dated Apr. 17, 2024.

Declaration of Kathryn S. Kayali in IPR2023-01358 dated Apr. 16, 2024.

Declaration of Kathryn S. Kayali in IPR2023-01359 dated Apr. 26, 2024.

Declaration of Kevin S. Prussia in IPR2023-01358 dated Apr. 17, 2024.

Declaration of Kevin S. Prussia in IPR2023-01359 dated Apr. 17, 2024.

Declaration of L. Ross Pierce in IPR2023-01358 dated Aug. 30, 2024 (Redacted public version).

Declaration of L. Ross Pierce in IPR2023-01359 dated Aug. 30, 2024. (Redacted public version).

Declaration of M. Ryan Meuth in IPR2023-01358 dated Apr. 25, 2024.

Declaration of M. Ryan Meuth in IPR2023-01359 dated Apr. 25, 2024.

Declaration of Philp Krause, M.D., in IPR2023-01359 dated May 28, 2024. (public redacted version of Ex. 2200).

Declaration of Philp Krause, M.D., in IPR2023-01358 dated May 28, 2024. (public redacted version of Ex. 2200).

Declaration of Thomas H. L. Selby in IPR2023-01358 dated Apr. 26, 2024.

Declaration of Thomas H. L. Selby in IPR2023-01359 dated Apr. 26, 2024.

Declaration of W. Christopher Bakewell in IPR2023-01358 dated Aug. 30, 2024.

Declaration of W. Christopher Bakewell in IPR2023-01359 dated Aug. 30, 2024.

Declaration of Warren C.W. Chan, Ph.D. in IPR2023-01358 dated May 29, 2024.

Declaration of Warren C.W. Chan, Ph.D. in IPR2023-01359 dated May 29, 2024.

Defendants' Supplemental Patent-Related Disclosures, Exhibit C—Invalidity Contentions of U.S. Pat. No. 10,702,600 (Jun. 27, 2023)(Excerpts).

Defendants' Patent-Related Disclosures, Exhibit D—Invalidity Contentions of U.S. Pat. No. 10,933,127, 2023, pp. 1-5 (Excerpts).

Defendants' Patent-Related Disclosures, Exhibit C—Invalidity Contentions of U.S. Pat. No. 10,702,600 (Mar. 31, 2023) (Excerpts).

Defendants' Supplemental Patent-Related Disclosures, Exhibit D—Invalidity Contentions of U.S. Pat. No. 10,933,127, 2023, pp. 1-110 (Excerpts).

Deming et al., "Vaccine Efficacy in Senescent Mice Challenged with Recombinant SARS-CoV Bearing Epidemic and Zoonotic Spike Variants," 2006, vol. 3, Issue 12, PLOS Medicine.

Deposition of Daniel O. Griffin, M.D., Ph.D. (May 22, 2024), in *BioNTech SE and Pfizer Inc. v. Moderna TX, Inc.*, in IPR2023-01358 and -01359.

Deposition of Daniel O. Griffin, M.D., Ph.D. (Sep. 20, 2024) in *BioNTech SE and Pfizer Inc. v. Moderna TX, Inc.*, in IPR2023-01358 and -01359.

Deposition of Deborah Fuller, Ph.D., dated Aug. 2, 2024.

Deposition of James E. Malackowski dated Jul. 22, 2024 and Errata Sheet dated Aug. 27, 2024.

Deposition of James J. Moon, Ph. D. (May 10, 2024), in *BioNTech SE and Pfizer Inc. v. Moderna TX, Inc.*, in IPR2023-01358 and -01359.

Deposition of James J. Moon, Ph.D. (Sep. 13, 2024) in *BioNTech SE and Pfizer Inc. v. Moderna TX, Inc.*, in IPR2023-01358 and -01359.

Deposition of L. Ross Pierce, M.D., (Oct. 4, 2024) in *BioNTech SE and Pfizer Inc. v. Moderna TX, Inc.*, in IPR2023-01358. (redacted version of Ex. 2256).

Deposition of L. Ross Pierce, M.D., (Oct. 4, 2024) in *BioNTech SE and Pfizer Inc. v. Moderna TX, Inc.*, in IPR2023-01359. (redacted version of Ex. 2256).

Deposition of Philip Krause, M.D., dated Aug. 14, 2024 (redacted).

Deposition of W. Christopher Bakewell (Oct. 3, 2024) in *BioNTech SE and Pfizer Inc. v. Moderna TX, Inc.*, in IPR2023-01358 and -01359.

Deposition of Warren C.W. Chan Ph.D., dated Jul. 30, 2024.

Division of Microbiology and Infectious Diseases (DMID) Protocol, Phase 1, Open-Label, Dose-Ranging Study of the Safety and Immunogenicity of 2019-nCoV Vaccine (mRNA-1273) in Healthy Adults, 2020, pp. 1-10 (Excerpts).

Division of Microbiology and Infectious Diseases (DMID) Protocol, "Phase I, Open-Label, Dose-Ranging Study of the Safety and Immunogenicity of 2019-nCoV Vaccine (mRNA-1273) in Healthy Adults," 2020, pp. 1-12 (Excerpts).

Division of Microbiology and Infectious Diseases (DMID) Protocol (Feb. 14, 2020); MOD_000477491-MOD_000477561 at MOD_000477491-MOD_000477498; MOD_000477504-MOD_000477509 (Excerpts).

Division of Microbiology and Infectious Diseases (DMID) Protocol (Mar. 30, 2020); MOD_000477134-MOD_000477212 at MOD_000477134-MOD_000477139, MOD_000477148-MOD_000477154 (Excerpts).

Division of Microbiology and Infectious Diseases (DMID) Protocol (May 20, 2020); MOD_000471975-MOD_000472059 at MOD_000471975-MOD_000471983; MOD_000471990-MOD_000471992; MOD_000471994-MOD_000471997 (Excerpts).

Division of Microbiology and Infectious Diseases (DMID) Protocol, "Phase I, Open-Label, Dose-Ranging Study of the Safety and Immunogenicity of 2019-nCoV Vaccine (mRNA-1273) in Healthy Adults," 2020, pp. 1-8 (Excerpts).

Drummond et al., "The Evolutionary Consequences of Erroneous Protein Synthesis," 2009, Nature Reviews Genetics (10 Pages).

European Medicines Agency Assessment Report for COVID-19 Vaccine Moderna, 2021, pp. 1-169.

European Medicines Agency, Assessment Report for Spikevax (Sep. 14, 2023).

European Medicines Agency, Nuvaxovid (available at <https://www.ema.europa.eu/en/medicines/human/EPAR/nuvaxovid>).

FDA, E6(R2) Good Clinical Practice: Integrated Addendum to ICH E6(R1) (Mar. 2018).

FDA, Guidance for Industry E6 Good Clinical Practice: Consolidated Guidance (Apr. 1996).

Felgner et al., "Lipofection: A highly efficient, lipid-mediated DNA-transfection procedure," 1987, 84 Proc. Natl. Acad. Sci. USA (5 Pages).

Fuller, Deborah, The Conversation, "How mRNA and DNA vaccines could soon treat cancers, HIV, autoimmune disorders and genetic diseases," 2022, pp. 1-5.

Gandhi et al., "Immunization of HIV-1-Infected Persons with Autologous Dendritic Cells Transfected with mRNA Encoding HIV-1 Gag and Nef: Results of a Randomized, Placebo-Controlled Clinical Trial, 71 J," 2016, Acquired Immune Deficiency Syndrome, Basic Translational Science.

(56)

References Cited**OTHER PUBLICATIONS**

- Gao et al., "Nonviral gene delivery: What we know and what is next," 2007, The AAPS Journal (13 Pages).
- Gilleron et al., "Image-Based Analysis of Lipid Nanoparticle-Mediated siRNA Delivery, Intracellular Trafficking and Endosomal Escape," 2013, Nature Biotech. 638.
- Ginn et al., "Gene Therapy Clinical Trials Worldwide to 2017: An Update," 2018, 20 J. Gene Med, Article e3015.
- Gorbalenya, A., E., "The Species Severe Acute Respiratory Syndrome-related Coronavirus: Classifying 2019-nCoV and Naming it SARS-CoV-2, 5," 2020, Nature Microbiology, pp. 536-544.
- Griffin et al., "Co-infection with Hepatitis C Virus Increases Mortality in HIV-1 Infected Patients Through Increased Liver-Related Deaths Rather Than by Increasing Malignancy Related Deaths," 2016, vol. 7, Issue 4, J. Aids and Clin. Res. (3 Pages).
- Griffin et al., "Cytokine storm of a different flavour: the different cytokine signature of SARS-CoV-2 the cause of COVID-19 from the original SARS outbreak," Journal of Global Antimicrobial Resistance, pp. 1-3.
- Griffin et al., "Human B1 cells in umbilical cord and adult peripheral blood express the novel phenotype CD20 CD27 CD43 CD70-," 2011, vol. 208, Issue 1, J. Exp. Med. (14 Pages).
- Griffin et al., "A small CD11b human B1 cell subpopulation stimulates T cells and is expanded in lupus, 2011, vol. 208, Issue 13, J. Exp. Med (8 Pages).
- Hattinger, E., et al. "Prophylactic mRNA Vaccination against Allergy Confers Long-Term Memory Responses and Persistent Protection in Mice," 2015, Journal of Immunology Research 2015, Article 797421.
- Hekele et al., "Rapidly produced SAM? vaccine against H7N9 influenza is immunogenic in mice," 2013, vol. 2, Issue 8, Emerging Microbes and Infections (7 Pages).
- Heyes et al., "Cationic lipid saturation influences intracellular delivery of encapsulated nucleic acids," 2005, 107 J.. Controlled Release, pp. 276-287.
- Hoerr et al., "In vivo application of RNA leads to induction of specific cytotoxic T lymphocytes and antibodies," 2000, 30 Eur. J. Immunol.
- Holtkamp et al., "Modification of antigen-encoding RNA increases stability, translational efficacy, and T-cell stimulatory capacity of dendritic cells," 2006, vol. 108, Issue 13, Gene Therapy.
- Hu et al., "Cytosolic delivery mediated via electrostatic surface binding of protein, virus, or siRNA cargos to pH-responsive core-shell gel particles," 2009, vol. 10, Issue 4, Biomacromolecules (10 Pages).
- Huang, "In Vivo Delivery of RNAi with Lipid-Based Nanoparticles," 2011, 13 Ann. Rev. Biomed. Eng'g.
- International Publication No. WO 2012/006378 A1 (Asserted in *GlaxoSmithKline Biologicals SA et al v. Pfizer Inc et al*, C.A. No. 24-512 (GBW) (D. Del.)).
- Japanese Package Insert for Spikevax Intramuscular Injection (Monovalent: Omicron XBB.1.5).
- Jaume et al., "Anti-Severe Acute Respiratory Syndrome Coronavirus Spike Antibodies Trigger Infection of Human Immune Cells via a pH- and Cysteine Protease-Independent FcγR Pathway," 2011, Journal of Virology, pp. 10582-10597.
- Jeffs et al., "A Scalable, Extrusion-Free Method for Efficient Liposomal Encapsulation of Plasmid DNA," 2005, 22 Pharmaceutical Research (11 Pages).
- Jesse H. Erasmus et al., "An Alphavirus-derived replicon RNA vaccine induces SARS-CoV-2 neutralizing antibody and T cell responses in mice and nonhuman primates," 2020, Sci. Transl. Med. (11 Pages).
- Johansson et al., "Intradermal Electroporation of Naked Replicon RNA Elicits Strong Immune Responses," 2012, PLoS ONE, Article e29732.
- Kam et al., "Antibodies against trimeric S glycoprotein protect hamsters against SARS-CoV challenge despite their capacity to mediate FcγRII-dependent entry into B cells in vitro," 2007, Vaccine (13 Pages).
- Kariko et al., "Modified Uridines Are the Key to a Successful Message," 2021, Nature Reviews Immunology (2 Pages).
- Kariko et al., "Naturally occurring nucleoside modifications suppress the immunostimulatory activity of RNA: Implication for therapeutic RNA development," Curr. Opin. Drug Discov. Devel., pp. 523-532.
- Kariko et al., "Suppression of RNA recognition by Toll-like receptors: The impact of nucleoside modification and the evolutionary origin of RNA," 2005, Immunity (8 Pages).
- Kariko et al., "Increased erythropoiesis in mice injected with submicrogram quantities of pseudouridine-containing mRNA encoding erythropoietin," 20 Molecular Therapy 948 (2012) (Ex. 1028).
- Kim et al., "Generation and Characterization of DNA Vaccines Targeting the Nucleocapsid Protein of Severe Acute Respiratory Syndrome Coronavirus," 2004, vol. 78 J, Issue 9, Virology, pp. 4638-4645.
- King et al., "Virus Taxonomy: Classification and Nomenclature of Viruses," 2012, Elsevier Academic Press (50 Pages).
- Kranz et al., "Systemic RNA Delivery to Dendritic Cells Exploits Antiviral Defense for Cancer Immunotherapy," 2016, Nature, vol. 534, pp. 396-411.
- Kreiter et al., "Intranodal Vaccination with Naked Antigen-Encoding RNA Elicits Potent Prophylactic and Therapeutic Antitumoral Immunity," 2010, Cancer Research, pp. 9301-9040.
- Kutzler et al., "DNA Vaccines: Ready for Prime Time?," 2008, Nature Reviews Genetics, vol. 9, pp. 776-788.
- Langer, R., "New methods of drug delivery," 1990, Science, 1527-1533.
- Leal et al., "Phase I clinical trial of an intranodally administered mRNA-based therapeutic vaccine against HIV-1 infection," 2018, AIDS, vol. 32, Issue 17, Pages.
- Leung et al., "Chapter Four—Lipid nanoparticles for short interfering RNA delivery," 2014, in Advances in Genetic, vol. 88 (40 Pages).
- Liang et al., "Efficient Targeting and Activation of Antigen-Presenting Cells In Vivo after Modified mRNA Vaccine Administration in Rhesus Macaques," 2017, 25 Mol Therapy, pp. 2635-2647.
- Lingshu Wang et al., "Evaluation of candidate vaccine approaches for MERS-CoV," 2015, Nature Communications (11 Pages).
- Liu et al., "The Membrane Protein of Severe Acute Respiratory Syndrome Coronavirus Acts as a Dominant Immunogen Revealed by a Clustering Region of Novel Functionally and Structurally Defined Cytotoxic T-Lymphocyte Epitopes," 2010, vol. 202 J, Infectious Diseases, pp. 1171-1180.
- Lu et al., "Immunogenicity of DNA Vaccines Expressing Human Immunodeficiency Virus Type 1 Envelope Glycoprotein With and Without Deletions in the V1/2 and V3 Regions," AIDS Research & Human Retroviruses, pp. 151-155.
- Marshall et al., "Ch. 79: Regulation and Testing of Vaccines," 2018, Plotkin's Vaccines, 7th edition, pp. 1547-1565.
- Martin et al., "A SARS DNA Vaccine Induces Neutralizing Antibody and Cellular Immune Responses in Healthy Adults in a Phase I Clinical Trial," 2008, Vaccine (6 Pages).
- Matthew G. Stanton & Kerry E. Murphy-Benenato, "Messenger RNA as a Novel Therapeutic Approach," 2017, Topics in Medicinal Chemistry RNA Therapeutics, pp. 1-2 (Excerpts).
- Moderna Form 10-K for the fiscal year ended Dec. 31, 2018.
- Moderna Form 10-K for the fiscal year ended Dec. 31, 2022.
- Moderna Form 10-Q for the quarterly period ended Jun. 30, 2024.
- Moderna Form 8-K, Mar. 16, 2020.
- Moderna, Inc. 2020 Form 10-K.
- Moderna, Inc. 2023 Form 10-K.
- Moderna, Inc., Patents, <https://www.modernatx.com/patents> (3 Pages).
- Moderna, Science & Technology Day dated May 17, 2022.
- Moderna, Vaccines and Related Biological Products Advisory Committee Presentation: Moderna COVID-19 Variant Vaccines (Jun. 15, 2023); MOD_000767652-MOD_000767686.
- Nakayama et al., "Comparison of Current Regulatory Status for Gene-Based Vaccines in the U.S., Europe and Japan," 2015, Vaccines, pp. 186-202.
- National Human Genome Research Institute, "Double Helix," 2023, NIH, <https://www.genome.gov/geneticsglossary/Double-Helix> (2 Pages).

(56)

References Cited**OTHER PUBLICATIONS**

- National Human Genome Research Institute, Central Dogma, 2023, NIH, <https://www.genome.gov/geneticsglossary/Central-Dogma> (4 Pages).
- ONPATRO Prescribing Label (available at https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/210922s000lbl.pdf).
- Openshaw et al., "Immune Responses and Disease Enhancement During Respiratory Syncytial Virus Infection," 2005, *Clinical Microbiology Reviews*, vol. 18, Issue 13, pp. 541-555.
- Package Insert for Spikevax (COVID-19 Vaccine, mRNA) – 2023-2024 Formula (45 Pages).
- Package Insert for Spikevax (COVID-19 Vaccine, mRNA) – 2023-2024 Formula (Apr. 2024).
- Paoletti, E., "Applications of Pox Virus Vectors to Vaccination: An Update," 1996, *Proceedings of the Natl. Acad. of Sci.*, pp. 11349-11353.
- Paramasivam et al., "Endosomal Escape of Delivered mRNA from Endosomal Recycling Tubules Visualized at the Nanoscale," 221 *J. Cell Biology* e202110137.
- Park et al., "Non-Viral COVID-19 Vaccine Delivery Systems," *Advanced Drug Delivery Reviews*, pp. 137-151.
- Pascolo, S., "Vaccination with Messenger RNA (mRNA)," 2008, *Handbook of Experimental Pharmacology*, pp. 221-235.
- Patent Sublicense Agreement Between Cellscript, LLC and ModernaTx, Inc.
- Plotkin, S., A., "Vaccines: Past, Present and Future, 11 *Nature Medicine Supplement* 55," 2005, *Historical Perspective, Nature Medicine*, pp. S5-S11.
- Regla-Nava et al., "Severe Acute Respiratory Syndrome Coronaviruses with Mutations in the E Protein are Attenuated and Promising Vaccine Candidates," 2015, vol. 89, *Journal of Virology* (18 Pages).
- Reply Declaration of Daniel Griffin in IPR2023-01358 dated Aug. 30, 2024. (Redacted public version).
- Reply Declaration of Daniel Griffin in IPR2023-01359 dated Aug. 30, 2024. (Redacted public version).
- Reply Declaration of James Moon in IPR2023-01358 dated Aug. 30, 2024. (Redacted public version).
- Reply Declaration of James Moon in IPR2023-01359 dated Aug. 30, 2024. (Redacted public version).
- Sahdev et al., *Biomaterials for Nanoparticle Vaccine Delivery Systems*, Pharmaceutical Research, 2014, pp. 2563-2582.
- Sahin et al., "mRNA-based therapeutics Developing a new class of drugs," 2014, *Nature Reviews Drug Discovery*.
- Semple et al., "Efficient encapsulation of antisense oligonucleotides in lipid vesicles using ionizable aminolipids: Formation of novel small multilamellar vesicle structures," 2001, *Biochimica Et Biophysica Acta*, pp. 152-166.
- Semple et al., "Rational design of cationic lipids for siRNA delivery," 2010, *Nature Biotechnology*, pp. 172-178.
- Shaw et al., "The Path to an RSV Vaccine," 2013, *Current Opinion in Virology*, pp. 332-342.
- Slimko et al., "Codon optimization of *Caenorhabditis elegans* GluCl ion channel genes for mammalian cells dramatically improves expression levels," 2003, *Journal of Neuroscience Methods*.
- Song et al., "Middle East Respiratory Syndrome Coronavirus Spike Protein Delivered by Modified Vaccinia Virus Ankara Efficiently Induces Virus-Neutralizing Antibodies," 2013, 87 *J. of Virology*, pp. 1-5.
- Steitz et al., "Effective Induction of Anti-Melanoma Immunity Following Genetic Vaccination with Synthetic mRNA Coding for the Fusion Protein EGFP-TRP2," 2006, *Cancer Immunology*, pp. 246-253.
- Stenler et al., "Safety and efficacy of DNA vaccines: Plasmids vs. minicircles," 2014, vol. 10, Issue 5, *Human Vaccines & Immunotherapeutics*.
- Tenchov et al., "Lipid nanoparticles 2014 From liposomes to mRNA vaccine delivery, a landscape of research diversity and advancement," 2021, *ACS Nano* (34 Pages).
- Updated Curriculum Vitae of Daniel O. Griffin, Ph.D.
- Vennema et al., "Early Death After Feline Infectious Peritonitis Virus Challenge due to Recombinant Vaccinia Virus Immunization," 1990, *Journal of Virology*, pp. 1407-1409.
- Vergati et al., "Strategies for Cancer Vaccine Development," 2010, *Journal of Biomedicine and Biotechnology*, (13 Pages).
- Wan et al., "Comparison of Immunogenicity Between Codon Optimized HIV-1 Thailand Subtype B gp140 and gp145 Vaccines," 2007, *Vaccine*, vol. 25, pp. 4949-4959.
- Weide et al., "Direct Injection of Protamine-protected mRNA: Results of a Phase 1/2 Vaccination Trial in Metastatic Melanoma Patients," 2009, *Immunotherapy* (11 Pages).
- Weide et al., "Results of the First Phase I/II Clinical Vaccination Trial with Direct Injection of mRNA," 2008, *J. Immunotherapy*, pp. 180-188.
- Withoff et al., "Replication-defective Recombinant Semliki Forest Virus Encoding GM-CSF as a Vector System for Rapid and Facile Generation of Autologous Human Tumor Cell Vaccines," 2001, *Gene Therapy*, pp. 1515-1523.
- Wolff et al., "Direct gene transfer into mouse muscle in vivo," 247 *Science* 1465 (1990) (Ex. 1013).
- Yang et al., "Evasion of Antibody Neutralization in Emerging Severe Acute Respiratory Syndrome Coronaviruses," 2005, *PNAS*, vol. 102, Issue 3, pp. 797-801.
- Yoo et al., "Synthesis and Processing of the Haemagglutinin-Esterase Glycoprotein of Bovine Coronavirus Encoded in the E3 Region of Adenovirus," vol. 73, *J. Gen. Virology*, pp. 2591-2600.
- Zhou et al., "Current RNA-Based Therapeutics in Clinical Trials," 2019, *Current Gene Therapy*, pp. 172-196.
- Final Written Decision, IPR2023-01358 dated Mar. 5, 2025.
- Final Written Decision, IPR2023-01359 dated Mar. 5, 2025.
- Alberer et al., "Safety and Immunogenicity of a mRNA Rabies Vaccine in Healthy Adults: An Open-label, Nonrandomised, Prospective, First-in-human Phase 1 Clinical Trial," 2017, *The Lancet*, pp. 1-10.
- Angus Liu, AstraZeneca withdraws US Covid vaccine application, shifts focus to antibody treatments, *Fierce Pharma* (Nov. 10, 2022).
- Berkeley Lovelace Jr. & Noah Higgins-Dunn, Dr. Anthony Fauci says chance of coronavirus vaccine being highly effective is 'not great', *CNBC.com* (last updated Aug. 8, 2020), <https://www.cnbc.com/2020/08/07/coronavirus-vaccine-dr-fauci-says-chances-of-it-being-highly-effective-is-not-great.html>.
- Ernsting et al., "Factors controlling the pharmacokinetics, biodistribution and intratumoral penetration of nanoparticles. *J Control Release*," Dec. 28, 2013;172(3):782-94. doi: 10.1016/j.jconrel.2013.09.013. Epub Sep. 25, 2013.
- Fraiser Kansteiner, It's the end of the line for J&J's COVID shot in the US, *CDC says, Fiercepharma* (May 16, 2023).
- Karin Bok, Accelerated COVID-19 vaccine development: milestones, lessons, and prospects, 54 *Immunology* 1636 (2021).
- Monika Kumari et al., A critical overview of current progress for COVID-19: development of vaccines, antiviral drugs, and therapeutic antibodies, *J. Biomed Sci* 29(68) (2022).
- Prescribing Information for Comirnaty (Pfizer).
- Prescribing Information for Spikevax (Moderna).
- Press Release, Novavax, U.S. FDA Grants Emergency Use Authorization for Novavax COVID-19 Vaccine, Adjuvanted for Individuals Aged 18 and Over (Jul. 13, 2022).
- Szebeni et al., "Activation of complement by therapeutic liposomes and other lipid excipient-based therapeutic products: prediction and prevention," 2011, *Adv Drug Deliv Rev.*, Sep. 16;63(12):1020-30. doi: 10.1016/j.addr.2011.06.017. Epub Jul. 14, 2011.
- Szebeni et al., "Complement activation as a bioequivalence issue relevant to the development of generic liposomes and other nanoparticulate drugs," *Biochem Biophys Res Commun*. Dec. 18, 2015;468(3):490-7. doi: 10.1016/j.bbrc.2015.06.177. Epub Jul. 14, 2015. PMID: 26182876.
- Szebeni J. Complement activation-related pseudoallergy: a stress reaction in blood triggered by nanomedicines and biologicals. *Mol Immunol*. Oct. 2014;61(2): 163-73. doi: 10.1016/j.molimm.2014.06.038. Epub Aug. 12, 2014.
- Walter et al., "Evaluation of the BNT162b2 COVID-19 Vaccine in Children 5 to 11 Years of Age," *New England J. Med.*, 2022; 386: 35-46 (published online Nov. 9, 2021).

Fig. 1

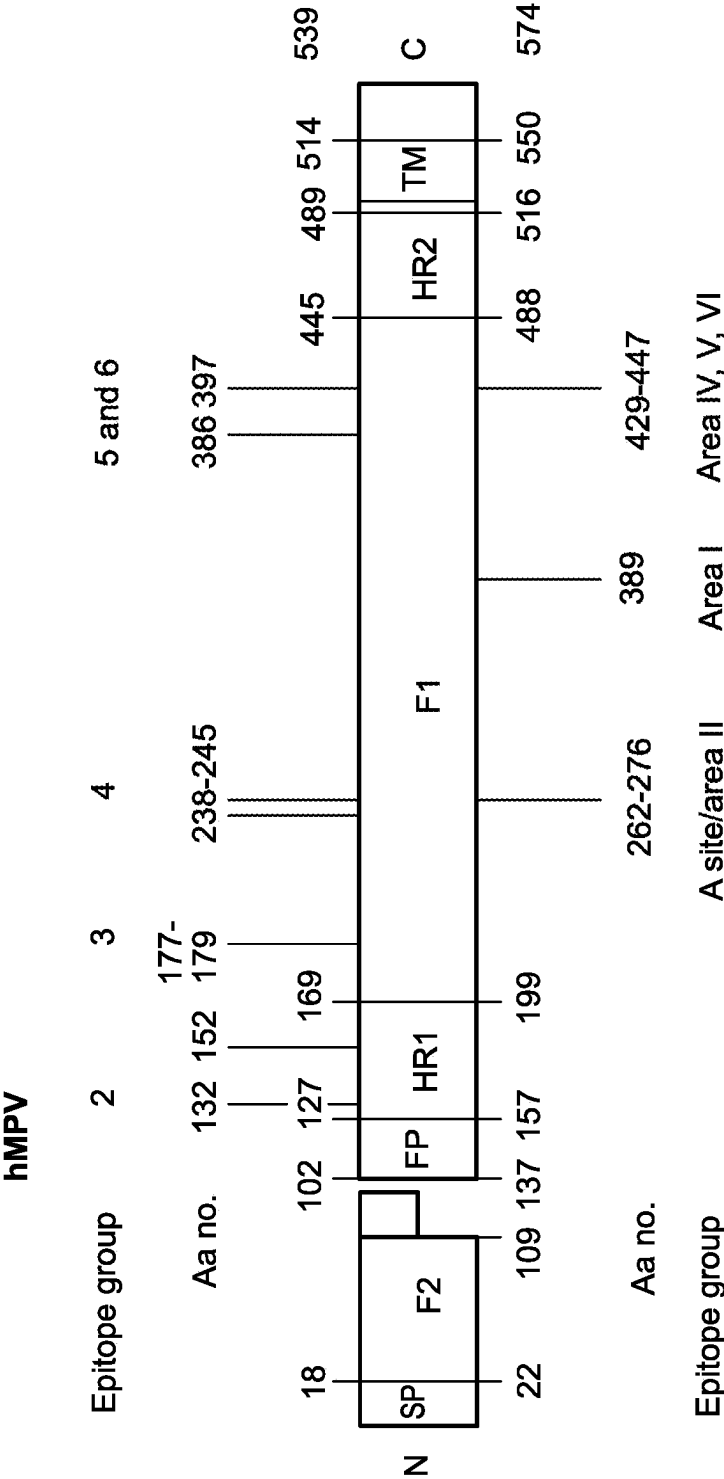


Fig. 2A

Day 0 serum titration

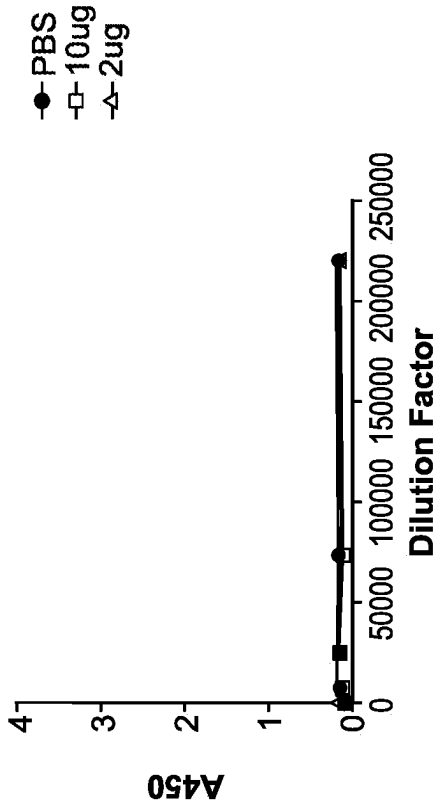


Fig. 2B

Day 14 serum titration

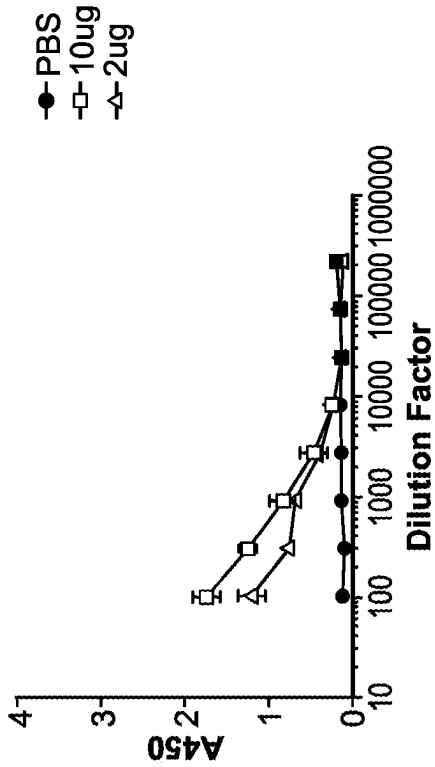


Fig. 2C

Day 35 serum titration

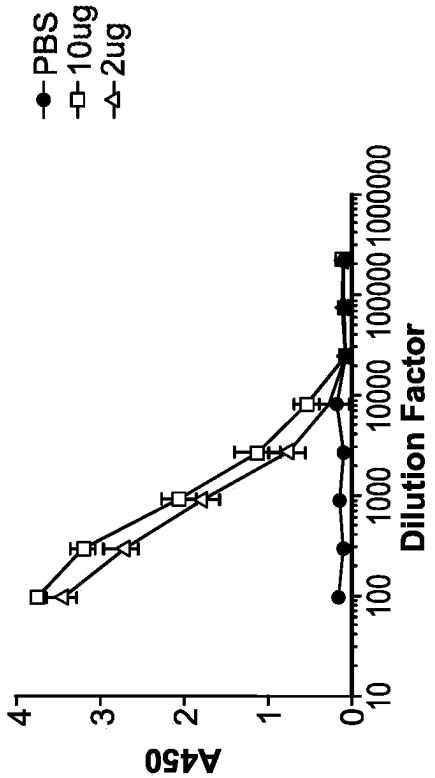


Fig. 3A

Mouse IgG2a - hMPV F specific



Fig. 3B

Mouse IgG1 - hMPV F specific

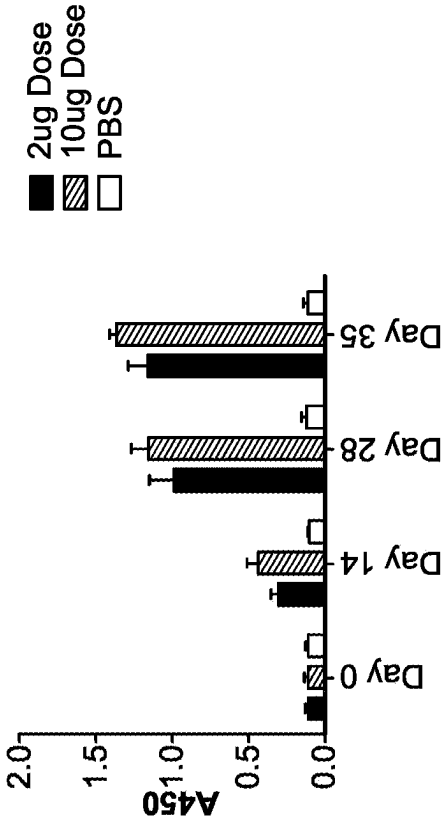


Fig. 3C

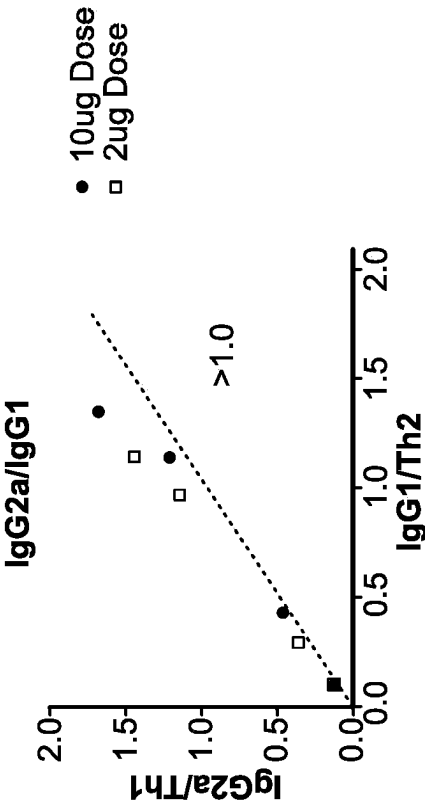
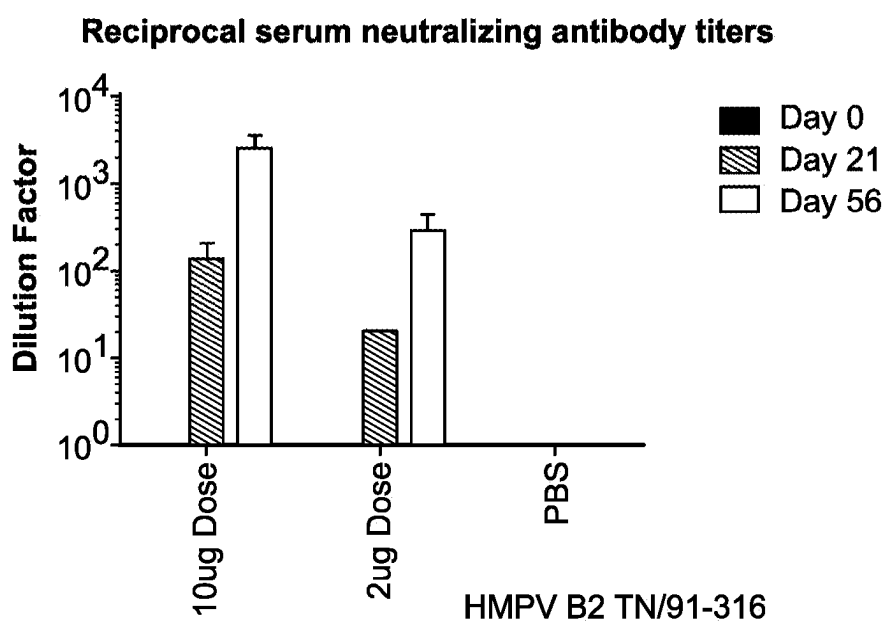
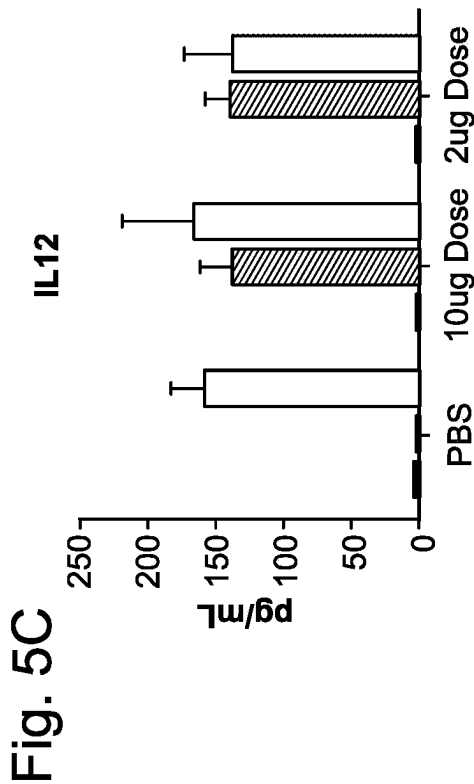
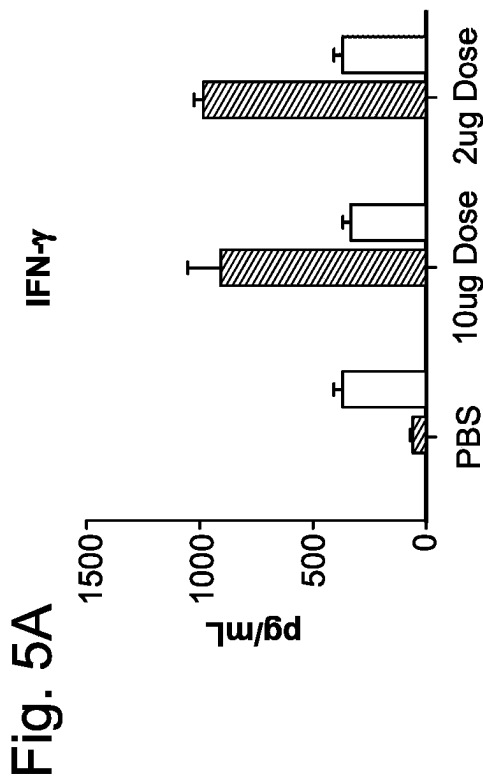
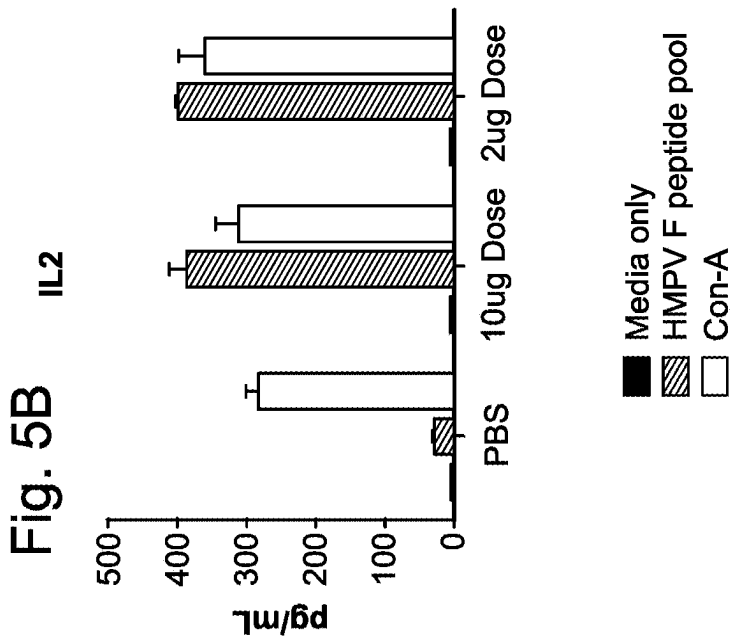
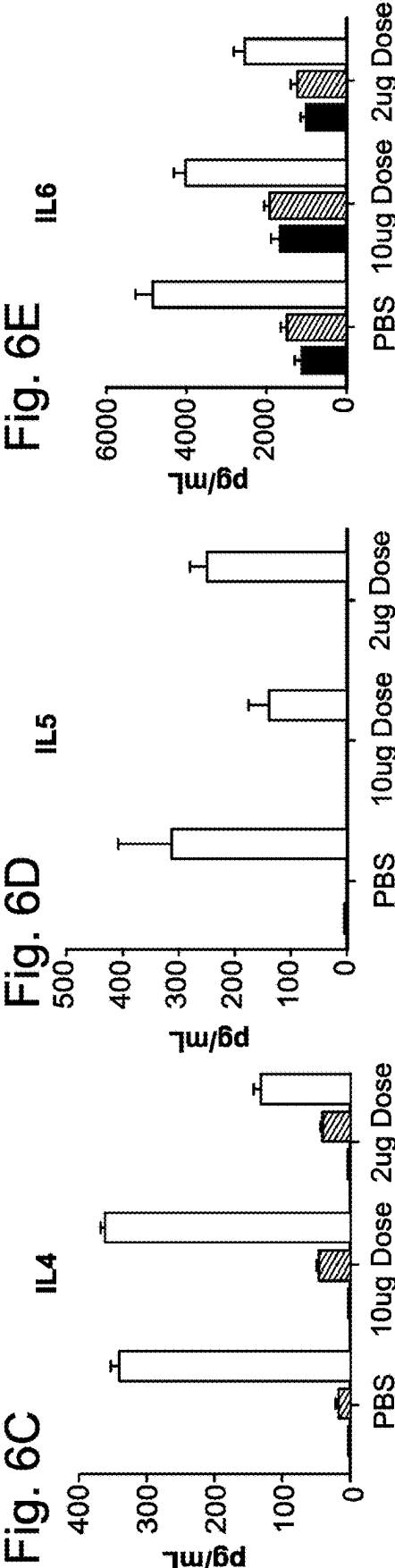
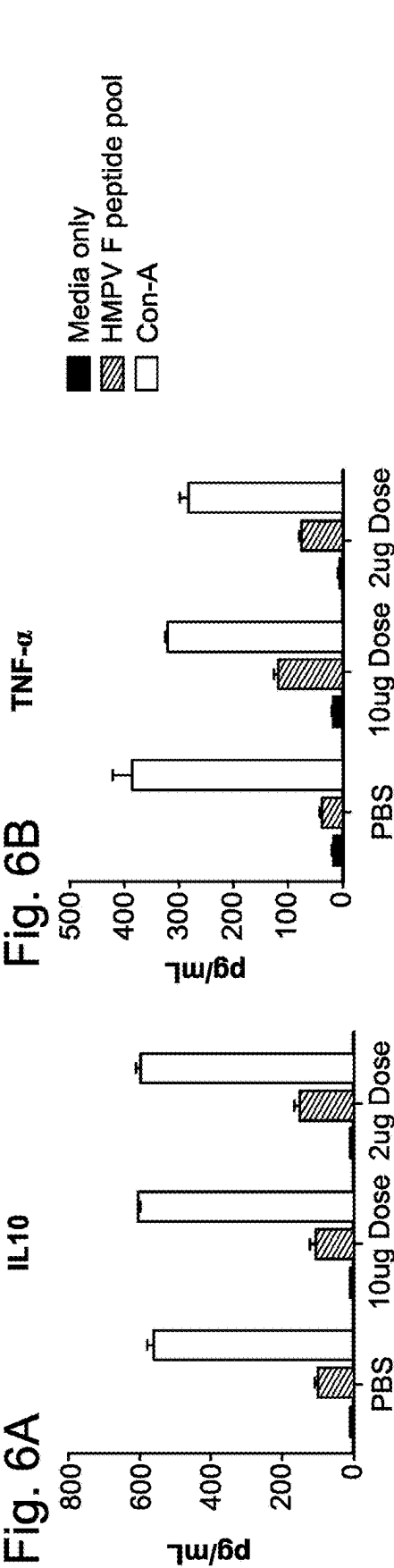
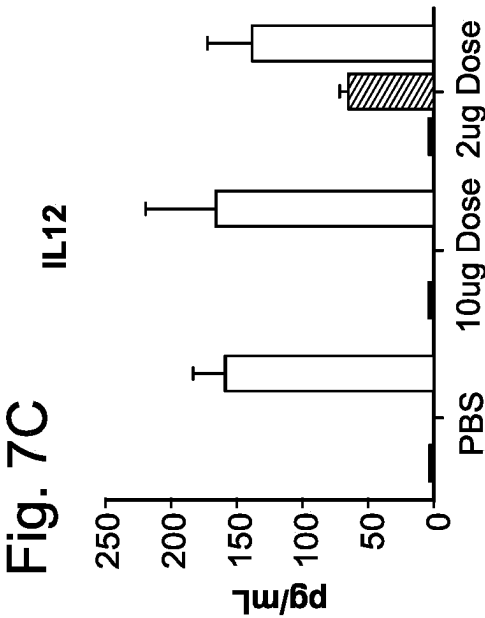
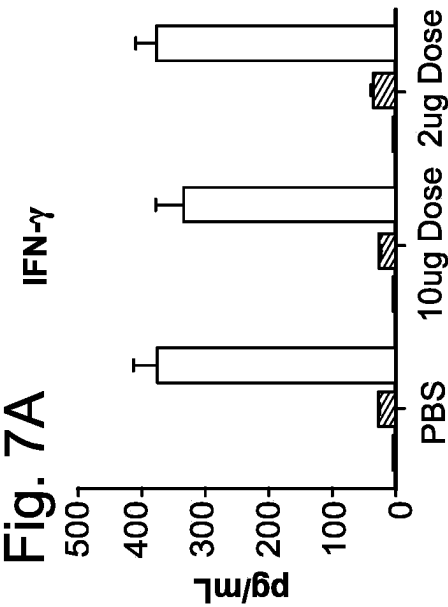
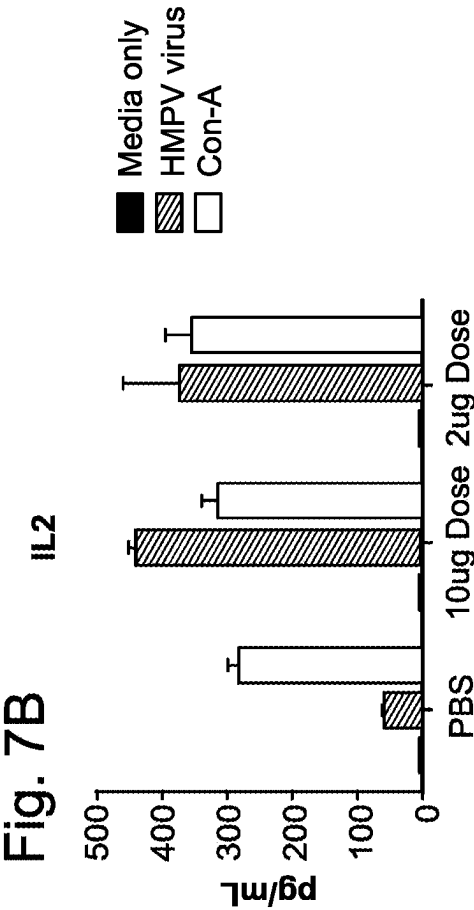


Fig. 4









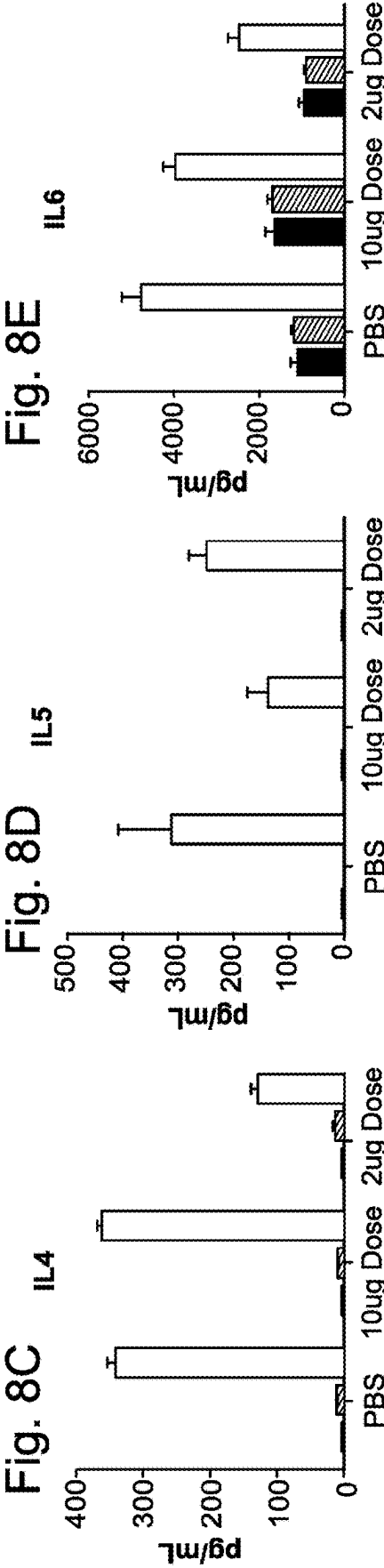
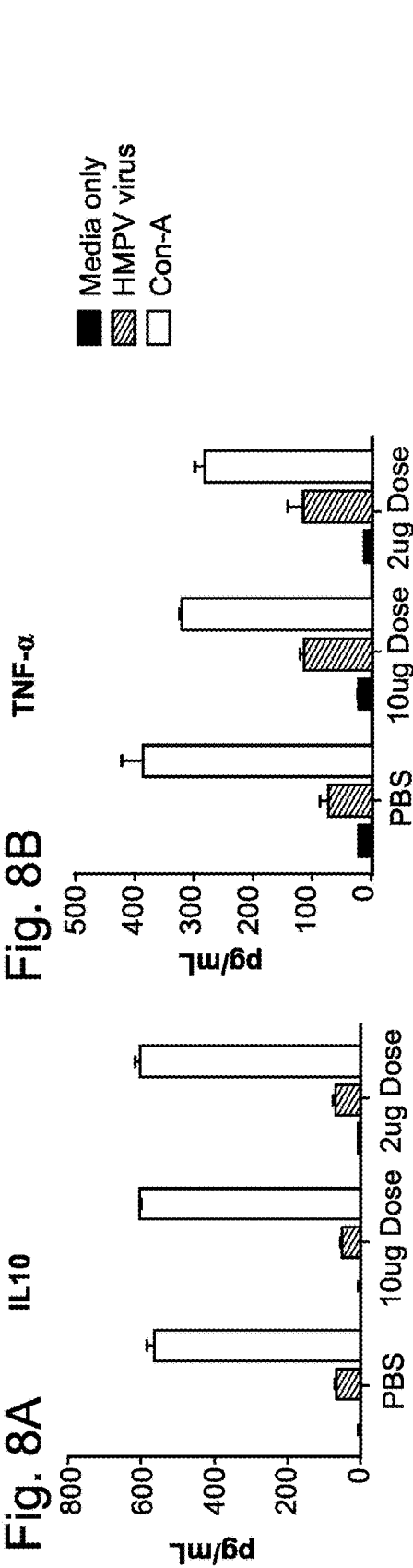


Fig. 9A

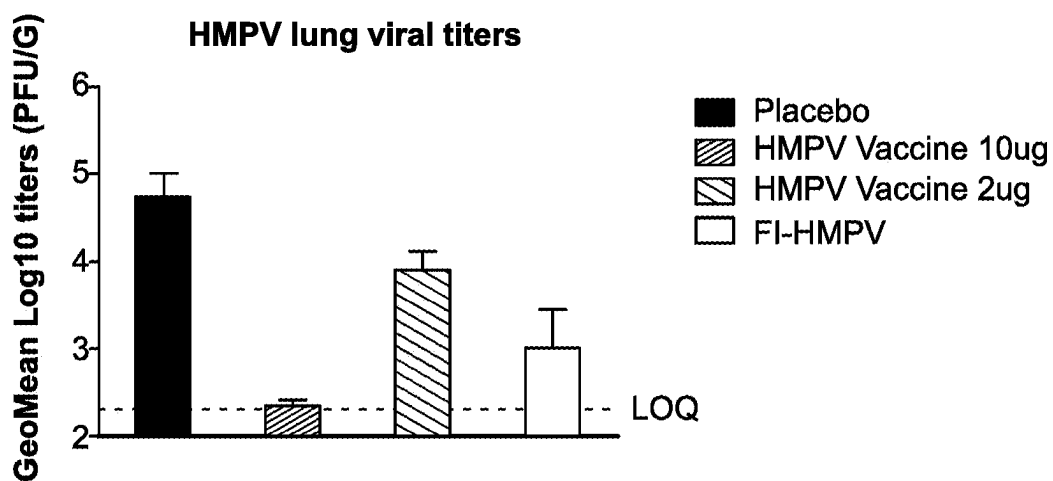


Fig. 9B

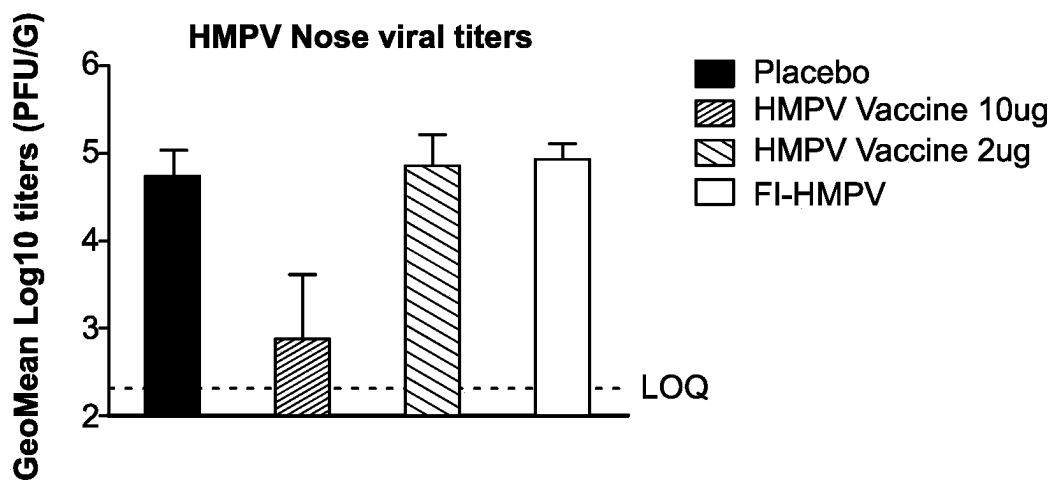


Fig. 10

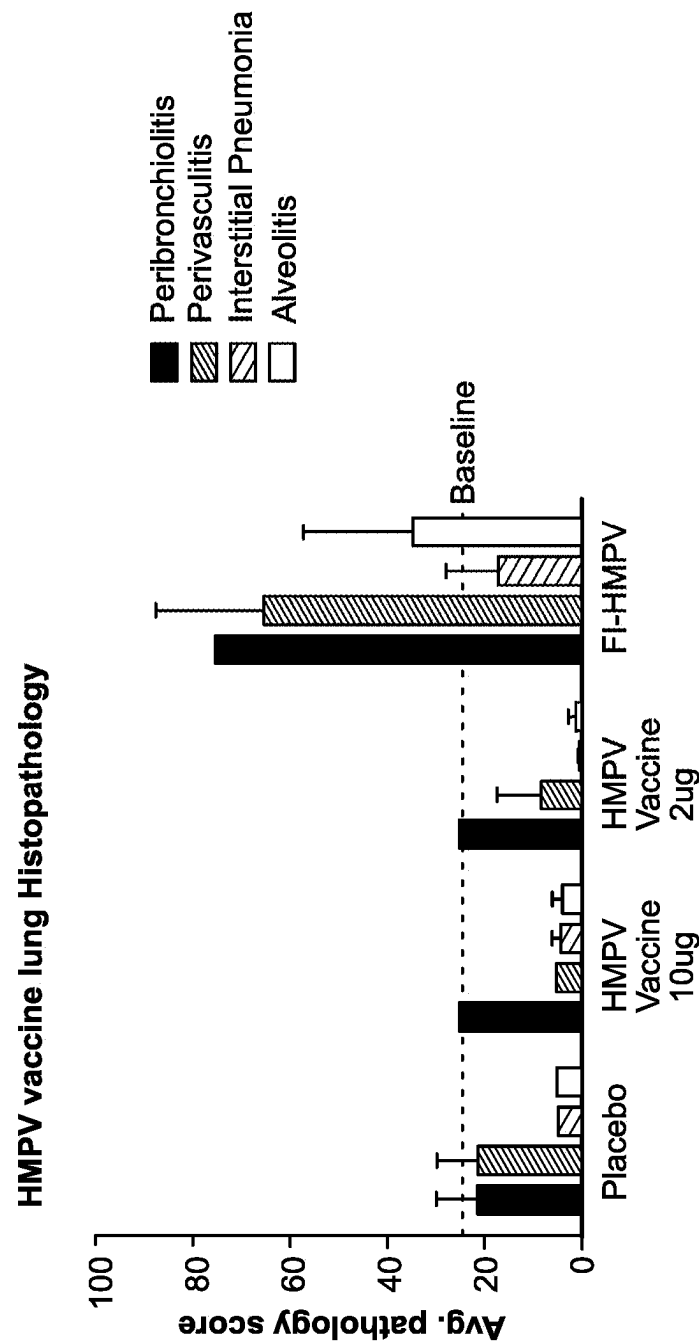
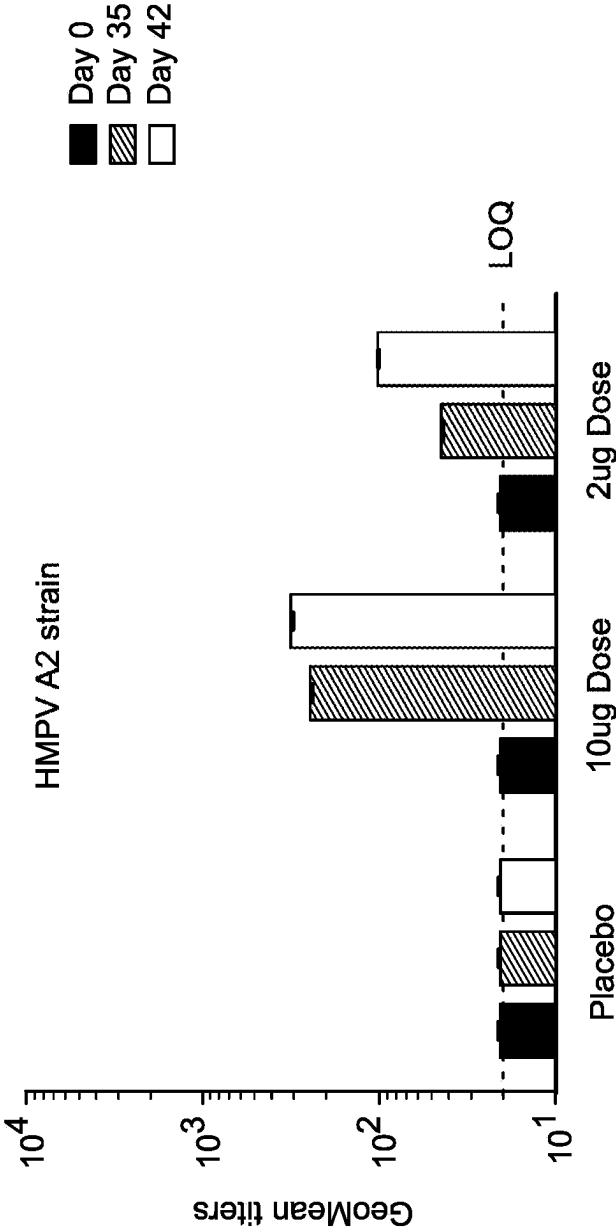


Fig. 11

HMPV neutralization antibody titers in cotton rats



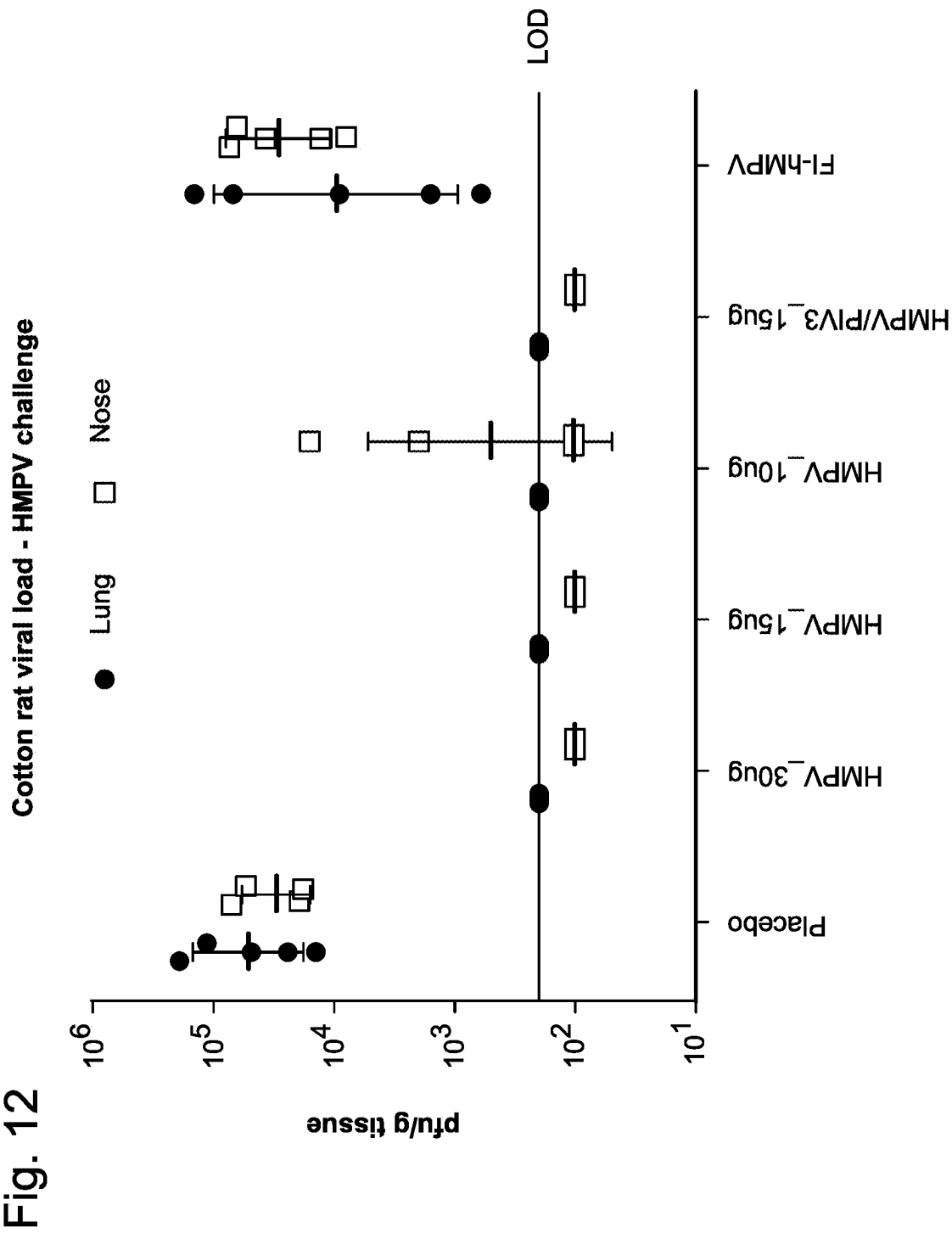


Fig. 13

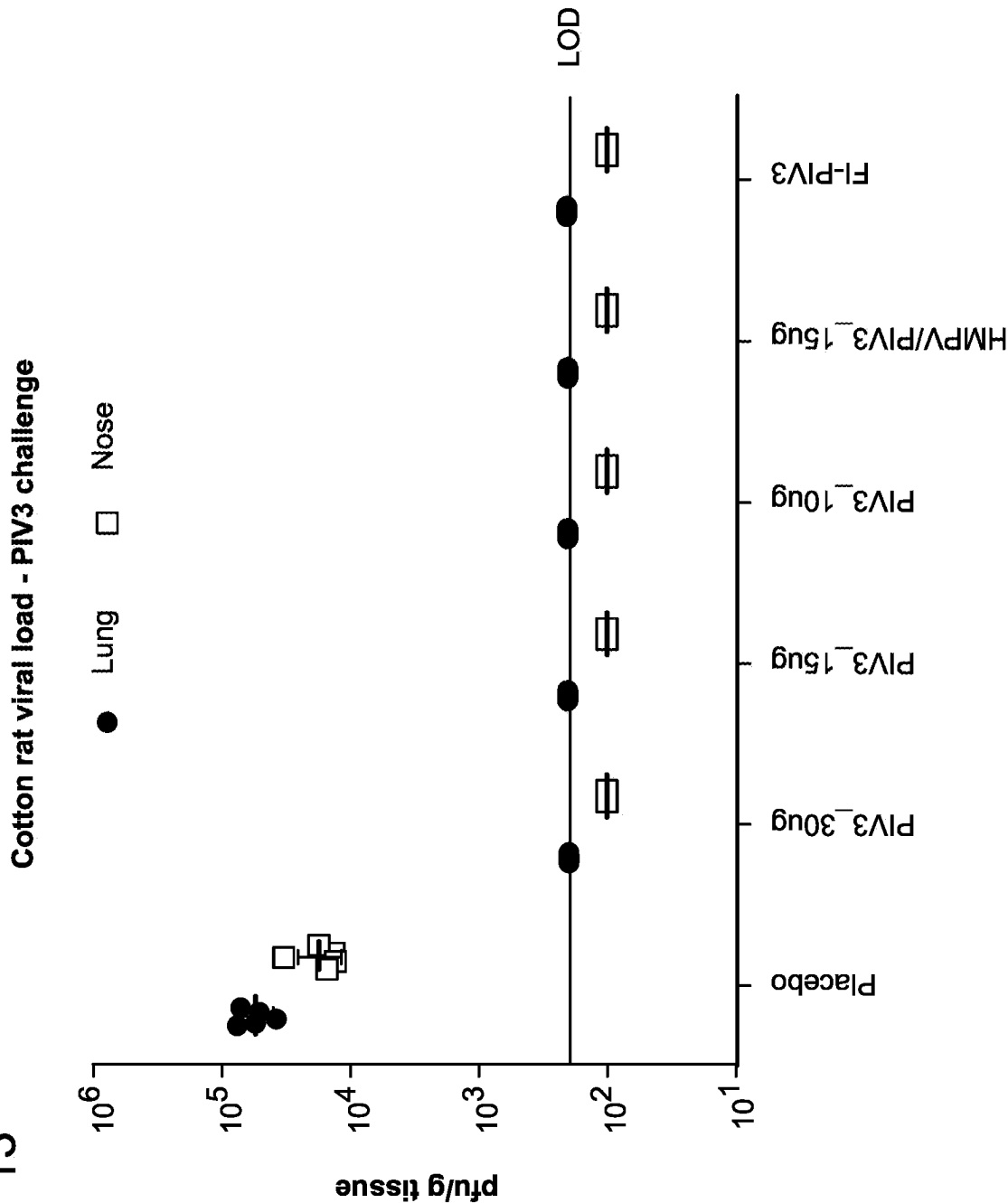


Fig. 14

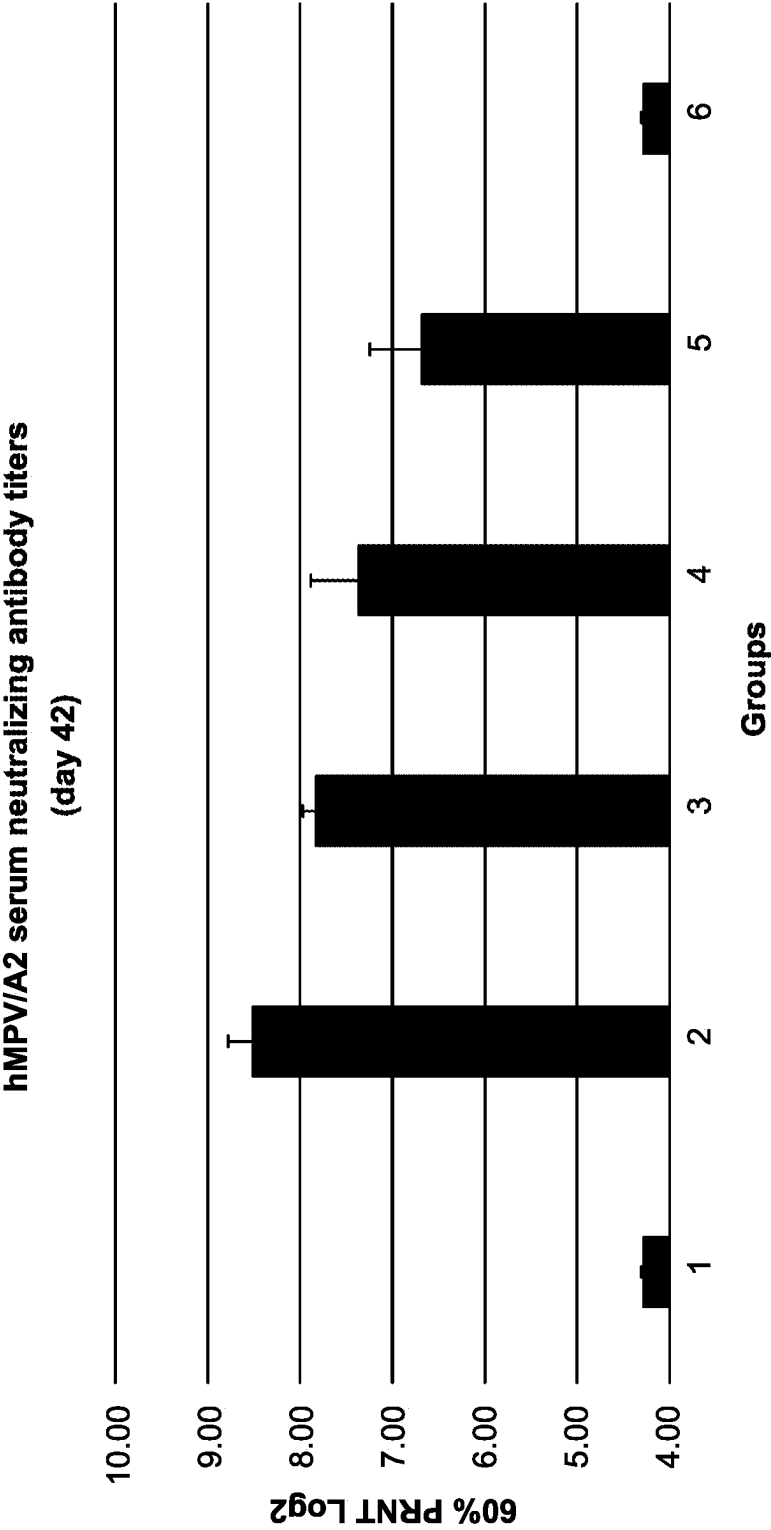


Fig. 15

PIV3 serum neutralizing antibody titers
(day 42)

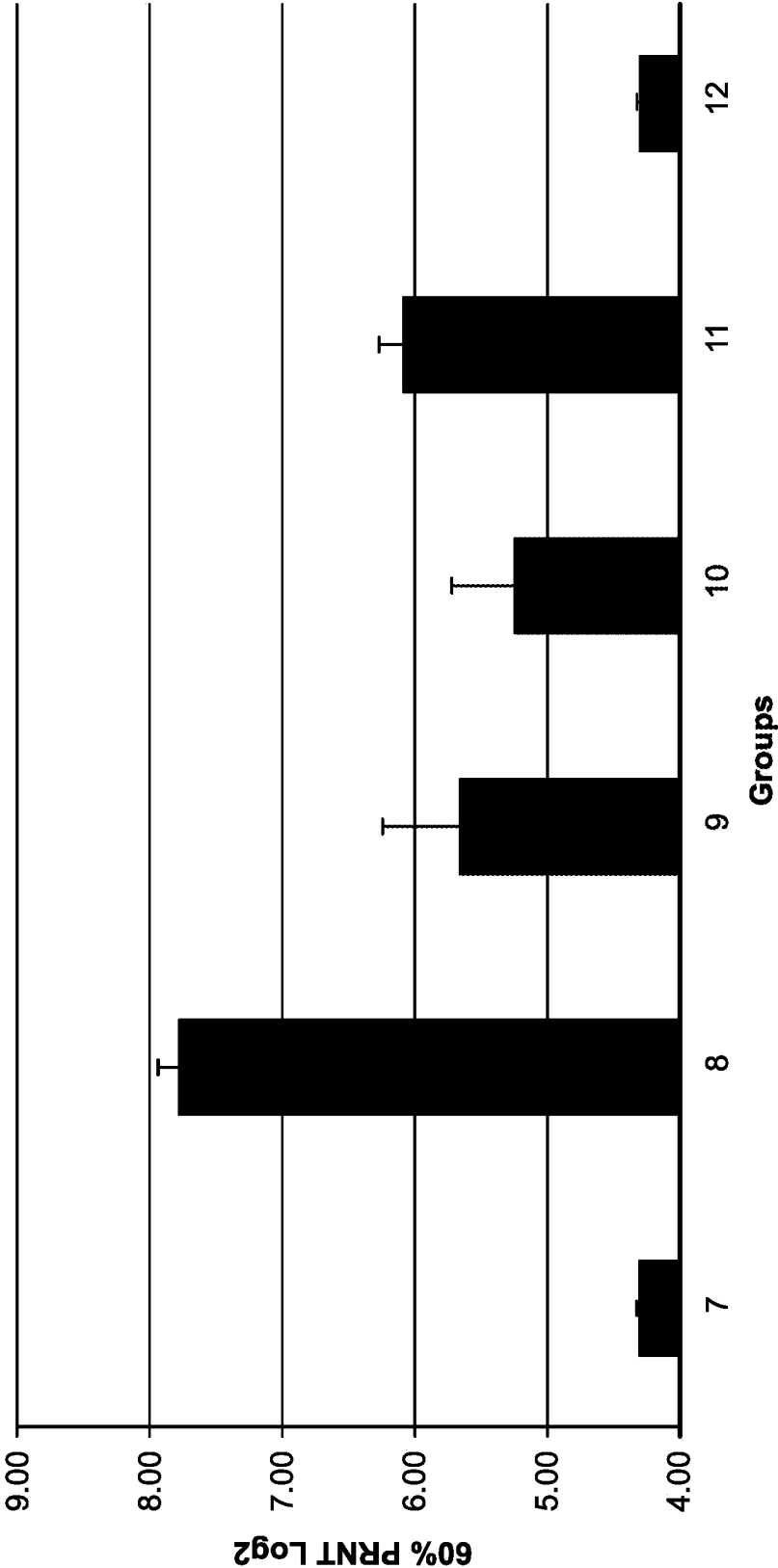


Fig. 16
Cotton rat lung histopathology

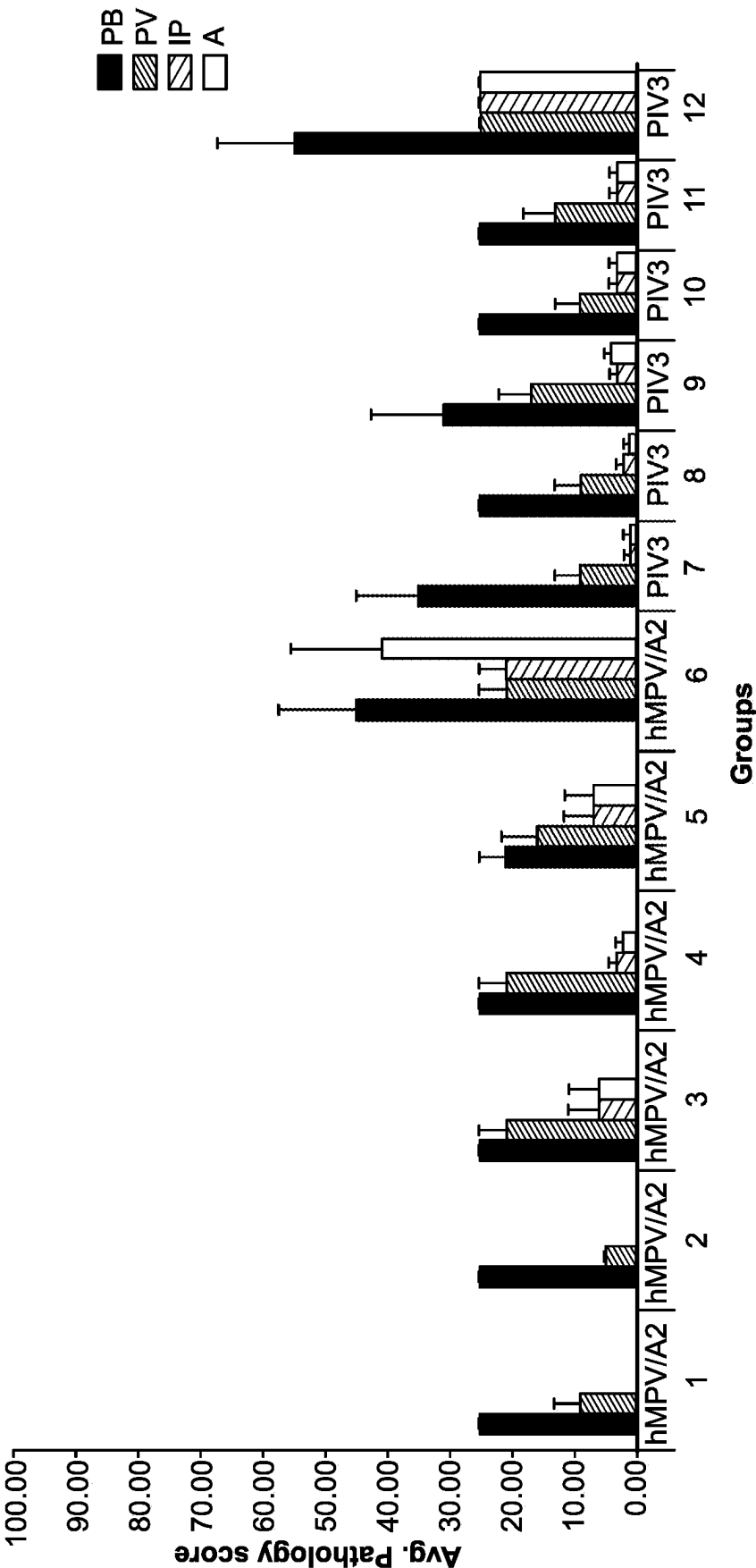


Fig. 17

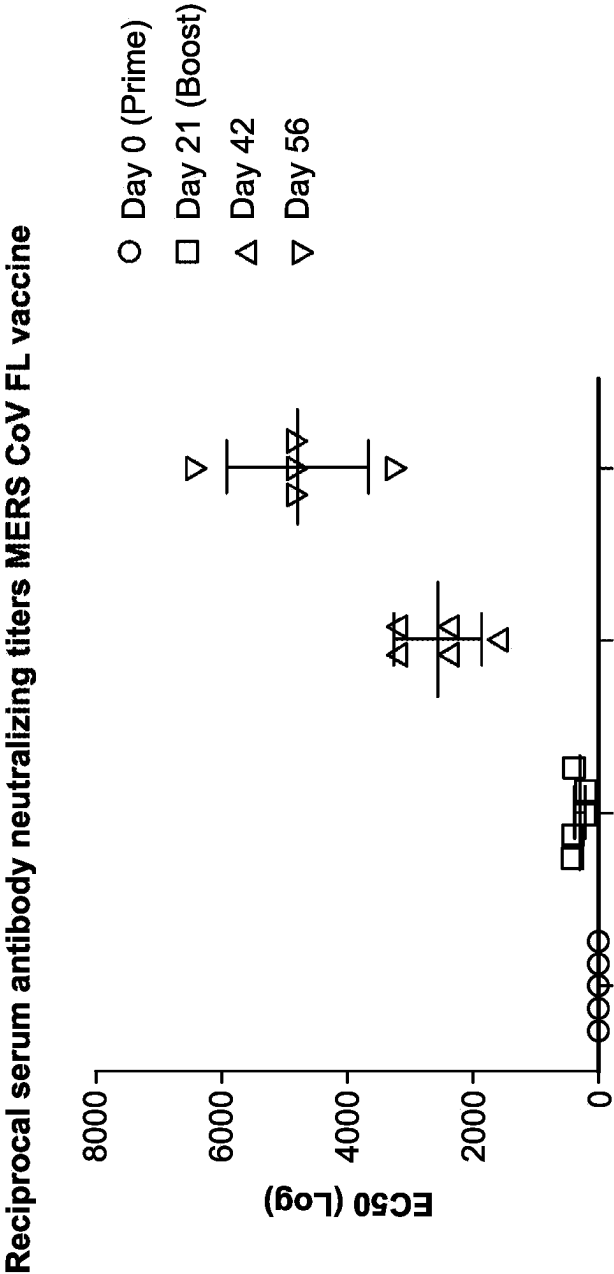


Fig. 18

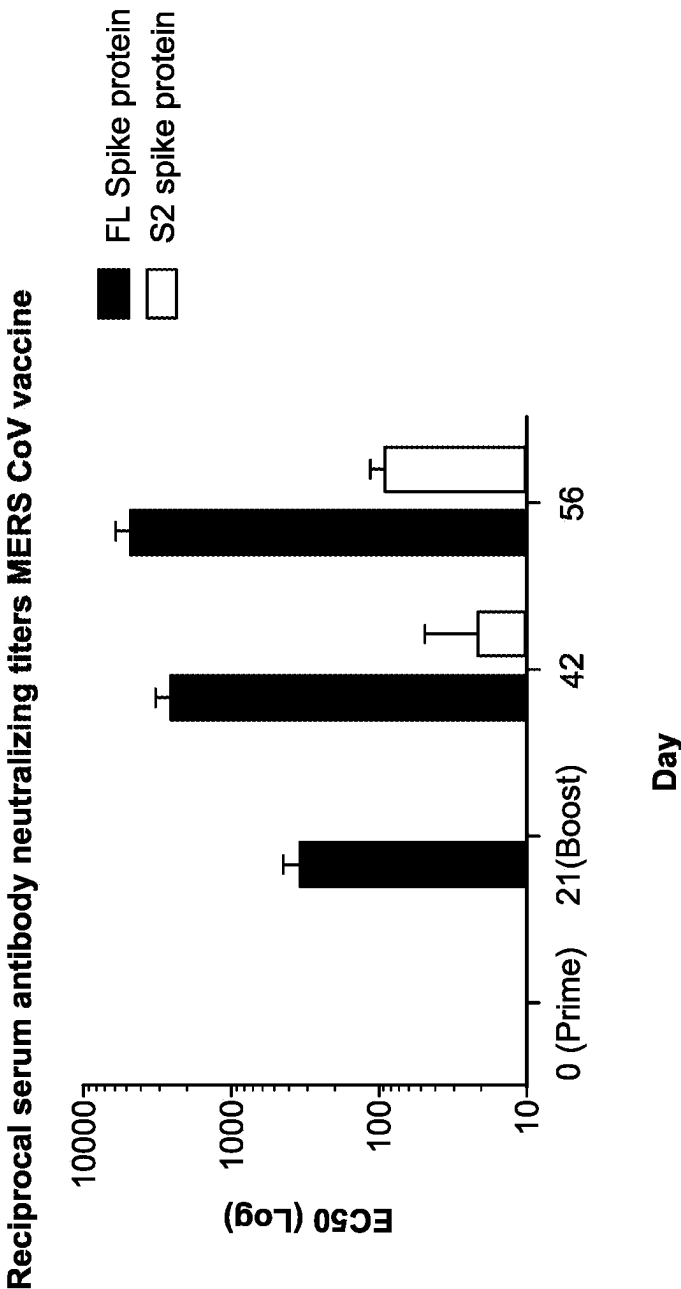


Fig. 19A

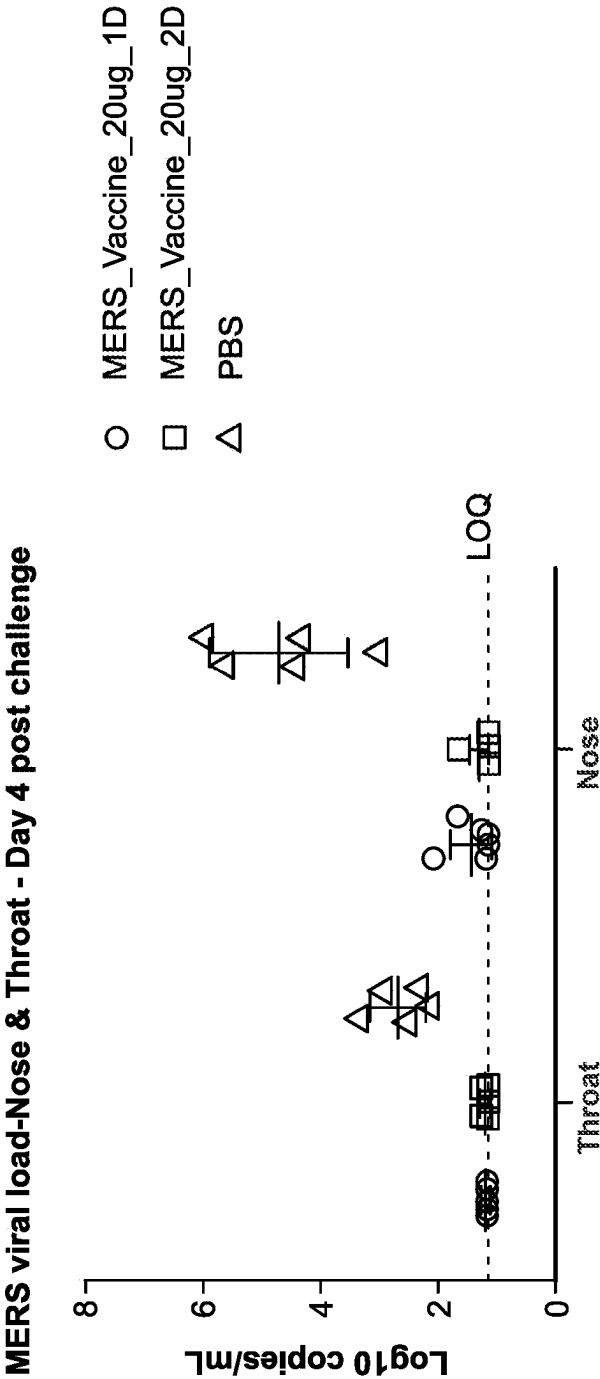


Fig. 19B

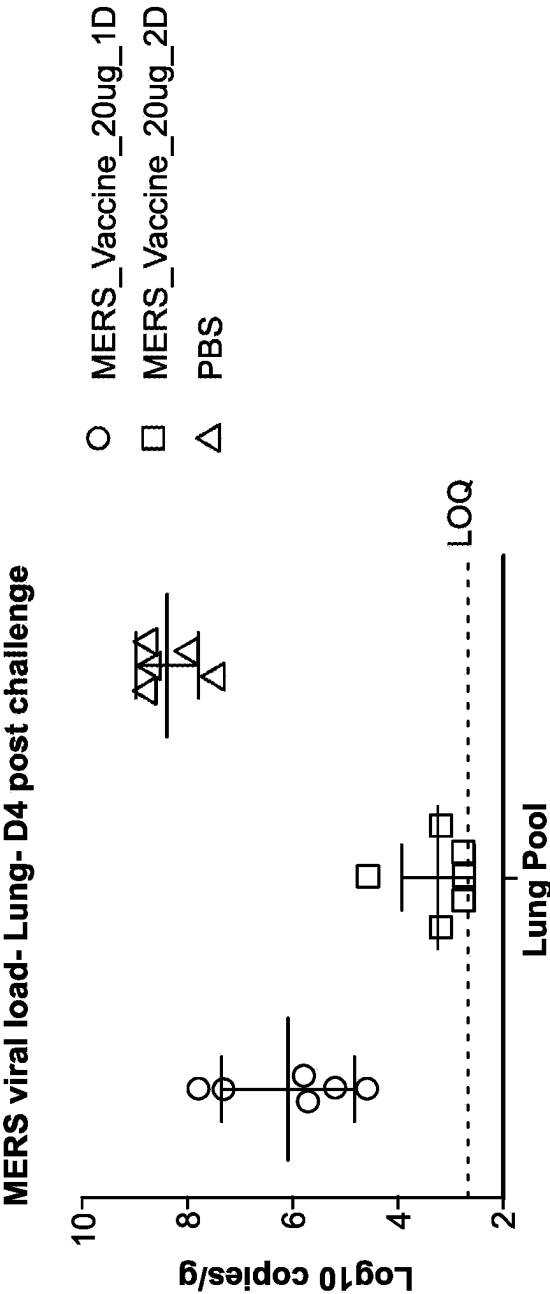


Fig. 19C

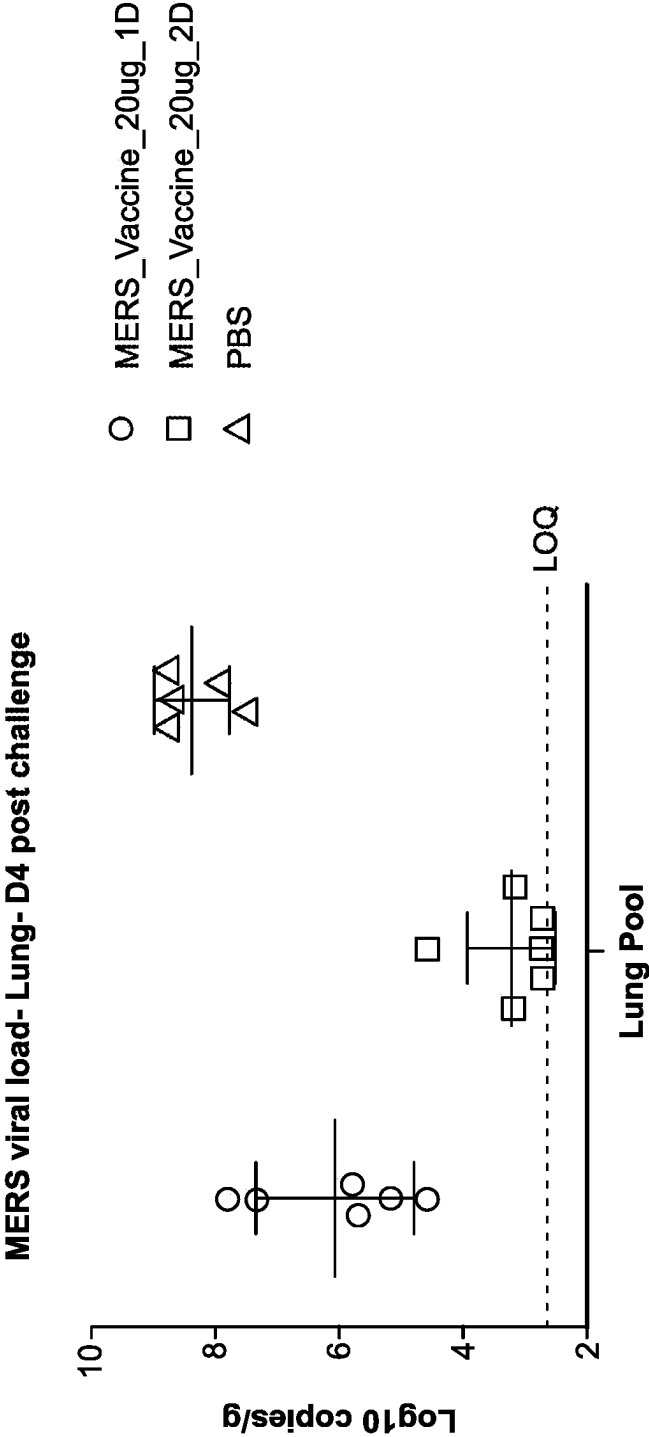


Fig. 20A

MERS-CoV RNA loads in lungs

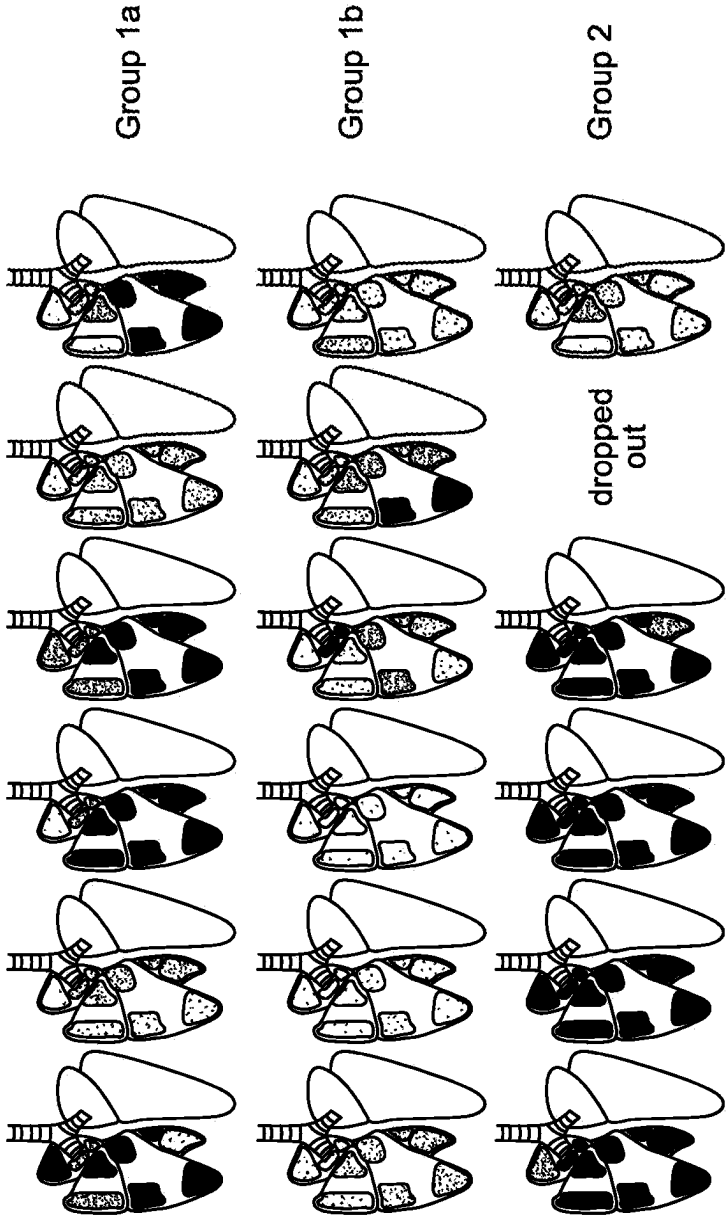
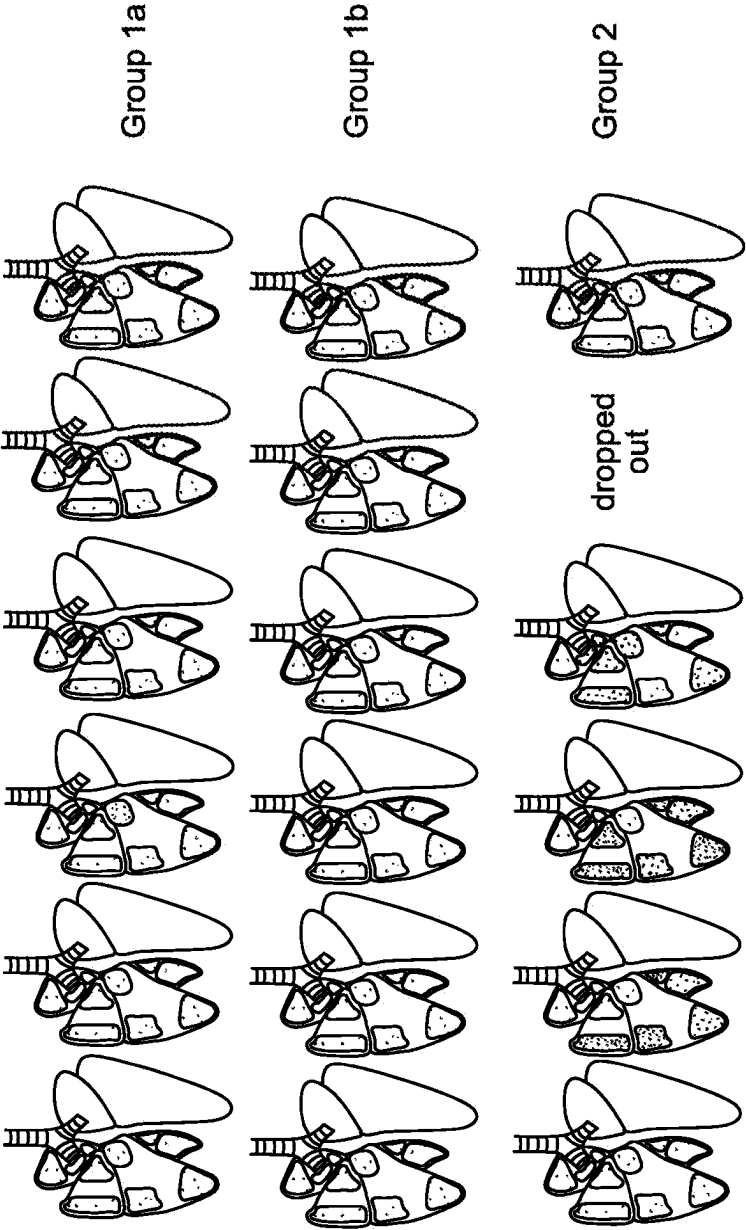


Fig. 20B

MERS-CoV replication in lungs



Legend	TCID50 Range
	negative
	1 - 2
	2 - 3
	3 - 4
	> 4

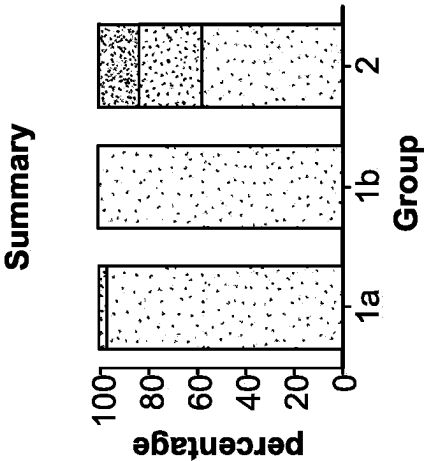
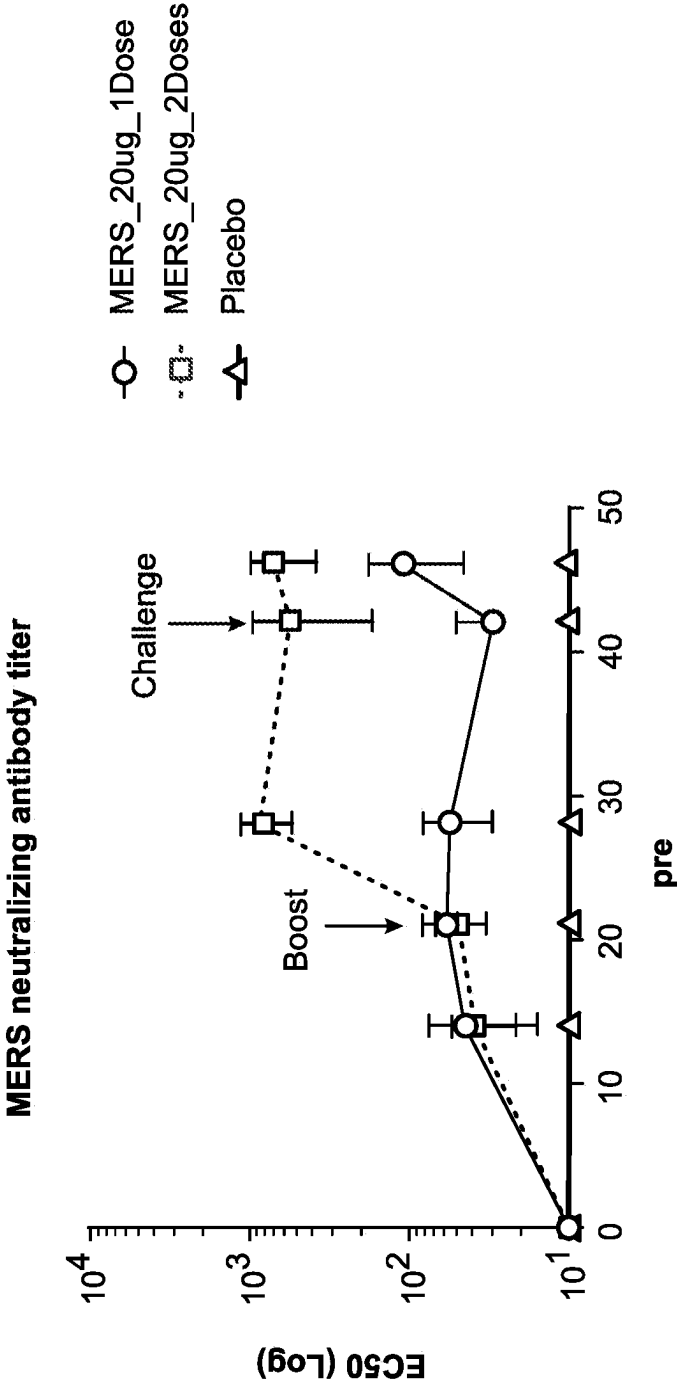


Fig. 21



BETACORONA VIRUS MRNA VACCINES

RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/762,384, U.S. application Ser. No. 18/762,372 (now U.S. Pat. No. 12,208,288), and U.S. application Ser. No. 18/762,296, each filed Jul. 2, 2024, and which each is a continuation of U.S. application Ser. No. 18/528,323 filed Dec. 4, 2023, which is a continuation of U.S. application Ser. No. 16/897,734, now U.S. Pat. No. 11,872,278, filed Jun. 10, 2020, which is a continuation of U.S. application Ser. No. 16/368,270, now U.S. Pat. No. 10,702,599, filed Mar. 28, 2019, which is a continuation of U.S. application Ser. No. 16/040,981, now U.S. Pat. No. 10,272,150, filed Jul. 20, 2018, which is a continuation of U.S. application Ser. No. 15/674,599, now U.S. Pat. No. 10,064,934, filed Aug. 11, 2017, which is a continuation of international application number PCT/US2016/058327, filed Oct. 21, 2016, which claims the benefit under 35 U.S.C. § 119 (e) of U.S. provisional application No. 62/244,802, filed Oct. 22, 2015, U.S. provisional application No. 62/247,297, filed Oct. 28, 2015, U.S. provisional application No. 62/244,946, filed Oct. 22, 2015, U.S. provisional application No. 62/247,362, filed Oct. 28, 2015, U.S. provisional application No. 62/244,813, filed Oct. 22, 2015, U.S. provisional application No. 62/247,394, filed Oct. 28, 2015, U.S. provisional application No. 62/244,837, filed Oct. 22, 2015, U.S. provisional application No. 62/247,483, filed Oct. 28, 2015, and U.S. provisional application No. 62/245,031, filed Oct. 22, 2015, each of which is incorporated by reference herein in its entirety.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

The contents of the electronic sequence listing (M137870053US11-SEQ-JXV.xml; Size: 387,115 bytes; and Date of Creation: Nov. 29, 2023) is herein incorporated by reference in its entirety.

BACKGROUND

Respiratory disease is a medical term that encompasses pathological conditions affecting the organs and tissues that make gas exchange possible in higher organisms, and includes conditions of the upper respiratory tract, trachea, bronchi, bronchioles, alveoli, pleura and pleural cavity, and the nerves and muscles of breathing. Respiratory diseases range from mild and self-limiting, such as the common cold, to life-threatening entities like bacterial pneumonia, pulmonary embolism, acute asthma and lung cancer. Respiratory disease is a common and significant cause of illness and death around the world. In the US, approximately 1 billion “common colds” occur each year. Respiratory conditions are among the most frequent reasons for hospital stays among children.

The human metapneumovirus (hMPV) is a negative-sense, single-stranded RNA virus of the genus Pneumovirinae and of the family Paramyxoviridae and is closely related to the avian metapneumovirus (AMPV) subgroup C. It was isolated for the first time in 2001 in the Netherlands by using the RAP-PCR (RNA arbitrarily primed PCR) technique for identification of unknown viruses growing in cultured cells. hMPV is second only to RSV as an important cause of viral lower respiratory tract illness (LRI) in young children. The

seasonal epidemiology of hMPV appears to be similar to that of RSV, but the incidence of infection and illness appears to be substantially lower.

Parainfluenza virus type 3 (PIV3), like hMPV, is also a negative-sense, single-stranded sense RNA virus of the genus Pneumovirinae and of the family Paramyxoviridae and is a major cause of ubiquitous acute respiratory infections of infancy and early childhood. Its incidence peaks around 4-12 months of age, and the virus is responsible for 3-10% of hospitalizations, mainly for bronchiolitis and pneumonia. PIV3 can be fatal, and in some instances is associated with neurologic diseases, such as febrile seizures. It can also result in airway remodeling, a significant cause of morbidity. In developing regions of the world, infants and young children are at the highest risk of mortality, either from primary PIV3 viral infection or a secondary consequences, such as bacterial infections. Human parainfluenza viruses (hPIV) types 1, 2 and 3 (hPIV1, hPIV2 and hPIV3, respectively), also like hMPV, are second only to RSV as important causes of viral LRI in young children.

RSV, too, is a negative-sense, single-stranded RNA virus of the genus Pneumovirinae and of the family Paramyxoviridae. Symptoms in adults typically resemble a sinus infection or the common cold, although the infection may be asymptomatic. In older adults (e.g., >60 years), RSV infection may progress to bronchiolitis or pneumonia. Symptoms in children are often more severe, including bronchiolitis and pneumonia. It is estimated that in the United States, most children are infected with RSV by the age of three. The RSV virion consists of an internal nucleocapsid comprised of the viral RNA bound to nucleoprotein (N), phosphoprotein (P), and large polymerase protein (L). The nucleocapsid is surrounded by matrix protein (M) and is encapsulated by a lipid bilayer into which the viral fusion (F) and attachment (G) proteins as well as the small hydrophobic protein (SH) are incorporated. The viral genome also encodes two non-structural proteins (NS1 and NS2), which inhibit type I interferon activity as well as the M-2 protein.

The continuing health problems associated with hMPV, PIV3 and RSV are of concern internationally, reinforcing the importance of developing effective and safe vaccine candidates against these virus.

Despite decades of research, no vaccines currently exist (Sato and Wright, *Pediatr. Infect. Dis. J.* 2008; 27(10 Suppl): S123-5). Recombinant technology, however, has been used to target the formation of vaccines for hPIV-1, 2 and 3 serotypes, for example, and has taken the form of several live-attenuated intranasal vaccines. Two vaccines in particular were found to be immunogenic and well tolerated against hPIV-3 in phase I trials. hPIV1 and hPIV2 vaccine candidates remain less advanced (Durbin and Karron, *Clinical infectious diseases: an official publication of the Infectious Diseases Society of America* 2003; 37(12):1668-77).

Measles virus (MeV), like hMPV, PIV3 and RSV, is a negative-sense, single-stranded RNA virus that is the cause of measles, an infection of the respiratory system. MeV is of the genus *Morbillivirus* within the family Paramyxoviridae. Humans are the natural hosts of the virus; no animal reservoirs are known to exist. Symptoms of measles include fever, cough, runny nose, red eyes and a generalized, maculopapular, erythematous rash. The virus is highly contagious and is spread by coughing

In addition to hMPV, PIV, RSV and MeV, betacoronaviruses are known to cause respiratory illnesses. Betacoronaviruses (BetaCoVs) are one of four genera of coronaviruses of the subfamily Coronavirinae in the family Coronaviridae, of the order Nidovirales. They are envel-

3

oped, positive-sense, single-stranded RNA viruses of zoonotic origin. The coronavirus genera are each composed of varying viral lineages, with the betacoronavirus genus containing four such lineages. The BetaCoVs of the greatest clinical importance concerning humans are OC43 and HKU1 of the A lineage, SARS-CoV of the B lineage, and MERS-CoV of the C lineage. MERS-CoV is the first betacoronavirus belonging to lineage C that is known to infect humans.

The Middle East respiratory syndrome coronavirus (MERS-CoV), or EMC/2012 (HCoV-EMC/2012), initially referred to as novel coronavirus 2012 or simply novel coronavirus, was first reported in 2012 after genome sequencing of a virus isolated from sputum samples from a person who fell ill during a 2012 outbreak of a new flu. As of July 2015, MERS-CoV cases have been reported in over 21 countries. The outbreaks of MERS-CoV have raised serious concerns world-wide, reinforcing the importance of developing effective and safe vaccine candidates against MERS-CoV.

Severe acute respiratory syndrome (SARS) emerged in China in 2002 and spread to other countries before brought under control. Because of a concern for reemergence or a deliberate release of the SARS coronavirus, vaccine development was initiated.

Deoxyribonucleic acid (DNA) vaccination is one technique used to stimulate humoral and cellular immune responses to foreign antigens, such as hMPV antigens and/or PIV antigens and/or RSV antigens. The direct injection of genetically engineered DNA (e.g., naked plasmid DNA) into a living host results in a small number of its cells directly producing an antigen, resulting in a protective immunological response. With this technique, however, comes potential problems, including the possibility of insertional mutagenesis, which could lead to the activation of oncogenes or the inhibition of tumor suppressor genes.

SUMMARY

Provided herein are ribonucleic acid (RNA) vaccines that build on the knowledge that RNA (e.g., messenger RNA (mRNA)) can safely direct the body's cellular machinery to produce nearly any protein of interest, from native proteins to antibodies and other entirely novel protein constructs that can have therapeutic activity inside and outside of cells. The RNA (e.g., mRNA) vaccines of the present disclosure may be used to induce a balanced immune response against hMPV, PIV, RSV, MeV, and/or BetaCoV (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1), or any combination of two or more of the foregoing viruses, comprising both cellular and humoral immunity, without risking the possibility of insertional mutagenesis, for example. hMPV, PIV, RSV, MeV, BetaCoV (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1) and combinations thereof are referred to herein as "respiratory viruses." Thus, the term "respiratory virus RNA vaccines" encompasses hMPV RNA vaccines, PIV RNA vaccines, RSV RNA vaccines, MeV RNA vaccines, BetaCoV RNA vaccines, and any combination of two or more of hMPV RNA vaccines, PIV RNA vaccines, RSV RNA vaccines, MeV RNA vaccines, and BetaCoV RNA vaccines.

The RNA (e.g., mRNA) vaccines may be utilized in various settings depending on the prevalence of the infection or the degree or level of unmet medical need. The RNA (e.g., mRNA) vaccines may be utilized to treat and/or prevent a

4

hMPV, PIV, RSV, MeV, a BetaCoV (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH, HCoV-HKU1), or any combination of two or more of the foregoing viruses, of various genotypes, strains, and isolates. The RNA (e.g., mRNA) vaccines have superior properties in that they produce much larger antibody titers and produce responses earlier than commercially available anti-viral therapeutic treatments. While not wishing to be bound by theory, it is believed that the RNA (e.g., mRNA) vaccines, as mRNA polynucleotides, are better designed to produce the appropriate protein conformation upon translation as the RNA (e.g., mRNA) vaccines co-opt natural cellular machinery. Unlike traditional vaccines, which are manufactured ex vivo and may trigger unwanted cellular responses, RNA (e.g., mRNA) vaccines are presented to the cellular system in a more native fashion.

In some aspects the invention is a respiratory virus vaccine, comprising at least one RNA polynucleotide having an open reading frame encoding at least one respiratory virus antigenic polypeptide, formulated in a cationic lipid nanoparticle.

Surprisingly, in some aspects, it has also been shown that efficacy of mRNA vaccines can be significantly enhanced when combined with a flagellin adjuvant, in particular, when one or more antigen-encoding mRNAs is combined with an mRNA encoding flagellin.

RNA (e.g., mRNA) vaccines combined with the flagellin adjuvant (e.g., mRNA-encoded flagellin adjuvant) have superior properties in that they may produce much larger antibody titers and produce responses earlier than commercially available vaccine formulations. While not wishing to be bound by theory, it is believed that the RNA (e.g., mRNA) vaccines, for example, as mRNA polynucleotides, are better designed to produce the appropriate protein conformation upon translation, for both the antigen and the adjuvant, as the RNA (e.g., mRNA) vaccines co-opt natural cellular machinery. Unlike traditional vaccines, which are manufactured ex vivo and may trigger unwanted cellular responses, RNA (e.g., mRNA) vaccines are presented to the cellular system in a more native fashion.

Some embodiments of the present disclosure provide RNA (e.g., mRNA) vaccines that include at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one antigenic polypeptide or an immunogenic fragment thereof (e.g., an immunogenic fragment capable of inducing an immune response to the antigenic polypeptide) and at least one RNA (e.g., mRNA polynucleotide) having an open reading frame encoding a flagellin adjuvant.

In some embodiments, at least one flagellin polypeptide (e.g., encoded flagellin polypeptide) is a flagellin protein. In some embodiments, at least one flagellin polypeptide (e.g., encoded flagellin polypeptide) is an immunogenic flagellin fragment. In some embodiments, at least one flagellin polypeptide and at least one antigenic polypeptide are encoded by a single RNA (e.g., mRNA) polynucleotide. In other embodiments, at least one flagellin polypeptide and at least one antigenic polypeptide are each encoded by a different RNA polynucleotide.

In some embodiments at least one flagellin polypeptide has at least 80%, at least 85%, at least 90%, or at least 95% identity to a flagellin polypeptide having a sequence identified by any one of SEQ ID NO: 54-56.

Provided herein, in some embodiments, is a ribonucleic acid (RNA) (e.g., mRNA) vaccine, comprising at least one (e.g., at least 2, 3, 4 or 5) RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one (e.g., at

5

least 2, 3, 4 or 5) hMPV, PIV, RSV, MeV, or a BetaCoV (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH, HCoV-HKU1) antigenic polypeptide, or any combination of two or more of the foregoing antigenic polypeptides. Herein, use of the term “antigenic polypeptide” encompasses immunogenic fragments of the antigenic polypeptide (an immunogenic fragment that induces (or is capable of inducing) an immune response to hMPV, PIV, RSV, MeV, or a BetaCoV), unless otherwise stated.

Also provided herein, in some embodiments, is a RNA (e.g., mRNA) vaccine comprising at least one (e.g., at least 2, 3, 4 or 5) RNA polynucleotide having an open reading frame encoding at least one (e.g., at least 2, 3, 4 or 5) hMPV, PIV, RSV, MeV, and/or a BetaCoV (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH, HCoV-HKU1) antigenic polypeptide or an immunogenic fragment thereof, linked to a signal peptide.

Further provided herein, in some embodiments, is a nucleic acid (e.g., DNA) encoding at least one (e.g., at least 2, 3, 4 or 5) hMPV, PIV, RSV, MeV, and/or a BetaCoV (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH, HCoV-HKU1) RNA (e.g., mRNA) polynucleotide.

Further still, provided herein, in some embodiments, is a method of inducing an immune response in a subject, the method comprising administering to the subject a vaccine comprising at least one (e.g., at least 2, 3, 4 or 5) RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one (e.g., at least 2, 3, 4 or 5) hMPV, PIV, RSV, MeV, and/or a BetaCoV (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH, HCoV-HKU1) antigenic polypeptide, or any combination of two or more of the foregoing antigenic polypeptides.

hMPV/PIV3/RSV

In some embodiments, a RNA (e.g., mRNA) vaccine comprises at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one hMPV, PIV3 or RSV antigenic polypeptide. In some embodiments, at least one antigenic polypeptide is a hMPV, PIV3 or RSV polypeptide. In some embodiments, at least one antigenic polypeptide is major surface glycoprotein G or an immunogenic fragment thereof.

In some embodiments, at least one antigenic polypeptide is Fusion (F) glycoprotein (e.g., Fusion glycoprotein F0, F1 or F2) or an immunogenic fragment thereof. In some embodiments, at least one antigenic polypeptide is major surface glycoprotein G or an immunogenic fragment thereof and F glycoprotein or an immunogenic fragment thereof. In some embodiments, the antigenic polypeptide is nucleoprotein (N) or an immunogenic fragment thereof, phosphoprotein (P) or an immunogenic fragment thereof, large polymerase protein (L) or an immunogenic fragment thereof, matrix protein (M) or an immunogenic fragment thereof, small hydrophobic protein (SH) or an immunogenic fragment thereof of nonstructural protein1 (NS1) or an immunogenic fragment thereof, or nonstructural protein 2 (NS2) and an immunogenic fragment thereof.

In some embodiments, at least one hMPV antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 5-8 (Table 3; see also amino acid sequences of Table 4). In some embodiments, the amino acid sequence of the hMPV antigenic polypeptide is, or is a fragment of, or is a homolog or variant having at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) identity to, the amino acid

6

sequence identified by any one of SEQ ID NO: 5-8 (Table 3; see also amino acid sequences of Table 4).

In some embodiments, at least one hMPV antigenic polypeptide is encoded by a nucleic acid sequence identified by any one of SEQ ID NO: 1-4 (Table 2).

In some embodiments, at least one hMPV RNA (e.g., mRNA) polynucleotide is encoded by a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 1-4 (Table 2). In some embodiments, at least one hMPV RNA (e.g., mRNA) polynucleotide comprises a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 57-60 (Table 2).

In some embodiments, at least one antigenic polypeptide is obtained from hMPV strain CAN98-75 (CAN75) or the hMPV strain CAN97-83 (CAN83).

In some embodiments, at least one PIV3 antigenic polypeptide comprises hemagglutinin-neuraminidase, Fusion (F) glycoprotein, matrix protein (M), nucleocapsid protein (N), viral replicase (L), non-structural V protein, or an immunogenic fragment thereof.

In some embodiments, at least one PIV3 antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 12-1.3 (Table 6; see also amino acid sequences of Table 7). In some embodiments, the amino acid sequence of the PIV3 antigenic polypeptide is, or is a fragment of, or is a homolog or variant having at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) identity to, the amino acid sequence identified by any one of SEQ ID NO: 12-13 (Table 6; see also amino acid sequences of Table 7).

In some embodiments, at least one PIV3 antigenic polypeptide is encoded by a nucleic acid sequence identified by any one of SEQ ID NO: 9-12 (Table 5; see also nucleic acid sequences of Table 7).

In some embodiments, at least one PIV3 RNA (e.g., mRNA) polynucleotide is encoded by a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 9-12 (Table 5; see also nucleic acid sequences of Table 7). In some embodiments, at least one PIV3 RNA (e.g., mRNA) polynucleotide comprises a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 61-64 (Table 5).

In some embodiments, at least one antigenic polypeptide is obtained from PIV3 strain HPIV3/*Homo sapiens*/PER/FLA4815/2008.

In some embodiments, at least one RSV antigenic polypeptide comprises at least one antigenic polypeptide that comprises glycoprotein G, glycoprotein F, or an immunogenic fragment thereof. In some embodiments, at least one RSV antigenic polypeptide comprises at least one antigenic polypeptide that comprises glycoprotein F and at least one or at least two antigenic polypeptide selected from G, M, N, P, L, SH, M2, NS1 and NS2.

MeV

In some embodiments, a RNA (e.g., mRNA) vaccine comprises at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one MeV antigenic polypeptide. In some embodiments, at least one antigenic polypeptide is a hemagglutinin (HA) protein or an immunogenic fragment thereof. The HA protein may be from MeV strain D3 or B8, for example. In some embodiments, at least one antigenic polypeptide is a Fusion (F) protein or an immunogenic fragment thereof. The F protein may be from MeV strain D3 or B8, for example. In some embodiments, a MeV RNA (e.g., mRNA) vaccine comprises a least one RNA polynucleotide encoding a HA

protein and a F protein. The HA and F proteins may be from MeV strain D3 or B8, for example.

In some embodiments, at least one MeV antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 47-50 (Table 14). In some embodiments, the amino acid sequence of the MeV antigenic polypeptide is, or is a fragment of, or is a homolog or variant having at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) identity to, the amino acid sequence identified by any one of SEQ ID NO: 47-50 (Table 14).

In some embodiments, at least one MeV antigenic polypeptide is encoded by a nucleic acid sequence of SEQ ID NO: 35-46 (Table 13).

In some embodiments, at least one MeV RNA (e.g., mRNA) polynucleotide is encoded by a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 35-46 (Table 13). In some embodiments, at least one MeV RNA (e.g., mRNA) polynucleotide comprises a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 69-80 (Table 13).

In some embodiments, at least one antigenic polypeptide is obtained from MeV strain B3/B3.1, C2, D4, D6, D7, D8, G3, H1, Moraten, Rubeovax, MVi/New Jersey.USA/45.05, MVi/Texas.USA/4.07, AIK-C, MVi/New York.USA/26.09/3, MVi/California.USA/16.03, MVi/Virginia.USA/15.09, MVi/California.USA/8.04, or MVi/Pennsylvania.USA/20.09.

BetaCoV

In some embodiments, a RNA (e.g., mRNA) vaccine comprises at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one BetaCoV antigenic polypeptide. In some embodiments, the BetaCoV is MERS-CoV. In some embodiments, the BetaCoV is SARS-CoV. In some embodiments, the BetaCoV is HCoV-OC43. In some embodiments, the BetaCoV is HCoV-229E. In some embodiments, the BetaCoV is HCoV-NL63. In some embodiments, the BetaCoV is HCoV-HKU1. In some embodiments, at least one antigenic polypeptide is a betacoronavirus structural protein. For example, a betacoronavirus structural protein may be spike protein (S), envelope protein (E), nucleocapsid protein (N), membrane protein (M) or an immunogenic fragment thereof. In some embodiments, a betacoronavirus structural protein is a spike protein (S). In some embodiments, a betacoronavirus structural protein is a S1 subunit or a S2 subunit of spike protein (S) or an immunogenic fragment thereof.

BetaCoV RNA (e.g., mRNA) polynucleotides of the vaccines provided herein may encode viral protein components of betacoronaviruses, for example, accessory proteins, replicase proteins and the like are encompassed by the present disclosure. RNA (e.g., mRNA) vaccines may include RNA polynucleotides encoding at least one accessory protein (e.g., protein 3, protein 4a, protein 4b, protein 5), at least one replicase protein (e.g., protein 1a, protein 1b), or a combination of at least one accessory protein and at least one replicase protein. The present disclosure also encompasses RNA (e.g., mRNA) vaccines comprising RNA (e.g., mRNA) polynucleotides encoding an accessory protein and/or a replicase protein in combination with at least one structural protein. Due to their surface expression properties, vaccines featuring RNA polynucleotides encoding structural proteins are believed to have preferred immunogenic activity and, hence, may be most suitable for use in the vaccines of the present disclosure.

Some embodiments of the present disclosure provide betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-

OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH, HCoV-HKU1 or a combination thereof) vaccines that include at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH, HCoV-HKU1) antigenic polypeptide. Also provided herein are pan-betacoronavirus vaccines. Thus, a betacoronavirus vaccine comprising a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding any one, two, three or four of MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, and HCoV-HKU1, for example, may be effective against any one of, any combination of, or all of: MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1. Other betacoronaviruses are encompassed by the present disclosure.

In some embodiments, at least one antigenic polypeptide is a MERS-CoV structural protein. For example, a MERS-CoV structural protein may be spike protein (S), envelope protein (E), nucleocapsid protein (N), membrane protein (M) or an immunogenic fragment thereof. In some embodiments, the MERS-CoV structural protein is a spike protein (S) (see, e.g., Coleman C M et al. *Vaccine* 2014; 32:3169-74, incorporated herein by reference). In some embodiments, the MERS-CoV structural protein is a S1 subunit or a S2 subunit of spike protein (S) or an immunogenic fragment thereof (Li J et al. *Viral Immunol* 2013; 26(2):126-32; He Y et al. *Biochem Biophys Res Commun* 2004; 324(2):773-81, each of which is incorporated herein by reference).

In some embodiments, at least one MERS-CoV antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 24-28 or 33 (Table 11). In some embodiments, the amino acid sequence of the MERS-CoV antigenic polypeptide is, or is a fragment of, or is a homolog or variant having at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) identity to, the amino acid sequence identified by any one of SEQ ID NO: 24-28 or 33 (Table 11).

In some embodiments, at least one MERS-CoV antigenic polypeptide is encoded by a nucleic acid sequence identified by any one of SEQ ID NO: 20-23 (Table 10).

In some embodiments, at least one MERS-CoV RNA (e.g., mRNA) polynucleotide is encoded by a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 20-23 (Table 10). In some embodiments, at least one MERS-CoV RNA (e.g., mRNA) polynucleotide comprises a nucleic acid sequence, or a fragment of a nucleotide sequence, identified by any one of SEQ ID NO: 65-68 (Table 10).

In some embodiments, at least one antigenic polypeptide is obtained from MERS-CoV strain Riyadh_14_2013, 2cEMC/2012, or Hasa_1_2013.

In some embodiments, at least one antigenic polypeptide is a SARS-CoV structural protein. For example, a SARS-CoV structural protein may be spike protein (S), envelope protein (E), nucleocapsid protein (N), membrane protein (M) or an immunogenic fragment thereof. In some embodiments, the SARS-CoV structural protein is a spike protein (S). In some embodiments, the SARS-CoV structural protein is a S1 subunit or a S2 subunit of spike protein (S) or an immunogenic fragment thereof.

In some embodiments, at least one SARS-CoV antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 29, 32 or 34 (Table 11). In some embodiments, the amino acid sequence of the SARS-CoV antigenic polypeptide is, or is a fragment of, or is a homolog or variant having at least 80% (e.g., 85%, 90%, 95%, 98%,

99%) identity to, the amino acid sequence identified by any one of SEQ ID NO: 29, 32 or 34 (Table 11).

In some embodiments, at least one antigenic polypeptide is a HCoV-OC43 structural protein. For example, a HCoV-OC43 structural protein may be spike protein (S), envelope protein (E), nucleocapsid protein (N), membrane protein (M) or an immunogenic fragment thereof. In some embodiments, the HCoV-OC43 structural protein is a spike protein (S). In some embodiments, the HCoV-OC43 structural protein is a S1 subunit or a S2 subunit of spike protein (S) or an immunogenic fragment thereof.

In some embodiments, at least one HCoV-OC43 antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 30 (Table 11). In some embodiments, the amino acid sequence of the HCoV-OC43 antigenic polypeptide is, or is a fragment of, or is a homolog or variant having at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) identity to, the amino acid sequence identified by any one of SEQ ID NO: 30 (Table 11).

In some embodiments, an antigenic polypeptide is a HCoV-HKU1 structural protein. For example, a HCoV-HKU1 structural protein may be spike protein (S), envelope protein (E), nucleocapsid protein (N), membrane protein (M) or an immunogenic fragment thereof. In some embodiments, the HCoV-HKU1 structural protein is a spike protein (S). In some embodiments, the HCoV-HKU1 structural protein is a S1 subunit or a S2 subunit of spike protein (S) or an immunogenic fragment thereof.

In some embodiments, at least one HCoV-HKU1 antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 31 (Table 11). In some embodiments, the amino acid sequence of the HCoV-HKU1 antigenic polypeptide is, or is a fragment of, or is a homolog or variant having at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) identity to, the amino acid sequence identified by any one of SEQ ID NO: 31 (Table 11).

In some embodiments, an open reading frame of a RNA (e.g., mRNA) vaccine is codon-optimized. In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and 14; see also amino acid sequences of Tables 4, 7, 12 and 15) and is codon optimized mRNA.

In some embodiments, a RNA (e.g., mRNA) vaccine further comprising an adjuvant.

Tables 4, 7, 12 and 15 provide National Center for Biotechnology Information (NCBI) accession numbers of interest. It should be understood that the phrase "an amino acid sequence of Tables 4, 7, 12 and 15" refers to an amino acid sequence identified by one or more NCBI accession numbers listed in Tables 4, 7, 12 and 15. Each of the amino acid sequences, and variants having greater than 95% identity or greater than 98% identity to each of the amino acid sequences encompassed by the accession numbers of Tables 4, 7, 12 and 15 are included within the constructs (polynucleotides/polypeptides) of the present disclosure.

In some embodiments, at least one mRNA polynucleotide is encoded by a nucleic acid having a sequence identified by any one of SEQ ID NO: 1-4, 9-12, 20-23, or 35-46 (Tables 2, 5, 10 and 13; see also nucleic acid sequences of Table 7) and having less than 80% identity to wild-type mRNA sequence. In some embodiments, at least one mRNA polynucleotide is encoded by a nucleic acid having a sequence identified by any one of SEQ ID NO: 1-4, 9-12, 20-23, or 35-46 (Tables 2, 5, 10 and 13; see also nucleic acid sequences of Table 7) and having less than 75%, 85% or 95% identity to a wild-type mRNA sequence. In some

embodiments, at least one mRNA polynucleotide is encoded by a nucleic acid having a sequence identified by any one of SEQ ID NO: 1-4, 9-12, 20-23, or 35-46 (Tables 2, 5, 10 and 13; see also nucleic acid sequences of Table 7) and having less than 50-80%, 60-80%, 40-80%, 30-80%, 70-80%, 75-80% or 78-80% identity to wild-type mRNA sequence. In some embodiments, at least one mRNA polynucleotide is encoded by a nucleic acid having a sequence identified by any one of SEQ ID NO: 1-4, 9-12, 20-23, or 35-46 (Tables 2, 5, 10 and 13; see also nucleic acid sequences of Table 7) and having less than 40-85%, 50-85%, 60-85%, 30-85%, 70-85%, 75-85% or 80-85% identity to wild-type mRNA sequence. In some embodiments, at least one mRNA polynucleotide is encoded by a nucleic acid having a sequence identified by any one of SEQ ID NO: 1-4, 9-12, 20-23, or 35-46 (Tables 2, 5, 10 and 13; see also nucleic acid sequences of Table 7) and having less than 40-90%, 50-90%, 60-90%, 30-90%, 70-90%, 75-90%, 80-90%, or 85-90% identity to wild-type mRNA sequence.

In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and 14; see also amino acid sequences of Tables 4, 7, 12 and 15) and having at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) identity to wild-type mRNA sequence, but does not include wild-type mRNA sequence.

In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and 14; see also amino acid sequences of Tables 4, 7, 12 and 15) and has less than 95%, 90%, 85%, 80% or 75% identity to wild-type mRNA sequence. In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and 14; see also amino acid sequences of Tables 4, 7, 12 and 15) and has 30-80%, 40-80%, 50-80%, 60-80%, 70-80%, 75-80% or 78-80%, 30-85%, 40-85%, 50-85%, 60-85%, 70-85%, 75-85% or 78-85%, 30-90%, 40-90%, 50-90%, 60-90%, 70-90%, 75-90%, 80-90% or 85-90% identity to wild-type mRNA sequence.

In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and 14; see also amino acid sequences of Tables 4, 7, 12 and 15). In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having 95%-99% identity to an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and 14; see also amino acid sequences of Tables 4, 7, 12 and 15).

In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and 14; see also amino acid sequences of Tables 4, 7, 12 and 15) and having membrane fusion activity. In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide having 95%-99% identity to an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, or 47-50 (Tables 3, 6, 11 and

11

14; see also amino acid sequences of Tables 4, 7, 12 and 15) and having membrane fusion activity.

In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide (e.g., at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides) that attaches to cell receptors.

In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide (e.g., at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides) that causes fusion of viral and cellular membranes.

In some embodiments, at least one RNA polynucleotide encodes at least one antigenic polypeptide (e.g., at least one

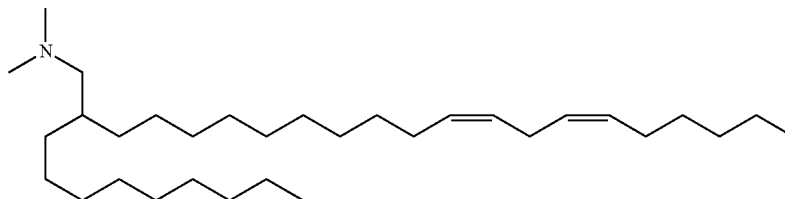
12

dihydrouridine, 2-thio-pseudouridine, 4-methoxy-2-thio-pseudouridine, 4-methoxy-pseudouridine, 4-thio-1-methyl-pseudouridine, 4-thio-pseudouridine, 5-aza-uridine, dihydropseudouridine, 5-methoxyuridine and 2'-O-methyl uridine. In some embodiments, the chemical modification is in the 5-position of the uracil. In some embodiments, the chemical modification is a N1-methylpseudouridine. In some embodiments, the chemical modification is a N1-ethylpseudouridine.

In some embodiments, a lipid nanoparticle comprises a cationic lipid, a PEG-modified lipid, a sterol and a non-cationic lipid. In some embodiments, a cationic lipid is an ionizable cationic lipid and the non-cationic lipid is a neutral lipid, and the sterol is a cholesterol. In some embodiments, a cationic lipid is selected from the group consisting of 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), (12Z,15Z)-N,N-dimethyl-2-nonylhenicosa-12,15-dien-1-amine (L608), and N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]heptadecan-8-amine (L530).

In some embodiments, the lipid is

(L608)



hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides) that is responsible for binding of the virus to a cell being infected.

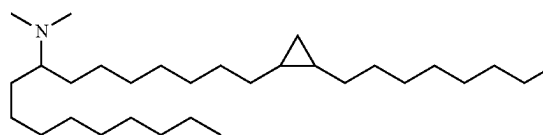
Some embodiments of the present disclosure provide a vaccine that includes at least one ribonucleic acid (RNA) (e.g., mRNA) polynucleotide having an open reading frame encoding at least one antigenic polypeptide (e.g., at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides), at least one 5' terminal cap and at least one chemical modification, formulated within a lipid nanoparticle.

In some embodiments, a 5' terminal cap is 7mG(5')ppp (5')NmpNp.

In some embodiments, at least one chemical modification is selected from pseudouridine, N1-methylpseudouridine, N1-ethylpseudouridine, 2-thiouridine, 4'-thiouridine, 5-methylcytosine, 5-methyluridine, 2-thio-1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-pseudouridine, 2-thio-5-aza-uridine, 2-thio-dihydropseudouridine, 2-thio-

In some embodiments, the lipid is

(L530)



In some embodiments, a lipid nanoparticle comprises compounds of Formula (I) and/or Formula (II), discussed below.

In some embodiments, a respiratory virus RNA (e.g., mRNA) vaccine is formulated in a lipid nanoparticle that comprises a compound selected from Compounds 3, 18, 20, 25, 26, 29, 30, 60, 108-112 and 122, described below.

Some embodiments of the present disclosure provide a vaccine that includes at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one antigenic polypeptide (e.g., at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides), wherein at least 80% (e.g., 85%, 90%, 95%, 98%, 99%) of the uracil in the open

reading frame have a chemical modification, optionally wherein the vaccine is formulated in a lipid nanoparticle (e.g., a lipid nanoparticle comprises a cationic lipid, a PEG-modified lipid, a sterol and a non-cationic lipid).

In some embodiments, 100% of the uracil in the open reading frame have a chemical modification. In some embodiments, a chemical modification is in the 5-position of the uracil. In some embodiments, a chemical modification is a N1-methyl pseudouridine. In some embodiments, 100% of the uracil in the open reading frame have a N1-methyl pseudouridine in the 5-position of the uracil.

In some embodiments, an open reading frame of a RNA (e.g., mRNA) polynucleotide encodes at least two antigenic polypeptides (e.g., at least two hMPV antigenic polypeptides, at least two PIV3 antigenic polypeptides, at least two RSV antigenic polypeptides, at least two MeV antigenic polypeptides, or at least two BetaCoV antigenic polypeptides. e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides). In some embodiments, the open reading frame encodes at least five or at least ten antigenic polypeptides. In some embodiments, the open reading frame encodes at least 100 antigenic polypeptides. In some embodiments, the open reading frame encodes 2-100 antigenic polypeptides.

In some embodiments, a vaccine comprises at least two RNA (e.g., mRNA) polynucleotides, each having an open reading frame encoding at least one antigenic polypeptide (e.g., at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides). In some embodiments, the vaccine comprises at least five or at least ten RNA (e.g., mRNA) polynucleotides, each having an open reading frame encoding at least one antigenic polypeptide or an immunogenic fragment thereof. In some embodiments, the vaccine comprises at least 100 RNA (e.g., mRNA) polynucleotides, each having an open reading frame encoding at least one antigenic polypeptide. In some embodiments, the vaccine comprises 2-100 RNA (e.g., mRNA) polynucleotides, each having an open reading frame encoding at least one antigenic polypeptide.

In some embodiments, at least one antigenic polypeptide (e.g., at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides) is fused to a signal peptide. In some embodiments, the signal peptide is selected from: a HulgGk signal peptide (METPAQLLFLLLLWLPDITG; SEQ ID NO: 15); IgE heavy chain epsilon-1 signal peptide (MDWTWIL-FLVAAATRVHS; SEQ ID NO: 16); Japanese encephalitis PRM signal sequence (MLGSNSGQRVVFITILLLVAPAYS; SEQ ID NO: 17); VSVg protein signal sequence (MKCLLYLAFLFIGVNCA; SEQ ID NO: 18) and Japanese encephalitis JEV signal sequence (MWLVSLAIVTACAGA; SEQ ID NO: 19).

In some embodiments, the signal peptide is fused to the N-terminus of at least one antigenic polypeptide. In some

embodiments, a signal peptide is fused to the C-terminus of at least one antigenic polypeptide.

In some embodiments, at least one antigenic polypeptide (e.g., at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, or at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides) comprises a mutated N-linked glycosylation site.

Also provided herein is a RNA (e.g., mRNA) vaccine of any one of the foregoing paragraphs (e.g., a hMPV vaccine, a PIV3 vaccine, a RSV vaccine, a MeV vaccine, or a BetaCoV vaccine, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing vaccines), formulated in a nanoparticle (e.g., a lipid nanoparticle).

In some embodiments, the nanoparticle has a mean diameter of 50-200 nm. In some embodiments, the nanoparticle is a lipid nanoparticle. In some embodiments, the lipid nanoparticle comprises a cationic lipid, a PEG-modified lipid, a sterol and a non-cationic lipid. In some embodiments, the lipid nanoparticle comprises a molar ratio of about 20-60% cationic lipid, 0.5-15% PEG-modified lipid, 25-55% sterol, and 25% non-cationic lipid. In some embodiments, the cationic lipid is an ionizable cationic lipid and the non-cationic lipid is a neutral lipid, and the sterol is a cholesterol. In some embodiments, the cationic lipid is selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1.3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319).

In some embodiments, a lipid nanoparticle comprises compounds of Formula (I) and/or Formula (II), as discussed below.

In some embodiments, a lipid nanoparticle comprises Compounds 3, 18, 20, 25, 26, 29, 30, 60, 108-112, or 122, as discussed below.

In some embodiments, the nanoparticle has a polydispersity value of less than 0.4 (e.g., less than 0.3, 0.2 or 0.1).

In some embodiments, the nanoparticle has a net neutral charge at a neutral pH value.

In some embodiments, the respiratory virus vaccine is multivalent.

Some embodiments of the present disclosure provide methods of inducing an antigen specific immune response in a subject, comprising administering to the subject any of the RNA (e.g., mRNA) vaccine as provided herein in an amount effective to produce an antigen-specific immune response. In some embodiments, the RNA (e.g., mRNA) vaccine is a hMPV vaccine, a PIV3 vaccine, a RSV vaccine, a MeV vaccine, or a BetaCoV vaccine, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1 vaccines. In some embodiments, the RNA (e.g., mRNA) vaccine is a combination vaccine comprising a combination of any two or more of the foregoing vaccines.

In some embodiments, an antigen-specific immune response comprises a T cell response or a B cell response.

In some embodiments, a method of producing an antigen-specific immune response comprises administering to a subject a single dose (no booster dose) of a RNA (e.g., mRNA) vaccine of the present disclosure. In some embodi-

ments, the RNA (e.g., mRNA) vaccine is a hMPV vaccine, a PIV3 vaccine, a RSV vaccine, a MeV vaccine, or a BetaCoV vaccine, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1 vaccines. In some embodiments, the RNA (e.g., mRNA) vaccine is a combination vaccine comprising a combination of any two or more of the foregoing vaccines.

In some embodiments, a method further comprises administering to the subject a second (booster) dose of a RNA (e.g., mRNA) vaccine. Additional doses of a RNA (e.g., mRNA) vaccine may be administered.

In some embodiments, the subjects exhibit a seroconversion rate of at least 80% (e.g., at least 85%, at least 90%, or at least 95%) following the first dose or the second (booster) dose of the vaccine. Seroconversion is the time period during which a specific antibody develops and becomes detectable in the blood. After seroconversion has occurred, a virus can be detected in blood tests for the antibody. During an infection or immunization, antigens enter the blood, and the immune system begins to produce antibodies in response. Before seroconversion, the antigen itself may or may not be detectable, but antibodies are considered absent. During seroconversion, antibodies are present but not yet detectable. Any time after seroconversion, the antibodies can be detected in the blood, indicating a prior or current infection.

In some embodiments, a RNA (e.g., mRNA) vaccine is administered to a subject by intradermal or intramuscular injection.

Some embodiments, of the present disclosure provide methods of inducing an antigen specific immune response in a subject, including administering to a subject a RNA (e.g., mRNA) vaccine in an effective amount to produce an antigen specific immune response in a subject. Antigen-specific immune responses in a subject may be determined, in some embodiments, by assaying for antibody titer (for titer of an antibody that binds to a hMPV, PIV3, RSV, MeV and/or BetaCoV antigenic polypeptide) following administration to the subject of any of the RNA (e.g., mRNA) vaccines of the present disclosure. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased by at least 1 log relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased by 1-3 log relative to a control.

In some embodiments, the anti-antigenic polypeptide antibody titer produced in a subject is increased at least 2 times relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased at least 5 times relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased at least 10 times relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased 2-10 times relative to a control.

In some embodiments, the control is an anti-antigenic polypeptide antibody titer produced in a subject who has not been administered a RNA (e.g., mRNA) vaccine of the present disclosure. In some embodiments, the control is an anti-antigenic polypeptide antibody titer produced in a subject who has been administered a live attenuated or inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine (see, e.g., Ren J. et al. *J of Gen. Virol.* 2015; 96: 1515-1520), or wherein the control is an anti-antigenic polypeptide antibody titer produced in a subject who has been administered a recombinant or purified hMPV, PIV3, RSV, MeV

and/or BetaCoV protein vaccine. In some embodiments, the control is an anti-antigenic polypeptide antibody titer produced in a subject who has been administered a hMPV, PIV3, RSV, MeV and/or BetaCoV virus-like particle (VLP) vaccine (see, e.g., Cox R G et al., *J Virol.* 2014 June; 88(11): 6368-6379).

A RNA (e.g., mRNA) vaccine of the present disclosure is administered to a subject in an effective amount (an amount effective to induce an immune response). In some embodiments, the effective amount is a dose equivalent to an at least 2-fold, at least 4-fold, at least 10-fold, at least 100-fold, at least 1000-fold reduction in the standard of care dose of a recombinant hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, wherein the anti-antigenic polypeptide antibody titer produced in the subject is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a recombinant hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, a purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, a live attenuated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine, an inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine, or a hMPV, PIV3, RSV, MeV and/or BetaCoV VLP vaccine. In some embodiments, the effective amount is a dose equivalent to 2-1000-fold reduction in the standard of care dose of a recombinant hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, wherein the anti-antigenic polypeptide antibody titer produced in the subject is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a recombinant hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, a purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, a live attenuated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine, an inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine, or a hMPV, PIV3, RSV, MeV and/or BetaCoV VLP vaccine.

In some embodiments, the control is an anti-antigenic polypeptide antibody titer produced in a subject who has been administered a virus-like particle (VLP) vaccine comprising structural proteins of hMPV, PIV3, RSV, MeV and/or BetaCoV.

In some embodiments, the RNA (e.g., mRNA) vaccine is formulated in an effective amount to produce an antigen specific immune response in a subject.

In some embodiments, the effective amount is a total dose of 25 µg to 1000 µg, or 50 µg to 1000 µg. In some embodiments, the effective amount is a total dose of 100 g. In some embodiments, the effective amount is a dose of 25 µg administered to the subject a total of two times. In some embodiments, the effective amount is a dose of 100 µg administered to the subject a total of two times. In some embodiments, the effective amount is a dose of 400 µg administered to the subject a total of two times. In some embodiments, the effective amount is a dose of 500 µg administered to the subject a total of two times.

In some embodiments, the efficacy (or effectiveness) of a RNA (e.g., mRNA) vaccine is greater than 60%. In some embodiments, the RNA (e.g., mRNA) polynucleotide of the vaccine at least one hMPV antigenic polypeptide, at least one PIV3 antigenic polypeptide, at least one RSV antigenic polypeptide, at least one MeV antigenic polypeptide, at least one BetaCoV antigenic polypeptide, e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1, or any combination of two or more of the foregoing antigenic polypeptides.

Vaccine efficacy may be assessed using standard analyses (see, e.g., Weinberg et al., *J Infect Dis.* 2010 Jun. 1; 201(11):1607-10). For example, vaccine efficacy may be measured by double-blind, randomized, clinical controlled trials. Vaccine efficacy may be expressed as a proportionate reduction in disease attack rate (AR) between the unvaccinated (ARU) and vaccinated (ARV) study cohorts and can be calculated from the relative risk (RR) of disease among the vaccinated group with use of the following formulas:

$$\text{Efficacy} = (\text{ARU} - \text{ARV}) / \text{ARU} \times 100; \text{ and}$$

$$\text{Efficacy} = (1 - \text{RR}) \times 100.$$

Likewise, vaccine effectiveness may be assessed using standard analyses (see, e.g., Weinberg et al., *J Infect Dis.* 2010 Jun. 1; 201(11):1607-10). Vaccine effectiveness is an assessment of how a vaccine (which may have already proven to have high vaccine efficacy) reduces disease in a population. This measure can assess the net balance of benefits and adverse effects of a vaccination program, not just the vaccine itself, under natural field conditions rather than in a controlled clinical trial. Vaccine effectiveness is proportional to vaccine efficacy (potency) but is also affected by how well target groups in the population are immunized, as well as by other non-vaccine-related factors that influence the 'real-world' outcomes of hospitalizations, ambulatory visits, or costs. For example, a retrospective case control analysis may be used, in which the rates of vaccination among a set of infected cases and appropriate controls are compared. Vaccine effectiveness may be expressed as a rate difference, with use of the odds ratio (OR) for developing infection despite vaccination:

$$\text{Effectiveness} = (1 - \text{OR}) \times 100.$$

In some embodiments, the efficacy (or effectiveness) of a RNA (e.g., mRNA) vaccine is at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, or at least 90%.

In some embodiments, the vaccine immunizes the subject against hMPV, PIV3, RSV, MeV, BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1), or any combination of two or more of the foregoing viruses for up to 2 years. In some embodiments, the vaccine immunizes the subject against hMPV, PIV3, RSV, MeV, BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1), or any combination of two or more of the foregoing viruses for more than 2 years, more than 3 years, more than 4 years, or for 5-10 years.

In some embodiments, the subject is about 5 years old or younger. For example, the subject may be between the ages of about 1 year and about 5 years (e.g., about 1, 2, 3, 5 or 5 years), or between the ages of about 6 months and about 1 year (e.g., about 6, 7, 8, 9, 10, 11 or 12 months). In some embodiments, the subject is about 12 months or younger (e.g., 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2 months or 1 month). In some embodiments, the subject is about 6 months or younger.

In some embodiments, the subject was born full term (e.g., about 37-42 weeks). In some embodiments, the subject was born prematurely, for example, at about 36 weeks of gestation or earlier (e.g., about 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26 or 25 weeks). For example, the subject may have been born at about 32 weeks of gestation or earlier. In some

embodiments, a RNA (e.g., mRNA) vaccine may be administered later in life, for example, at the age of about 6 months to about 5 years, or older.

In some embodiments, the subject is pregnant (e.g., in the first, second or third trimester) when administered a RNA (e.g., mRNA) vaccine. Viruses such as hMPV, PIV3 and RSV causes infections of the lower respiratory tract, mainly in infants and young children. One-third of RSV related deaths, for example, occur in the first year of life, with 99 percent of these deaths occurring in low-resource countries. It's so widespread in the United States that nearly all children become infected with the virus before their second birthdays.

Thus, the present disclosure provides RNA (e.g., mRNA) vaccines for maternal immunization to improve mother-to-child transmission of protection against the virus.

In some embodiments, the subject is a young adult between the ages of about 20 years and about 50 years (e.g., about 20, 25, 30, 35, 40, 45 or 50 years old).

In some embodiments, the subject is an elderly subject about 60 years old, about 70 years old, or older (e.g., about 60, 65, 70, 75, 80, 85 or 90 years old).

In some embodiments, the subject has a chronic pulmonary disease (e.g., chronic obstructive pulmonary disease (COPD) or asthma). Two forms of COPD include chronic bronchitis, which involves a long-term cough with mucus, and emphysema, which involves damage to the lungs over time. Thus, a subject administered a RNA (e.g., mRNA) vaccine may have chronic bronchitis or emphysema.

In some embodiments, the subject has been exposed to hMPV, PIV3, RSV, MeV, BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1), or any combination of two or more of the foregoing viruses; the subject is infected with hMPV, PIV3, RSV, MeV, BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1), or any combination of two or more of the foregoing viruses; or subject is at risk of infection by hMPV, PIV3, RSV, MeV, BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1), or any combination of two or more of the foregoing viruses.

In some embodiments, the subject is immunocompromised (has an impaired immune system, e.g., has an immune disorder or autoimmune disorder).

In some embodiments the nucleic acid vaccines described herein are chemically modified. In other embodiments the nucleic acid vaccines are unmodified.

Yet other aspects provide compositions for and methods of vaccinating a subject comprising administering to the subject a nucleic acid vaccine comprising one or more RNA polynucleotides having an open reading frame encoding a first respiratory virus antigenic polypeptide, wherein the RNA polynucleotide does not include a stabilization element, and wherein an adjuvant is not coformulated or co-administered with the vaccine.

In other aspects the invention is a composition for or method of vaccinating a subject comprising administering to the subject a nucleic acid vaccine comprising one or more RNA polynucleotides having an open reading frame encoding a first antigenic polypeptide wherein a dosage of between 10 µg/kg and 400 µg/kg of the nucleic acid vaccine is administered to the subject. In some embodiments the dosage of the RNA polynucleotide is 1-5 µg, 5-10 µg, 10-15 µg, 15-20 µg, 10-25 µg, 20-25 µg, 20-50 µg, 30-50 µg, 40-50 µg, 40-60 µg, 60-80 µg, 60-100 µg, 50-100 µg, 80-120 µg,

40-120 µg, 40-150 µg, 50-150 µg, 50-200 µg, 80-200 µg, 100-200 µg, 120-250 µg, 150-250 µg, 180-280 µg, 200-300 µg, 50-300 µg, 80-300 µg, 100-300 µg, 40-300 µg, 50-350 µg, 100-350 µg, 200-350 µg, 300-350 µg, 320-400 µg, 40-380 µg, 40-100 µg, 100-400 µg, 200-400 µg, or 300-400 µg per dose. In some embodiments, the nucleic acid vaccine is administered to the subject by intradermal or intramuscular injection. In some embodiments, the nucleic acid vaccine is administered to the subject on day zero. In some embodiments, a second dose of the nucleic acid vaccine is administered to the subject on day twenty one.

In some embodiments, a dosage of 25 micrograms of the RNA polynucleotide is included in the nucleic acid vaccine administered to the subject. In some embodiments, a dosage of 100 micrograms of the RNA polynucleotide is included in the nucleic acid vaccine administered to the subject. In some embodiments, a dosage of 50 micrograms of the RNA polynucleotide is included in the nucleic acid vaccine administered to the subject. In some embodiments, a dosage of 75 micrograms of the RNA polynucleotide is included in the nucleic acid vaccine administered to the subject. In some embodiments, a dosage of 150 micrograms of the RNA polynucleotide is included in the nucleic acid vaccine administered to the subject. In some embodiments, a dosage of 400 micrograms of the RNA polynucleotide is included in the nucleic acid vaccine administered to the subject. In some embodiments, a dosage of 200 micrograms of the RNA polynucleotide is included in the nucleic acid vaccine administered to the subject. In some embodiments, the RNA polynucleotide accumulates at a 100 fold higher level in the local lymph node in comparison with the distal lymph node. In other embodiments the nucleic acid vaccine is chemically modified and in other embodiments the nucleic acid vaccine is not chemically modified.

Aspects of the invention provide a nucleic acid vaccine comprising one or more RNA polynucleotides having an open reading frame encoding a first antigenic polypeptide, wherein the RNA polynucleotide does not include a stabilization element, and a pharmaceutically acceptable carrier or excipient, wherein an adjuvant is not included in the vaccine. In some embodiments, the stabilization element is a histone stem-loop. In some embodiments, the stabilization element is a nucleic acid sequence having increased GC content relative to wild type sequence.

Aspects of the invention provide nucleic acid vaccines comprising one or more RNA polynucleotides having an open reading frame encoding a first antigenic polypeptide, wherein the RNA polynucleotide is present in the formulation for in vivo administration to a host, which confers an antibody titer superior to the criterion for seroprotection for the first antigen for an acceptable percentage of human subjects. In some embodiments, the antibody titer produced by the mRNA vaccines of the invention is a neutralizing antibody titer. In some embodiments the neutralizing antibody titer is greater than a protein vaccine. In other embodiments the neutralizing antibody titer produced by the mRNA vaccines of the invention is greater than an adjuvanted protein vaccine. In yet other embodiments the neutralizing antibody titer produced by the mRNA vaccines of the invention is 1,000-10,000, 1,200-10,000, 1,400-10,000, 1,500-10,000, 1,000-5,000, 1,000-4,000, 1,800-10,000, 2,000-10,000, 2,000-5,000, 2,000-3,000, 2,000-4,000, 3,000-5,000, 3,000-4,000, or 2,000-2,500. A neutralization titer is typically expressed as the highest serum dilution required to achieve a 50% reduction in the number of plaques.

Also provided are nucleic acid vaccines comprising one or more RNA polynucleotides having an open reading frame

encoding a first antigenic polypeptide, wherein the RNA polynucleotide is present in a formulation for in vivo administration to a host for eliciting a longer lasting high antibody titer than an antibody titer elicited by an mRNA vaccine having a stabilizing element or formulated with an adjuvant and encoding the first antigenic polypeptide. In some embodiments, the RNA polynucleotide is formulated to produce a neutralizing antibodies within one week of a single administration. In some embodiments, the adjuvant is selected from a cationic peptide and an immunostimulatory nucleic acid. In some embodiments, the cationic peptide is protamine.

Aspects provide nucleic acid vaccines comprising one or more RNA polynucleotides having an open reading frame comprising at least one chemical modification or optionally no nucleotide modification, the open reading frame encoding a first antigenic polypeptide, wherein the RNA polynucleotide is present in the formulation for in vivo administration to a host such that the level of antigen expression in the host significantly exceeds a level of antigen expression produced by an mRNA vaccine having a stabilizing element or formulated with an adjuvant and encoding the first antigenic polypeptide.

Other aspects provide nucleic acid vaccines comprising one or more RNA polynucleotides having an open reading frame comprising at least one chemical modification or optionally no nucleotide modification, the open reading frame encoding a first antigenic polypeptide, wherein the vaccine has at least 10 fold less RNA polynucleotide than is required for an unmodified mRNA vaccine to produce an equivalent antibody titer. In some embodiments, the RNA polynucleotide is present in a dosage of 25-100 micrograms.

Aspects of the invention also provide a unit of use vaccine, comprising between 10 ug and 400 ug of one or more RNA polynucleotides having an open reading frame comprising at least one chemical modification or optionally no nucleotide modification, the open reading frame encoding a first antigenic polypeptide, and a pharmaceutically acceptable carrier or excipient, formulated for delivery to a human subject. In some embodiments, the vaccine further comprises a cationic lipid nanoparticle.

Aspects of the invention provide methods of creating, maintaining or restoring antigenic memory to a respiratory virus strain in an individual or population of individuals comprising administering to said individual or population an antigenic memory booster nucleic acid vaccine comprising (a) at least one RNA polynucleotide, said polynucleotide comprising at least one chemical modification or optionally no nucleotide modification and two or more codon-optimized open reading frames, said open reading frames encoding a set of reference antigenic polypeptides, and (b) optionally a pharmaceutically acceptable carrier or excipient. In some embodiments, the vaccine is administered to the individual via a route selected from the group consisting of intramuscular administration, intradermal administration and subcutaneous administration. In some embodiments, the administering step comprises contacting a muscle tissue of the subject with a device suitable for injection of the composition. In some embodiments, the administering step comprises contacting a muscle tissue of the subject with a device suitable for injection of the composition in combination with electroporation.

Aspects of the invention provide methods of vaccinating a subject comprising administering to the subject a single dosage of between 25 ug/kg and 400 ug/kg of a nucleic acid vaccine comprising one or more RNA polynucleotides hav-

ing an open reading frame encoding a first antigenic polypeptide in an effective amount to vaccinate the subject.

Other aspects provide nucleic acid vaccines comprising one or more RNA polynucleotides having an open reading frame comprising at least one chemical modification, the open reading frame encoding a first antigenic polypeptide, wherein the vaccine has at least 10 fold less RNA polynucleotide than is required for an unmodified mRNA vaccine to produce an equivalent antibody titer. In some embodiments, the RNA polynucleotide is present in a dosage of 25-100 micrograms.

Other aspects provide nucleic acid vaccines comprising an LNP formulated RNA polynucleotide having an open reading frame comprising no nucleotide modifications (unmodified), the open reading frame encoding a first antigenic polypeptide, wherein the vaccine has at least 10 fold less RNA polynucleotide than is required for an unmodified mRNA vaccine not formulated in a LNP to produce an equivalent antibody titer. In some embodiments, the RNA polynucleotide is present in a dosage of 25-100 micrograms.

The data presented in the Examples demonstrate significant enhanced immune responses using the formulations of the invention. Both chemically modified and unmodified RNA vaccines are useful according to the invention. Surprisingly, in contrast to prior art reports that it was preferable to use chemically unmodified mRNA formulated in a carrier for the production of vaccines, it is described herein that chemically modified mRNA-LNP vaccines required a much lower effective mRNA dose than unmodified mRNA, i.e., tenfold less than unmodified mRNA when formulated in carriers other than LNP. Both the chemically modified and unmodified RNA vaccines of the invention produce better immune responses than mRNA vaccines formulated in a different lipid carrier.

In other aspects the invention encompasses a method of treating an elderly subject age 60 years or older comprising administering to the subject a nucleic acid vaccine comprising one or more RNA polynucleotides having an open reading frame encoding a respiratory virus antigenic polypeptide in an effective amount to vaccinate the subject.

In other aspects the invention encompasses a method of treating a young subject age 17 years or younger comprising administering to the subject a nucleic acid vaccine comprising one or more RNA polynucleotides having an open reading frame encoding a respiratory virus antigenic polypeptide in an effective amount to vaccinate the subject.

In other aspects the invention encompasses a method of treating an adult subject comprising administering to the subject a nucleic acid vaccine comprising one or more RNA polynucleotides having an open reading frame encoding a respiratory virus antigenic polypeptide in an effective amount to vaccinate the subject.

In some aspects the invention is a method of vaccinating a subject with a combination vaccine including at least two nucleic acid sequences encoding respiratory antigens wherein the dosage for the vaccine is a combined therapeutic dosage wherein the dosage of each individual nucleic acid encoding an antigen is a sub therapeutic dosage. In some embodiments, the combined dosage is 25 micrograms of the RNA polynucleotide in the nucleic acid vaccine administered to the subject. In some embodiments, the combined dosage is 100 micrograms of the RNA polynucleotide in the nucleic acid vaccine administered to the subject. In some embodiments, the combined dosage is 50 micrograms of the RNA polynucleotide in the nucleic acid vaccine administered to the subject. In some embodiments, the combined dosage is 75 micrograms of the RNA polynucleotide in the

nucleic acid vaccine administered to the subject. In some embodiments, the combined dosage is 150 micrograms of the RNA polynucleotide in the nucleic acid vaccine administered to the subject. In some embodiments, the combined dosage is 400 micrograms of the RNA polynucleotide in the nucleic acid vaccine administered to the subject. In some embodiments, the sub therapeutic dosage of each individual nucleic acid encoding an antigen is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 micrograms. In other embodiments the nucleic acid vaccine is chemically modified and in other embodiments the nucleic acid vaccine is not chemically modified.

The RNA polynucleotide is one of SEQ ID NO: 1-4, 9-12, 20-23, 35-46, 57-61, and 64-80 and includes at least one chemical modification. In other embodiments the RNA polynucleotide is one of SEQ ID NO: 1-4, 9-12, 20-23, 35-46, 57-61, and 64-80 and does not include any nucleotide modifications, or is unmodified. In yet other embodiments the at least one RNA polynucleotide encodes an antigenic protein of any of SEQ ID NO: 5-8, 12-13, 24-34, and 47-50 and includes at least one chemical modification. In other embodiments the RNA polynucleotide encodes an antigenic protein of any of SEQ ID NO: 5-8, 12-13, 24-34, and 47-50 and does not include any nucleotide modifications, or is unmodified.

In preferred aspects, vaccines of the invention (e.g., LNP-encapsulated mRNA vaccines) produce prophylactically- and/or therapeutically-efficacious levels, concentrations and/or titers of antigen-specific antibodies in the blood or serum of a vaccinated subject. As defined herein, the term antibody titer refers to the amount of antigen-specific antibody produces in a subject, e.g., a human subject. In exemplary embodiments, antibody titer is expressed as the inverse of the greatest dilution (in a serial dilution) that still gives a positive result. In exemplary embodiments, antibody titer is determined or measured by enzyme-linked immunosorbent assay (ELISA). In exemplary embodiments, antibody titer is determined or measured by neutralization assay, e.g., by microneutralization assay. In certain aspects, antibody titer measurement is expressed as a ratio, such as 1:40, 1:100, etc.

In exemplary embodiments of the invention, an efficacious vaccine produces an antibody titer of greater than 1:40, greater than 1:100, greater than 1:400, greater than 1:1000, greater than 1:2000, greater than 1:3000, greater than 1:4000, greater than 1:500, greater than 1:6000, greater than 1:7500, greater than 1:10000. In exemplary embodiments, the antibody titer is produced or reached by 10 days following vaccination, by 20 days following vaccination, by 30 days following vaccination, by 40 days following vaccination, or by 50 or more days following vaccination. In exemplary embodiments, the titer is produced or reached following a single dose of vaccine administered to the subject. In other embodiments, the titer is produced or reached following multiple doses, e.g., following a first and a second dose (e.g., a booster dose.)

In exemplary aspects of the invention, antigen-specific antibodies are measured in units of µg/ml or are measured in units of IU/L (International Units per liter) or mIU/ml (milli International Units per ml). In exemplary embodiments of the invention, an efficacious vaccine produces >0.5 g/ml, >0.1 µg/ml, >0.2 µg/ml, >0.35 µg/ml, >0.5 µg/ml, >1 µg/ml, >2 µg/ml, >5 µg/ml or >10 µg/ml. In exemplary embodiments of the invention, an efficacious vaccine produces >10 mIU/ml, >20 mIU/ml, >50 mIU/ml, >100 mIU/ml, >200 mIU/ml, >500 mIU/ml or >1000 mIU/ml. In exemplary embodiments, the antibody level or concentration is pro-

duced or reached by 10 days following vaccination, by 20 days following vaccination, by 30 days following vaccination, by 40 days following vaccination, or by 50 or more days following vaccination. In exemplary embodiments, the level or concentration is produced or reached following a single dose of vaccine administered to the subject. In other embodiments, the level or concentration is produced or reached following multiple doses, e.g., following a first and a second dose (e.g., a booster dose.) In exemplary embodiments, antibody level or concentration is determined or measured by enzyme-linked immunosorbent assay (ELISA). In exemplary embodiments, antibody level or concentration is determined or measured by neutralization assay, e.g., by microneutralization assay.

The details of various embodiments of the disclosure are set forth in the description below. Other features, objects, and advantages of the disclosure will be apparent from the description and from the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing and other objects, features and advantages will be apparent from the following description of particular embodiments of the disclosure, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles of various embodiments of the disclosure.

FIG. 1 shows a schematic of one example of a RNA (e.g. mRNA) vaccine construct of the present disclosure. The construct depicts a human metapneumovirus and human respiratory syncytial virus full length fusion protein obtained from wild-type strains (*The Journal of General Virology*. 2008; 89(Pt 12):3113-3118, incorporated herein by reference).

FIGS. 2A-2C are graphs showing the levels of anti-hMPV fusion protein-specific antibodies in the serum of mice immunized with hMPV mRNA vaccines on day 0 (FIG. 2A), day 14 (FIG. 2B) and day 35 (FIG. 2C) post immunization. The mice were immunized with a single dose (2 µg or 10 µg) on day 0 and were given a boost dose (2 µg or 10 µg) on day 21, hMPV fusion protein-specific antibodies were detected at up to 1:10000 dilution of serum on day 35 for both doses.

FIGS. 3A-3C are graphs showing the result of IgG isotyping in the serum of mice immunized with hMPV mRNA vaccines. The levels of hMPV fusion protein-specific IgG2a (FIG. 3A) and IgG1 (FIG. 3B) antibodies in the serum are measured by ELISA. FIG. 3C shows that hMPV fusion protein mRNA vaccine induced a mixed Th1/Th2 cytokine response with a Th1 bias.

FIG. 4 is a graph showing in vitro neutralization of a hMPV B2 strain (TN/91-316) using the sera of mice immunized with a mRNA vaccine encoding hMPV fusion protein. Mouse serum obtained from mice receiving a 10 µg or a 2 µg dose contained hMPV-neutralizing antibodies.

FIGS. 5A-5C are graphs showing a Th1 cytokine response induced by a hMPV fusion peptide pool (15-mers-50 (overlap)) in splenocytes isolated from mice immunized with the hMPV mRNA vaccines. Virus-free media was used as a negative control and Concanavalin A (ConA, a positive control for splenocyte stimulation) was included. The cytokines tested included IFN-γ (FIG. 5A), IL-2 (FIG. 5B) and IL12 (FIG. 5C).

FIGS. 6A-6E are graphs showing the Th2 cytokine response induced by a hMPV fusion peptide pool (15-mers-50) in splenocytes isolated from mice immunized with the

hMPV mRNA vaccines. Virus-free media was used as a negative control and Concanavalin A was also included. The cytokines tested included IL-10 (FIG. 6A), TNF-α (FIG. 6B), IL4 (FIG. 6C), IL-5 (FIG. 6D) and IL-6 (FIG. 6E).

FIGS. 7A-7C are graphs showing the Th1 response induced by inactivated hMPV virus in splenocytes isolated from mice immunized with hMPV mRNA vaccines. Virus-free media was used as a negative control and Concanavalin A was included. The cytokines tested included IFN-γ (FIG. 7A), IL-2 (FIG. 7B) and IL12 (FIG. 7C).

FIGS. 8A-8E are graphs showing the Th2 response induced by inactivated hMPV virus in splenocytes isolated from mice immunized with the hMPV mRNA vaccines. Virus-free media was used as a negative control and Concanavalin A was included. The cytokines tested include IL-10 (FIG. 8A), TNF-α (FIG. 8B), IL4 (FIG. 8C), IL-5 (FIG. 8D) and IL-6 (FIG. 8E).

FIGS. 9A-9B are graphs showing the results of cotton rat challenge experiments. Two different doses of the hMPV mRNA vaccines were used (2 µg or 10 µg doses) to immunize the cotton rats before challenge. The hMPV mRNA vaccines reduced the viral titer in the lung and nose of the cotton rat, with the 10 µg dose being more effective in reducing viral titer. Use of a 10 µg dose resulted in 100% protection in the lung and a ~2 log reduction in nose viral titer. Use of a 2 µg dose resulted in a 1 log reduction in lung viral titer and no reduction in nose viral titer. The vaccine was administered on Day 0, and a boost was administered on Day 21.

FIG. 10 is a graph showing the lung histopathology of cotton rats that received hMPV mRNA vaccines. Pathology associated with vaccine-enhanced disease was not observed in immunized groups.

FIG. 11 is a graph showing hMPV neutralization antibody titers in cotton rats that received hMPV mRNA vaccines (2 µg or 10 µg doses) on days 35 and 42 post immunization.

FIG. 12 is a graph showing the lung and nose viral load in cotton rats challenged with a hMPV/A2 strain after immunization with the indicated mRNA vaccines (hMPV mRNA vaccine or hMPV/PIV mRNA combination vaccine). Vaccinated cotton rats showed reduced lung and nose viral loads after challenge, compared to control.

FIG. 13 is a graph showing the lung and nose viral load in cotton rats challenged with PIV3 strain after immunization with indicated mRNA vaccines (PIV mRNA vaccine or hMPV/PIV combination vaccine). Vaccinated cotton rats showed reduced lung and nose viral loads after challenge, compared to control.

FIG. 14 is a graph showing hMPV neutralizing antibody titers in cotton rats that received different dosages of hMPV mRNA vaccines or hMPV/PIV combination mRNA vaccines on day 42 post immunization. The dosages of the vaccine are indicated in Table 9.

FIG. 15 is a graph showing PIV3 neutralizing antibody titers in cotton rats that received different dosages of PIV mRNA vaccines or hMPV/PIV combination mRNA vaccines on day 42 post immunization. The dosages of the vaccine are indicated in Table 9.

FIG. 16 is a graph showing the lung histopathology score of cotton rats immunized with hMPV mRNA vaccines, PIV mRNA vaccines or hMPV/PIV combination mRNA vaccines as indicated in Table 9. Low occurrence of alevolitis and interstitial pneumonia was observed, indicating no antibody-dependent enhancement (ADE) of hMPV associated diseases.

FIG. 17 is a graph showing the reciprocal MERS-CoV neutralizing antibody titers in mice immunized with beta-

coronavirus mRNA vaccine encoding the MERS-CoV full-length Spike protein, on days 0, 21, 42, and 56 post immunization.

FIG. 18 is a graph showing the reciprocal MERS-CoV neutralizing antibody titers in mice immunized with beta-coronavirus mRNA vaccine encoding either the MERS-CoV full-length Spike protein, or the S2 subunit of the Spike protein. The full length spike protein induced a stronger immune response compared to the S2 subunit alone.

FIGS. 19A-19C are graphs showing the viral load in the nose and throat, the bronchoalveolar lavage (BAL), or the lungs of New Zealand white rabbits 4 days post challenge with MERS-CoV. The New Zealand white rabbits were immunized with one 20 µg-dose (on day 0) or two 20 µg-doses (on day 0 and 21) of MERS-CoV mRNA vaccine encoding the full-length Spike protein before challenge. FIG. 19A shows that two doses of MERS-CoV mRNA vaccine resulted in a 3 log reduction of viral load in the nose and led to complete protection in the throat of the New Zealand white rabbits. FIG. 19B shows that two doses of MERS-CoV mRNA vaccine resulted in a 4 log reduction of viral load in the BAL of the New Zealand white rabbits. FIG. 19C show one dose of MERS-CoV mRNA vaccine resulted in a 2 log reduction of viral load, while two doses of MERS-CoV mRNA vaccine resulted in an over 4 log reduction of viral load in the lungs of the New Zealand white rabbits.

FIGS. 20A-20B are images and graphs showing viral load or replicating virus detected by PCR in the lungs of New Zealand white rabbits 4 days post challenge with MERS-CoV. The New Zealand white rabbits were immunized with a single 20 µg dose (on day 0, Group 1a) of MERS-CoV mRNA vaccine encoding the full-length Spike protein, two 20 µg doses (on day 0 and 21, Group 1b) of MERS-CoV mRNA vaccine encoding the full-length Spike protein, or placebo (Group 2) before challenge. FIG. 20A shows that two doses of 20 µg a MERS-CoV mRNA vaccine reduced over 99% (2 log) of viruses in the lungs of New Zealand white rabbits. FIG. 20B shows that the group of New Zealand white rabbits that received 2 doses of 20 µg MERS-CoV mRNA vaccine did not have any detectable replicating MERS-CoV virus in their lungs.

FIG. 21 is a graph showing the MERS-CoV neutralizing antibody titers in New Zealand white rabbits immunized with MERS-CoV mRNA vaccine encoding the full-length Spike protein. The results show that two doses of 20 µg MERS-CoV mRNA vaccine induced a significant amount of neutralizing antibodies against MERS-CoV (EC₅₀ between 500-1000). The MERS-CoV mRNA vaccine induced antibody titer is 3-5 fold better than any other vaccines tested in the same model.

DETAILED DESCRIPTION

The present disclosure provides, in some embodiments, vaccines that comprise RNA (e.g., mRNA) polynucleotides encoding a human metapneumovirus (hMPV) antigenic polypeptide, a parainfluenza virus type 3 (PIV3) antigenic polypeptide, a respiratory syncytial virus (RSV) antigenic polypeptide, a measles virus (MeV) antigenic polypeptide, or a betacoronavirus antigenic polypeptide (e.g., Middle East respiratory syndrome coronavirus (MERS-CoV), SARS-CoV, human coronavirus (HCoV)—OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH (New Haven) and HCoV-HKU1) (see, e.g., Esper F. et al. *Emerging Infectious Diseases*, 12(5), 2006; and Pyrc K. et al. *Journal of Virology*, 81(7):3051-57, 2007, the contents of each of

which is here incorporated by reference in their entirety). The present disclosure also provides, in some embodiments, combination vaccines that comprise at least one RNA (e.g., mRNA) polynucleotide encoding at least two antigenic polypeptides selected from hMPV antigenic polypeptides, PIV3 antigenic polypeptides, RSV antigenic polypeptides, MeV antigenic polypeptides and BetaCoV antigenic polypeptides. Also provided herein are methods of administering the RNA (e.g., mRNA) vaccines, methods of producing the RNA (e.g., mRNA) vaccines, compositions (e.g., pharmaceutical compositions) comprising the RNA (e.g., mRNA) vaccines, and nucleic acids (e.g., DNA) encoding the RNA (e.g., mRNA) vaccines. In some embodiments, a RNA (e.g., mRNA) vaccine comprises an adjuvant, such as a flagellin adjuvant, as provided herein.

The RNA (e.g., mRNA) vaccines (e.g., hMPV, PIV3, RSV, MeV, BetaCoV RNA vaccines and combinations thereof), in some embodiments, may be used to induce a balanced immune response, comprising both cellular and humoral immunity, without many of the risks associated with DNA vaccination.

The entire contents of International Application No. PCT/US2015/02740 is incorporated herein by reference.

Human Metapneumovirus (hMPV)

hMPV shares substantial homology with respiratory syncytial virus (RSV) in its surface glycoproteins. hMPV fusion protein (F) is related to other paramyxovirus fusion proteins and appears to have homologous regions that may have similar functions. The hMPV fusion protein amino acid sequence contains features characteristic of other paramyxovirus F proteins, including a putative cleavage site and potential N-linked glycosylation sites. Paramyxovirus fusion proteins are synthesized as inactive precursors (F0) that are cleaved by host cell proteases into the biologically fusion-active F1 and F2 domains (see, e.g., Cseke G. et al. *Journal of Virology* 2007; 81(2):698-707, incorporated herein by reference). hMPV has one putative cleavage site, in contrast to the two sites established for RSV F, and only shares 34% amino acid sequence identity with RSV F. F2 is extracellular and disulfide linked to F1. Fusion proteins are type I glycoproteins existing as trimers, with two 4-3 heptad repeat domains at the N- and C-terminal regions of the protein (HR1 and HR2), which form coiled-coil alpha-helices. These coiled coils become apposed in an antiparallel fashion when the protein undergoes a conformational change into the fusogenic state. There is a hydrophobic fusion peptide N proximal to the N-terminal heptad repeat, which is thought to insert into the target cell membrane, while the association of the heptad repeats brings the transmembrane domain into close proximity, inducing membrane fusion (see, e.g., Baker, K A et al. *Mol. Cell* 1999; 3:309-319). This mechanism has been proposed for a number of different viruses, including RSV, influenza virus, and human immunodeficiency virus. Fusion proteins are major antigenic determinants for all known paramyxoviruses and for other viruses that possess similar fusion proteins such as human immunodeficiency virus, influenza virus, and Ebola virus.

In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding hMPV fusion protein (F). In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding a F1 or F2 subunit of a hMPV F protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding hMPV glycoprotein (G). In some embodiments, a hMPV vaccine of the present disclosure

comprises a RNA (e.g., mRNA) polynucleotide encoding hMPV matrix protein (M). In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding hMPV phosphoprotein (P). In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding hMPV nucleoprotein (N). In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding hMPV SH protein (SH).

In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein, M protein, P protein, N protein and SH protein.

In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and G protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and M protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and P protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and N protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and SH protein.

In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and M protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and P protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and N protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and SH protein.

In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and M protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and P protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and N protein. In some embodiments, a hMPV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and SH protein.

A hMPV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one hMPV antigenic polypeptide identified by any one of SEQ ID NO: 5-8 (Table 3; see also amino acid sequences of Table 4).

A hMPV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide encoded by a nucleic acid (e.g., DNA) identified by any one of SEQ ID NO: 1-4 (Table 2).

The present disclosure is not limited by a particular strain of hMPV. The strain of hMPV used in a vaccine may be any strain of hMPV. Non-limiting examples of strains of hMPV for use as provide herein include the CAN98-75 (CAN75) and the CAN97-83 (CAN83) hMPV strains (Skiadopoulos M H et al. *J Virol.* 20014; 78(13):6927-37, incorporated herein by reference), a hMPV A1, A2, B1 or B2 strain (see, e.g., de Graaf M et al. *The Journal of General Virology*

2008; 89:975-83; Peret T C T et al. *The Journal of Infectious Disease* 2002; 185:1660-63, incorporated herein by reference), a hMPV isolate TN/92-4 (e.g., SEQ ID NO: 1 and 5), a hMPV isolate NL/1/99 (e.g., SEQ ID NO: 2 and 6), or a hMPV isolate PER/CFI0497/2010/B (e.g., SEQ ID NO: 3 and 7).

In some embodiments, at least one hMPV antigenic polypeptide is obtained from a hMPV A1, A2, B1 or B2 strain (see, e.g., de Graaf M et al. *The Journal of General Virology* 2008; 89:975-83; Peret T C T et al. *The Journal of Infectious Disease* 2002; 185:1660-63, incorporated herein by reference). In some embodiments, at least one antigenic polypeptide is obtained from the CAN98-75 (CAN75) hMPV strain. In some embodiments, at least one antigenic polypeptide is obtained from the CAN97-83 (CAN83) hMPV strain. In some embodiments, at least one antigenic polypeptide is obtained from hMPV isolate TN/92-4 (e.g., SEQ ID NO: 1 and 5). In some embodiments, at least one antigenic polypeptide is obtained from hMPV isolate NL/1/99 (e.g., SEQ ID NO: 2 and 6). In some embodiments, at least one antigenic polypeptide is obtained from hMPV isolate PER/CFI0497/2010/B (e.g., SEQ ID NO: 3 and 7).

In some embodiments, hMPV vaccines comprise RNA (e.g., mRNA) polynucleotides encoding a hMPV antigenic polypeptides having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with hMPV F protein and having F protein activity.

A protein is considered to have F protein activity if, for example, the protein acts to fuse the viral envelope and host cell plasma membrane, mediates viral entry into a host cell via an interaction with arginine-glycine-aspartate RGD-binding integrins, or a combination thereof (see, e.g., Cox R G et al. *J Virol.* 2012; 88(22):12148-60, incorporated herein by reference).

In some embodiments, hMPV vaccines comprise RNA (e.g., mRNA) polynucleotides encoding hMPV antigenic polypeptides having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with hMPV G protein and having G protein activity.

A protein is considered to have G protein activity if, for example, the protein acts to modulate (e.g., inhibit) hMPV-induced cellular (immune) responses (see, e.g., Bao X et al. *PLoS Pathog.* 2008; 4(5):e1000077, incorporated herein by reference).

Human Parainfluenza Virus Type 3 (PIV3)

Parainfluenza viruses belong to the family Paramyxoviridae. These are enveloped viruses with a negative-sense single-stranded RNA genome. Parainfluenza viruses belong to the subfamily Paramyxoviridae, which is subdivided into three genera: Respirovirus (PIV-1, PIV-3, and Sendai virus (SeV)), Rubulavirus (PIV-2, PIV-4 and mumps virus) and *Morbillivirus* (measles virus, rinderpest virus and canine distemper virus (CDV)). Their genome, a ~15 500 nucleotide-long negative-sense RNA molecule, encodes two envelope glycoproteins, the hemagglutinin-neuraminidase (HN), the fusion protein (F or F0), which is cleaved into F1 and F2 subunits, a matrix protein (M), a nucleocapsid protein (N) and several nonstructural proteins including the viral replicase (L). All parainfluenza viruses, except for PIV-1, express a non-structural V protein that blocks IFN signaling in the infected cell and acts therefore as a virulence factor (see, e.g., Nishio M et al. *J Virol.* 2008; 82(13):6130-38).

PIV3 hemagglutinin-neuraminidase (HN), a structural protein, is found on the viral envelope, where it is necessary for attachment and cell entry. It recognizes and binds to sialic acid-containing receptors on the host cell's surface. As a neuroaminidase, HN removes sialic acid from virus par-

ticles, preventing self-aggregation of the virus, and promoting the efficient spread of the virus. Furthermore, HN promotes the activity of the fusion (F or F0) protein, contributing to the penetration of the host cell's surface.

PIV3 fusion protein (PIV3 F) is located on the viral envelope, where it facilitates the viral fusion and cell entry. The F protein is initially inactive, but proteolytic cleavage leads to its active forms, F1 and F2, which are linked by disulfide bonds. This occurs when the HN protein binds its receptor on the host cell's surface. During early phases of infection, the F glycoprotein mediates penetration of the host cell by fusion of the viral envelope to the plasma membrane. In later stages of the infection, the F protein facilitates the fusion of the infected cells with neighboring uninfected cells, which leads to the formation of a syncytium and spread of the infection.

PIV3 matrix protein (M) is found within the viral envelope and assists with viral assembly. It interacts with the nucleocapsid and envelope glycoproteins, where it facilitates the budding of progeny viruses through its interactions with specific sites on the cytoplasmic tail of the viral glycoproteins and nucleocapsid. It also plays a role in transporting viral components to the budding site.

PIV3 phosphoprotein (P) and PIV3 large polymerase protein (L) are found in the nucleocapsid where they form part of the RNA polymerase complex. The L protein, a viral RNA-dependent RNA polymerase, facilitates genomic transcription, while the host cell's ribosomes translate the viral mRNA into viral proteins.

PIV3 V is a non-structural protein that blocks IFN signaling in the infected cell, therefore acting as a virulence factor.

PIV3 nucleoprotein (N) encapsidates the genome in a ratio of 1 N per 6 ribonucleotides, protecting it from nucleases. The nucleocapsid (NC) has a helical structure. The encapsidated genomic RNA is termed the NC and serves as template for transcription and replication. During replication, encapsidation by PIV3 N is coupled to RNA synthesis and all replicative products are resistant to nucleases. PIV3 N homo-multimerizes to form the nucleocapsid and binds to viral genomic RNA. PIV3 N binds the P protein and thereby positions the polymerase on the template.

In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding PIV3 fusion protein (F). In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding a F1 or F2 subunit of a PIV3 F protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding PIV3 hemagglutinin-neuraminidase (HN) (see, e.g., van Wyke Coelingh K L et al. *J Virol.* 1987; 61(5):1473-77, incorporated herein by reference). In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding PIV3 matrix protein (M). In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding PIV3 phosphoprotein (P). In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding PIV3 nucleoprotein (N).

In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, HN protein, M protein, P protein, and N protein.

In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and HN protein. In some embodiments,

a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and M protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and P protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and N protein.

In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HN protein and M protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HN protein and P protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HN protein and N protein.

In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, HN protein and M protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, HN protein and P protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, HN protein and N protein.

A PIV3 vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one PIV3 antigenic polypeptide identified by any one of SEQ ID NO: 12-13 (Table 6; see also amino acid sequences of Table 7).

A PIV3 vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide encoded by a nucleic acid (e.g., DNA) identified by any one of SEQ ID NO: 9-12 (Table 5; see also nucleic acid sequences of Table 7).

The present disclosure is not limited by a particular strain of PIV3. The strain of PIV3 used in a vaccine may be any strain of PIV3. A non-limiting example of a strain of PIV3 for use as provide herein includes HPIV3/*Homo sapiens*/PER/FLA4815/2008.

In some embodiments, PIV3 vaccines comprise RNA (e.g., mRNA) polynucleotides encoding a PIV3 antigenic polypeptides having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with PIV3 F protein and having F protein activity.

In some embodiments, PIV3 vaccines comprise RNA (e.g., mRNA) polynucleotides encoding PIV3 antigenic polypeptides having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with PIV3 hemagglutinin-neuraminidase (HN) and having hemagglutinin-neuraminidase activity.

A protein is considered to have hemagglutinin-neuraminidase activity if, for example, it is capable of both receptor binding and receptor cleaving. Such proteins are major surface glycoproteins that have functional sites for cell attachment and for neuraminidase activity. They are able to cause red blood cells to agglutinate and to cleave the glycosidic linkages of neuraminic acids, so they have the potential to both bind a potential host cell and then release the cell if necessary, for example, to prevent self-aggregation of the virus.

In some embodiments, PIV3 vaccines comprise RNA (e.g., mRNA) polynucleotides encoding PIV3 antigenic polypeptides having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with PIV3 HN, F (e.g., F, F1 or F2), M, N, L or V and having HN, F (e.g., F, F1 or F2), M, N, L or V activity, respectively.

Respiratory Syncytial Virus (RSV)

RSV is a negative-sense, single-stranded RNA virus of the genus *Pneumovirinae*. The virus is present in at least two antigenic subgroups, known as Group A and Group B, primarily resulting from differences in the surface G glycoproteins. Two RSV surface glycoproteins—G and F—mediate attachment with and attachment to cells of the respiratory epithelium. F surface glycoproteins mediate coalescence of neighboring cells. This results in the formation of syncytial cells. RSV is the most common cause of bronchiolitis. Most infected adults develop mild cold-like symptoms such as congestion, low-grade fever, and wheezing. Infants and small children may suffer more severe symptoms such as bronchiolitis and pneumonia. The disease may be transmitted among humans via contact with respiratory secretions.

The genome of RSV encodes at least three surface glycoproteins, including F, G, and SH, four nucleocapsid proteins, including L, P, N, and M2, and one matrix protein, M. Glycoprotein F directs viral penetration by fusion between the virion and the host membrane. Glycoprotein G is a type II transmembrane glycoprotein and is the major attachment protein. SH is a short integral membrane protein. Matrix protein M is found in the inner layer of the lipid bilayer and assists virion formation. Nucleocapsid proteins L, P, N, and M2 modulate replication and transcription of the RSV genome. It is thought that glycoprotein G tethers and stabilizes the virus particle at the surface of bronchial epithelial cells, while glycoprotein F interacts with cellular glycosaminoglycans to mediate fusion and delivery of the RSV virion contents into the host cell (Krzyzaniak M A et al. *PLoS Pathog* 2013; 9(4)).

In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding L protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding P protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding N protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding M2 protein. In some embodiments, a PIV3 vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding M protein.

In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein, L protein, P protein, N protein, M2 protein and M protein.

In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and G protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and L protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and P protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and N protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and M2 protein. In some embodiments,

a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and M protein.

In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and L protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and P protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and N protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and M2 protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding G protein and M protein.

In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and L protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and P protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and N protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and M2 protein. In some embodiments, a RSV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein, G protein and M protein.

The present disclosure is not limited by a particular strain of RSV. The strain of RSV used in a vaccine may be any strain of RSV.

In some embodiments, RSV vaccines comprise RNA (e.g., mRNA) polynucleotides encoding a RSV antigenic polypeptides having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with RSV F protein and having F protein activity.

In some embodiments, RSV vaccines comprise RNA (e.g., mRNA) polynucleotides encoding RSV antigenic polypeptides having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with RSV G protein and having G protein activity.

A protein is considered to have G protein activity if, for example, the protein acts to modulate (e.g., inhibit) hMPV-induced cellular (immune) responses (see, e.g., Bao X et al. *PLoS Pathog*. 2008; 4(5):e1000077, incorporated herein by reference).

Measles Virus (MeV)

Molecular epidemiologic investigations and virologic surveillance contribute notably to the control and prevention of measles. Nearly half of measles-related deaths worldwide occur in India, yet virologic surveillance data are incomplete for many regions of the country. Previous studies have documented the presence of measles virus genotypes D4, D7, and D8 in India, and genotypes D5, D9, D11, H1, and G3 have been detected in neighboring countries. Recently, MeV genotype B3 was detected in India (Kuttiatt V S et al. *Emerg Infect Dis*. 2014; 20(10): 1764-66).

The glycoprotein complex of paramyxoviruses mediates receptor binding and membrane fusion. In particular, the MeV fusion (F) protein executes membrane fusion, after receptor binding by the hemagglutinin (HA) protein (Muhlebach M D et al. *Journal of Virology* 2008; 82(22):11437-45). The MeV P gene codes for three proteins: P, an essential polymerase cofactor, and V and C, which have multiple functions but are not strictly required for viral propagation

in cultured cells. V shares the amino-terminal domain with P but has a zinc-binding carboxyl-terminal domain, whereas C is translated from an overlapping reading frame. The MeV C protein is an infectivity factor. During replication, the P protein binds incoming monomeric nucleocapsid (N) proteins with its amino-terminal domain and positions them for assembly into the nascent ribonucleocapsid. The P protein amino-terminal domain is natively unfolded (Deveaux P et al. *Journal of Virology* 2004; 78(21):11632-40).

In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding P protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding V protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding C protein.

In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein, F protein, P protein, V protein and C protein.

In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein and F protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein and P protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein and V protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein and C protein.

In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and P protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and V protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding F protein and C protein.

In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein, F protein and P protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein, F protein and V protein. In some embodiments, a MeV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding HA protein, F protein and C protein.

In some embodiments, MeV vaccines comprise RNA (e.g., mRNA) encoding a MeV antigenic polypeptide having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with MeV HA protein and having MeV HA protein activity.

In some embodiments, MeV vaccines comprise RNA (e.g., mRNA) encoding a MeV antigenic polypeptide having at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity with MeV F protein and having MeV F protein activity.

A protein is considered to have HA protein activity if the protein mediates receptor binding and/or membrane fusion. MeV F protein executes membrane fusion, after receptor binding by the MeV HA protein.

A MeV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one MeV antigenic polypeptide identified by any one of SEQ ID NO: 47-50 (Table 14; see also amino acid sequences of Table 15).

A MeV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide identified by any one of SEQ ID NO: 37, 40, 43, 46 (Table 13).

A MeV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide encoded by a nucleic acid (e.g., DNA) identified by any one of SEQ ID NO: 35, 36, 38, 39, 41, 42, 44 and 45 (Table 13).

The present disclosure is not limited by a particular strain of MeV. The strain of MeV used in a vaccine may be any strain of MeV. Non-limiting examples of strains of MeV for use as provide herein include B3/B3.1, C2, D4, D6, D7, D8, G3, H1, Moraten, Rubeovax, MVi/New Jersey.USA/45.05, MVi/Texas.USA/4.07, AIK-C, MVi/New York.USA/26.09/3, MVi/California.USA/16.03, MVi/Virginia.USA/15.09, MVi/California.USA/8.04, and MVi/Pennsylvania.USA/20.09.

MeV proteins may be from MeV genotype D4, D5, D7, D8, D9, D11, H1, G3 or B3. In some embodiments, a MeV HA protein or a MeV F protein is from MeV genotype D8. In some embodiments, a MeV HA protein or a MeV F protein is from MeV genotype B3. Betacoronaviruses (BetaCoV)

MERS-CoV. MERS-CoV is a positive-sense, single-stranded RNA virus of the genus Betacoronavirus. The genomes are phylogenetically classified into two clades, clade A and clade B. It has a strong tropism for non-ciliated bronchial epithelial cells, evades the innate immune response and antagonizes interferon (IFN) production in infected cells. Dipeptidyl peptidase 4 (DPP4, also known as CD26) has been identified as a functional cellular receptor for MERS-CoV. Its enzymatic activity is not required for infection, although its amino acid sequence is highly conserved across species and is expressed in the human bronchial epithelium and kidneys. Most infected individuals develop severe acute respiratory illnesses, including fever, cough, and shortness of breath, and the virus can be fatal. The disease may be transmitted among humans, generally among those in close contact.

The genome of MERS-CoV encodes at least four unique accessory proteins, such as 3, 4a, 4b and 5, two replicase proteins (open reading frame 1a and 1b), and four major structural proteins, including spike (S), envelope (E), nucleocapsid (N), and membrane (M) proteins (Almazan F et al. *M Bio* 2013; 4(5):e00650-13). The accessory proteins play nonessential roles in MERS-CoV replication, but they are likely structural proteins or interferon antagonists, modulating in vivo replication efficiency and/or pathogenesis, as in the case of SARS-CoV (Almazan F et al. *M Bio* 2013; 4(5):e00650-13; Totura A L et al. *Curr Opin Virol* 2012; 2(3):264-75; Scobey T et al. *Proc Natl Acad Sci USA* 2013; 110(40):16157-62). The other proteins of MERS-CoV maintain different functions in virus replication. The E protein, for example, involves in virulence, and deleting the E-coding gene results in replication-competent and propagation-defective viruses or attenuated viruses (Almazan F et al. *M Bio* 2013; 4(5):e00650-13). The S protein is particularly essential in mediating virus binding to cells expressing receptor dipeptidyl peptidase-4 (DPP4) through receptor-binding domain (RBD) in the S1 subunit, whereas the S2 subunit subsequently mediates virus entry via fusion of the virus and target cell membranes (Li F. *J Virol* 2015; 89(4): 1954-64; Raj V S et al. *Nature* 2013; 495(7440):251-4).

In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding the S1 subunit of the S protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding the S2 subunit of the S protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding E protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding N protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding M protein.

In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), E protein, N protein and M protein.

In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2) and E protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2) and N protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2) and M protein.

In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), E protein and M protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), E protein and N protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), M protein and N protein. In some embodiments, a MERS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding E protein, M protein and N protein.

A MERS-CoV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one MERS-CoV antigenic polypeptide identified by any one of SEQ ID NO: 24-38 or 33 (Table 11; see also amino acid sequences of Table 12).

A MERS-CoV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide encoded by a nucleic acid (e.g., DNA) identified by any one of SEQ ID NO: 20-23 (Table 10).

The present disclosure is not limited by a particular strain of MERS-CoV. The strain of MERS-CoV used in a vaccine may be any strain of MERS-CoV. Non-limiting examples of strains of MERS-CoV for use as provide herein include Riyadh_14_2013, and 2cEMC/2012. Hasi_1_2013.

SARS-CoV. The genome of SARS-CoV includes of a single, positive-strand RNA that is approximately 29,700 nucleotides long. The overall genome organization of SARS-CoV is similar to that of other coronaviruses. The reference genome includes 13 genes, which encode at least 14 proteins. Two large overlapping reading frames (ORFs) encompass 71% of the genome. The remainder has 12 potential ORFs, including genes for structural proteins S (spike), E (small envelope), M (membrane), and N (nucleocapsid). Other potential ORFs code for unique putative

SARS-CoV-specific polypeptides that lack obvious sequence similarity to known proteins. A detailed analysis of the SARS-CoV genome has been published in *J Mol Biol* 2003; 331: 991-1004.

In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), E protein, N protein and M protein.

In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2) and E protein. In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2) and N protein. In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2) and M protein.

In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), E protein and M protein. In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), E protein and N protein. In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding S protein (S, S1 and/or S2), M protein and N protein. In some embodiments, a SARS-CoV vaccine of the present disclosure comprises a RNA (e.g., mRNA) polynucleotide encoding E protein, M protein and N protein.

A SARS-CoV vaccine may comprise, for example, at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one SARS-CoV antigenic polypeptide identified by any one of SEQ ID NO: 29, 32 or 34 (Table 11; see also amino acid sequences of Table 12).

The present disclosure is not limited by a particular strain of SARS-CoV. The strain of SARS-CoV used in a vaccine may be any strain of SARS-CoV.

HCoV-OC43.

Human coronavirus OC43 is an enveloped, positive-sense, single-stranded RNA virus in the species Betacoronavirus-1 (genus Betacoronavirus, subfamily Coronavirinae, family Coronaviridae, order Nidovirales). Four HCoV-OC43 genotypes (A to D), have been identified with genotype D most likely arising from recombination. The complete genome sequencing of two genotype C and D strains and bootscan analysis shows recombination events between genotypes B and C in the generation of genotype D. Of 29 strains identified, none belong to the more ancient genotype A. Along with HCoV-229E, a species in the *Alphacoronavirus* genus, HCoV-OC43 are among the known viruses that cause the common cold. Both viruses can cause severe lower respiratory tract infections, including pneumonia in infants, the elderly, and immunocompromised individuals such as those undergoing chemotherapy and those with HIV-AIDS.

HCoV-HKU1.

Human coronavirus HKU1 (HCoV-HKU1) is a positive-sense, single-stranded RNA virus with the H4E gene, which distinguishes it as a group 2, or betacoronavirus. It was discovered in January 2005 in two patients in Hong Kong. The genome of HCoV-HKU1 is a 29,926-nucleotide, polyadenylated RNA. The GC content is 32%, the lowest among all known coronaviruses. The genome organization is the same as that of other group II coronaviruses, with the characteristic gene order 1a, 1b, HE, S, E, M, and N.

Furthermore, accessory protein genes are present between the S and E genes (ORF4) and at the position of the N gene (ORF8). The TRS is presumably located within the AAUC-UAAAC sequence, which precedes each ORF except E. As in sialodacryoadenitis virus and mouse hepatitis virus (MHV), translation of the E protein possibly occurs via an internal ribosomal entry site. The 3' untranslated region contains a predicted stem-loop structure immediately downstream of the N ORF (nucleotide position 29647 to 29711). Further downstream, a pseudoknot structure is present at nucleotide position 29708 to 29760. Both RNA structures are conserved in group II coronaviruses and are critical for virus replication. HCoV-NL63.

The RNA genome of human coronavirus NL63 (HCoV-NL63) is 27,553 nucleotides, with a poly(A) tail (FIG. 1). With a GC content of 34%, HCoV-NL63 has one of the lowest GC contents of the coronaviruses, for which GC content ranges from 32 to 42%. Untranslated regions of 286 and 287 nucleotides are present at the 5' and 3' termini, respectively. Genes predicted to encode the S, E, M, and N proteins are found in the 3' part of the HCoV-NL63 genome. The HE gene, which is present in some group II coronaviruses, is absent, and there is only a single, monocistronic accessory protein ORF (ORF3) located between the S and E genes. Subgenomic mRNAs are generated for all ORFs (S, ORF3, E, M, and N), and the core sequence of the TRS of HCoV-NL63 is defined as AACUAAA. This sequence is situated upstream of every ORF except for the E ORF, which contains the suboptimal core sequence AACUAUA. Interestingly, a 13-nucleotide sequence with perfect homology to the leader sequence is situated upstream of the suboptimal E TRS. Annealing of this 13-nucleotide sequence to the leader sequence may act as a compensatory mechanism for the disturbed leader-TRS/body-TRS interaction. HCoV-229E.

Human coronavirus 229E (HCoV-229E) is a single-stranded, positive-sense, RNA virus species in the *Alpha-coronavirus* genus of the subfamily Coronavirinae, in the family Coronaviridae, of the order Nidovirales. Along with Human coronavirus OC43, it is responsible for the common cold. HCoV-NL63 and HCoV-229E are two of the four human coronaviruses that circulate worldwide. These two viruses are unique in their relationship towards each other. Phylogenetically, the viruses are more closely related to each other than to any other human coronavirus, yet they only share 65% sequence identity. Moreover, the viruses use different receptors to enter their target cell. HCoV-NL63 is associated with croup in children, whereas all signs suggest that the virus probably causes the common cold in healthy adults. HCoV-229E is a proven common cold virus in healthy adults, so it is probable that both viruses induce comparable symptoms in adults, even though their mode of infection differs (HCoV-NL63 and HCoV-229E are two of the four human coronaviruses that circulate worldwide. These two viruses are unique in their relationship towards each other. Phylogenetically, the viruses are more closely related to each other than to any other human coronavirus, yet they only share 65% sequence identity. Moreover, the viruses use different receptors to enter their target cell. HCoV-NL63 is associated with croup in children, whereas all signs suggest that the virus probably causes the common cold in healthy adults. HCoV-229E is a proven common cold virus in healthy adults, so it is probable that both viruses induce comparable symptoms in adults, even though their mode of infection differs (Dijkman R. et al. *J Formos Med*

Assoc. 2009 April; 108(4):270-9, the contents of which is incorporated herein by reference in their entirety).

Combination Vaccines

Embodiments of the present disclosure also provide combination RNA (e.g., mRNA) vaccines. A "combination RNA (e.g., mRNA) vaccine" of the present disclosure refers to a vaccine comprising at least one (e.g., at least 2, 3, 4, or 5) RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a combination of any two or more (or all of) antigenic polypeptides selected from hMPV antigenic polypeptides, PIV3 antigenic polypeptides, RSV antigenic polypeptides, MeV antigenic polypeptides, and BetaCoV antigenic polypeptides (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a PIV3 antigenic polypeptide, a RSV antigenic polypeptide, a MeV antigenic polypeptide, and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide and a PIV3 antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide and a RSV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide and a MeV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide and a BetaCoV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a PIV3 antigenic polypeptide and a RSV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a PIV3 antigenic polypeptide and a MeV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a PIV3 antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a RSV antigenic polypeptide and a MeV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a RSV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a MeV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV,

SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a PIV3 antigenic polypeptide, a RSV antigenic polypeptide and a MeV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a PIV3 antigenic polypeptide, a RSV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a PIV3 antigenic polypeptide, a MeV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a RSV antigenic polypeptide, a MeV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a PIV3 antigenic polypeptide, a RSV antigenic polypeptide, a MeV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a PIV3 antigenic polypeptide and a RSV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a PIV3 antigenic polypeptide and a MeV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a PIV3 antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a RSV antigenic polypeptide and a MeV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a RSV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a hMPV antigenic polypeptide, a MeV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g.,

selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a PIV3 antigenic polypeptide, a RSV antigenic polypeptide and a MeV antigenic polypeptide.

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a PIV3 antigenic polypeptide, a RSV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

In some embodiments, a combination RNA (e.g., mRNA) vaccine comprises a RNA (e.g., mRNA) polynucleotide encoding a RSV antigenic polypeptide, a MeV antigenic polypeptide and a BetaCoV antigenic polypeptide (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1).

Other combination respiratory virus RNA (e.g., mRNA) vaccines are encompassed by the present disclosure.

It has been discovered that the mRNA vaccines described herein are superior to current vaccines in several ways. First, the lipid nanoparticle (LNP) delivery is superior to other formulations including a protamine base approach described in the literature and no additional adjuvants are to be necessary. The use of LNPs enables the effective delivery of chemically modified or unmodified mRNA vaccines. Additionally it has been demonstrated herein that both modified and unmodified LNP formulated mRNA vaccines were superior to conventional vaccines by a significant degree. In some embodiments the mRNA vaccines of the invention are superior to conventional vaccines by a factor of at least 10 fold, 20 fold, 40 fold, 50 fold, 100 fold, 500 fold or 1,000 fold.

Although attempts have been made to produce functional RNA vaccines, including mRNA vaccines and self-replicating RNA vaccines, the therapeutic efficacy of these RNA vaccines have not yet been fully established. Quite surprisingly, the inventors have discovered, according to aspects of the invention a class of formulations for delivering mRNA vaccines in vivo that results in significantly enhanced, and in many respects synergistic, immune responses including enhanced antigen generation and functional antibody production with neutralization capability. These results can be achieved even when significantly lower doses of the mRNA are administered in comparison with mRNA doses used in other classes of lipid based formulations. The formulations of the invention have demonstrated significant unexpected in vivo immune responses sufficient to establish the efficacy of functional mRNA vaccines as prophylactic and therapeutic agents. Additionally, self-replicating RNA vaccines rely on viral replication pathways to deliver enough RNA to a cell to produce an immunogenic response. The formulations of the invention do not require viral replication to produce enough protein to result in a strong immune response. Thus, the mRNA of the invention are not self-replicating RNA and do not include components necessary for viral replication.

The invention involves, in some aspects, the surprising finding that lipid nanoparticle (LNP) formulations significantly enhance the effectiveness of mRNA vaccines, including chemically modified and unmodified mRNA vaccines. The efficacy of mRNA vaccines formulated in LNP was examined in vivo using several distinct antigens. The results presented herein demonstrate the unexpected superior effi-

cacy of the mRNA vaccines formulated in LNP over other commercially available vaccines.

In addition to providing an enhanced immune response, the formulations of the invention generate a more rapid immune response with fewer doses of antigen than other vaccines tested. The mRNA-LNP formulations of the invention also produce quantitatively and qualitatively better immune responses than vaccines formulated in a different carriers.

The data described herein demonstrate that the formulations of the invention produced significant unexpected improvements over existing antigen vaccines. Additionally, the mRNA-LNP formulations of the invention are superior to other vaccines even when the dose of mRNA is lower than other vaccines. Mice immunized with either 10 µg or 2 µg doses of an hMPV fusion protein mRNA LNP vaccine or a PIV3 mRNA LNP vaccine produced neutralizing antibodies which for instance, successfully neutralized the hMPV B2 virus. A 10 µg dose of mRNA vaccine protected 100% of mice from lethal challenge and drastically reduced the viral titer after challenge (~2 log reduction).

Two 20 µg doses of MERS-CoV mRNA LNP vaccine significantly reduced viral load and induced significant amount of neutralizing antibodies against MERS-CoV (EC₅₀ between 500-1000). The MERS-CoV mRNA vaccine induced antibody titer was 3-5 fold better than any other vaccines tested in the same model.

The LNP used in the studies described herein has been used previously to deliver siRNA in various animal models as well as in humans. In view of the observations made in association with the siRNA delivery of LNP formulations, the fact that LNP is useful in vaccines is quite surprising. It has been observed that therapeutic delivery of siRNA formulated in LNP causes an undesirable inflammatory response associated with a transient IgM response, typically leading to a reduction in antigen production and a compromised immune response. In contrast to the findings observed with siRNA, the LNP-mRNA formulations of the invention are demonstrated herein to generate enhanced IgG levels, sufficient for prophylactic and therapeutic methods rather than transient IgM responses.

Nucleic Acids/Polynucleotides

Respiratory virus vaccines, as provided herein, comprise at least one (one or more) ribonucleic acid (RNA) (e.g., mRNA) polynucleotide having an open reading frame encoding at least one antigenic polypeptide selected from hMPV, PIV3, RSV, MeV and BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1) antigenic polypeptides. The term “nucleic acid” includes any compound and/or substance that comprises a polymer of nucleotides (nucleotide monomer). These polymers are referred to as polynucleotides. Thus, the terms “nucleic acid” and “polynucleotide” are used interchangeably.

Nucleic acids may be or may include, for example, ribonucleic acids (RNAs), deoxyribonucleic acids (DNAs), threose nucleic acids (TNAs), glycol nucleic acids (GNAs), peptide nucleic acids (PNAs), locked nucleic acids (LNAs), including LNA having a β-D-ribo configuration, α-LNA having an α-L-ribo configuration (a diastereomer of LNA), 2'-amino-LNA having a 2'-amino functionalization, and 2'-amino-α-LNA having a 2'-amino functionalization), ethylene nucleic acids (ENAs), cyclohexenyl nucleic acids (CeNA) or chimeras or combinations thereof.

In some embodiments, polynucleotides of the present disclosure function as messenger RNA (mRNA). “Messenger RNA” (mRNA) refers to any polynucleotide that

encodes a (at least one) polypeptide (a naturally-occurring, non-naturally-occurring, or modified polymer of amino acids) and can be translated to produce the encoded polypeptide in vitro, in vivo, in situ or ex vivo. The skilled artisan will appreciate that, except where otherwise noted, polynucleotide sequences set forth in the instant application will recite “T”s in a representative DNA sequence but where the sequence represents RNA (e.g., mRNA), the “T”s would be substituted for “U”s. Thus, any of the RNA polynucleotides encoded by a DNA identified by a particular sequence identification number may also comprise the corresponding RNA (e.g., mRNA) sequence encoded by the DNA, where each “T” of the DNA sequence is substituted with “U.”

The basic components of an mRNA molecule typically include at least one coding region, a 5' untranslated region (UTR), a 3' UTR, a 5' cap and a poly-A tail. Polynucleotides of the present disclosure may function as mRNA but can be distinguished from wild-type mRNA in their functional and/or structural design features, which serve to overcome existing problems of effective polypeptide expression using nucleic-acid based therapeutics.

In some embodiments, a RNA polynucleotide of an RNA (e.g., mRNA) vaccine encodes 2-10, 2-9, 2-8, 2-7, 2-6, 2-5, 2-4, 2-3, 3-10, 3-9, 3-8, 3-7, 3-6, 3-5, 3-4, 4-10, 4-9, 4-8, 4-7, 4-6, 4-5, 5-10, 5-9, 5-8, 5-7, 5-6, 6-10, 6-9, 6-8, 6-7, 7-10, 7-9, 7-8, 8-10, 8-9 or 9-10 antigenic polypeptides. In some embodiments, a RNA (e.g., mRNA) polynucleotide of a respiratory virus vaccine encodes at least 10, 20, 30, 40, 50, 60, 70, 80, 90 or 100 antigenic polypeptides. In some embodiments, a RNA (e.g., mRNA) polynucleotide of a respiratory virus vaccine encodes at least 100 or at least 200 antigenic polypeptides. In some embodiments, a RNA polynucleotide of an respiratory virus vaccine encodes 1-10, 5-15, 10-20, 15-25, 20-30, 25-35, 30-40, 35-45, 40-50, 1-50, 1-100, 2-50 or 2-100 antigenic polypeptides.

Polynucleotides of the present disclosure, in some embodiments, are codon optimized. Codon optimization methods are known in the art and may be used as provided herein. Codon optimization, in some embodiments, may be used to match codon frequencies in target and host organisms to ensure proper folding; bias GC content to increase mRNA stability or reduce secondary structures; minimize tandem repeat codons or base runs that may impair gene construction or expression; customize transcriptional and translational control regions; insert or remove protein trafficking sequences; remove/add post translation modification sites in encoded protein (e.g. glycosylation sites); add, remove or shuffle protein domains; insert or delete restriction sites; modify ribosome binding sites and mRNA degradation sites; adjust translational rates to allow the various domains of the protein to fold properly; or to reduce or eliminate problem secondary structures within the polynucleotide. Codon optimization tools, algorithms and services are known in the art—non-limiting examples include services from GeneArt (Life Technologies), DNA2.0 (Menlo Park CA) and/or proprietary methods. In some embodiments, the open reading frame (ORF) sequence is optimized using optimization algorithms.

In some embodiments, a codon optimized sequence shares less than 95% sequence identity, less than 90% sequence identity, less than 85% sequence identity, less than 80% sequence identity, or less than 75% sequence identity to a naturally-occurring or wild-type sequence (e.g., a naturally-occurring or wild-type mRNA sequence encoding a polypeptide or protein of interest (e.g., an antigenic protein or antigenic polypeptide)).

In some embodiments, a codon-optimized sequence shares between 65% and 85% (e.g., between about 67% and about 85%, or between about 67% and about 80%) sequence identity to a naturally-occurring sequence or a wild-type sequence (e.g., a naturally-occurring or wild-type mRNA sequence encoding a polypeptide or protein of interest (e.g., an antigenic protein or polypeptide)). In some embodiments, a codon-optimized sequence shares between 65% and 75%, or about 80% sequence identity to a naturally-occurring sequence or wild-type sequence (e.g., a naturally-occurring or wild-type mRNA sequence encoding a polypeptide or protein of interest (e.g., an antigenic protein or polypeptide)).

In some embodiments a codon-optimized RNA (e.g., mRNA) may, for instance, be one in which the levels of G/C are enhanced. The G/C-content of nucleic acid molecules may influence the stability of the RNA. RNA having an increased amount of guanine (G) and/or cytosine (C) residues may be functionally more stable than nucleic acids containing a large amount of adenine (A) and thymine (T) or uracil (U) nucleotides. WO02/098443 discloses a pharmaceutical composition containing an mRNA stabilized by sequence modifications in the translated region. Due to the degeneracy of the genetic code, the modifications work by substituting existing codons for those that promote greater RNA stability without changing the resulting amino acid. The approach is limited to coding regions of the RNA.

Antigens/Antigenic Polypeptides

In some embodiments, an antigenic polypeptide (e.g., a hMPV, PIV3, RSV, MeV or BetaCoV antigenic polypeptide) is longer than 25 amino acids and shorter than 50 amino acids. Polypeptides include gene products, naturally occurring polypeptides, synthetic polypeptides, homologs, orthologs, paralogs, fragments and other equivalents, variants, and analogs of the foregoing. A polypeptide may be a single molecule or may be a multi-molecular complex such as a dimer, trimer or tetramer. Polypeptides may also comprise single chain polypeptides or multichain polypeptides, such as antibodies or insulin, and may be associated or linked to each other. Most commonly, disulfide linkages are found in multichain polypeptides. The term "polypeptide" may also apply to amino acid polymers in which at least one amino acid residue is an artificial chemical analogue of a corresponding naturally-occurring amino acid.

A "polypeptide variant" is a molecule that differs in its amino acid sequence relative to a native sequence or a reference sequence. Amino acid sequence variants may possess substitutions, deletions, insertions, or a combination of any two or three of the foregoing, at certain positions within the amino acid sequence, as compared to a native sequence or a reference sequence. Ordinarily, variants possess at least 50% identity to a native sequence or a reference sequence. In some embodiments, variants share at least 80% identity or at least 90% identity with a native sequence or a reference sequence.

In some embodiments "variant mimics" are provided. A "variant mimic" contains at least one amino acid that would mimic an activated sequence. For example, glutamate may serve as a mimic for phospho-threonine and/or phospho-serine. Alternatively, variant mimics may result in deactivation or in an inactivated product containing the mimic. For example, phenylalanine may act as an inactivating substitution for tyrosine, or alanine may act as an inactivating substitution for serine.

"Orthologs" refers to genes in different species that evolved from a common ancestral gene by speciation. Normally, orthologs retain the same function in the course of

evolution. Identification of orthologs is important for reliable prediction of gene function in newly sequenced genomes.

"Analog" is meant to include polypeptide variants that differ by one or more amino acid alterations, for example, substitutions, additions or deletions of amino acid residues that still maintain one or more of the properties of the parent or starting polypeptide.

The present disclosure provides several types of compositions that are polynucleotide or polypeptide based, including variants and derivatives. These include, for example, substitutional, insertional, deletion and covalent variants and derivatives. The term "derivative" is synonymous with the term "variant" and generally refers to a molecule that has been modified and/or changed in any way relative to a reference molecule or a starting molecule.

As such, polynucleotides encoding peptides or polypeptides containing substitutions, insertions and/or additions, deletions and covalent modifications with respect to reference sequences, in particular the polypeptide sequences disclosed herein, are included within the scope of this disclosure. For example, sequence tags or amino acids, such as one or more lysines, can be added to peptide sequences (e.g., at the N-terminal or C-terminal ends). Sequence tags can be used for peptide detection, purification or localization. Lysines can be used to increase peptide solubility or to allow for biotinylation. Alternatively, amino acid residues located at the carboxy and amino terminal regions of the amino acid sequence of a peptide or protein may optionally be deleted providing for truncated sequences. Certain amino acids (e.g., C-terminal residues or N-terminal residues) alternatively may be deleted depending on the use of the sequence, as for example, expression of the sequence as part of a larger sequence that is soluble, or linked to a solid support.

"Substitutional variants" when referring to polypeptides are those that have at least one amino acid residue in a native or starting sequence removed and a different amino acid inserted in its place at the same position. Substitutions may be single, where only one amino acid in the molecule has been substituted, or they may be multiple, where two or more (e.g., 3, 4 or 5) amino acids have been substituted in the same molecule.

As used herein the term "conservative amino acid substitution" refers to the substitution of an amino acid that is normally present in the sequence with a different amino acid of similar size, charge, or polarity. Examples of conservative substitutions include the substitution of a non-polar (hydrophobic) residue such as isoleucine, valine and leucine for another non-polar residue. Likewise, examples of conservative substitutions include the substitution of one polar (hydrophilic) residue for another such as between arginine and lysine, between glutamine and asparagine, and between glycine and serine. Additionally, the substitution of a basic residue such as lysine, arginine or histidine for another, or the substitution of one acidic residue such as aspartic acid or glutamic acid for another acidic residue are additional examples of conservative substitutions. Examples of non-conservative substitutions include the substitution of a non-polar (hydrophobic) amino acid residue such as isoleucine, valine, leucine, alanine, methionine for a polar (hydrophilic) residue such as cysteine, glutamine, glutamic acid or lysine and/or a polar residue for a non-polar residue.

"Features" when referring to polypeptide or polynucleotide are defined as distinct amino acid sequence-based or nucleotide-based components of a molecule respectively. Features of the polypeptides encoded by the polynucleotides

include surface manifestations, local conformational shape, folds, loops, half-loops, domains, half-domains, sites, termini and any combination(s) thereof.

As used herein when referring to polypeptides the term “domain” refers to a motif of a polypeptide having one or more identifiable structural or functional characteristics or properties (e.g., binding capacity, serving as a site for protein-protein interactions).

As used herein when referring to polypeptides the terms “site” as it pertains to amino acid based embodiments is used synonymously with “amino acid residue” and “amino acid side chain.” As used herein when referring to polynucleotides the terms “site” as it pertains to nucleotide based embodiments is used synonymously with “nucleotide.” A site represents a position within a peptide or polypeptide or polynucleotide that may be modified, manipulated, altered, derivatized or varied within the polypeptide-based or polynucleotide-based molecules.

As used herein the terms “termini” or “terminus” when referring to polypeptides or polynucleotides refers to an extremity of a polypeptide or polynucleotide respectively. Such extremity is not limited only to the first or final site of the polypeptide or polynucleotide but may include additional amino acids or nucleotides in the terminal regions. Polypeptide-based molecules may be characterized as having both an N-terminus (terminated by an amino acid with a free amino group (NH₂)) and a C-terminus (terminated by an amino acid with a free carboxyl group (COOH)). Proteins are in some cases made up of multiple polypeptide chains brought together by disulfide bonds or by non-covalent forces (multimers, oligomers). These proteins have multiple N- and C-termini. Alternatively, the termini of the polypeptides may be modified such that they begin or end, as the case may be, with a non-polypeptide based moiety such as an organic conjugate.

As recognized by those skilled in the art, protein fragments, functional protein domains, and homologous proteins are also considered to be within the scope of polypeptides of interest. For example, provided herein is any protein fragment (meaning a polypeptide sequence at least one amino acid residue shorter than a reference polypeptide sequence but otherwise identical) of a reference protein having a length of 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 or longer than 100 amino acids. In another example, any protein that includes a stretch of 20, 30, 40, 50, or 100 (contiguous) amino acids that are 40%, 50%, 60%, 70%, 80%, 90%, 95%, or 100% identical to any of the sequences described herein can be utilized in accordance with the disclosure. In some embodiments, a polypeptide includes 2, 3, 4, 5, 6, 7, 8, 9, 10, or more mutations as shown in any of the sequences provided herein or referenced herein. In another example, any protein that includes a stretch of 20, 30, 40, 50, or 100 amino acids that are greater than 80%, 90%, 95%, or 100% identical to any of the sequences described herein, wherein the protein has a stretch of 5, 10, 15, 20, 25, or 30 amino acids that are less than 80%, 75%, 70%, 65% to 60% identical to any of the sequences described herein can be utilized in accordance with the disclosure.

Polypeptide or polynucleotide molecules of the present disclosure may share a certain degree of sequence similarity or identity with the reference molecules (e.g., reference polypeptides or reference polynucleotides), for example, with art-described molecules (e.g., engineered or designed molecules or wild-type molecules). The term “identity,” as known in the art, refers to a relationship between the sequences of two or more polypeptides or polynucleotides, as determined by comparing the sequences. In the art,

identity also means the degree of sequence relatedness between two sequences as determined by the number of matches between strings of two or more amino acid residues or nucleic acid residues. Identity measures the percent of identical matches between the smaller of two or more sequences with gap alignments (if any) addressed by a particular mathematical model or computer program (e.g., “algorithms”). Identity of related peptides can be readily calculated by known methods. “% identity” as it applies to polypeptide or polynucleotide sequences is defined as the percentage of residues (amino acid residues or nucleic acid residues) in the candidate amino acid or nucleic acid sequence that are identical with the residues in the amino acid sequence or nucleic acid sequence of a second sequence after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent identity. Methods and computer programs for the alignment are well known in the art. Identity depends on a calculation of percent identity but may differ in value due to gaps and penalties introduced in the calculation. Generally, variants of a particular polynucleotide or polypeptide have at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% but less than 100% sequence identity to that particular reference polynucleotide or polypeptide as determined by sequence alignment programs and parameters described herein and known to those skilled in the art. Such tools for alignment include those of the BLAST suite (Stephen F. Altschul, et al. (1997). “Gapped BLAST and PSI-BLAST: a new generation of protein database search programs,” *Nucleic Acids Res.* 25:3389-3402). Another popular local alignment technique is based on the Smith-Waterman algorithm (Smith, T. F. & Waterman, M. S. (1981) “Identification of common molecular subsequences.” *J. Mol. Biol.* 147:195-197). A general global alignment technique based on dynamic programming is the Needleman-Wunsch algorithm (Needleman, S. B. & Wunsch, C. D. (1970) “A general method applicable to the search for similarities in the amino acid sequences of two proteins.” *J. Mol. Biol.* 48:443-453). More recently, a Fast Optimal Global Sequence Alignment Algorithm (FOGSAA) was developed that purportedly produces global alignment of nucleotide and protein sequences faster than other optimal global alignment methods, including the Needleman-Wunsch algorithm. Other tools are described herein, specifically in the definition of “identity” below.

As used herein, the term “homology” refers to the overall relatedness between polymeric molecules, e.g. between nucleic acid molecules (e.g. DNA molecules and/or RNA molecules) and/or between polypeptide molecules. Polymeric molecules (e.g. nucleic acid molecules (e.g. DNA molecules and/or RNA molecules) and/or polypeptide molecules) that share a threshold level of similarity or identity determined by alignment of matching residues are termed homologous. Homology is a qualitative term that describes a relationship between molecules and can be based upon the quantitative similarity or identity. Similarity or identity is a quantitative term that defines the degree of sequence match between two compared sequences. In some embodiments, polymeric molecules are considered to be “homologous” to one another if their sequences are at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% identical or similar. The term “homologous” necessarily refers to a comparison between at least two sequences (polynucleotide or polypeptide sequences). Two polynucleotide sequences are considered homologous if the polypeptides they encode are at least 50%, 60%, 70%, 80%, 90%, 95%, or even 99% for at least one stretch of at

least 20 amino acids. In some embodiments, homologous polynucleotide sequences are characterized by the ability to encode a stretch of at least 4-5 uniquely specified amino acids. For polynucleotide sequences less than 60 nucleotides in length, homology is determined by the ability to encode a stretch of at least 4-5 uniquely specified amino acids. Two protein sequences are considered homologous if the proteins are at least 50%, 60%, 70%, 80%, or 90% identical for at least one stretch of at least 20 amino acids.

Homology implies that the compared sequences diverged in evolution from a common origin. The term "homolog" refers to a first amino acid sequence or nucleic acid sequence (e.g., gene (DNA or RNA) or protein sequence) that is related to a second amino acid sequence or nucleic acid sequence by descent from a common ancestral sequence. The term "homolog" may apply to the relationship between genes and/or proteins separated by the event of speciation or to the relationship between genes and/or proteins separated by the event of genetic duplication. "Orthologs" are genes (or proteins) in different species that evolved from a common ancestral gene (or protein) by speciation. Typically, orthologs retain the same function in the course of evolution. "Paralogs" are genes (or proteins) related by duplication within a genome. Orthologs retain the same function in the course of evolution, whereas paralogs evolve new functions, even if these are related to the original one.

The term "identity" refers to the overall relatedness between polymeric molecules, for example, between polynucleotide molecules (e.g. DNA molecules and/or RNA molecules) and/or between polypeptide molecules. Calculation of the percent identity of two polynucleic acid sequences, for example, can be performed by aligning the two sequences for optimal comparison purposes (e.g., gaps can be introduced in one or both of a first and a second nucleic acid sequences for optimal alignment and non-identical sequences can be disregarded for comparison purposes). In certain embodiments, the length of a sequence aligned for comparison purposes is at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or 100% of the length of the reference sequence. The nucleotides at corresponding nucleotide positions are then compared. When a position in the first sequence is occupied by the same nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which needs to be introduced for optimal alignment of the two sequences. The comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. For example, the percent identity between two nucleic acid sequences can be determined using methods such as those described in Computational Molecular Biology, Lesk, A. M., ed., Oxford University Press, New York, 1988; Biocomputing: Informatics and Genome Projects, Smith, D. W., ed., Academic Press, New York, 1993; Sequence Analysis in Molecular Biology, von Heinje, G., Academic Press, 1987; Computer Analysis of Sequence Data, Part I, Griffin, A. M., and Griffin, H. G., eds., Humana Press, New Jersey, 1994; and Sequence Analysis Primer, Gribskov, M. and Devereux, J., eds., M Stockton Press, New York, 1991; each of which is incorporated herein by reference. For example, the percent identity between two nucleic acid sequences can be determined using the algorithm of Meyers and Miller (CABTOS, 1989, 4:11-17), which has been incorporated into the ALIGN

program (version 2.0) using a PAM120 weight residue table, a gap length penalty of 12 and a gap penalty of 4. The percent identity between two nucleic acid sequences can, alternatively, be determined using the GAP program in the GCG software package using an NWSgapdna.CMP matrix. Methods commonly employed to determine percent identity between sequences include, but are not limited to those disclosed in Carillo, H., and Lipman, D., *SIAM J Applied Math.*, 48:1073 (1988); incorporated herein by reference. Techniques for determining identity are codified in publicly available computer programs. Exemplary computer software to determine homology between two sequences include, but are not limited to, GCG program package, Devereux, J., et al., *Nucleic Acids Research*, 12(1), 387 (1984)), BLASTP, BLASTN, and FASTA Altschul, S. F. et al., *J. Molec. Biol.*, 215, 403 (1990)).

Multiprotein and Multicomponent Vaccines

The present disclosure encompasses respiratory virus vaccines comprising multiple RNA (e.g., mRNA) polynucleotides, each encoding a single antigenic polypeptide, as well as respiratory virus vaccines comprising a single RNA polynucleotide encoding more than one antigenic polypeptide (e.g., as a fusion polypeptide). Thus, a vaccine composition comprising a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a first antigenic polypeptide and a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a second antigenic polypeptide encompasses (a) vaccines that comprise a first RNA polynucleotide encoding a first antigenic polypeptide and a second RNA polynucleotide encoding a second antigenic polypeptide, and (b) vaccines that comprise a single RNA polynucleotide encoding a first and second antigenic polypeptide (e.g., as a fusion polypeptide). RNA (e.g., mRNA) vaccines of the present disclosure, in some embodiments, comprise 2-10 (e.g., 2, 3, 4, 5, 6, 7, 8, 9 or 10), or more, RNA polynucleotides having an open reading frame, each of which encodes a different antigenic polypeptide (or a single RNA polynucleotide encoding 2-10, or more, different antigenic polypeptides). The antigenic polypeptides may be selected from hMPV, PIV3, RSV, MEV and BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1) antigenic polypeptides.

In some embodiments, a respiratory virus vaccine comprises a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a viral capsid protein, a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a viral premembrane/membrane protein, and a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a viral envelope protein. In some embodiments, a respiratory virus vaccine comprises a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a viral fusion (F) protein and a RNA polynucleotide having an open reading frame encoding a viral major surface glycoprotein (G protein). In some embodiments, a vaccine comprises a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a viral F protein. In some embodiments, a vaccine comprises a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a viral G protein. In some embodiments, a vaccine comprises a RNA (e.g., mRNA) polynucleotide having an open reading frame encoding a HN protein.

In some embodiments, a multicomponent vaccine comprises at least one RNA (e.g., mRNA) polynucleotide encoding at least one antigenic polypeptide fused to a signal peptide (e.g., any one of SEQ ID NO: 15-19). The signal peptide may be fused at the N-terminus or the C-terminus of

an antigenic polypeptide. An antigenic polypeptide fused to a signal peptide may be selected from hMPV, PIV3, RSV, MEV and BetaCoV (e.g., selected from MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and HCoV-HKU1) antigenic polypeptides.

Signal Peptides

In some embodiments, antigenic polypeptides encoded by respiratory virus RNA (e.g., mRNA) polynucleotides comprise a signal peptide. Signal peptides, comprising the N-terminal 15-60 amino acids of proteins, are typically needed for the translocation across the membrane on the secretory pathway and, thus, universally control the entry of most proteins both in eukaryotes and prokaryotes to the secretory pathway. Signal peptides generally include three regions: an N-terminal region of differing length, which usually comprises positively charged amino acids; a hydrophobic region; and a short carboxy-terminal peptide region. In eukaryotes, the signal peptide of a nascent precursor protein (pre-protein) directs the ribosome to the rough endoplasmic reticulum (ER) membrane and initiates the transport of the growing peptide chain across it for processing. ER processing produces mature proteins, wherein the signal peptide is cleaved from precursor proteins, typically by a ER-resident signal peptidase of the host cell, or they remain uncleaved and function as a membrane anchor. A signal peptide may also facilitate the targeting of the protein to the cell membrane. The signal peptide, however, is not responsible for the final destination of the mature protein. Secretory proteins devoid of additional address tags in their sequence are by default secreted to the external environment. During recent years, a more advanced view of signal peptides has evolved, showing that the functions and immunodominance of certain signal peptides are much more versatile than previously anticipated.

Respiratory virus vaccines of the present disclosure may comprise, for example, RNA (e.g., mRNA) polynucleotides encoding an artificial signal peptide, wherein the signal peptide coding sequence is operably linked to and is in frame with the coding sequence of the antigenic polypeptide. Thus, respiratory virus vaccines of the present disclosure, in some embodiments, produce an antigenic polypeptide comprising an antigenic polypeptide (e.g., hMPV, PIV3, RSV, MeV or BetaCoV) fused to a signal peptide. In some embodiments, a signal peptide is fused to the N-terminus of the antigenic polypeptide. In some embodiments, a signal peptide is fused to the C-terminus of the antigenic polypeptide.

In some embodiments, the signal peptide fused to the antigenic polypeptide is an artificial signal peptide. In some embodiments, an artificial signal peptide fused to the antigenic polypeptide encoded by the RNA (e.g., mRNA) vaccine is obtained from an immunoglobulin protein, e.g., an IgE signal peptide or an IgG signal peptide. In some embodiments, a signal peptide fused to the antigenic polypeptide encoded by a RNA (e.g., mRNA) vaccine is an Ig heavy chain epsilon-1 signal peptide (IgE HC SP) having the sequence of: MDWTWILFLVAAATRVHS (SEQ ID NO: 16). In some embodiments, a signal peptide fused to the antigenic polypeptide encoded by the (e.g., mRNA) RNA (e.g., mRNA) vaccine is an IgGk chain V-III region HAH signal peptide (IgGk SP) having the sequence of MET-PAQLFLLLLWLPDTTG (SEQ ID NO: 15). In some embodiments, the signal peptide is selected from: Japanese encephalitis PRM signal sequence (MLG-SNSGQRVVFITLLLLVAPAYS; SEQ ID NO: 17), VSVg protein signal sequence (MKCLLYLAFLFIGVNCA; SEQ ID NO: 18) and Japanese encephalitis JEV signal sequence (MWLVSLAIVTACAGA; SEQ ID NO: 19).

In some embodiments, the antigenic polypeptide encoded by a RNA (e.g., mRNA) vaccine comprises an amino acid sequence identified by any one of SEQ ID NO: 5-8, 12-13, 24-34, 47-50 or 54-56 (Tables 3, 6, 11, 14 or 17; see also amino acid sequences of Tables 4, 7, 12 or 15) fused to a signal peptide identified by any one of SEQ ID NO: 15-19 (Table 8). The examples disclosed herein are not meant to be limiting and any signal peptide that is known in the art to facilitate targeting of a protein to ER for processing and/or targeting of a protein to the cell membrane may be used in accordance with the present disclosure.

A signal peptide may have a length of 15-60 amino acids. For example, a signal peptide may have a length of 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, or 60 amino acids. In some embodiments, a signal peptide has a length of 20-60, 25-60, 30-60, 35-60, 40-60, 45-60, 50-60, 55-60, 15-55, 20-55, 25-55, 30-55, 35-55, 40-55, 45-55, 50-55, 15-50, 20-50, 25-50, 30-50, 35-50, 40-50, 45-50, 15-45, 20-45, 25-45, 30-45, 35-45, 40-45, 15-40, 20-40, 25-40, 30-40, 35-40, 15-35, 20-35, 25-35, 30-35, 15-30, 20-30, 25-30, 15-25, 20-25, or 15-20 amino acids.

A signal peptide is typically cleaved from the nascent polypeptide at the cleavage junction during ER processing. The mature antigenic polypeptide produced by a respiratory virus RNA (e.g., mRNA) vaccine of the present disclosure typically does not comprise a signal peptide.

Chemical Modifications

Respiratory virus vaccines of the present disclosure, in some embodiments, comprise at least RNA (e.g. mRNA) polynucleotide having an open reading frame encoding at least one antigenic polypeptide that comprises at least one chemical modification.

The terms “chemical modification” and “chemically modified” refer to modification with respect to adenosine (A), guanosine (G), uridine (U), thymidine (T) or cytidine (C) ribonucleosides or deoxyribonucleosides in at least one of their position, pattern, percent or population. Generally, these terms do not refer to the ribonucleotide modifications in naturally occurring 5'-terminal mRNA cap moieties. With respect to a polypeptide, the term “modification” refers to a modification relative to the canonical set 20 amino acids. Polypeptides, as provided herein, are also considered “modified” if they contain amino acid substitutions, insertions or a combination of substitutions and insertions.

Polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides), in some embodiments, comprise various (more than one) different modifications. In some embodiments, a particular region of a polynucleotide contains one, two or more (optionally different) nucleoside or nucleotide modifications. In some embodiments, a modified RNA polynucleotide (e.g., a modified mRNA polynucleotide), introduced to a cell or organism, exhibits reduced degradation in the cell or organism, respectively, relative to an unmodified polynucleotide. In some embodiments, a modified RNA polynucleotide (e.g., a modified mRNA polynucleotide), introduced into a cell or organism, may exhibit reduced immunogenicity in the cell or organism, respectively (e.g., a reduced innate response).

Modifications of polynucleotides include, without limitation, those described herein. Polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) may comprise modifications that are naturally-occurring, non-naturally-occurring or the polynucleotide may comprise a combination of naturally-occurring and non-naturally-occurring modifications. Polynucleotides may include any useful

modification, for example, of a sugar, a nucleobase, or an internucleoside linkage (e.g., to a linking phosphate, to a phosphodiester linkage or to the phosphodiester backbone).

Polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides), in some embodiments, comprise non-natural modified nucleotides that are introduced during synthesis or post-synthesis of the polynucleotides to achieve desired functions or properties. The modifications may be present on an internucleotide linkages, purine or pyrimidine bases, or sugars. The modification may be introduced with chemical synthesis or with a polymerase enzyme at the terminal of a chain or anywhere else in the chain. Any of the regions of a polynucleotide may be chemically modified.

The present disclosure provides for modified nucleosides and nucleotides of a polynucleotide (e.g., RNA polynucleotides, such as mRNA polynucleotides). A “nucleoside” refers to a compound containing a sugar molecule (e.g., a pentose or ribose) or a derivative thereof in combination with an organic base (e.g., a purine or pyrimidine) or a derivative thereof (also referred to herein as “nucleobase”). A nucleotide” refers to a nucleoside, including a phosphate group. Modified nucleotides may be synthesized by any useful method, such as, for example, chemically, enzymatically, or recombinantly, to include one or more modified or non-natural nucleosides. Polynucleotides may comprise a region or regions of linked nucleosides. Such regions may have variable backbone linkages. The linkages may be standard phosphodiester linkages, in which case the polynucleotides would comprise regions of nucleotides.

Modified nucleotide base pairing encompasses not only the standard adenosine-thymine, adenosine-uracil, or guanosine-cytosine base pairs, but also base pairs formed between nucleotides and/or modified nucleotides comprising non-standard or modified bases, wherein the arrangement of hydrogen bond donors and hydrogen bond acceptors permits hydrogen bonding between a non-standard base and a standard base or between two complementary non-standard base structures. One example of such non-standard base pairing is the base pairing between the modified nucleotide inosine and adenine, cytosine or uracil. Any combination of base/sugar or linker may be incorporated into polynucleotides of the present disclosure.

Modifications of polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) that are useful in the vaccines of the present disclosure include, but are not limited to the following: 2-methylthio-N6-(cis-hydroxyisopentenyl)adenosine; 2-methylthio-N6-methyladenosine; 2-methylthio-N6-threonyl carbamoyl-adenosine; N6-glycylcarbamoyl-adenosine; N6-isopentenyladenosine; N6-methyladenosine; N6-threonylcarbamoyl-adenosine; 1,2'-O-dimethyladenosine; 1-methyladenosine; 2'-O-methyladenosine; 2'-O-ribosyladenosine (phosphate); 2-methyladenosine; 2-methylthio-N6 isopentenyladenosine; 2-methylthio-N6-hydroxynorvalyl carbamoyl-adenosine; 2'-O-methyladenosine; 2'-O-ribosyladenosine (phosphate); Isopentenyladenosine; N6-(cis-hydroxyisopentenyl)adenosine; N6,2'-O-dimethyladenosine; N6,2'-O-dimethyladenosine; N6,N6,2'-O-trimethyladenosine; N6,N6-dimethyladenosine; N6-acetyladenosine; N6-hydroxynorvalylcarbamoyl-adenosine; N6-methyl-N6-threonylcarbamoyl-adenosine; 2-methyladenosine; 2-methylthio-N6-isopentenyladenosine; 7-deaza-adenosine; N1-methyl-adenosine; N6, N6 (dimethyl)adenine; N6-cis-hydroxy-isopentenyl-adenosine; α -thio-adenosine; 2 (amino)adenine; 2 (aminopropyl)adenine; 2 (methylthio) N6 (isopentenyl)adenine; 2-(alkyl)adenine; 2-(aminoalkyl)adenine; 2-(aminopropyl)adenine; 2-(halo)adenine; 2-(halo)

adenine; 2-(propyl)adenine; 2'-Amino-2'-deoxy-ATP; 2'-Azido-2'-deoxy-ATP; 2'-Deoxy-2'-a-aminoadenosine TP; 2'-Deoxy-2'-a-azidoadenosine TP; 6 (alkyl)adenine; 6 (methyl)adenine; 6-(alkyl)adenine; 6-(methyl)adenine; 7 (deaza)adenine; 8 (alkenyl)adenine; 8 (alkynyl)adenine; 8 (amino)adenine; 8 (thioalkyl)adenine; 8-(alkenyl)adenine; 8-(alkyl)adenine; 8-(alkynyl)adenine; 8-(amino)adenine; 8-(halo)adenine; 8-(hydroxyl)adenine; 8-(thioalkyl)adenine; 8-(thiol)adenine; 8-azido-adenosine; aza adenosine; deaza adenosine; N6 (methyl)adenine; N6-(isopentyl)adenine; 7-deaza-8-aza-adenosine; 7-methyladenine; 1-Deazaadenosine TP; 2'Fluoro-N6-Bz-deoxyadenosine TP; 2'-OMe-2'-Amino-ATP; 2'O-methyl-N6-Bz-deoxyadenosine TP; 2'-a-Ethynyladenosine TP; 2-aminoadenine; 2-Aminoadenosine TP; 2-Amino-ATP; 2'-a-Trifluoromethyladenosine TP; 2-Azidoadenosine TP; 2'-b-Ethynyladenosine TP; 2-Bromoadenosine TP; 2'-b-Trifluoromethyladenosine TP; 2-Chloroadenosine TP; 2'-Deoxy-2',2'-difluoroadenosine TP; 2'-Deoxy-2'-a-mercaptopadenosine TP; 2'-Deoxy-2'-a-thiomethoxyadenosine TP; 2'-Deoxy-2'-b-aminoadenosine TP; 2'-Deoxy-2'-b-azidoadenosine TP; 2'-Deoxy-2'-b-bromoadenosine TP; 2'-Deoxy-2'-b-chloroadenosine TP; 2'-Deoxy-2'-b-fluoroadenosine TP; 2'-Deoxy-2'-b-iodoadenosine TP; 2'-Deoxy-2'-b-mercaptopadenosine TP; 2'-Deoxy-2'-b-thiomethoxyadenosine TP; 2-Fluoroadenosine TP; 2-Iodoadenosine TP; 2-Mercaptopadenosine TP; 2-methoxyadenine; 2-methylthio-adenine; 2-Trifluoromethyladenosine TP; 3-Deaza-3-bromoadenosine TP; 3-Deaza-3-chloroadenosine TP; 3-Deaza-3-fluoroadenosine TP; 3-Deaza-3-iodoadenosine TP; 3-Deazaadenosine TP; 4'-Azidoadenosine TP; 4'-Carbocyclic adenosine TP; 4'-Ethynyladenosine TP; 5'-Homo-adenosine TP; 8-Aza-ATP; 8-bromo-adenosine TP; 8-Trifluoromethyladenosine TP; 9-Deazaadenosine TP; 2-aminopurine; 7-deaza-2,6-diaminopurine; 7-deaza-8-aza-2,6-diaminopurine; 7-deaza-8-aza-2-aminopurine; 2,6-diaminopurine; 7-deaza-8-aza-adenine, 7-deaza-2-aminopurine; 2-thiocytidine; 3-methylcytidine; 5-formylcytidine; 5-hydroxymethylcytidine; 5-methylcytidine; N4-acetylcytidine; 2'-O-methylcytidine; 2'-O-methylcytidine; 5,2'-O-dimethylcytidine; 5-formyl-2'-O-methylcytidine; Lysidine; N4,2'-O-dimethylcytidine; N4-acetyl-2'-O-methylcytidine; N4-methylcytidine; N4,N4-Dimethyl-2'-OMe-Cytidine TP; 4-methylcytidine; 5-aza-cytidine; Pseudo-iso-cytidine; pyrrolo-cytidine; α -thio-cytidine; 2-(thio)cytosine; 2'-Amino-2'-deoxy-CTP; 2'-Azido-2'-deoxy-CTP; 2'-Deoxy-2'-a-aminocytidine TP; 2'-Deoxy-2'-a-azidocytidine TP; 3 (deaza) 5 (aza)cytosine; 3 (methyl)cytosine; 3-(alkyl)cytosine; 3-(deaza) 5 (aza)cytosine; 3-(methyl)cytidine; 4,2'-O-dimethyleytidine; 5 (halo)cytosine; 5 (methyl)cytosine; 5 (propynyl)cytosine; 5 (trifluoromethyl)cytosine; 5-(alkyl)cytosine; 5-(alkynyl)cytosine; 5-(halo)cytosine; 5-(propynyl)cytosine; 5-(trifluoromethyl)cytosine; 5-bromo-cytidine; 5-iodo-cytidine; 5-propynyl cytosine; 6-(azo)cytosine; 6-aza-cytidine; aza cytosine; deaza cytosine; N4 (acetyl)cytosine; 1-methyl-1-deaza-pseudoisocytidine; 1-methyl-pseudoisocytidine; 2-methoxy-5-methyl-cytidine; 2-methoxy-cytidine; 2-thio-5-methyl-cytidine; 4-methoxy-1-methyl-pseudoisocytidine; 4-methoxy-pseudoisocytidine; 4-thio-1-methyl-1-deaza-pseudoisocytidine; 4-thio-1-methyl-pseudoisocytidine; 4-thio-pseudoisocytidine; 5-azazebularine; 5-methyl-zebularine; pyrrolo-pseudoisocytidine; Zebularine; (E)-5-(2-Bromo-vinyl)cytidine TP; 2,2'-anhydro-cytidine TP hydrochloride; 2'Fluor-N4-Bz-cytidine TP; 2'Fluoro-N4-Acetyl-cytidine TP; 2'-O-Methyl-N4-Acetyl-cytidine TP; 2'O-methyl-N4-Bz-cytidine TP; 2'-a-Ethynylcytidine TP; 2'-a-Trifluoromethylcytidine TP; 2'-b-Ethynylcytidine TP; 2'-b-Trifluoromethylcytidine TP;

2'-Deoxy-2',2'-difluorocytidine TP; 2'-Deoxy-2'-a-mercaptopcytidine TP; 2'-Deoxy-2'-a-thiomethoxycytidine TP; 2'-Deoxy-2'-b-aminocytidine TP; 2'-Deoxy-2'-b-azidocytidine TP; 2'-Deoxy-2'-b-bromocytidine TP; 2'-Deoxy-2'-b-chlorocytidine TP; 2'-Deoxy-2'-b-fluorocytidine TP; 2'-Deoxy-2'-b-iodocytidine TP; 2'-Deoxy-2'-b-mercaptopcytidine TP; 2'-Deoxy-2'-b-thiomethoxycytidine TP; 2'-O-Methyl-5-(1-propynyl)cytidine TP; 3'-Ethynylcytidine TP; 4'-Azidocytidine TP; 4'-Carbocyclic cytidine TP; 4'-Ethynylcytidine TP; 5-(1-Propynyl)ara-cytidine TP; 5-(2-Chloro-phenyl)-2-thiocytidine TP; 5-(4-Amino-phenyl)-2-thiocytidine TP; 5-Aminoallyl-CTP; 5-Cyanocytidine TP; 5-Ethynylara-cytidine TP; 5-Ethynylcytidine TP; 5'-Homo-cytidine TP; 5-Methoxycytidine TP; 5-Trifluoromethyl-Cytidine TP; N4-Amino-cytidine TP; N4-Benzoyl-cytidine TP; Pseudosocytidine; 7-methylguanosine; N2,2'-O-dimethylguanosine; N2-methylguanosine; Wyosine; 1,2'-O-dimethylguanosine; 1-methylguanosine; 2'-O-methylguanosine; 2'-O-ribosylguanosine (phosphate); 2'-O-methylguanosine; 2'-O-ribosylguanosine (phosphate); 7-aminomethyl-7-deazaguanosine; 7-cyano-7-deazaguanosine; Archaeosine; Methylwyosine; N2,7-dimethylguanosine; N2,N2,2'-O-trimethylguanosine; N2,N2,7-trimethylguanosine; N2,N2-dimethylguanosine; N2,7,2'-O-trimethylguanosine; 6-thioguanosine; 7-deaza-guanosine; 8-oxo-guanosine; N1-methyl-guanosine; α -thio-guanosine; 2 (propyl)guanine; 2-(alkyl)guanine; 2'-Amino-2'-deoxy-GTP; 2'-Azido-2'-deoxy-GTP; 2'-Deoxy-2'-a-aminoguanosine TP; 2'-Deoxy-2'-a-azidoguanosine TP; 6 (methyl)guanine; 6-(alkyl)guanine; 6-(methyl)guanine; 6-methyl-guanosine; 7 (alkyl)guanine; 7 (deaza)guanine; 7 (methyl)guanine; 7-(alkyl)guanine; 7-(deaza)guanine; 7-(methyl)guanine; 8 (alkyl)guanine; 8 (alkynyl)guanine; 8 (halo)guanine; 8 (thioalkyl)guanine; 8-(alkenyl)guanine; 8-(alkyl)guanine; 8-(alkynyl)guanine; 8-(amino)guanine; 8-(halo)guanine; 8-(hydroxyl)guanine; 8-(thioalkyl)guanine; 8-(thiol)guanine; aza guanine; deaza guanine; N (methyl)guanine; N-(methyl)guanine; 1-methyl-6-thio-guanosine; 6-methoxy-guanosine; 6-thio-7-deaza-8-aza-guanosine; 6-thio-7-deaza-guanosine; 6-thio-7-methyl-guanosine; 7-deaza-8-aza-guanosine; 7-methyl-8-oxoguanosine; N2,N2-dimethyl-6-thio-guanosine; N2-methyl-6-thio-guanosine; 1-Me-GTP; 2'Fluoro-N2-isobutylguanosine TP; 2'O-methyl-N2-isobutyl-guanosine TP; 2'-a-Ethynylguanosine TP; 2'-a-Trifluoromethylguanosine TP; 2'-b-Ethynylguanosine TP; 2'-b-Trifluoromethylguanosine TP; 2'-Deoxy-2',2'-difluoroguanosine TP; 2'-Deoxy-2'-a-mercaptopguanosine TP; 2'-Deoxy-2'-a-thiomethoxyguanosine TP; 2'-Deoxy-2'-b-aminoguanosine TP; 2'-Deoxy-2'-b-azidoguanosine TP; 2'-Deoxy-2'-b-bromoguanosine TP; 2'-Deoxy-2'-b-chloroguanosine TP; 2'-Deoxy-2'-b-fluoroguanosine TP; 2'-Deoxy-2'-b-iodoguanosine TP; 2'-Deoxy-2'-b-mercaptopguanosine TP; 2'-Deoxy-2'-b-thiomethoxyguanosine TP; 4'-Azidoguanosine TP; 4'-Carbocyclic guanosine TP; 4'-Ethynylguanosine TP; 5'-Homo-guanosine TP; 8-bromo-guanosine TP; 9-Deazaguanosine TP; N2-isobutyl-guanosine TP; 1-methylinosine; Inosine; 1,2'-O-dimethylinosine; 2'-O-methylinosine; 7-methylinosine; 2'-O-methylinosine; Epoxyqueuosine; galactosyl-queuosine; Mannosylqueuosine; Queuosine; allyamino-thymidine; aza thymidine; deaza thymidine; deoxy-thymidine; 2'-O-methyluridine; 2-thiouridine; 3-methyluridine; 5-carboxymethyluridine; 5-hydroxyuridine; 5-methyluridine; 5-taurinomethyl-2-thiouridine; 5-taurinomethyluridine; Dihydrouridine; Pseudouridine; (3-(3-amino-3-carboxypropyl)uridine; 1-methyl-3-(3-amino-5-carboxypropyl)pseudouridine; 1-methylpseudouridine; 1-methyl-pseudouridine; 2'-O-methyluridine; 2'-O-methylp-

seudouridine; 2'-O-methyluridine; 2-thio-2'-O-methyluridine; 3-(3-amino-3-carboxypropyl)uridine; 3,2'-O-dimethyluridine; 3-Methyl-pseudo-Uridine TP; 4-thiouridine; 5-(carboxyhydroxymethyl)uridine; 5-(carboxyhydroxymethyl)uridine methyl ester; 5,2'-O-dimethyluridine; 5,6-dihydro-uridine; 5-aminomethyl-2-thiouridine; 5-carbamoylmethyl-2'-O-methyluridine; 5-carbamoylmethyluridine; 5-carboxyhydroxymethyluridine; 5-carboxyhydroxymethyluridine methyl ester; 5-carboxymethylaminomethyl-2'-O-methyluridine; 5-carboxymethylaminomethyl-2-thiouridine; 5-carboxymethylaminomethyl-2-thiouridine; 5-carboxymethylaminomethyluridine; 5-Carbamoylmethyluridine TP; 5-methoxycarbonylmethyl-2'-O-methyluridine; 5-methoxycarbonylmethyl-2-thiouridine; 5-methoxycarbonylmethyluridine; 5-methoxyuridine; 5-methyl-2-thiouridine; 5-methylaminomethyl-2-selenouridine; 5-methylaminomethyl-2-thiouridine; 5-methylaminomethyluridine; 5-Methylidihydrouridine; 5-Oxyacetic acid-Uridine TP; 5-Oxyacetic acid-methyl ester-Uridine TP; N1-methylpseudo-uridine; uridine 5-oxyacetic acid; uridine 5-oxyacetic acid methyl ester; 3-(3-Amino-3-carboxypropyl)-Uridine TP; 5-(iso-Pentenylaminomethyl)-2-thiouridine TP; 5-(iso-Pentenylaminomethyl)-2'-O-methyluridine TP; 5-(iso-Pentenylaminomethyl)uridine TP; 5-propynyl uracil; α -thio-uridine; 1 (aminoalkylamino-carbonylethyl)enyl)-2 (thio)-pseudouracil; 1 (aminoalkylaminocarbonylethyl)enyl)-2,4-(dithio)pseudouracil; 1 (aminoalkylaminocarbonylethyl)enyl)-4 (thio)pseudouracil; 1 (aminoalkylaminocarbonylethyl)enyl)-pseudouracil; 1 (aminocarbonylethyl)enyl)-2(thio)-pseudouracil; 1 (aminocarbonylethyl)enyl)-2,4-(dithio)pseudouracil; 1 (aminocarbonylethyl)enyl)-4 (thio)pseudouracil; 1 (aminocarbonylethyl)enyl)-pseudouracil; 1 substituted 2(thio)-pseudouracil; 1 substituted 2,4-(dithio)pseudouracil; 1 substituted 4 (thio)pseudouracil; 1 substituted pseudouracil; 1-(aminoalkylamino-carbonylethyl)enyl)-2-(thio)-pseudouracil; 1-Methyl-3-(3-amino-3-carboxypropyl)pseudouridine TP; 1-Methyl-3-(3-amino-3-carboxypropyl)pseudo-UTP; 1-Methyl-pseudo-UTP; 2 (thio)pseudouracil; 2' deoxy uridine; 2' fluorouridine; 2-(thio)uracil; 2,4-(dithio)pseudouracil; 2' methyl, 2'amino, 2'azido, 2'fluoro-guanosine; 2'-Amino-2'-deoxy-UTP; 2'-Azido-2'-deoxy-UTP; 2'-Azido-deoxyuridine TP; 2'-O-methylpseudouridine; 2' deoxy uridine; 2' fluorouridine; 2'-Deoxy-2'- α -aminouridine TP; 2'-Deoxy-2'-a-azidouridine TP; 2-methylpseudouridine; 3 (3 amino-3 carboxypropyl)uracil; 4 (thio)pseudouracil; 4-(thio)pseudouracil; 4-(thio)uracil; 4-thiouracil; 5 (1,3-diazole-1-alkyl)uracil; 5 (2-aminopropyl)uracil; 5 (aminoalkyl)uracil; 5 (dimethylaminoalkyl)uracil; 5 (guanidiniumalkyl)uracil; 5 (methoxycarbonylmethyl)-2-(thio)uracil; 5 (methoxycarbonylmethyl)uracil; 5 (methyl) 2 (thio)uracil; 5 (methyl) 2,4 (dithio)uracil; 5 (methyl) 4 (thio)uracil; 5 (methylaminomethyl)-2 (thio)uracil; 5 (methylaminomethyl)-2,4 (dithio)uracil; 5 (methylaminomethyl)-4 (thio)uracil; 5 (propynyl)uracil; 5 (trifluoromethyl)uracil; 5-(2-aminopropyl)uracil; 5-(alkyl)-2-(thio)pseudouracil; 5-(alkyl)-2,4 (dithio)pseudouracil; 5-(alkyl)-4 (thio)pseudouracil; 5-(alkyl)pseudouracil; 5-(alkyl)uracil; 5-(alkynyl)uracil; 5-(allylamino)uracil; 5-(cyanoalkyl)uracil; 5-(dialkylaminoalkyl)uracil; 5-(dimethylaminoalkyl)uracil; 5-(guanidiniumalkyl)uracil; 5-(halo)uracil; 5-(1,3-diazole-1-alkyl)uracil; 5-(methoxy)uracil; 5-(methoxycarbonylmethyl)-2-(thio)uracil; 5-(methoxycarbonylmethyl)uracil; 5-(methyl) 2(thio)uracil; 5-(methyl) 2,4 (dithio)uracil; 5-(methyl) 4 (thio)uracil; 5-(methyl)-2-(thio)pseudouracil; 5-(methyl)-2,4 (dithio)pseudouracil;

55

5-(methyl)-4 (thio)pseudouracil; 5-(methyl)pseudouracil; 5-(methyaminomethyl)-2 (thio)uracil; 5-(methyaminomethyl)-2,4(dithio)uracil; 5-(methyaminomethyl)-4-(thio)uracil; 5-(propynyl)uracil; 5-(trifluoromethyl)uracil; 5-aminoallyl-uridine; 5-bromo-uridine; 5-iodo-uridine; 5-uracil; 6 (azo)uracil; 6-(azo)uracil; 6-aza-uridine; allyamino-uracil; aza uracil; deaza uracil; N3 (methyl)uracil; Pseudo-UTP-1-2-ethanoic acid; Pseudouracil; 4-Thio-pseudo-UTP; 1-carboxymethyl-pseudouridine; 1-methyl-1-deaza-pseudouridine; 1-propynyl-uridine; 1-taurinomethyl-1-methyl-uridine; 1-taurinomethyl-4-thio-uridine; 1-taurinomethyl-pseudouridine; 2-methoxy-4-thio-pseudouridine; 2-thio-1-methyl-1-deaza-pseudouridine; 2-thio-1-methyl-pseudouridine; 2-thio-5-aza-uridine; 2-thio-dihydropseudouridine; 2-thio-dihydrouridine; 2-thio-pseudouridine; 4-methoxy-2-thio-pseudouridine; 4-methoxy-pseudouridine; 4-thio-1-methyl-pseudouridine; 4-thio-pseudouridine; 5-aza-uridine; Dihydropseudouridine; ($\pm$)1-(2-Hydroxypropyl)pseudouridine TP; (2R)-1-(2-Hydroxypropyl)pseudouridine TP; (2S)-1-(2-Hydroxypropyl)pseudouridine TP; (E)-5-(2-Bromo-vinyl)ara-uridine TP; (E)-5-(2-Bromo-vinyl)uridine TP; (Z)-5-(2-Bromo-vinyl)ara-uridine TP; (Z)-5-(2-Bromo-vinyl)uridine TP; 1-(2,2,2-Trifluoroethyl)-pseudo-UTP; 1-(2,2,3,3,3-Pentafluoropropyl)pseudouridine TP; 1-(2,2-Diethoxyethyl)pseudouridine TP; 1-(2,4,6-Trimethylbenzyl)pseudouridine TP; 1-(2,4,6-Trimethyl-benzyl)pseudo-UTP; 1-(2,4,6-Trimethyl-phenyl)pseudo-UTP; 1-(2-Amino-2-carboxyethyl)pseudo-UTP; 1-(2-Amino-ethyl)pseudo-UTP; 1-(2-Hydroxyethyl)pseudouridine TP; 1-(2-Methoxyethyl)pseudouridine TP; 1-(3,4-Bis-trifluoromethoxybenzyl)pseudouridine TP; 1-(3,4-Dimethoxybenzyl)pseudouridine TP; 1-(3-Amino-3-carboxypropyl)pseudo-UTP; 1-(3-Amino-propyl)pseudo-UTP; 1-(3-Cyclopropyl-prop-2-ynyl)pseudouridine TP; 1-(4-Amino-4-carboxybutyl)pseudo-UTP; 1-(4-Amino-benzyl)pseudo-UTP; 1-(4-Amino-butyl)pseudo-UTP; 1-(4-Amino-phenyl)pseudo-UTP; 1-(4-Azidobenzyl)pseudouridine TP; 1-(4-Bromobenzyl)pseudouridine TP; 1-(4-Chlorobenzyl)pseudouridine TP; 1-(4-Fluorobenzyl)pseudouridine TP; 1-(4-Iodobenzyl)pseudouridine TP; 1-(4-Methanesulfonylbenzyl)pseudouridine TP; 1-(4-Methoxybenzyl)pseudouridine TP; 1-(4-Methoxy-benzyl)pseudo-UTP; 1-(4-Methoxy-phenyl)pseudo-UTP; 1-(4-Methylbenzyl)pseudouridine TP; 1-(4-Methyl-benzyl)pseudo-UTP; 1-(4-Nitrobenzyl)pseudouridine TP; 1-(4-Nitro-benzyl)pseudo-UTP; 1-(4-Nitro-phenyl)pseudo-UTP; 1-(4-Thiomethoxybenzyl)pseudouridine TP; 1-(4-Trifluoromethoxybenzyl)pseudouridine TP; 1-(4-Trifluoromethylbenzyl)pseudouridine TP; 1-(5-Amino-pentyl)pseudo-UTP; 1-(6-Amino-hexyl)pseudo-UTP; 1,6-Dimethyl-pseudo-UTP; 1-[3-(2-[2-(2-Aminoethoxy)-ethoxy]-ethoxy)-ethoxy]-propionylpseudouridine TP; 1-{3-[2-(2-Aminoethoxy)-ethoxy]-propionyl}pseudouridine TP; 1-Acetylpsudouridine TP; 1-Alkyl-6-(1-propynyl)-pseudo-UTP; 1-Alkyl-6-(2-propynyl)-pseudo-UTP; 1-Alkyl-6-allyl-pseudo-UTP; 1-Alkyl-6-ethynyl-pseudo-UTP; 1-Alkyl-6-homoallyl-pseudo-UTP; 1-Alkyl-6-vinyl-pseudo-UTP; 1-Allylpseudouridine TP; 1-Aminomethyl-pseudo-UTP; 1-Benzoylpseudouridine TP; 1-Benzoyloxymethylpseudouridine TP; 1-Benzyl-pseudo-UTP; 1-Biotinyl-PEG2-pseudouridine TP; 1-Biotinypseudouridine TP; 1-Butyl-pseudo-UTP; 1-Cyanomethylpseudouridine TP; 1-Cyclobutylmethyl-pseudo-UTP; 1-Cyclobutyl-pseudo-UTP; 1-Cycloheptyl-pseudo-UTP; 1-Cyclohexylmethyl-pseudo-UTP; 1-Cyclohexyl-pseudo-UTP; 1-Cyclooctylmethyl-pseudo-UTP; 1-Cyclooctyl-pseudo-UTP; 1-Cyclopentylmethyl-pseudo-UTP; 1-Cyclo-

56

pentyl-pseudo-UTP; 1-Cyclopropylmethyl-pseudo-UTP; 1-Cyclopropyl-pseudo-UTP; 1-Ethyl-pseudo-UTP; 1-Hexyl-pseudo-UTP; 1-Homoallylpseudouridine TP; 1-Hydroxymethylpseudouridine TP; 1-iso-propyl-pseudo-UTP; 1-Me-2-thio-pseudo-UTP; 1-Me-4-thio-pseudo-UTP; 1-Me-alpha-thio-pseudo-UTP; 1-Methanesulfonylmethylpseudouridine TP; 1-Methoxymethylpseudouridine TP; 1-Methyl-6-(2,2,2-Trifluoroethyl)pseudo-UTP; 1-Methyl-6-(4-morpholino)-pseudo-UTP; 1-Methyl-6-(4-thiomorpholino)-pseudo-UTP; 1-Methyl-6-(substituted phenyl)pseudo-UTP; 1-Methyl-6-amino-pseudo-UTP; 1-Methyl-6-azido-pseudo-UTP; 1-Methyl-6-bromo-pseudo-UTP; 1-Methyl-6-butyl-pseudo-UTP; 1-Methyl-6-chloro-pseudo-UTP; 1-Methyl-6-cyano-pseudo-UTP; 1-Methyl-6-dimethylamino-pseudo-UTP; 1-Methyl-6-ethoxy-pseudo-UTP; 1-Methyl-6-ethylcarboxylate-pseudo-UTP; 1-Methyl-6-ethyl-pseudo-UTP; 1-Methyl-6-fluoro-pseudo-UTP; 1-Methyl-6-formyl-pseudo-UTP; 1-Methyl-6-hydroxyamino-pseudo-UTP; 1-Methyl-6-hydroxy-pseudo-UTP; 1-Methyl-6-iodo-pseudo-UTP; 1-Methyl-6-iso-propyl-pseudo-UTP; 1-Methyl-6-methoxy-pseudo-UTP; 1-Methyl-6-methylamino-pseudo-UTP; 1-Methyl-6-phenyl-pseudo-UTP; 1-Methyl-6-propyl-pseudo-UTP; 1-Methyl-6-tert-butyl-pseudo-UTP; 1-Methyl-6-trifluoromethoxy-pseudo-UTP; 1-Methyl-6-trifluoromethyl-pseudo-UTP; 1-Morpholinomethylpseudouridine TP; 1-Pentyl-pseudo-UTP; 1-Phenyl-pseudo-UTP; 1-Pivaloylpseudouridine TP; 1-Propargylpseudouridine TP; 1-Propyl-pseudo-UTP; 1-propynyl-pseudouridine; 1-p-tolyl-pseudo-UTP; 1-tert-Butyl-pseudo-UTP; 1-Thiomethoxymethylpseudouridine TP; 1-Thiomorpholinomethylpseudouridine TP; 1-Trifluoroacetylpsudouridine TP; 1-Trifluoromethyl-pseudo-UTP; 1-Vinylpseudouridine TP; 2,2'-anhydro-uridine TP; 2'-bromo-deoxyuridine TP; 2'-F-5-Methyl-2'-deoxy-UTP; 2'-OMe-5-Me-UTP; 2'-OMe-pseudo-UTP; 2'-a-Ethynyluridine TP; 2'-a-Trifluoromethyluridine TP; 2'-b-Ethynyluridine TP; 2'-b-Trifluoromethyluridine TP; 2'-Deoxy-2',2'-difluorouridine TP; 2'-Deoxy-2'-a-mercaptopuridine TP; 2'-Deoxy-2'-a-thiomethoxyuridine TP; 2'-Deoxy-2'-b-aminouridine TP; 2'-Deoxy-2'-b-azidouridine TP; 2'-Deoxy-2'-b-bromouridine TP; 2'-Deoxy-2'-b-chlorouridine TP; 2'-Deoxy-2'-b-fluorouridine TP; 2'-Deoxy-2'-b-iodouridine TP; 2'-Deoxy-2'-b-mercaptopuridine TP; 2'-Deoxy-2'-b-thiomethoxyuridine TP; 2-methoxy-4-thio-uridine; 2-methoxyuridine; 2'-O-Methyl-5-(1-propynyl)uridine TP; 3-Alkyl-pseudo-UTP; 4'-Azidouridine TP; 4'-Carbocyclic uridine TP; 4'-Ethynyluridine TP; 5-(1-Propynyl)ara-uridine TP; 5-(2-Furanyl)uridine TP; 5-Cyanouridine TP; 5-Dimethylaminouridine TP; 5'-Homo-uridine TP; 5-iodo-2'-fluoro-deoxyuridine TP; 5-Phenylethynyluridine TP; 5-Tri-deuteromethyl-6-deuterouridine TP; 5-Trifluoromethyl-Uridine TP; 5-Vinylarauridine TP; 6-(2,2,2-Trifluoroethyl)-pseudo-UTP; 6-(4-Morpholino)-pseudo-UTP; 6-(4-Thiomorpholino)-pseudo-UTP; 6-(Substituted-Phenyl)-pseudo-UTP; 6-Amino-pseudo-UTP; 6-Azido-pseudo-UTP; 6-Bromo-pseudo-UTP; 6-Butyl-pseudo-UTP; 6-Chloro-pseudo-UTP; 6-Cyano-pseudo-UTP; 6-Dimethylamino-pseudo-UTP; 6-Ethoxy-pseudo-UTP; 6-Ethylcarboxylate-pseudo-UTP; 6-Ethyl-pseudo-UTP; 6-Fluoro-pseudo-UTP; 6-Formyl-pseudo-UTP; 6-Hydroxyamino-pseudo-UTP; 6-Hydroxy-pseudo-UTP; 6-Iodo-pseudo-UTP; 6-iso-Propyl-pseudo-UTP; 6-Methoxy-pseudo-UTP; 6-Methylamino-pseudo-UTP; 6-Methyl-pseudo-UTP; 6-Phenyl-pseudo-UTP; 6-Propyl-pseudo-UTP; 6-tert-Butyl-pseudo-UTP; 6-Trifluoromethoxy-pseudo-UTP; 6-Trifluoromethyl-pseudo-UTP; Alpha-thio-pseudo-UTP; Pseudouridine 1-(4-methylbenzenesulfonic

acid) TP; Pseudouridine 1-(4-methylbenzoic acid) TP; Pseudouridine TP 1-[3-(2-ethoxy)]propionic acid; Pseudouridine TP 1-[3-{2-(2-[2-(2-ethoxy)-ethoxy]-ethoxy)-ethoxy}]propionic acid; Pseudouridine TP 1-[3-{2-(2-[2-(2-ethoxy)-ethoxy]-ethoxy)-ethoxy}]propionic acid; Pseudouridine TP 1-[3-{2-(2-[2-ethoxy]-ethoxy)-ethoxy}]propionic acid; Pseudouridine TP 1-[3-{2-(2-ethoxy)-ethoxy}]propionic acid; Pseudouridine TP 1-methylphosphonic acid; Pseudouridine TP 1-methylphosphonic acid diethyl ester; Pseudo-UTP-N1-3-propionic acid; Pseudo-UTP-N1-4-butanoic acid; Pseudo-UTP-N1-5-pentanoic acid; Pseudo-UTP-N1-6-hexanoic acid; Pseudo-UTP-N1-7-heptanoic acid; Pseudo-UTP-N1-methyl-p-benzoic acid; Pseudo-UTP-N1-p-benzoic acid; Wybutosine; Hydroxywybutosine; Isowyosine; Peroxywybutosine; undermodified hydroxywybutosine; 4-demethylwyosine; 2,6-(diamino)purine; 1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl; 1,3-(diaz)-2-(oxo)-phenanthiazin-1-yl; 1,3-(diaz)-2-(oxo)-phenoxazin-1-yl; 1,3,5-(triaz)-2,6-(diox)-naphthalene; 2 (amino)purine; 2,4,5-(trimethyl)phenyl; 2' methyl, 2' amino, 2' azido, 2' fluoro-cytidine; 2' methyl, 2' amino, 2' azido, 2' fluoro-uridine; 2'-amino-2'-deoxyribose; 2-amino-6-Chloro-purine; 2-aza-inosinyl; 2'-azido-2'-deoxyribose; 2'fluoro-2'-deoxyribose; 2'-fluoro-modified bases; 2'-O-methyl-ribose; 2-oxo-7-aminopyridopyrimidin-3-yl; 2-oxo-pyridopyrimidine-3-yl; 2-pyridinone; 3 nitropyrrole; 3-(methyl)-7-(propynyl) isocarbostyryl; 3-(methyl)isocarbostyryl; 4-(fluoro)-6-(methyl)benzimidazole; 4-(methyl)benzimidazole; 4-(methyl)indolyl; 4,6-(dimethyl)indolyl; 5 nitroindole; 5 substituted pyrimidines; 5-(methyl)isocarbostyryl; 5-nitroindole; 6-(aza)pyrimidine; 6-(azo)thymine; 6-(methyl)-7-(aza)indolyl; 6-chloro-purine; 6-phenyl-pyrrolo-pyrimidin-2-on-3-yl; 7-(aminoalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenanthiazin-1-yl; 7-(aminoalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl; 7-(aminoalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenoxazin-1-yl; 7-(aminoalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenanthiazin-1-yl; 7-(aminoalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenoxazin-1-yl; 7-(aza)indolyl; 7-(guanidiniumalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl; 7-(guanidiniumalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenanthiazin-1-yl; 7-(guanidiniumalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl; 7-(guanidiniumalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenanthiazin-1-yl; 7-(guanidiniumalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenoxazin-1-yl; 7-(propynyl)isocarbostyryl; 7-(propynyl)isocarbostyryl, propynyl-7-(aza)indolyl; 7-deaza-inosinyl; 7-substituted 1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl; 7-substituted 1,3-(diaz)-2-(oxo)-phenoxazin-1-yl; 9-(methyl)-imidizopyridinyl; Aminoindolyl; Anthracenyl; bis-ortho-(aminoalkylhydroxy)-6-phenyl-pyrrolo-pyrimidin-2-on-3-yl; bis-ortho-substituted-6-phenyl-pyrrolo-pyrimidin-2-on-3-yl; Difluorotolyl; Hypoxanthine; Imidizopyridinyl; Inosinyl; Isocarbostyryl; Isoguanisine; N2-substituted purines; N6-methyl-2-amino-purine; N6-substituted purines; N-alkylated derivative; Napthalenyl; Nitrobenzimidazolyl; Nitroimidazolyl; Nitroindazolyl; Nitropyrazolyl; Nubularine; O6-substituted purines; O-alkylated derivative; ortho-(aminoalkylhydroxy)-6-phenyl-pyrrolo-pyrimidin-2-on-3-yl; ortho-substituted-6-phenyl-pyrrolo-pyrimidin-2-on-3-yl; Oxoformycin TP; para-(aminoalkylhydroxy)-6-phenyl-pyrrolo-pyrimidin-2-on-3-yl; para-substituted-6-phenyl-pyrrolo-pyrimidin-2-on-3-yl; Pentacenyl; Phenanthracenyl; Phenyl; propynyl-7-(aza)indolyl; Pyrenyl; pyridopyrimidin-3-yl; pyridopyrimidin-3-yl,

2-oxo-7-amino-pyridopyrimidin-3-yl; pyrrolo-pyrimidin-2-on-3-yl; Pyrrolopyrimidinyl; Pyrrolopyrizinyl; Stilbenzyl; substituted 1,2,4-triazoles; Tetracenyl; Tubercidine; Xanthine; Xanthosine-5'-TP; 2-thio-zebularine; 5-aza-2-thio-zebularine; 7-deaza-2-amino-purine; pyridin-4-one ribonucleoside; 2-Amino-riboside-TP; Formycin A TP; Formycin B TP; Pyrrolosine TP; 2'-OH-ara-adenosine TP; 2'-OH-ara-cytidine TP; 2'-OH-ara-uridine TP; 2'-OH-ara-guanosine TP; 5-(2-carbomethoxyvinyl)uridine TP; and N6-(19-Amino-pentaaxanoadecyl)adenosine TP.

In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) include a combination of at least two (e.g., 2, 3, 4 or more) of the aforementioned modified nucleobases.

In some embodiments, modified nucleobases in polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) are selected from the group consisting of pseudouridine (ψ), N1-methylpseudouridine ($m^1\psi$), N1-ethylpseudouridine, 2-thiouridine, 4'-thiouridine, 5-methylcytosine, 2-thio-1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-pseudouridine, 2-thio-5-aza-uridine, 2-thio-dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-pseudouridine, 4-methoxy-2-thio-pseudouridine, 4-methoxy-pseudouridine, 4-thio-1-methyl-pseudouridine, 4-thio-pseudouridine, 5-aza-uridine, dihydropseudouridine, 5-methoxyuridine and 2'-O-methyl uridine. In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) include a combination of at least two (e.g., 2, 3, 4 or more) of the aforementioned modified nucleobases.

In some embodiments, modified nucleobases in polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) are selected from the group consisting of 1-methyl-pseudouridine ($m^1\psi$), 5-methoxy-uridine (mo^5U), 5-methyl-cytidine (m^5C), pseudouridine (ψ), α -thio-guanosine and α -thio-adenosine. In some embodiments, polynucleotides include a combination of at least two (e.g., 2, 3, 4 or more) of the aforementioned modified nucleobases.

In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise pseudouridine (ψ) and 5-methyl-cytidine (m^5C). In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise 1-methyl-pseudouridine ($m^1\psi$). In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise 1-methyl-pseudouridine ($m^1\psi$) and 5-methyl-cytidine (m^5C). In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise 2-thiouridine (s^2U). In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise 5-methoxy-uridine (mo^5U) and 5-methyl-cytidine (m^5C). In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise methoxy-uridine (mo^5U). In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise 2'-O-methyl uridine. In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise 2'-O-methyl uridine and 5-methyl-cytidine (m^5C). In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) comprise N6-methyl-adenosine (m^6A). In some embodiments, polynucleotides (e.g.,

RNA polynucleotides, such as mRNA polynucleotides) comprise N6-methyl-adenosine (m⁶A) and 5-methyl-cytidine (m⁵C).

In some embodiments, polynucleotides (e.g., RNA polynucleotides, such as mRNA polynucleotides) are uniformly modified (e.g., fully modified, modified throughout the entire sequence) for a particular modification. For example, a polynucleotide can be uniformly modified with 5-methyl-cytidine (m⁵C), meaning that all cytosine residues in the mRNA sequence are replaced with 5-methyl-cytidine (m⁵C). Similarly, a polynucleotide can be uniformly modified for any type of nucleoside residue present in the sequence by replacement with a modified residue such as those set forth above.

Exemplary nucleobases and nucleosides having a modified cytosine include N4-acetyl-cytidine (ac4C), 5-methyl-cytidine (m5C), 5-halo-cytidine (e.g., 5-iodo-cytidine), 5-hydroxymethyl-cytidine (hm5C), 1-methyl-pseudoisocytidine, 2-thio-cytidine (s2C), and 2-thio-5-methyl-cytidine.

In some embodiments, a modified nucleobase is a modified uridine. Exemplary nucleobases and In some embodiments, a modified nucleobase is a modified cytosine, nucleosides having a modified uridine include 5-cyano uridine, and 4'-thio uridine.

In some embodiments, a modified nucleobase is a modified adenine. Exemplary nucleobases and nucleosides having a modified adenine include 7-deaza-adenine, 1-methyl-adenosine (m1A), 2-methyl-adenine (m2A), and N6-methyl-adenosine (m6A).

In some embodiments, a modified nucleobase is a modified guanine. Exemplary nucleobases and nucleosides having a modified guanine include inosine (I), 1-methyl-inosine (m1I), wyosine (imG), methylwyosine (mimG), 7-deaza-guanosine, 7-cyano-7-deaza-guanosine (preQ0), 7-aminomethyl-7-deaza-guanosine (preQ1), 7-methyl-guanosine (m7G), 1-methyl-guanosine (m1G), 8-oxo-guanosine, 7-methyl-8-oxo-guanosine.

The polynucleotides of the present disclosure may be partially or fully modified along the entire length of the molecule. For example, one or more or all or a given type of nucleotide (e.g., purine or pyrimidine, or any one or more or all of A, G, U, C) may be uniformly modified in a polynucleotide of the disclosure, or in a given predetermined sequence region thereof (e.g., in the mRNA including or excluding the poly(A) tail). In some embodiments, all nucleotides X in a polynucleotide of the present disclosure (or in a given sequence region thereof) are modified nucleotides, wherein X may any one of nucleotides A, G, U, C, or any one of the combinations A+G, A+U, A+C, G+U, G+C, U+C, A+G+U, A+G+C, G+U+C or A+G+C.

The polynucleotide may contain from about 1% to about 100% modified nucleotides (either in relation to overall nucleotide content, or in relation to one or more types of nucleotide, i.e., any one or more of A, G, U or C) or any intervening percentage (e.g., from 1% to 20%, from 1% to 25%, from 1% to 50%, from 1% to 60%, from 1% to 70%, from 1% to 80%, from 1% to 90%, from 1% to 95%, from 10% to 20%, from 10% to 25%, from 10% to 50%, from 10% to 60%, from 10% to 70%, from 10% to 80%, from 10% to 90%, from 10% to 95%, from 10% to 100%, from 20% to 25%, from 20% to 50%, from 20% to 60%, from 20% to 70%, from 20% to 80%, from 20% to 90%, from 20% to 95%, from 20% to 100%, from 50% to 60%, from 50% to 70%, from 50% to 80%, from 50% to 90%, from 50% to 95%, from 50% to 100%, from 70% to 80%, from 70% to 90%, from 70% to 95%, from 70% to 100%, from 80% to 90%, from 80% to 95%, from 80% to 100%, from

90% to 95%, from 90% to 100%, and from 95% to 100%). Any remaining percentage is accounted for by the presence of unmodified A, G, U, or C.

The polynucleotides may contain at a minimum 1% and at maximum 100% modified nucleotides, or any intervening percentage, such as at least 5% modified nucleotides, at least 10% modified nucleotides, at least 25% modified nucleotides, at least 50% modified nucleotides, at least 80% modified nucleotides, or at least 90% modified nucleotides. For example, the polynucleotides may contain a modified pyrimidine such as a modified uracil or cytosine. In some embodiments, at least 5%, at least 10%, at least 25%, at least 50%, at least 80%, at least 90% or 100% of the uracil in the polynucleotide is replaced with a modified uracil (e.g., a 5-substituted uracil). The modified uracil can be replaced by a compound having a single unique structure, or can be replaced by a plurality of compounds having different structures (e.g., 2, 3, 4 or more unique structures). In some embodiments, at least 5%, at least 10%, at least 25%, at least 50%, at least 80%, at least 90% or 100% of the cytosine in the polynucleotide is replaced with a modified cytosine (e.g., a 5-substituted cytosine). The modified cytosine can be replaced by a compound having a single unique structure, or can be replaced by a plurality of compounds having different structures (e.g., 2, 3, 4 or more unique structures).

Thus, in some embodiments, the RNA (e.g., mRNA) vaccines comprise a 5'UTR element, an optionally codon optimized open reading frame, and a 3'UTR element, a poly(A) sequence and/or a polyadenylation signal wherein the RNA is not chemically modified.

In some embodiments, the modified nucleobase is a modified uracil. Exemplary nucleobases and nucleosides having a modified uracil include pseudouridine (ψ), pyridin-4-one ribonucleoside, 5-aza-uridine, 6-aza-uridine, 2-thio-5-aza-uridine, 2-thio-uridine (s²U), 4-thio-uridine (s⁴U), 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxy-uridine (ho⁵U), 5-aminoallyl-uridine, 5-halo-uridine (e.g., 5-iodo-uridine or 5-bromo-uridine), 3-methyl-uridine (m³U), 5-methoxy-uridine (mo⁵U), uridine 5-oxyacetic acid (cmo⁵U), uridine 5-oxyacetic acid methyl ester (mcmo⁵U), 5-carboxymethyl-uridine (cm⁵U), 1-carboxymethyl-pseudouridine, 5-carboxyhydroxymethyl-uridine (chm⁵U), 5-carboxyhydroxymethyl-uridine methyl ester (mchm⁵U), 5-methoxycarbonylmethyl-uridine (mcm⁵U), 5-methoxycarbonylmethyl-2-thio-uridine (mcm⁵s²U), 5-aminomethyl-2-thio-uridine (nm⁵s²U), 5-methylaminomethyl-uridine (mm⁵U), 5-methylaminomethyl-2-thio-uridine (mm⁵s²U), 5-methylaminomethyl-2-seleno-uridine (mm⁵se²U), 5-carbamoylmethyl-uridine (ncm⁵U), 5-carboxymethylaminomethyl-uridine (cmnm⁵U), 5-carboxymethylaminomethyl-2-thio-uridine (cmnm⁵s²U), 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyl-uridine (tm⁵U), 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uridine (tm⁵s²U), 1-taurinomethyl-4-thio-pseudouridine, 5-methyl-uridine (m⁵U, i.e., having the nucleobase deoxythymine), 1-methyl-pseudouridine (m¹ψ), 5-methyl-2-thio-uridine (m⁵s²U), 1-methyl-4-thio-pseudouridine (m¹s⁴ψ), 4-thio-1-methyl-pseudouridine, 3-methyl-pseudouridine (m³ψ), 2-thio-1-methyl-pseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine (D), dihydropseudouridine, 5,6-dihydrouridine, 5-methyl-dihydrouridine (m⁵D), 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxy-uridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, N1-methyl-pseudouridine, 3-(3-amino-3-carboxypropyl)uridine (acp³ψ), 1-methyl-3-(3-amino-3-carboxypropyl)pseudouri-

61

dine (acp³ψ), 5-(isopentenylaminomethyl)uridine (inm⁵U), 5-(isopentenylaminomethyl)-2-thio-uridine (inm⁵s²U), α-thio-uridine, 2'-O-methyl-uridine (Um), 5,2'-O-dimethyl-uridine (m⁵Um), 2'-O-methyl-pseudouridine (ψm), 2-thio-2'-O-methyl-uridine (s²Um), 5-methoxycarbonylmethyl-2'-O-methyl-uridine (mcm⁵Um), 5-carbamoylmethyl-2'-O-methyl-uridine (ncm⁵Um), 5-carboxymethylaminomethyl-2'-O-methyl-uridine (cmnm⁵Um), 3,2'-O-dimethyl-uridine (m³Um), and 5-(isopentenylaminomethyl)-2'-O-methyl-uridine (inm⁵Um), 1-thio-uridine, deoxythymidine, 2'-F-ara-uridine, 2'-F-uridine, 2'-OH-ara-uridine, 5-(2-carbomethoxyvinyl) uridine, and 5-[3-(1-E-propenylamino)] uridine.

In some embodiments, the modified nucleobase is a modified cytosine. Exemplary nucleobases and nucleosides having a modified cytosine include 5-aza-cytidine, 6-aza-cytidine, pseudoisocytidine, 3-methyl-cytidine (m³C), N4-acetyl-cytidine (ac⁴C), 5-formyl-cytidine (f⁵C), N4-methyl-cytidine (m⁴C), 5-methyl-cytidine (m⁵C), 5-halo-cytidine (e.g., 5-iodo-cytidine), 5-hydroxymethyl-cytidine (hm⁵C), 1-methyl-pseudoisocytidine, pyrrolo-cytidine, pyrrolo-pseudoisocytidine, 2-thio-cytidine (s²C), 2-thio-5-methyl-cytidine, 4-thio-pseudoisocytidine, 4-thio-1-methyl-pseudoisocytidine, 4-thio-1-methyl-1-deaza-pseudoisocytidine, 1-methyl-1-deaza-pseudoisocytidine, zebularine, 5-aza-zebularine, 5-methyl-zebularine, 5-aza-2-thio-zebularine, 2-thio-zebularine, 2-methoxy-cytidine, 2-methoxy-5-methyl-cytidine, 4-methoxy-pseudoisocytidine, 4-methoxy-1-methyl-pseudoisocytidine, lysidine (k₂C), α-thio-cytidine, 2'-O-methyl-cytidine (Cm), 5,2'-O-dimethyl-cytidine (m⁵Cm), N4-acetyl-2'-O-methyl-cytidine (ac⁴Cm), N4,2'-O-dimethyl-cytidine (m⁴Cm), 5-formyl-2'-O-methyl-cytidine (f⁵Cm), N4,N4,2'-O-trimethyl-cytidine (m⁴₂Cm), 1-thio-cytidine, 2'-F-ara-cytidine, 2'-F-cytidine, and 2'-OH-ara-cytidine.

In some embodiments, the modified nucleobase is a modified adenine. Exemplary nucleobases and nucleosides having a modified adenine include 2-amino-purine, 2, 6-diaminopurine, 2-amino-6-halo-purine (e.g., 2-amino-6-chloro-purine), 6-halo-purine (e.g., 6-chloro-purine), 2-amino-6-methyl-purine, 8-azido-adenosine, 7-deaza-adenine, 7-deaza-8-aza-adenine, 7-deaza-2-amino-purine, 7-deaza-8-aza-2-amino-purine, 7-deaza-2,6-diaminopurine, 7-deaza-8-aza-2,6-diaminopurine, 1-methyl-adenosine (m¹A), 2-methyl-adenine (m²A), N6-methyl-adenosine (m⁶A), 2-methylthio-N6-methyl-adenosine (ms²m⁶A), N6-isopentenyl-adenosine (i⁶A), 2-methylthio-N6-isopentenyl-adenosine (ms²i⁶A), N6-(cis-hydroxyisopentenyl)adenosine (io⁶A), 2-methylthio-N6-(cis-hydroxyisopentenyl)adenosine (ms²io⁶A), N6-glycylcarbamoyl-adenosine (g⁶A), N6-threonylcarbamoyl-adenosine (t⁶A), N6-methyl-N6-threonylcarbamoyl-adenosine (m⁶t⁶A), 2-methylthio-N6-threonylcarbamoyl-adenosine (ms²g⁶A), N6,N6-dimethyl-adenosine (m⁶₂A), N6-hydroxynorvalylcarbamoyl-adenosine (hn⁶A), 2-methylthio-N6-hydroxynorvalylcarbamoyl-adenosine (ms²hn⁶A), N6-acetyl-adenosine (ac⁶A), 7-methyl-adenine, 2-methylthio-adenine, 2-methoxy-adenine, α-thio-adenosine, 2'-O-methyl-adenosine (Am), N6,2'-O-dimethyl-adenosine (m⁶Am), N6,N6,2'-O-trimethyl-adenosine (m⁶2Am), 1,2'-O-dimethyl-adenosine (m¹Am), 2'-O-ribosyladenosine (phosphate) (Ar(p)), 2-amino-N6-methyl-purine, 1-thio-adenosine, 8-azido-adenosine, 2'-F-ara-adenosine, 2'-F-adenosine, 2'-OH-ara-adenosine, and N6-(19-amino-pentaoxonanadecyl)-adenosine.

In some embodiments, the modified nucleobase is a modified guanine. Exemplary nucleobases and nucleosides

62

having a modified guanine include inosine (I), 1-methyl-inosine (m¹I), wyosine (imG), methylwyosine (mimG), 4-demethyl-wyosine (imG-14), isowyosine (imG2), wybutosine (yW), peroxywybutosine (o₂yW), hydroxywybutosine (OhyW), undermodified hydroxywybutosine (OhyW*), 7-deaza-guanosine, queuosine (Q), epoxyqueuosine (oQ), galactosyl-queuosine (galQ), mannosyl-queuosine (manQ), 7-cyano-7-deaza-guanosine (preQ₀), 7-aminomethyl-7-deaza-guanosine (preQ₁), archaeosine (G⁺), 7-deaza-8-aza-guanosine, 6-thio-guanosine, 6-thio-7-deaza-guanosine, 6-thio-7-deaza-8-aza-guanosine, 7-methyl-guanosine (m⁷G), 6-thio-7-methyl-guanosine, 7-methyl-inosine, 6-methoxy-guanosine, 1-methyl-guanosine (m¹G), N2-methyl-guanosine (m²G), N2,N2-dimethyl-guanosine (m²₂G), N2,7-dimethyl-guanosine (m²⁷G), N2, N2,7-dimethyl-guanosine (m^{2,2,7}G), 8-oxo-guanosine, 7-methyl-8-oxo-guanosine, 1-methyl-6-thio-guanosine, N2-methyl-6-thio-guanosine, N2,N2-dimethyl-6-thio-guanosine, α-thio-guanosine, 2'-O-methyl-guanosine (Gm), N2-methyl-2'-O-methyl-guanosine (m²Gm), N2,N2-dimethyl-2'-O-methyl-guanosine (m²₂Gm), 1-methyl-2'-O-methyl-guanosine (m²Gm), N2,7-dimethyl-2'-O-methyl-guanosine (m^{2,7}Gm), 2'-O-methyl-inosine (Im), 1,2'-O-dimethyl-inosine (m¹Im), 2'-O-ribosylguanosine (phosphate) (Gr(p)), 1-thio-guanosine, 06-methyl-guanosine, 2'-F-ara-guanosine, and 2'-F-guanosine.

N-Linked Glycosylation Site Mutants

N-linked glycans of viral proteins play important roles in modulating the immune response. Glycans can be important for maintaining the appropriate antigenic conformations, shielding potential neutralization epitopes, and may alter the proteolytic susceptibility of proteins. Some viruses have putative N-linked glycosylation sites. Deletion or modification of an N-linked glycosylation site may enhance the immune response. Thus, the present disclosure provides, in some embodiments, RNA (e.g., mRNA) vaccines comprising nucleic acids (e.g., mRNA) encoding antigenic polypeptides that comprise a deletion or modification at one or more N-linked glycosylation sites.

In Vitro Transcription of RNA (e.g., mRNA)

Respiratory virus vaccines of the present disclosure comprise at least one RNA polynucleotide, such as a mRNA (e.g., modified mRNA). mRNA, for example, is transcribed in vitro from template DNA, referred to as an "in vitro transcription template." In some embodiments, an in vitro transcription template encodes a 5' untranslated (UTR) region, contains an open reading frame, and encodes a 3' UTR and a polyA tail. The particular nucleic acid sequence composition and length of an in vitro transcription template will depend on the mRNA encoded by the template.

A "5' untranslated region" (5'UTR) refers to a region of an mRNA that is directly upstream (i.e., 5') from the start codon (i.e., the first codon of an mRNA transcript translated by a ribosome) that does not encode a polypeptide.

A "3' untranslated region" (3'UTR) refers to a region of an mRNA that is directly downstream (i.e., 3') from the stop codon (i.e., the codon of an mRNA transcript that signals a termination of translation) that does not encode a polypeptide.

An "open reading frame" is a continuous stretch of DNA beginning with a start codon (e.g., methionine (ATG)), and ending with a stop codon (e.g., TAA, TAG or TGA) and encodes a polypeptide.

A "polyA tail" is a region of mRNA that is downstream, e.g., directly downstream (i.e., 3'), from the 3' UTR that contains multiple, consecutive adenosine monophosphates. A polyA tail may contain 10 to 300 adenosine monophosphates.

phates. For example, a polyA tail may contain 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290 or 300 adenosine monophosphates. In some embodiments, a polyA tail contains 50 to 250 adenosine monophosphates. In a relevant biological setting (e.g., in cells, in vivo) the poly(A) tail functions to protect mRNA from enzymatic degradation, e.g., in the cytoplasm, and aids in transcription termination, export of the mRNA from the nucleus and translation.

In some embodiments, a polynucleotide includes 200 to 3,000 nucleotides. For example, a polynucleotide may include 200 to 500, 200 to 1000, 200 to 1500, 200 to 3000, 500 to 1000, 500 to 1500, 500 to 2000, 500 to 3000, 1000 to 1500, 1000 to 2000, 1000 to 3000, 1500 to 3000, or 2000 to 3000 nucleotides.

Flagellin Adjuvants

Flagellin is an approximately 500 amino acid monomeric protein that polymerizes to form the flagella associated with bacterial motion. Flagellin is expressed by a variety of flagellated bacteria (*Salmonella typhimurium* for example) as well as non-flagellated bacteria (such as *Escherichia coli*). Sensing of flagellin by cells of the innate immune system (dendritic cells, macrophages, etc.) is mediated by the Toll-like receptor 5 (TLR5) as well as by Nod-like receptors (NLRs) Ipaf and Naip5. TLRs and NLRs have been identified as playing a role in the activation of innate immune response and adaptive immune response. As such, flagellin provides an adjuvant effect in a vaccine.

The nucleotide and amino acid sequences encoding known flagellin polypeptides are publicly available in the NCBI GenBank database. The flagellin sequences from *S. typhimurium*, *H. pylori*, *V. cholera*, *S. marcesens*, *S. flexneri*, *T. pallidum*, *L. pneumophila*, *B. burgdorferi*, *C. difficile*, *R. meliloti*, *A. tumefaciens*, *R. lupini*, *B. clarridgeiae*, *P. mirabilis*, *B. subtilis*, *L. monocytogenes*, *P. aeruginosa*, and *E. coli*, among others are known.

A flagellin polypeptide, as used herein, refers to a full length flagellin protein, immunogenic fragments thereof, and peptides having at least 50% sequence identity to a flagellin protein or immunogenic fragments thereof. Exemplary flagellin proteins include flagellin from *Salmonella typhi* (UniPro Entry number: Q56086), *Salmonella typhimurium* (A0A0C9DG09), *Salmonella enteritidis* (A0A0C9BAB7), and *Salmonella choleraesuis* (Q6V2X8), and SEQ ID NO: 54-56 (Table 17). In some embodiments, the flagellin polypeptide has at least 60%, 70%, 75%, 80%, 90%, 95%, 97%, 98%, or 99% sequence identity to a flagellin protein or immunogenic fragments thereof.

In some embodiments, the flagellin polypeptide is an immunogenic fragment. An immunogenic fragment is a portion of a flagellin protein that provokes an immune response. In some embodiments, the immune response is a TLR5 immune response. An example of an immunogenic fragment is a flagellin protein in which all or a portion of a hinge region has been deleted or replaced with other amino acids. For example, an antigenic polypeptide may be inserted in the hinge region. Hinge regions are the hypervariable regions of a flagellin. Hinge regions of a flagellin are also referred to as "D3 domain or region," "propeller domain or region," "hypervariable domain or region" and "variable domain or region." "At least a portion of a hinge region," as used herein, refers to any part of the hinge region of the flagellin, or the entirety of the hinge region. In other embodiments an immunogenic fragment of flagellin is a 20, 25, 30, 35, or 40 amino acid C-terminal fragment of flagellin.

The flagellin monomer is formed by domains DO through D3. DO and D1, which form the stem, are composed of tandem long alpha helices and are highly conserved among different bacteria. The D1 domain includes several stretches of amino acids that are useful for TLR5 activation. The entire D1 domain or one or more of the active regions within the domain are immunogenic fragments of flagellin. Examples of immunogenic regions within the D1 domain include residues 88-114 and residues 411-431 (in *Salmonella typhimurium* FliC flagellin). Within the 13 amino acids in the 88-100 region, at least 6 substitutions are permitted between *Salmonella* flagellin and other flagellins that still preserve TLR5 activation. Thus, immunogenic fragments of flagellin include flagellin like sequences that activate TLR5 and contain a 13 amino acid motif that is 53% or more identical to the *Salmonella* sequence in 88-100 of FliC (LQRVRELAVQSAN; SEQ ID NO: 84).

In some embodiments, the RNA (e.g., mRNA) vaccine includes an RNA that encodes a fusion protein of flagellin and one or more antigenic polypeptides. A "fusion protein" as used herein, refers to a linking of two components of the construct. In some embodiments, a carboxy-terminus of the antigenic polypeptide is fused or linked to an amino terminus of the flagellin polypeptide. In other embodiments, an amino-terminus of the antigenic polypeptide is fused or linked to a carboxy-terminus of the flagellin polypeptide. The fusion protein may include, for example, one, two, three, four, five, six or more flagellin polypeptides linked to one, two, three, four, five, six or more antigenic polypeptides. When two or more flagellin polypeptides and/or two or more antigenic polypeptides are linked such a construct may be referred to as a "multimer."

Each of the components of a fusion protein may be directly linked to one another or they may be connected through a linker. For instance, the linker may be an amino acid linker. The amino acid linker encoded for by the RNA (e.g., mRNA) vaccine to link the components of the fusion protein may include, for instance, at least one member selected from the group consisting of a lysine residue, a glutamic acid residue, a serine residue and an arginine residue. In some embodiments the linker is 1-30, 1-25, 1-25, 5-10, 5, 15, or 5-20 amino acids in length.

In other embodiments the RNA (e.g., mRNA) vaccine includes at least two separate RNA polynucleotides, one encoding one or more antigenic polypeptides and the other encoding the flagellin polypeptide. The at least two RNA polynucleotides may be co-formulated in a carrier such as a lipid nanoparticle.

Broad Spectrum RNA (e.g., mRNA) Vaccines

There may be situations where persons are at risk for infection with more than one strain of hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1). RNA (e.g., mRNA) therapeutic vaccines are particularly amenable to combination vaccination approaches due to a number of factors including, but not limited to, speed of manufacture, ability to rapidly tailor vaccines to accommodate perceived geographical threat, and the like. Moreover, because the vaccines utilize the human body to produce the antigenic protein, the vaccines are amenable to the production of larger, more complex antigenic proteins, allowing for proper folding, surface expression, antigen presentation, etc. in the human subject. To protect against more than one strain of hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1),

a combination vaccine can be administered that includes RNA (e.g., mRNA) encoding at least one antigenic polypeptide protein (or antigenic portion thereof) of a first respiratory virus and further includes RNA encoding at least one antigenic polypeptide protein (or antigenic portion thereof) of a second respiratory virus. RNA (e.g., mRNA) can be co-formulated, for example, in a single lipid nanoparticle (LNP) or can be formulated in separate LNPs for co-administration.

Methods of Treatment

Provided herein are compositions (e.g., pharmaceutical compositions), methods, kits and reagents for prevention and/or treatment of respiratory diseases/infections in humans and other mammals. Respiratory virus RNA (e.g., mRNA) vaccines can be used as therapeutic or prophylactic agents, alone or in combination with other vaccine(s). They may be used in medicine to prevent and/or treat respiratory disease/infection. In exemplary aspects, the RNA (e.g., mRNA) vaccines of the present disclosure are used to provide prophylactic protection from hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1). Prophylactic protection from hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) can be achieved following administration of a RNA (e.g., mRNA) vaccine of the present disclosure. Respiratory virus RNA (e.g., mRNA) vaccines of the present disclosure may be used to treat or prevent viral "co-infections" containing two or more respiratory infections. Vaccines can be administered once, twice, three times, four times or more, but it is likely sufficient to administer the vaccine once (optionally followed by a single booster). It is possible, although less desirable, to administer the vaccine to an infected individual to achieve a therapeutic response. Dosing may need to be adjusted accordingly.

A method of eliciting an immune response in a subject against hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) is provided in aspects of the present disclosure. The method involves administering to the subject a respiratory virus RNA (e.g., mRNA) vaccine comprising at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) antigenic polypeptide thereof, thereby inducing in the subject an immune response specific to hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) antigenic polypeptide or an immunogenic fragment thereof, wherein anti-antigenic polypeptide antibody titer in the subject is increased following vaccination relative to anti-antigenic polypeptide antibody titer in a subject vaccinated with a prophylactically effective dose of a traditional vaccine against hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1). An "anti-antigenic polypeptide antibody" is a serum antibody that binds specifically to the antigenic polypeptide.

In some embodiments, a RNA (e.g., mRNA) vaccine (e.g., a hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1

RNA vaccine) capable of eliciting an immune response is administered intramuscularly via a composition including a compound according to Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IId) or (IIe) (e.g., Compound 3, 18, 20, 25, 26, 29, 30, 60, 108-112, or 122).

A prophylactically effective dose is a therapeutically effective dose that prevents infection with the virus at a clinically acceptable level. In some embodiments the therapeutically effective dose is a dose listed in a package insert for the vaccine. A traditional vaccine, as used herein, refers to a vaccine other than the RNA (e.g., mRNA) vaccines of the present disclosure. For instance, a traditional vaccine includes but is not limited to live/attenuated microorganism vaccines, killed/inactivated microorganism vaccines, subunit vaccines, protein antigen vaccines, DNA vaccines, VLP vaccines, etc. In exemplary embodiments, a traditional vaccine is a vaccine that has achieved regulatory approval and/or is registered by a national drug regulatory body, for example the Food and Drug Administration (FDA) in the United States or the European Medicines Agency (EMA).

In some embodiments the anti-antigenic polypeptide antibody titer in the subject is increased 1 log to 10 log following vaccination relative to anti-antigenic polypeptide antibody titer in a subject vaccinated with a prophylactically effective dose of a traditional vaccine against hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1).

In some embodiments the anti-antigenic polypeptide antibody titer in the subject is increased 1 log, 2 log, 3 log, 5 log or 10 log following vaccination relative to anti-antigenic polypeptide antibody titer in a subject vaccinated with a prophylactically effective dose of a traditional vaccine against hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1).

A method of eliciting an immune response in a subject against hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) is provided in other aspects of the disclosure. The method involves administering to the subject a respiratory virus RNA (e.g., mRNA) vaccine comprising at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) antigenic polypeptide or an immunogenic fragment thereof, thereby inducing in the subject an immune response specific to hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) antigenic polypeptide or an immunogenic fragment thereof, wherein the immune response in the subject is equivalent to an immune response in a subject vaccinated with a traditional vaccine against the hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) at 2 times to 100 times the dosage level relative to the RNA (e.g., mRNA) vaccine.

In some embodiments, the immune response in the subject is equivalent to an immune response in a subject vaccinated with a traditional vaccine at 2, 3, 4, 5, 10, 50, 100 times the dosage level relative to the hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV,

HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) RNA (e.g., mRNA) vaccine.

In some embodiments the immune response in the subject is equivalent to an immune response in a subject vaccinated with a traditional vaccine at 10-100 times, or 100-1000 times, the dosage level relative to the hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) RNA (e.g., mRNA) vaccine.

In some embodiments the immune response is assessed by determining [protein] antibody titer in the subject.

Some aspects of the present disclosure provide a method of eliciting an immune response in a subject against a In some embodiments the immune response in the subject is equivalent to an immune response in a subject vaccinated with a traditional vaccine at 2, 3, 4, 5, 10, 50, 100 times the dosage level relative to the hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) RNA (e.g., mRNA) vaccine by administering to the subject a respiratory virus RNA (e.g., mRNA) vaccine comprising at least one RNA (e.g., mRNA) polynucleotide having an open reading frame encoding at least one hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) antigenic polypeptide, thereby inducing in the subject an immune response specific to the antigenic polypeptide or an immunogenic fragment thereof, wherein the immune response in the subject is induced 2 days to 10 weeks earlier relative to an immune response induced in a subject vaccinated with a prophylactically effective dose of a traditional vaccine against the hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1). In some embodiments, the immune response in the subject is induced in a subject vaccinated with a prophylactically effective dose of a traditional vaccine at 2 times to 100 times the dosage level relative to the RNA (e.g., mRNA) vaccine.

In some embodiments, the immune response in the subject is induced 2 days earlier, or 3 days earlier, relative to an immune response induced in a subject vaccinated with a prophylactically effective dose of a traditional vaccine.

In some embodiments the immune response in the subject is induced 1 week, 2 weeks, 3 weeks, 5 weeks, or 10 weeks earlier relative to an immune response induced in a subject vaccinated with a prophylactically effective dose of a traditional vaccine.

Also provided herein is a method of eliciting an immune response in a subject against hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1) by administering to the subject a respiratory virus RNA (e.g., mRNA) vaccine having an open reading frame encoding a first antigenic polypeptide, wherein the RNA polynucleotide does not include a stabilization element, and wherein an adjuvant is not co-formulated or co-administered with the vaccine.

Therapeutic and Prophylactic Compositions

Provided herein are compositions (e.g., pharmaceutical compositions), methods, kits and reagents for prevention, treatment or diagnosis of hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH

and/or HCoV-HKU1) in humans and other mammals, for example. Respiratory virus RNA (e.g. mRNA) vaccines can be used as therapeutic or prophylactic agents. They may be used in medicine to prevent and/or treat infectious disease.

In some embodiments, the respiratory RNA (e.g., mRNA) vaccines of the present disclosure are used in the priming of immune effector cells, for example, to activate peripheral blood mononuclear cells (PBMCs) ex vivo, which are then infused (re-infused) into a subject.

In some embodiments, respiratory virus vaccine containing RNA (e.g., mRNA) polynucleotides as described herein can be administered to a subject (e.g., a mammalian subject, such as a human subject), and the RNA (e.g., mRNA) polynucleotides are translated in vivo to produce an antigenic polypeptide.

The respiratory virus RNA (e.g., mRNA) vaccines may be induced for translation of a polypeptide (e.g., antigen or immunogen) in a cell, tissue or organism. In some embodiments, such translation occurs in vivo, although such translation may occur ex vivo, in culture or in vitro. In some embodiments, the cell, tissue or organism is contacted with an effective amount of a composition containing a respiratory virus RNA (e.g., mRNA) vaccine that contains a polynucleotide that has at least one a translatable region encoding an antigenic polypeptide.

An "effective amount" of an respiratory virus RNA (e.g. mRNA) vaccine is provided based, at least in part, on the target tissue, target cell type, means of administration, physical characteristics of the polynucleotide (e.g., size, and extent of modified nucleosides) and other components of the vaccine, and other determinants. In general, an effective amount of the respiratory virus RNA (e.g., mRNA) vaccine composition provides an induced or boosted immune response as a function of antigen production in the cell, preferably more efficient than a composition containing a corresponding unmodified polynucleotide encoding the same antigen or a peptide antigen. Increased antigen production may be demonstrated by increased cell transfection (the percentage of cells transfected with the RNA, e.g., mRNA, vaccine), increased protein translation from the polynucleotide, decreased nucleic acid degradation (as demonstrated, for example, by increased duration of protein translation from a modified polynucleotide), or altered antigen specific immune response of the host cell.

In some embodiments, RNA (e.g. mRNA) vaccines (including polynucleotides their encoded polypeptides) in accordance with the present disclosure may be used for treatment of hMPV, PIV3, RSV, MeV and/or BetaCoV (including MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH and/or HCoV-HKU1).

Respiratory RNA (e.g. mRNA) vaccines may be administered prophylactically or therapeutically as part of an active immunization scheme to healthy individuals or early in infection during the incubation phase or during active infection after onset of symptoms. In some embodiments, the amount of RNA (e.g., mRNA) vaccine of the present disclosure provided to a cell, a tissue or a subject may be an amount effective for immune prophylaxis.

Respiratory virus RNA (e.g. mRNA) vaccines may be administered with other prophylactic or therapeutic compounds. As a non-limiting example, a prophylactic or therapeutic compound may be an adjuvant or a booster. As used herein, when referring to a prophylactic composition, such as a vaccine, the term "booster" refers to an extra administration of the prophylactic (vaccine) composition. A booster (or booster vaccine) may be given after an earlier adminis-

tration of the prophylactic composition. The time of administration between the initial administration of the prophylactic composition and the booster may be, but is not limited to, 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, 10 minutes, 15 minutes, 20 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 9 hours, 10 hours, 11 hours, 12 hours, 13 hours, 14 hours, 15 hours, 16 hours, 17 hours, 18 hours, 19 hours, 20 hours, 21 hours, 22 hours, 23 hours, 1 day, 36 hours, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 10 days, 2 weeks, 3 weeks, 1 month, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 1 year, 18 months, 2 years, 3 years, 4 years, 5 years, 6 years, 7 years, 8 years, 9 years, 10 years, 11 years, 12 years, 13 years, 14 years, 15 years, 16 years, 17 years, 18 years, 19 years, 20 years, 25 years, 30 years, 35 years, 40 years, 45 years, 50 years, 55 years, 60 years, 65 years, 70 years, 75 years, 80 years, 85 years, 90 years, 95 years or more than 99 years. In some embodiments, the time of administration between the initial administration of the prophylactic composition and the booster may be, but is not limited to, 1 week, 2 weeks, 3 weeks, 1 month, 2 months, 3 months, 6 months or 1 year.

In some embodiments, respiratory virus RNA (e.g. mRNA) vaccines may be administered intramuscularly or intradermally, similarly to the administration of inactivated vaccines known in the art.

Respiratory virus RNA (e.g. mRNA) vaccines may be utilized in various settings depending on the prevalence of the infection or the degree or level of unmet medical need. As a non-limiting example, the RNA (e.g., mRNA) vaccines may be utilized to treat and/or prevent a variety of respiratory infections. RNA (e.g., mRNA) vaccines have superior properties in that they produce much larger antibody titers and produce responses early than commercially available anti-viral agents/compositions.

Provided herein are pharmaceutical compositions including respiratory virus RNA (e.g. mRNA) vaccines and RNA (e.g. mRNA) vaccine compositions and/or complexes optionally in combination with one or more pharmaceutically acceptable excipients.

Respiratory virus RNA (e.g. mRNA) vaccines may be formulated or administered alone or in conjunction with one or more other components. For instance, hMPV/PIV3/RSV RNA (e.g., mRNA) vaccines (vaccine compositions) may comprise other components including, but not limited to, adjuvants.

In some embodiments, respiratory virus (e.g. mRNA) vaccines do not include an adjuvant (they are adjuvant free).

Respiratory virus RNA (e.g. mRNA) vaccines may be formulated or administered in combination with one or more pharmaceutically-acceptable excipients. In some embodiments, vaccine compositions comprise at least one additional active substances, such as, for example, a therapeutically-active substance, a prophylactically-active substance, or a combination of both. Vaccine compositions may be sterile, pyrogen-free or both sterile and pyrogen-free. General considerations in the formulation and/or manufacture of pharmaceutical agents, such as vaccine compositions, may be found, for example, in Remington: The Science and Practice of Pharmacy 21 st ed., Lippincott Williams & Wilkins, 2005 (incorporated herein by reference in its entirety).

In some embodiments, respiratory virus RNA (e.g. mRNA) vaccines are administered to humans, human patients or subjects. For the purposes of the present disclo-

sure, the phrase "active ingredient" generally refers to the RNA (e.g., mRNA) vaccines or the polynucleotides contained therein, for example, RNA polynucleotides (e.g., mRNA polynucleotides) encoding antigenic polypeptides.

Formulations of the respiratory virus vaccine compositions described herein may be prepared by any method known or hereafter developed in the art of pharmacology. In general, such preparatory methods include the step of bringing the active ingredient (e.g., mRNA polynucleotide) into association with an excipient and/or one or more other accessory ingredients, and then, if necessary and/or desirable, dividing, shaping and/or packaging the product into a desired single- or multi-dose unit.

Relative amounts of the active ingredient, the pharmaceutically acceptable excipient, and/or any additional ingredients in a pharmaceutical composition in accordance with the disclosure will vary, depending upon the identity, size, and/or condition of the subject treated and further depending upon the route by which the composition is to be administered. By way of example, the composition may comprise between 0.1% and 100%, e.g., between 0.5 and 50%, between 1-30%, between 5-80%, at least 80% (w/w) active ingredient.

Respiratory virus RNA (e.g. mRNA) vaccines can be formulated using one or more excipients to: (1) increase stability; (2) increase cell transfection; (3) permit the sustained or delayed release (e.g., from a depot formulation); (4) alter the biodistribution (e.g., target to specific tissues or cell types); (5) increase the translation of encoded protein in vivo; and/or (6) alter the release profile of encoded protein (antigen) in vivo. In addition to traditional excipients such as any and all solvents, dispersion media, diluents, or other liquid vehicles, dispersion or suspension aids, surface active agents, isotonic agents, thickening or emulsifying agents, preservatives, excipients can include, without limitation, lipidoids, liposomes, lipid nanoparticles, polymers, lipoplexes, core-shell nanoparticles, peptides, proteins, cells transfected with respiratory virus RNA (e.g. mRNA) vaccines (e.g., for transplantation into a subject), hyaluronidase, nanoparticle mimics and combinations thereof.

Stabilizing Elements

Naturally-occurring eukaryotic mRNA molecules have been found to contain stabilizing elements, including, but not limited to untranslated regions (UTR) at their 5'-end (5'UTR) and/or at their 3'-end (3'UTR), in addition to other structural features, such as a 5'-cap structure or a 3'-poly(A) tail. Both the 5'UTR and the 3'UTR are typically transcribed from the genomic DNA and are elements of the premature mRNA. Characteristic structural features of mature mRNA, such as the 5'-cap and the 3'-poly(A) tail are usually added to the transcribed (premature) mRNA during mRNA processing. The 3'-poly(A) tail is typically a stretch of adenine nucleotides added to the 3'-end of the transcribed mRNA. It can comprise up to about 400 adenine nucleotides. In some embodiments the length of the 3'-poly(A) tail may be an essential element with respect to the stability of the individual mRNA.

In some embodiments the RNA (e.g., mRNA) vaccine may include one or more stabilizing elements. Stabilizing elements may include for instance a histone stem-loop. A stem-loop binding protein (SLBP), a 32 kDa protein has been identified. It is associated with the histone stem-loop at the 3'-end of the histone messages in both the nucleus and the cytoplasm. Its expression level is regulated by the cell cycle; it peaks during the S-phase, when histone mRNA levels are also elevated. The protein has been shown to be essential for efficient 3'-end processing of histone pre-

mRNA by the U7 snRNP. SLBP continues to be associated with the stem-loop after processing, and then stimulates the translation of mature histone mRNAs into histone proteins in the cytoplasm. The RNA binding domain of SLBP is conserved through metazoa and protozoa; its binding to the histone stem-loop depends on the structure of the loop. The minimum binding site includes at least three nucleotides 5' and two nucleotides 3' relative to the stem-loop.

In some embodiments, the RNA (e.g., mRNA) vaccines include a coding region, at least one histone stem-loop, and optionally, a poly(A) sequence or polyadenylation signal. The poly(A) sequence or polyadenylation signal generally should enhance the expression level of the encoded protein. The encoded protein, in some embodiments, is not a histone protein, a reporter protein (e.g. Luciferase, GFP, EGFP, P-Galactosidase, EGFP), or a marker or selection protein (e.g. alpha-Globin, Galactokinase and Xanthine:guanine phosphoribosyl transferase (GPT)).

In some embodiments, the combination of a poly(A) sequence or polyadenylation signal and at least one histone stem-loop, even though both represent alternative mechanisms in nature, acts synergistically to increase the protein expression beyond the level observed with either of the individual elements. It has been found that the synergistic effect of the combination of poly(A) and at least one histone stem-loop does not depend on the order of the elements or the length of the poly(A) sequence.

In some embodiments, the RNA (e.g., mRNA) vaccine does not comprise a histone downstream element (HDE). "Histone downstream element" (HDE) includes a purine-rich polynucleotide stretch of approximately 15 to 20 nucleotides 3' of naturally occurring stem-loops, representing the binding site for the U7 snRNA, which is involved in processing of histone pre-mRNA into mature histone mRNA. Ideally, the inventive nucleic acid does not include an intron.

In some embodiments, the RNA (e.g., mRNA) vaccine may or may not contain an enhancer and/or promoter sequence, which may be modified or unmodified or which may be activated or inactivated. In some embodiments, the histone stem-loop is generally derived from histone genes, and includes an intramolecular base pairing of two neighbored partially or entirely reverse complementary sequences separated by a spacer, including (e.g., consisting of) a short sequence, which forms the loop of the structure. The unpaired loop region is typically unable to base pair with either of the stem loop elements. It occurs more often in RNA, as is a key component of many RNA secondary structures, but may be present in single-stranded DNA as well. Stability of the stem-loop structure generally depends on the length, number of mismatches or bulges, and base composition of the paired region. In some embodiments, wobble base pairing (non-Watson-Crick base pairing) may result. In some embodiments, the at least one histone stem-loop sequence comprises a length of 15 to 45 nucleotides.

In other embodiments the RNA (e.g., mRNA) vaccine may have one or more AU-rich sequences removed. These sequences, sometimes referred to as AURES are destabilizing sequences found in the 3'UTR. The AURES may be removed from the RNA (e.g., mRNA) vaccines. Alternatively the AURES may remain in the RNA (e.g., mRNA) vaccine.

Nanoparticle Formulations

In some embodiments, respiratory virus RNA (e.g. mRNA) vaccines are formulated in a nanoparticle. In some embodiments, respiratory virus RNA (e.g. mRNA) vaccines are formulated in a lipid nanoparticle. In some embodi-

ments, respiratory virus RNA (e.g. mRNA) vaccines are formulated in a lipid-polycation complex, referred to as a cationic lipid nanoparticle. As a non-limiting example, the polycation may include a cationic peptide or a polypeptide such as, but not limited to, polylysine, polyornithine and/or polyarginine. In some embodiments, respiratory virus RNA (e.g., mRNA) vaccines are formulated in a lipid nanoparticle that includes a non-cationic lipid such as, but not limited to, cholesterol or dioleoyl phosphatidylethanolamine (DOPE).

A lipid nanoparticle formulation may be influenced by, but not limited to, the selection of the cationic lipid component, the degree of cationic lipid saturation, the nature of the PEGylation, ratio of all components and biophysical parameters such as size. In one example by Semple et al. (*Nature Biotech.* 2010 28:172-176), the lipid nanoparticle formulation is composed of 57.1% cationic lipid, 7.1% dipalmitoylphosphatidylcholine, 34.3% cholesterol, and 1.4% PEG-c-DMA. As another example, changing the composition of the cationic lipid can more effectively deliver siRNA to various antigen presenting cells (Basha et al. *Mol Ther.* 2011 19:2186-2200).

In some embodiments, lipid nanoparticle formulations may comprise 35 to 45% cationic lipid, 40% to 50% cationic lipid, 50% to 60% cationic lipid and/or 55% to 65% cationic lipid. In some embodiments, the ratio of lipid to RNA (e.g., mRNA) in lipid nanoparticles may be 5:1 to 20:1, 10:1 to 25:1, 15:1 to 30:1 and/or at least 30:1.

In some embodiments, the ratio of PEG in the lipid nanoparticle formulations may be increased or decreased and/or the carbon chain length of the PEG lipid may be modified from C14 to C18 to alter the pharmacokinetics and/or biodistribution of the lipid nanoparticle formulations. As a non-limiting example, lipid nanoparticle formulations may contain 0.5% to 3.0%, 1.0% to 3.5%, 1.5% to 4.0%, 2.0% to 4.5%, 2.5% to 5.0% and/or 3.0% to 6.0% of the lipid molar ratio of PEG-c-DOMG (R-3-[(ω-methoxy-poly(ethyleneglycol)2000)carbamoyl]-1,2-dimyristyloxypropyl-3-amine) (also referred to herein as PEG-DOMG) as compared to the cationic lipid, DSPC and cholesterol. In some embodiments, the PEG-c-DOMG may be replaced with a PEG lipid such as, but not limited to, PEG-DSG (1,2-Distearoyl-sn-glycerol, methoxypolyethylene glycol), PEG-DMG (1,2-Dimyristoyl-sn-glycerol) and/or PEG-DPG (1,2-Dipalmitoyl-sn-glycerol, methoxypolyethylene glycol). The cationic lipid may be selected from any lipid known in the art such as, but not limited to, DLin-MC3-DMA, DLin-DMA, C12-200 and DLin-KC2-DMA.

In some embodiments, an respiratory virus RNA (e.g. mRNA) vaccine formulation is a nanoparticle that comprises at least one lipid. The lipid may be selected from, but is not limited to, DLin-DMA, DLin-K-DMA, 98N12-5, C12-200, DLin-MC3-DMA, DLin-KC2-DMA, DODMA, PLGA, PEG, PEG-DMG, PEGylated lipids and amino alcohol lipids. In some embodiments, the lipid may be a cationic lipid such as, but not limited to, DLin-DMA, DLin-D-DMA, DLin-MC3-DMA, DLin-KC2-DMA, DODMA and amino alcohol lipids. The amino alcohol cationic lipid may be the lipids described in and/or made by the methods described in U.S. Patent Publication No. US20130150625, herein incorporated by reference in its entirety. As a non-limiting example, the cationic lipid may be 2-amino-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-2-[[[(9Z,2Z)-octadeca-9,12-dien-1-yloxy]methyl]propan-1-ol (Compound 1 in US20130150625); 2-amino-3-[(9Z)-octadec-9-en-1-yloxy]-2-[[[(9Z)-octadec-9-en-1-yloxy]methyl]propan-1-ol (Compound 2 in US20130150625); 2-amino-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-2-[(octyloxy)methyl]propan-1-ol

(Compound 3 in US20130150625); and 2-(dimethylamino)-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-2-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]methyl]propan-1-ol (Compound 4 in US20130150625); or any pharmaceutically acceptable salt or stereoisomer thereof.

Lipid nanoparticle formulations typically comprise a lipid, in particular, an ionizable cationic lipid, for example, 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), or di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), and further comprise a neutral lipid, a sterol and a molecule capable of reducing particle aggregation, for example a PEG or PEG-modified lipid.

In some embodiments, a lipid nanoparticle formulation consists essentially of (i) at least one lipid selected from the group consisting of 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319); (ii) a neutral lipid selected from DSPC, DPPC, POPC, DOPE and SM; (iii) a sterol, e.g., cholesterol; and (iv) a PEG-lipid, e.g., PEG-DMG or PEG-cDMA, in a molar ratio of 20-60% cationic lipid:5-25% neutral lipid:25-55% sterol; 0.5-15% PEG-lipid.

In some embodiments, a lipid nanoparticle formulation includes 25% to 75% on a molar basis of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), e.g., 35 to 65%, 45 to 65%, 60%, 57.5%, 50% or 40% on a molar basis.

In some embodiments, a lipid nanoparticle formulation includes 0.5% to 15% on a molar basis of the neutral lipid, e.g., 3 to 12%, 5 to 10% or 15%, 10%, or 7.5% on a molar basis. Examples of neutral lipids include, without limitation, DSPC, POPC, DPPC, DOPE and SM. In some embodiments, the formulation includes 5% to 50% on a molar basis of the sterol (e.g., 15 to 45%, 20 to 40%, 40%, 38.5%, 35%, or 31% on a molar basis. A non-limiting example of a sterol is cholesterol. In some embodiments, a lipid nanoparticle formulation includes 0.5% to 20% on a molar basis of the PEG or PEG-modified lipid (e.g., 0.5 to 10%, 0.5 to 5%, 1.5%, 0.5%, 1.5%, 3.5%, or 5% on a molar basis. In some embodiments, a PEG or PEG modified lipid comprises a PEG molecule of an average molecular weight of 2,000 Da. In some embodiments, a PEG or PEG modified lipid comprises a PEG molecule of an average molecular weight of less than 2,000, for example around 1,500 Da, around 1,000 Da, or around 500 Da. Non-limiting examples of PEG-modified lipids include PEG-distearoyl glycerol (PEG-DMG) (also referred herein as PEG-C14 or C14-PEG), PEG-cDMA (further discussed in Reyes et al. *J. Controlled Release*, 107, 276-287 (2005) the contents of which are herein incorporated by reference in their entirety).

In some embodiments, lipid nanoparticle formulations include 25-75% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 0.5-15% of the neutral lipid, 5-50% of the sterol, and 0.5-20% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 35-65% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-

DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 3-12% of the neutral lipid, 15-45% of the sterol, and 0.5-10% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 45-65% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 5-10% of the neutral lipid, 25-40% of the sterol, and 0.5-10% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 60% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 7.5% of the neutral lipid, 31% of the sterol, and 1.5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 50% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 10% of the neutral lipid, 38.5% of the sterol, and 1.5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 50% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 10% of the neutral lipid, 35% of the sterol, 4.5% or 5% of the PEG or PEG-modified lipid, and 0.5% of the targeting lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 40% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 15% of the neutral lipid, 40% of the sterol, and 5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 57.2% of a cationic lipid selected from 2,2-dilinoylel-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoylel-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 7.1% of the neutral lipid, 34.3% of the sterol, and 1.4% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations include 57.5% of a cationic lipid selected from the PEG lipid is PEG-cDMA (PEG-cDMA is further discussed in Reyes et al. (*J. Controlled Release*, 107, 276-287 (2005), the contents of which are herein incorporated by reference in their entirety), 7.5% of the neutral lipid, 31.5% of the sterol, and 3.5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulations consists essentially of a lipid mixture in molar ratios of 20-70% cationic lipid:5-45% neutral lipid:20-55% cholesterol: 0.5-15% PEG-modified lipid. In some embodiments, lipid nanoparticle formulations consists essentially of a lipid

mixture in a molar ratio of 20-60% cationic lipid:5-25% neutral lipid:25-55% cholesterol: 0.5-15% PEG-modified lipid.

In some embodiments, the molar lipid ratio is 50/10/38.5/1.5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG, PEG-DSG or PEG-DPG), 57.2/7.1134.3/1.4 (mol % cationic lipid/neutral lipid, e.g., DPPC/Chol/PEG-modified lipid, e.g., PEG-cDMA), 40/15/40/5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG), 50/10/35/4.5/0.5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DSG), 50/10/35/5 (cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG), 40/10/40/10 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG or PEG-cDMA), 35/15/40/10 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG or PEG-cDMA) or 52/13/30/5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG or PEG-cDMA).

Non-limiting examples of lipid nanoparticle compositions and methods of making them are described, for example, in Semple et al. (2010) *Nat. Biotechnol.* 28:172-176; Jayarama et al. (2012), *Angew. Chem. Int. Ed.*, 51: 8529-8533; and Maier et al. (2013) *Molecular Therapy* 21, 1570-1578 (the contents of each of which are incorporated herein by reference in their entirety).

In some embodiments, lipid nanoparticle formulations may comprise a cationic lipid, a PEG lipid and a structural lipid and optionally comprise a non-cationic lipid. As a non-limiting example, a lipid nanoparticle may comprise 40-60% of cationic lipid, 5-15% of a non-cationic lipid, 1-2% of a PEG lipid and 30-50% of a structural lipid. As another non-limiting example, the lipid nanoparticle may comprise 50% cationic lipid, 10% non-cationic lipid, 1.5% PEG lipid and 38.5% structural lipid. As yet another non-limiting example, a lipid nanoparticle may comprise 55% cationic lipid, 10% non-cationic lipid, 2.5% PEG lipid and 32.5% structural lipid. In some embodiments, the cationic lipid may be any cationic lipid described herein such as, but not limited to, DLin-KC2-DMA, DLin-MC3-DMA and L319.

In some embodiments, the lipid nanoparticle formulations described herein may be 4 component lipid nanoparticles. The lipid nanoparticle may comprise a cationic lipid, a non-cationic lipid, a PEG lipid and a structural lipid. As a non-limiting example, the lipid nanoparticle may comprise 40-60% of cationic lipid, 5-15% of a non-cationic lipid, 1-2% of a PEG lipid and 30-50% of a structural lipid. As another non-limiting example, the lipid nanoparticle may comprise 50% cationic lipid, 10% non-cationic lipid, 1.5% PEG lipid and 38.5% structural lipid. As yet another non-limiting example, the lipid nanoparticle may comprise 55% cationic lipid, 10% non-cationic lipid, 2.5% PEG lipid and 32.5% structural lipid. In some embodiments, the cationic lipid may be any cationic lipid described herein such as, but not limited to, DLin-KC2-DMA, DLin-MC3-DMA and L319.

In some embodiments, the lipid nanoparticle formulations described herein may comprise a cationic lipid, a non-cationic lipid, a PEG lipid and a structural lipid. As a non-limiting example, the lipid nanoparticle comprise 50% of the cationic lipid DLin-KC2-DMA, 10% of the non-cationic lipid DSPC, 1.5% of the PEG lipid PEG-DOMG and 38.5% of the structural lipid cholesterol. As a non-limiting example, the lipid nanoparticle comprise 50% of the cationic lipid DLin-MC3-DMA, 10% of the non-cationic

lipid DSPC, 1.5% of the PEG lipid PEG-DOMG and 38.5% of the structural lipid cholesterol. As a non-limiting example, the lipid nanoparticle comprise 50% of the cationic lipid DLin-MC3-DMA, 10% of the non-cationic lipid DSPC, 1.5% of the PEG lipid PEG-DMG and 38.5% of the structural lipid cholesterol. As yet another non-limiting example, the lipid nanoparticle comprise 55% of the cationic lipid L319, 10% of the non-cationic lipid DSPC, 2.5% of the PEG lipid PEG-DMG and 32.5% of the structural lipid cholesterol.

Relative amounts of the active ingredient, the pharmaceutically acceptable excipient, and/or any additional ingredients in a vaccine composition may vary, depending upon the identity, size, and/or condition of the subject being treated and further depending upon the route by which the composition is to be administered. For example, the composition may comprise between 0.1% and 99% (w/w) of the active ingredient. By way of example, the composition may comprise between 0.1% and 100%, e.g., between 0.5 and 50%, between 1-30%, between 5-80%, at least 80% (w/w) active ingredient.

In some embodiments, the respiratory virus RNA (e.g. mRNA) vaccine composition may comprise the polynucleotide described herein, formulated in a lipid nanoparticle comprising MC3, Cholesterol, DSPC and PEG2000-DMG, the buffer trisodium citrate, sucrose and water for injection. As a non-limiting example, the composition comprises: 2.0 mg/mL of drug substance (e.g., polynucleotides encoding H10N8 hMPV), 21.8 mg/mL of MC3, 10.1 mg/mL of cholesterol, 5.4 mg/mL of DSPC, 2.7 mg/mL of PEG2000-DMG, 5.16 mg/mL of trisodium citrate, 71 mg/mL of sucrose and 1.0 nL of water for injection.

In some embodiments, a nanoparticle (e.g., a lipid nanoparticle) has a mean diameter of 10-500 nm, 20-400 nm, 30-300 nm, 40-200 nm. In some embodiments, a nanoparticle (e.g., a lipid nanoparticle) has a mean diameter of 50-150 nm, 50-200 nm, 80-100 nm or 80-200 nm.

Liposomes, Lipoplexes, and Lipid Nanoparticles

The RNA (e.g., mRNA) vaccines of the disclosure can be formulated using one or more liposomes, lipoplexes, or lipid nanoparticles. In some embodiments, pharmaceutical compositions of RNA (e.g., mRNA) vaccines include liposomes. Liposomes are artificially-prepared vesicles which may primarily be composed of a lipid bilayer and may be used as a delivery vehicle for the administration of nutrients and pharmaceutical formulations. Liposomes can be of different sizes such as, but not limited to, a multilamellar vesicle (MLV) which may be hundreds of nanometers in diameter and may contain a series of concentric bilayers separated by narrow aqueous compartments, a small unilamellar vesicle (SUV) which may be smaller than 50 nm in diameter, and a large unilamellar vesicle (LUV) which may be between 50 and 500 nm in diameter. Liposome design may include, but is not limited to, opsonins or ligands in order to improve the attachment of liposomes to unhealthy tissue or to activate events such as, but not limited to, endocytosis. Liposomes may contain a low or a high pH in order to improve the delivery of the pharmaceutical formulations.

The formation of liposomes may depend on the physicochemical characteristics such as, but not limited to, the pharmaceutical formulation entrapped and the liposomal ingredients, the nature of the medium in which the lipid vesicles are dispersed, the effective concentration of the entrapped substance and its potential toxicity, any additional processes involved during the application and/or delivery of the vesicles, the optimization size, polydispersity and the shelf-life of the vesicles for the intended application, and the

batch-to-batch reproducibility and possibility of large-scale production of safe and efficient liposomal products.

In some embodiments, pharmaceutical compositions described herein may include, without limitation, liposomes such as those formed from 1,2-dioleoyloxy-N,N-dimethylaminopropane (DODMA) liposomes, DiLa2 liposomes from Marina Biotech (Bothell, WA), 1,2-dilinoleoyloxy-3-dimethylaminopropane (DLin-DMA), 2,2-dilinoleyl-4-(2-dimethylaminoethyl)-[1,3]-dioxolane (DLin-KC2-DMA), and MC3 (US20100324120; herein incorporated by reference in its entirety) and liposomes which may deliver small molecule drugs such as, but not limited to, DOXIL® from Janssen Biotech, Inc. (Horsham, PA).

In some embodiments, pharmaceutical compositions described herein may include, without limitation, liposomes such as those formed from the synthesis of stabilized plasmid-lipid particles (SPLP) or stabilized nucleic acid lipid particle (SNALP) that have been previously described and shown to be suitable for oligonucleotide delivery in vitro and in vivo (see Wheeler et al. *Gene Therapy*. 1999 6:271-281; Zhang et al. *Gene Therapy*. 1999 6:1438-1447; Jeffs et al. *Pharm Res*. 2005 22:362-372; Morrissey et al., *Nat Biotechnol*. 2005 2:1002-1007; Zimmermann et al., *Nature*. 2006 441:111-114; Heyes et al. *J Contr Rel*. 2005 107:276-287; Semple et al. *Nature Biotech*. 2010 28:172-176; Judge et al. *J Clin Invest*. 2009 119:661-673; deFougerolles Hum Gene Ther. 2008 19:125-132; U.S. Patent Publication No US20130122104; all of which are incorporated herein in their entireties). The original manufacture method by Wheeler et al. was a detergent dialysis method, which was later improved by Jeffs et al. and is referred to as the spontaneous vesicle formation method. The liposome formulations are composed of 3 to 4 lipid components in addition to the polynucleotide. As an example a liposome can contain, but is not limited to, 55% cholesterol, 20% distearylphosphatidyl choline (DSPC), 10% PEG-S-DSG, and 15% 1,2-dioleoyloxy-N,N-dimethylaminopropane (DODMA), as described by Jeffs et al. As another example, certain liposome formulations may contain, but are not limited to, 48% cholesterol, 20% DSPC, 2% PEG-c-DMA, and 30% cationic lipid, where the cationic lipid can be 1,2-distearoyloxy-N,N-dimethylaminopropane (DSDMA), DODMA, DLin-DMA, or 1,2-dilinolenyloxy-3-dimethylaminopropane (DLinDMA), as described by Heyes et al.

In some embodiments, liposome formulations may comprise from about 25.0% cholesterol to about 40.0% cholesterol, from about 30.0% cholesterol to about 45.0% cholesterol, from about 35.0% cholesterol to about 50.0% cholesterol and/or from about 48.5% cholesterol to about 60% cholesterol. In some embodiments, formulations may comprise a percentage of cholesterol selected from the group consisting of 28.5%, 31.5%, 33.5%, 36.5%, 37.0%, 38.5%, 39.0% and 43.5%. In some embodiments, formulations may comprise from about 5.0% to about 10.0% DSPC and/or from about 7.0% to about 15.0% DSPC.

In some embodiments, the RNA (e.g., mRNA) vaccine pharmaceutical compositions may be formulated in liposomes such as, but not limited to, DiLa2 liposomes (Marina Biotech, Bothell, WA), SMARTICLES® (Marina Biotech, Bothell, WA), neutral DOPC (1,2-dioleoyl-sn-glycero-3-phosphocholine) based liposomes (e.g., siRNA delivery for ovarian cancer (Landen et al. *Cancer Biology & Therapy* 2006 5(12):1708-1713; herein incorporated by reference in its entirety) and hyaluronan-coated liposomes (Quiet Therapeutics, Israel).

In some embodiments, the cationic lipid may be a low molecular weight cationic lipid such as those described in

U.S. Patent Application No. 20130090372, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the RNA (e.g., mRNA) vaccines may be formulated in a lipid vesicle, which may have crosslinks between functionalized lipid bilayers.

In some embodiments, the RNA (e.g., mRNA) vaccines may be formulated in a lipid-polycation complex. The formation of the lipid-polycation complex may be accomplished by methods known in the art and/or as described in U.S. Pub. No. 20120178702, herein incorporated by reference in its entirety. As a non-limiting example, the polycation may include a cationic peptide or a polypeptide such as, but not limited to, polylysine, polyornithine and/or polyarginine. In some embodiments, the RNA (e.g., mRNA) vaccines may be formulated in a lipid-polycation complex, which may further include a non-cationic lipid such as, but not limited to, cholesterol or dioleoyl phosphatidylethanolamine (DOPE).

In some embodiments, the ratio of PEG in the lipid nanoparticle (LNP) formulations may be increased or decreased and/or the carbon chain length of the PEG lipid may be modified from C14 to C18 to alter the pharmacokinetics and/or biodistribution of the LNP formulations. As a non-limiting example, LNP formulations may contain from about 0.5% to about 3.0%, from about 1.0% to about 3.5%, from about 1.5% to about 4.0%, from about 2.0% to about 4.5%, from about 2.5% to about 5.0% and/or from about 3.0% to about 6.0% of the lipid molar ratio of PEG-c-DOMG (R-3-[(w-methoxy-poly(ethyleneglycol) 2000)carbamoyl]-1,2-dimyristyloxypropyl-3-amine) (also referred to herein as PEG-DOMG) as compared to the cationic lipid, DSPC and cholesterol. In some embodiments, the PEG-c-DOMG may be replaced with a PEG lipid such as, but not limited to, PEG-DSG (1,2-Distearoyl-sn-glycerol, methoxypolyethylene glycol), PEG-DMG (1,2-Dimyristoyl-sn-glycerol) and/or PEG-DPG (1,2-Dipalmitoyl-sn-glycerol, methoxypolyethylene glycol). The cationic lipid may be selected from any lipid known in the art such as, but not limited to, DLin-MC3-DMA, DLin-DMA, C12-200 and DLin-KC2-DMA.

In some embodiments, the RNA (e.g., mRNA) vaccines may be formulated in a lipid nanoparticle.

In some embodiments, the RNA (e.g., mRNA) vaccine formulation comprising the polynucleotide is a nanoparticle which may comprise at least one lipid. The lipid may be selected from, but is not limited to, DLin-DMA, DLin-K-DMA, 98N12-5, C12-200, DLin-MC3-DMA, DLin-KC2-DMA, DODMA, PLGA, PEG, PEG-DMG, PEGylated lipids and amino alcohol lipids. In another aspect, the lipid may be a cationic lipid such as, but not limited to, DLin-DMA, DLin-D-DMA, DLin-MC3-DMA, DLin-KC2-DMA, DODMA and amino alcohol lipids. The amino alcohol cationic lipid may be the lipids described in and/or made by the methods described in U.S. Patent Publication No. US20130150625, herein incorporated by reference in its entirety. As a non-limiting example, the cationic lipid may be 2-amino-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-2-[(9Z,2Z)-octadeca-9,12-dien-1-yloxy]methyl]propan-1-ol (Compound 1 in US20130150625); 2-amino-3-[(9Z)-octadec-9-en-1-yloxy]-2-[(9Z)-octadec-9-en-1-yloxy]methyl]propan-1-ol (Compound 2 in US20130150625); 2-amino-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-2-[(octyloxy)methyl]propan-1-ol (Compound 3 in US20130150625); and 2-(dimethylamino)-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-2-[(9Z,12Z)-octadeca-9,12-

dien-1-yloxy]methyl}propan-1-ol (Compound 4 in US20130150625); or any pharmaceutically acceptable salt or stereoisomer thereof.

Lipid nanoparticle formulations typically comprise a lipid, in particular, an ionizable cationic lipid, for example, 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), or di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), and further comprise a neutral lipid, a sterol and a molecule capable of reducing particle aggregation, for example a PEG or PEG-modified lipid.

In some embodiments, the lipid nanoparticle formulation consists essentially of (i) at least one lipid selected from the group consisting of 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319); (ii) a neutral lipid selected from DSPC, DPPC, POPC, DOPE and SM; (iii) a sterol, e.g., cholesterol; and (iv) a PEG-lipid, e.g., PEG-DMG or PEG-cDMA, in a molar ratio of about 20-60% cationic lipid:5-25% neutral lipid:25-55% sterol; 0.5-15% PEG-lipid.

In some embodiments, the formulation includes from about 25% to about 75% on a molar basis of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), e.g., from about 35 to about 65%, from about 45 to about 65%, about 60%, about 57.5%, about 50% or about 40% on a molar basis.

In some embodiments, the formulation includes from about 0.5% to about 15% on a molar basis of the neutral lipid e.g., from about 3 to about 12%, from about 5 to about 10% or about 15%, about 10%, or about 7.5% on a molar basis. Examples of neutral lipids include, but are not limited to, DSPC, POPC, DPPC, DOPE and SM. In some embodiments, the formulation includes from about 5% to about 50% on a molar basis of the sterol (e.g., about 15 to about 45%, about 20 to about 40%, about 40%, about 38.5%, about 35%, or about 31% on a molar basis. An exemplary sterol is cholesterol. In some embodiments, the formulation includes from about 0.5% to about 20% on a molar basis of the PEG or PEG-modified lipid (e.g., about 0.5 to about 10%, about 0.5 to about 5%, about 1.5%, about 0.5%, about 1.5%, about 3.5%, or about 5% on a molar basis. In some embodiments, the PEG or PEG modified lipid comprises a PEG molecule of an average molecular weight of 2,000 Da. In other embodiments, the PEG or PEG modified lipid comprises a PEG molecule of an average molecular weight of less than 2,000, for example around 1,500 Da, around 1,000 Da, or around 500 Da. Examples of PEG-modified lipids include, but are not limited to, PEG-distearoyl glycerol (PEG-DMG) (also referred herein as PEG-C14 or C14-PEG), PEG-cDMA (further discussed in Reyes et al. *J. Controlled Release*, 107, 276-287 (2005) the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the formulations of the present disclosure include 25-75% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 0.5-15% of the neutral lipid, 5-50% of the sterol, and 0.5-20% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include 35-65% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 3-12% of the neutral lipid, 15-45% of the sterol, and 0.5-10% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include 45-65% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), 5-10% of the neutral lipid, 25-40% of the sterol, and 0.5-10% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include about 60% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), about 7.5% of the neutral lipid, about 31% of the sterol, and about 1.5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include about 50% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), about 10% of the neutral lipid, about 38.5% of the sterol, and about 1.5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include about 50% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), about 10% of the neutral lipid, about 35% of the sterol, about 4.5% or about 5% of the PEG or PEG-modified lipid, and about 0.5% of the targeting lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include about 40% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), about 15% of the neutral lipid, about 40% of the sterol, and about 5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include about 57.2% of a cationic lipid selected from 2,2-dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319), about 7.1% of the neutral lipid, about 34.3% of the sterol, and about 1.4% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, the formulations of the present disclosure include about 57.5% of a cationic lipid selected from the PEG lipid is PEG-cDMA (PEG-cDMA is further discussed in Reyes et al. (*J. Controlled Release*, 107, 276-287 (2005), the contents of which are herein incorporated by reference in their entirety), about 7.5% of the neutral lipid,

about 31.5% of the sterol, and about 3.5% of the PEG or PEG-modified lipid on a molar basis.

In some embodiments, lipid nanoparticle formulation consists essentially of a lipid mixture in molar ratios of about 20-70% cationic lipid:5-45% neutral lipid:20-55% cholesterol:0.5-15% PEG-modified lipid; more preferably in a molar ratio of about 20-60% cationic lipid:5-25% neutral lipid:25-55% cholesterol: 0.5-15% PEG-modified lipid.

In some embodiments, the molar lipid ratio is approximately 50/10/38.5/1.5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG, PEG-DSG or PEG-DPG), 57.2/7.1134.3/1.4 (mol % cationic lipid/neutral lipid, e.g., DPPC/Chol/PEG-modified lipid, e.g., PEG-cDMA), 40/15/40/5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG), 50/10/35/4.5/0.5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DSG), 50/10/35/5 (cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG), 40/10/40/10 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG or PEG-cDMA), 35/15/40/10 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG or PEG-cDMA) or 52/13/30/5 (mol % cationic lipid/neutral lipid, e.g., DSPC/Chol/PEG-modified lipid, e.g., PEG-DMG or PEG-cDMA).

Examples of lipid nanoparticle compositions and methods of making same are described, for example, in Semple et al. (2010) *Nat. Biotechnol.* 28:172-176; Jayarama et al. (2012), *Angew. Chem. Int. Ed.*, 51: 8529-8533; and Maier et al. (2013) *Molecular Therapy* 21, 1570-1578 (the contents of each of which are incorporated herein by reference in their entirety).

In some embodiments, the lipid nanoparticle formulations described herein may comprise a cationic lipid, a PEG lipid and a structural lipid and optionally comprise a non-cationic lipid. As a non-limiting example, the lipid nanoparticle may comprise about 40-60% of cationic lipid, about 5-15% of a non-cationic lipid, about 1-2% of a PEG lipid and about 30-50% of a structural lipid. As another non-limiting example, the lipid nanoparticle may comprise about 50% cationic lipid, about 10% non-cationic lipid, about 1.5% PEG lipid and about 38.5% structural lipid. As yet another non-limiting example, the lipid nanoparticle may comprise about 55% cationic lipid, about 10% non-cationic lipid, about 2.5% PEG lipid and about 32.5% structural lipid. In some embodiments, the cationic lipid may be any cationic lipid described herein such as, but not limited to, DLin-KC2-DMA, DLin-MC3-DMA and L319.

In some embodiments, the lipid nanoparticle formulations described herein may be 4 component lipid nanoparticles. The lipid nanoparticle may comprise a cationic lipid, a non-cationic lipid, a PEG lipid and a structural lipid. As a non-limiting example, the lipid nanoparticle may comprise about 40-60% of cationic lipid, about 5-15% of a non-cationic lipid, about 1-2% of a PEG lipid and about 30-50% of a structural lipid. As another non-limiting example, the lipid nanoparticle may comprise about 50% cationic lipid, about 10% non-cationic lipid, about 1.5% PEG lipid and about 38.5% structural lipid. As yet another non-limiting example, the lipid nanoparticle may comprise about 55% cationic lipid, about 10% non-cationic lipid, about 2.5% PEG lipid and about 32.5% structural lipid. In some embodiments, the cationic lipid may be any cationic lipid described herein such as, but not limited to, DLin-KC2-DMA, DLin-MC3-DMA and L319.

In some embodiments, the lipid nanoparticle formulations described herein may comprise a cationic lipid, a non-

cationic lipid, a PEG lipid and a structural lipid. As a non-limiting example, the lipid nanoparticle comprise about 50% of the cationic lipid DLin-KC2-DMA, about 10% of the non-cationic lipid DSPC, about 1.5% of the PEG lipid PEG-DOMG and about 38.5% of the structural lipid cholesterol. As a non-limiting example, the lipid nanoparticle comprise about 50% of the cationic lipid DLin-MC3-DMA, about 10% of the non-cationic lipid DSPC, about 1.5% of the PEG lipid PEG-DOMG and about 38.5% of the structural lipid cholesterol. As a non-limiting example, the lipid nanoparticle comprise about 50% of the cationic lipid DLin-MC3-DMA, about 10% of the non-cationic lipid DSPC, about 1.5% of the PEG lipid PEG-DMG and about 38.5% of the structural lipid cholesterol. As yet another non-limiting example, the lipid nanoparticle comprise about 55% of the cationic lipid L319, about 10% of the non-cationic lipid DSPC, about 2.5% of the PEG lipid PEG-DMG and about 32.5% of the structural lipid cholesterol.

As a non-limiting example, the cationic lipid may be selected from (20Z,23Z)—N,N-dimethylnonacos-20,23-dien-10-amine, (17Z,20Z)—N,N-dimethylhexacos-17,20-dien-9-amine, (1Z,19Z)—N,N-dimethylpentacos-1,6,19-dien-8-amine, (13Z,16Z)—N,N-dimethyldocos-13,16-dien-5-amine, (12Z,15Z)—N,N-dimethylhenicos-12,15-dien-4-amine, (14Z,17Z)—N,N-dimethyltricos-14,17-dien-6-amine, (15Z,18Z)—N,N-dimethyltetracos-15,18-dien-7-amine, (18Z,21Z)—N,N-dimethylheptacos-18,21-dien-10-amine, (15Z,18Z)—N,N-dimethyltetracos-15,18-dien-5-amine, (14Z,17Z)—N,N-dimethyltricos-14,17-dien-4-amine, (19Z,22Z)—N,N-dimethyloctacos-19,22-dien-9-amine, (18Z,21 Z)—N,N-dimethylheptacos-18,21-dien-8-amine, (17Z,20Z)—N,N-dimethylhexacos-17,20-dien-7-amine, (16Z,19Z)—N,N-dimethylpentacos-16,19-dien-6-amine, (22Z,25Z)—N,N-dimethylhentriacont-22,25-dien-10-amine, (21 Z,24Z)—N,N-dimethyltriacont-21,24-dien-9-amine, (18Z)—N,N-dimethylheptacos-18-en-10-amine, (17Z)—N,N-dimethylhexacos-17-en-9-amine, (19Z,22Z)—N,N-dimethyloctacos-19,22-dien-7-amine, N,N-dimethylheptacos-10-amine, (20Z,23Z)—N-ethyl-N-methylnonacos-20,23-dien-10-amine, 1-[(11Z,14Z)-1-nonylicos-11,14-dien-1-yl]pyrrolidine, (20Z)—N,N-dimethylheptacos-20-en-10-amine, (15Z)—N,N-dimethylheptacos-15-en-10-amine, (14Z)—N,N-dimethylnonacos-14-en-10-amine, (17Z)—N,N-dimethylnonacos-17-en-10-amine, (24Z)—N,N-dimethyltriacont-24-en-10-amine, (20Z)—N,N-dimethylnonacos-20-en-10-amine, (22Z)—N,N-dimethylhentriacont-22-en-10-amine, (16Z)—N,N-dimethylpentacos-16-en-8-amine, (12Z,15Z)—N,N-dimethyl-2-nonylhenicos-12,15-dien-1-amine, (13Z,16Z)—N,N-dimethyl-3-nonyldocos-13,16-dien-1-amine, N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]heptadecan-8-amine, 1-[(1 S,2R)-2-hexylcyclopropyl]-N,N-dimethylnonadecan-10-amine, N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]nonadecan-10-amine, N,N-dimethyl-21-[(1S,2R)-2-octylcyclopropyl]henicosan-10-amine, N,N-dimethyl-1-[(1S,2S)-2-[(1R,2R)-2-pentylcyclopropyl]methyl]cyclopropyl]nonadecan-10-amine, N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]hexadecan-8-amine, N,N-dimethyl-1-[(1R,2S)-2-undecylcyclopropyl]tetradecan-5-amine, N,N-dimethyl-3-{7-[(1S,2R)-2-octylcyclopropyl]heptyl}dodecan-1-amine, 1-[(1R,2S)-2-heptylcyclopropyl]-N,N-dimethyloctadecan-9-amine, 1-[(1S,2R)-2-decylcyclopropyl]-N,N-dimethylpentadecan-6-amine, N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]pentadecan-8-amine, R—N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-(octyloxy)propan-2-amine, S—N,N-dimethyl-1-

[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-(octyloxy)propan-2-amine, 1-{2-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-1-[(octyloxy)methyl]ethyl}pyrrolidine, (2S)—N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-[(5Z)-oct-5-en-1-yloxy]propan-2-amine, 1-{2-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-1-[(octyloxy)methyl]ethyl}azetidine, (2S)-1-(hexyloxy)-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, (2S)-1-(heptyloxy)-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, N,N-dimethyl-1-(nonyloxy)-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, N,N-dimethyl-1-[(9Z)-octadec-9-en-1-yloxy]-3-(octyloxy)propan-2-amine; (2S)—N,N-dimethyl-1-[(6Z,9Z,12Z)-octadeca-6,9,12-trien-1-yloxy]-3-(octyloxy)propan-2-amine, (2S)-1-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethyl-3-(pentyloxy)propan-2-amine, (2S)-1-(hexyloxy)-3-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethylpropan-2-amine, 1-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, 1-[(13Z,16Z)-docosa-13,16-dien-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, (2S)-1-[(13Z,16Z)-docosa-13,16-dien-1-yloxy]-3-(hexyloxy)-N,N-dimethylpropan-2-amine, 1-1(13Z)-docos-13-en-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, 1-[(9Z)-hexadec-9-en-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, (2R)—N,N-dimethyl-H(1-metoyloctyl)oxy]-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, (2R)-1-[(3,7-dimethyloctyl)oxy]-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, N,N-dimethyl-1-(octyloxy)-3-[(8-[(1S,2S)-2-[(1R,2R)-2-pentylcyclopropyl]methyl]cyclopropyl]octyl]oxy]propan-2-amine, N,N-dimethyl-1-[[8-(2-oclylcyclopropyl)octyl]oxy]-3-(octyloxy)propan-2-amine and (11E,20Z,23Z)—N,N-dimethylnonacos-11,20,2-trien-10-amine or a pharmaceutically acceptable salt or stereoisomer thereof.

In some embodiments, the LNP formulations of the RNA (e.g., mRNA) vaccines may contain PEG-c-DOMG at 3% lipid molar ratio. In some embodiments, the LNP formulations of the RNA (e.g., mRNA) vaccines may contain PEG-c-DOMG at 1.5% lipid molar ratio.

In some embodiments, the pharmaceutical compositions of the RNA (e.g., mRNA) vaccines may include at least one of the PEGylated lipids described in International Publication No. WO2012099755, the contents of which are herein incorporated by reference in its entirety.

In some embodiments, the LNP formulation may contain PEG-DMG 2000 (1,2-dimyristoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000]). In some embodiments, the LNP formulation may contain PEG-DMG 2000, a cationic lipid known in the art and at least one other component. In some embodiments, the LNP formulation may contain PEG-DMG 2000, a cationic lipid known in the art, DSPC and cholesterol. As a non-limiting example, the LNP formulation may contain PEG-DMG 2000, DLinDMA, DSPC and cholesterol. As another non-limiting example the LNP formulation may contain PEG-DMG 2000, DLinDMA, DSPC and cholesterol in a molar ratio of 2:40:10:48 (see e.g., Geall et al., Nonviral delivery of self-amplifying RNA (e.g., mRNA) vaccines, PNAS 2012; PMID: 22908294, the contents of each of which are herein incorporated by reference in its entirety).

The lipid nanoparticles described herein may be made in a sterile environment.

In some embodiments, the LNP formulation may be formulated in a nanoparticle such as a nucleic acid-lipid particle. As a non-limiting example, the lipid particle may

comprise one or more active agents or therapeutic agents; one or more cationic lipids comprising from about 50 mol % to about 85 mol % of the total lipid present in the particle; one or more non-cationic lipids comprising from about 13 mol % to about 49.5 mol % of the total lipid present in the particle; and one or more conjugated lipids that inhibit aggregation of particles comprising from about 0.5 mol % to about 2 mol % of the total lipid present in the particle.

The nanoparticle formulations may comprise a phosphate conjugate. The phosphate conjugate may increase in vivo circulation times and/or increase the targeted delivery of the nanoparticle. As a non-limiting example, the phosphate conjugates may include a compound of any one of the formulas described in International Application No. WO2013033438, the contents of which are herein incorporated by reference in its entirety.

The nanoparticle formulation may comprise a polymer conjugate. The polymer conjugate may be a water soluble conjugate. The polymer conjugate may have a structure as described in U.S. Patent Application No. 20130059360, the contents of which are herein incorporated by reference in its entirety. In some embodiments, polymer conjugates with the polynucleotides of the present disclosure may be made using the methods and/or segmented polymeric reagents described in U.S. Patent Application No. 20130072709, the contents of which are herein incorporated by reference in its entirety. In some embodiments, the polymer conjugate may have pendant side groups comprising ring moieties such as, but not limited to, the polymer conjugates described in U.S. Patent Publication No. US20130196948, the contents which are herein incorporated by reference in its entirety.

The nanoparticle formulations may comprise a conjugate to enhance the delivery of nanoparticles of the present disclosure in a subject. Further, the conjugate may inhibit phagocytic clearance of the nanoparticles in a subject. In one aspect, the conjugate may be a “self” peptide designed from the human membrane protein CD47 (e.g., the “self” particles described by Rodriguez et al. (*Science* 2013 339, 971-975), herein incorporated by reference in its entirety). As shown by Rodriguez et al., the self peptides delayed macrophage-mediated clearance of nanoparticles which enhanced delivery of the nanoparticles. In another aspect, the conjugate may be the membrane protein CD47 (e.g., see Rodriguez et al. *Science* 2013 339, 971-975, herein incorporated by reference in its entirety). Rodriguez et al. showed that, similarly to “self” peptides, CD47 can increase the circulating particle ratio in a subject as compared to scrambled peptides and PEG coated nanoparticles.

In some embodiments, the RNA (e.g., mRNA) vaccines of the present disclosure are formulated in nanoparticles which comprise a conjugate to enhance the delivery of the nanoparticles of the present disclosure in a subject. The conjugate may be the CD47 membrane or the conjugate may be derived from the CD47 membrane protein, such as the “self” peptide described previously. In some embodiments, the nanoparticle may comprise PEG and a conjugate of CD47 or a derivative thereof. In some embodiments, the nanoparticle may comprise both the “self” peptide described above and the membrane protein CD47.

In some embodiments, a “self” peptide and/or CD47 protein may be conjugated to a virus-like particle or pseudovirion, as described herein for delivery of the RNA (e.g., mRNA) vaccines of the present disclosure.

In some embodiments, RNA (e.g., mRNA) vaccine pharmaceutical compositions comprising the polynucleotides of the present disclosure and a conjugate that may have a degradable linkage. Non-limiting examples of conjugates

include an aromatic moiety comprising an ionizable hydrogen atom, a spacer moiety, and a water-soluble polymer. As a non-limiting example, pharmaceutical compositions comprising a conjugate with a degradable linkage and methods for delivering such pharmaceutical compositions are described in U.S. Patent Publication No. US20130184443, the contents of which are herein incorporated by reference in their entirety.

The nanoparticle formulations may be a carbohydrate nanoparticle comprising a carbohydrate carrier and a RNA (e.g., mRNA) vaccine. As a non-limiting example, the carbohydrate carrier may include, but is not limited to, an anhydride-modified phytoglycogen or glycogen-type material, phytoglycogen octenyl succinate, phytoglycogen beta-dextrin, anhydride-modified phytoglycogen beta-dextrin. (See e.g., International Publication No. WO2012109121; the contents of which are herein incorporated by reference in their entirety).

Nanoparticle formulations of the present disclosure may be coated with a surfactant or polymer in order to improve the delivery of the particle. In some embodiments, the nanoparticle may be coated with a hydrophilic coating such as, but not limited to, PEG coatings and/or coatings that have a neutral surface charge. The hydrophilic coatings may help to deliver nanoparticles with larger payloads such as, but not limited to, RNA (e.g., mRNA) vaccines within the central nervous system. As a non-limiting example nanoparticles comprising a hydrophilic coating and methods of making such nanoparticles are described in U.S. Patent Publication No. US20130183244, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the lipid nanoparticles of the present disclosure may be hydrophilic polymer particles. Non-limiting examples of hydrophilic polymer particles and methods of making hydrophilic polymer particles are described in U.S. Patent Publication No. US20130210991, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the lipid nanoparticles of the present disclosure may be hydrophobic polymer particles.

Lipid nanoparticle formulations may be improved by replacing the cationic lipid with a biodegradable cationic lipid which is known as a rapidly eliminated lipid nanoparticle (reLNP). Ionizable cationic lipids, such as, but not limited to, DLinDMA, DLin-KC2-DMA, and DLin-MC3-DMA, have been shown to accumulate in plasma and tissues over time and may be a potential source of toxicity. The rapid metabolism of the rapidly eliminated lipids can improve the tolerability and therapeutic index of the lipid nanoparticles by an order of magnitude from a 1 mg/kg dose to a 10 mg/kg dose in rat. Inclusion of an enzymatically degraded ester linkage can improve the degradation and metabolism profile of the cationic component, while still maintaining the activity of the reLNP formulation. The ester linkage can be internally located within the lipid chain or it may be terminally located at the terminal end of the lipid chain. The internal ester linkage may replace any carbon in the lipid chain.

In some embodiments, the internal ester linkage may be located on either side of the saturated carbon.

In some embodiments, an immune response may be elicited by delivering a lipid nanoparticle which may include a nanospecies, a polymer and an immunogen. (U.S. Publication No. 20120189700 and International Publication No. WO2012099805; each of which is herein incorporated by reference in their entirety). The polymer may encapsulate the nanospecies or partially encapsulate the nanospecies.

The immunogen may be a recombinant protein, a modified RNA and/or a polynucleotide described herein. In some embodiments, the lipid nanoparticle may be formulated for use in a vaccine such as, but not limited to, against a pathogen.

Lipid nanoparticles may be engineered to alter the surface properties of particles so the lipid nanoparticles may penetrate the mucosal barrier. Mucus is located on mucosal tissue such as, but not limited to, oral (e.g., the buccal and esophageal membranes and tonsil tissue), ophthalmic, gastrointestinal (e.g., stomach, small intestine, large intestine, colon, rectum), nasal, respiratory (e.g., nasal, pharyngeal, tracheal and bronchial membranes), genital (e.g., vaginal, cervical and urethral membranes). Nanoparticles larger than 10-200 nm which are preferred for higher drug encapsulation efficiency and the ability to provide the sustained delivery of a wide array of drugs have been thought to be too large to rapidly diffuse through mucosal barriers. Mucus is continuously secreted, shed, discarded or digested and recycled so most of the trapped particles may be removed from the mucosa tissue within seconds or within a few hours. Large polymeric nanoparticles (200 nm-500 nm in diameter) which have been coated densely with a low molecular weight polyethylene glycol (PEG) diffused through mucus only 4 to 6-fold lower than the same particles diffusing in water (Lai et al. PNAS 2007 104(5):1482-487; Lai et al. *Adv Drug Deliv Rev.* 2009 61(2): 158-171; each of which is herein incorporated by reference in their entirety). The transport of nanoparticles may be determined using rates of permeation and/or fluorescent microscopy techniques including, but not limited to, fluorescence recovery after photobleaching (FRAP) and high resolution multiple particle tracking (MPT). As a non-limiting example, compositions which can penetrate a mucosal barrier may be made as described in U.S. Pat. No. 8,241,670 or International Patent Publication No. WO2013110028, the contents of each of which are herein incorporated by reference in its entirety.

The lipid nanoparticle engineered to penetrate mucus may comprise a polymeric material (i.e. a polymeric core) and/or a polymer-vitamin conjugate and/or a tri-block co-polymer. The polymeric material may include, but is not limited to, polyamines, polyethers, polyamides, polyesters, polycarbonates, polyureas, polycarbonates, poly(styrenes), polyimides, polysulfones, polyurethanes, polyacetylenes, polyethylenes, polyethylenimines, polyisocyanates, polyacrylates, polymethacrylates, polyacrylonitriles, and polyarylates. The polymeric material may be biodegradable and/or biocompatible. Non-limiting examples of biocompatible polymers are described in International Patent Publication No. WO2013116804, the contents of which are herein incorporated by reference in their entirety. The polymeric material may additionally be irradiated. As a non-limiting example, the polymeric material may be gamma irradiated (see e.g., International App. No. WO201282165, herein incorporated by reference in its entirety). Non-limiting examples of specific polymers include poly(caprolactone) (PCL), ethylene vinyl acetate polymer (EVA), poly(lactic acid) (PLA), poly(L-lactic acid) (PLLA), poly(glycolic acid) (PGA), poly(lactic acid-co-glycolic acid) (PLGA), poly(L-lactic acid-co-glycolic acid) (PLLGA), poly(D,L-lactide) (PDLA), poly(L-lactide) (PLLA), poly(D,L-lactide-co-caprolactone), poly(D,L-lactide-co-caprolactone-co-glycolide), poly(D,L-lactide-co-PEO-co-D,L-lactide), poly(D,L-lactide-co-PPO-co-D,L-lactide), polyalkyl cyanoacrylate, polyurethane, poly-L-lysine (PLL), hydroxypropyl methacrylate (HPMA), polyethyleneglycol, poly-L-glutamic acid, poly(hydroxy

acids), polyanhydrides, polyorthoesters, poly(ester amides), polyamides, poly(ester ethers), polycarbonates, polyalkylenes such as polyethylene and polypropylene, polyalkylene glycols such as poly(ethylene glycol) (PEG), polyalkylene oxides (PEO), polyalkylene terephthalates such as poly(ethylene terephthalate), polyvinyl alcohols (PVA), polyvinyl ethers, polyvinyl esters such as poly(vinyl acetate), polyvinyl halides such as poly(vinyl chloride) (PVC), polyvinylpyrrolidone, polysiloxanes, polystyrene (PS), polyurethanes, derivatized celluloses such as alkyl celluloses, hydroxyalkyl celluloses, cellulose ethers, cellulose esters, nitro celluloses, hydroxypropylcellulose, carboxymethylcellulose, polymers of acrylic acids, such as poly(methyl(meth)acrylate) (PMMA), poly(ethyl(meth)acrylate), poly(butyl(meth)acrylate), poly(isobutyl(meth)acrylate), poly(hexyl(meth)acrylate), poly(isodecyl(meth)acrylate), poly(lauryl(meth)acrylate), poly(phenyl(meth)acrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate) and copolymers and mixtures thereof, polydioxanone and its copolymers, polyhydroxyalkanoates, polypropylene fumarate, polyoxymethylene, poloxamers, poly(ortho)esters, poly(butyric acid), poly(valeric acid), poly(lactide-co-caprolactone), PEG-PLGA-PEG and trimethylene carbonate, polyvinylpyrrolidone. The lipid nanoparticle may be coated or associated with a co-polymer such as, but not limited to, a block co-polymer (such as a branched polyether-polyamide block copolymer described in International Publication No. WO2013012476, herein incorporated by reference in its entirety), and (poly(ethylene glycol))-(poly(propylene oxide))-(poly(ethylene glycol)) triblock copolymer (see e.g., U.S. Publication 20120121718 and U.S. Publication 20100003337 and U.S. Pat. No. 8,263,665, the contents of each of which is herein incorporated by reference in their entirety). The co-polymer may be a polymer that is generally regarded as safe (GRAS) and the formation of the lipid nanoparticle may be in such a way that no new chemical entities are created. For example, the lipid nanoparticle may comprise poloxamers coating PLGA nanoparticles without forming new chemical entities which are still able to rapidly penetrate human mucus (Yang et al. *Angew. Chem. Int. Ed.* 2011 50:2597-2600; the contents of which are herein incorporated by reference in their entirety). A non-limiting scalable method to produce nanoparticles which can penetrate human mucus is described by Xu et al. (see, e.g., *J Control Release* 2013, 170(2):279-86; the contents of which are herein incorporated by reference in their entirety).

The vitamin of the polymer-vitamin conjugate may be vitamin E. The vitamin portion of the conjugate may be substituted with other suitable components such as, but not limited to, vitamin A, vitamin E, other vitamins, cholesterol, a hydrophobic moiety, or a hydrophobic component of other surfactants (e.g., sterol chains, fatty acids, hydrocarbon chains and alkylene oxide chains).

The lipid nanoparticle engineered to penetrate mucus may include surface altering agents such as, but not limited to, polynucleotides, anionic proteins (e.g., bovine serum albumin), surfactants (e.g., cationic surfactants such as for example dimethyldioctadecyl-ammonium bromide), sugars or sugar derivatives (e.g., cyclodextrin), nucleic acids, polymers (e.g., heparin, polyethylene glycol and poloxamer), mucolytic agents (e.g., N-acetylcysteine, mugwort, bromelain, papain, clerodendrum, acetylcysteine, bromhexine, carbocisteine, eprazinone, mesna, ambroxol, sobrerol, domidol, letosteine, stepronin, tiopronin, gelsolin, thymosin β 4 domase alfa, neltexine, erdosteine) and various DNases including rhDNase. The surface altering agent may be

embedded or enmeshed in the particle's surface or disposed (e.g., by coating, adsorption, covalent linkage, or other process) on the surface of the lipid nanoparticle. (see e.g., U.S. Publication 20100215580 and U.S. Publication 20080166414 and US20130164343; the contents of each of which are herein incorporated by reference in their entirety).

In some embodiments, the mucus penetrating lipid nanoparticles may comprise at least one polynucleotide described herein. The polynucleotide may be encapsulated in the lipid nanoparticle and/or disposed on the surface of the particle. The polynucleotide may be covalently coupled to the lipid nanoparticle. Formulations of mucus penetrating lipid nanoparticles may comprise a plurality of nanoparticles. Further, the formulations may contain particles which may interact with the mucus and alter the structural and/or adhesive properties of the surrounding mucus to decrease mucoadhesion, which may increase the delivery of the mucus penetrating lipid nanoparticles to the mucosal tissue.

In some embodiments, the mucus penetrating lipid nanoparticles may be a hypotonic formulation comprising a mucosal penetration enhancing coating. The formulation may be hypotonic for the epithelium to which it is being delivered. Non-limiting examples of hypotonic formulations may be found in International Patent Publication No. WO2013110028, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, in order to enhance the delivery through the mucosal barrier the RNA (e.g., mRNA) vaccine formulation may comprise or be a hypotonic solution. Hypotonic solutions were found to increase the rate at which muco-inert particles such as, but not limited to, mucus-penetrating particles, were able to reach the vaginal epithelial surface (see e.g., Ensign et al. *Biomaterials* 2013 34(28): 6922-9, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the RNA (e.g., mRNA) vaccine is formulated as a lipoplex, such as, without limitation, the ATUPLEX™ system, the DACC system, the DBTC system and other siRNA-lipoplex technology from Silence Therapeutics (London, United Kingdom), STEMFECT™ from STEMGENT® (Cambridge, MA), and polyethylenimine (PET) or protamine-based targeted and non-targeted delivery of nucleic acids (Aleku et al. *Cancer Res.* 2008 68:9788-9798; Strumberg et al. *Int J Clin Pharmacol Ther* 2012 50:76-78; Santel et al., *Gene Ther* 2006 13:1222-1234; Santel et al., *Gene Ther* 2006 13:1360-1370; Gutbier et al., *Pulm Pharmacol. Ther.* 2010 23:334-344; Kaufmann et al. *Microvasc Res* 2010 80:286-293; Weide et al. *J Immunother.* 2009 32:498-507; Weide et al. *J Immunother.* 2008 31:180-188; Pascolo Expert Opin. Biol. Ther. 4:1285-1294; Fotin-Mleczek et al., 2011 *J. Immunother.* 34:1-15; Song et al., *Nature Biotechnol.* 2005, 23:709-717; Peer et al., *Proc Natl Acad Sci USA.* 2007 6; 104:4095-4100; deFougerolles *Hum Gene Ther.* 2008 19:125-132, the contents of each of which are incorporated herein by reference in their entirety).

In some embodiments, such formulations may also be constructed or compositions altered such that they passively or actively are directed to different cell types in vivo, including but not limited to hepatocytes, immune cells, tumor cells, endothelial cells, antigen presenting cells, and leukocytes (Akinc et al. *Mol Ther.* 2010 18:1357-1364; Song et al., *Nat Biotechnol.* 2005 23:709-717; Judge et al., *J Clin Invest.* 2009 119:661-673; Kaufmann et al., *Microvasc Res* 2010 80:286-293; Santel et al., *Gene Ther* 2006 13:1222-1234; Santel et al., *Gene Ther* 2006 13:1360-1370; Gutbier et al., *Pulm Pharmacol. Ther.* 2010 23:334-344; Basha et al., *Mol. Ther.* 2011 19:2186-2200; Fenske and

Cullis, *Expert Opin Drug Deliv.* 2008 5:25-44; Peer et al., *Science.* 2008 319:627-630; Peer and Lieberman, *Gene Ther.* 2011 18:1127-1133, the contents of each of which are incorporated herein by reference in their entirety). One example of passive targeting of formulations to liver cells includes the DLin-DMA, DLin-KC2-DMA and DLin-MC3-DMA-based lipid nanoparticle formulations, which have been shown to bind to apolipoprotein E and promote binding and uptake of these formulations into hepatocytes *in vivo* (Akinc et al. *Mol Ther.* 2010 18:1357-1364, the contents of which are incorporated herein by reference in their entirety). Formulations can also be selectively targeted through expression of different ligands on their surface as exemplified by, but not limited by, folate, transferrin, N-acetylgalactosamine (GalNAc), and antibody targeted approaches (Kolhatkar et al., *Curr Drug Discov Technol.* 2011 8:197-206; Musacchio and Torchilin, *Front Biosci.* 2011 16:1388-1412; Yu et al., *Mol Membr Biol.* 2010 27:286-298; Patil et al., *Crit Rev Ther Drug Carrier Syst.* 2008 25:1-61; Benoit et al., *Biomacromolecules.* 2011 12:2708-2714; Zhao et al., *Expert Opin Drug Deliv.* 2008 5:309-319; Akinc et al., *Mol Ther.* 2010 18:1357-1364; Srinivasan et al., *Methods Mol Biol.* 2012 820:105-116; Ben-Arie et al., *Methods Mol Biol.* 2012 757:497-507; Peer 2010 *J Control Release.* 20:63-68; Peer et al., *Proc Natl Acad Sci USA.* 2007 104:4095-4100; Kim et al., *Methods Mol Biol.* 2011 721:339-353; Subramanya et al., *Mol Ther.* 2010 18:2028-2037; Song et al., *Nat Biotechnol.* 2005 23:709-717; Peer et al., *Science.* 2008 319:627-630; Peer and Lieberman, *Gene Ther.* 2011 18:1127-1133, the contents of each of which are incorporated herein by reference in their entirety).

In some embodiments, the RNA (e.g., mRNA) vaccine is formulated as a solid lipid nanoparticle. A solid lipid nanoparticle (SLN) may be spherical with an average diameter between 10 to 1000 nm. SLN possess a solid lipid core matrix that can solubilize lipophilic molecules and may be stabilized with surfactants and/or emulsifiers. In some embodiments, the lipid nanoparticle may be a self-assembly lipid-polymer nanoparticle (see Zhang et al., *ACS Nano*, 2008, 2 (8), pp 1696-1702; the contents of which are herein incorporated by reference in their entirety). As a non-limiting example, the SLN may be the SLN described in International Patent Publication No. WO2013105101, the contents of which are herein incorporated by reference in their entirety. As another non-limiting example, the SLN may be made by the methods or processes described in International Patent Publication No. WO2013105101, the contents of which are herein incorporated by reference in their entirety.

Liposomes, lipoplexes, or lipid nanoparticles may be used to improve the efficacy of polynucleotides directed protein production as these formulations may be able to increase cell transfection by the RNA (e.g., mRNA) vaccine; and/or increase the translation of encoded protein. One such example involves the use of lipid encapsulation to enable the effective systemic delivery of polyplex plasmid DNA (Heyes et al., *Mol Ther.* 2007 15:713-720; the contents of which are incorporated herein by reference in their entirety). The liposomes, lipoplexes, or lipid nanoparticles may also be used to increase the stability of the polynucleotide.

In some embodiments, the RNA (e.g., mRNA) vaccines of the present disclosure can be formulated for controlled release and/or targeted delivery. As used herein, "controlled release" refers to a pharmaceutical composition or compound release profile that conforms to a particular pattern of release to effect a therapeutic outcome. In some embodiments, the RNA (e.g., mRNA) vaccines may be encapsu-

lated into a delivery agent described herein and/or known in the art for controlled release and/or targeted delivery. As used herein, the term "encapsulate" means to enclose, surround or encase. As it relates to the formulation of the compounds of the disclosure, encapsulation may be substantial, complete or partial. The term "substantially encapsulated" means that at least greater than 50, 60, 70, 80, 85, 90, 95, 96, 97, 98, 99, 99.9, 99.9 or greater than 99.999% of the pharmaceutical composition or compound of the disclosure may be enclosed, surrounded or encased within the delivery agent. "Partially encapsulation" means that less than 10, 10, 20, 30, 40 50 or less of the pharmaceutical composition or compound of the disclosure may be enclosed, surrounded or encased within the delivery agent. Advantageously, encapsulation may be determined by measuring the escape or the activity of the pharmaceutical composition or compound of the disclosure using fluorescence and/or electron micrograph. For example, at least 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 85, 90, 95, 96, 97, 98, 99, 99.9, 99.99 or greater than 99.99% of the pharmaceutical composition or compound of the disclosure are encapsulated in the delivery agent.

In some embodiments, the controlled release formulation may include, but is not limited to, tri-block co-polymers. As a non-limiting example, the formulation may include two different types of tri-block co-polymers (International Pub. No. WO2012131104 and WO2012131106, the contents of each of which are incorporated herein by reference in their entirety).

In some embodiments, the RNA (e.g., mRNA) vaccines may be encapsulated into a lipid nanoparticle or a rapidly eliminated lipid nanoparticle and the lipid nanoparticles or a rapidly eliminated lipid nanoparticle may then be encapsulated into a polymer, hydrogel and/or surgical sealant described herein and/or known in the art. As a non-limiting example, the polymer, hydrogel or surgical sealant may be PLGA, ethylene vinyl acetate (EVAc), poloxamer, GELSITE® (Nanotherapeutics, Inc. Alachua, FL), HYL-ENEX® (Halozyme Therapeutics, San Diego CA), surgical sealants such as fibrinogen polymers (Ethicon Inc. Cornelia, GA), TISSELL® (Baxter International, Inc Deerfield, IL), PEG-based sealants, and COSEAL® (Baxter International, Inc Deerfield, IL).

In some embodiments, the lipid nanoparticle may be encapsulated into any polymer known in the art which may form a gel when injected into a subject. As another non-limiting example, the lipid nanoparticle may be encapsulated into a polymer matrix which may be biodegradable.

In some embodiments, the RNA (e.g., mRNA) vaccine formulation for controlled release and/or targeted delivery may also include at least one controlled release coating.

Controlled release coatings include, but are not limited to, OPADRY®, polyvinylpyrrolidone/vinyl acetate copolymer, polyvinylpyrrolidone, hydroxypropyl methylcellulose, hydroxypropyl cellulose, hydroxyethyl cellulose, EUDRAGIT RL®, EUDRAGIT RS® and cellulose derivatives such as ethylcellulose aqueous dispersions (AQUACOAT® and SURELEASE®).

In some embodiments, the RNA (e.g., mRNA) vaccine controlled release and/or targeted delivery formulation may comprise at least one degradable polyester which may contain polycationic side chains. Degradable polyesters include, but are not limited to, poly(serine ester), poly(L-lactide-co-L-lysine), poly(4-hydroxy-L-proline ester), and combinations thereof. In some embodiments, the degradable polyesters may include a PEG conjugation to form a PEGylated polymer.

In some embodiments, the RNA (e.g., mRNA) vaccine controlled release and/or targeted delivery formulation comprising at least one polynucleotide may comprise at least one PEG and/or PEG related polymer derivatives as described in U.S. Pat. No. 8,404,222, the contents of which are incorporated herein by reference in their entirety.

In some embodiments, the RNA (e.g., mRNA) vaccine controlled release delivery formulation comprising at least one polynucleotide may be the controlled release polymer system described in US20130130348, the contents of which are incorporated herein by reference in their entirety.

In some embodiments, the RNA (e.g., mRNA) vaccines of the present disclosure may be encapsulated in a therapeutic nanoparticle, referred to herein as "therapeutic nanoparticle RNA (e.g., mRNA) vaccines." Therapeutic nanoparticles may be formulated by methods described herein and known in the art such as, but not limited to, International Pub Nos. WO2010005740, WO2010030763, WO2010005721, WO2010005723, WO2012054923, U.S. Publication Nos. US20110262491, US20100104645, US20100087337, US20100068285, US20110274759, US20100068286, US20120288541, US20130123351 and US20130230567 and U.S. Pat. Nos. 8,206,747, 8,293,276, 8,318,208 and 8,318,211; the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, therapeutic polymer nanoparticles may be identified by the methods described in US Pub No. US20120140790, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the therapeutic nanoparticle RNA (e.g., mRNA) vaccine may be formulated for sustained release. As used herein, "sustained release" refers to a pharmaceutical composition or compound that conforms to a release rate over a specific period of time. The period of time may include, but is not limited to, hours, days, weeks, months and years. As a non-limiting example, the sustained release nanoparticle may comprise a polymer and a therapeutic agent such as, but not limited to, the polynucleotides of the present disclosure (see International Pub No. 2010075072 and US Pub No. US20100216804, US20110217377 and US20120201859, the contents of each of which are incorporated herein by reference in their entirety). In another non-limiting example, the sustained release formulation may comprise agents which permit persistent bioavailability such as, but not limited to, crystals, macromolecular gels and/or particulate suspensions (see U.S. Patent Publication No US20130150295, the contents of each of which are incorporated herein by reference in their entirety).

In some embodiments, the therapeutic nanoparticle RNA (e.g., mRNA) vaccines may be formulated to be target specific. As a non-limiting example, the therapeutic nanoparticles may include a corticosteroid (see International Pub. No. WO2011084518, the contents of which are incorporated herein by reference in their entirety). As a non-limiting example, the therapeutic nanoparticles may be formulated in nanoparticles described in International Pub No. WO2008121949, WO2010005726, WO2010005725, WO2011084521 and US Pub No. US20100069426, US20120004293 and US20100104655, the contents of each of which are incorporated herein by reference in their entirety.

In some embodiments, the nanoparticles of the present disclosure may comprise a polymeric matrix. As a non-limiting example, the nanoparticle may comprise two or more polymers such as, but not limited to, polyethylenes, polycarbonates, polyanhydrides, polyhydroxyacids, poly-

propylfumerates, polycaprolactones, polyamides, polyacetals, polyethers, polyesters, poly(orthoesters), polycyanoacrylates, polyvinyl alcohols, polyurethanes, polyphosphazenes, polyacrylates, polymethacrylates, polycyanoacrylates, polyureas, polystyrenes, polyamines, polylysine, poly(ethylene imine), poly(serine ester), poly(L-lactide-co-L-lysine), poly(4-hydroxy-L-proline ester) or combinations thereof.

In some embodiments, the therapeutic nanoparticle comprises a diblock copolymer. In some embodiments, the diblock copolymer may include PEG in combination with a polymer such as, but not limited to, polyethylenes, polycarbonates, polyanhydrides, polyhydroxyacids, polypropylfumerates, polycaprolactones, polyamides, polyacetals, polyethers, polyesters, poly(orthoesters), polycyanoacrylates, polyvinyl alcohols, polyurethanes, polyphosphazenes, polyacrylates, polymethacrylates, polycyanoacrylates, polyureas, polystyrenes, polyamines, polylysine, poly(ethylene imine), poly(serine ester), poly(L-lactide-co-L-lysine), poly(4-hydroxy-L-proline ester) or combinations thereof. In yet another embodiment, the diblock copolymer may be a high-X diblock copolymer such as those described in International Patent Publication No. WO2013120052, the contents of which are incorporated herein by reference in their entirety.

As a non-limiting example the therapeutic nanoparticle comprises a PLGA-PEG block copolymer (see U.S. Publication No. US20120004293 and U.S. Pat. No. 8,236,330, each of which is herein incorporated by reference in their entirety). In another non-limiting example, the therapeutic nanoparticle is a stealth nanoparticle comprising a diblock copolymer of PEG and PLA or PEG and PLGA (see U.S. Pat. No. 8,246,968 and International Publication No. WO2012166923, the contents of each of which are herein incorporated by reference in their entirety). In yet another non-limiting example, the therapeutic nanoparticle is a stealth nanoparticle or a target-specific stealth nanoparticle as described in U.S. Patent Publication No. US20130172406, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the therapeutic nanoparticle may comprise a multiblock copolymer (see e.g., U.S. Pat. Nos. 8,263,665 and 8,287,910 and U.S. Patent Pub. No. US20130195987, the contents of each of which are herein incorporated by reference in their entirety).

In yet another non-limiting example, the lipid nanoparticle comprises the block copolymer PEG-PLGA-PEG (see e.g., the thermosensitive hydrogel (PEG-PLGA-PEG) was used as a TGF-beta1 gene delivery vehicle in Lee et al. Thermosensitive Hydrogel as a Tgf-β1 Gene Delivery Vehicle Enhances Diabetic Wound Healing. Pharmaceutical Research, 2003 20(12): 1995-2000; as a controlled gene delivery system in Li et al. Controlled Gene Delivery System Based on Thermosensitive Biodegradable Hydrogel. Pharmaceutical Research 2003 20(6):884-888; and Chang et al., Non-ionic amphiphilic biodegradable PEG-PLGA-PEG copolymer enhances gene delivery efficiency in rat skeletal muscle. J Controlled Release. 2007 118:245-253, the contents of each of which are herein incorporated by reference in their entirety). The RNA (e.g., mRNA) vaccines of the present disclosure may be formulated in lipid nanoparticles comprising the PEG-PLGA-PEG block copolymer.

In some embodiments, the therapeutic nanoparticle may comprise a multiblock copolymer (see e.g., U.S. Pat. Nos. 8,263,665 and 8,287,910 and U.S. Patent Pub. No.

US20130195987, the contents of each of which are herein incorporated by reference in their entirety).

In some embodiments, the block copolymers described herein may be included in a polyion complex comprising a non-polymeric micelle and the block copolymer. (see e.g., U.S. Publication No. 20120076836, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the therapeutic nanoparticle may comprise at least one acrylic polymer. Acrylic polymers include but are not limited to, acrylic acid, methacrylic acid, acrylic acid and methacrylic acid copolymers, methyl methacrylate copolymers, ethoxyethyl methacrylates, cyanoethyl methacrylate, amino alkyl methacrylate copolymer, poly (acrylic acid), poly(methacrylic acid), polycyanoacrylates and combinations thereof.

In some embodiments, the therapeutic nanoparticles may comprise at least one poly(vinyl ester) polymer. The poly (vinyl ester) polymer may be a copolymer such as a random copolymer. As a non-limiting example, the random copolymer may have a structure such as those described in International Application No. WO2013032829 or U.S. Patent Publication No US20130121954, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, the poly(vinyl ester) polymers may be conjugated to the polynucleotides described herein.

In some embodiments, the therapeutic nanoparticle may comprise at least one diblock copolymer. The diblock copolymer may be, but it not limited to, a poly(lactic) acid-poly (ethylene)glycol copolymer (see, e.g., International Patent Publication No. WO2013044219, the contents of which are herein incorporated by reference in their entirety). As a non-limiting example, the therapeutic nanoparticle may be used to treat cancer (see International publication No. WO2013044219, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the therapeutic nanoparticles may comprise at least one cationic polymer described herein and/or known in the art.

In some embodiments, the therapeutic nanoparticles may comprise at least one amine-containing polymer such as, but not limited to polylysine, polyethylene imine, poly(amido-amine) dendrimers, poly(beta-amino esters) (see, e.g., U.S. Pat. No. 8,287,849, the contents of which are herein incorporated by reference in their entirety) and combinations thereof.

In some embodiments, the nanoparticles described herein may comprise an amine cationic lipid such as those described in International Patent Application No. WO2013059496, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the cationic lipids may have an amino-amine or an amino-amide moiety.

In some embodiments, the therapeutic nanoparticles may comprise at least one degradable polyester which may contain polycationic side chains. Degradable polyesters include, but are not limited to, poly(serine ester), poly(L-lactide-co-L-lysine), poly(4-hydroxy-L-proline ester), and combinations thereof. In some embodiments, the degradable polyesters may include a PEG conjugation to form a PEGylated polymer.

In some embodiments, the synthetic nanocarriers may contain an immunostimulatory agent to enhance the immune response from delivery of the synthetic nanocarrier. As a non-limiting example, the synthetic nanocarrier may comprise a Th1 immunostimulatory agent, which may enhance a Th1-based response of the immune system (see International Pub No. WO2010123569 and U.S. Publication No.

US20110223201, the contents of each of which are herein incorporated by reference in their entirety).

In some embodiments, the synthetic nanocarriers may be formulated for targeted release. In some embodiments, the synthetic nanocarrier is formulated to release the polynucleotides at a specified pH and/or after a desired time interval. As a non-limiting example, the synthetic nanoparticle may be formulated to release the RNA (e.g., mRNA) vaccines after 24 hours and/or at a pH of 4.5 (see International Publication Nos. WO2010138193 and WO2010138194 and US Pub Nos. US20110020388 and US20110027217, each of which is herein incorporated by reference in their entirety).

In some embodiments, the synthetic nanocarriers may be formulated for controlled and/or sustained release of the polynucleotides described herein. As a non-limiting example, the synthetic nanocarriers for sustained release may be formulated by methods known in the art, described herein and/or as described in International Pub No. WO2010138192 and US Pub No. 20100303850, each of which is herein incorporated by reference in their entirety.

In some embodiments, the RNA (e.g., mRNA) vaccine may be formulated for controlled and/or sustained release wherein the formulation comprises at least one polymer that is a crystalline side chain (CYSC) polymer. CYSC polymers are described in U.S. Pat. No. 8,399,007, herein incorporated by reference in its entirety.

In some embodiments, the synthetic nanocarrier may be formulated for use as a vaccine. In some embodiments, the synthetic nanocarrier may encapsulate at least one polynucleotide which encode at least one antigen. As a non-limiting example, the synthetic nanocarrier may include at least one antigen and an excipient for a vaccine dosage form (see International Publication No. WO2011150264 and U.S. Publication No. US20110293723, the contents of each of which are herein incorporated by reference in their entirety). As another non-limiting example, a vaccine dosage form may include at least two synthetic nanocarriers with the same or different antigens and an excipient (see International Publication No. WO2011150249 and U.S. Publication No. US20110293701, the contents of each of which are herein incorporated by reference in their entirety). The vaccine dosage form may be selected by methods described herein, known in the art and/or described in International Publication No. WO2011150258 and U.S. Publication No. US20120027806, the contents of each of which are herein incorporated by reference in their entirety).

In some embodiments, the synthetic nanocarrier may comprise at least one polynucleotide which encodes at least one adjuvant. As non-limiting example, the adjuvant may comprise dimethyldioctadecylammonium-bromide, dimethyldioctadecylammonium-chloride, dimethyldioctadecylammonium-phosphate or dimethyldioctadecylammonium-acetate (DDA) and an apolar fraction or part of said apolar fraction of a total lipid extract of a *mycobacterium* (see, e.g., U.S. Pat. No. 8,241,610, the content of which is herein incorporated by reference in its entirety). In some embodiments, the synthetic nanocarrier may comprise at least one polynucleotide and an adjuvant. As a non-limiting example, the synthetic nanocarrier comprising and adjuvant may be formulated by the methods described in International Publication No. WO2011150240 and U.S. Publication No. US20110293700, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, the synthetic nanocarrier may encapsulate at least one polynucleotide that encodes a peptide, fragment or region from a virus. As a non-limiting example, the synthetic nanocarrier may include, but is not

limited to, any of the nanocarriers described in International Publication No. WO2012024621, WO201202629, WO2012024632 and U.S. Publication No. US20120064110, US20120058153 and US20120058154, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, the synthetic nanocarrier may be coupled to a polynucleotide which may be able to trigger a humoral and/or cytotoxic T lymphocyte (CTL) response (see, e.g., International Publication No. WO2013019669, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the RNA (e.g., mRNA) vaccine may be encapsulated in, linked to and/or associated with zwitterionic lipids. Non-limiting examples of zwitterionic lipids and methods of using zwitterionic lipids are described in U.S. Patent Publication No. US20130216607, the contents of which are herein incorporated by reference in their entirety. In some aspects, the zwitterionic lipids may be used in the liposomes and lipid nanoparticles described herein.

In some embodiments, the RNA (e.g., mRNA) vaccine may be formulated in colloid nanocarriers as described in U.S. Patent Publication No. US20130197100, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the nanoparticle may be optimized for oral administration. The nanoparticle may comprise at least one cationic biopolymer such as, but not limited to, chitosan or a derivative thereof. As a non-limiting example, the nanoparticle may be formulated by the methods described in U.S. Publication No. 20120282343, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, LNPs comprise the lipid KL52 (an amino-lipid disclosed in U.S. Application Publication No. 2012/0295832, the contents of which are herein incorporated by reference in their entirety. Activity and/or safety (as measured by examining one or more of ALT/AST, white blood cell count and cytokine induction, for example) of LNP administration may be improved by incorporation of such lipids. LNPs comprising KL52 may be administered intravenously and/or in one or more doses. In some embodiments, administration of LNPs comprising KL52 results in equal or improved mRNA and/or protein expression as compared to LNPs comprising MC3.

In some embodiments, RNA (e.g., mRNA) vaccine may be delivered using smaller LNPs. Such particles may comprise a diameter from below 0.1 μm up to 100 nm such as, but not limited to, less than 0.1 μm , less than 1.0 μm , less than 5 μm , less than 10 μm , less than 15 μm , less than 20 μm , less than 25 μm , less than 30 μm , less than 35 μm , less than 40 μm , less than 50 μm , less than 55 μm , less than 60 μm , less than 65 μm , less than 70 μm , less than 75 μm , less than 80 μm , less than 85 μm , less than 90 μm , less than 95 μm , less than 100 μm , less than 125 μm , less than 150 μm , less than 175 μm , less than 200 μm , less than 225 μm , less than 250 μm , less than 275 μm , less than 300 μm , less than 325 μm , less than 350 μm , less than 375 μm , less than 400 μm , less than 425 μm , less than 450 μm , less than 475 μm , less than 500 μm , less than 525 μm , less than 550 μm , less than 575 μm , less than 600 μm , less than 625 μm , less than 650 μm , less than 675 μm , less than 700 μm , less than 725 μm , less than 750 μm , less than 775 μm , less than 800 μm , less than 825 μm , less than 850 μm , less than 875 μm , less than 900 μm , less than 925 μm , less than 950 μm , less than 975 μm , or less than 1000 μm .

In some embodiments, RNA (e.g., mRNA) vaccines may be delivered using smaller LNPs, which may comprise a diameter from about 1 nm to about 100 nm, from about 1 nm to about 10 nm, about 1 nm to about 20 nm, from about 1 nm to about 30 nm, from about 1 nm to about 40 nm, from about 1 nm to about 50 nm, from about 1 nm to about 60 nm, from about 1 nm to about 70 nm, from about 1 nm to about 80 nm, from about 1 nm to about 90 nm, from about 5 nm to about 100 nm, from about 5 nm to about 10 nm, about 5 nm to about 20 nm, from about 5 nm to about 30 nm, from about 5 nm to about 40 nm, from about 5 nm to about 50 nm, from about 5 nm to about 60 nm, from about 5 nm to about 70 nm, from about 5 nm to about 80 nm, from about 5 nm to about 90 nm, about 10 to about 50 nm, from about 20 to about 50 nm, from about 30 to about 50 nm, from about 40 to about 50 nm, from about 20 to about 60 nm, from about 30 to about 60 nm, from about 40 to about 60 nm, from about 20 to about 70 nm, from about 30 to about 70 nm, from about 40 to about 70 nm, from about 50 to about 70 nm, from about 60 to about 70 nm, from about 20 to about 80 nm, from about 30 to about 80 nm, from about 40 to about 80 nm, from about 50 to about 80 nm, from about 60 to about 80 nm, from about 20 to about 90 nm, from about 30 to about 90 nm, from about 40 to about 90 nm, from about 50 to about 90 nm, from about 60 to about 90 nm and/or from about 70 to about 90 nm.

In some embodiments, such LNPs are synthesized using methods comprising microfluidic mixers. Examples of microfluidic mixers may include, but are not limited to, a slit interdigital micromixer including, but not limited to those manufactured by *Microinova* (Allerheiligen bei Wildon, Austria) and/or a staggered herringbone micromixer (SHM) (Zhigaltsev, T. V. et al., Bottom-up design and synthesis of limit size lipid nanoparticle systems with aqueous and triglyceride cores using millisecond microfluidic mixing have been published (Langmuir. 2012. 28:3633-40; Bellevue, N. M. et al., Microfluidic synthesis of highly potent limit-size lipid nanoparticles for in vivo delivery of siRNA. Molecular Therapy-Nucleic Acids. 2012. 1:e37; Chen, D. et al., Rapid discovery of potent siRNA-containing lipid nanoparticles enabled by controlled microfluidic formulation. J Am Chem Soc. 2012. 134(16):6948-51, the contents of each of which are herein incorporated by reference in their entirety). In some embodiments, methods of LNP generation comprising SHM, further comprise the mixing of at least two input streams wherein mixing occurs by microstructure-induced chaotic advection (MICA). According to this method, fluid streams flow through channels present in a herringbone pattern causing rotational flow and folding the fluids around each other. This method may also comprise a surface for fluid mixing wherein the surface changes orientations during fluid cycling. Methods of generating LNPs using SHM include those disclosed in U.S. Application Publication Nos. 2004/0262223 and 2012/0276209, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, the RNA (e.g., mRNA) vaccine of the present disclosure may be formulated in lipid nanoparticles created using a micromixer such as, but not limited to, a Slit Interdigital Microstructured Mixer (SIMM-V2) or a Standard Slit Interdigital Micro Mixer (SSIMM) or Caterpillar (CPMM) or Impinging-jet (IJMM) from the Institut für Mikrotechnik Mainz GmbH, Mainz Germany).

In some embodiments, the RNA (e.g., mRNA) vaccines of the present disclosure may be formulated in lipid nanoparticles created using microfluidic technology (see, e.g., Whitesides, George M. The Origins and the Future of Micro-

fluidics. *Nature*, 2006 442: 368-373; and Abraham et al. Chaotic Mixer for Microchannels. *Science*, 2002 295: 647-651; each of which is herein incorporated by reference in its entirety). As a non-limiting example, controlled microfluidic formulation includes a passive method for mixing streams of steady pressure-driven flows in micro channels at a low Reynolds number (see, e.g., Abraham et al. Chaotic Mixer for Microchannels. *Science*, 2002 295: 647-651, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the RNA (e.g., mRNA) vaccines of the present disclosure may be formulated in lipid nanoparticles created using a micromixer chip such as, but not limited to, those from Harvard Apparatus (Holliston, MA) or Dolomite Microfluidics (Royston, UK). A micromixer chip can be used for rapid mixing of two or more fluid streams with a split and recombine mechanism.

In some embodiments, the RNA (e.g., mRNA) vaccines of the disclosure may be formulated for delivery using the drug encapsulating microspheres described in International Patent Publication No. WO2013063468 or U.S. Pat. No. 8,440,614, the contents of each of which are herein incorporated by reference in their entirety. The microspheres may comprise a compound of the formula (I), (II), (III), (IV), (V) or (VI) as described in International Patent Publication No. WO2013063468, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the amino acid, peptide, polypeptide, lipids (APPL) are useful in delivering the RNA (e.g., mRNA) vaccines of the disclosure to cells (see International Patent Publication No. WO2013063468, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the RNA (e.g., mRNA) vaccines of the disclosure may be formulated in lipid nanoparticles having a diameter from about 10 to about 100 nm such as, but not limited to, about 10 to about 20 nm, about 10 to about 30 nm, about 10 to about 40 nm, about 10 to about 50 nm, about 10 to about 60 nm, about 10 to about 70 nm, about 10 to about 80 nm, about 10 to about 90 nm, about 20 to about 30 nm, about 20 to about 40 nm, about 20 to about 50 nm, about 20 to about 60 nm, about 20 to about 70 nm, about 20 to about 80 nm, about 20 to about 90 nm, about 20 to about 100 nm, about 30 to about 40 nm, about 30 to about 50 nm, about 30 to about 60 nm, about 30 to about 70 nm, about 30 to about 80 nm, about 30 to about 90 nm, about 30 to about 100 nm, about 40 to about 50 nm, about 40 to about 60 nm, about 40 to about 70 nm, about 40 to about 80 nm, about 40 to about 90 nm, about 40 to about 100 nm, about 50 to about 60 nm, about 50 to about 70 nm, about 50 to about 80 nm, about 50 to about 90 nm, about 50 to about 100 nm, about 60 to about 70 nm, about 60 to about 80 nm, about 60 to about 90 nm, about 60 to about 100 nm, about 70 to about 80 nm, about 70 to about 90 nm, about 70 to about 100 nm, about 80 to about 90 nm, about 80 to about 100 nm and/or about 90 to about 100 nm.

In some embodiments, the lipid nanoparticles may have a diameter from about 10 to 500 nm.

In some embodiments, the lipid nanoparticle may have a diameter greater than 100 nm, greater than 150 nm, greater than 200 nm, greater than 250 nm, greater than 300 nm, greater than 350 nm, greater than 400 nm, greater than 450 nm, greater than 500 nm, greater than 550 nm, greater than 600 nm, greater than 650 nm, greater than 700 nm, greater than 750 nm, greater than 800 nm, greater than 850 nm, greater than 900 nm, greater than 950 nm or greater than 1000 nm.

In some embodiments, the lipid nanoparticle may be a limit size lipid nanoparticle described in International Patent Publication No. WO2013059922, the contents of which are herein incorporated by reference in their entirety. The limit size lipid nanoparticle may comprise a lipid bilayer surrounding an aqueous core or a hydrophobic core; where the lipid bilayer may comprise a phospholipid such as, but not limited to, diacylphosphatidylcholine, a diacylphosphatidylethanolamine, a ceramide, a sphingomyelin, a dihydro-sphingomyelin, a cephalin, a cerebroside, a C8-C20 fatty acid diacylphosphatidylcholine, and 1-palmitoyl-2-oleoyl phosphatidylethanolamine (POPC). In some embodiments, the limit size lipid nanoparticle may comprise a polyethylene glycol-lipid such as, but not limited to, DLPE-PEG, DMPE-PEG, DPPC-PEG and DSPE-PEG.

In some embodiments, the RNA (e.g., mRNA) vaccines may be delivered, localized and/or concentrated in a specific location using the delivery methods described in International Patent Publication No. WO2013063530, the contents of which are herein incorporated by reference in their entirety. As a non-limiting example, a subject may be administered an empty polymeric particle prior to, simultaneously with or after delivering the RNA (e.g., mRNA) vaccines to the subject. The empty polymeric particle undergoes a change in volume once in contact with the subject and becomes lodged, embedded, immobilized or entrapped at a specific location in the subject.

In some embodiments, the RNA (e.g., mRNA) vaccines may be formulated in an active substance release system (see, e.g., U.S. Patent Publication No. US20130102545, the contents of which are herein incorporated by reference in their entirety). The active substance release system may comprise 1) at least one nanoparticle bonded to an oligonucleotide inhibitor strand which is hybridized with a catalytically active nucleic acid and 2) a compound bonded to at least one substrate molecule bonded to a therapeutically active substance (e.g., polynucleotides described herein), where the therapeutically active substance is released by the cleavage of the substrate molecule by the catalytically active nucleic acid.

In some embodiments, the RNA (e.g., mRNA) vaccines may be formulated in a nanoparticle comprising an inner core comprising a non-cellular material and an outer surface comprising a cellular membrane. The cellular membrane may be derived from a cell or a membrane derived from a virus. As a non-limiting example, the nanoparticle may be made by the methods described in International Patent Publication No. WO2013052167, the contents of which are herein incorporated by reference in their entirety. As another non-limiting example, the nanoparticle described in International Patent Publication No. WO2013052167, the contents of which are herein incorporated by reference in their entirety, may be used to deliver the RNA (e.g., mRNA) vaccines described herein.

In some embodiments, the RNA (e.g., mRNA) vaccines may be formulated in porous nanoparticle-supported lipid bilayers (protocells). Protocells are described in International Patent Publication No. WO2013056132, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the RNA (e.g., mRNA) vaccines described herein may be formulated in polymeric nanoparticles as described in or made by the methods described in U.S. Pat. Nos. 8,420,123 and 8,518,963 and European Patent No. EP2073848B1, the contents of each of which are herein incorporated by reference in their entirety. As a non-limiting example, the polymeric nanoparticle may have a high glass

transition temperature such as the nanoparticles described in or nanoparticles made by the methods described in U.S. Pat. No. 8,518,963, the contents of which are herein incorporated by reference in their entirety. As another non-limiting example, the polymer nanoparticle for oral and parenteral formulations may be made by the methods described in European Patent No. EP2073848B1, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the RNA (e.g., mRNA) vaccines described herein may be formulated in nanoparticles used in imaging. The nanoparticles may be liposome nanoparticles such as those described in U.S. Patent Publication No. US20130129636, herein incorporated by reference in its entirety. As a non-limiting example, the liposome may comprise gadolinium(III)2-[4,7-bis-carboxymethyl-10-[(N, N-distearylamidomethyl-N'-amido-methyl)-1,4,7,10-tetra-azacyclododec-1-yl]-acetic acid and a neutral, fully saturated phospholipid component (see, e.g., U.S. Patent Publication No. US20130129636, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the nanoparticles which may be used in the present disclosure are formed by the methods described in U.S. Patent Application No. US20130130348, the contents of which are herein incorporated by reference in their entirety.

The nanoparticles of the present disclosure may further include nutrients such as, but not limited to, those which deficiencies can lead to health hazards from anemia to neural tube defects (see, e.g., the nanoparticles described in International Patent Publication No. WO2013072929, the contents of which are herein incorporated by reference in their entirety). As a non-limiting example, the nutrient may be iron in the form of ferrous, ferric salts or elemental iron, iodine, folic acid, vitamins or micronutrients.

In some embodiments, the RNA (e.g., mRNA) vaccines of the present disclosure may be formulated in a swellable nanoparticle. The swellable nanoparticle may be, but is not limited to, those described in U.S. Pat. No. 8,440,231, the contents of which are herein incorporated by reference in their entirety. As a non-limiting embodiment, the swellable nanoparticle may be used for delivery of the RNA (e.g., mRNA) vaccines of the present disclosure to the pulmonary system (see, e.g., U.S. Pat. No. 8,440,231, the contents of which are herein incorporated by reference in their entirety).

The RNA (e.g., mRNA) vaccines of the present disclosure may be formulated in polyanhydride nanoparticles such as, but not limited to, those described in U.S. Pat. No. 8,449,916, the contents of which are herein incorporated by reference in their entirety.

The nanoparticles and microparticles of the present disclosure may be geometrically engineered to modulate macrophage and/or the immune response. In some embodiments, the geometrically engineered particles may have varied shapes, sizes and/or surface charges in order to incorporate the polynucleotides of the present disclosure for targeted delivery such as, but not limited to, pulmonary delivery (see, e.g., International Publication No. WO2013082111, the contents of which are herein incorporated by reference in their entirety). Other physical features the geometrically engineering particles may have include, but are not limited to, fenestrations, angled arms, asymmetry and surface roughness, charge which can alter the interactions with cells and tissues. As a non-limiting example, nanoparticles of the present disclosure may be made by the methods described in International Publication No. WO2013082111 the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the nanoparticles of the present disclosure may be water soluble nanoparticles such as, but not limited to, those described in International Publication No. WO2013090601, the contents of which are herein incorporated by reference in their entirety. The nanoparticles may be inorganic nanoparticles which have a compact and zwitterionic ligand in order to exhibit good water solubility. The nanoparticles may also have small hydrodynamic diameters (HD), stability with respect to time, pH, and salinity and a low level of non-specific protein binding.

In some embodiments the nanoparticles of the present disclosure may be developed by the methods described in U.S. Patent Publication No. US20130172406, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the nanoparticles of the present disclosure are stealth nanoparticles or target-specific stealth nanoparticles such as, but not limited to, those described in U.S. Patent Publication No. US20130172406, the contents of which are herein incorporated by reference in their entirety. The nanoparticles of the present disclosure may be made by the methods described in U.S. Patent Publication No. US20130172406, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the stealth or target-specific stealth nanoparticles may comprise a polymeric matrix. The polymeric matrix may comprise two or more polymers such as, but not limited to, polyethylenes, polycarbonates, polyanhydrides, polyhydroxyacids, polypropylfumerates, polycaprolactones, polyamides, polyacetals, polyethers, polyesters, poly(orthoesters), polycyanoacrylates, polyvinyl alcohols, polyurethanes, polyphosphazenes, polyacrylates, polymethacrylates, polycyanoacrylates, polyureas, polystyrenes, polyamines, polyesters, polyanhydrides, polyethers, polyurethanes, polymethacrylates, polyacrylates, polycyanoacrylates or combinations thereof.

In some embodiments, the nanoparticle may be a nanoparticle-nucleic acid hybrid structure having a high density nucleic acid layer. As a non-limiting example, the nanoparticle-nucleic acid hybrid structure may be made by the methods described in U.S. Patent Publication No. US20130171646, the contents of which are herein incorporated by reference in their entirety. The nanoparticle may comprise a nucleic acid such as, but not limited to, polynucleotides described herein and/or known in the art.

At least one of the nanoparticles of the present disclosure may be embedded in in the core a nanostructure or coated with a low density porous 3-D structure or coating which is capable of carrying or associating with at least one payload within or on the surface of the nanostructure. Non-limiting examples of the nanostructures comprising at least one nanoparticle are described in International Patent Publication No. WO2013123523, the contents of which are herein incorporated by reference in their entirety.

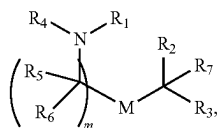
In some embodiments the RNA (e.g., mRNA) vaccine may be associated with a cationic or polycationic compounds, including protamine, nucleoline, spermine or spermidine, or other cationic peptides or proteins, such as poly-L-lysine (PLL), polyarginine, basic polypeptides, cell penetrating peptides (CPPs), including HIV-binding peptides, HIV-1 Tat (HIV), Tat-derived peptides, Penetratin, VP²² derived or analog peptides, Pestivirus Erns, HSV, VP²² (Herpes simplex), MAP, KALA or protein transduction domains (PTDs), PpT620, prolin-rich peptides, arginine-rich peptides, lysine-rich peptides, MPG-peptide(s), Pep-1, L-oligomers, Calcitonin peptide(s), Antennapedia-derived peptides (particularly from *Drosophila antennapedia*),

101

pAntp, pIsl, FGF, Lactoferrin, Transportan, Buforin-2, Bac715-24, SynB, SynB(1), pVEC, hCT-derived peptides, SAP, histones, cationic polysaccharides, for example chitosan, polybrene, cationic polymers, e.g. polyethyleneimine (PEI), cationic lipids, e.g. DOTMA: [1-(2,3-sioleyloxy) propyl]-N,N,N-trimethylammonium chloride, DMRIE, di-C14-amidine, DOTIM, SAINT, DC-Chol, BGTC, CTAP, DOPC, DODAP, DOPE: Dioleoyl phosphatidylethanolamine, DOSPA, DODAB, DOIC, DMEPC, DOGS: Dioctadecylamidoglycylspermin, DIMRI: Dimyristoxypropyl dimethyl hydroxyethyl ammonium bromide, DOTAP: diolcoyloxy-3-(trimethylammonio)propane, DC-6-14: O,O-ditetradecanoyl-N-.alpha.-trimethylammonioacetyl)diethanolamine chloride, CLIP 1: rac-[(2,3-dioctadecyloxypropyl) (2-hydroxyethyl)]-dimethylammonium chloride, CLIP6: rac-[2(2,3-dihexadecyloxypropyloxymethyloxy)ethyl]-trimethylammonium, CLIP9: rac-[2(2,3-dihexadecyloxypropyloxysuccinyloxy)ethyl]-trimethylammonium, oligo-fectamine, or cationic or polycationic polymers, e.g. modified polyaminoacids, such as beta-aminoacid-polymers or reversed polyamides, etc., modified polyethylenes, such as PVP (poly(N-ethyl-4-vinylpyridinium bromide)), etc., modified acrylates, such as pDMAEMA (poly(dimethylaminoethyl methylacrylate)), etc., modified amidoamines such as pAMAM (poly(amidoamine)), etc., modified polybetaminoester (PBAE), such as diamine end modified 1,4 butanediol diacrylate-co-5-amino-1-pentanol polymers, etc., dendrimers, such as polypropylamine dendrimers or pAMAM based dendrimers, etc., polyimine(s), such as PEI: poly(ethyleneimine), poly(propyleneimine), etc., polyallylamine, sugar backbone based polymers, such as cyclodextrin based polymers, dextran based polymers, chitosan, etc., silan backbone based polymers, such as PMOXA-PDMS copolymers, etc., blockpolymers consisting of a combination of one or more cationic blocks (e.g. selected from a cationic polymer as mentioned above) and of one or more hydrophilic or hydrophobic blocks (e.g. polyethyleneglycole), etc.

In other embodiments the RNA (e.g., mRNA) vaccine is not associated with a cationic or polycationic compounds.

In some embodiments, a nanoparticle comprises compounds of Formula (I):



or a salt or isomer thereof, wherein:

R₁ is selected from the group consisting of C₅₋₃₀alkyl, C₅₋₂₀ alkenyl, —R*YR", —YR", and —R"M'R";

R₂ and R₃ are independently selected from the group consisting of H, C₁₋₁₄ alkyl, C₂₋₁₄ alkenyl, —R*YR", —YR", and —R*OR", or R₂ and R₃, together with the atom to which they are attached, form a heterocycle or carbocycle;

R₄ is selected from the group consisting of a C₃₋₆ carbocycle, —(CH₂)_nQ, —(CH₂)_nCHQR, —CHQR, —CQ(R)₂, and unsubstituted C₁₋₆ alkyl, where Q is selected from a carbocycle, heterocycle, —OR, —O(CH₂)_nN(R)₂, —C(O)OR, —OC(O)R, —CX₃, —CX₂H, —CXH₂, —CN, —N(R)₂, —C(O)N(R)₂, —N(R)C(O)R, —N(R)S(O)₂R, —N(R)C(O)N(R)₂, —N(R)C(S)N(R)₂, —N(R)R₈, —O(CH₂)_nOR, —N(R)

102

C(=NR₉)N(R)₂, —N(R)C(=CHR₉)N(R)₂, —OC(O)N(R)₂, —N(R)C(O)OR, —N(OR)C(O)R, —N(OR)S(O)₂R, —N(OR)C(O)OR, —N(OR)C(O)N(R)₂, —N(OR)C(S)N(R)₂, —N(OR)C(=NR₉)N(R)₂, —N(OR)C(=CHR₉)N(R)₂, —C(=NR₉)N(R)₂, —C(=NR₉)R, —C(O)N(R)O R, and C(R)N(R)₂C(O)OR, and each n is independently selected from 1, 2, 3, 4, and 5;

each R₅ is independently selected from the group consisting of C1-3 alkyl, C2-3 alkenyl, and H;

each R₆ is independently selected from the group consisting of C1-3 alkyl, C₂₋₃ alkenyl, and H;

M and M' are independently selected from —C(O)O—, —OC(O)—, —C(O)N(R')—, —N(R')C(O)—, —C(O)—, —C(S)—, —C(S)S—, —SC(S)—, —CH(OH)—, —P(O)(OR')O—, —S(O)₂—, —S—S—, an aryl group, and a heteroaryl group;

R₇ is selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

R₈ is selected from the group consisting of C3-6 carbocycle and heterocycle;

R₉ is selected from the group consisting of H, CN, NO₂, C₁₋₆ alkyl, —OR, —S(O)₂R, —S(O)₂N(R)₂, C₂₋₆ alkenyl, C₃₋₆ carbocycle and heterocycle;

each R is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

each R' is independently selected from the group consisting of C₁₋₁₁ alkyl, C₂₋₁₈ alkenyl, —R*YR", —YR", and H;

each R" is independently selected from the group consisting of C₃₋₁₄ alkyl and C₃₋₁₄ alkenyl;

each R* is independently selected from the group consisting of C₁₋₁₂ alkyl and C₂₋₁₂ alkenyl;

each Y is independently a C₃₋₆ carbocycle;

each X is independently selected from the group consisting of F, Cl, Br, and I; and m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13.

In some embodiments, a subset of compounds of Formula (I) includes those in which when R₄ is —(CH₂)_nQ, —(CH₂)_nCHQR, —CHQR, or —CQ(R)₂, then (i) Q is not —N(R)₂ when n is 1, 2, 3, 4 or 5, or (ii) Q is not 5, 6, or 7-membered heterocycloalkyl when n is 1 or 2.

In some embodiments, another subset of compounds of Formula (I) includes those in which

(I) R₁ is selected from the group consisting of C₅₋₃₀alkyl, C₅₋₂₀ alkenyl, —R*YR", —YR", and —R"M'R";

R₂ and R₃ are independently selected from the group consisting of H, C₁₋₁₄ alkyl, C₂₋₁₄ alkenyl, —R*YR", —YR", and —R*OR", or R₂ and R₃, together with the atom to which they are attached, form a heterocycle or carbocycle;

R₄ is selected from the group consisting of a C₃₋₆ carbocycle, —(CH₂)_nQ, —(CH₂)_nCHQR, —CHQR, —CQ(R)₂, and unsubstituted C₁₋₆ alkyl, where Q is selected from a C₃₋₆ carbocycle, a 5- to 14-membered heteroaryl having one or more heteroatoms selected from N, O, and S, —OR, —O(CH₂)_nN(R)₂, —C(O)OR, —OC(O)R, —CX₃, —CX₂H, —CXH₂, —CN, —C(O)N(R)₂, —N(R)C(O)R, —N(R)S(O)₂R, —N(R)C(O)N(R)₂, —N(R)C(S)N(R)₂, —CRN(R)₂C(O)OR, —N(R)R₈, —O(CH₂)_nOR, —N(R)C(=NR₉)N(R)₂, —N(R)C(=CHR₉)N(R)₂, —OC(O)N(R)₂, —N(R)C(O)OR, —N(OR)C(O)R, —N(OR)S(O)₂R, —N(OR)C(O)OR, —N(OR)C(O)N(R)₂, —N(OR)C(S)N(R)₂, —N(OR)C(=NR₉)N(R)₂, —N(OR)C(=CHR₉)N(R)₂, —C(=NR₉)N(R)₂, —C(=NR₉)R, —C(O)N(R)O R, and a 5- to 14-membered heterocycloalkyl having one or

103

more heteroatoms selected from N, O, and S which is substituted with one or more substituents selected from oxo (=O), OH, amino, mono- or di-alkylamino, and C₁₋₃ alkyl, and each n is independently selected from 1, 2, 3, 4, and 5;

each R₅ is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

each R₆ is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

M and M' are independently selected from —C(O)O—, —OC(O)—, —C(O)N(R')—, —N(R')C(O)—, —C(O)—, —C(S)—, —C(S)S—, —SC(S)—, —CH(OH)—, —P(O)(OR')O—, —S(O)₂—, —S—S—, an aryl group, and a heteroaryl group;

R₇ is selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

R₈ is selected from the group consisting of C₃₋₆ carbocycle and heterocycle;

R₉ is selected from the group consisting of H, CN, NO₂, C₁₋₆ alkyl, —OR, —S(O)₂R, —S(O)₂N(R)₂, C₂₋₆ alkenyl, C₃₋₆ carbocycle and heterocycle;

each R is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

each R' is independently selected from the group consisting of C₁₋₁₈ alkyl, C₂₋₁₈ alkenyl, —R*YR", —YR", and H;

each R" is independently selected from the group consisting of C₃₋₁₄ alkyl and C₃₋₁₄ alkenyl;

each R* is independently selected from the group consisting of C₁₋₁₂ alkyl and C₂₋₁₂ alkenyl;

each Y is independently a C₃₋₆ carbocycle;

each X is independently selected from the group consisting of F, Cl, Br, and I; and

m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13, or salts or isomers thereof.

In some embodiments, another subset of compounds of Formula (I) includes those in which

R₁ is selected from the group consisting of C₅₋₃₀ alkyl, C₅₋₂₀ alkenyl, —R*YR", —YR", and —R"M'R';

R₂ and R₃ are independently selected from the group consisting of H, C₁₋₁₄ alkyl, C₂₋₁₄ alkenyl, —R*YR", —YR", and —R*OR", or R₂ and R₃, together with the atom to which they are attached, form a heterocycle or carbocycle;

R₄ is selected from the group consisting of a C₃₋₆ carbocycle, —(CH₂)_nQ, —(CH₂)_nCHQR, —CHQR, —CQ(R)₂, and unsubstituted C₁₋₆ alkyl, where Q is selected from a C₃₋₆ carbocycle, a 5- to 14-membered heterocycle having one or more heteroatoms selected from N, O, and S, —OR,

—O(CH₂)_nN(R)₂, —C(O)OR, —OC(O)R, —CX₃, —CX₂H, —CXH₂, —CN, —C(O)N(R)₂, —N(R)C(O)R, —N(R)S(O)₂R, —N(R)C(O)N(R)₂, —N(R)C(S)N(R)₂, —CRN(R)₂C(O)OR, —N(R)R₈, —O(CH₂)_nOR, —N(R)C(=NR₉)N(R)₂, —N(R)C(=CHR₉)N(R)₂, —OC(O)N(R)₂, —N(R)C(O)OR, —N(OR)C(O)R, —N(OR)S(O)₂R, —N(OR)C(O)OR, —N(OR)C(O)N(R)₂, —N(OR)C(S)N(R)₂, —N(OR)C(=NR₉)N(R)₂, —N(OR)C(=CHR₉)N(R)₂, —C(=NR₉)R, —C(O)N(R)OR, and —C(=NR₉)N(R)₂, and each n is independently selected from 1, 2, 3, 4, and 5; and when Q is a 5- to 14-membered heterocycle and

(i) R₄ is —(CH₂)_nQ in which n is 1 or 2, or (ii) R₄ is —(CH₂)_nCHQR in which n is 1, or (iii) R₄ is —CHQR, and —CQ(R)₂, then Q is either a 5- to 14-membered heteroaryl or 8- to 14-membered heterocycloalkyl;

each R₅ is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

104

each R₆ is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

M and M' are independently selected from —C(O)O—, —OC(O)—, —C(O)N(R')—, —N(R')C(O)—, —C(O)—, —C(S)—, —C(S)S—, —SC(S)—, —CH(OH)—, —P(O)(OR')O—, —S(O)₂—, —S—S—, an aryl group, and a heteroaryl group;

R₇ is selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

R₈ is selected from the group consisting of C₃₋₆ carbocycle and heterocycle;

R₉ is selected from the group consisting of H, CN, NO₂, C₁₋₆ alkyl, —OR, —S(O)₂R, —S(O)₂N(R)₂, C₂₋₆ alkenyl, C₃₋₆ carbocycle and heterocycle;

each R is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

each R' is independently selected from the group consisting of C₁₋₁₈ alkyl, C₂₋₁₈ alkenyl, —R*YR", —YR", and H;

each R" is independently selected from the group consisting of C₃₋₁₄ alkyl and C₃₋₁₄ alkenyl;

each R* is independently selected from the group consisting of C₁₋₁₂ alkyl and C₂₋₁₂ alkenyl;

each Y is independently a C₃₋₆ carbocycle;

each X is independently selected from the group consisting of F, Cl, Br, and I; and

m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13, or salts or isomers thereof.

In some embodiments, another subset of compounds of Formula (I) includes those in which

R₁ is selected from the group consisting of C₅₋₃₀ alkyl, C₅₋₂₀ alkenyl, —R*YR", —YR", and —R"M'R';

R₂ and R₃ are independently selected from the group consisting of H, C₁₋₁₄ alkyl, C₂₋₁₄ alkenyl, —R*YR", —YR", and —R*OR", or R₂ and R₃, together with the atom to which they are attached, form a heterocycle or carbocycle;

R₄ is selected from the group consisting of a C₃₋₆ carbocycle, —(CH₂)_nQ, —(CH₂)_nCHQR, —CHQR, —CQ(R)₂, and unsubstituted C₁₋₆ alkyl, where Q is selected from a C₃₋₆ carbocycle, a 5- to 14-membered heteroaryl having one or more heteroatoms selected from N, O, and S, —OR,

—O(CH₂)_nN(R)₂, —C(O)OR, —OC(O)R, —CX₃, —CX₂H, —CXH₂, —CN, —C(O)N(R)₂, —N(R)C(O)R, —N(R)S(O)₂R, —N(R)C(O)N(R)₂, —N(R)C(S)N(R)₂, —CRN(R)₂C(O)OR, —N(R)R₈, —O(CH₂)_nOR, —N(R)C(=NR₉)N(R)₂, —N(R)C(=CHR₉)N(R)₂, —OC(O)N(R)₂, —N(R)C(O)OR, —N(OR)C(O)R, —N(OR)S(O)₂R, —N(OR)C(O)OR, —N(OR)C(O)N(R)₂, —N(OR)C(S)N(R)₂, —N(OR)C(=NR₉)N(R)₂, —N(OR)C(=CHR₉)N(R)₂, —C(=NR₉)R, —C(O)N(R)OR, and —C(=NR₉)N(R)₂, and each n is independently selected from 1, 2, 3, 4, and 5;

each R₅ is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

each R₆ is independently selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

M and M' are independently selected from —C(O)O—, —OC(O)—, —C(O)N(R')—, —N(R')C(O)—, —C(O)—, —C(S)—, —C(S)S—, —SC(S)—, —CH(OH)—, —P(O)(OR')O—, —S(O)₂—, —S—S—, an aryl group, and a heteroaryl group;

R₇ is selected from the group consisting of C₁₋₃ alkyl, C₂₋₃ alkenyl, and H;

R₈ is selected from the group consisting of C₃₋₆ carbocycle and heterocycle;

105

R_9 is selected from the group consisting of H, CN, NO_2 , C_{1-6} alkyl, $-\text{OR}$, $-\text{S}(\text{O})_2\text{R}$, $-\text{S}(\text{O})_2\text{N}(\text{R})_2$, C_{2-6} alkenyl, C_{3-6} carbocycle and heterocycle; each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H; each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and H; each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl; each R^* is independently selected from the group consisting of C_{1-12} alkyl and C_{2-12} alkenyl; each Y is independently a C_{3-6} carbocycle; each X is independently selected from the group consisting of F, Cl, Br, and I; and m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13, or salts or isomers thereof.

In some embodiments, another subset of compounds of Formula (I) includes those in which

R_1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}''\text{M}'\text{R}'$; R_2 and R_3 are independently selected from the group consisting of H, C_{2-14} alkyl, C_{2-14} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}^*\text{OR}''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;

R_4 is $-(\text{CH}_2)_n\text{Q}$ or $-(\text{CH}_2)_n\text{CHQR}$, where Q is $-\text{N}(\text{R})_2$, and n is selected from 3, 4, and 5;

each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

M and M' are independently selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-\text{C}(\text{O})\text{N}(\text{R}')-$, $-\text{N}(\text{R}')\text{C}(\text{O})-$, $-\text{C}(\text{O})-$, $-\text{C}(\text{S})-$, $-\text{C}(\text{S})\text{S}-$, $-\text{SC}(\text{S})-$, $-\text{CH}(\text{OH})-$, $-\text{P}(\text{O})(\text{OR}')\text{O}-$, $-\text{S}(\text{O})_2-$, $-\text{S}-\text{S}-$, an aryl group, and a heteroaryl group;

R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and H;

each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl;

each R^* is independently selected from the group consisting of C_{1-12} alkyl and C_{1-12} alkenyl;

each Y is independently a C_{3-6} carbocycle;

each X is independently selected from the group consisting of F, Cl, Br, and I; and

m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13, or salts or isomers thereof.

In some embodiments, another subset of compounds of Formula (I) includes those in which

R_1 is selected from the group consisting of C_{5-30} alkyl, C_{5-20} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}''\text{M}'\text{R}'$;

R_2 and R_3 are independently selected from the group consisting of C_{1-14} alkyl, C_{2-14} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and $-\text{R}^*\text{OR}''$, or R_2 and R_3 , together with the atom to which they are attached, form a heterocycle or carbocycle;

R_4 is selected from the group consisting of $-(\text{CH}_2)_n\text{Q}$, $-(\text{CH}_2)_n\text{CHQR}$, $-\text{CHQR}$, and $-\text{CQ}(\text{R})_2$, where Q is $-\text{N}(\text{R})_2$, and n is selected from 1, 2, 3, 4, and 5;

each R_5 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

106

each R_6 is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

M and M' are independently selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-\text{C}(\text{O})\text{N}(\text{R}')-$, $-\text{N}(\text{R}')\text{C}(\text{O})-$, $-\text{C}(\text{O})-$, $-\text{C}(\text{S})-$, $-\text{C}(\text{S})\text{S}-$, $-\text{SC}(\text{S})-$, $-\text{CH}(\text{OH})-$, $-\text{P}(\text{O})(\text{OR}')\text{O}-$, $-\text{S}(\text{O})_2-$, $-\text{S}-\text{S}-$, an aryl group, and a heteroaryl group;

R_7 is selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

each R is independently selected from the group consisting of C_{1-3} alkyl, C_{2-3} alkenyl, and H;

each R' is independently selected from the group consisting of C_{1-18} alkyl, C_{2-18} alkenyl, $-\text{R}^*\text{YR}''$, $-\text{YR}''$, and H;

each R'' is independently selected from the group consisting of C_{3-14} alkyl and C_{3-14} alkenyl;

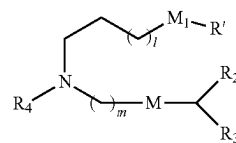
each R^* is independently selected from the group consisting of C_{1-12} alkyl and C_{1-12} alkenyl;

each Y is independently a C_{3-6} carbocycle;

each X is independently selected from the group consisting of F, Cl, Br, and I; and

m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13, or salts or isomers thereof.

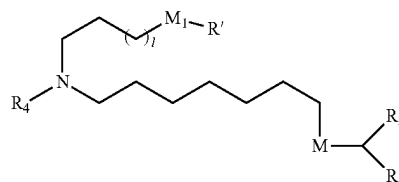
In some embodiments, a subset of compounds of Formula (I) includes those of Formula (IA):



(IA)

or a salt or isomer thereof, wherein l is selected from 1, 2, 3, 4, and 5; m is selected from 5, 6, 7, 8, and 9; M_1 is a bond or M' ; R_4 is unsubstituted C_{1-3} alkyl, or $-(\text{CH}_2)_n\text{Q}$, in which Q is OH, $-\text{NHC}(\text{S})\text{N}(\text{R})_2$, $-\text{NHC}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{R}$, $-\text{N}(\text{R})\text{S}(\text{O})_2\text{R}$, $-\text{N}(\text{R})\text{R}_8$, $-\text{NHC}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{NHC}(=\text{CHR}_9)\text{N}(\text{R})_2$, $-\text{OC}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{OR}$, heteroaryl or heterocycloalkyl; M and M' are independently selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-\text{C}(\text{O})\text{N}(\text{R}')-$, $-\text{P}(\text{O})(\text{OR}')\text{O}-$, $-\text{S}-\text{S}-$, an aryl group, and a heteroaryl group; and R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, and C_{2-14} alkenyl.

In some embodiments, a subset of compounds of Formula (I) includes those of Formula (II):



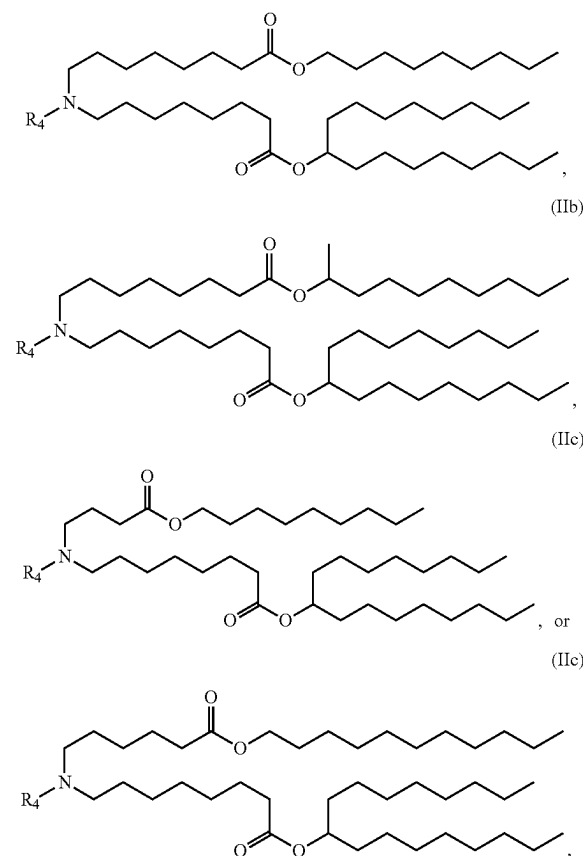
(II)

or a salt or isomer thereof, wherein l is selected from 1, 2, 3, 4, and 5; M_1 is a bond or M' ; R_4 is unsubstituted C_{1-3} alkyl, or $-(\text{CH}_2)_n\text{Q}$, in which n is 2, 3, or 4, and Q is OH, $-\text{NHC}(\text{S})\text{N}(\text{R})_2$, $-\text{NHC}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{R}$, $-\text{N}(\text{R})\text{S}(\text{O})_2\text{R}$, $-\text{N}(\text{R})\text{R}_8$, $-\text{NHC}(=\text{NR}_9)\text{N}(\text{R})_2$, $-\text{NHC}(=\text{CHR}_9)\text{N}(\text{R})_2$, $-\text{OC}(\text{O})\text{N}(\text{R})_2$, $-\text{N}(\text{R})\text{C}(\text{O})\text{OR}$, heteroaryl or heterocycloalkyl; M and M' are independently selected

107

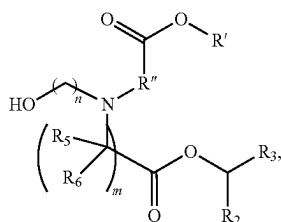
from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, $-\text{C}(\text{O})\text{N}(\text{R}')-$, $-\text{P}(\text{O})(\text{OR}')\text{O}-$, $-\text{S}-\text{S}-$, an aryl group, and a heteroaryl group; and R_2 and R_3 are independently selected from the group consisting of H, C_{1-14} alkyl, and C_{2-14} alkenyl.

In some embodiments, a subset of compounds of Formula (I) includes those of Formula (IIa), (IIb), (IIc), or (IIe):



or a salt or isomer thereof, wherein R_4 is as described herein.

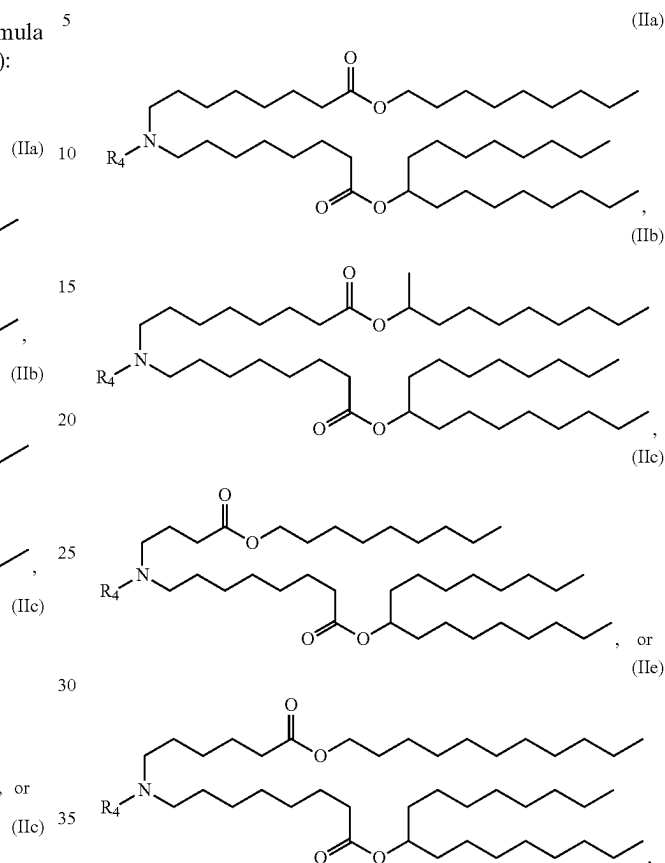
In some embodiments, a subset of compounds of Formula (T) includes those of Formula (IIId):



or a salt or isomer thereof, wherein n is 2, 3, or 4; and m , R' , R'' , and R_2 through R_6 are as described herein. For example, each of R_2 and R_3 may be independently selected from the group consisting of C_{5-14} alkyl and C_{5-14} alkenyl.

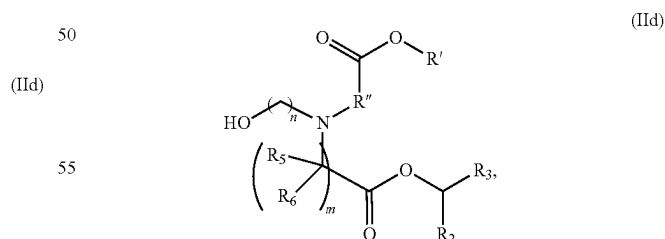
108

In some embodiments, a subset of compounds of Formula (I) includes those of Formula (IIa), (IIb), (IIc), or (IIe):



or a salt or isomer thereof, wherein R_4 is as described herein.

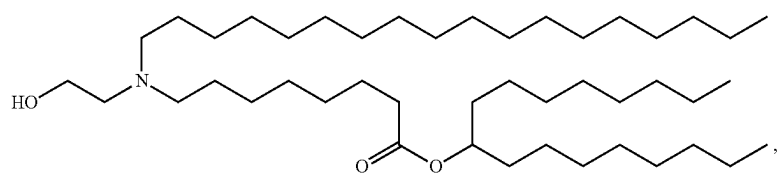
In some embodiments, a subset of compounds of Formula (I) includes those of Formula (IIId):



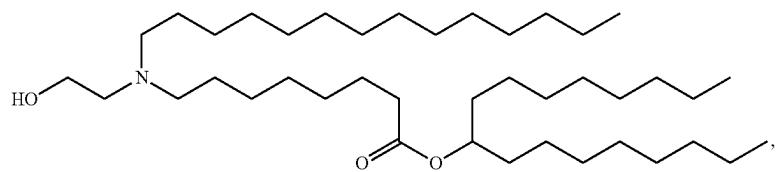
or a salt or isomer thereof, wherein n is 2, 3, or 4; and m , R' , R'' , and R_2 through R_6 are as described herein. For example, each of R_2 and R_3 may be independently selected from the group consisting of C_{5-14} alkyl and C_{5-14} alkenyl.

In some embodiments, the compound of Formula (I) is selected from the group consisting of:

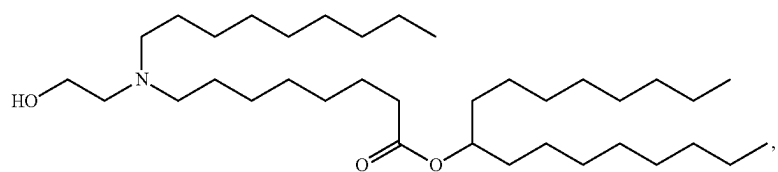
109



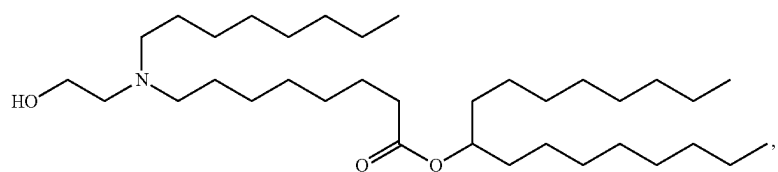
(Compound 1)



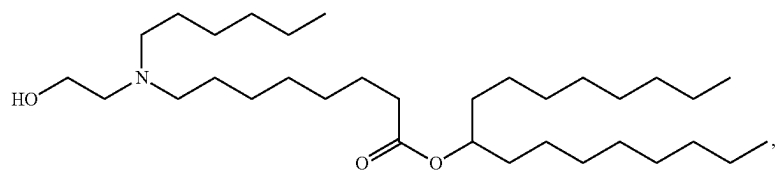
(Compound 2)



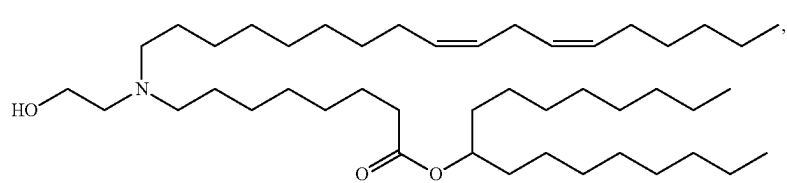
(Compound 3)



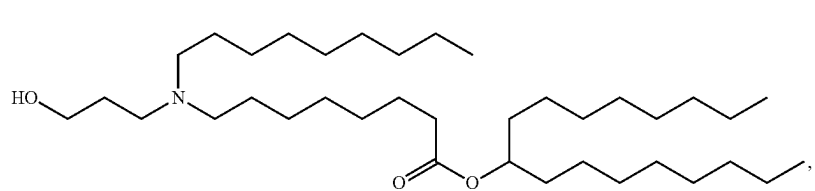
(Compound 4)



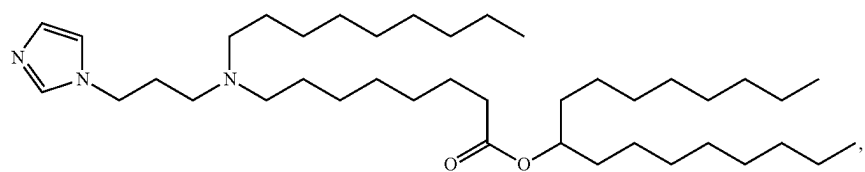
(Compound 5)



(Compound 6)



(Compound 7)



(Compound 8)

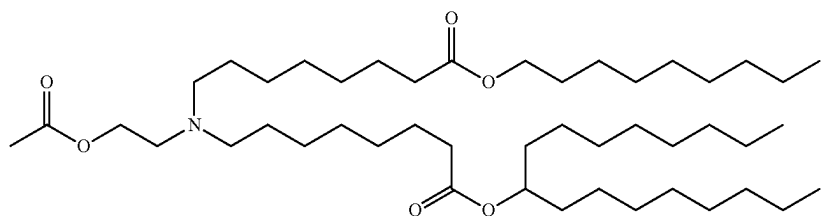
110

111

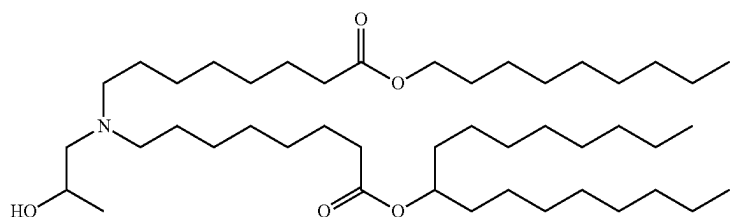
112

-continued

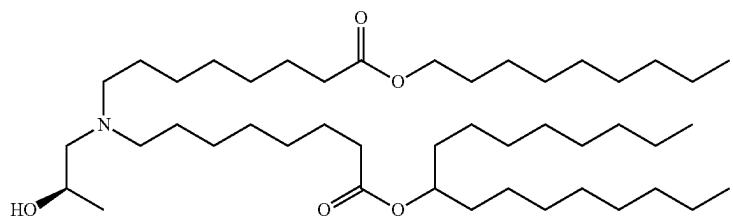
(Compound 9)



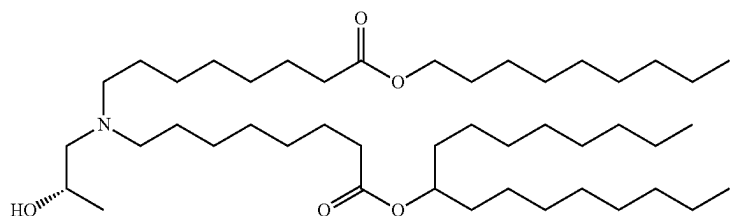
(Compound 10)



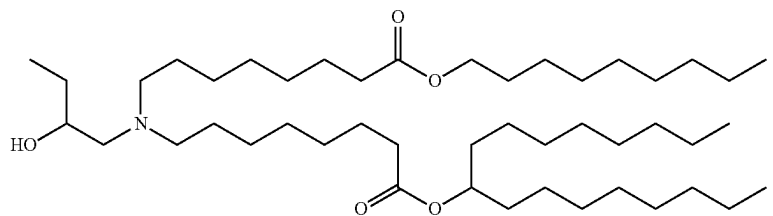
(Compound 11)



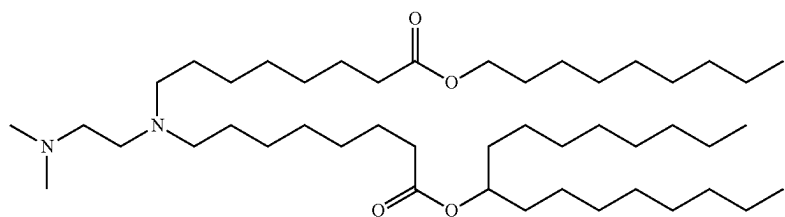
(Compound 12)



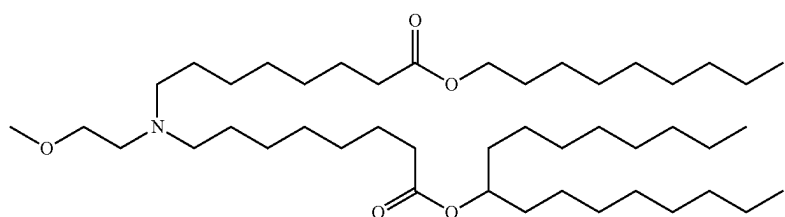
(Compound 13)



(Compound 14)



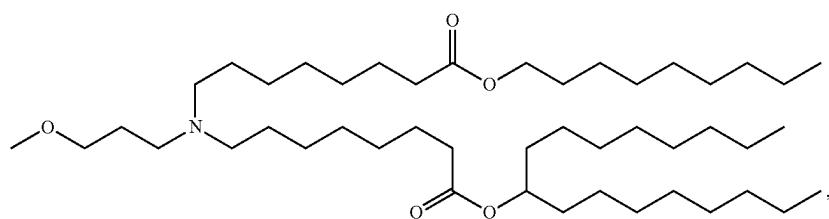
(Compound 15)



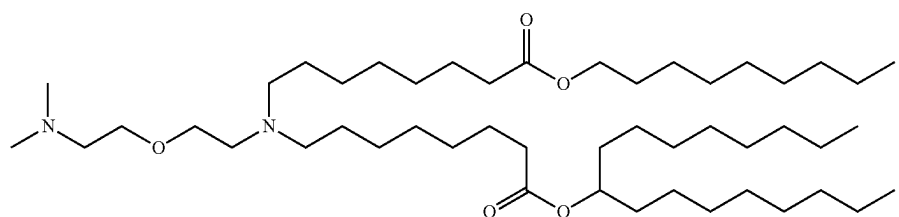
113

114

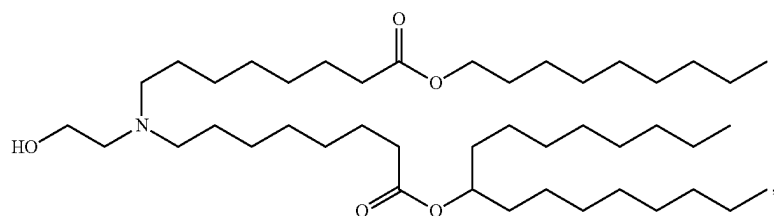
-continued



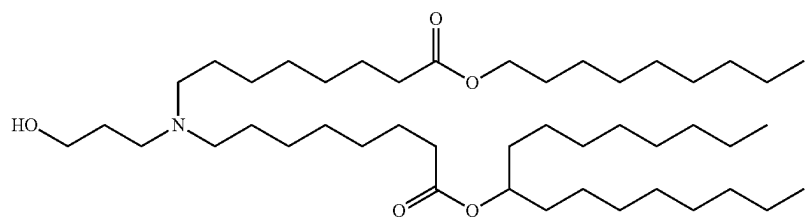
(Compound 16)



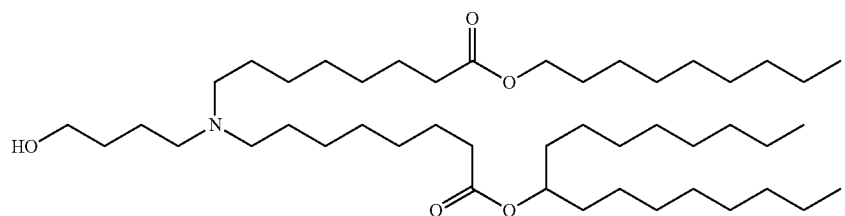
(Compound 17)



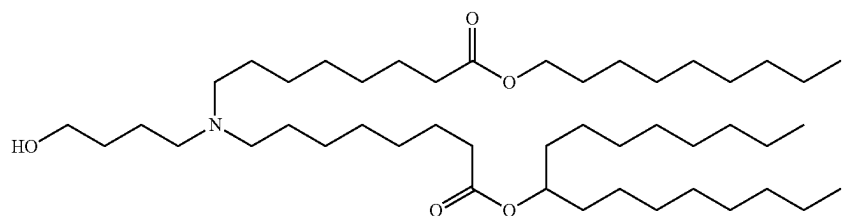
(Compound 18)



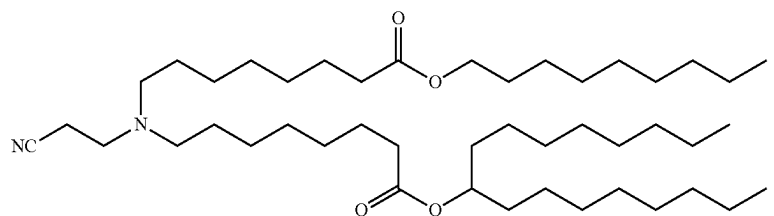
(Compound 19)



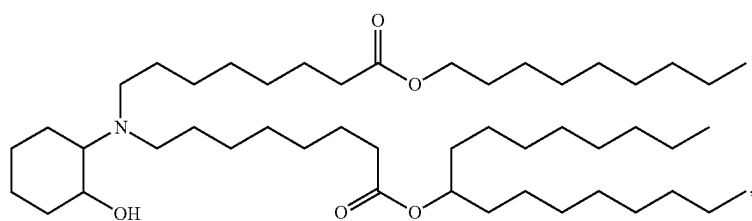
(Compound 20)



(Compound 21)



(Compound 22)

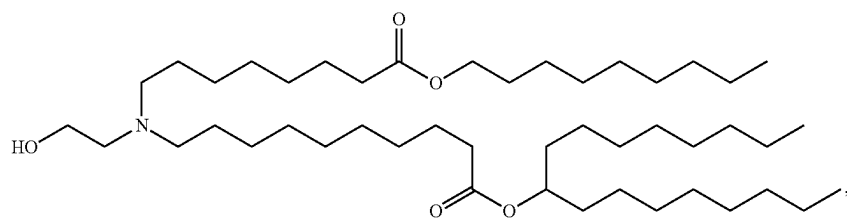


115

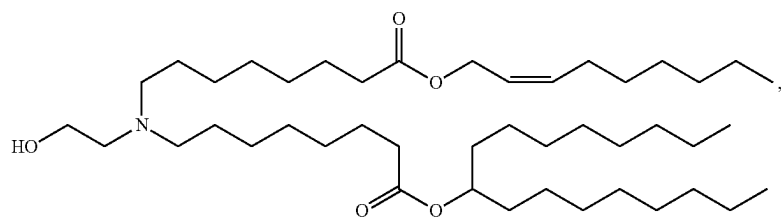
-continued

116

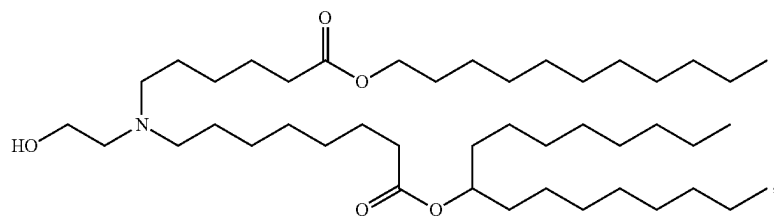
(Compound 23)



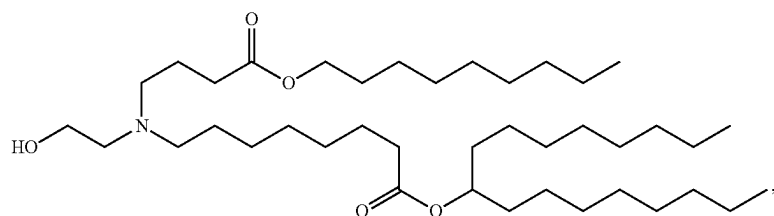
(Compound 24)



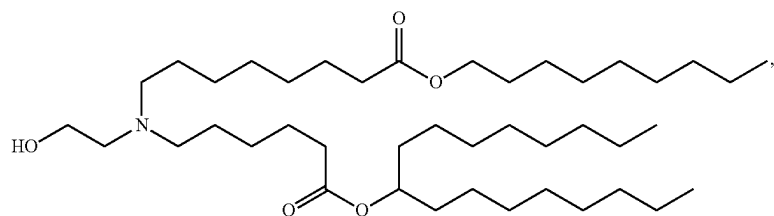
(Compound 25)



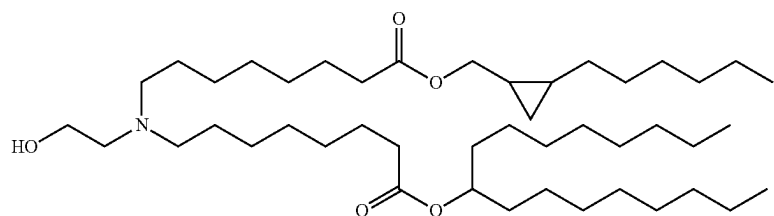
(Compound 26)



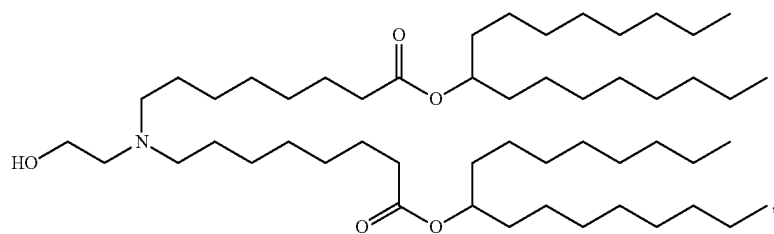
(Compound 27)



(Compound 28)



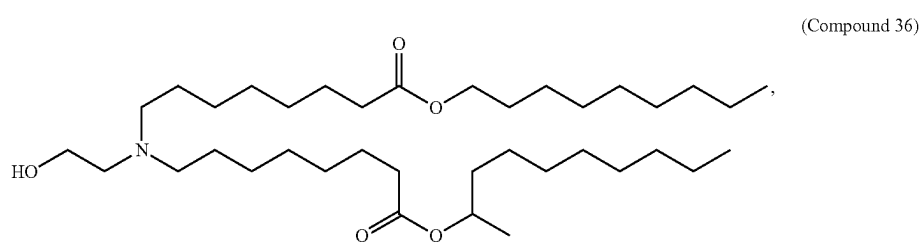
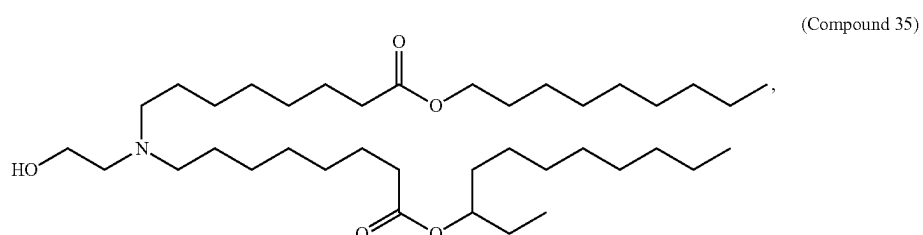
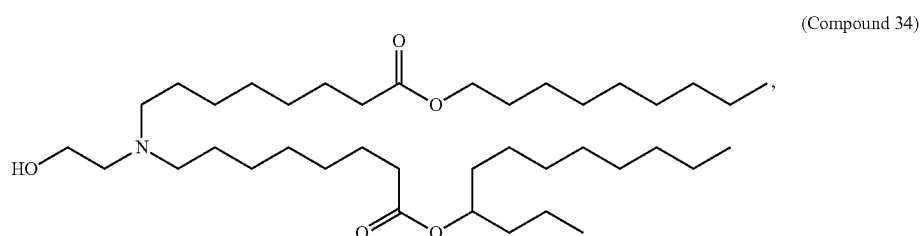
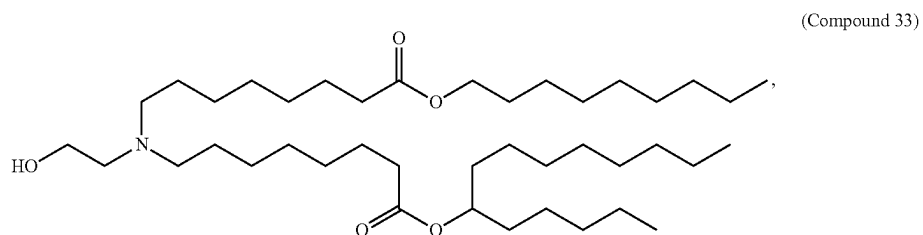
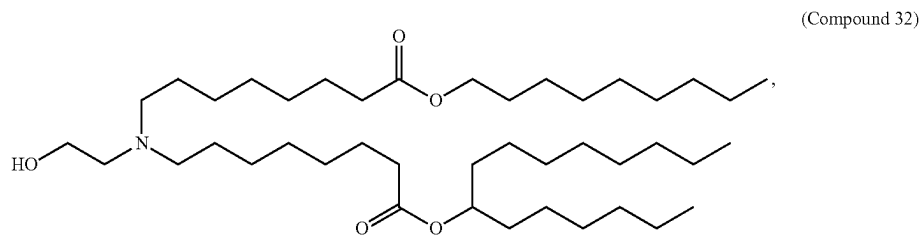
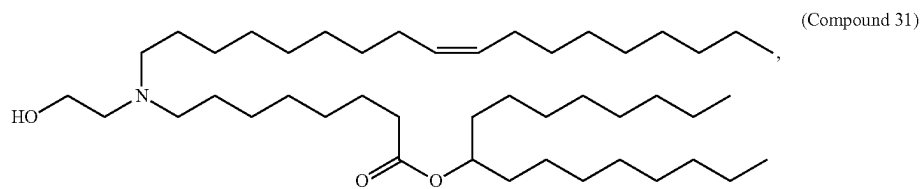
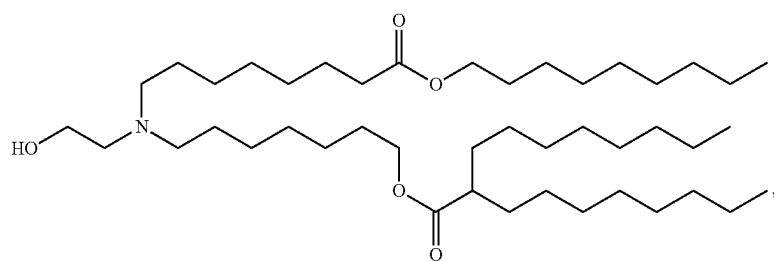
(Compound 29)



117

118

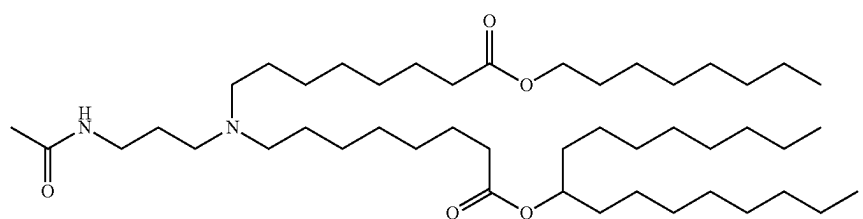
-continued



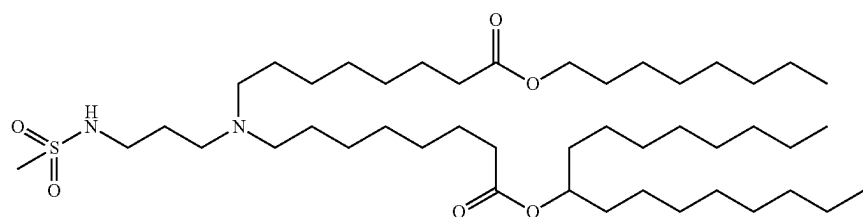
119

120

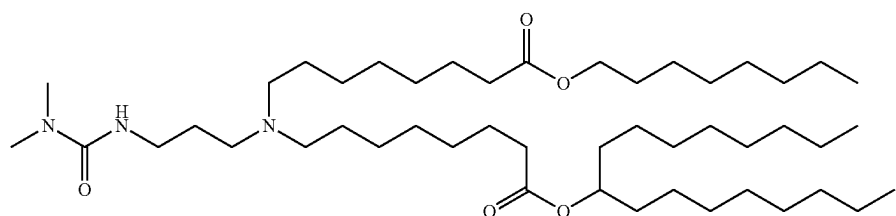
-continued



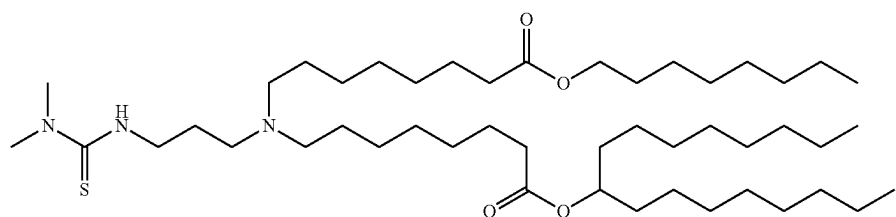
(Compound 37)



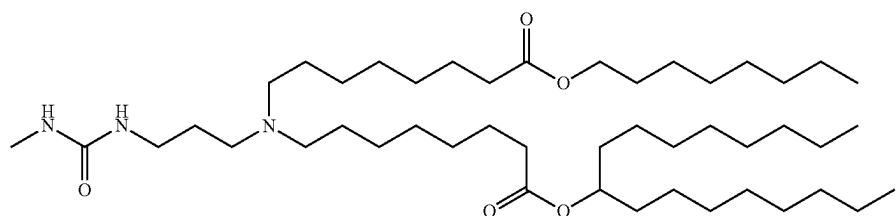
(Compound 38)



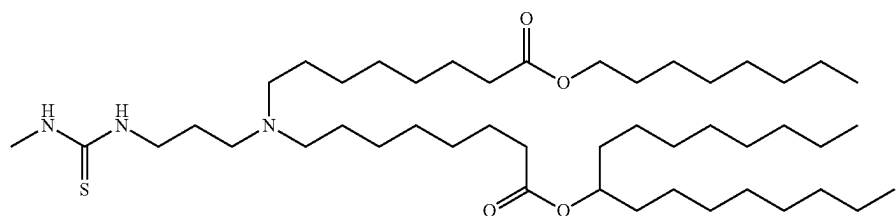
(Compound 39)



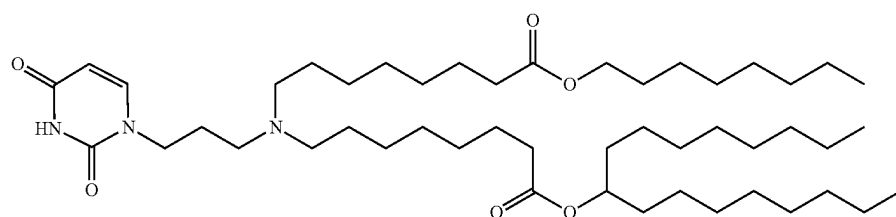
(Compound 40)



(Compound 41)

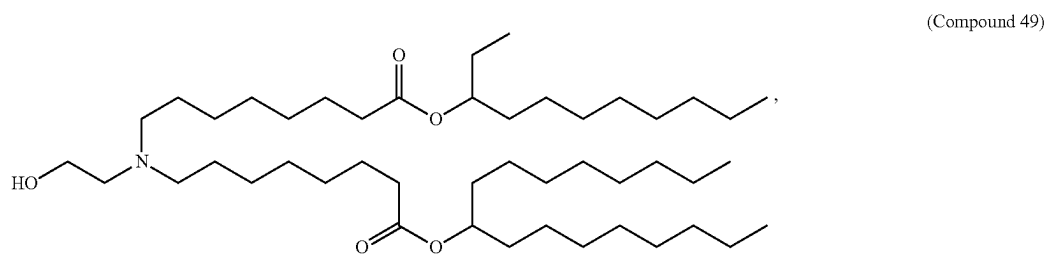
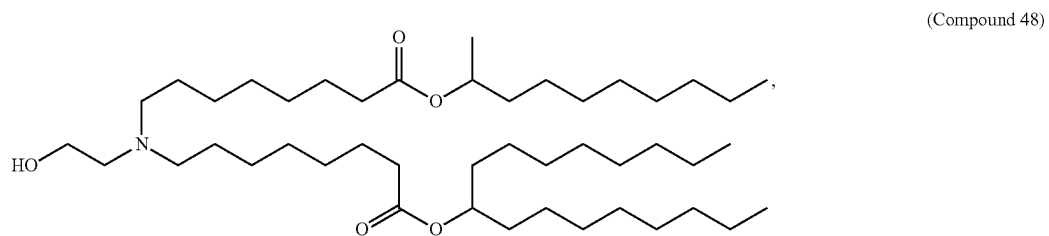
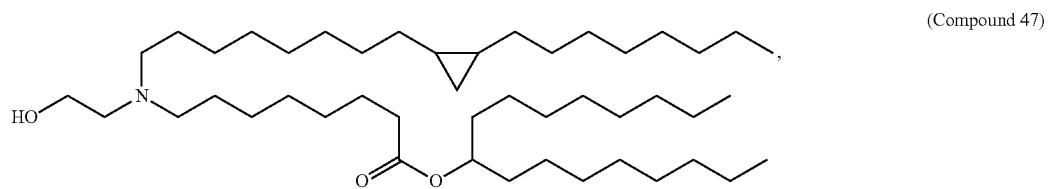
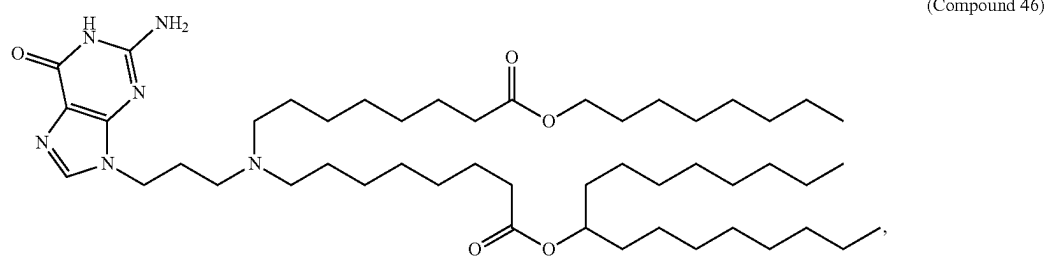
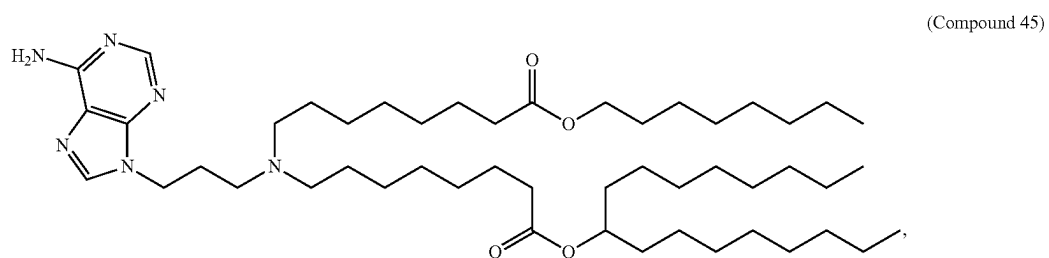
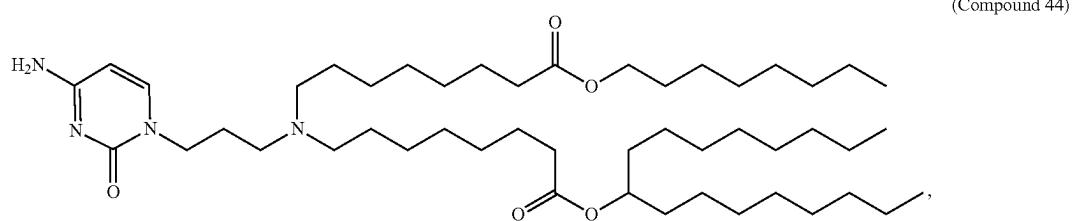


(Compound 42)



(Compound 43)

-continued

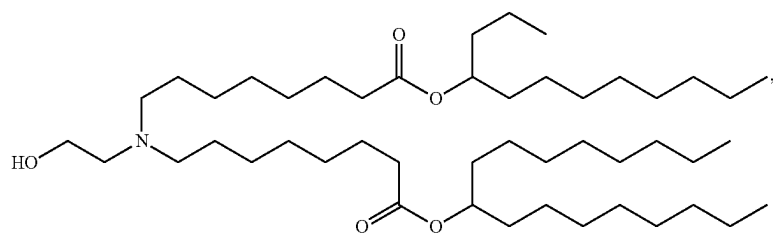


123

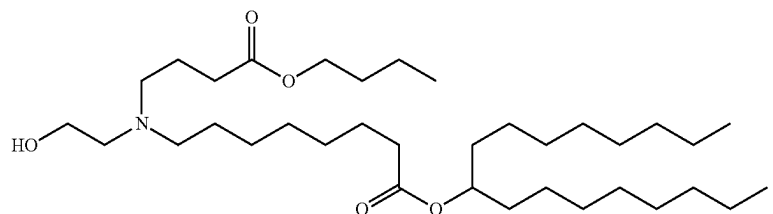
124

-continued

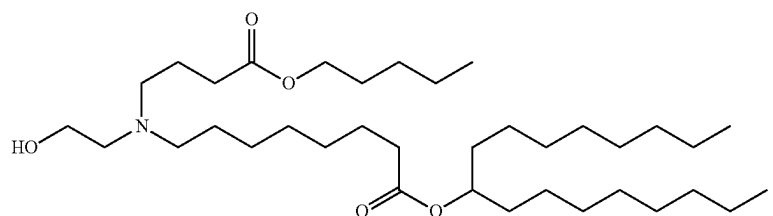
(Compound 50)



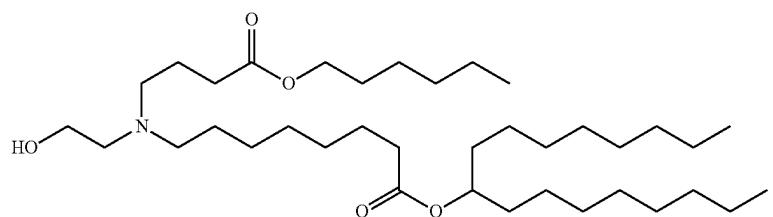
(Compound 51)



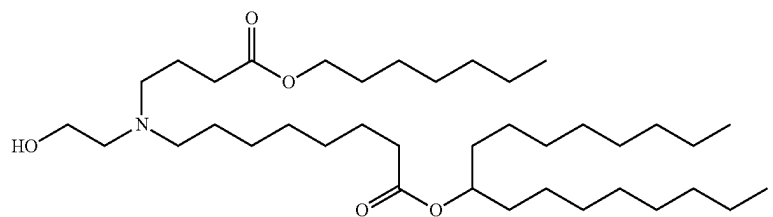
(Compound 52)



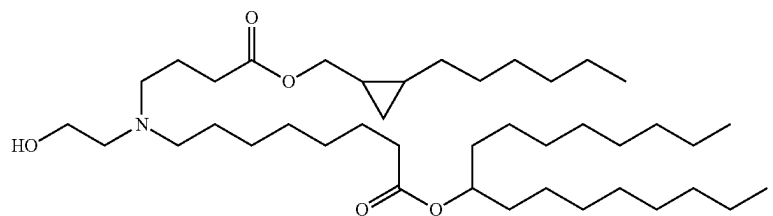
(Compound 53)



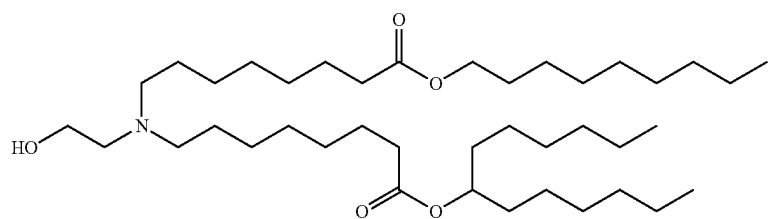
(Compound 54)



(Compound 55)



(Compound 56)

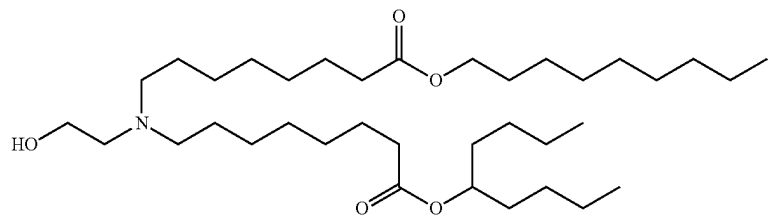


125

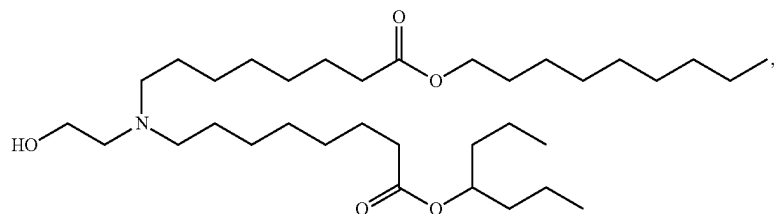
126

-continued

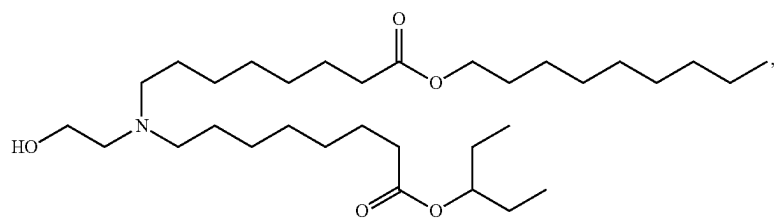
(Compound 57)



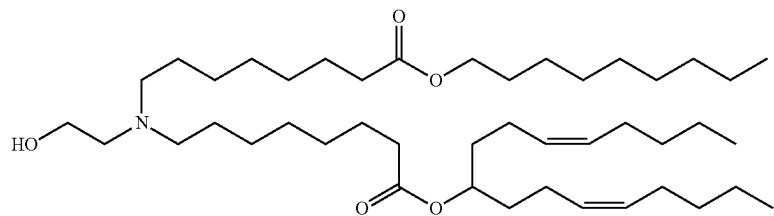
(Compound 58)



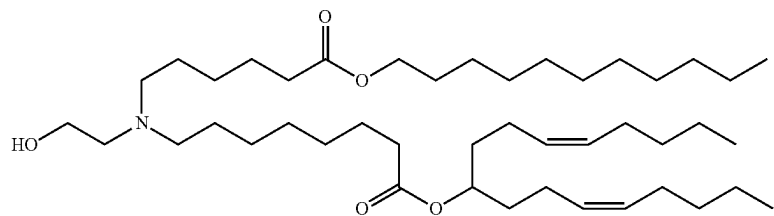
(Compound 59)



(Compound 60)

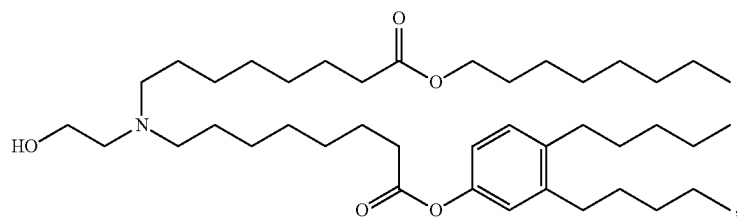


(Compound 61)



In further embodiments, the compound of Formula (I) is selected from the group consisting of:

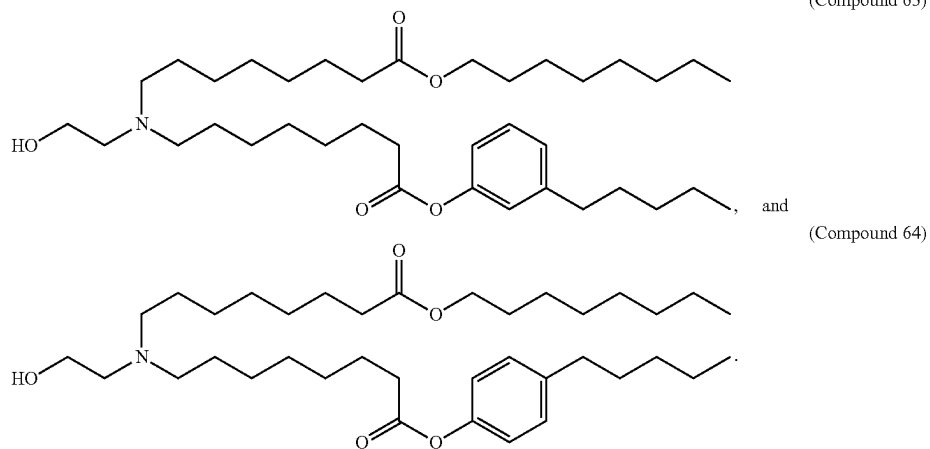
(Compound 62)



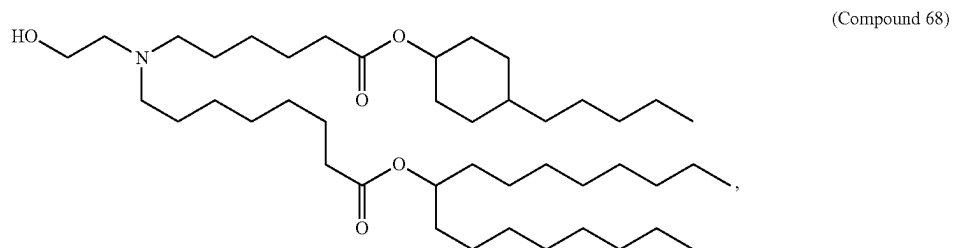
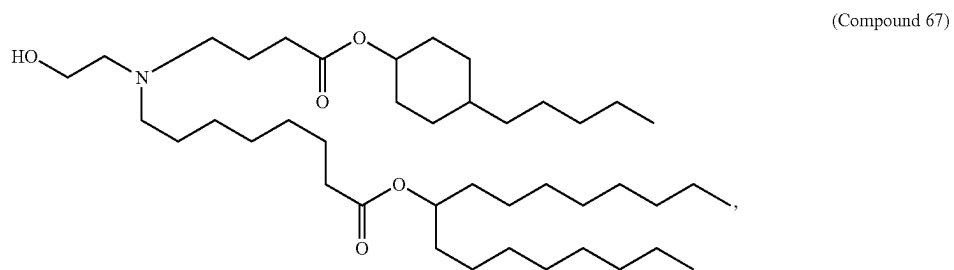
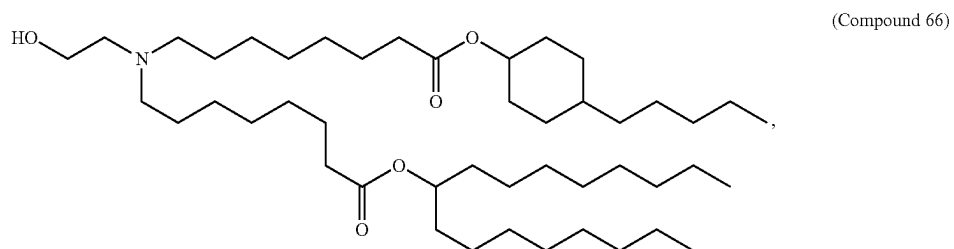
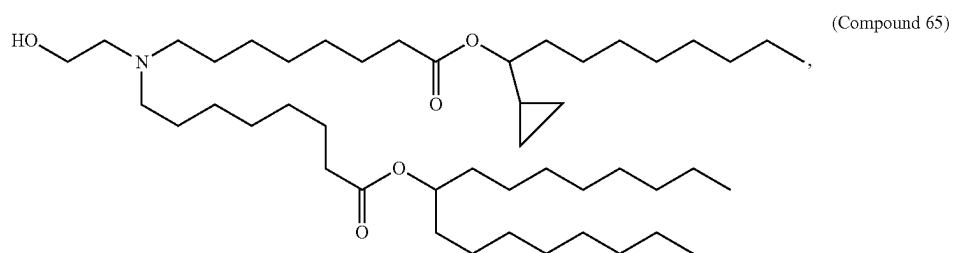
127

128

-continued



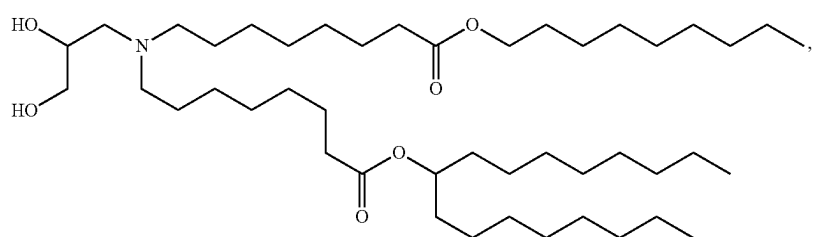
In some embodiments, the compound of Formula (I) is selected from the group consisting of:



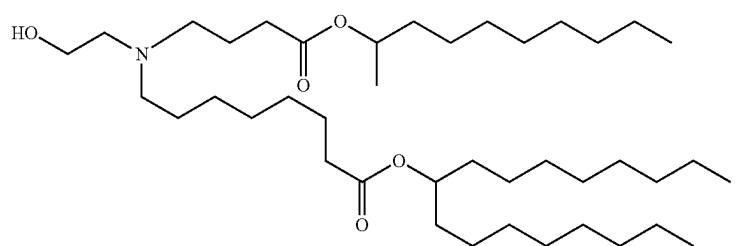
129

130

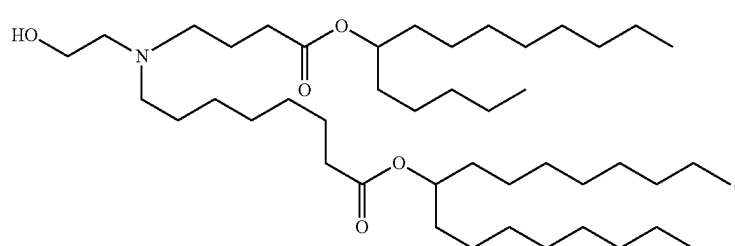
-continued



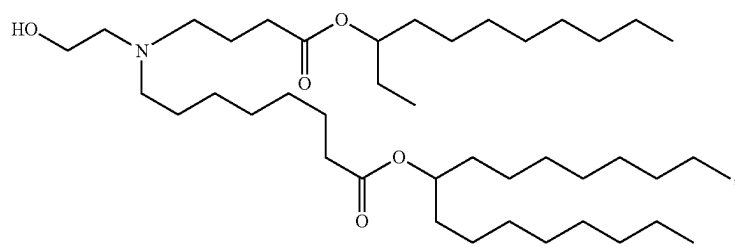
(Compound 69)



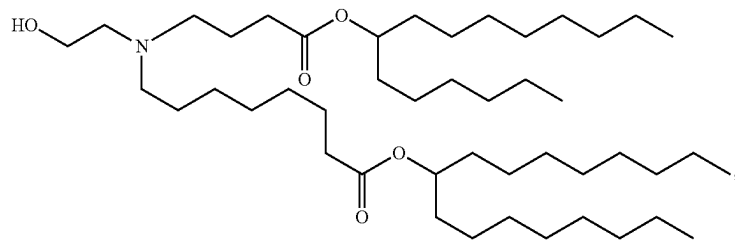
(Compound 70)



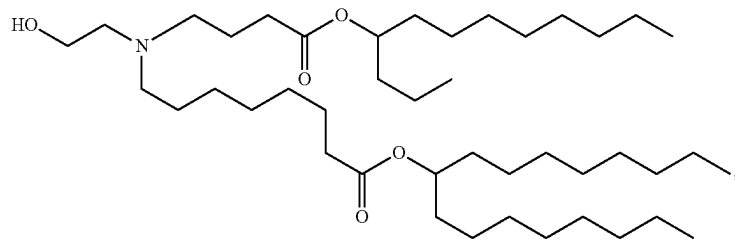
(Compound 71)



(Compound 72)



(Compound 73)

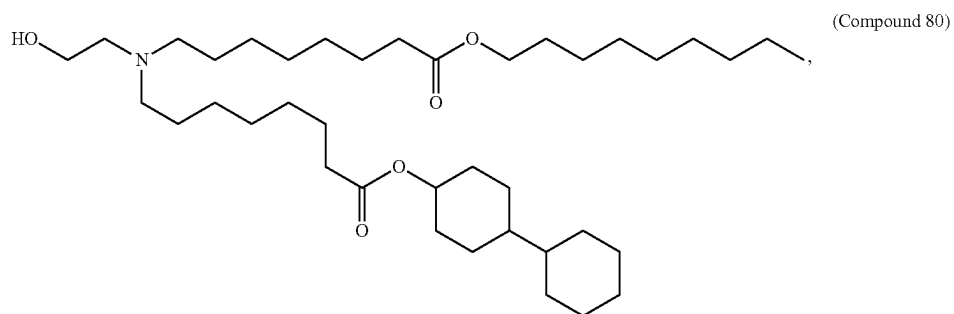
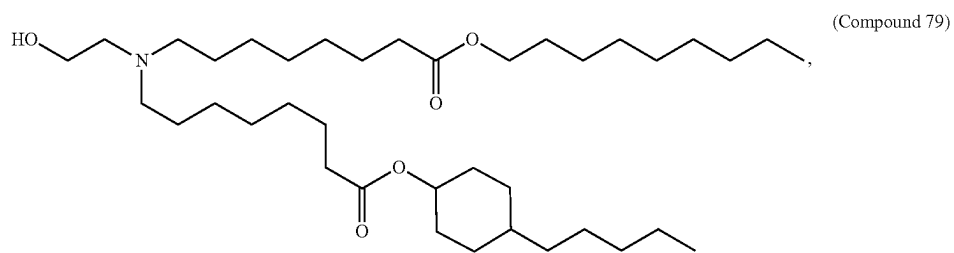
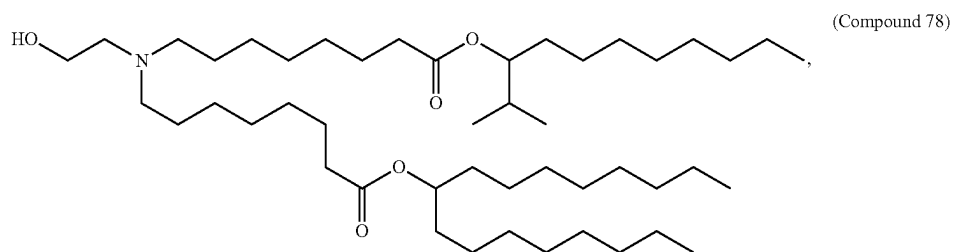
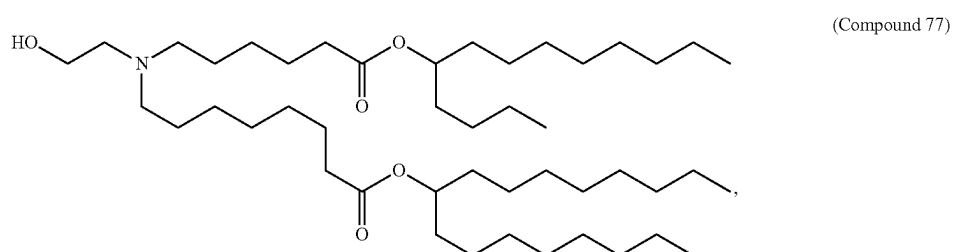
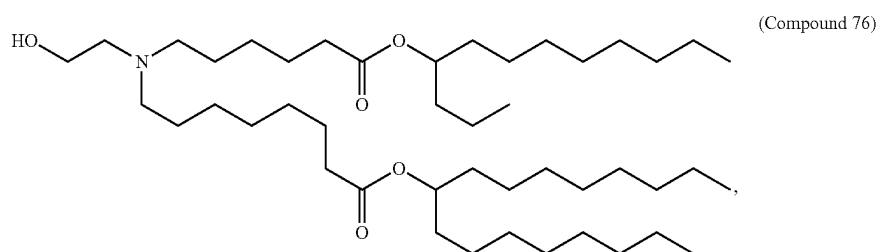
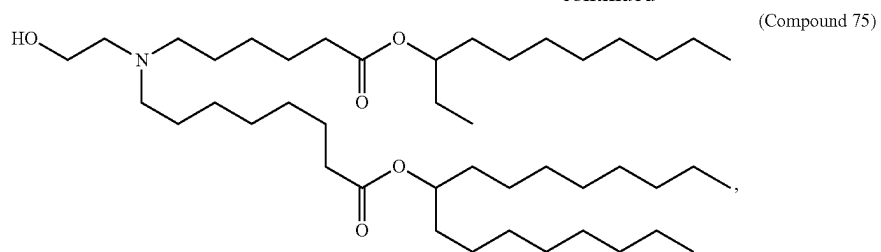


(Compound 74)

131

132

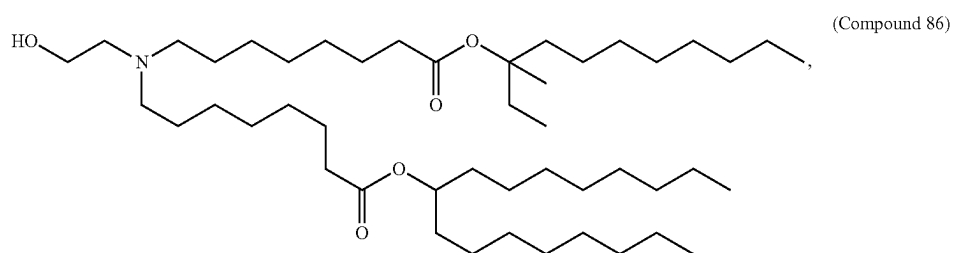
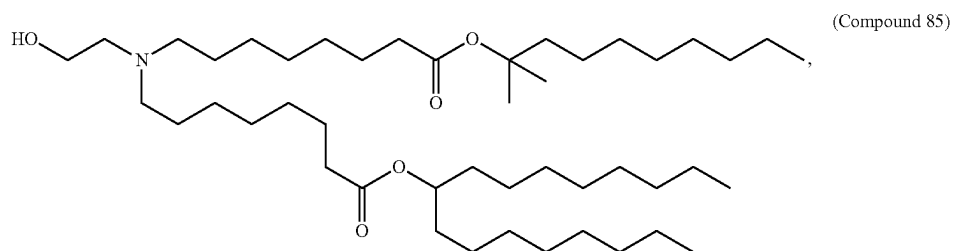
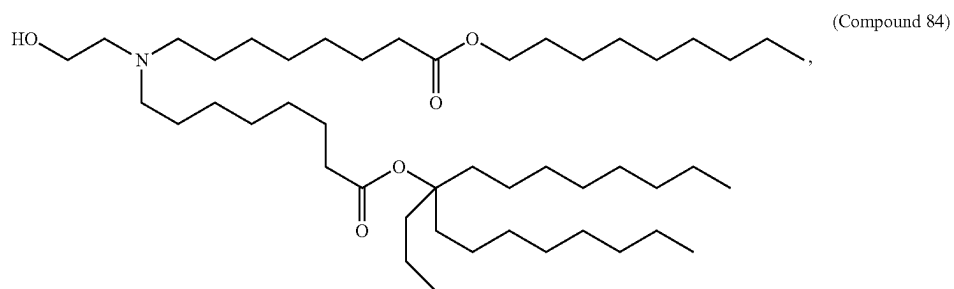
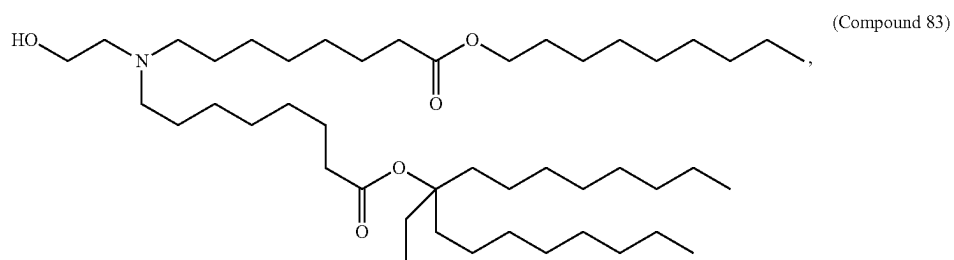
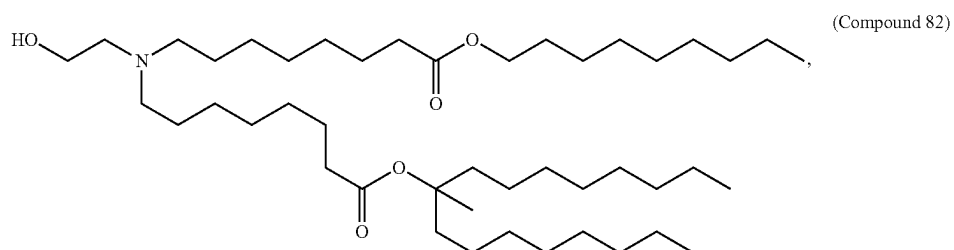
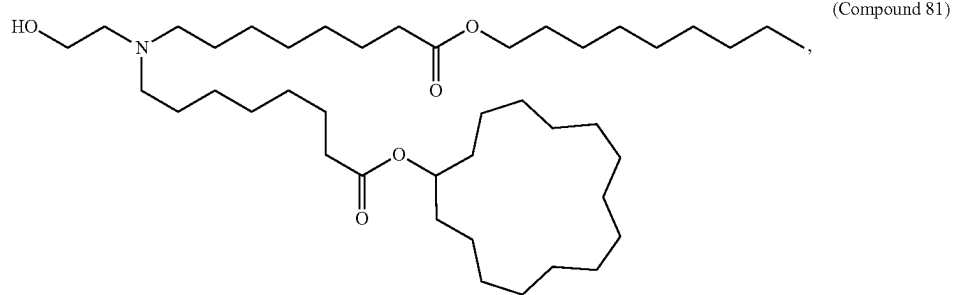
-continued



133

134

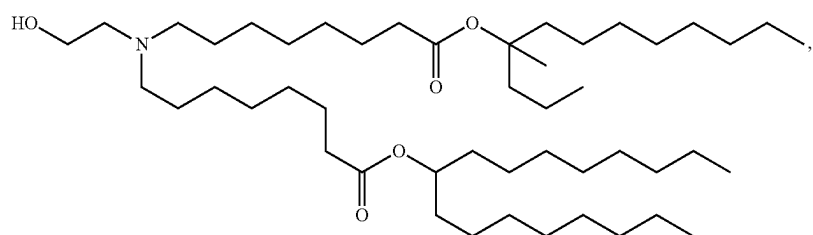
-continued



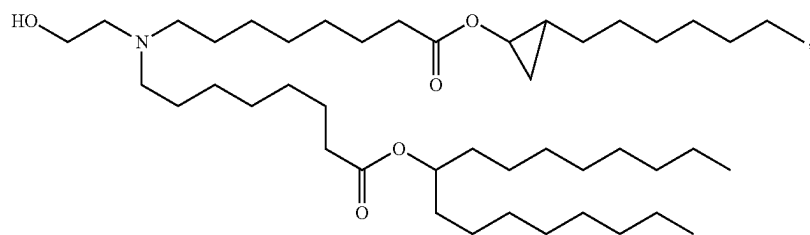
135

136

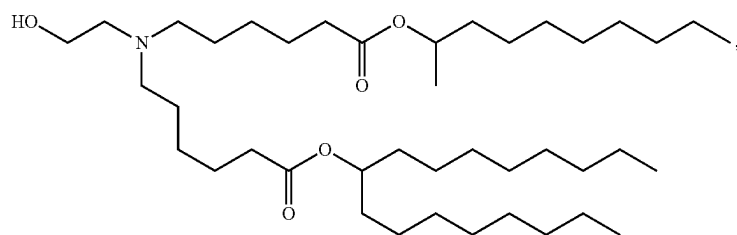
-continued



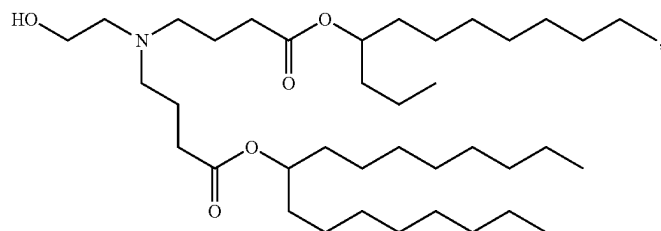
(Compound 87)



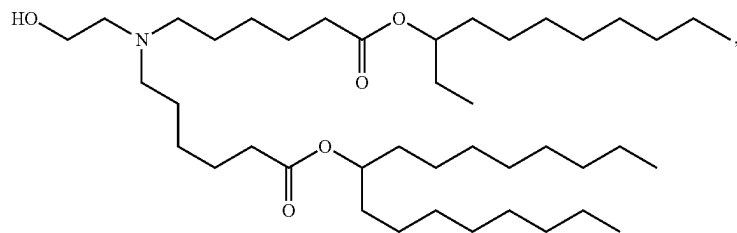
(Compound 88)



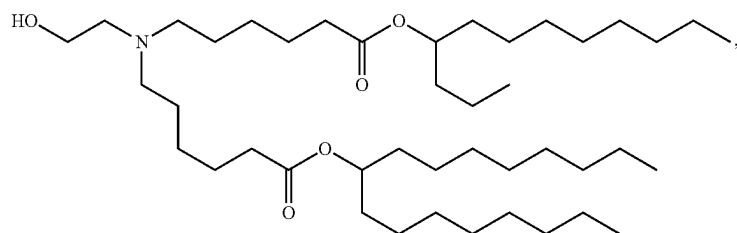
(Compound 89)



(Compound 90)



(Compound 91)

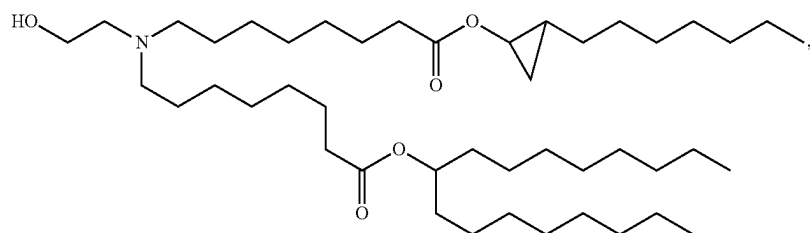


(Compound 92)

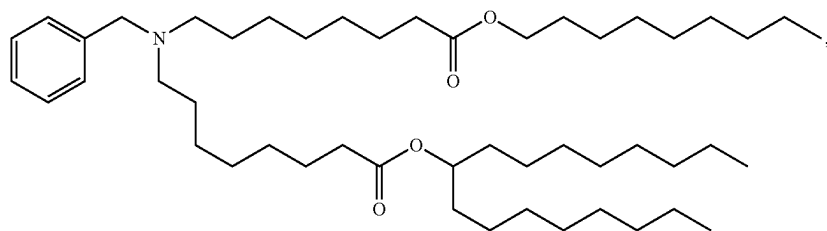
137

138

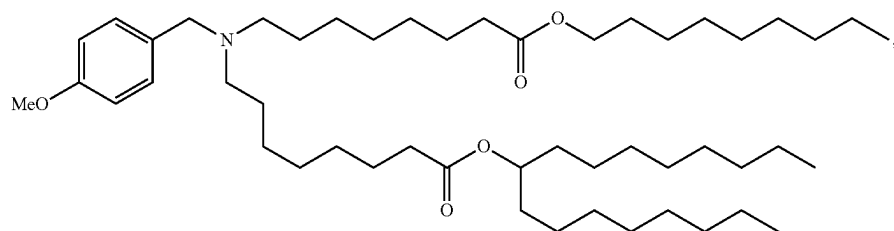
-continued



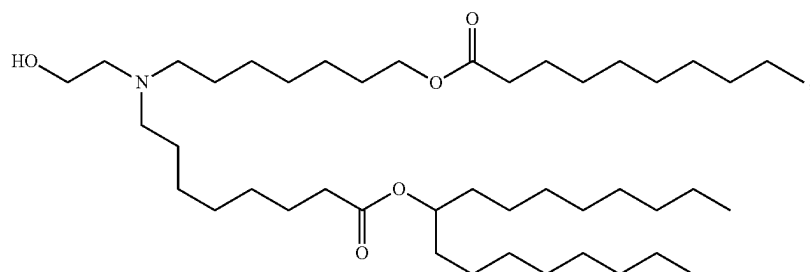
(Compound 93)



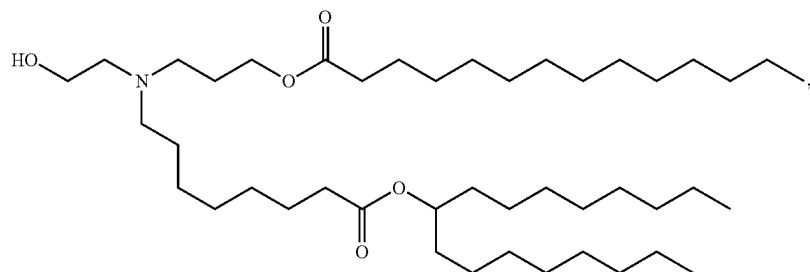
(Compound 94)



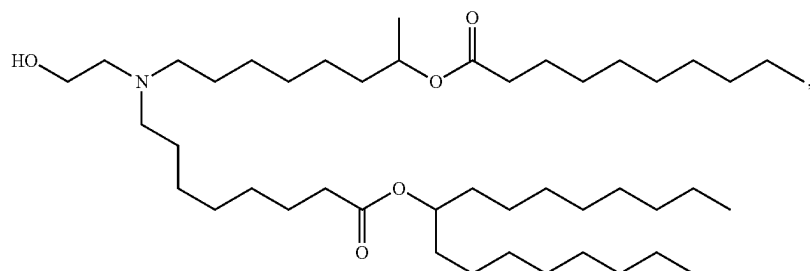
(Compound 95)



(Compound 96)



(Compound 97)

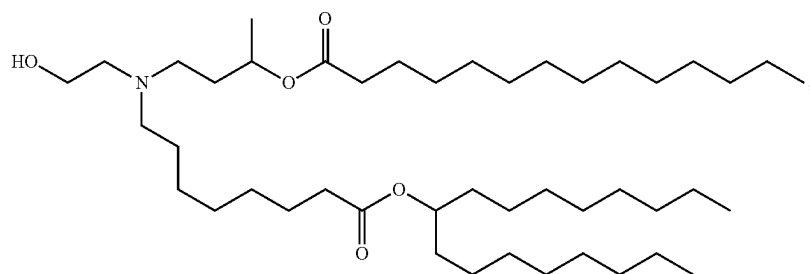


(Compound 98)

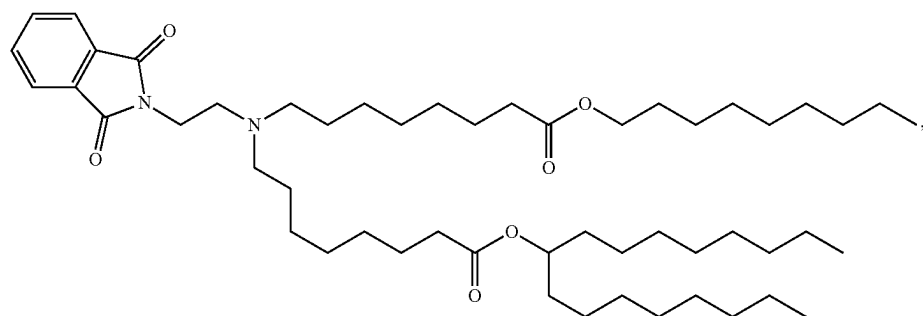
139

140

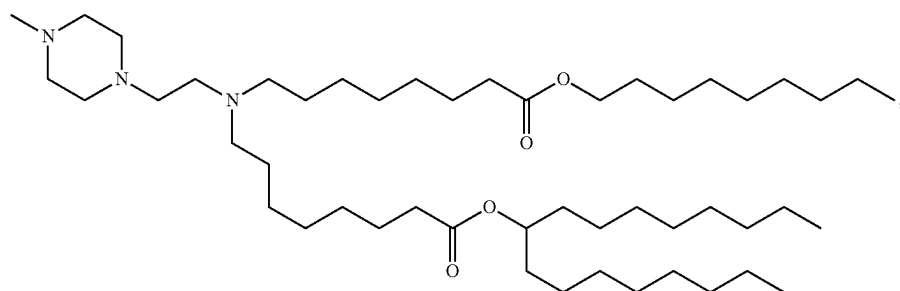
-continued



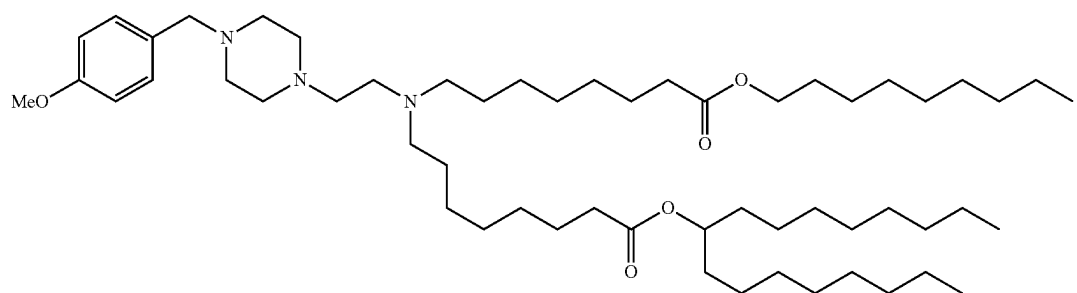
(Compound 99)



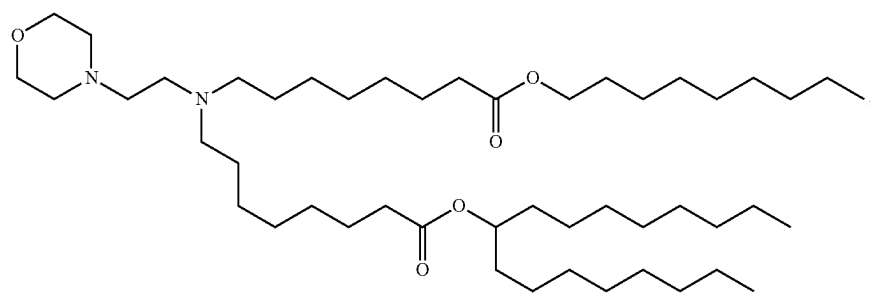
(Compound 100)



(Compound 101)



(Compound 102)

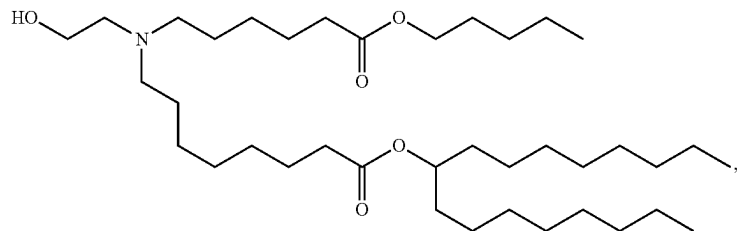


(Compound 103)

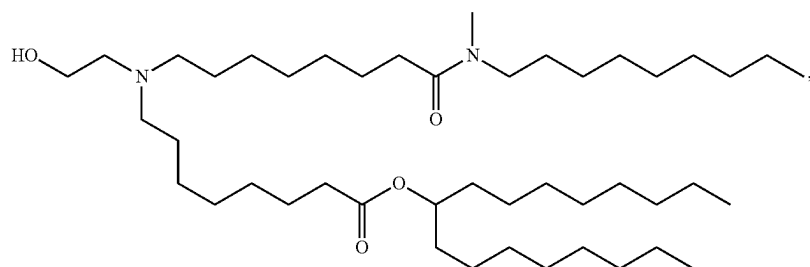
141

142

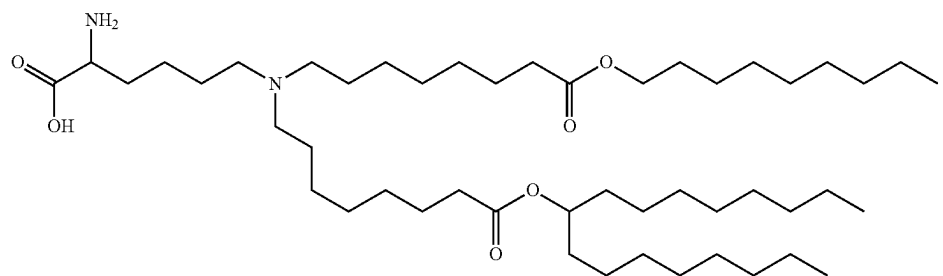
-continued



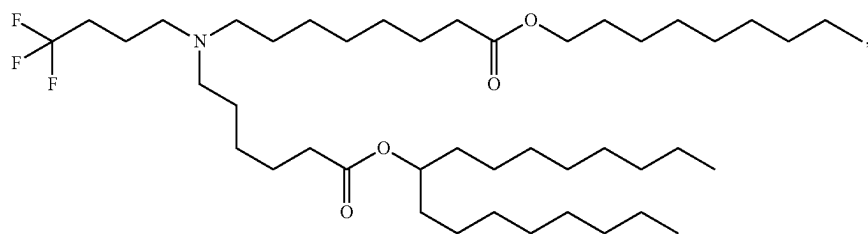
(Compound 104)



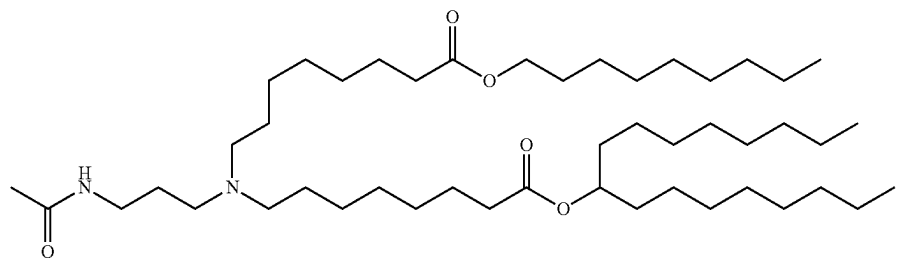
(Compound 105)



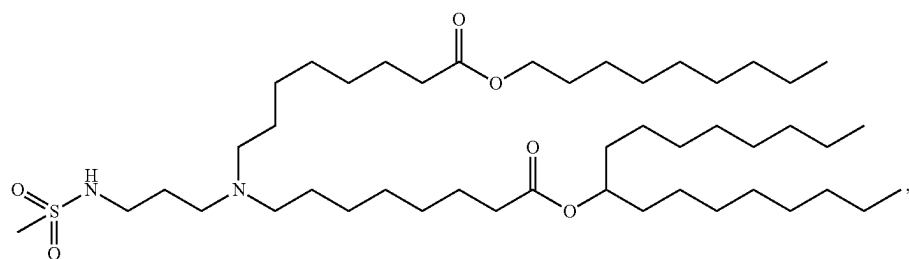
(Compound 106)



(Compound 107)

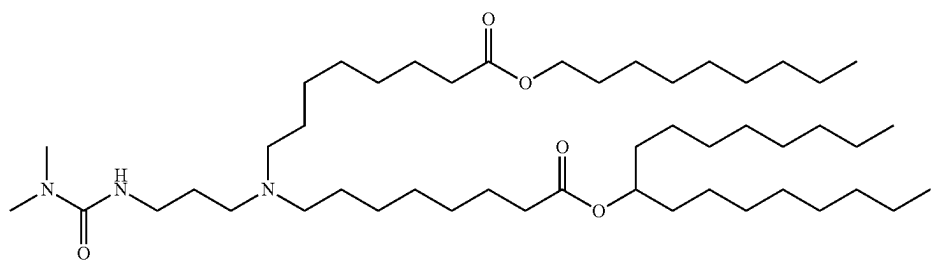


(Compound 108)

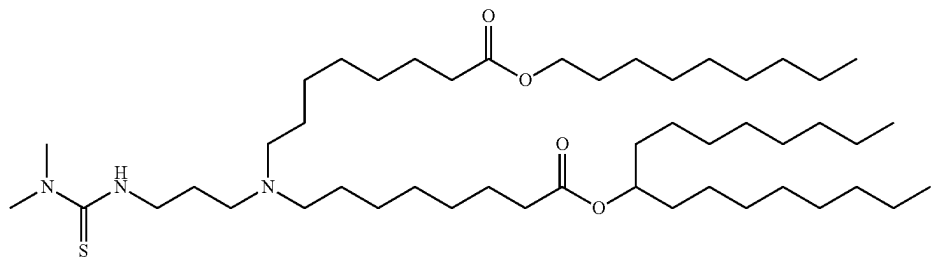


(Compound 109)

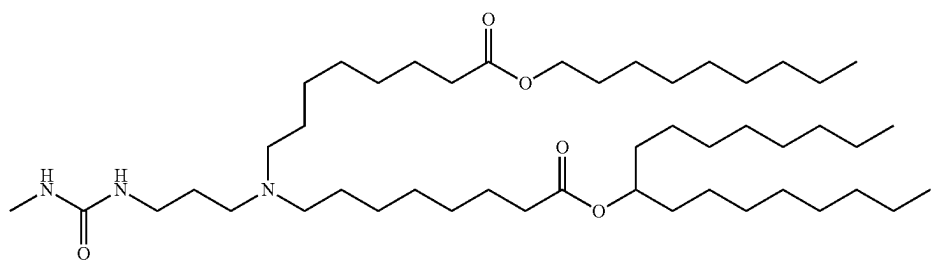
(Compound 110)



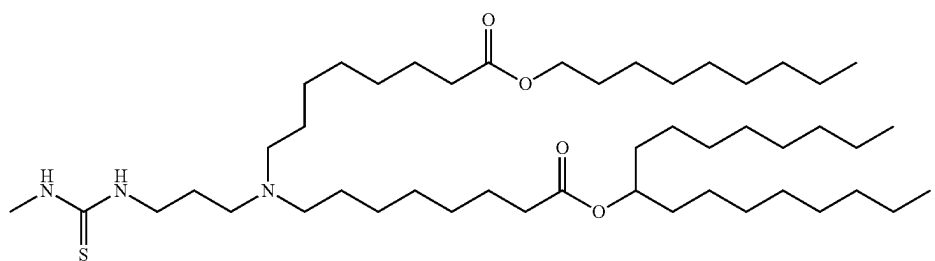
(Compound 111)



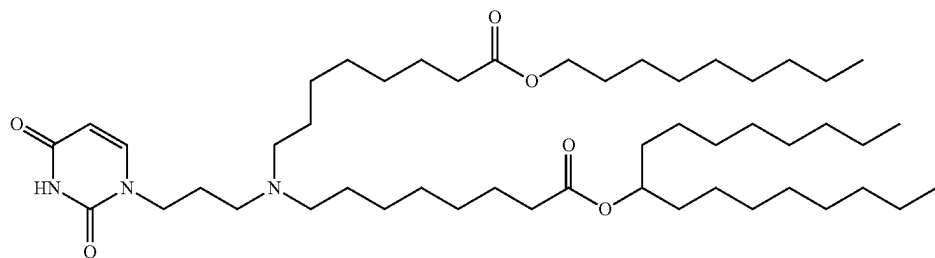
(Compound 112)



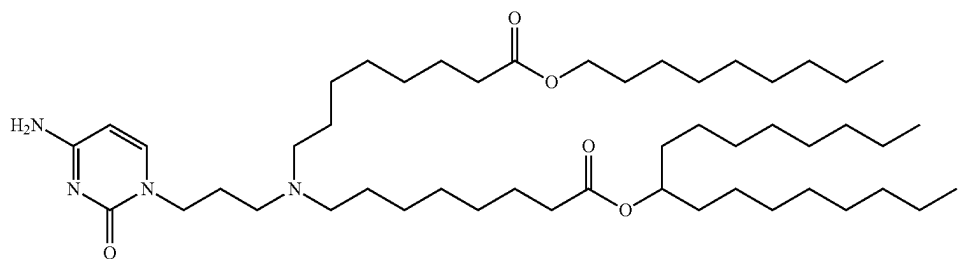
(Compound 113)



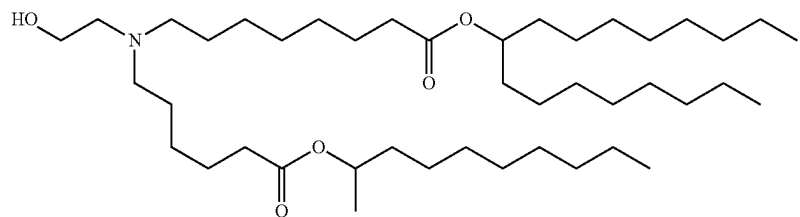
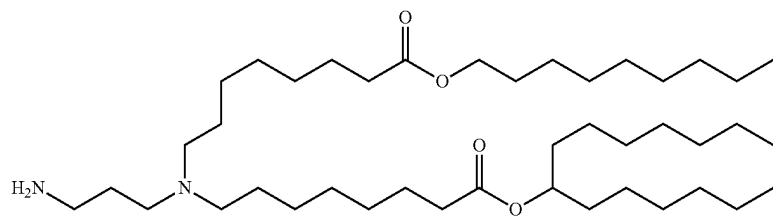
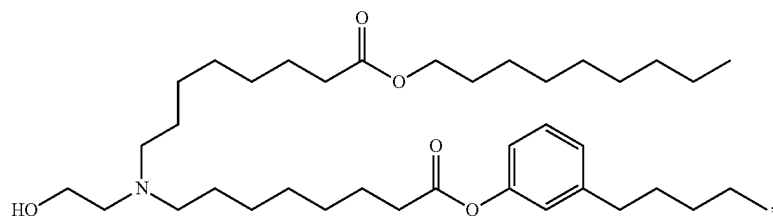
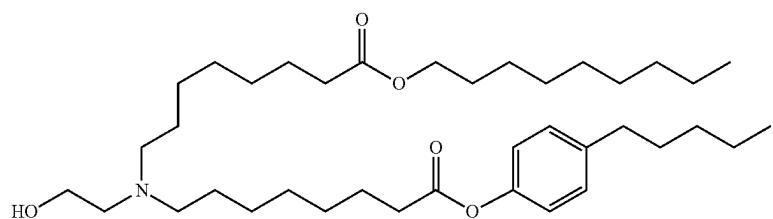
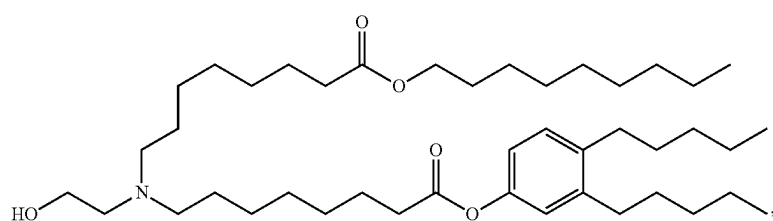
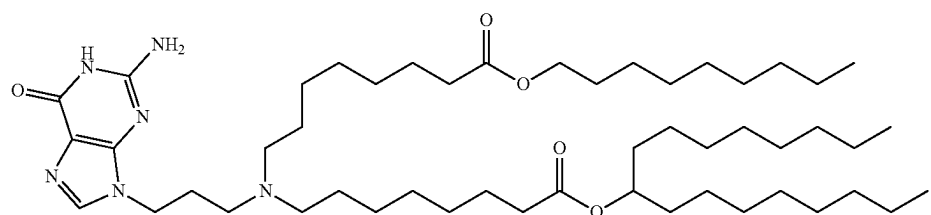
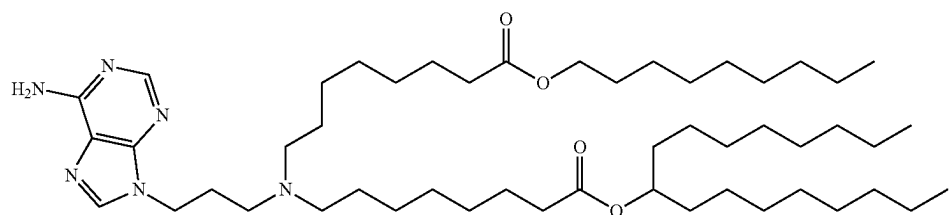
(Compound 114)



(Compound 115)



-continued

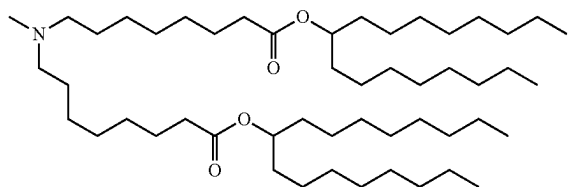
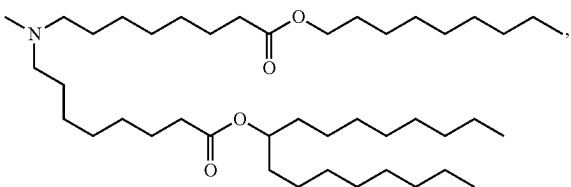


147

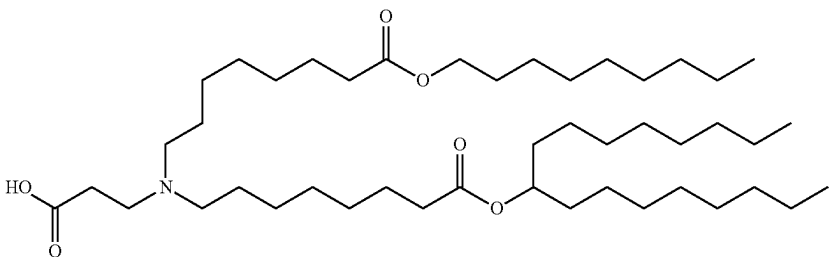
148

-continued
(Compound 123)

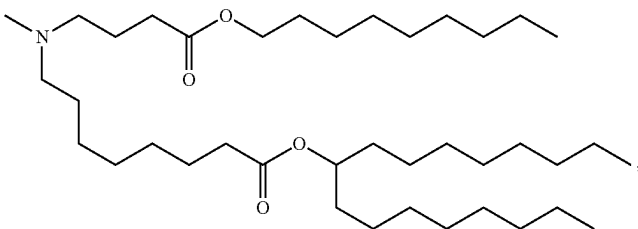
(Compound 124)



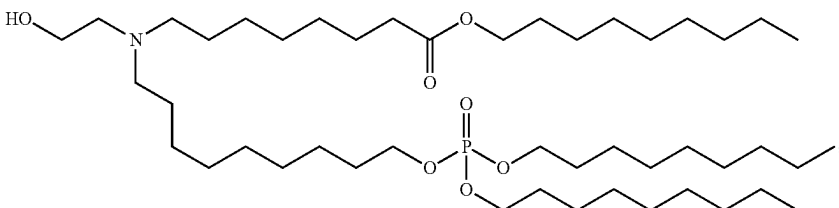
(Compound 125)



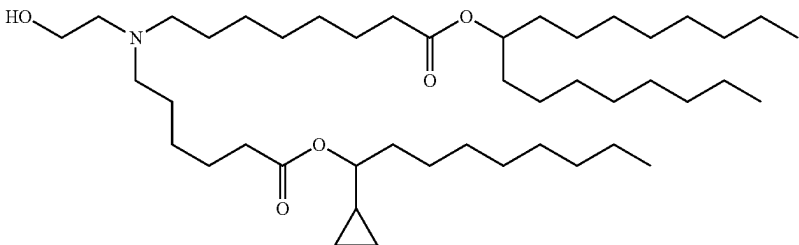
(Compound 126)



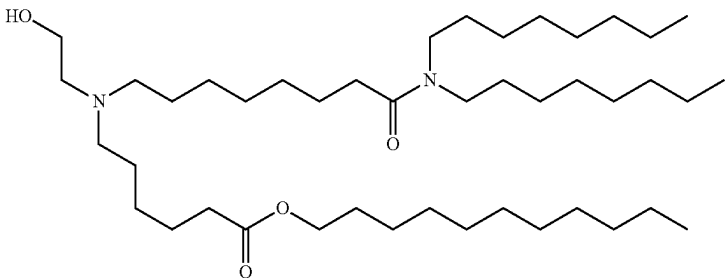
(Compound 127)



(Compound 128)



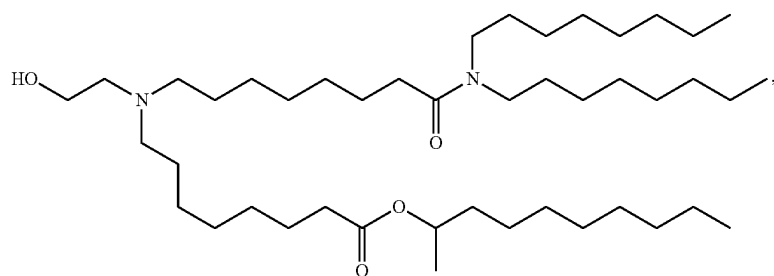
(Compound 129)



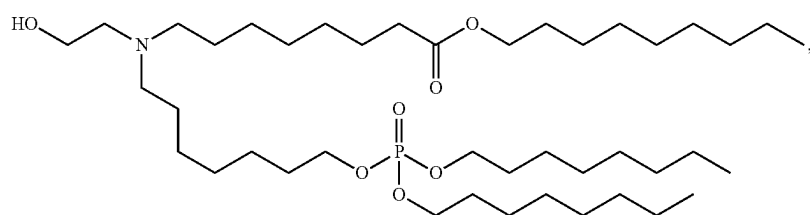
149

150

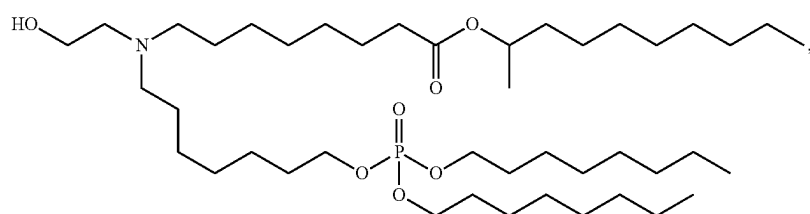
-continued



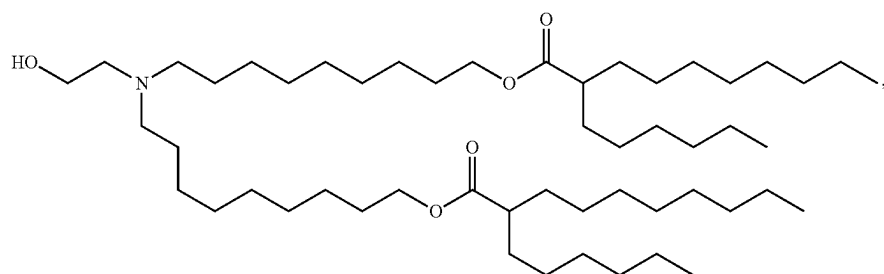
(Compound 130)



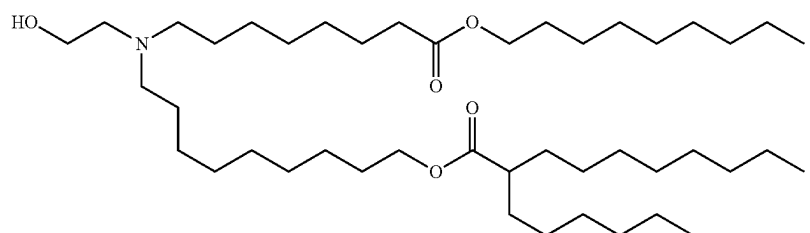
(Compound 131)



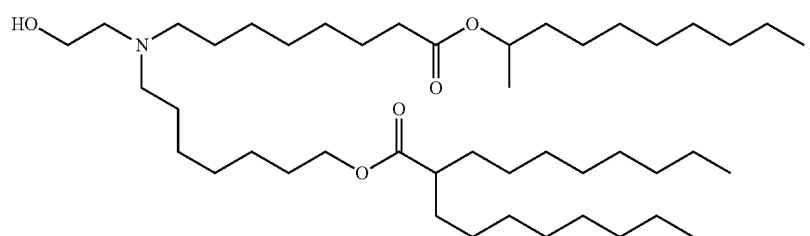
(Compound 132)



(Compound 133)



(Compound 134)

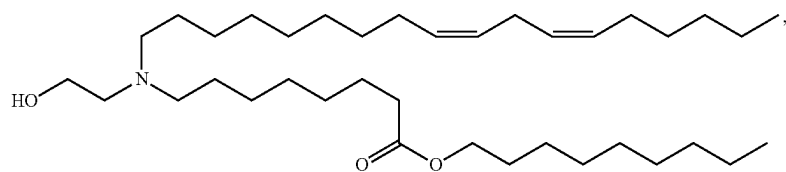


(Compound 135)

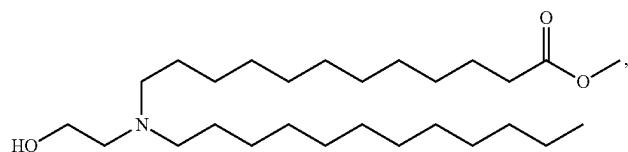
151

152

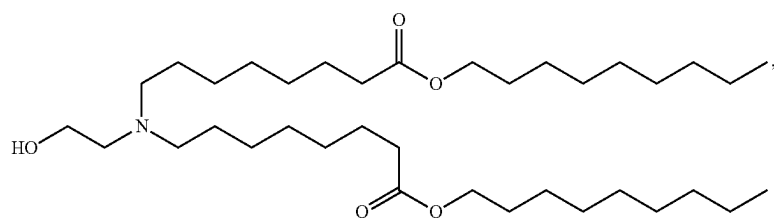
-continued



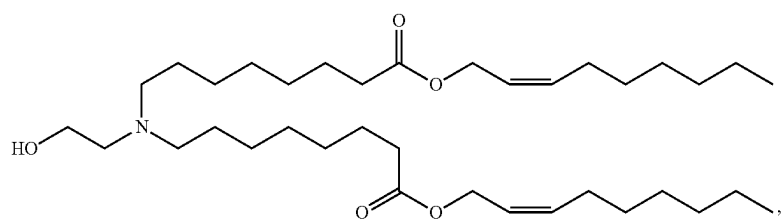
(Compound 136)



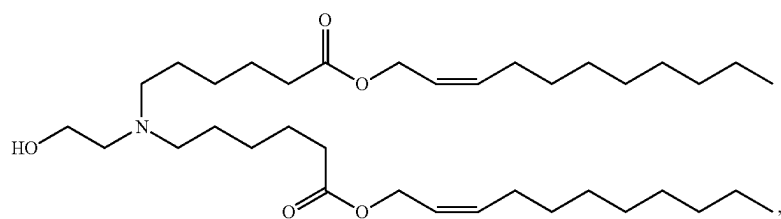
(Compound 137)



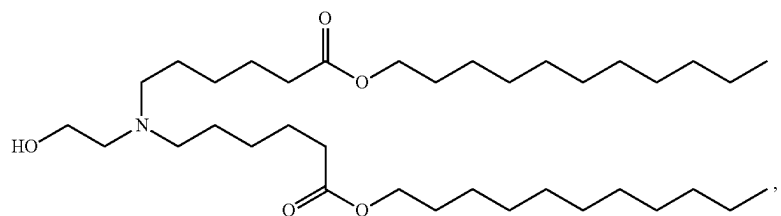
(Compound 138)



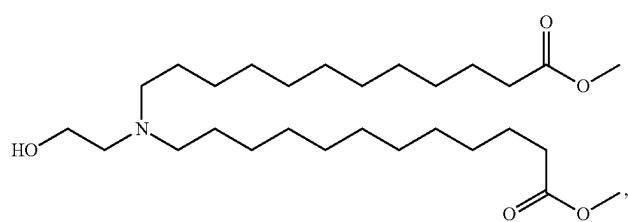
(Compound 139)



(Compound 140)



(Compound 141)

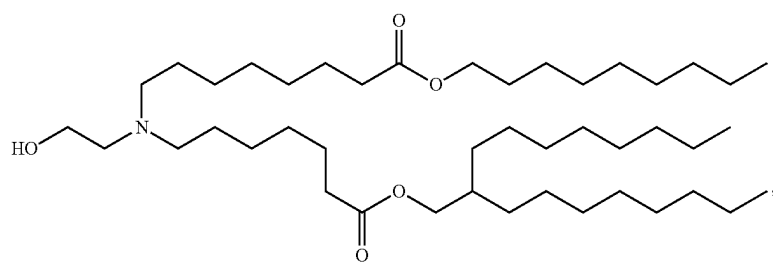


(Compound 142)

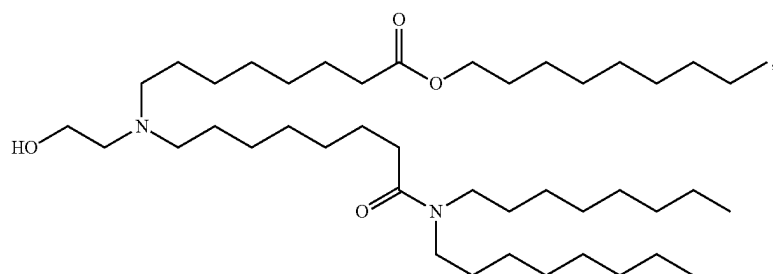
153

154

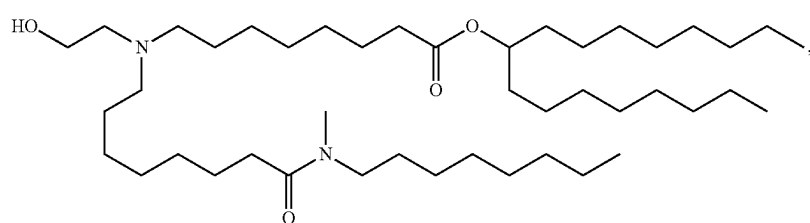
-continued



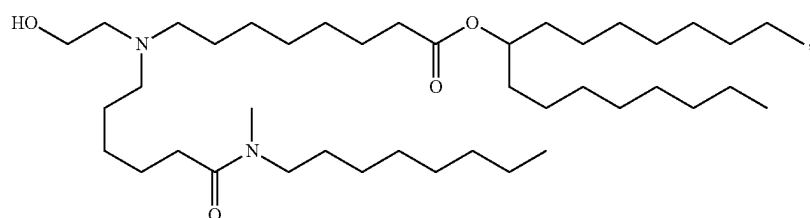
(Compound 143)



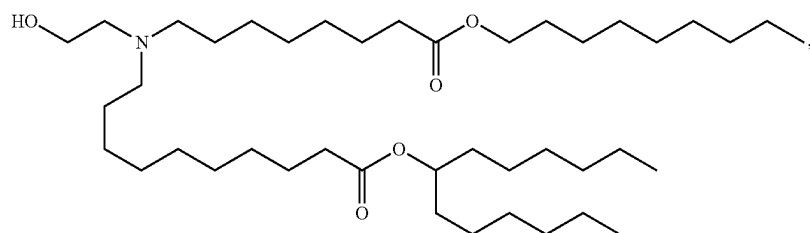
(Compound 144)



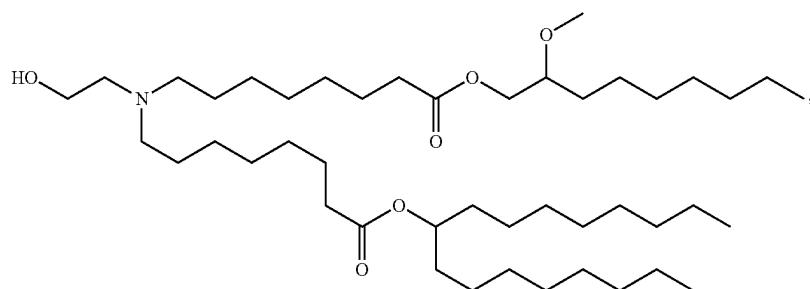
(Compound 145)



(Compound 146)



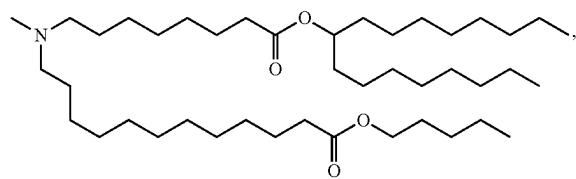
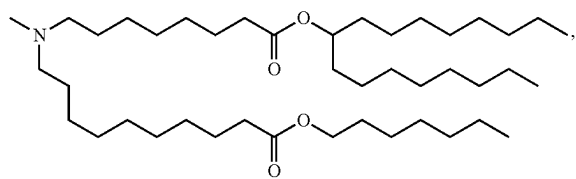
(Compound 147)



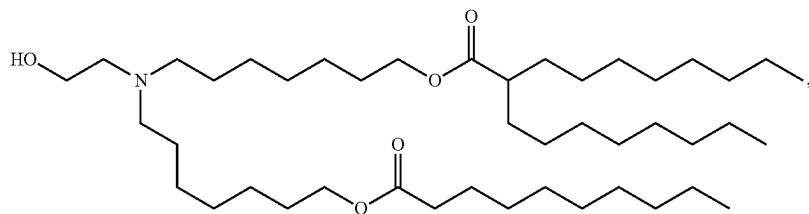
(Compound 148)

(Compound 149)

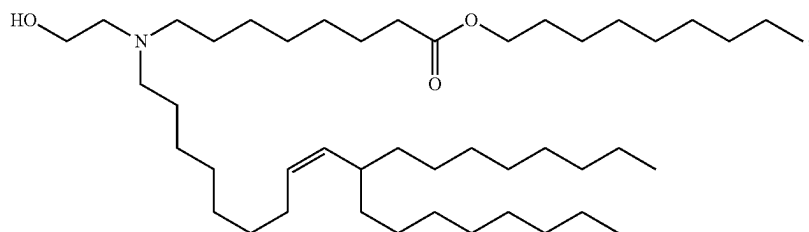
(Compound 150)



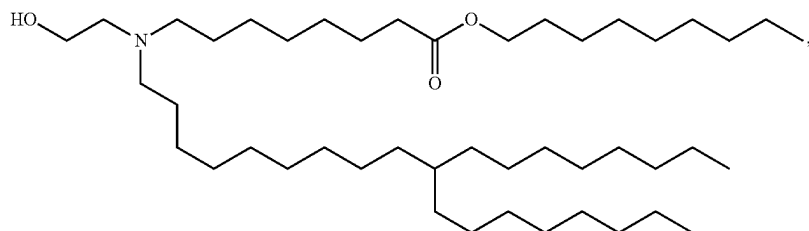
(Compound 151)



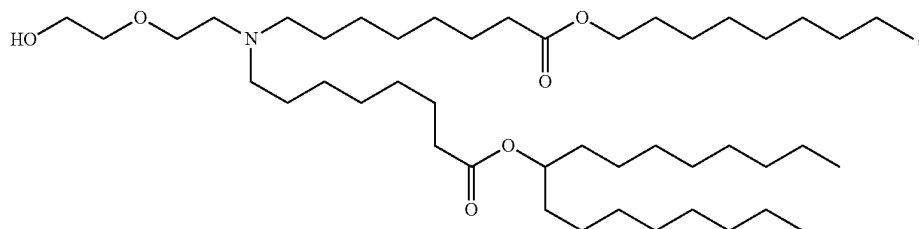
(Compound 152)



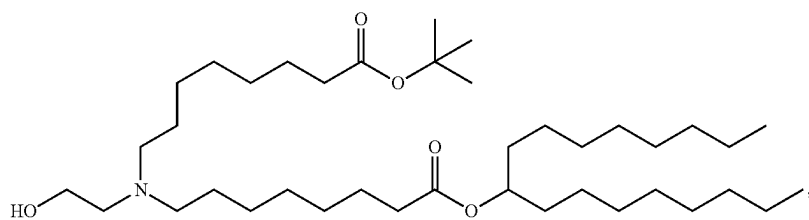
(Compound 153)



(Compound 154)



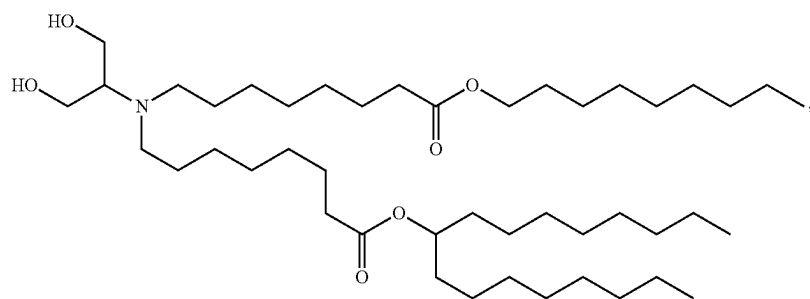
(Compound 155)



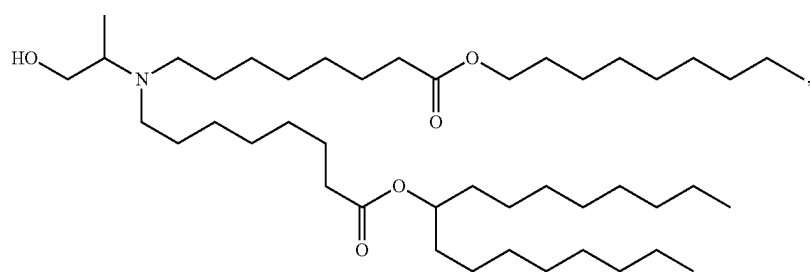
157

158

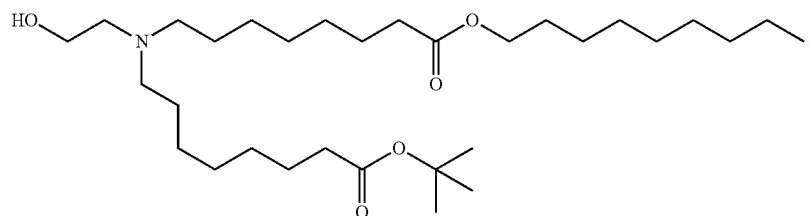
-continued



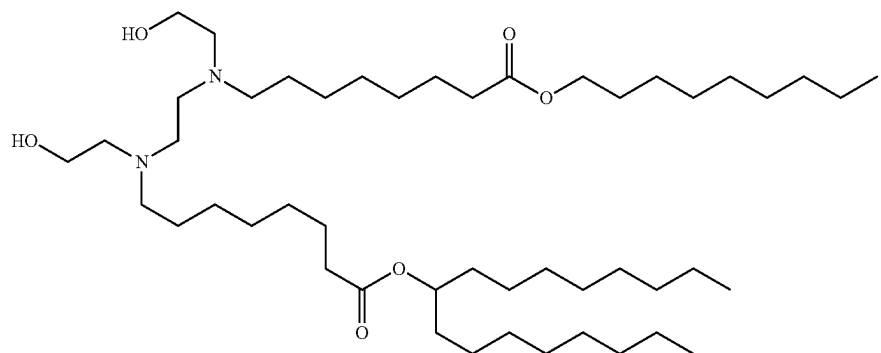
(Compound 156)



(Compound 157)



(Compound 158)



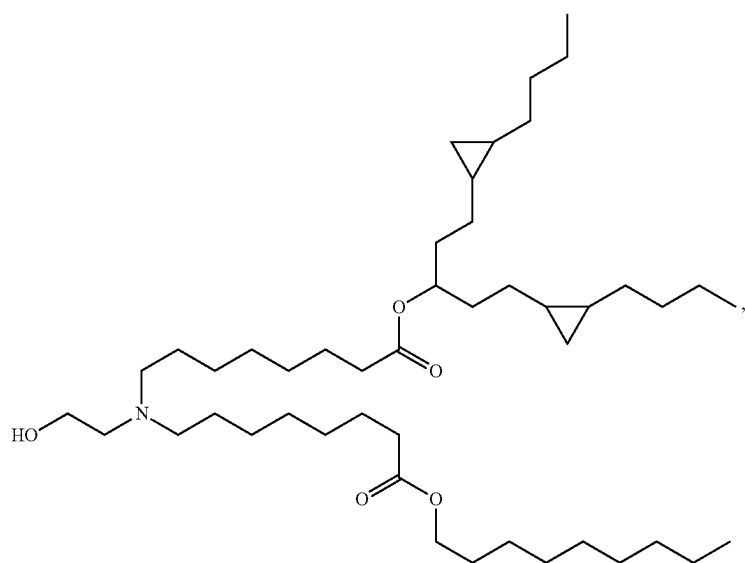
(Compound 159)

159

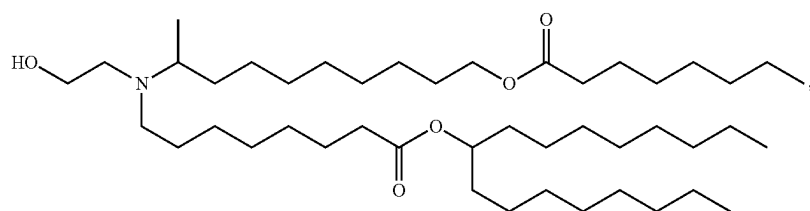
-continued

160

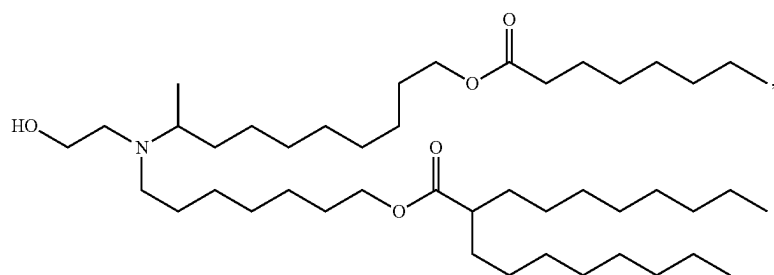
(Compound 160)



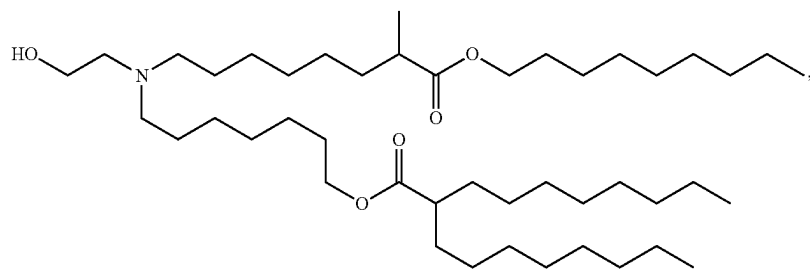
(Compound 161)



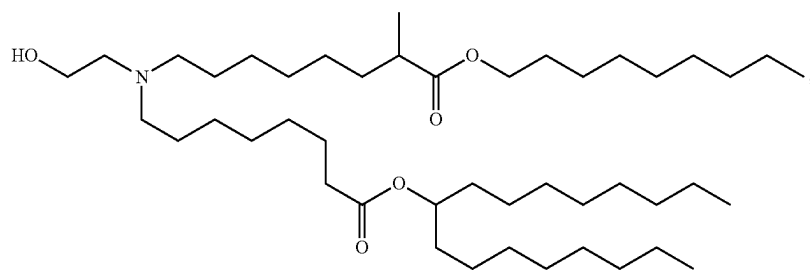
(Compound 162)



(Compound 163)



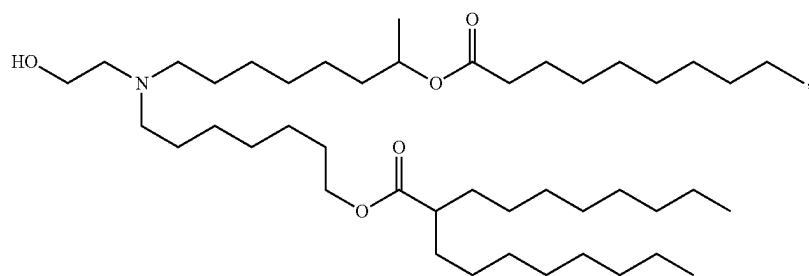
(Compound 164)



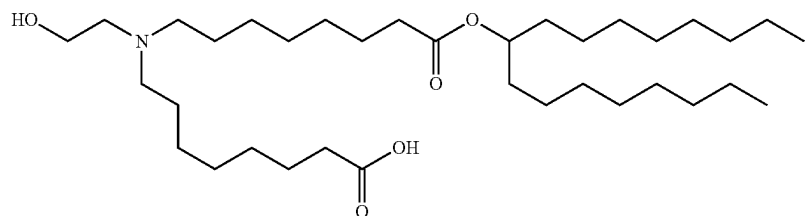
161

162

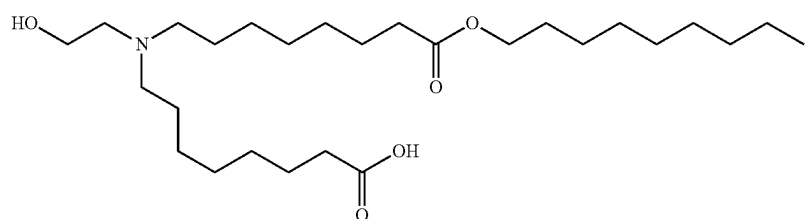
-continued



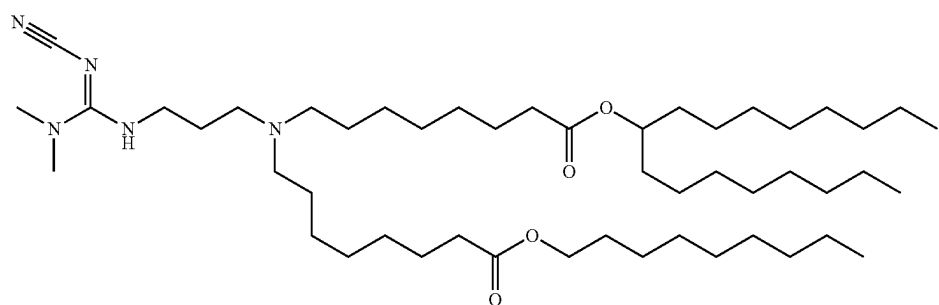
(Compound 165)



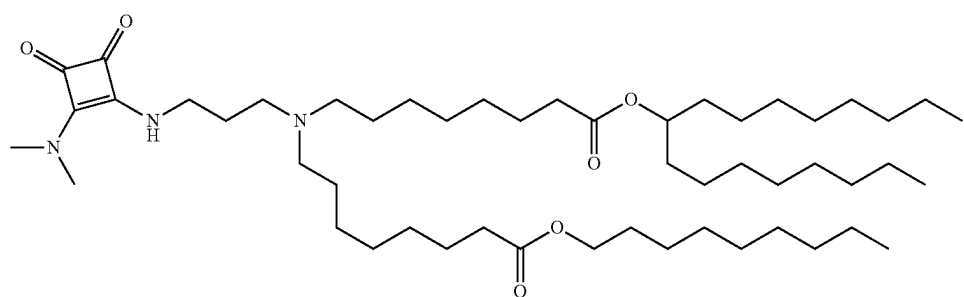
(Compound 166)



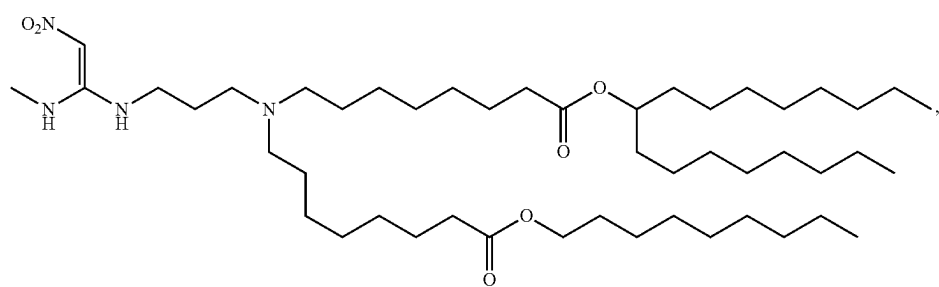
(Compound 167)



(Compound 168)

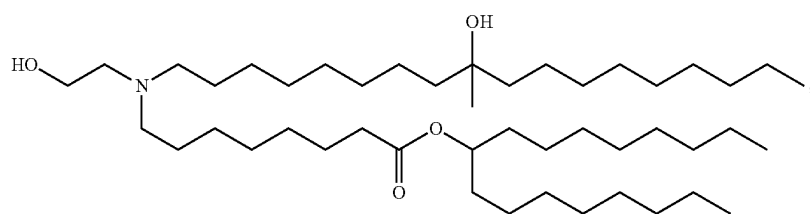


(Compound 169)

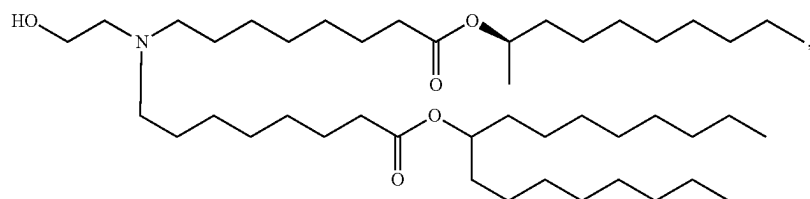


(Compound 170)

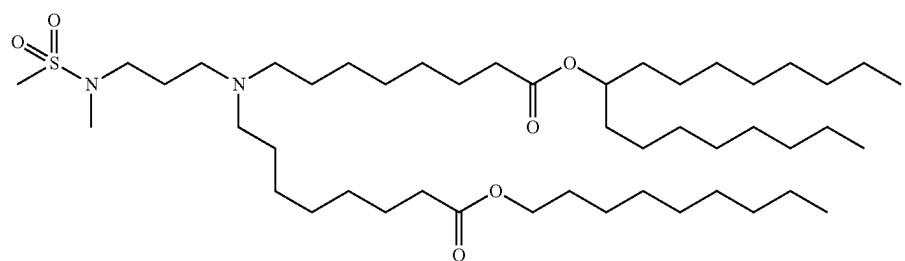
-continued



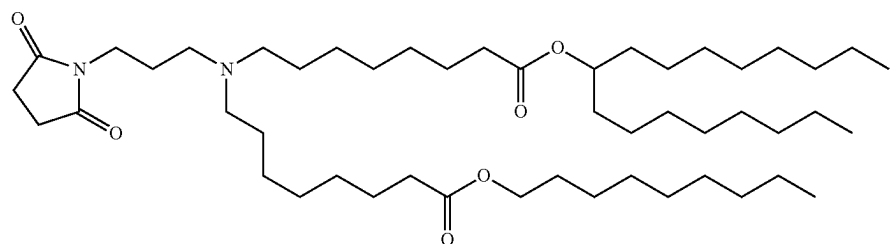
(Compound 171)



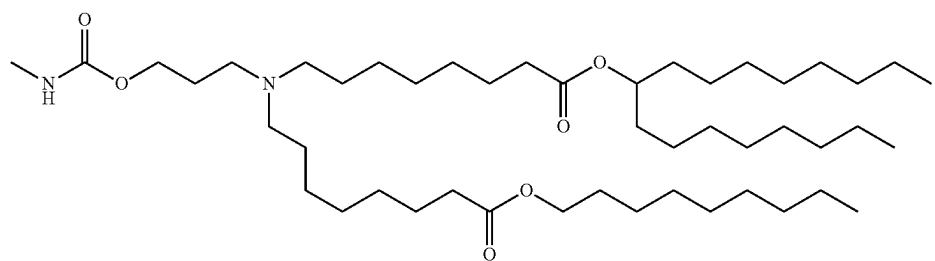
(Compound 172)



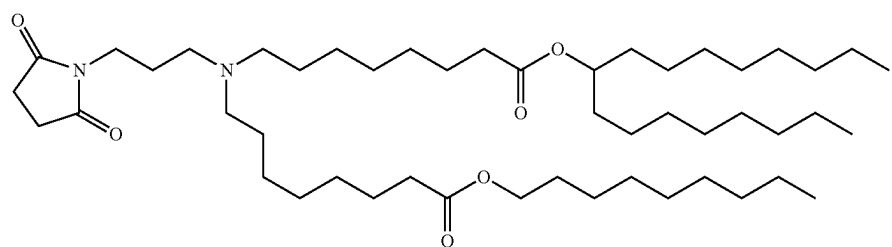
(Compound 173)



(Compound 174)



(Compound 175)

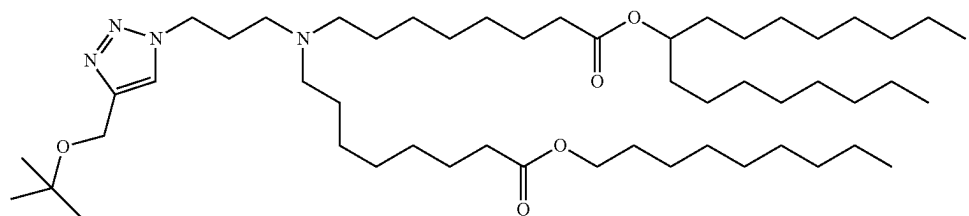


(Compound 176)

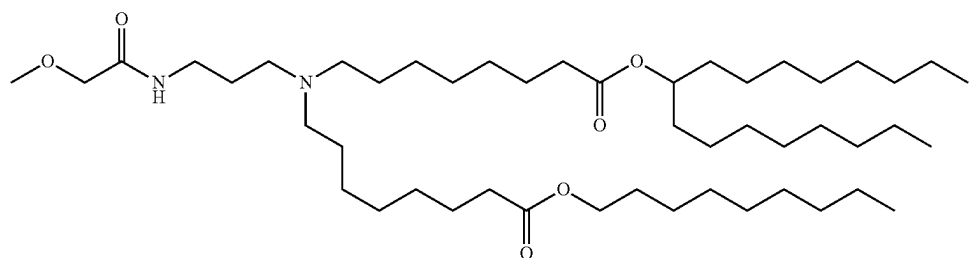
165

166

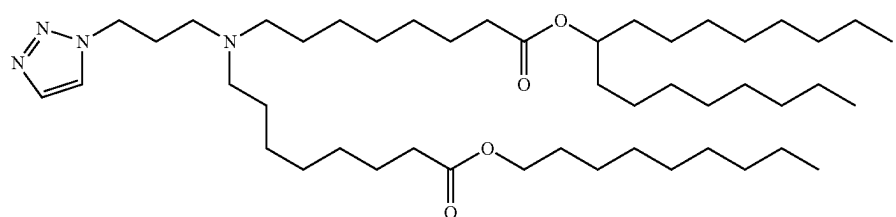
-continued



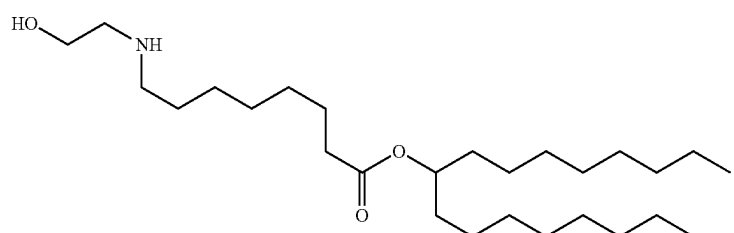
(Compound 177)



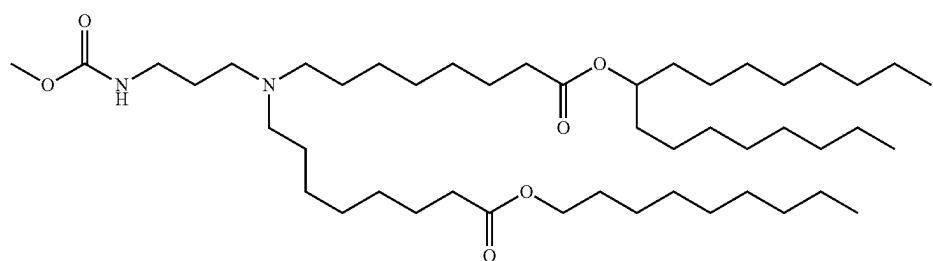
(Compound 178)



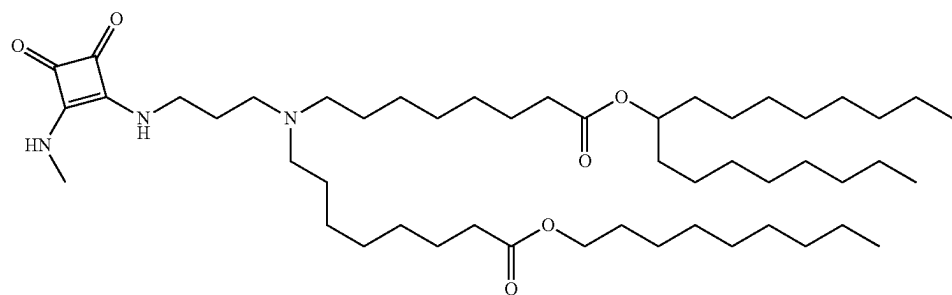
(Compound 179)



(Compound 180)



(Compound 181)

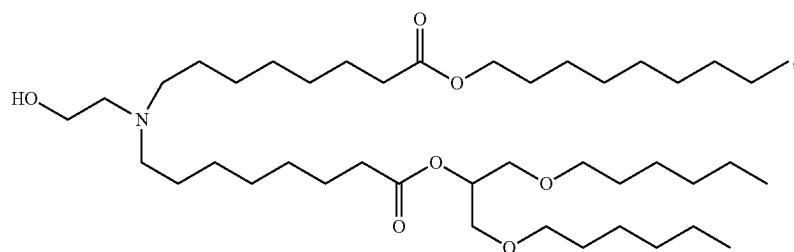


(Compound 182)

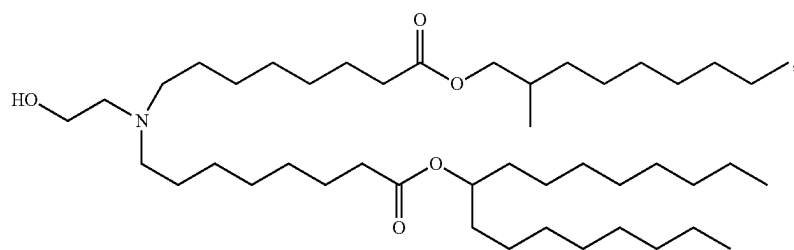
167

168

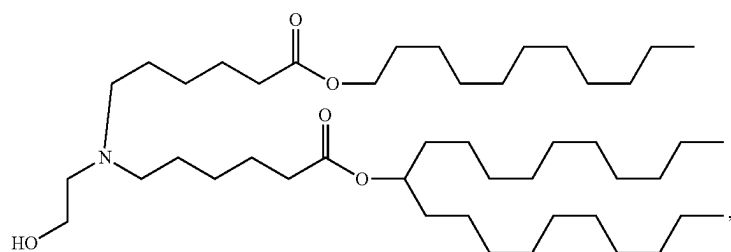
-continued



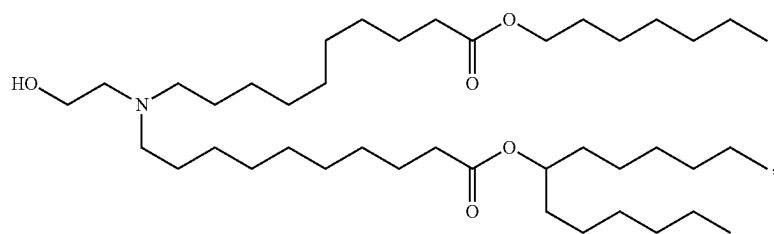
(Compound 183)



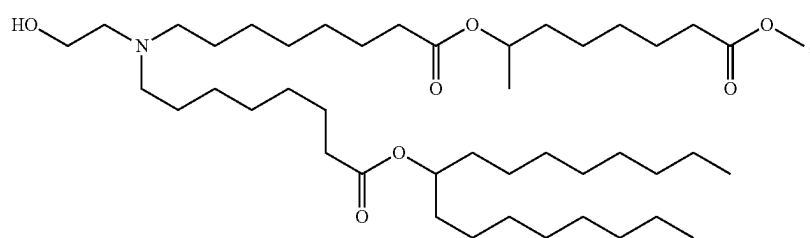
(Compound 184)



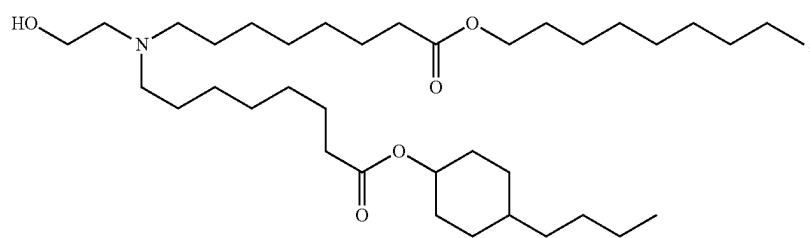
(Compound 185)



(Compound 186)



(Compound 187)

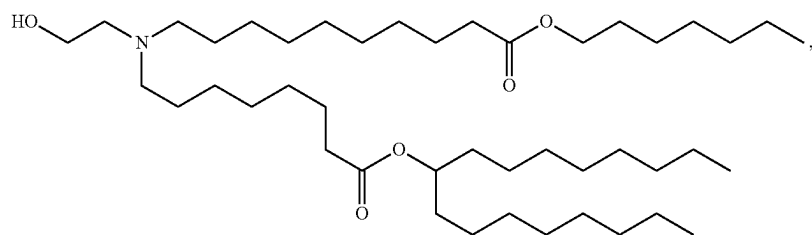


(Compound 188)

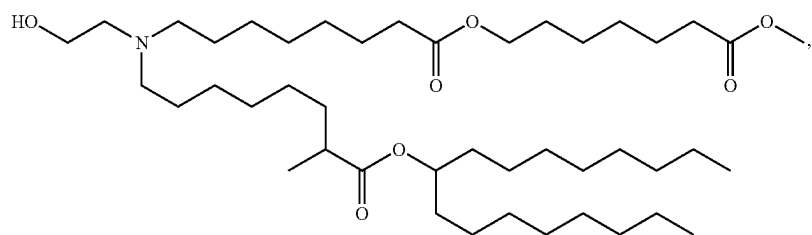
169

170

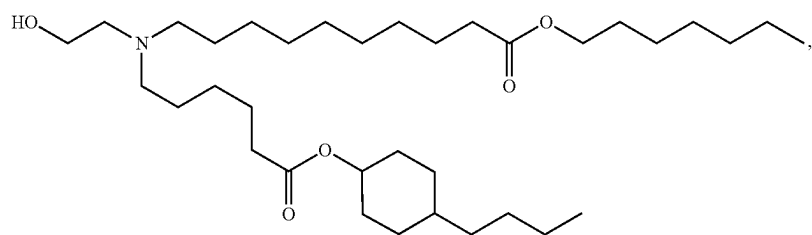
-continued



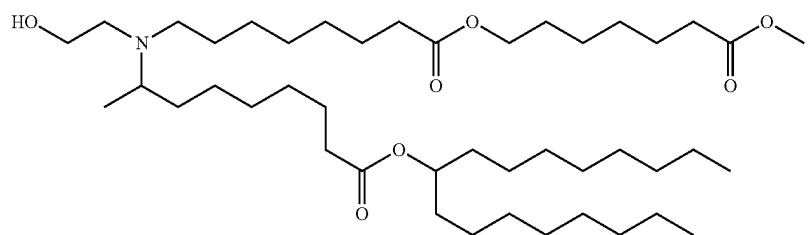
(Compound 189)



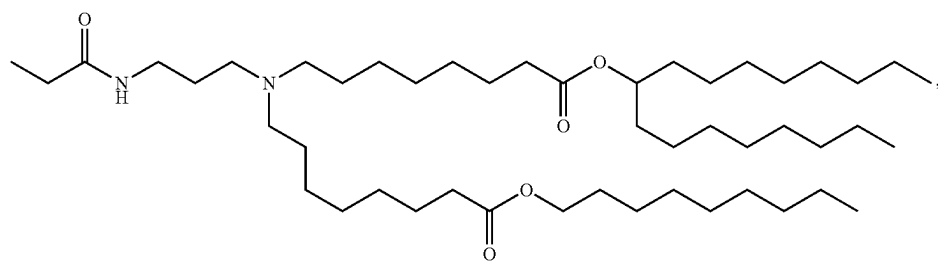
(Compound 190)



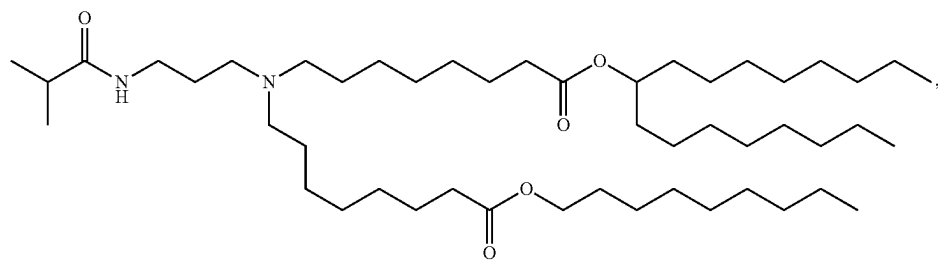
(Compound 191)



(Compound 192)



(Compound 193)

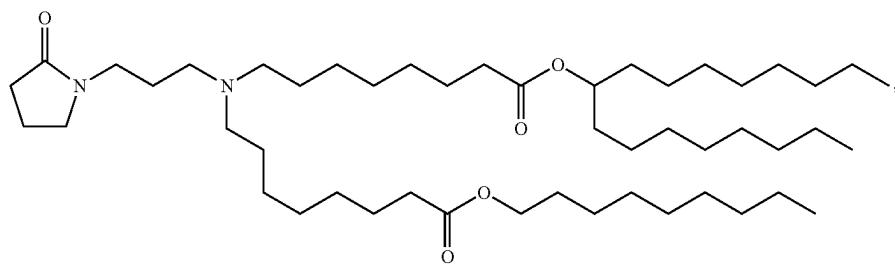


(Compound 194)

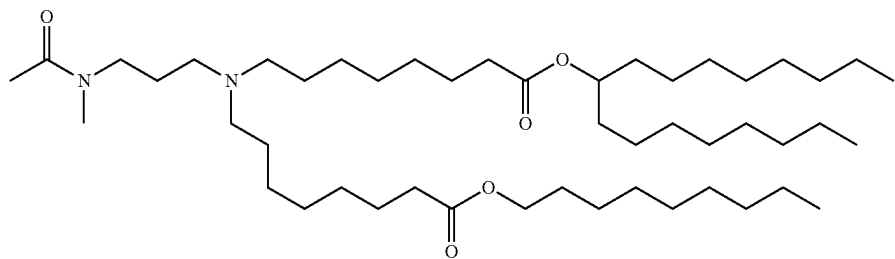
171

172

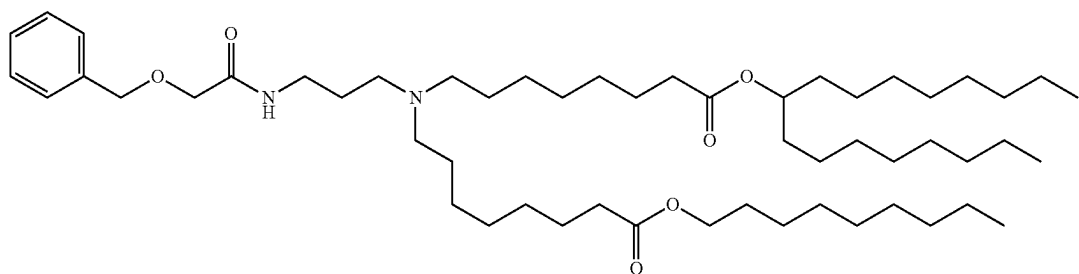
(Compound 195)



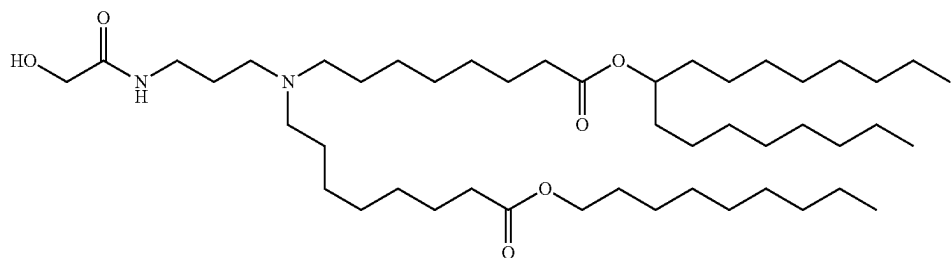
(Compound 196)



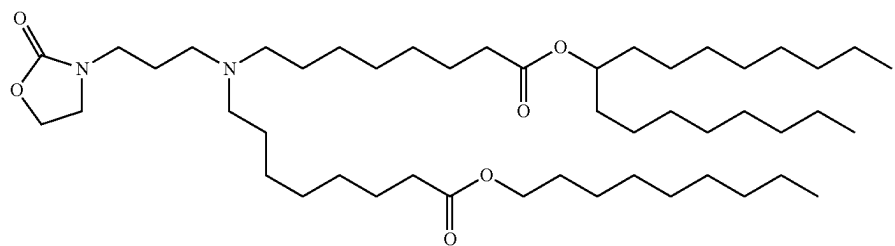
(Compound 197)



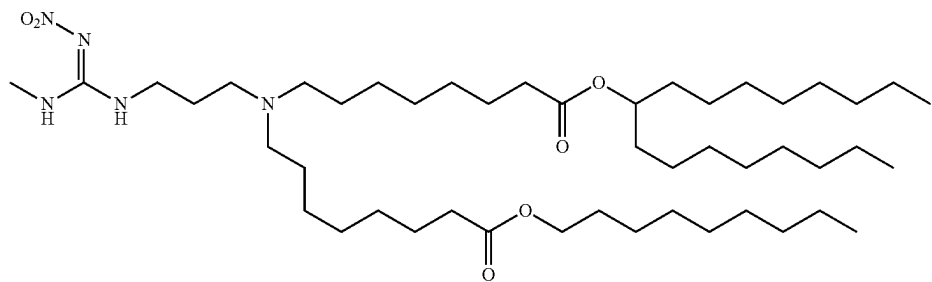
(Compound 198)



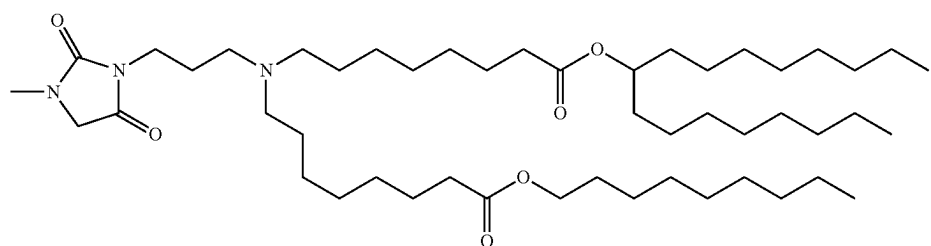
(Compound 199)



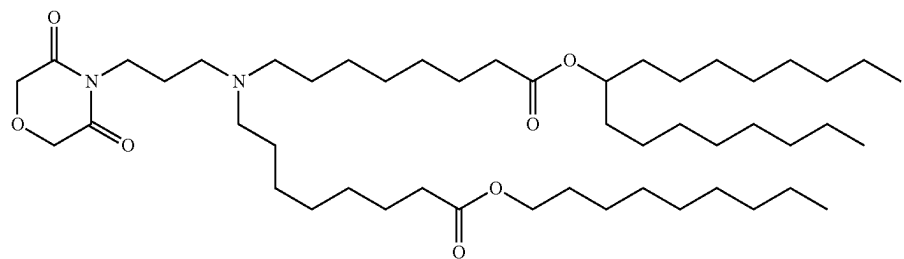
(Compound 200)



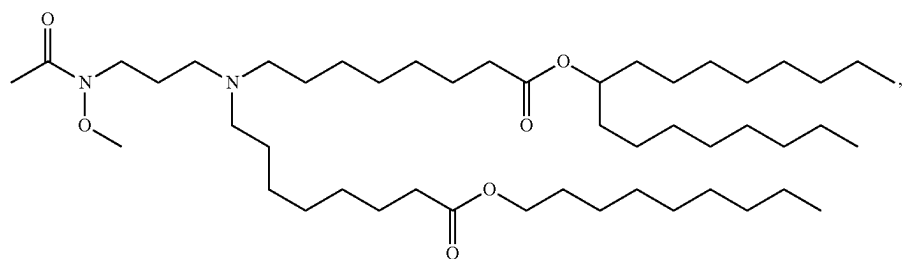
-continued



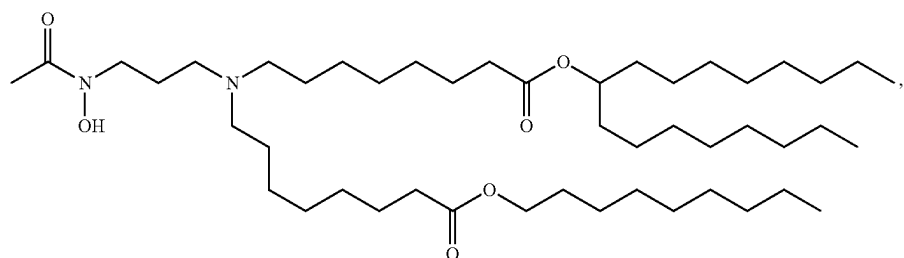
(Compound 201)



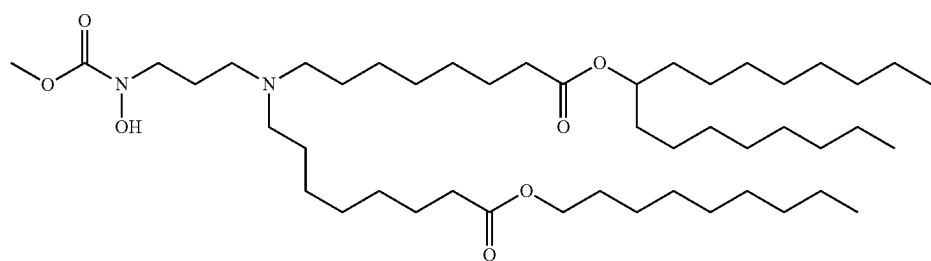
(Compound 202)



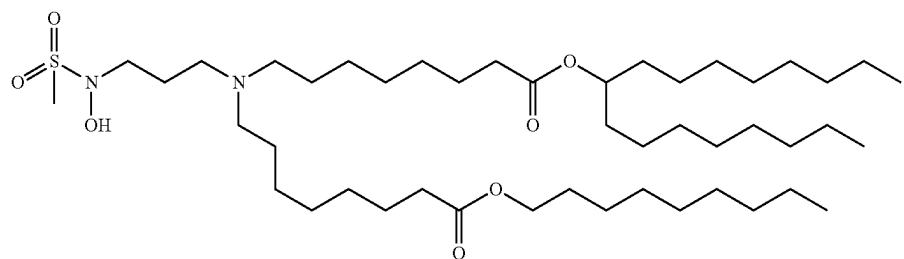
(Compound 203)



(Compound 204)



(Compound 205)

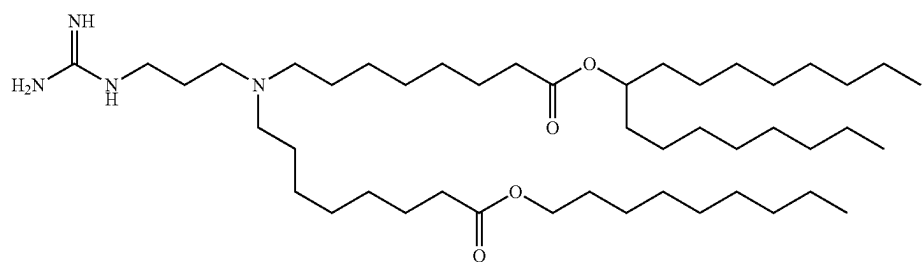


(Compound 206)

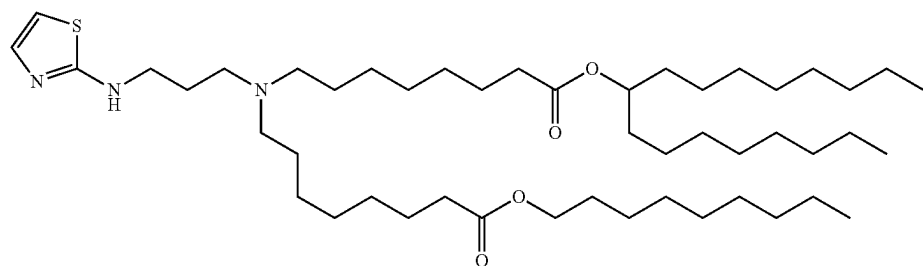
175

176

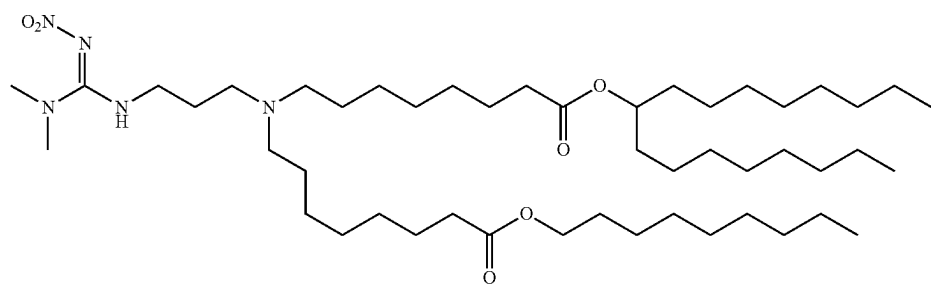
-continued



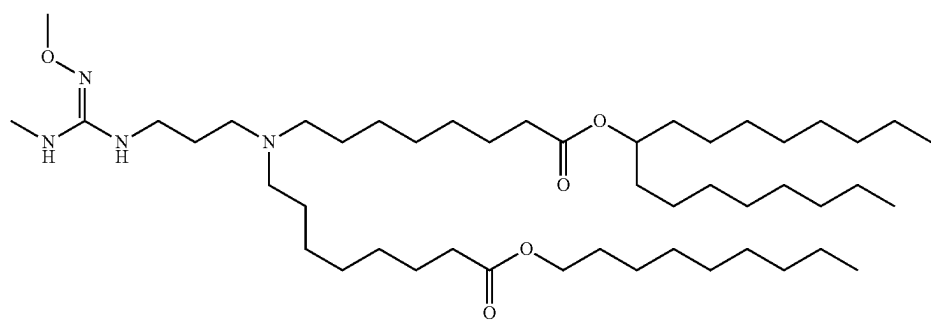
(Compound 207)



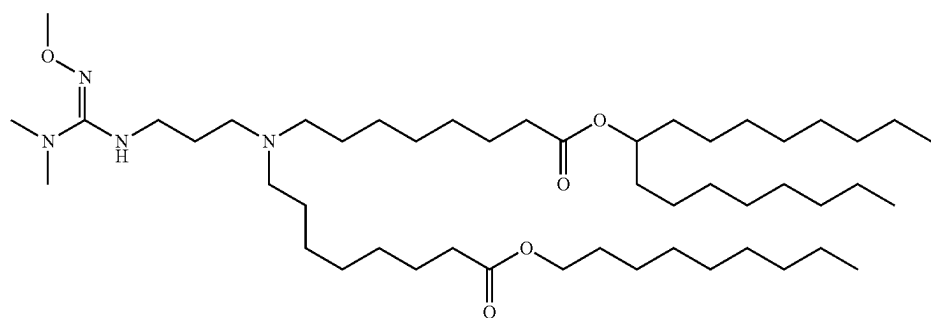
(Compound 208)



(Compound 209)



(Compound 210)

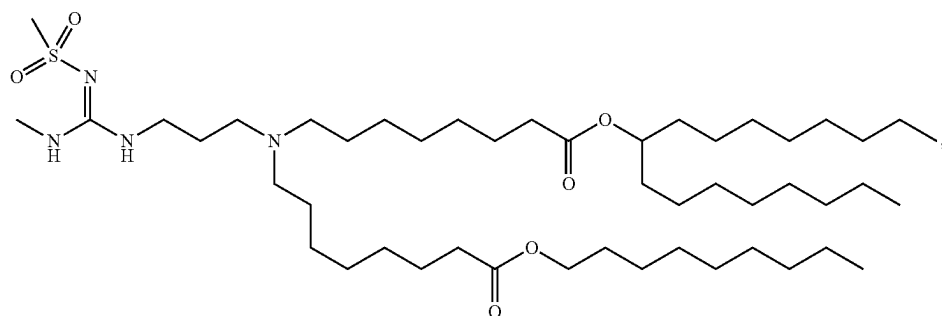


(Compound 211)

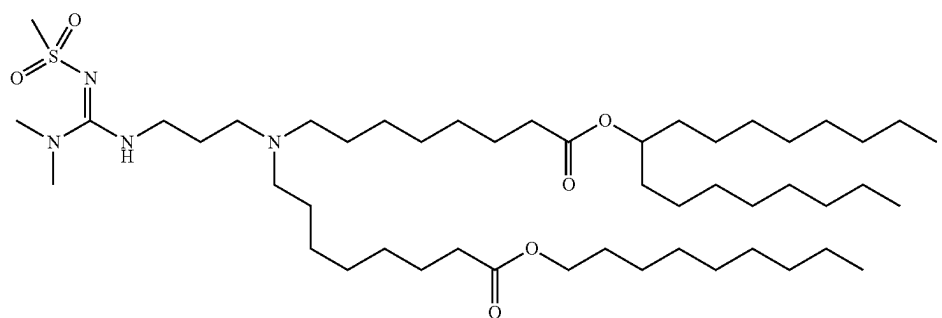
177

178

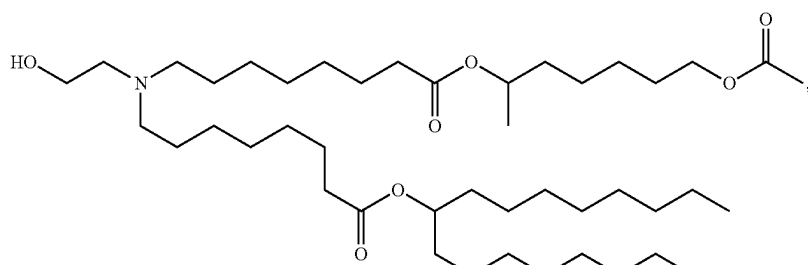
-continued



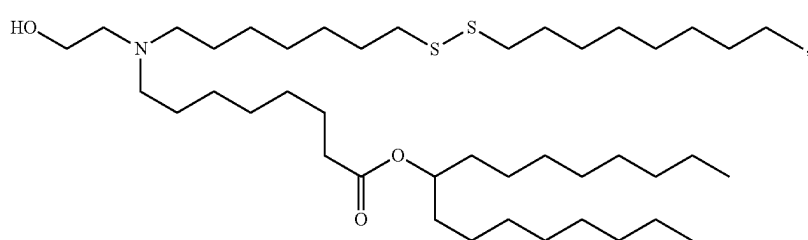
(Compound 212)



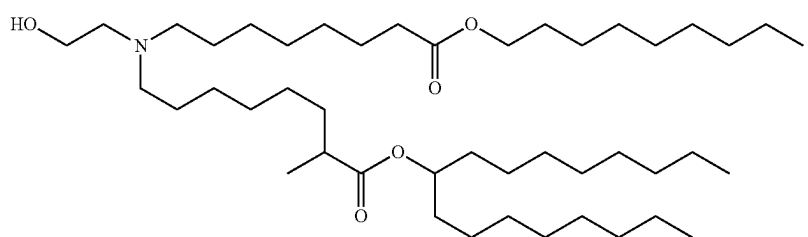
(Compound 213)



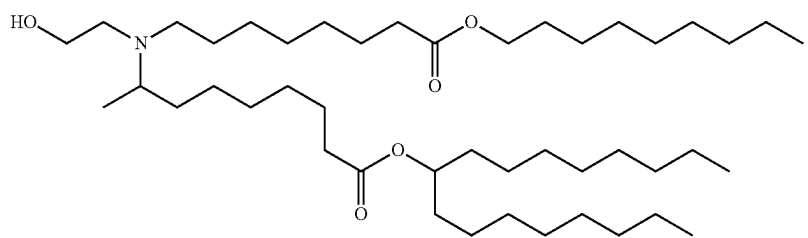
(Compound 214)



(Compound 215)



(Compound 216)

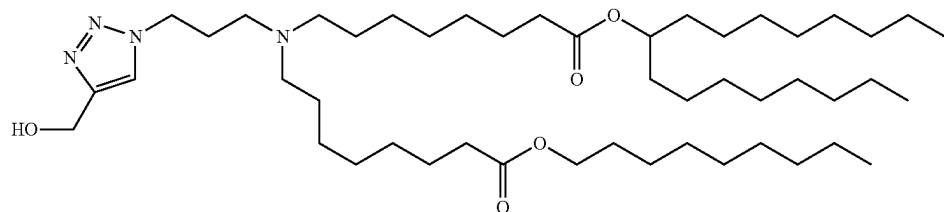


(Compound 217)

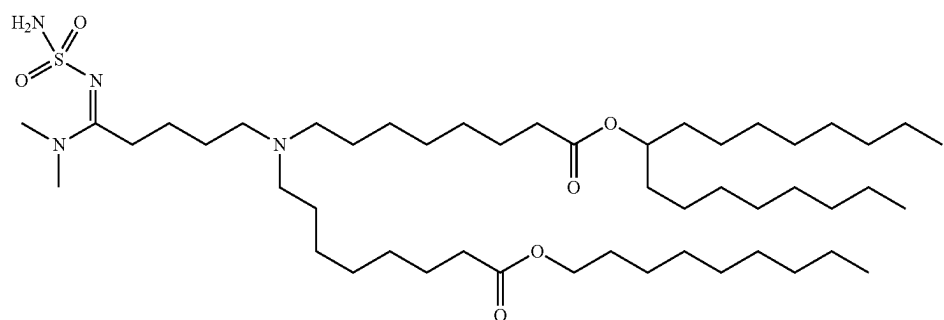
179

180

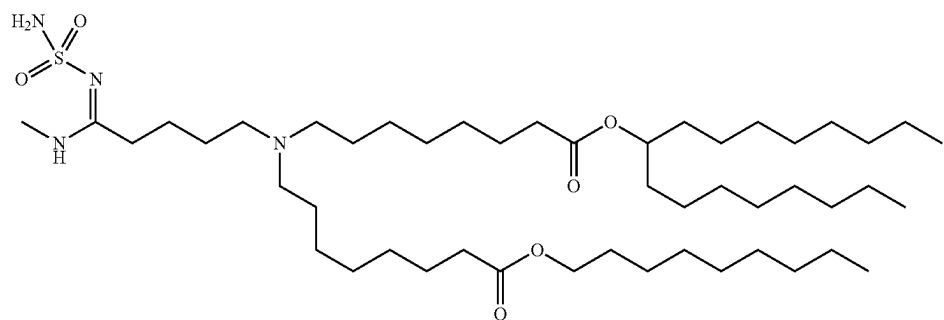
-continued



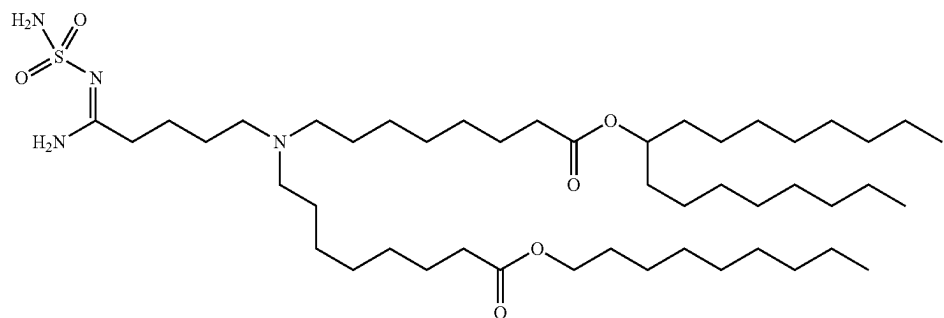
(Compound 218)



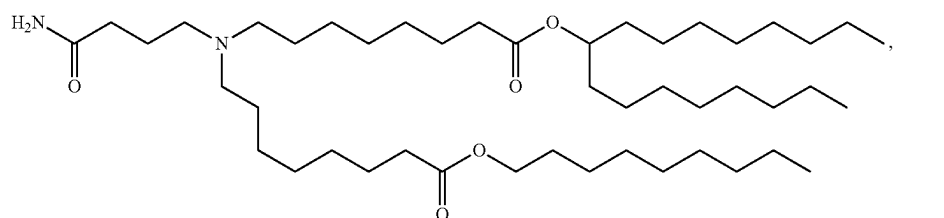
(Compound 219)



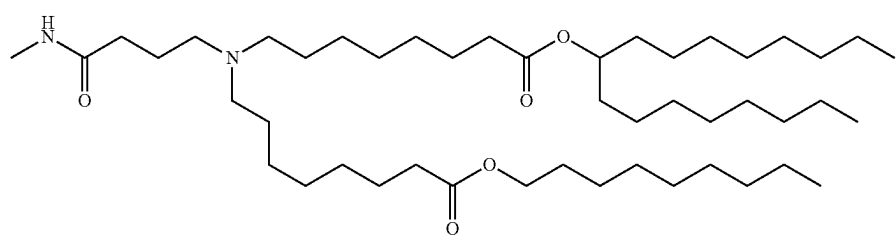
(Compound 220)



(Compound 221)

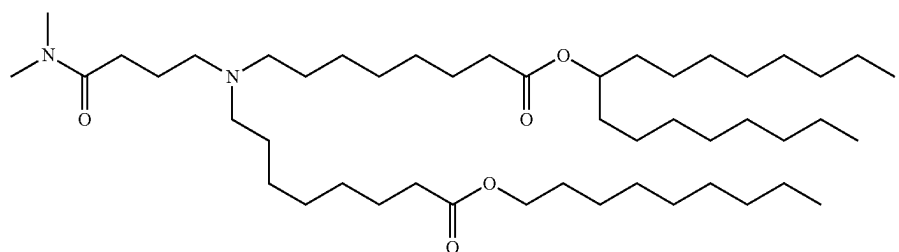


(Compound 222)

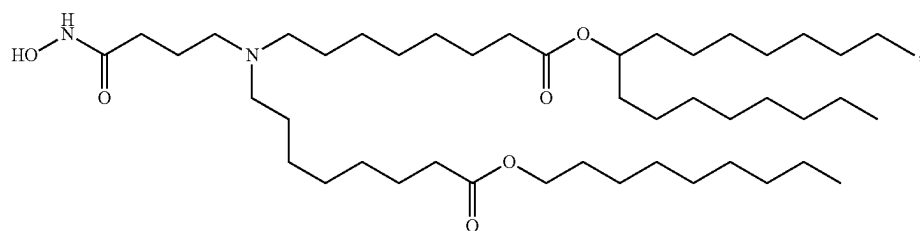


(Compound 223)

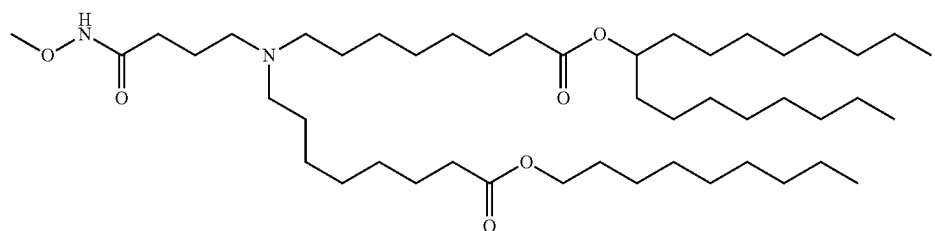
-continued



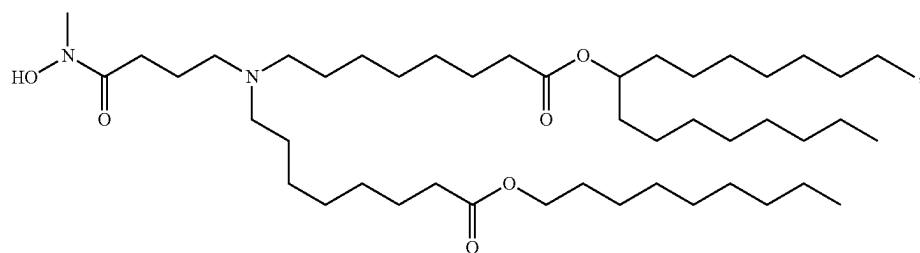
(Compound 224)



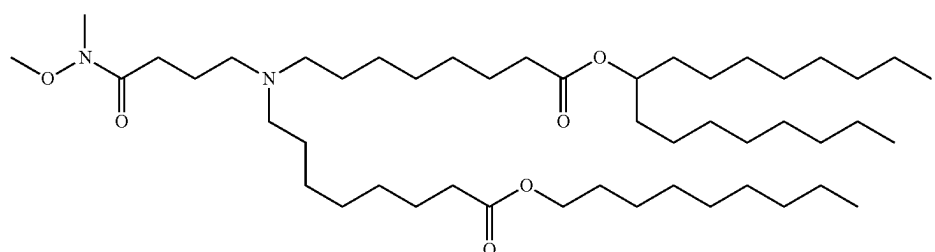
(Compound 225)



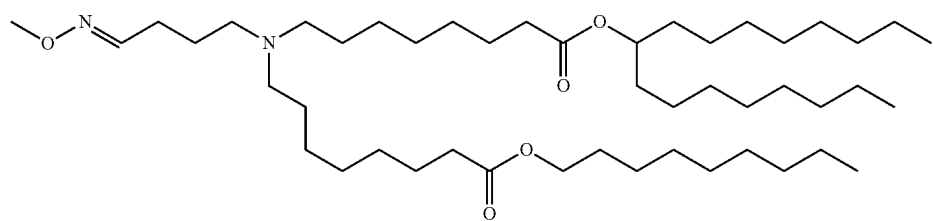
(Compound 226)



(Compound 227)



(Compound 228)



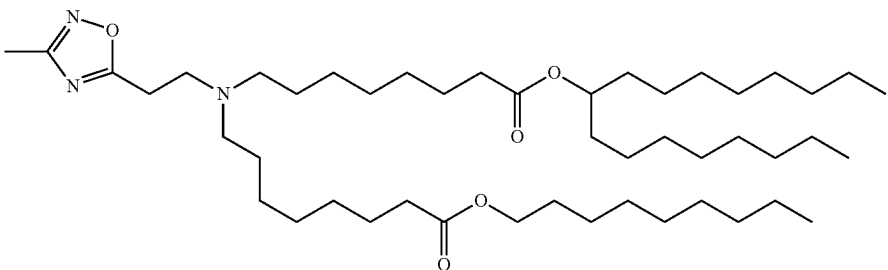
(Compound 229)

183

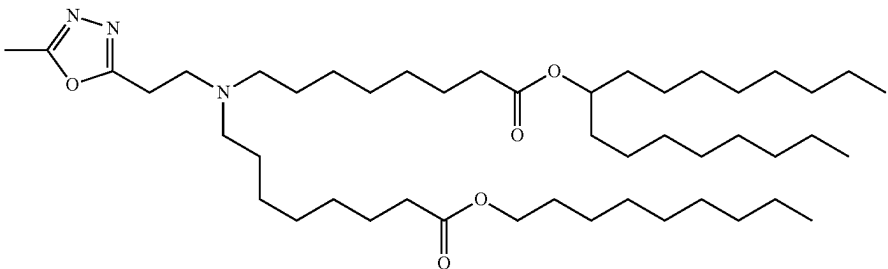
184

-continued

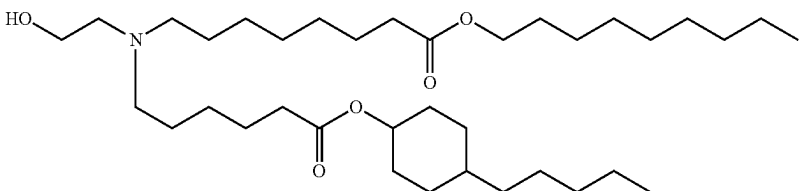
(Compound 230)



(Compound 231)



(Compound 232)

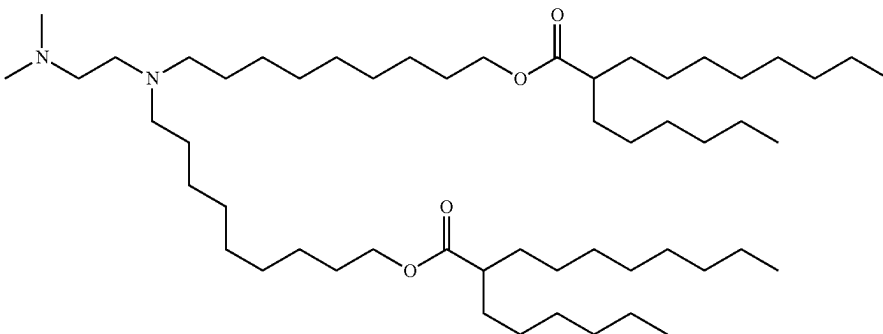


and salts and isomers thereof.

In some embodiments, a nanoparticle comprises the following compound:

(e.g., about -5°C. , -10°C. , -15°C. , -20°C. , -25°C. , -30°C. , -40°C. , -50°C. , -60°C. , -70°C. , -80°C. , -90°C. , -130°C. or -150°C.). For example, the pharmaceutical

(Compound 233)



or salts and isomers thereof.

In some embodiments, the disclosure features a nanoparticle composition including a lipid component comprising a compound as described herein (e.g., a compound according to Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IId) or (Ile)).

In some embodiments, the disclosure features a pharmaceutical composition comprising a nanoparticle composition according to the preceding embodiments and a pharmaceutically acceptable carrier. For example, the pharmaceutical composition is refrigerated or frozen for storage and/or shipment (e.g., being stored at a temperature of 4°C. or lower, such as a temperature between about -150°C. and about 0°C. or between about -80°C. and about -20°C.

composition is a solution that is refrigerated for storage and/or shipment at, for example, about -20°C. , -30°C. , -40°C. , -50°C. , -60°C. , -70°C. , or -80°C.

In some embodiments, the disclosure provides a method of delivering a therapeutic and/or prophylactic (e.g., RNA, such as mRNA) to a cell (e.g., a mammalian cell). This method includes the step of administering to a subject (e.g., a mammal, such as a human) a nanoparticle composition including (i) a lipid component including a phospholipid (such as a polyunsaturated lipid), a PEG lipid, a structural lipid, and a compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IId) or (Ile) and (ii) a therapeutic and/or prophylactic, in which administering involves contacting the cell with the

nanoparticle composition, whereby the therapeutic and/or prophylactic is delivered to the cell.

In some embodiments, the disclosure provides a method of producing a polypeptide of interest in a cell (e.g., a mammalian cell). The method includes the step of contacting the cell with a nanoparticle composition including (i) a lipid component including a phospholipid (such as a polyunsaturated lipid), a PEG lipid, a structural lipid, and a compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IID) or (IIE) and (ii) an mRNA encoding the polypeptide of interest, whereby the mRNA is capable of being translated in the cell to produce the polypeptide.

In some embodiments, the disclosure provides a method of treating a disease or disorder in a mammal (e.g., a human) in need thereof. The method includes the step of administering to the mammal a therapeutically effective amount of a nanoparticle composition including (i) a lipid component including a phospholipid (such as a polyunsaturated lipid), a PEG lipid, a structural lipid, and a compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IID) or (IIE) and (ii) a therapeutic and/or prophylactic (e.g., an mRNA). In some embodiments, the disease or disorder is characterized by dysfunctional or aberrant protein or polypeptide activity. For example, the disease or disorder is selected from the group consisting of rare diseases, infectious diseases, cancer and proliferative diseases, genetic diseases (e.g., cystic fibrosis), autoimmune diseases, diabetes, neurodegenerative diseases, cardio- and reno-vascular diseases, and metabolic diseases.

In some embodiments, the disclosure provides a method of delivering (e.g., specifically delivering) a therapeutic and/or prophylactic to a mammalian organ (e.g., a liver, spleen, lung, or femur). This method includes the step of administering to a subject (e.g., a mammal) a nanoparticle composition including (i) a lipid component including a phospholipid, a PEG lipid, a structural lipid, and a compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IID) or (IIE) and (ii) a therapeutic and/or prophylactic (e.g., an mRNA), in which administering involves contacting the cell with the nanoparticle composition, whereby the therapeutic and/or prophylactic is delivered to the target organ (e.g., a liver, spleen, lung, or femur).

In some embodiments, the disclosure features a method for the enhanced delivery of a therapeutic and/or prophylactic (e.g., an mRNA) to a target tissue (e.g., a liver, spleen, lung, or femur). This method includes administering to a subject (e.g., a mammal) a nanoparticle composition, the composition including (i) a lipid component including a compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IID) or (IIE), a phospholipid, a structural lipid, and a PEG lipid; and (ii) a therapeutic and/or prophylactic, the administering including contacting the target tissue with the nanoparticle composition, whereby the therapeutic and/or prophylactic is delivered to the target tissue.

In some embodiments, the disclosure features a method of lowering immunogenicity comprising introducing the nanoparticle composition of the disclosure into cells, wherein the nanoparticle composition reduces the induction of the cellular immune response of the cells to the nanoparticle composition, as compared to the induction of the cellular immune response in cells induced by a reference composition which comprises a reference lipid instead of a compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IID) or (IIE). For example, the cellular immune response is an innate immune response, an adaptive immune response, or both.

The disclosure also includes methods of synthesizing a compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IID) or (IIE) and methods of making a nanoparticle composition

including a lipid component comprising the compound of Formula (I), (IA), (II), (IIa), (IIb), (IIc), (IID) or (IIE).

Modes of Vaccine Administration

Respiratory virus RNA (e.g., mRNA) vaccines may be administered by any route which results in a therapeutically effective outcome. These include, but are not limited, to intradermal, intramuscular, and/or subcutaneous administration. The present disclosure provides methods comprising administering RNA (e.g., mRNA) vaccines to a subject in need thereof. The exact amount required will vary from subject to subject, depending on the species, age, and general condition of the subject, the severity of the disease, the particular composition, its mode of administration, its mode of activity, and the like. Respiratory virus RNA (e.g., mRNA) vaccines compositions are typically formulated in dosage unit form for ease of administration and uniformity of dosage. It will be understood, however, that the total daily usage of RNA (e.g., mRNA) vaccine compositions may be decided by the attending physician within the scope of sound medical judgment. The specific therapeutically effective, prophylactically effective, or appropriate imaging dose level for any particular patient will depend upon a variety of factors including the disorder being treated and the severity of the disorder; the activity of the specific compound employed; the specific composition employed; the age, body weight, general health, sex and diet of the patient; the time of administration, route of administration, and rate of excretion of the specific compound employed; the duration of the treatment; drugs used in combination or coincidental with the specific compound employed; and like factors well known in the medical arts.

In some embodiments, respiratory virus RNA (e.g., mRNA) vaccines compositions may be administered at dosage levels sufficient to deliver 0.0001 mg/kg to 100 mg/kg, 0.001 mg/kg to 0.05 mg/kg, 0.005 mg/kg to 0.05 mg/kg, 0.001 mg/kg to 0.005 mg/kg, 0.05 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 50 mg/kg, 0.1 mg/kg to 40 mg/kg, 0.5 mg/kg to 30 mg/kg, 0.01 mg/kg to 10 mg/kg, 0.1 mg/kg to 10 mg/kg, or 1 mg/kg to 25 mg/kg, of subject body weight per day, one or more times a day, per week, per month, etc. to obtain the desired therapeutic, diagnostic, prophylactic, or imaging effect (see, e.g., the range of unit doses described in International Publication No WO2013078199, the contents of which are herein incorporated by reference in their entirety). The desired dosage may be delivered three times a day, two times a day, once a day, every other day, every third day, every week, every two weeks, every three weeks, every four weeks, every 2 months, every three months, every 6 months, etc. In some embodiments, the desired dosage may be delivered using multiple administrations (e.g., two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, or more administrations). When multiple administrations are employed, split dosing regimens such as those described herein may be used. In exemplary embodiments, respiratory virus RNA (e.g., mRNA) vaccines compositions may be administered at dosage levels sufficient to deliver 0.0005 mg/kg to 0.01 mg/kg, e.g., about 0.0005 mg/kg to about 0.0075 mg/kg, e.g., about 0.0005 mg/kg, about 0.001 mg/kg, about 0.002 mg/kg, about 0.003 mg/kg, about 0.004 mg/kg or about 0.005 mg/kg.

In some embodiments, respiratory virus RNA (e.g., mRNA) vaccine compositions may be administered once or twice (or more) at dosage levels sufficient to deliver 0.025 mg/kg to 0.250 mg/kg, 0.025 mg/kg to 0.500 mg/kg, 0.025 mg/kg to 0.750 mg/kg, or 0.025 mg/kg to 1.0 mg/kg.

In some embodiments, respiratory virus RNA (e.g., mRNA) vaccine compositions may be administered twice

(e.g., Day 0 and Day 7, Day 0 and Day 14, Day 0 and Day 21, Day 0 and Day 28, Day 0 and Day 60, Day 0 and Day 90, Day 0 and Day 120, Day 0 and Day 150, Day 0 and Day 180, Day 0 and 3 months later, Day 0 and 6 months later, Day 0 and 9 months later, Day 0 and 12 months later, Day 0 and 18 months later, Day 0 and 2 years later, Day 0 and 5 years later, or Day 0 and 10 years later) at a total dose of or at dosage levels sufficient to deliver a total dose of 0.0100 mg, 0.025 mg, 0.050 mg, 0.075 mg, 0.100 mg, 0.125 mg, 0.150 mg, 0.175 mg, 0.200 mg, 0.225 mg, 0.250 mg, 0.275 mg, 0.300 mg, 0.325 mg, 0.350 mg, 0.375 mg, 0.400 mg, 0.425 mg, 0.450 mg, 0.475 mg, 0.500 mg, 0.525 mg, 0.550 mg, 0.575 mg, 0.600 mg, 0.625 mg, 0.650 mg, 0.675 mg, 0.700 mg, 0.725 mg, 0.750 mg, 0.775 mg, 0.800 mg, 0.825 mg, 0.850 mg, 0.875 mg, 0.900 mg, 0.925 mg, 0.950 mg, 0.975 mg, or 1.0 mg. Higher and lower dosages and frequency of administration are encompassed by the present disclosure. For example, a respiratory virus RNA (e.g., mRNA) vaccine composition may be administered three or four times.

In some embodiments, respiratory virus RNA (e.g., mRNA) vaccine compositions may be administered twice (e.g., Day 0 and Day 7, Day 0 and Day 14, Day 0 and Day 21, Day 0 and Day 28, Day 0 and Day 60, Day 0 and Day 90, Day 0 and Day 120, Day 0 and Day 150, Day 0 and Day 180, Day 0 and 3 months later, Day 0 and 6 months later, Day 0 and 9 months later, Day 0 and 12 months later, Day 0 and 18 months later, Day 0 and 2 years later, Day 0 and 5 years later, or Day 0 and 10 years later) at a total dose of or at dosage levels sufficient to deliver a total dose of 0.010 mg, 0.025 mg, 0.100 mg or 0.400 mg.

In some embodiments, the respiratory virus RNA (e.g., mRNA) vaccine for use in a method of vaccinating a subject is administered to the subject as a single dosage of between 10 µg/kg and 400 µg/kg of the nucleic acid vaccine (in an effective amount to vaccinate the subject). In some embodiments the RNA (e.g., mRNA) vaccine for use in a method of vaccinating a subject is administered to the subject as a single dosage of between 10 µg and 400 µg of the nucleic acid vaccine (in an effective amount to vaccinate the subject). In some embodiments, a respiratory virus RNA (e.g., mRNA) vaccine for use in a method of vaccinating a subject is administered to the subject as a single dosage of 25-1000 µg (e.g., a single dosage of mRNA encoding hMPV, PIV3, RSV, MeV and/or BetaCoV antigen). In some embodiments, a respiratory virus RNA (e.g., mRNA) vaccine is administered to the subject as a single dosage of 25, 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950 or 1000 µg. For example, a respiratory virus RNA (e.g., mRNA) vaccine may be administered to a subject as a single dose of 25-100, 25-500, 50-100, 50-500, 50-1000, 100-500, 100-1000, 250-500, 250-1000, or 500-1000 µg. In some embodiments, a respiratory virus RNA (e.g., mRNA) vaccine for use in a method of vaccinating a subject is administered to the subject as two dosages, the combination of which equals 25-1000 µg of the respiratory virus RNA (e.g., mRNA) vaccine.

A respiratory virus RNA (e.g. mRNA) vaccine pharmaceutical composition described herein can be formulated into a dosage form described herein, such as an intranasal, intratracheal, or injectable (e.g., intravenous, intraocular, intravitreal, intramuscular, intradermal, intracardiac, intraperitoneal, and subcutaneous).

Respiratory Virus RNA (e.g., mRNA) Vaccine Formulations and Methods of Use

Some aspects of the present disclosure provide formulations of the respiratory virus RNA (e.g., mRNA) vaccine,

wherein the RNA (e.g., mRNA) vaccine is formulated in an effective amount to produce an antigen specific immune response in a subject (e.g., production of antibodies specific to an hMPV, PIV3, RSV, MeV and/or BetaCoV antigenic polypeptide). "An effective amount" is a dose of an RNA (e.g., mRNA) vaccine effective to produce an antigen-specific immune response. Also provided herein are methods of inducing an antigen-specific immune response in a subject.

In some embodiments, the antigen-specific immune response is characterized by measuring an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide antibody titer produced in a subject administered a respiratory virus RNA (e.g., mRNA) vaccine as provided herein. An antibody titer is a measurement of the amount of antibodies within a subject, for example, antibodies that are specific to a particular antigen (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) or epitope of an antigen. Antibody titer is typically expressed as the inverse of the greatest dilution that provides a positive result. Enzyme-linked immunosorbent assay (ELISA) is a common assay for determining antibody titers, for example.

In some embodiments, an antibody titer is used to assess whether a subject has had an infection or to determine whether immunizations are required. In some embodiments, an antibody titer is used to determine the strength of an autoimmune response, to determine whether a booster immunization is needed, to determine whether a previous vaccine was effective, and to identify any recent or prior infections. In accordance with the present disclosure, an antibody titer may be used to determine the strength of an immune response induced in a subject by the respiratory virus RNA (e.g., mRNA) vaccine.

In some embodiments, an anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject is increased by at least 1 log relative to a control. For example, anti-antigenic polypeptide antibody titer produced in a subject may be increased by at least 1.5, at least 2, at least 2.5, or at least 3 log relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased by 1, 1.5, 2, 2.5 or 3 log relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased by 1-3 log relative to a control. For example, the anti-antigenic polypeptide antibody titer produced in a subject may be increased by 1-1.5, 1-2, 1-2.5, 1-3, 1.5-2, 1.5-2.5, 1.5-3, 2-2.5, 2-3, or 2.5-3 log relative to a control.

In some embodiments, the anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject is increased at least 2 times relative to a control. For example, the anti-antigenic polypeptide antibody titer produced in a subject may be increased at least 3 times, at least 4 times, at least 5 times, at least 6 times, at least 7 times, at least 8 times, at least 9 times, or at least 10 times relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is increased 2, 3, 4, 5, 6, 7, 8, 9, or 10 times relative to a control. In some embodiments, the anti-antigenic polypeptide antibody titer produced in a subject is increased 2-10 times relative to a control. For example, the anti-antigenic polypeptide antibody titer produced in a subject may be increased 2-10, 2-9, 2-8, 2-7, 2-6, 2-5, 2-4, 2-3, 3-10, 3-9, 3-8, 3-7, 3-6, 3-5, 3-4, 4-10, 4-9, 4-8, 4-7, 4-6, 4-5, 5-10, 5-9,

5-8, 5-7, 5-6, 6-10, 6-9, 6-8, 6-7, 7-10, 7-9, 7-8, 8-10, 8-9, or 9-10 times relative to a control.

A control, in some embodiments, is the anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject who has not been administered a respiratory virus RNA (e.g., mRNA) vaccine of the present disclosure. In some embodiments, a control is an anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject who has been administered a live attenuated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine. An attenuated vaccine is a vaccine produced by reducing the virulence of a viable (live). An attenuated virus is altered in a manner that renders it harmless or less virulent relative to live, unmodified virus. In some embodiments, a control is an anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject administered inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine. In some embodiments, a control is an anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject administered a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. Recombinant protein vaccines typically include protein antigens that either have been produced in a heterologous expression system (e.g., bacteria or yeast) or purified from large amounts of the pathogenic organism. In some embodiments, a control is an anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject who has been administered an hMPV, PIV3, RSV, MeV and/or BetaCoV virus-like particle (VLP) vaccine. For example, an hMPV VLP vaccine used as a control may be a hMPV VLPs, comprising (or consisting of) viral matrix (M) and fusion (F) proteins, generated by expressing viral proteins in suspension-adapted human embryonic kidney epithelial (293-F) cells (see, e.g., Cox R G et al., *J Virol.* 2014 June; 88(11): 6368-6379, the contents of which are herein incorporated by reference).

In some embodiments, an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a dose that is reduced compared to the standard of care dose of a recombinant hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. A "standard of care," as provided herein, refers to a medical or psychological treatment guideline and can be general or specific. "Standard of care" specifies appropriate treatment based on scientific evidence and collaboration between medical professionals involved in the treatment of a given condition. It is the diagnostic and treatment process that a physician/clinician should follow for a certain type of patient, illness or clinical circumstance. A "standard of care dose," as provided herein, refers to the dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, or a live attenuated or inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine, that a physician/clinician or other medical professional would administer to a subject to treat or prevent hMPV, PIV3, RSV, MeV and/or BetaCoV, or a hMPV-, PIV3-, RSV-, MeV- and/or BetaCoV-related condition, while following the standard of care guideline for treating or preventing hMPV, PIV3, RSV, MeV and/or BetaCoV, or a hMPV-, PIV3-, RSV-, MeV- and/or BetaCoV-related condition.

In some embodiments, the anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or

anti-BetaCoV antigenic polypeptide) antibody titer produced in a subject administered an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is equivalent to an anti-antigenic polypeptide (e.g., an anti-hMPV, anti-PIV3, anti-RSV, anti-MeV and/or anti-BetaCoV antigenic polypeptide) antibody titer produced in a control subject administered a standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine or a live attenuated or inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine.

In some embodiments, an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a dose equivalent to an at least 2-fold reduction in a standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. For example, an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine may be a dose equivalent to an at least 3-fold, at least 4-fold, at least 5-fold, at least 6-fold, at least 7-fold, at least 8-fold, at least 9-fold, or at least 10-fold reduction in a standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. In some embodiments, an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a dose equivalent to an at least 100-fold, at least 500-fold, or at least 1000-fold reduction in a standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. In some embodiments, an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a dose equivalent to a 2-, 3-, 4-, 5-, 6-, 7-, 8-, 9-, 10-, 20-, 50-, 100-, 250-, 500-, or 1000-fold reduction in a standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. In some embodiments, the anti-antigenic polypeptide antibody titer produced in a subject administered an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a recombinant or protein hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine or a live attenuated or inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine. In some embodiments, an effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a dose equivalent to a 2-fold to 1000-fold (e.g., 2-fold to 100-fold, 10-fold to 1000-fold) reduction in the standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine, wherein the anti-antigenic polypeptide antibody titer produced in the subject is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine or a live attenuated or inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine.

In some embodiments, the effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a dose equivalent to a 2 to 1000-, 2 to 900-, 2 to 800-, 2 to 700-, 2 to 600-, 2 to 500-, 2 to 400-, 2 to 300-, 2 to 200-, 2 to 100-, 2 to 90-, 2 to 80-, 2 to 70-, 2 to 60-, 2 to 50-, 2 to 40-, 2 to 30-, 2 to 20-, 2 to 10-, 2 to 9-, 2 to 8-, 2 to 7-, 2 to 6-, 2 to 5-, 2 to 4-, 2 to 3-, 3 to 1000-, 3 to 900-, 3 to 800-, 3 to 700-, 3 to 600-, 3 to 500-, 3 to 400-, 3 to 300-, 3 to 200-, 3 to 100-, 3 to 90-, 3 to 80-, 3 to 70-, 3 to 60-, 3 to 50-, 3 to 40-, 3 to 30-, 3 to 20-, 3 to 10-, 3 to 9-, 3 to 8-, 3 to 7-, 3 to 6-, 3 to 5-, 3 to 4-, 4 to 1000-, 4 to 900-, 4 to 800-, 4 to 700-, 4 to 600-, 4 to 500-, 4 to 400-, 4 to 300-, 4 to 200-, 4 to 100-, 4 to 90-, 4 to 80-, 4 to 70-, 4 to 60-, 4 to 50-, 4 to 40-, 4 to 30-, 4 to 20-, 4 to 10-, 4 to 9-, 4 to 8-, 4 to 7-, 4 to 6-, 4 to 5-, 4 to 4-, 5 to 1000-, 5 to 900-, 5 to 800-, 5 to 700-, 5 to

191

600-, 5 to 500-, 5 to 400-, 5 to 300-, 5 to 200-, 5 to 100-, 5 to 90-, 5 to 80-, 5 to 70-, 5 to 60-, 5 to 50-, 5 to 40-, 5 to 30-, 5 to 20-, 5 to 10-, 5 to 9-, 5 to 8-, 5 to 7-, 5 to 6-, 6 to 1000-, 6 to 900-, 6 to 800-, 6 to 700-, 6 to 600-, 6 to 500-, 6 to 400-, 6 to 300-, 6 to 200-, 6 to 100-, 6 to 90-, 6 to 80-, 6 to 70-, 6 to 60-, 6 to 50-, 6 to 40-, 6 to 30-, 6 to 20-, 6 to 10-, 6 to 9-, 6 to 8-, 6 to 7-, 7 to 1000-, 7 to 900-, 7 to 800-, 7 to 700-, 7 to 600-, 7 to 500-, 7 to 400-, 7 to 300-, 7 to 200-, 7 to 100-, 7 to 90-, 7 to 80-, 7 to 70-, 7 to 60-, 7 to 50-, 7 to 40-, 7 to 30-, 7 to 20-, 7 to 10-, 7 to 9-, 7 to 8-, 8 to 1000-, 8 to 900-, 8 to 800-, 8 to 700-, 8 to 600-, 8 to 500-, 8 to 400-, 8 to 300-, 8 to 200-, 8 to 100-, 8 to 90-, 8 to 80-, 8 to 70-, 8 to 60-, 8 to 50-, 8 to 40-, 8 to 30-, 8 to 20-, 8 to 10-, 8 to 9-, 9 to 1000-, 9 to 900-, 9 to 800-, 9 to 700-, 9 to 600-, 9 to 500-, 9 to 400-, 9 to 300-, 9 to 200-, 9 to 100-, 9 to 90-, 9 to 80-, 9 to 70-, 9 to 60-, 9 to 50-, 9 to 40-, 9 to 30-, 9 to 20-, 9 to 10-, 10 to 1000-, 10 to 900-, 10 to 800-, 10 to 700-, 10 to 600-, 10 to 500-, 10 to 400-, 10 to 300-, 10 to 200-, 10 to 100-, 10 to 90-, 10 to 80-, 10 to 70-, 10 to 60-, 10 to 50-, 10 to 40-, 10 to 30-, 10 to 20-, 20 to 1000-, 20 to 900-, 20 to 800-, 20 to 700-, 20 to 600-, 20 to 500-, 20 to 400-, 20 to 300-, 20 to 200-, 20 to 100-, 20 to 90-, 20 to 80-, 20 to 70-, 20 to 60-, 20 to 50-, 20 to 40-, 20 to 30-, 30 to 1000-, 30 to 900-, 30 to 800-, 30 to 700-, 30 to 600-, 30 to 500-, 30 to 400-, 30 to 300-, 30 to 200-, 30 to 100-, 30 to 90-, 30 to 80-, 30 to 70-, 30 to 60-, 30 to 50-, 30 to 40-, 40 to 1000-, 40 to 900-, 40 to 800-, 40 to 700-, 40 to 600-, 40 to 500-, 40 to 400-, 40 to 300-, 40 to 200-, 40 to 100-, 40 to 90-, 40 to 80-, 40 to 70-, 40 to 60-, 40 to 50-, 50 to 1000-, 50 to 900-, 50 to 800-, 50 to 700-, 50 to 600-, 50 to 500-, 50 to 400-, 50 to 300-, 50 to 200-, 50 to 100-, 50 to 90-, 50 to 80-, 50 to 70-, 50 to 60-, 60 to 1000-, 60 to 900-, 60 to 800-, 60 to 700-, 60 to 600-, 60 to 500-, 60 to 400-, 60 to 300-, 60 to 200-, 60 to 100-, 60 to 90-, 60 to 80-, 60 to 70-, 70 to 1000-, 70 to 900-, 70 to 800-, 70 to 700-, 70 to 600-, 70 to 500-, 70 to 400-, 70 to 300-, 70 to 200-, 70 to 100-, 70 to 90-, 70 to 80-, 80 to 1000-, 80 to 900-, 80 to 800-, 80 to 700-, 80 to 600-, 80 to 500-, 80 to 400-, 80 to 300-, 80 to 200-, 80 to 100-, 80 to 90-, 90 to 1000-, 90 to 900-, 90 to 800-, 90 to 700-, 90 to 600-, 90 to 500-, 90 to 400-, 90 to 300-, 90 to 200-, 90 to 100-, 100 to 1000-, 100 to 900-, 100 to 800-, 100 to 700-, 100 to 600-, 100 to 500-, 100 to 400-, 100 to 300-, 100 to 200-, 200 to 1000-, 200 to 900-, 200 to 800-, 200 to 700-, 200 to 600-, 200 to 500-, 200 to 400-, 200 to 300-, 300 to 1000-, 300 to 900-, 300 to 800-, 300 to 700-, 300 to 600-, 300 to 500-, 300 to 400-, 400 to 1000-, 400 to 900-, 400 to 800-, 400 to 700-, 400 to 600-, 400 to 500-, 500 to 1000-, 500 to 900-, 500 to 800-, 500 to 700-, 500 to 600-, 600 to 1000-, 600 to 900-, 600 to 800-, 600 to 700-, 700 to 1000-, 700 to 900-, 700 to 800-, 800 to 1000-, 800 to 900-, or 900 to 1000-fold reduction in the standard of care dose of a recombinant hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. In some embodiments, the anti-antigenic polypeptide antibody titer produced in the subject is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine or a live attenuated or inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine. In some embodiments, the effective amount is a dose equivalent to (or equivalent to an at least) 2-, 3-, 4-, 5-, 6-, 7-, 8-, 9-, 10-, 20-, 30-, 40-, 50-, 60-, 70-, 80-, 90-, 100-, 110-, 120-, 130-, 140-, 150-, 160-, 170-, 1280-, 190-, 200-, 210-, 220-, 230-, 240-, 250-, 260-, 270-, 280-, 290-, 300-, 310-, 320-, 330-, 340-, 350-, 360-, 370-, 380-, 390-, 400-, 410-, 420-, 430-, 440-, 450-, 4360-, 470-, 480-, 490-, 500-, 510-, 520-, 530-, 540-, 550-, 560-, 5760-, 580-, 590-, 600-, 610-, 620-, 630-,

192

640-, 650-, 660-, 670-, 680-, 690-, 700-, 710-, 720-, 730-, 740-, 750-, 760-, 770-, 780-, 790-, 800-, 810-, 820-, 830-, 840-, 850-, 860-, 870-, 880-, 890-, 900-, 910-, 920-, 930-, 940-, 950-, 960-, 970-, 980-, 990-, or 1000-fold reduction in the standard of care dose of a recombinant hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine. In some embodiments, an anti-antigenic polypeptide antibody titer produced in the subject is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a recombinant or purified hMPV, PIV3, RSV, MeV and/or BetaCoV protein vaccine or a live attenuated or inactivated hMPV, PIV3, RSV, MeV and/or BetaCoV vaccine.

In some embodiments, the effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a total dose of 50-1000 g. In some embodiments, the effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a total dose of 50-1000, 50-900, 50-800, 50-700, 50-600, 50-500, 50-400, 50-300, 50-200, 50-100, 50-90, 50-80, 50-70, 50-60, 60-1000, 60-900, 60-800, 60-700, 60-600, 60-500, 60-400, 60-300, 60-200, 60-100, 60-90, 60-80, 60-70, 70-1000, 70-900, 70-800, 70-700, 70-600, 70-500, 70-400, 70-300, 70-200, 70-100, 70-90, 70-80, 80-1000, 80-900, 80-800, 80-700, 80-600, 80-500, 80-400, 80-300, 80-200, 80-100, 80-90, 90-1000, 90-900, 90-800, 90-700, 90-600, 90-500, 90-400, 90-300, 90-200, 90-100, 100-1000, 100-900, 100-800, 100-700, 100-600, 100-500, 100-400, 100-300, 100-200, 200-1000, 200-900, 200-800, 200-700, 200-600, 200-500, 200-400, 200-300, 300-1000, 300-900, 300-800, 300-700, 300-600, 300-500, 300-400, 400-1000, 400-900, 400-800, 400-700, 400-600, 400-500, 500-1000, 500-900, 500-800, 500-700, 500-600, 600-1000, 600-900, 600-800, 600-700, 700-1000, 700-900, 700-800, 800-1000, 800-900, or 900-1000 g. In some embodiments, the effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a total dose of 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950 or 1000 g. In some embodiments, the effective amount is a dose of 25-500 g administered to the subject a total of two times. In some embodiments, the effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a dose of 25-500, 25-400, 25-300, 25-200, 25-100, 25-50, 50-500, 50-400, 50-300, 50-200, 50-100, 100-500, 100-400, 100-300, 100-200, 150-500, 150-400, 150-300, 150-200, 200-500, 200-400, 200-300, 250-500, 250-400, 250-300, 300-500, 300-400, 350-500, 350-400, 400-500 or 450-500 g administered to the subject a total of two times. In some embodiments, the effective amount of a respiratory virus RNA (e.g., mRNA) vaccine is a total dose of 25, 50, 100, 150, 200, 250, 300, 350, 400, 450, or 500 µg administered to the subject a total of two times.

EXAMPLES OF ADDITIONAL EMBODIMENTS OF THE DISCLOSURE

Additional embodiments of the present disclosure are encompassed by the following numbered paragraphs:

1. A respiratory virus vaccine, comprising: at least one ribonucleic acid (RNA) polynucleotide having an open reading frame encoding at least one, at least two, at least three, at least four or at least five antigenic polypeptides selected from human metapneumovirus (hMPV) antigenic polypeptides or immunogenic fragments thereof, human parainfluenza virus type 3 (PIV3) antigenic polypeptides or immunogenic fragments thereof, respiratory syncytial virus (RSV) antigenic polypeptides or immunogenic fragments thereof, measles virus (MeV) antigenic polypeptides or

at least one RNA polynucleotide having an open reading frame encoding a hMPV antigenic polypeptide or an immunogenic fragment thereof, a PIV3 antigenic polypeptide or an immunogenic fragment thereof, and a BetaCoV antigenic polypeptide or an immunogenic fragment thereof; or

36. The respiratory virus vaccine of paragraph 1, comprising:

at least two, three or four RNA polynucleotides, one having an open reading frame encoding a hMPV antigenic polypeptide or an immunogenic fragment thereof, one having an open reading frame encoding a PIV3 antigenic polypeptide or an immunogenic fragment thereof, one having an open reading frame encoding a RSV antigenic polypeptide or an immunogenic fragment thereof, and one having an open reading

at least one RNA polynucleotide having an open reading frame encoding a hMPV antigenic polypeptide or an immunogenic fragment thereof, a PIV3 antigenic polypeptide or an immunogenic fragment thereof, a RSV antigenic polypeptide or an immunogenic fragment thereof, a MeV antigenic polypeptide or an immunogenic fragment thereof, and a BetaCoV antigenic polypeptide or an immunogenic fragment thereof; or

at least two, three, four or five RNA polynucleotides, one having an open reading frame encoding a hMPV antigenic polypeptide or an immunogenic fragment thereof, one having an open reading frame encoding a PIV3 antigenic polypeptide or an immunogenic fragment thereof, one having an open reading frame encoding a RSV antigenic polypeptide or an immunogenic fragment thereof, one having an open reading frame encoding a MeV antigenic polypeptide or an immunogenic fragment thereof, and one having an open reading frame encoding a BetaCoV antigenic polypeptide or an immunogenic fragment thereof.

53. The respiratory virus vaccine of paragraph 52, wherein the hMPV antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 5-8 or an amino acid sequence having at least 90% or 95% identity to an amino acid sequence identified by any one of SEQ ID NO: 5-8, wherein the PIV3 antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 12-13 or an amino acid sequence having at least 90% or 95% identity to an amino acid sequence identified by any one of SEQ ID NO: 12-13, wherein the MeV antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 47-50 or an amino acid sequence having at least 90% or 95% identity to an amino acid sequence identified by any one of SEQ ID NO: 47-50, and/or wherein the BetaCoV antigenic polypeptide comprises an amino acid sequence identified by any one of SEQ ID NO: 24-34 or an amino acid sequence having at least 90% or 95% identity to an amino acid sequence identified by any one of SEQ ID NO: 24-34.

54. The vaccine of any one of paragraphs 1-53, wherein at least one RNA polynucleotide has less than 80% identity to wild-type mRNA sequence.

55. The vaccine of any one of paragraphs 1-53, wherein at least one RNA polynucleotide has at least 80% identity to wild-type mRNA sequence, but does not include wild-type mRNA sequence.

56. The vaccine of any one of paragraphs 1-55, wherein at least one antigenic polypeptide has membrane fusion activity, attaches to cell receptors, causes fusion of viral and cellular membranes, and/or is responsible for binding of the virus to a cell being infected.

57. The vaccine of any one of paragraphs 1-56, wherein at least one RNA polynucleotide comprises at least one chemical modification.

58. The vaccine of paragraph 57, wherein the chemical modification is selected from pseudouridine, N1-methylpseudouridine, N1-ethylpseudouridine, 2-thiouridine, 4'-thiouridine, 5-methylcytosine, 5-methyluridine, 2-thio-1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-pseudouridine, 2-thio-5-aza-uridine, 2-thio-dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-pseudouridine, 4-methoxy-2-thio-pseudouridine, 4-methoxy-pseudouridine, 4-thio-1-methyl-pseudouridine, 4-thio-pseudouridine, 5-aza-uridine, dihydropseudouridine, 5-methoxyuridine and 2'-O-methyluridine.

59. The vaccine of paragraph 57 or 58, wherein the chemical modification is in the 5-position of the uracil.

60. The vaccine of any one of paragraphs 57-59, wherein the chemical modification is a N1-methylpseudouridine or N1-ethylpseudouridine.

61. The vaccine of any one of paragraphs 57-60, wherein at least 80%, at least 90% or 100% of the uracil in the open reading frame have a chemical modification.

62. The vaccine of any one of paragraphs 1-61, wherein at least one RNA polynucleotide further encodes at least one 5' terminal cap, optionally wherein the 5' terminal cap is 7mG(5')ppp(5')NlmpNp.

63. The vaccine of any one of paragraphs 1-62, wherein at least one antigenic polypeptide or immunogenic fragment thereof is fused to a signal peptide selected from: a HuIgGk signal peptide (METPAQLLFLLLLWLPDITG; SEQ ID NO: 15); IgE heavy chain epsilon-1 signal peptide (MDWTWILFLVAAATRVHS; SEQ ID NO: 16); Japanese encephalitis PRM signal sequence (MLG-SNSGQRVVFITLLLLLVAPAYS; SEQ ID NO: 17); VSVg protein signal sequence (MKCLLYLAFLFIGVNCA; SEQ ID NO: 18) and Japanese encephalitis JEV signal sequence (MWLVSLAIVTACAGA; SEQ ID NO: 19).

64. The vaccine of paragraph 63, wherein the signal peptide is fused to the N-terminus or the C-terminus of at least one antigenic polypeptide.

65. The vaccine of any one of paragraphs 1-64, wherein the antigenic polypeptide or immunogenic fragment thereof comprises a mutated N-linked glycosylation site.

66. The vaccine of any one of paragraphs 1-65 formulated in a nanoparticle, optionally a lipid nanoparticle.

67. The vaccine of paragraph 66, wherein the lipid nanoparticle comprises a cationic lipid, a PEG-modified lipid, a sterol and a non-cationic lipid; optionally wherein the lipid nanoparticle carrier comprises a molar ratio of about 20-60% cationic lipid, 0.5-15% PEG-modified lipid, 25-55% sterol, and 25% non-cationic lipid; optionally wherein the cationic lipid is an ionizable cationic lipid and the non-cationic lipid is a neutral lipid, and the sterol is a cholesterol; and optionally wherein the cationic lipid is selected from 2,2-dilinoleyl-4-dimethylaminoethyl-1,3-dioxolane (DLin-KC2-DMA), dilinoleyl-methyl-4-dimethylaminobutyrate (DLin-MC3-DMA), and di((Z)-non-2-en-1-yl) 9-((4-(dimethylamino)butanoyl)oxy)heptadecanedioate (L319). Formula (II)

68. The vaccine of paragraph 66 or 67, wherein the nanoparticle (e.g., lipid nanoparticle) comprises a compound of Formula (I) and/or Formula (II), optionally Compound 3, 18, 20, 25, 26, 29, 30, 60, 108-112, or 122.

69. The vaccine of any one of paragraphs 1-68 further comprising an adjuvant, optionally a flagellin protein or peptide that optionally comprises an amino acid sequence identified by any one of SEQ ID NO: 54-56.

70. The vaccine of any one of paragraphs 1-69, wherein the open reading frame is codon-optimized.

71. The vaccine of any one of paragraphs 1-70 formulated in an effective amount to produce an antigen-specific immune response.

72. A method of inducing an immune response in a subject, the method comprising administering to the subject the vaccine of any one of paragraphs 1-71 in an amount effective to produce an antigen-specific immune response in the subject.

73. The method of paragraph 72, wherein the subject is administered a single dose of the vaccine, or wherein the subject is administered a first dose and then a booster dose of the vaccine.

74. The method of paragraph 72 or 73, wherein the vaccine is administered to the subject by intradermal injection or intramuscular injection.

75. The method of any one of paragraphs 72-74, wherein an anti-antigenic polypeptide antibody titer produced in the subject is increased by at least 1 log relative to a control,

and/or wherein the anti-antigenic polypeptide antibody titer produced in the subject is increased at least 2 times relative to a control.

76. The method of any one of paragraphs 72-75, wherein the control is an anti-antigenic polypeptide antibody titer produced in a subject who has not been administered a vaccine against the virus, and/or wherein the control is an anti-antigenic polypeptide antibody titer produced in a subject who has been administered a live attenuated vaccine or an inactivated vaccine against the virus, and/or wherein the control is an anti-antigenic polypeptide antibody titer produced in a subject who has been administered a recombinant protein vaccine or purified protein vaccine against the virus, and/or wherein the control is an anti-antigenic polypeptide antibody titer produced in a subject who has been administered a VLP vaccine against the virus.

77. The method of any one of paragraphs 72-76, wherein the effective amount is a dose equivalent to an at least 2-fold reduction in the standard of care dose of a recombinant protein vaccine or a purified protein vaccine against the virus, and wherein an anti-antigenic polypeptide antibody titer produced in the subject is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a recombinant protein vaccine or a purified protein vaccine against the virus, respectively; and/or wherein the effective amount is a dose equivalent to an at least 2-fold reduction in the standard of care dose of a live attenuated vaccine or an inactivated vaccine against the virus, and wherein an anti-antigenic polypeptide antibody titer produced in the subject is equivalent to an anti-antigenic polypeptide antibody titer produced in a control subject administered the standard of care dose of a VLP vaccine against the virus.

78. The method of any one of paragraphs 72-77, wherein the effective amount is a total dose of 50 g-1000 µg, optionally wherein the effective amount is a dose of 25 µg, 100 g, 400 µg, or 500 µg administered to the subject a total of two times.

79. The method of any one of paragraphs 72-78, wherein the efficacy of the vaccine against the virus is greater than 65%; and/or wherein the vaccine immunizes the subject against the virus for up to 2 years or wherein the vaccine immunizes the subject against the virus for more than 2 years.

80. The method of any one of paragraphs 72-79, wherein the subject has an age of about 5 years old or younger or wherein the subject has an age of about 60 years old or older; and/or wherein the subject has a chronic pulmonary disease; and/or the subject has been exposed to the virus, wherein the subject is infected with the virus, or wherein the subject is at risk of infection by the virus; and/or wherein the subject is immunocompromised.

81. The respiratory virus vaccine of any one of paragraphs 1-71, comprising at least one (e.g., at least two, at least three, at least four, or at least five) RNA polynucleotide having an open reading frame encoding at least one (e.g., at least two, at least three, at least four, or at least five) antigenic polypeptide selected from hMPV antigenic polypeptides (SEQ ID NO: 5-8), PIV3 antigenic polypeptides (SEQ ID

NO: 12-13), RSV antigenic polypeptides, MeV antigenic polypeptides (SEQ ID NO: 47-50) and BetaCoV antigenic polypeptides (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1; (SEQ ID NO: 24-34)), formulated in a cationic lipid nanoparticle

(a) having a molar ratio of about 20-60% cationic lipid, about 5-25% non-cationic lipid, about 25-55% sterol, and about 0.5-15% PEG-modified lipid, and/or

(b) comprising a compound of Formula (I) and/or Formula (II),

wherein the at least one (e.g., at least two, at least three, at least four, or at least five) RNA polynucleotide comprises at least one chemical modification.

82. The respiratory virus vaccine of any one of paragraphs 1-71, comprising at least one (e.g., at least two, at least three, at least four, or at least five) RNA polynucleotide having an open reading frame encoding at least one (e.g., at least two, at least three, at least four, or at least five) antigenic polypeptide selected from hMPV antigenic polypeptides (SEQ ID NO: 5-8), PIV3 antigenic polypeptides (SEQ ID NO: 12-13), RSV antigenic polypeptides, MeV antigenic polypeptides (SEQ ID NO: 47-50) and BetaCoV antigenic polypeptides (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1; (SEQ ID NO: 24-34)), formulated in a cationic lipid nanoparticle (a) having a molar ratio of about 20-60% cationic lipid, about 5-25% non-cationic lipid, about 25-55% sterol, and about 0.5-15% PEG-modified lipid, and/or (b) comprising at least one (e.g., at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, or 14) Compound selected from Compounds 3, 18, 20, 25, 26, 29, 30, 60, 108-112 and 122.

83. The respiratory virus vaccine of paragraphs 81 or 82, wherein the at least one antigenic polypeptide is selected from hMPV antigenic polypeptides (e.g., SEQ ID NO: 5-8).

84. The respiratory virus vaccine of any one of paragraphs 81-83, wherein the at least one antigenic polypeptide is selected from PIV3 antigenic polypeptides (e.g., SEQ ID NO: 12-13).

85. The respiratory virus vaccine of any one of paragraphs 81-84, wherein the at least one antigenic polypeptide is selected from RSV antigenic polypeptides.

86. The respiratory virus vaccine of any one of paragraphs 81-85, wherein the at least one antigenic polypeptide is selected from MeV antigenic polypeptides (e.g., SEQ ID NO: 47-50).

87. The respiratory virus vaccine of any one of paragraphs 81-86, wherein the at least one antigenic polypeptide is selected from BetaCoV antigenic polypeptides (e.g., SEQ ID NO: 24-34).

88. The respiratory virus vaccine of paragraph 87, wherein the BetaCoV antigenic polypeptides are MERS antigenic polypeptides.

89. The respiratory virus vaccine of paragraph 87, wherein the BetaCoV antigenic polypeptides are SARS antigenic polypeptides.

90. The respiratory virus vaccine of any one of paragraphs 81-89, wherein the at least one (e.g., at least two, at least three, at least four, or at least five) RNA polynucleotide comprises at least one chemical modification (e.g., selected from pseudouridine, N1-methylpseudouridine, N1-ethylpseudouridine, 2-thiouridine, 4'-thiouridine, 5-methylcytosine, 5-methyluridine, 2-thio-1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-pseudouridine, 2-thio-5-aza-uridine, 2-thio-dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-pseudouridine, 4-methoxy-2-thio-pseudouridine, 4-methoxy-pseudouridine, 4-thio-1-methyl-pseudouridine,

4-thio-pseudouridine, 5-aza-uridine, dihydropseudouridine, 5-methoxyuridine and 2'-O-methyl uridine).

91. A respiratory virus vaccine, comprising:

at least one messenger ribonucleic acid (mRNA) polynucleotide having a 5' terminal cap, an open reading frame encoding at least one respiratory virus antigenic polypeptide, and a 3' polyA tail.

92. The vaccine of paragraph 91, wherein the at least one mRNA polynucleotide comprises a sequence identified by any one of SEQ ID NO: 57-80.

93. The vaccine of paragraph 91 or 92, wherein the 5' terminal cap is or comprises 7mG(5')ppp(5')NlmpNp.

94. The vaccine of any one of paragraphs 91-93, wherein 100% of the uracil in the open reading frame is modified to include N1-methyl pseudouridine at the 5-position of the uracil.

95. The vaccine of any one of paragraphs 91-94, wherein the vaccine is formulated in a lipid nanoparticle comprising: DLin-MC3-DMA; cholesterol; 1,2-Distearoyl-sn-glycero-3-phosphocholine (DSPC); and polyethylene glycol (PEG) 2000-DMG.

96. The vaccine of paragraph 95, wherein the lipid nanoparticle further comprises trisodium citrate buffer, sucrose and water.

97. A respiratory syncytial virus (RSV) vaccine, comprising:

at least one messenger ribonucleic acid (mRNA) polynucleotide having a 5' terminal cap 7mG(5')ppp(5')NlmpNp, a sequence identified by any one of SEQ ID NO: 57-80 and a 3' polyA tail, formulated in a lipid nanoparticle comprising DLin-MC3-DMA, cholesterol, 1,2-Distearoyl-sn-glycero-3-phosphocholine (DSPC), and polyethylene glycol (PEG)2000-DMG, wherein the uracil nucleotides of the sequence identified by any one of SEQ ID NO: 57-80 are modified to include N1-methyl pseudouridine at the 5-position of the uracil nucleotide.

This disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of "including," "comprising," or "having," "containing," "involving," and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

EXAMPLES

Example 1: Manufacture of Polynucleotides

According to the present disclosure, the manufacture of polynucleotides and/or parts or regions thereof may be accomplished utilizing the methods taught in International Publication WO2014/152027, entitled "Manufacturing Methods for Production of RNA Transcripts," the contents of which is incorporated herein by reference in its entirety.

Purification methods may include those taught in International Publication WO2014/152030 and International Publication WO2014/152031, each of which is incorporated herein by reference in its entirety.

Detection and characterization methods of the polynucleotides may be performed as taught in International Publication WO2014/144039, which is incorporated herein by reference in its entirety.

Characterization of the polynucleotides of the disclosure may be accomplished using polynucleotide mapping, reverse transcriptase sequencing, charge distribution analysis, detection of RNA impurities, or any combination of two or more of the foregoing. "Characterizing" comprises determining the RNA transcript sequence, determining the purity of the RNA transcript, or determining the charge heterogeneity of the RNA transcript, for example. Such methods are taught in, for example, International Publication WO2014/144711 and International Publication WO2014/144767, the content of each of which is incorporated herein by reference in its entirety.

Example 2: Chimeric Polynucleotide Synthesis

According to the present disclosure, two regions or parts of a chimeric polynucleotide may be joined or ligated using triphosphate chemistry. A first region or part of 100 nucleotides or less is chemically synthesized with a 5' monophosphate and terminal 3'desOH or blocked OH, for example. If the region is longer than 80 nucleotides, it may be synthesized as two strands for ligation.

If the first region or part is synthesized as a non-positionally modified region or part using in vitro transcription (IVT), conversion the 5'monophosphate with subsequent capping of the 3' terminus may follow.

Monophosphate protecting groups may be selected from any of those known in the art.

The second region or part of the chimeric polynucleotide may be synthesized using either chemical synthesis or IVT methods. IVT methods may include an RNA polymerase that can utilize a primer with a modified cap. Alternatively, a cap of up to 130 nucleotides may be chemically synthesized and coupled to the IVT region or part.

For ligation methods, ligation with DNA T4 ligase, followed by treatment with DNase should readily avoid concatenation.

The entire chimeric polynucleotide need not be manufactured with a phosphate-sugar backbone. If one of the regions or parts encodes a polypeptide, then such region or part may comprise a phosphate-sugar backbone.

Ligation is then performed using any known click chemistry, orthoclick chemistry, solulink, or other bioconjugate chemistries known to those in the art.

Synthetic Route

The chimeric polynucleotide may be made using a series of starting segments. Such segments include:

- (a) a capped and protected 5' segment comprising a normal 3'OH (SEG. 1)
- (b) a 5' triphosphate segment, which may include the coding region of a polypeptide and a normal 3'OH (SEG. 2)
- (c) a 5' monophosphate segment for the 3' end of the chimeric polynucleotide (e.g., the tail) comprising cordycepin or no 3'OH (SEG. 3)

After synthesis (chemical or IVT), segment 3 (SEG. 3) may be treated with cordycepin and then with pyrophosphatase to create the 5' monophosphate.

Segment 2 (SEG. 2) may then be ligated to SEG. 3 using RNA ligase. The ligated polynucleotide is then purified and treated with pyrophosphatase to cleave the diphosphate. The treated SEG.2-SEG. 3 construct may then be purified and SEG. 1 is ligated to the 5' terminus. A further purification step of the chimeric polynucleotide may be performed.

209

Where the chimeric polynucleotide encodes a polypeptide, the ligated or joined segments may be represented as: 5'UTR (SEG. 1), open reading frame or ORF (SEG. 2) and 3'UTR+PolyA (SEG. 3).

The yields of each step may be as much as 90-95%.

Example 3: PCR for cDNA Production

PCR procedures for the preparation of cDNA may be performed using 2xKAPA HIFI™ HotStart ReadyMix by Kapa Biosystems (Woburn, MA). This system includes 2xKAPA ReadyMix 12.5 µl; Forward Primer (10 µM) 0.75 µl; Reverse Primer (10 µM) 0.75 µl; Template cDNA 100 ng; and dH₂O diluted to 25.0 µl. The reaction conditions may be at 95° C. for 5 min. The reaction may be performed for 25 cycles of 98° C. for 20 sec, then 58° C. for 30 sec, then 72° C. for 45 sec, then 72° C. for 5 min, then 4° C. to termination.

The reaction may be cleaned up using Invitrogen's PURELINK™ PCR Micro Kit (Carlsbad, CA) per manufacturer's instructions (up to 5 µg). Larger reactions may require a cleanup using a product with a larger capacity. Following the cleanup, the cDNA may be quantified using the NANODROPT™ and analyzed by agarose gel electrophoresis to confirm that the cDNA is the expected size. The cDNA may then be submitted for sequencing analysis before proceeding to the in vitro transcription reaction.

Example 4: In Vitro Transcription (IVT)

The in vitro transcription reaction generates RNA polynucleotides. Such polynucleotides may comprise a region or part of the polynucleotides of the disclosure, including chemically modified RNA (e.g., mRNA) polynucleotides. The chemically modified RNA polynucleotides can be uniformly modified polynucleotides. The in vitro transcription reaction utilizes a custom mix of nucleotide triphosphates (NTPs). The NTPs may comprise chemically modified NTPs, or a mix of natural and chemically modified NTPs, or natural NTPs.

A typical in vitro transcription reaction includes the following:

1)	Template cDNA	1.0 µg
2)	10x transcription buffer (400 mM Tris-HCl pH 8.0, 190 mM MgCl ₂ , 50 mM DTT, 10 mM Spermidine)	2.0 µl
3)	Custom NTPs (25 mM each)	0.2 µl
4)	RNase Inhibitor	20 U
5)	T7 RNA polymerase	3000 U
6)	dH ₂ O	up to 20.0 µl, and
7)	Incubation at 37° C. for 3 hr-5 hrs.	

The crude IVT mix may be stored at 4° C. overnight for cleanup the next day. 1 U of RNase-free DNase may then be used to digest the original template. After 15 minutes of incubation at 37° C., the mRNA may be purified using Ambion's MEGACLEAR™ Kit (Austin, TX) following the manufacturer's instructions. This kit can purify up to 500 µg of RNA. Following the cleanup, the RNA polynucleotide may be quantified using the NanoDrop and analyzed by agarose gel electrophoresis to confirm the RNA polynucleotide is the proper size and that no degradation of the RNA has occurred.

Example 5: Enzymatic Capping

Capping of a RNA polynucleotide is performed as follows where the mixture includes: IVT RNA 60 µg-180 µg and

210

dH₂O up to 72 µl. The mixture is incubated at 65° C. for 5 minutes to denature RNA, and then is transferred immediately to ice.

The protocol then involves the mixing of 10x Capping Buffer (0.5 M Tris-HCl (pH 8.0), 60 mM KCl, 12.5 mM MgCl₂) (10.0 µl); 20 mM GTP (5.0 µl); 20 mM S-Adenosyl Methionine (2.5 µl); RNase Inhibitor (100 U); 2'-O-Methyltransferase (400U); Vaccinia capping enzyme (Guanylyl transferase) (40 U); dH₂O (Up to 28 µl); and incubation at 37° C. for 30 minutes for 60 µg RNA or up to 2 hours for 180 µg of RNA.

The RNA polynucleotide may then be purified using Ambion's MEGACLEAR™ Kit (Austin, TX) following the manufacturer's instructions. Following the cleanup, the RNA may be quantified using the NANODROPT™ (ThermoFisher, Waltham, MA) and analyzed by agarose gel electrophoresis to confirm the RNA polynucleotide is the proper size and that no degradation of the RNA has occurred. The RNA polynucleotide product may also be sequenced by running a reverse-transcription-PCR to generate the cDNA for sequencing.

Example 6: PolyA Tailing Reaction

Without a poly-T in the cDNA, a poly-A tailing reaction must be performed before cleaning the final product. This is done by mixing capped IVT RNA (100 µl); RNase Inhibitor (20 U); 10x Tailing Buffer (0.5 M Tris-HCl (pH 8.0), 2.5 M NaCl, 100 mM MgCl₂) (12.0 µl); 20 mM ATP (6.0 µl); Poly-A Polymerase (20 U); dH₂O up to 123.5 µl and incubation at 37° C. for 30 min. If the poly-A tail is already in the transcript, then the tailing reaction may be skipped and proceed directly to cleanup with Ambion's MEGACLEAR™ kit (Austin, TX) (up to 500 g). Poly-A Polymerase may be a recombinant enzyme expressed in yeast.

It should be understood that the processivity or integrity of the polyA tailing reaction may not always result in an exact size polyA tail. Hence, polyA tails of approximately between 40-200 nucleotides, e.g., about 40, 50, 60, 70, 80, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 150-165, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164 or 165 are within the scope of the present disclosure.

Example 7: Natural 5' Caps and 5' Cap Analogues

5'-capping of polynucleotides may be completed concomitantly during the in vitro-transcription reaction using the following chemical RNA cap analogs to generate the 5'-guanosine cap structure according to manufacturer protocols: 3'-O-Me-m7G(5')ppp(5') G [the ARCA cap]; G(5')ppp(5')A; G(5')ppp(5')G; m7G(5')ppp(5')A; m7G(5')ppp(5')G (New England BioLabs, Ipswich, MA). 5'-capping of modified RNA may be completed post-transcriptionally using a Vaccinia Virus Capping Enzyme to generate the "Cap 0" structure: m7G(5')ppp(5')G (New England BioLabs, Ipswich, MA). Cap 1 structure may be generated using both Vaccinia Virus Capping Enzyme and a 2'-O methyltransferase to generate: m7G(5')ppp(5')G-2'-O-methyl. Cap 2 structure may be generated from the Cap 1 structure followed by the 2'-O-methylation of the 5'-antepenultimate nucleotide using a 2'-O methyl-transferase. Cap 3 structure may be generated from the Cap 2 structure followed by the 2'-O-methylation of the 5'-preantepenultimate nucleotide using a 2'-O methyl-transferase. Enzymes are preferably derived from a recombinant source.

211

When transfected into mammalian cells, the modified mRNAs have a stability of between 12-18 hours or more than 18 hours, e.g., 24, 36, 48, 60, 72 or greater than 72 hours.

Example 8: Capping Assays

Protein Expression Assay

Polynucleotides (e.g., mRNA) encoding a polypeptide, containing any of the caps taught herein, can be transfected into cells at equal concentrations. The amount of protein secreted into the culture medium can be assayed by ELISA at 6, 12, 24 and/or 36 hours post-transfection. Synthetic polynucleotides that secrete higher levels of protein into the medium correspond to a synthetic polynucleotide with a higher translationally-competent cap structure.

Purity Analysis Synthesis

RNA (e.g., mRNA) polynucleotides encoding a polypeptide, containing any of the caps taught herein can be compared for purity using denaturing Agarose-Urea gel electrophoresis or HPLC analysis. RNA polynucleotides with a single, consolidated band by electrophoresis correspond to the higher purity product compared to polynucleotides with multiple bands or streaking bands. Chemically modified RNA polynucleotides with a single HPLC peak also correspond to a higher purity product. The capping reaction with a higher efficiency provides a more pure polynucleotide population.

Cytokine Analysis

RNA (e.g., mRNA) polynucleotides encoding a polypeptide, containing any of the caps taught herein can be transfected into cells at multiple concentrations. The amount of pro-inflammatory cytokines, such as TNF-alpha and IFN-beta, secreted into the culture medium can be assayed by ELISA at 6, 12, 24 and/or 36 hours post-transfection. RNA polynucleotides resulting in the secretion of higher levels of pro-inflammatory cytokines into the medium correspond to a polynucleotides containing an immune-activating cap structure.

Capping Reaction Efficiency

RNA (e.g., mRNA) polynucleotides encoding a polypeptide, containing any of the caps taught herein can be analyzed for capping reaction efficiency by LC-MS after nuclease treatment. Nuclease treatment of capped polynucleotides yield a mixture of free nucleotides and the capped 5'-5'-triphosphate cap structure detectable by LC-MS. The amount of capped product on the LC-MS spectra can be expressed as a percent of total polynucleotide from the reaction and correspond to capping reaction efficiency. The cap structure with a higher capping reaction efficiency has a higher amount of capped product by LC-MS.

Example 9: Agarose Gel Electrophoresis of Modified RNA or RT PCR Products

Individual RNA polynucleotides (200-400 ng in a 20 µl volume) or reverse transcribed PCR products (200-400 ng) may be loaded into a well on a non-denaturing 1.2% Agarose E-Gel (Invitrogen, Carlsbad, CA) and run for 12-15 minutes, according to the manufacturer protocol.

Example 10: Nanodrop Modified RNA Quantification and UV Spectral Data

Chemically modified RNA polynucleotides in TE buffer (1 µl) are used for Nanodrop UV absorbance readings to

212

quantitate the yield of each polynucleotide from an chemical synthesis or in vitro transcription reaction.

Example 11: Formulation of Modified mRNA Using Lipidoids

RNA (e.g., mRNA) polynucleotides may be formulated for in vitro experiments by mixing the polynucleotides with the lipidoid at a set ratio prior to addition to cells. In vivo formulation may require the addition of extra ingredients to facilitate circulation throughout the body. To test the ability of these lipidoids to form particles suitable for in vivo work, a standard formulation process used for siRNA-lipidoid formulations may be used as a starting point. After formation of the particle, polynucleotide is added and allowed to integrate with the complex. The encapsulation efficiency is determined using a standard dye exclusion assays.

Example 12: Immunogenicity Study

The instant study is designed to test the immunogenicity in mice of candidate hMPV vaccines comprising a mRNA polynucleotide encoding Fusion (F) glycoprotein, major surface glycoprotein G, or a combination thereof, obtained from hMPV.

Mice are immunized intravenously (IV), intramuscularly (IM), or intradermally (ID) with candidate vaccines. Candidate vaccines are chemically modified or unmodified. A total of four immunizations are given at 3-week intervals (i.e., at weeks 0, 3, 6, and 9), and sera are collected after each immunization until weeks 33-51. Serum antibody titers against Fusion (F) glycoprotein or major surface glycoprotein (G) protein are determined by ELISA. Sera collected from each mouse during weeks 10-16 are pooled, and total IgG purified. Purified antibodies are used for immunoelectron microscopy, antibody-affinity testing, and in vitro protection assays.

Example 13: hMPV Rodent Challenge

The instant study is designed to test the efficacy in cotton rats of candidate hMPV vaccines against a lethal challenge using an hMPV vaccine comprising mRNA encoding Fusion (F) glycoprotein, major surface glycoprotein G, or a combination of both antigens obtained from hMPV. Cotton rats are challenged with a lethal dose of the hMPV.

Animals are immunized intravenously (IV), intramuscularly (IM), or intradermally (ID) at week 0 and week 3 with candidate hMPV vaccines with and without adjuvant. Candidate vaccines are chemically modified or unmodified. The animals are then challenged with a lethal dose of hMPV on week 7 via IV, IM or ID. Endpoint is day 13 post infection, death or euthanasia. Animals displaying severe illness as determined by >30% weight loss, extreme lethargy or paralysis are euthanized. Body temperature and weight are assessed and recorded daily.

In experiments where a lipid nanoparticle (LNP) formulation is used, the formulation may include a cationic lipid, non-cationic lipid, PEG lipid and structural lipid in the ratios 50:10:1.5:38.5. The cationic lipid is DLin-KC2-DMA (50 mol %) or DLin-MC3-DMA (50 mol %), the non-cationic lipid is DSPC (10 mol %), the PEG lipid is PEG-DOMG (1.5 mol %) and the structural lipid is cholesterol (38.5 mol %), for example.

Example 14: Immunogenicity of hMPV mRNA Vaccine in BALB/c Mice

The instant study was designed to test the immunogenicity in BALB/c mice of hMPV vaccines comprising an

mRNA polynucleotide encoding the hMPV Fusion (F) glycoprotein. The mRNA polynucleotide encodes the full-length fusion protein and comprises the wild-type nucleotide sequence obtained from the hMPV A2a strain. Mice were divided into 3 groups (n=8 for each group) and immunized intramuscularly (IM) with PBS, a 10 µg dose of mRNA vaccines encoding hMPV fusion protein, or a 2 µg dose of mRNA vaccines encoding hMPV fusion protein. A total of two immunizations were given at 3-week intervals (i.e., at weeks 0, and 3 weeks), and sera were collected after each immunization according to the schedule described in Table 1. Serum antibody titers against hMPV fusion glycoprotein were determined by ELISA and antibodies were detected in the sera collected on day 14 onward. Both vaccine doses tested induced comparable levels of immune response in mice (FIGS. 2A-2C).

Additionally, mice sera were used for IgG isotyping (FIGS. 3A-3C). Both hMPV fusion protein-specific IgG1 and IgG2a were detected in mice sera. hMPV fusion protein mRNA vaccine also induced Th1 and Th2 cytokine responses, with a Th1 bias.

Sera from mice immunized with either 10 µg or 2 µg doses of the hMPV fusion protein mRNA vaccine contain neutralizing antibodies. The ability of these antibodies to neutralize hMPV B2 strain was also tested. The antibody-containing sera successfully neutralized the hMPV B2 virus (FIG. 4).

Example 15: T-Cell Stimulation

The instant study was designed to test T-cell stimulation in the splenocytes of mice immunized with mRNA vaccines encoding hMPV fusion protein, as described herein. Immunization of BALB/c mice was performed as described in Example 14. The splenocytes for each group were pooled and split into two parts. One part of splenocytes from each group of mice was stimulated with hMPV-free media, Concanavalin A or a hMPV fusion protein peptide pool comprising 15-mers (15 amino acids long); while the other part of splenocytes from each group of mice was stimulated with hMPV-free media, Concanavalin A or inactivated hMPV virus. Secreted mouse cytokines were measured using the Meso Scale Discovery (MSD) assay.

Cytokines specific to Th1 or Th2 responses were measured. For Th1 response, IFN-γ, IL2 and IL12 were detected from splenocytes stimulated with the hMPV fusion protein peptide pool at a level comparable to that of Concanavalin A (FIGS. 5A-5C). For a Th2 response, the hMPV fusion protein peptide pool induced the secretion of detectable IL10, TNF-α, IL4 and IL6, but not IL5, while Concanavalin A stimulated the secretion of all the above-mentioned Th2 cytokines (FIGS. 6A-6E) at a much higher level.

In contrast, inactivated hMPV virus only induced the secretion of IL2 in the Th1 response comparable to that of Concanavalin A (FIGS. 7A-7C). For the Th2 response, the inactivated hMPV virus induced the secretion of detectable IL10, TNF-α, IL4 and IL6, but not IL5, while Concanavalin A stimulated the secretion of all the above-mentioned Th2 cytokines (FIGS. 8A-8E) at a much higher level.

Example 16: hMPV Rodent Challenge in Cotton Rats Immunized with mRNA Vaccine Encoding hMPV Fusion Protein

The instant study was designed to test the efficacy in cotton rats of hMPV vaccines against a lethal challenge. mRNA vaccines encoding hMPV fusion protein were used. The mRNA polynucleotide encodes a full-length fusion

protein and comprises the wild-type nucleotide sequence obtained from the hMPV A2a strain.

Cotton rats were immunized intramuscularly (IM) at week 0 and week 3 with the mRNA vaccines encoding hMPV fusion protein with either 2 µg or 10 µg doses for each immunization. The animals were then challenged with a lethal dose of hMPV in week 7 post initial immunization via IV, TM or ID. The endpoint was day 13 post infection, death or euthanasia. Viral titers in the noses and lungs of the cotton rats were measured. The results (FIGS. 9A and 9B) show that a 10 µg dose of mRNA vaccine protected the cotton mice 100% in the lung and drastically reduced the viral titer in the nose after challenge (~2 log reduction). Moreover, a 2 µg dose of mRNA vaccine showed a 1 log reduction in lung viral titer in the cotton mice challenged.

Further, the histopathology of the lungs of the cotton mice immunized and challenged showed no pathology associated with vaccine-enhanced disease (FIG. 10).

Example 17: Immunogenicity Study

The instant study is designed to test the immunogenicity in mice of candidate PIV3 vaccines comprising a mRNA polynucleotide encoding hemagglutinin-neuraminidase or fusion protein (F or F0) obtained from PIV3.

Mice are immunized intravenously (IV), intramuscularly (IM), or intradermally (ID) with candidate vaccines. Candidate vaccines are chemically modified or unmodified. A total of four immunizations are given at 3-week intervals (i.e., at weeks 0, 3, 6, and 9), and sera are collected after each immunization until weeks 33-51. Serum antibody titers against hemagglutinin-neuraminidase or fusion protein (F or F0) are determined by ELISA. Sera collected from each mouse during weeks 10-16 are, optionally, pooled, and total IgGs are purified. Purified antibodies are used for immunoelectron microscopy, antibody-affinity testing, and in vitro protection assays.

Example 18: PIV3 Rodent Challenge

The instant study is designed to test the efficacy in cotton rats of candidate PIV3 vaccines against a lethal challenge using a PIV3 vaccine comprising mRNA encoding hemagglutinin-neuraminidase or fusion protein (F or F0) obtained from PIV3. Cotton rats are challenged with a lethal dose of the PIV3.

Animals are immunized intravenously (IV), intramuscularly (IM), or intradermally (ID) at week 0 and week 3 with candidate PIV3 vaccines with and without adjuvant. Candidate vaccines are chemically modified or unmodified. The animals are then challenged with a lethal dose of PIV3 on week 7 via IV, IM or ID. Endpoint is day 13 post infection, death or euthanasia. Animals displaying severe illness as determined by >30% weight loss, extreme lethargy or paralysis are euthanized. Body temperature and weight are assessed and recorded daily.

In experiments where a lipid nanoparticle (LNP) formulation is used, the formulation may include a cationic lipid, non-cationic lipid, PEG lipid and structural lipid in the ratios 50:10:1.5:38.5. The cationic lipid is DLin-KC2-DMA (50 mol %) or DLin-MC3-DMA (50 mol %), the non-cationic lipid is DSPC (10 mol %), the PEG lipid is PEG-DOMG (1.5 mol %) and the structural lipid is cholesterol (38.5 mol %), for example.

Example 19: hMPV/PIV Cotton Rat Challenge

The instant study was designed to test the efficacy in cotton rats of candidate hMPV mRNA vaccines, PIV3

215

mRNA vaccines, or hMPV/PIV combination mRNA vaccines against a lethal challenge using PIV3 strain or hMPV/A2 strain. The study design is shown in Table 9.

Cotton rats of 10-12 weeks old were divided into 12 groups (n=5), and each group was vaccinated with mRNA vaccines indicated in Table 9. The PIV3 vaccine comprises mRNA encoding hemagglutinin-neuraminidase or fusion protein (F or F0) obtained from PIV3. The hMPV mRNA vaccine encodes the full-length hMPV fusion protein. The hMPV/PIV combination mRNA vaccine is a mixture of the PIV3 vaccine and hMPV vaccine at a 1:1 ratio.

Cotton rats were immunized intramuscularly (IM) at week 0 and week 3 with candidate vaccines with the doses indicated in Table 9. Cotton rats immunized with hMPV mRNA vaccines or hMPV/PIV combination mRNA vaccines were challenged with a lethal dose of hMPV/A2 strain on week 7 via IM. Cotton rats immunized with PIV mRNA vaccines or hMPV/PIV combination mRNA vaccines were challenged with a lethal dose of PIV3 strain on week 7 via IM.

The endpoint was day 13 post infection, death or euthanasia. Animals displaying severe illness as determined by >30% weight loss, extreme lethargy or paralysis were euthanized. Body temperature and weight were assessed and recorded daily.

Lung and nose hMPV/A2 (FIG. 12) or PIV3 (FIG. 13) viral titers were assessed. Lung histopathology of the immunized and challenged cotton rat immunized and challenged were assessed to determine pathology associated with vaccine enhance disease. Neutralization antibody titers in the serum of immunized cotton rats on day 0 and 42 post immunization were assessed (FIG. 11).

hMPV/A2 (FIG. 14) or PIV3 (FIG. 15) neutralizing antibody titers in the serum samples of the immunized cotton rat 42 days post immunization were measured. All mRNA vaccines tested induced strong neutralizing antibodies cotton rats. Lung histopathology of the immunized cotton rats were also evaluated (FIG. 16). Low occurrence of alveolitis and interstitial pneumonia was observed, indicating no antibody-dependent enhancement (ADE) of hMPV or PIV associated diseases.

Example 20: Betacoronavirus Immunogenicity Study

The instant study is designed to test the immunogenicity in rabbits of candidate betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1 or a combination thereof) vaccines comprising a mRNA polynucleotide encoding the spike (S) protein, the S1 subunit (S1) of the spike protein, or the S2 subunit (S2) of the spike protein obtained from a betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1).

Rabbits are vaccinated on week 0 and 3 via intravenous (IV), intramuscular (IM), or intradermal (ID) routes. One group remains unvaccinated and one is administered inactivated betacoronavirus. Serum is collected from each rabbit on weeks 1, 3 (pre-dose) and 5. Individual bleeds are tested for anti-S, anti-S1 or anti-S2 activity via a virus neutralization assay from all three time points, and pooled samples from week 5 only are tested by Western blot using inactivated betacoronavirus (e.g., inactivated MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1).

216

In experiments where a lipid nanoparticle (LNP) formulation is used, the formulation may include a cationic lipid, non-cationic lipid, PEG lipid and structural lipid in the ratios 50:10:1.5:38.5. The cationic lipid is DLin-KC2-DMA (50 mol %) or DLin-MC3-DMA (50 mol %), the non-cationic lipid is DSPC (10 mol %), the PEG lipid is PEG-DOMG (1.5 mol %) and the structural lipid is cholesterol (38.5 mol %), for example.

Example 21: Betacoronavirus Challenge

The instant study is designed to test the efficacy in rabbits of candidate betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-HKU1 or a combination thereof) vaccines against a lethal challenge using a betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-HKU1 or a combination thereof) vaccine comprising mRNA encoding the spike (S) protein, the S1 subunit (S1) of the spike protein, or the S2 subunit (S2) of the spike protein obtained from betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1). Rabbits are challenged with a lethal dose (10×LD₅₀; ~100 plaque-forming units; PFU) of betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1).

The animals used are 6-8 week old female rabbits in groups of 10. Rabbits are vaccinated on weeks 0 and 3 via an IM, ID or IV route of administration. Candidate vaccines are chemically modified or unmodified. Rabbit serum is tested for microneutralization (see Example 14). Rabbits are then challenged with ~1 LD₅₀ of betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1) on week 7 via an IN, IM, ID or IV route of administration. Endpoint is day 13 post infection, death or euthanasia. Animals displaying severe illness as determined by >30% weight loss, extreme lethargy or paralysis are euthanized. Body temperature and weight are assessed and recorded daily.

Example 22: Microneutralization Assay

Nine serial 2-fold dilutions (1:50-1:12,800) of rabbit serum are made in 50 µl virus growth medium (VGM) with trypsin in 96 well microtiter plates. Fifty microliters of virus containing ~50 pfu of betacoronavirus (e.g., MERS-CoV, SARS-CoV, HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-NL, HCoV-NH or HCoV-HKU1) is added to the serum dilutions and allowed to incubate for 60 minutes at room temperature (RT). Positive control wells of virus without sera and negative control wells without virus or sera are included in triplicate on each plate. While the serum-virus mixtures incubate, a single cell suspension of Madin-Darby Canine-Kidney cells are prepared by trypsinizing (Gibco 0.5% bovine pancrease trypsin in EDTA) a confluent monolayer and suspended cells are transferred to a 50 ml centrifuge tube, topped with sterile PBS and gently mixed. The cells are then pelleted at 200 g for 5 minutes, supernatant aspirated and cells resuspended in PBS. This procedure is repeated once and the cells are resuspended at a concentration of 3×10⁵/ml in VGM with porcine trypsin. Then, 100 µl of cells are added to the serum-virus mixtures and the plates incubated at 35° C. in CO₂ for 5 days. The plates are fixed with 80% acetone in phosphate buffered saline (PBS) for 15 minutes at RT, air dried and then blocked for 30 minutes containing PBS with 0.5% gelatin and 2% FCS. An

217

antibody to the S proteins, S1 protein or S2 protein is diluted in PBS with 0.5% gelatin/2% FCS/0.5% Tween 20 and incubated at RT for 2 hours. Wells are washed and horseradish peroxidase-conjugated goat anti-mouse IgG added, followed by another 2 hour incubation. After washing, 0-phenylenediamine dihydrochloride is added and the neutralization titer is defined as the titer of serum that reduced color development by 50% compared to the positive control wells.

Example 23: MERS CoV Vaccine Immunogenicity Study in Mice

The instant study was designed to test the immunogenicity in mice of candidate MERS-CoV vaccines comprising a mRNA polynucleotide encoding the full-length Spike (S) protein, or the S2 subunit (S2) of the Spike protein obtained from MERS-CoV.

Mice were vaccinated with a 10 µg dose of MERS-CoV mRNA vaccine encoding either the full-length MERS-CoV Spike (S) protein, or the S2 subunit (S2) of the Spike protein on days 0 and 21. Sera were collected from each mice on days 0, 21, 42, and 56. Individual bleeds were tested for anti-S, anti-S2 activity via a virus neutralization assay from all four time points.

As shown in FIG. 17, the MERS-CoV vaccine encoding the full-length S protein induced strong immune response after the boost dose on day 21. Further, full-length S protein vaccine generated much higher neutralizing antibody titers as compared to S2 alone (FIG. 18).

Example 24: MERS CoV Vaccine Immunogenicity Study in New Zealand White Rabbits

The instant study was designed to test the immunogenicity of candidate MERS-CoV mRNA vaccines encoding the full-length Spike (S) protein. The New Zealand white rabbits used in this study weighed about 4-5 kg. The rabbits were divided into three groups (Group 1a, Group 1b, and Group 2, n=8). Rabbits in Group 1a were immunized intramuscularly (IM) with one 20 µg dose of the MERS-CoV mRNA vaccine encoding the full-length Spike protein on day 0. Rabbits in Group 1b were immunized intramuscularly (IM) with one 20 µg dose of the MERS-CoV mRNA vaccine encoding the full-length Spike protein on day 0, and again on day 21 (booster dose). Group 2 received placebo (PBS). The immunized rabbits were then challenged and samples were collected 4 days after challenge. The viral loads in the lungs, bronchoalveolar lavage (Bal), nose, and throat of the rabbits were determined, e.g., via quantitative PCR. Replicating virus in the lung tissues of the rabbits were also detected. Lung histopathology were evaluated and the neutralizing antibody titers in serum samples of the rabbits were determined.

Two 20 µg doses of MERS-CoV mRNA vaccine resulted in a 3 log reduction of viral load in the nose and led to complete protection in the throat of the New Zealand white rabbits (FIG. 19A). Two 20 µg doses of MERS-CoV mRNA vaccine also resulted in a 4 log reduction of viral load in the BAL of the New Zealand white rabbits (FIG. 19B). One 20 µg dose of MERS-CoV mRNA vaccine resulted in a 2 log reduction of viral load, while two 20 µg doses of MERS-CoV mRNA vaccine resulted in an over 4 log reduction of viral load in the lungs of the New Zealand white rabbits (FIG. 19C).

Quantitative PCR results show that two 20 µg doses of MERS-CoV mRNA vaccine reduced over 99% (2 log) of

218

viruses in the lungs of New Zealand white rabbits (FIG. 20A). No replicating virus were detected in the lungs (FIG. 20B).

Further, as shown in FIG. 21, two 20 µg doses of MERS-CoV mRNA vaccine induced significant amount of neutralizing antibodies against MERS-CoV (EC₅₀ between 500-1000).

The MERS-CoV mRNA vaccine induced antibody titer is 3-5 fold better than any other vaccines tested in the same model.

Example 25: Immunogenicity Study

The instant study is designed to test the immunogenicity in mice of candidate MeV vaccines comprising a mRNA polynucleotide encoding MeV hemagglutinin (HA) protein, MeV Fusion (F) protein or a combination of both.

Mice are immunized intravenously (IV), intramuscularly (IM), or intradermally (ID) with candidate vaccines. Up to three immunizations are given at 3-week intervals (i.e., at weeks 0, 3, 6, and 9), and sera are collected after each immunization until weeks 33-51. Serum antibody titers against MeV HA protein or MeV F protein are determined by ELISA.

Example 26: MeV Rodent Challenge

The instant study is designed to test the efficacy in transgenic mice of candidate MeV vaccines against a lethal challenge using a MeV vaccine comprising mRNA encoding MeV HA protein or MeV F protein. The transgenic mice express human receptor CD46 or signaling lymphocyte activation molecule (SLAM) (also referred to as CD150). Humans are the only natural host for MeV infection, thus transgenic lines are required for this study. CD46 is a complement regulatory protein that protects host tissue from complement deposition by binding to complement components C3b and C4b. Its expression on murine fibroblast and lymphoid cell lines renders these otherwise refractory cells permissive for MeV infection, and the expression of CD46 on primate cells parallels the clinical tropism of MeV infection in humans and nonhuman primates (Rall G F et al. PNAS USA 1997; 94(9):4659-63). SLAM is a type 1 membrane glycoprotein belonging to the immunoglobulin superfamily. It is expressed on the surface of activated lymphocytes, macrophages, and dendritic cells and is thought to play an important role in lymphocyte signaling. SLAM is a receptor for both wild-type and vaccine MeV strains (Sellin C I et al. J Virol. 2006; 80(13):6420-29).

CD46 or SLAM/CD150 transgenic mice are challenged with a lethal dose of the MeV. Animals are immunized intravenously (IV), intramuscularly (IM), or intradermally (ID) at week 0 and week 3 with candidate MeV vaccines with and without adjuvant. The animals are then challenged with a lethal dose of MeV on week 7 via IV, TM or TD. Endpoint is day 13 post infection, death or euthanasia. Animals displaying severe illness as determined by >30% weight loss, extreme lethargy or paralysis are euthanized. Body temperature and weight are assessed and recorded daily.

In experiments where a lipid nanoparticle (LNP) formulation is used, the formulation may include a cationic lipid, non-cationic lipid, PEG lipid and structural lipid in the ratios 50:10:1.5:38.5. The cationic lipid is DLin-KC2-DMA (50 mol %), the non-cationic lipid is DSPC (10 mol %), the PEG lipid is PEG-DOMG (1.5 mol %) and the structural lipid is cholesterol (38.5 mol %), for example.

TABLE 1

hMPV Immunogenicity studies bleeding schedule										
Animal groups			Day							
(n = 8)	vaccine		-2	0	7	14	21	28	35	56
Placebo	Group	PBS	Pre-Bleed	Prime	Bleeds	Bleeds	Bleeds/Boost	Bleeds	Bleeds	Harvest Spleens/Terminal Bleeds
10 µg Dose	Group 1 (n = 8) (IM)	10 µg								
2 µg Dose	Group 2 (n = 8) (IM)	2 µg								
	Group 3 (n = 8) (IM)									

Total n = 24

15

Each of the sequences described herein encompasses a chemically modified sequence or an unmodified sequence which includes no nucleotide modifications.

TABLE 2

Description	Sequence	SEQ ID NO :
hMPV Nucleic Acid Sequences		
gi 122891979 gb EF051124.1 Human metapneumovirus isolate TN/92-4 fusion protein gene, complete genome	ATGAGCTGGAAGGTGGTGATTATCTTCAGCCTGCTGATTA CACCTCAACACGGCCTGAAGGAGAGCTACCTGGAAGAGA GCTGCTCCACCATCACCGAGGGCTACCTGAGCGTGCTGC GGACCGGCTGGTACACCAACGTGTTACCCCTGGAGGTGG GCGACGTGGAGAACCTGACCTGCAGCGACGGCCCTAGCC TGATCAAGACCGAGCTGGACCTGACCAAGAGCGCTCTGA GAGAGCTGAAGACCGTGTCCGCCGACCACTGGCCAGAG AGGAACAGATCGAGAACCCTCGGCAGAGCAGATTCTGTGC TGGGCGCCATCGCTCTGGGAGTCGCCGCTGCCGCTGCAG TGACAGCTGGAGTGGCCATTGCTAAGACCATCAGACTGG AAAGCGAGGTGACAGCCATCAACAATGCCCTGAAGAAG ACCAACGAGGCGGTGAGCACCTGGGCAATGGAGTGAGA GTGCTGGCCACAGCGTGGCGGAGCTGAAGGACTTCGTG AGCAAGAACCTGACCAGAGCCATCAACAAGAACAAGTG CGACATCGATGACCTGAAGATGGCCGTGAGCTTCTCCCA GTTCAACAGACGGTTCCTGAACGTGGTGAGACAGTTCTC CGACAACGCTGGAATCACACCTGCCATTAGCCTGGACCT GATGACCGACGCGAGCTGGCTAGAGCCGTGCCCAACAT GCCCACCAGCGTGGCCAGATCAAGCTGATGCTGGAGAA CAGAGCCATGGTGGCGAGAAAGGGCTTCGGCATCCTGAT TGGGGTGATGGAAGCTCCGTGATCTACATGGTGACGCT GCCCATCTTCGGCGTGATCGACACACCTGCTGGATCGTG AAGGCCGCTCCTAGCTGCTCCGAGAAGAAAGGAACTAT GCCTGTCTGCTGAGAGAGGACCAGGGCTGGTACTGCCAG AACGCCGGAAGCACAGTGTACTATCCCAACGAGAAGGAC TGCAGAGCCAGAGGCGACACGTGTTCTGCGACACCGCT GCCGGAATCAACGTGGCCGAGCAGAGCAAGGAGTGCAA CATCAACATCAGCACAACCACTACCCCTGCAAGGTGAG CACCGGACGGCACCCCATCAGCATGGTGGCTCTGAGCCC TCTGGGCGCTCTGGTGGCTGCTATAAGGGCGTGTCTGT AGCATCGGCAGCAATCGGGTGGGCATCATCAAGCAGCTG AACAAGGGATGCTCTACATCACCAACCAGGACGCCGAC ACCGTGACCATCGACAACACCGTGTACCAGCTGAGCAAG GTGGAGGGCGAGCAGCACGTGATCAAGGGCAGACCCGT GAGCTCCAGCTTCGACCCCATCAAGTTCCTGAGGACCA GTTCAACGTGGCCCTGGACCAGGTGTTTGAGAACATCGA GAACAGCCAGGCCCTGGTGGACCAGAGCAACAGAATCCT GTCCAGCGCTGAGAAGGGCAACACCGGCTTCATCATTGT GATCATTCTGATCGCCGTGCTGGGCAGCTCCATGATCCTG GTGAGCATCTTCATCATTATCAAGAAGACCAAGAAACCC ACCGGAGCCCCCTCTGAGCTGAGCGGCGTGACCAACAAT GGCTTCATTCCCCACAACCTGA	1
gb AY525843.1 : 3065-4684 Human metapneumovirus isolate NL/1/99, complete genome	ATGCTTTGGAAAGTGATGATCATCTTTCGTTACTCATAA CACCCAGCACGGGCTAAAGGAGAGTTATTTGGAAGAAT CATGTAGTACTATACTGAGGGATACCTCAGTGTTTTAAG AACAGGCTGGTACATAATGTCTTACATTAGAAGTTGGT GATGTTGAAAATCTTACATGTACTGATGGACCTAGCTTAA TCAAAACAGAACTTGATCTAACAAAAGTGCTTTAAGGG AACTCAAAACAGTCTCTGTGATCAGTTGGCGAGAGAGG	2

TABLE 2-continued

Description	Sequence	SEQ ID NO :
gb KJ627414.1 : 3015-4634 Human metapneumovirus strain hMPV/ <i>Homo sapiens</i> /PER/ CFI0497/2010/B, complete genome	AGCAAATTGAAAATCCAGACAATCAAGATTTGTCTTAG GTGCGATAGCTCTCGGAGTTGCTACAGCAGCAGCAGTCA CAGCAGGCATTGCAATAGCCAAAACCATAAGGCTTGAGA GTGAGGTGAATGCAATTAAAGGTGCTCTCAAACAACTA ATGAAGCAGTATCCACATTAGGGAATGGTGTGCGGGTCC TAGCCACTGCAGTGAGAGAGCTAAAGAATTGTGAGCA AAAACCTGACTAGTGCATCAACAGGAACAAATGTGACA TTGCTGATCTGAAGATGGCTGTCAGCTTCAGTCAATTCAA CAGAAGATTCTTAAATGTTGTGCGCAGTTTTTCAGACAAT GCAGGGATAACACCAGCAATATCATTGGACCTGATGACT GATGCTGAGTTGGCCAGAGCTGTATCATACATGCCAACA TCTGCAGGGCAGATAAAACTGATGTTGGAGAACC CGCA ATGGTAAGGAGAAAAGGATTGGAATCCTGATAGGGGTC TACGGAAGCTCTGTGATTTACATGGTTCAATTGCCGATCT TTGGTGT CATAGATACACCTTGTGGATCATCAAGGCAGC TCCCTCTTGCTCAGAAAAACGGGAATTATGCTTGCTC CTAAGAGAGGATCAAGGGTGGTATTGTAAAAATGCAGGA TCTACTGTTTACTACCCAAATGAAAAGACTGCGAAACA AGAGGTGATCATGTTTTTTGTGACACAGCAGCAGGGATC AATGTTGCTGAGCAATCAAGAGAATGCAACATCAACATA TCTACTACCAACTACCCATGCAAAGTCAGCACAGGAAGA CACCCCTATAAGCATGGTTGCACTATCACCTCTCGGTGCTT TGGTGGCTTGCTATAAAGGGTAAGCTGCTCGATTGGCA GCAATTGGGT TGGATCATCAAACAATTACCCAAAGGCTGCTCATACAT AACCACAGGATGCAGACACTGTAACAATTGACAATAC CGTGATCAACTAAGCAAAGTTGAAGGTGAACAGCATGT AATAAAAGGGAGACCAGTTTCAAGCAGTTTTGATCCAAT CAAGTTTCTGAGGATCAGTTCAATGTTGCGCTTGATCAA GTCTTCGAAAGCATTGAGAACAGTCAGGCACCTAGTGGAC CAGTCAAAACAAATTTCTAAACAGTGCAGAAAAGGAAA CACTGGTTTCATTATCGTAGTAATTTTGGTTGCTGTTCTTG GTCTAACCATGATTTTCAGTGAGCATCATCATATAATCAA GAAAAACAAGGAAGCCACAGGAGCACCTCCAGAGCTGA ATGGTGTCAACACGGCGGTTTCATACCACATAGTTA	3
	ATGTCTTGAAAGTGATGATTATCATTTTCGTTACTCATAA CACCTCAGCATGGACTAAAAGAAAGTTATTTAGAAGAAT CATGTAGTACTATAACTGAAGGATATCTCAGTGTTTTAAG AACAGGTTGGTACACCAATGTCTTTACATTAGAAGTTGGT GATGTTGAAAACTCTTACATGTACTGATGGACCTAGCTTAA TCAAACAGAACTTGACCTAACCAAAAGTGCTTTAAGAG AACTCAAAACAGTTTCTGCTGATCAGTTAGCGAGAGAAG AACAAATTGAAAATCCAGACAATCAAGGTTTGTCCTAG GTGCAATAGCTCTGGAGTTGCCACAGCAGCAGCAGTCA CAGCAGGCATTGCAATAGCCAAAACATAAGGCTTGAGA GTGAAGTGAATGCAATCAAAGGTGCTCTCAAACAAACA ATGAGGCAGTATCAACACTAGGAAATGGAGTGC GGCTCC TAGCCACTGCAGTAAGAGAGCTGAAAGAATTGTGAGCA AAAACCTGACTAGTGCATCAACAAGACAAGTGTGACA TTGCTGATTTGAAGATGGCTGTCAGCTTCAGTCAGTTCAA CAGAAGATTCTTAAATGTTGTGCGCAGTTTTTCAGACAAT GCAGGGATAACACCAGCAATATCATTGGACCTGATGAAT GATGCTGAGCTGGCCAGAGCTGTATCATACATGCCAACA TCTGCAGGACAGATAAACTAATGTTAGAGAACCCTGCA ATGGTGAGGAGAAAAGGATTGGAATCTTGATAGGGGTC TACGGAAGCTCTGTGATTTACATGGTCCAGCTGCCGATCT TTGGTGTCTATAAATACACCTTGTGGATAATCAAGGCAGC TCCCTCTTGTTTCAAAAAAGATGGAATTTATGCTTGCTC CTAAGAGAGGATCAAGGGTGGTATTGTAAAAATGCAGGA TCCACTGTTTACTACCCAAATGAAAAGACTGCGAAACA AGAGGTGATCATGTTTTTTGTGACACAGCAGCAGGGATC AATGTTGCTGAGCAATCAAGAGAATGCAACATCAACATA TCTACCACCAACTACCCATGCAAAGTCAGCACAGGAAGA CACCCATCAGCATGGTTGCACTATCACCTCTCGGTGCTT TGGTAGCTTGCTACAAAGGGTTAGCTGCTCGACTGGCA GTAATCAGGTTGGAATAATCAAACAACTACCTAAAGGCT GCTCATACATACTAACCAGGACGCAGACACTGTAACAA TTGACAACACTGTGTATCAACTAAGCAAAGTTGAGGGTG AACAGCATGTAATAAAAGGGAGACCAGTTTCAAGCAGTT TTGATCCAATCAGGTTTCTGAGGATCAGTTCAATGTTGC GCTTGATCAAGTCTTTGAAAGCATTGAAACAGTCAAGC ACTAGTGGACCAAGTCAACAAAATCTGAACAGTGCAGA AAAAGGAAACACTGGT	

TABLE 2-continued

Description	Sequence	SEQ ID NO:
	TTCATTATTGTAATAATTTTGATTGCTGTTCTTGGGTTAAC CATGATTTTCAGTGAGCATCATCATATAATCAAAAAAAC AAGGAAGCCACAGGGGCACCTCCGGAGCTGAATGGTGT TACCAACGCGGTTTCATACCGCATAGTTAG	
gb KJ723483.1 : 5586-7310 Human respiratory syncytial virus strain RSVA/ <i>Homo sapiens</i> /USA/84I- 215A-01/1984, complete genome	ATGGAGTTGCCAATCCTCAAAACAAATGCAATTACCACA ATCCTTGCTGCAGTCACACTCTGTTTCGCTTCCAGTCAAA ACATCACTGAAGAATTTTATCAATCAACATGCAGTGACAG TTAGCAAAGGCTATCTTAGTGCTCTAAGAAGTGGTTGGTA TACTAGTGTTATAACTATAGAATTAAAGTAATATCAAGGA AAATAAGTGTAAATGGAACAGATGCTAAGGTAAATTTGAT AAAACAAGAATTAGATAAATATAAAAAATGCTGTAAACAGA ATTGCAGTTGCTCATGCAAGCACACAGCAGCCAAACAA TCGAGCCAGAAGAGAACTACCAAGGTTTATGAATTATAC ACTCAATAATACCAAAAAATACCAATGTAACATTAAGCAA GAAAAGGAAAAGAAGATTTCTTGGCTTTTGTAGGTGTT GGATCTGCAATCGCCAGTGGCATTGCTGTATCTAAGGTCC TGCACCTAGAAGGGGAAGTGAACAAAATCAAAAGTGTCT TACTATCCACAACCAAGGCTGTAGTCAGCTTATCAATG GAGTTAGTGCTTAAACCAGCAAGTGTAGACCTCAAAA ACTATATAGATAAAACAGTTGTTACCTATTGTGAACAAGC AAAGCTGCAGCATATCAAACATTGAAACTGTGATAGAGT TCCAACAAAAGAACAACAGACTACTAGAGATTACCAGGG AATTTAGTGTTAATGCAGGTGTAACACCTGTAAGCAC TTATATGTTAACTAATAGTGAATTATTATCATTAATCAAT GATATGCCTATAACAAATGATCAGAAAAAGTTAATGTCC AACAAATGTTCAAATAGTTAGACAGCAAGTTACTCTATC ATGTCCATAATAAAGGAGGAAGTCTTAGCATATGTAGTA CAATTACCACTATATGGTGAATAGATACACCTGTTGGA AACTGCACACATCCCTCTATGTACAACCAACACAAAGG AAGGGTCCAACATCTGCTTAACAAGAACCAGCAGAGGAT GGTATTGTGACAATGCAGGATCAGTATCTTCTTCCCACA AGCTGAAACATGTAAGTTCAATCGAATCGGGTATTTTGT GACACAAATGAACAGTTTAACTTACCAAGTGAAGTAAAT CTCTGCAACATTGACATATTCAACCCCAATATGATTGCA AAATTATGACTTCAAAAAACAGATGTAAGCAGCTCCGTTA TCACATCTCTAGGAGCCATTGTGTATGCTATGGCAAAAC TAAATGTACAGCATCCAATAAAAAATCGTGGGATCATAAA GACATTTTCTAACGGGTGTGATTATGTATCAAATAAGGG GGTGGATACTGTGTCTGTAGGTAATCATTATATTATGTA AATAAGCAAGAAGGCAAAAGTCTCTATGTAAGGTGAA CCAATAATAAATTTCTATGACCCATTAGTGTCCCTCTG ATGAATTTGATGCATCAATATCTCAAGTCAATGAGAAGA TTAACCAGAGCCTAGCATTTATTCGTAATCCGATGAATT ATTACATAATGTAAATGCTGGTAAATCCACCACAAATAT CATGATAACTACTATAATTATAGTGATTATAGTAATATTG TTATCATTAATTGCAGTTGGACTGCTCTATCTGCAAGG CCAGAAGCACACCAGTCACTAAGTAAGGATCAACTGA GTGGTATAAATAATATTGCATTTAGTAAGTGA	4
hMPV mRNA Sequences		
gi 122891979 gb EF051124.1 Human metapneumovirus isolate TN/92-4 fusion protein gene, complete genome	AUGAGCUGGAAGGUGUGAUUAUCUUCAGCCUGCUGAU UACACCUCACACGGCCUGAAGGAGAGCUACCUGGAAG AGAGCUGCUGCCACCAUCACCGAGGGCUACUGAGCGUG CUGCGGACCGGUGUGUACACCAACGUGUUCACCCUGGA GGUGGGCGACGUGGAGAACUGACCUGCAGCGACGGCC CUAGCCUGAUCAAGACCGAGCUGGACCUGACCAAGAGC GCUUGAGAGAGCUGAAGACCUGUCCGCGGACAGCU GGCCAGAGAGGAACAGAU CGAGAACCUCGGCAGAGCA GAUUCGUGCUGGGCGCAUCGUCUGGGAGUCGCGCGU GCCGUGCAGUGACAGCUGGAGUGGCCAUUGCUAAGAC CAUCAGACUGGAAAGCGAGGUGACAGCCAUAACA AUG CCUGAAGAAGACCAACGAGGCCUGAGCACCUGGGC AAUGGAGUGAGAGUGCUGGCCACAGCCUGCGGGAGCU GAAGGACUUCGUGAGCAAGAACCUGACCAGAGCCAUA ACAAGAACAAGUGCGACAUCGAUGACCUGAAGAUGGCC GUGAGCUUCUCCAGUUAACAGACGGUUCUGAACGU GGUGAGACAGUUCUCCGACAACGUGGAAUACACCCUG CCAUAAGCCUGGACUGAUGACCAGCGCCAGCUGGCU AGAGCCGUGCCCAACAUGCCACAGCGCUGGCCAGAU CAAGCUGAUGCUGGAGAACAGAGCCAUGGUGCGGAGAA AGGGCUUCGGCAUCCUGAUUGGGUGUUAUGGAAGCUCC GUGAUCUACAUGGUGCAGCUGCCAUUCUGCGCGUGAU CGACACACCCUGCUGGAUCGUGAAGGCCGCUCCUAGCU GCUCCGAGAAGAAAGGAACUAUGCUGUCUGCUGAGA	57

TABLE 2-continued

Description	Sequence	SEQ ID NO :
	GAGGACCAGGGCUGGUACUGCCAGAACGCCGGAAGCAC AGUGUACUAUCCCAACGAGAAGGACUGCGAGACCAGAG GCGACCACGUGUUCUGCGACACCGCUGCCGGAUUAAC GUGGCCGAGCAGAGCAAGGAGUGCAACAUCAACAUCA CACAACCAACUACCCUGCAAGGUGAGCACCCGGACGGC ACCCAUACAGCAUGGUGGCUUGAGCCUUGGGCGCU CUGGUGGCCUGCUAUAAGGGCUGUCCUGUAGCAUCGG CAGCAAUCCGGUGGGCAUCAACAGCAGCUGAACAAAG GAUGCUCCUACAUCAACCAACAGGACGCCGACACCGUG ACCAUCGACAACACCGUGUACAGCUGAGCAAGGUGGA GGGCGAGCAGCAGUGAUCAAGGGCAGACCCUGAGCU CCAGCUUCGACCCCAUCAAGUCCUGAGGACAGUUC AACGUGGCCUGGACCCAGGUGUUUGAGAACAUCCGAGAA CAGCCAGGCCUGGUGGACAGAGCAACAGAAUCCUGU CCAGCGCUGAGAAGGGCAACACCGCUUCAUCAUUGUG AUCAUUCUGAUCGCCGUGCUGGGCAGCUCUAUGAUCCU GGUGAGCAUCUUAUCAUUAUCAAGAAGACCAAGAAAC CCACCGAGGCCCCUCCUGAGCUGAGCGCGUGACCAAC AAUGGCUUCAUCCCCACACUGA	
gb AY525843.1 : 3065-4684 Human metapneumovirus isolate NL/1/99, complete genome	AUGUCUUGGAAAGUGAUGAUCAUUAUUCGUUACUCAU AACACCCACGACGGGCUAAAGGAGAUUAUUGGAAG AAUCAUGUAGUACUAUAACUGAGGGAUACCUAGUGUU UUAAGAACAGGCUUGUACACUAAGUUCUUAUCAUAGA AGUUGGUGAUGUUGAAAUCUUAUAGUUAUCUGAUGGA CCUAGCUAAUCAAACAGAACUUGAUCAACAAAAG UGCUUUAAGGGAACUCAAACAGUCUCUGCUGAUCAGU UGGCGAGAGAGGAGCAAAUUGAAAUCGCCAGACAUCA AGAUUUGUCUAGGUGCGAUAGCUCUCGGAGUUGCUAC AGCAGCAGCAGUCAAGCAGGCAUUGCAAUAGCCAAAA CCAUAAGGCUUGAGAGUGAGGUGAAUGCAAUUAAGG UGCUCUCAAACAAACUAAUGAAGCAGUAUCCCAUUAAG GGAAUGGUGUGCGGGUCCUAGCCACUGCAGUGAGAGAG CUAAAAGAAUUGUGAGCAAAACCUAGUAGUGCAAU CAACAGGAACAAUUGAGCAUUGCUGAUCUGAAGAUGG CUGUCAGCUUCAGUCAAUUCAAACAGAAUUAUUAUUA GUUGUGCGGCAGUUUUCAGACAUAUGCAGGGAUAACACC AGCAAUAUCAUUGGACCGAUGACUGAUGCUGAGUUGG CCAGAGCUGUAUCAUACAUGCCAACUUGCAGGGCAG AUAAAACUGAUGUUGGAGAACCGCGCAUUGGUAAGGAG AAAAGGAUUGGAUCCUGAUAGGGGUCUACGGAAGCU CUGUGAUUUAUAGGUUCAAUUGCCGAUCUUUGGUGUC AUAGAUAACCCUUGUUGGAUCAUACAGGCAGCUCUCCUC UUGCUAGAAAAAACCGGAAUUAUGCUUGCCUCCUAA GAGAGGAUCAAGGGUGGUUAUUGUAAAAUAGCAGGAUC UACUGUUUAUACCCAAUUGAAAAAGACUGCGAAACAA GAGGUGAUCAGUUUUUGUGACACAGCAGCAGGGAUUC AAUGUUGCUGAGCAAUCAAGAGAAUGCAACAUCAACAU AUCUACUACCAACUACCCAUUGCAAAGUCAGCACAGGAA GACACCCUAUAAGCAUGGUGCACAUCACCCUUCGUGU GCUUUGGUGGCUUGCUAUAAGGGGUAAGCUCUCGAU UGGCAGCAAUUGGGU UGGAAUCAUCAAACAAUUAACCAAGGCUGCUCUAACA UAACCAACAGGAUGCAGACACUGUAACAAUUGACAUA ACCGUGUAUCAACUAAGCAAGUUGAAGGUGAACAGCA UGUAAUAAAAGGGAGACAGUUUAAGCAGUUUUGAUC CAAUCAGUUUCCUGAGGAUCAGUUCAAUGUUGCGCUU GAUCAAGUCUUCGAAAGCAUUGAGAACAGUCAGGCACU AGUGGACCAGUCAAAACAAAUUCUAAACAGUGCAGAAA AAGGAAACACUGGUUUAUUAUCGUAAGUAAUUGGU UGCUGUUUUGGUCUAACCAUGAUUUCAGUGAGCAUCA UCAUCAUAUAAGAAAAAAGGAAGCCACAGGAGCA CCUCCAGAGCUGAAUGGUUCCCAACGGCGGUUUAU ACCACAUAUUAUAG	58
gb KJ627414.1 : 3015-4634 Human metapneumovirus strain hMPV/ <i>Homo sapiens</i> /PER/ CFI0497/2010/B, complete genome	AUGUCUUGGAAAGUGAUGAUUAUCAUUAUUCGUUACUCAU AACACCCACGACUGGACUAAAAGAAAGUUAUUAAGAA AAUCAUGUAGUACUAUAACUGAAGGAUAUCUAGUGUU UUAAGAACAGGUUGGUACCAUAGUUCUUAUCAUAGA AGUUGGUGAUGUUGAAAUCUUAUAGUUAUCUGAUGGA CCUAGCUAAUCAAACAGAACUUGACCUAACCAAAAG UGCUUUAAGAGAACUCAAACAGUUUCUGCUGAUCAGU UAGCGAGAGAAGAACAAUUGAAAUCGCCAGACAAUCA AGGUUUGUCUAGGUGCAAUAGCUCUUGGAGUUGCCAC AGCAGCAGCAGUCAAGCAGGCAUUGCAAUAGCCAAAA CUUAUAGGCUUGAGAGUGAAGUGAAUGCAAUCAAAGG	59

TABLE 2-continued

Description	Sequence	SEQ ID NO :
gb KJ723483.1 : 5586-7310 Human respiratory syncytial virus strain RSV A/Homo sapiens/USA/84I- 215A-01/1984, complete genome	UGCUCUCAAAACAACCAUAGGCGAGUAUCAACACUAG GAAAUUGGAGUGCGGGUCCUAGCCACUGCAGUAAGAGAG CUGAAAGAAUUGUGAGCAAAACUGACUAGUGCGAU CAACAAGAAACAAGUGUGACAUUGCGAUUUGAAGAUUG CUGUCAGCUUCAGUCAGUUAACAAGAGAUUCCUAAAU GUUGUGCGGCAGUUUUCAGACAAUGCAGGGAUAACACC AGCAUAUCAUUGGACCUGAUGAAUGAUGCUGAGCUGG CCAGAGCUGUAUCAUAUGCCAACAUUGCAGGACAG AUAAAACUAAUGUUAAGAGAACCUGCAUUGGUGAGGA GAAAGGAUUUGGAUUCUUGAUAGGGGUCUACGGAAG CUCUGUGAUUUACAUGGUCCAGCUGCCGAUCUUUGGUG UCAUAAAUAACCCUUGUUGGAUAAUCAAGGCAGCUC UCUUGUUCAGAAAAGAUUGGAAUUAUGCUUGCCUCCU AAGAGAGGAUCAAGGGUGUAUUGUAAAAUUGCAGGA UCCACUGUUUACUACCCAAUUGAAAGACUGCGAAAC AAGAGGUGAUCUAGUUUUUGUGACACAGCAGCAGGGA UCAUUGUUGCUGAGCAAUCAAGAGAAUGCAACAUCAAC AUAUCUACCACCAACUACCCAUUGCAAGUAGCAGCAGG AAGACACCCUAUCAGCAUGGUUGCAUAUACCCUCUCG GUGCUUUGGUAGCUUGCUACAAGGGGUUAGCUGCUCG ACUGGCAGUAAUCAGGUUGGAUUAUCAAACAACUACC UAAAGGCUGCUCAUAUCAUAACUAAACAGGACGCAGACA CUGUAACAUAUGACAACAUCUGUGUAUACAUAAGCAAA GUUGAGGGUGAACAGCAUGUAUAAUAAAGGGAGACCAG UUUCAAGCAGUUUGAUCCAUAUCAGGUUCCUGAGGAU CAGUUCAAUGUUGCGCUUGAUCAAGUCUUUGAAAGCAU UGAAACAGUCAAGCACUAGUGGACAGUCAACAAAAA UUCUGAACAGUGCAGAAAAGGAAACACUGGU UUCAUUAUUGUAUAAUUAUUGAUUGCUGUUCUUGGGU UAACCAUGAUUUUAGUGAGCAUCAUCAUAUAAUCAA AAAACAAGGAAGCCACAGGGGCACUCCGGAGCUGAA UGGUGUUAACACGGCGGUUCAUACCGCAUGAUUAG	
	AUGGAGUUGCCAAUCCUCAAAACAAUUGCAAUUAACCA AAUCCUUGCUGCAGUCACACUCUGUUUCGUUCCAGUC AAAACAUCACUGAAGAAUUAUCAAUCAAACUAGCAGU GCAGUUAGCAAAGGCUAUUCUAGUGCUUUAAGAACUGG UUGGUUAUCUAGUGUUAUAAUUAUAGAAUUAAGUAAU AUCAGGAAUUAAGUGUAAUGGAACAGAUUGCUAAGG UAAAAUUGAUAAACAAGAAUUAAGUAAUUAUAAAA UGCUGUAACAGAAUUGCAGUUGCUCUAGCAAAGCACAC CAGCAGCCAAACAUCGAGCCAGAAGAGAACUACCAAGG UUUUAGAAUUAUACACUCAUAAUACCAAAAUACCAA UGUAACAUUAAGCAAGAAAGGAAAGAAUUAUUCU GGCUUUUUGUUAGGUUUUGAUUCUGCAAUCGCCAGUGG CAUUGCUGUAUCUAAAGGUCCUGCACCUAGAAGGGGAAG UGAACAAAUCAAAAGUGCUUACUUAUCCACAACAAAG CGUUGAGUCAGCUUUAUCAAUUGGAGUUAUGUCUUAAC CAGCAAAGUGUAGACCUCAAAAACUUAUAGAUAAAC AGUUGUUAACCUAUGUGAACAAGCAAGCUGCAGCAUA UCAAAUAUUGAAACUGUGAUAGAGUUCACAAAAGAA CAACAGACUACUAGAGAUUAACAGGGAUUUAGUGUUA AUGCAGGUGUAACUACACUGUAAGCAUUAUAGUUA ACUAAUAGUGAAUUAUUAUCAUUAUCAAUGAUUAGCC UAUAACAAUAGAUUCAGAAAAAGUUAUGUCCAAACAAUG UUCAAAUAGUUAGACAGCAAGUUACUUAUUAUGUCC AUAUAAAGGAGGAAGUCUUAAGCAUAGUAGUACAUA UACCAUAUAGGUGUAUUAAGAUACACCCUGUUGGAA CUGCACACAUCUCCUUAUUGUACAACCAACACAAAGGA AGGUCACAAUCUGCUUUAACAAGAACCGACAGAGGAU GGUUAUUGGACAAUGCAGGAUCAGUAUCUUUCUCCCA CAAGCUGAAACAUUAAAGUUAUUAUCAAUUGCGGUUU UUGUGACACAAUGAACAGUUUAACAUAUACCAAGUGAAG UAAAUUCUGCAACAUUGACAUUAUUAACCCCAAUAU GAUUGCAAAAUUAUGAUUCAAUAAACAGAUUAAGCAG CUCCGUUAUCAUUCUAGGAGCCAUUGUGUCAUGCU AUGGCAAAACUAAUUGUACAGAUCCAAUAAAAUUCGU GGGAUCAUAAAGACAUUUUUAACGGGUGUGAUUAUG UAUCAAAUAAGGGGUGGAUACUGUGUCUGUAGGUAA UACAUUAUUAUUGUAAUUAAGCAAGAGGCAAAAGU CUCUAUGUAAAAGGUGAACCAUUAUAAUUAUUAUGA CCCAUUAGUGUUCUCCUUGAUGAAUUGAUGCAUCA UAUCUCAAGUCAUAGAGAAUUAACAGAGCCUAGCA UUUUUUCGUAAAUCCGAUGAAUUAUUAUUAUUAUGA AUGCUGUAAAUCCACCAAAUUAUUAUUAUUAUUAU AUAUUAUUAUGAUUAUUAUUAUUAUUAUUAUUAU	60

TABLE 2-continued

Description	Sequence	SEQ ID NO:
	UUGCAGUUGGACUGCUCCUAUACUGCAAGGCCAGAAGC ACACCAGUCACACUAAGUAAGGAUCAACUGAGUGGUU AAAUAAUUAUGCAUUUAGUAACUGA	

TABLE 3

hMPV Amino Acid Sequences	
---------------------------	--

Description	Sequence	SEQ ID NO:
-------------	----------	------------------

gi|122891979|gb|EF051124.1| Human metapneumovirus isolate TN/92-4 fusion protein gene, complete cds

MSWKVVIIFSLLITPQHGLKESY
LEESCSTITEGYLSVLRGTWYTN
VPTLEVGDVENLTCSGSLIKT
ELDLTKSALRELKTVSADQLARE
EQIENPRQSRFVLGAIALGVAAA
AAVTAGVAIAKTIRLESEVTAIN
NALKKTNEAVSTLGNQVRLATA
VRELKDFVSKNLTRAINKNKCDI
DDLKMAVSFSQFNRRFLNVVRQF
SDNAGITPAISLDLMTDAELARA
VPNMPTSAGQIKMLLENRAMVRR
KGFGLIGVYGSSVIYMQLPPIF
GVIDTPCWIIKAAPSCSEKKNY
ACLLREDQGWYCNAGSTVYYPN
EKDCETRGRDHVPCDTAAGINVAE
QSKENINISTTNYPCVKSTGRH
PISMVALSPLGALVACYKGVSCS
IGSNRVGIIKQLNKGCSYITNQD
ADTVTIDNTVYQLSKVEGEQHV
KGRPVSSSFDPIKFPEDQFNVAL
DQVFENIENSQALVDQSNRISS
AEKGTGFIIVILIAVLGSSMI
LVSIFIIKKTKKPTGAPPELNG
VTNNGFIPHN

gb|AY525843.1| : 3065-4684 Human metapneumovirus isolate NL/1/99, complete cds

MSWKVMIIFSLLITPQHGLKESY
LEESCSTITEGYLSVLRGTWYTN
VPTLEVGDVENLTCTDGPGLIKT
ELDLTKSALRELKTVSADQLARE
EQIENPRQSRFVLGAIALGVATA
AAVTAGIAIAKTIRLESEVNAIK
GALKQTNEAVSTLGNQVRLATA
VRELKEFVSKNLTSAINRNKCDI
ADLKMAVSFSQFNRRFLNVVRQF
SDNAGITPAISLDLMTDAELARA
VSYMPTAGQIKMLLENRAMVRR
KGFGLIGVYGSSVIYMQLPPIF
GVIDTPCWIIKAAPSCSEKKNY
ACLLREDQGWYCKNAGSTVYYPN
EKDCETRGRDHVPCDTAAGINVAE
QSRECNINISTTNYPCVKSTGRH
PISMVALSPLGALVACYKGVSCS
IGSNWVGIIKQLPKGCSYITNQD
ADTVTIDNTVYQLSKVEGEQHV
KGRPVSSSFDPIKFPEDQFNVAL
DQVFESIENSQALVDQSNKILNS
AEKGTGFIIVILIAVLGLTMI
SVSIIIIKKTRKPTGAPPELNG
VTNNGFIPHS

gb|KJ627414.1| : 3015-4634 Human metapneumovirus strain hMPV/Homo sapiens/PER/CFI0497/2010/B, complete cds

MSWKVMIIFSLLITPQHGLKESY
LEESCSTITEGYLSVLRGTWYTN
VPTLEVGDVENLTCTDGPGLIKT
ELDLTKSALRELKTVSADQLARE
EQIENPRQSRFVLGAIALGVATA
AAVTAGIAIAKTIRLESEVNAIK
GALKKTNEAVSTLGNQVRLATA
VRELKEFVSKNLTSAINKNKCDI
ADLKMAVSFSQFNRRFLNVVRQF
SDNAGITPAISLDLMDAELARA
VSYMPTAGQIKMLLENRAMVRR

TABLE 3-continued

hMPV Amino Acid Sequences	
---------------------------	--

Description	Sequence	SEQ ID NO:
-------------	----------	------------------

KGFGLIGVYGSSVIYMQLPPIF
GVINTPCWIIKAAPSCSEKKNY
ACLLREDQGWYCKNAGSTVYYPN
EKDCETRGRDHVPCDTAAGINVAE
QSRECNINISTTNYPCVKSTGRH
PISMVALSPLGALVACYKGVSCS
TGSNQVGIIKQLPKGCSYITNQD
ADTVTIDNTVYQLSKVEGEQHV
KGRPVSSSFDPIKFPEDQFNVAL
DQVFESIENSQALVDQSNKILNS
AEKGTGFIIVILIAVLGLTMI
SVSIIIIKKTRKPTGAPPELNG
VTNNGFIPHS

gb|KJ723483.1| : 5586-7310 Human respiratory syncytial virus strain RSVa/Homo sapiens/USA/84I-215A-01/1984, complete cds

MELPILKTNAITTILAAVTLCPA
SSQNIITEFYQSTCSAVSKGYLS
ALRTGWYTSVITIELSNIKENKC
NGTDAKVKLIKQELDKYKNAVTE
LQLLMQSTPAANNRARELPRFM
NYTLNNTKNTNVTLSKKRKRFL
GFLLGVGSAIASGIAVSKVLHLE
GEVNIKSALLSTNKAVVSLNSG
VSVLTSKVLDLKNYIDKQLLPPIV
NKQSCSISNIETVIEFQKNNRL
LEITREFSVNAGVTTTPVSTYMLT
NSELLSLINDMPTINDQKKLMSN
NVQIVRQQSYSIMSIIKEEVLAY
VVQLPLYGVIDTPCWKLHTSPLC
TTNTKEGSNICLTRDRGWYCDN
AGSVSFFPQAEETCKVQSNRVPCD
TMNSLTLPSEVNLNIDIFNPKY
DCKIMTSKTDVSSSVITSLGAIV
SCYGKTKCTASNNKRGIIKTFSN
GCDYVSNKGVDVTVSGNTLYYVN
KQEGKSLYVKGEPINFDPLVF
PSDEPDASISQVNEKINQSLAFI
RKSEDELLHNVNAGKSTTNIMIT
IIIVIIIVILLSLIAVGLLLYCKA
RSTPVTLSKQQLSGINNIAFSN

TABLE 4

hMPV NCBI Accession Numbers (Amino Acid Sequences)	
--	--

Virus	GenBank Accession
F [Human metapneumovirus] [Human metapneumovirus]	AEK26895.1
fusion glycoprotein [Human metapneumovirus]	ACJ53565.1
fusion glycoprotein [Human metapneumovirus]	ACJ53566.1
fusion glycoprotein [Human metapneumovirus]	ACJ53569.1
fusion protein [Human metapneumovirus]	AEZ52347.1
fusion glycoprotein [Human metapneumovirus]	ACJ53574.1
fusion glycoprotein [Human metapneumovirus]	AHV79473.1
fusion glycoprotein [Human metapneumovirus]	ACJ53570.1
fusion glycoprotein [Human metapneumovirus]	ACJ53567.1
fusion protein [Human metapneumovirus]	AAS22125.1
fusion glycoprotein [Human metapneumovirus]	AHV79795.1
fusion glycoprotein [Human metapneumovirus]	AHV79455.1

231

TABLE 4-continued

hMPV NCBI Accession Numbers (Amino Acid Sequences)	
Virus	GenBank Accession
fusion glycoprotein [Human metapneumovirus]	ACJ53568.1
fusion protein [Human metapneumovirus]	AAS22109.1
fusion glycoprotein [Human metapneumovirus]	AGU68417.1
fusion glycoprotein [Human metapneumovirus]	AGJ74228.1
fusion glycoprotein [Human metapneumovirus]	ACJ53575.1
fusion protein [Human metapneumovirus]	AAU25820.1
fusion glycoprotein [Human metapneumovirus]	AGU68377.1
fusion glycoprotein [Human metapneumovirus]	AGU68371.1
fusion glycoprotein [Human metapneumovirus]	AGJ74087.1
fusion glycoprotein [Human metapneumovirus]	ACJ53560.1
fusion glycoprotein [Human metapneumovirus]	AHV79858.1
fusion glycoprotein [Human metapneumovirus]	ACJ53577.1
fusion protein [Human metapneumovirus]	AAS22085.1
fusion protein [Human metapneumovirus]	AEZ52348.1
fusion glycoprotein [Human metapneumovirus]	AGJ74044.1
fusion glycoprotein [Human metapneumovirus]	ACJ53563.1
fusion glycoprotein precursor [Human metapneumovirus]	YP_012608.1
fusion glycoprotein [Human metapneumovirus]	AGJ74053.1
fusion protein [Human metapneumovirus]	BAM37562.1
fusion glycoprotein [Human metapneumovirus]	ACJ53561.1
fusion glycoprotein [Human metapneumovirus]	AGU68387.1
fusion [Human metapneumovirus]	AGL74060.1
fusion glycoprotein precursor [Human metapneumovirus]	AAV88364.1
fusion protein [Human metapneumovirus]	AAN52910.1
fusion protein [Human metapneumovirus]	AAN52915.1
fusion protein [Human metapneumovirus]	BAM37564.1
fusion glycoprotein precursor [Human metapneumovirus]	BAH59618.1
fusion protein [Human metapneumovirus]	AAQ90144.1
fusion glycoprotein [Human metapneumovirus]	AHV79446.1
fusion protein [Human metapneumovirus]	AEL87260.1
fusion glycoprotein [Human metapneumovirus]	AHV79867.1
fusion protein [Human metapneumovirus]	ABQ66027.2
fusion glycoprotein [Human metapneumovirus]	ACJ53621.1
fusion protein [Human metapneumovirus]	AAN52911.1
fusion glycoprotein [Human metapneumovirus]	AHV79536.1
fusion glycoprotein [Human metapneumovirus]	AGU68411.1
fusion protein [Human metapneumovirus]	AEZ52346.1
fusion protein [Human metapneumovirus]	AAN52913.1
fusion protein [Human metapneumovirus]	AAN52908.1
fusion glycoprotein [Human metapneumovirus]	ACJ53553.1
fusion glycoprotein [Human metapneumovirus]	ATY25727.1
fusion protein [Human metapneumovirus]	ABM67072.1
fusion protein [Human metapneumovirus]	AEZ52361.1
fusion protein [Human metapneumovirus]	AAS22093.1
fusion glycoprotein [Human metapneumovirus]	AGH27049.1
fusion protein [metapneumovirus]	AAK62968.2
fusion glycoprotein [Human metapneumovirus]	ACJ53556.1
fusion glycoprotein [Human metapneumovirus]	ACJ53620.1
fusion protein [Human metapneumovirus]	ABQ58820.1
F [Human metapneumovirus] [Human metapneumovirus]	AEK26886.1
fusion glycoprotein [Human metapneumovirus]	ACJ53619.1
fusion glycoprotein [Human metapneumovirus]	ACJ53555.1
fusion [Human metapneumovirus]	AGL74057.1
fusion protein [Human metapneumovirus]	ABD27850.1
fusion protein [Human metapneumovirus]	AEZ52349.1
fusion protein [Human metapneumovirus]	ABD27848.1
fusion protein [Human metapneumovirus]	ABD27846.1
fusion protein [Human metapneumovirus]	ABQ66021.1
fusion protein [Human metapneumovirus]	AFM57710.1
fusion protein [Human metapneumovirus]	AFM57709.1
fusion protein [Human metapneumovirus]	ABH05968.1
fusion protein [Human metapneumovirus]	AEZ52350.1
fusion protein [Human metapneumovirus]	AFM57712.1
fusion protein [Human metapneumovirus]	AEZ52364.1
fusion protein [Human metapneumovirus]	AAN52912.1
fusion protein [Human metapneumovirus]	AEZ52363.1
fusion [Human metapneumovirus]	AGL74059.1
fusion glycoprotein [Human metapneumovirus]	ACJ53583.1
fusion protein [Human metapneumovirus]	AEZ52356.1
fusion protein [Human metapneumovirus]	AEZ52353.1
fusion glycoprotein [Human metapneumovirus]	ACJ53581.1
fusion glycoprotein [Human metapneumovirus]	ACJ53578.1
fusion protein [Human metapneumovirus]	AAS22117.1
fusion protein [Human metapneumovirus]	BAN75965.1
fusion protein [Human metapneumovirus]	AGF92105.1

232

TABLE 4-continued

hMPV NCBI Accession Numbers (Amino Acid Sequences)	
Virus	GenBank Accession
fusion protein [Human metapneumovirus]	AAS22077.1
fusion protein [Human metapneumovirus]	AAN52909.1
fusion glycoprotein [Human metapneumovirus]	ACJ53586.1
fusion protein [Human metapneumovirus]	AAQ90145.1
fusion glycoprotein [Human metapneumovirus]	AGT75042.1
fusion [Human metapneumovirus]	AGL74058.1
fusion protein [Human metapneumovirus]	AEL87263.1
fusion glycoprotein [Human metapneumovirus]	AGH27057.1
fusion glycoprotein [Human metapneumovirus]	AHV79491.1
F [Human metapneumovirus] [Human metapneumovirus]	AEK26906.1
fusion glycoprotein [Human metapneumovirus]	ACJ53580.1
fusion protein [Human metapneumovirus]	AEZ52354.1
fusion protein [Human metapneumovirus]	AAN52914.1
G [Human metapneumovirus] [Human metapneumovirus]	AEK26901.1
glycoprotein [Human metapneumovirus]	AFI56738.1
glycoprotein [Human metapneumovirus]	AFI56739.1
glycoprotein [Human metapneumovirus]	AFI56745.1
G protein [Human metapneumovirus]	AAQ62718.1
G protein [Human metapneumovirus]	AAQ62719.1
attachment glycoprotein G [Human metapneumovirus]	AGH27104.1
G protein [Human metapneumovirus]	AAQ62729.1
G protein [Human metapneumovirus]	AAQ62728.1
glycoprotein [Human metapneumovirus]	AFI56753.1
glycoprotein [Human metapneumovirus]	AFI56746.1
glycoprotein [Human metapneumovirus]	AFI56750.1
glycoprotein [Human metapneumovirus]	AFI56747.1
G protein [Human metapneumovirus]	AAQ62721.1
glycoprotein [Human metapneumovirus]	AAT46573.1
glycoprotein [Human metapneumovirus]	AFI56748.1
glycoprotein [Human metapneumovirus]	AFI56736.1
glycoprotein [Human metapneumovirus]	AFI56749.1
attachment glycoprotein G [Human metapneumovirus]	AGH27131.1
attachment glycoprotein G [Human metapneumovirus]	AHV79558.1
glycoprotein [Human metapneumovirus]	AFI56740.1
glycoprotein [Human metapneumovirus]	AFI56743.1
glycoprotein [Human metapneumovirus]	AFI56744.1
attachment glycoprotein G [Human metapneumovirus]	AHV79790.1
attachment glycoprotein G [Human metapneumovirus]	AGH27122.1
attachment glycoprotein G [Human metapneumovirus]	AHV79763.1
attachment glycoprotein G [Human metapneumovirus]	AGZ48849.1
glycoprotein [Human metapneumovirus]	AFI56743.1
attachment glycoprotein G [Human metapneumovirus]	AHV79450.1
glycoprotein [Human metapneumovirus]	AFI56751.1
attachment glycoprotein [Human metapneumovirus]	AAS48482.1
attachment glycoprotein G [Human metapneumovirus]	AHV79889.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43050.1
glycoprotein [Human metapneumovirus]	AFI56754.1
attachment glycoprotein G [Human metapneumovirus]	AHV79601.1
glycoprotein [Human metapneumovirus]	AFI56752.1
attachment glycoprotein G [Human metapneumovirus]	AHV79871.1
G protein [Human metapneumovirus]	AEZ68099.1
attachment glycoprotein G [Human metapneumovirus]	AHV79817.1
attachment glycoprotein G [Human metapneumovirus]	AHV79943.1
attachment glycoprotein G [Human metapneumovirus]	BAN75968.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43045.1
attachment glycoprotein G [Human metapneumovirus]	AHV79628.1
attachment glycoprotein [Human metapneumovirus]	AFK49783.1
G protein [Human metapneumovirus]	AAQ62723.1
attachment glycoprotein [Human metapneumovirus]	ABD27839.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43046.1
G protein [Human metapneumovirus]	AAQ62717.1
glycoprotein [Human metapneumovirus]	AFI56742.1
attachment protein [Human metapneumovirus]	ABQ44522.1
glycoprotein [Human metapneumovirus]	AFI56735.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43065.1
G protein [Human metapneumovirus]	AAQ62724.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43075.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43062.1
glycoprotein [Human metapneumovirus]	AAT46579.1

TABLE 4-continued

hMPV NCBI Accession Numbers (Amino Acid Sequences)	
Virus	GenBank Accession
attachment surface glycoprotein [Human metapneumovirus]	AGW43064.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43054.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43042.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43078.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43067.1
G protein [Human metapneumovirus]	AAQ62722.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43063.1
glycoprotein [Human metapneumovirus]	AAT46571.1
glycoprotein [Human metapneumovirus]	AAT46578.1
attachment glycoprotein G [Human metapneumovirus]	AGJ74232.1
glycoprotein [Human metapneumovirus]	AAT46580.1
glycoprotein [Human metapneumovirus]	AAT46574.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43061.1
attachment glycoprotein [Human metapneumovirus]	AFK49791.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43047.1
glycoprotein [Human metapneumovirus]	ABC26386.1
attachment glycoprotein [Human metapneumovirus]	AAS48466.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43048.1
attachment glycoprotein G [Human metapneumovirus]	AGH27140.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43049.1
attachment glycoprotein G [Human metapneumovirus]	AGJ74082.1
attachment glycoprotein G [Human metapneumovirus]	AHV79442.1
attachment glycoprotein G [Human metapneumovirus]	AGJ74091.1
attachment glycoprotein G [Human metapneumovirus]	AHV79477.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43056.1
attachment protein [Human metapneumovirus]	ABQ44523.1
attachment glycoprotein G [Human metapneumovirus]	BAH59622.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43070.1
glycoprotein [Human metapneumovirus]	AAT46585.1
attachment glycoprotein G [Human metapneumovirus]	AGU68409.1
attachment glycoprotein G [Human metapneumovirus]	AGJ74223.1
attachment glycoprotein [Human metapneumovirus]	AAS22129.1
attachment glycoprotein G [Human metapneumovirus]	AGJ74048.1
G protein [Human metapneumovirus]	AAQ62725.1
glycoprotein [Human metapneumovirus]	ABC26384.1
attachment protein [Human metapneumovirus]	ABQ44525.1
attachment glycoprotein G [Human metapneumovirus]	YP_012612.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43071.1
attachment glycoprotein G [Human metapneumovirus]	AGJ74162.1
attachment glycoprotein G [Human metapneumovirus]	AGH27095.1
attachment glycoprotein G [Human metapneumovirus]	AHV79531.1
G protein [Human metapneumovirus]	AAQ62726.1
attachment glycoprotein [Human metapneumovirus]	AAS48465.1
attachment surface glycoprotein [Human metapneumovirus]	AGW43058.1
P [Human metapneumovirus] [Human metapneumovirus]	AEK26894.1
phosphoprotein [Human metapneumovirus]	AHV79631.1
phosphoprotein [Human metapneumovirus]	AHV79901.1
phosphoprotein [Human metapneumovirus]	AHV79570.1
phosphoprotein [Human metapneumovirus]	AGJ74076.1
phosphoprotein [Human metapneumovirus]	AAS22123.1
phosphoprotein [Human metapneumovirus]	ABB16895.1
phosphoprotein [Human metapneumovirus]	AHV79579.1
phosphoprotein [Human metapneumovirus]	AGJ74244.1
phosphoprotein [Human metapneumovirus]	AHV79856.1
phosphoprotein [Human metapneumovirus]	ACJ70113.1
phosphoprotein [Human metapneumovirus]	AGZ48843.1
phosphoprotein [Human metapneumovirus]	AHV79498.1
phosphoprotein [Human metapneumovirus]	AHV79480.1
phosphoprotein [Human metapneumovirus]	ABQ43382.1
phosphoprotein [Human metapneumovirus]	AAS22107.1
phosphoprotein [Human metapneumovirus]	ABB16898.1
phosphoprotein [Human metapneumovirus]	AGH27134.1

TABLE 4-continued

hMPV NCBI Accession Numbers (Amino Acid Sequences)	
Virus	GenBank Accession
phosphoprotein [Human metapneumovirus]	ABB16899.1
phosphoprotein [Human metapneumovirus]	AGH27098.1
phosphoprotein [Human metapneumovirus]	AAN52866.1
phosphoprotein [Human metapneumovirus]	AAS22083.1
phosphoprotein [Human metapneumovirus]	YP_012606.1
phosphoprotein [Human metapneumovirus]	AHV79973.1
phosphoprotein [Human metapneumovirus]	AHV79462.1
phosphoprotein [Human metapneumovirus]	AGJ74042.1
phosphoprotein [Human metapneumovirus]	AAV88362.1
P [Human metapneumovirus] [Human metapneumovirus]	AIL23591.1
phosphoprotein [Human metapneumovirus]	AHV79453.1
phosphoprotein [Human metapneumovirus]	AGJ74261.1
phosphoprotein [Human metapneumovirus]	AGH27116.1
phosphoprotein [Human metapneumovirus]	ABB16444.1
phosphoprotein [Human metapneumovirus]	ABB16445.1
phosphoprotein [Human metapneumovirus]	AHV79507.1
phosphoprotein [Human metapneumovirus]	BAH59616.1
phosphoprotein [Human metapneumovirus]	ABB16443.1
phosphoprotein [Human metapneumovirus]	ABQ43388.1
phosphoprotein [Human metapneumovirus]	ABQ43389.1
phosphoprotein [Human metapneumovirus]	ABQ43395.1
phosphoprotein [Human metapneumovirus]	ABQ43385.1
phosphoprotein [Human metapneumovirus]	AAP84042.1
phosphoprotein [Human metapneumovirus]	AAN52868.1
phosphoprotein [Human metapneumovirus]	AAP84041.1
phosphoprotein [Human metapneumovirus]	AGH27080.1
phosphoprotein [Human metapneumovirus]	ABQ43387.1
phosphoprotein [Human metapneumovirus]	AAS22099.1
phosphoprotein [Human metapneumovirus]	ABB16896.1
phosphoprotein [Human metapneumovirus]	AGJ74094.1
phosphoprotein [Human metapneumovirus]	AEZ68089.1
phosphoprotein [Human metapneumovirus]	ABK97002.1
phosphoprotein [Human metapneumovirus]	AAP13486.1
phosphoprotein [Human metapneumovirus]	AHV79444.1
phosphoprotein [Human metapneumovirus]	AHV79865.1
phosphoprotein [Human metapneumovirus]	AGJ74226.1
phosphoprotein [Human metapneumovirus]	ABQ43383.1
phosphoprotein [Human metapneumovirus]	AAN52863.1
phosphoprotein [Human metapneumovirus]	AHV79775.1
phosphoprotein [Human metapneumovirus]	AEZ68094.1
phosphoprotein [Human metapneumovirus]	AHV79883.1
phosphoprotein [Human metapneumovirus]	AEZ68092.1
phosphoprotein [Human metapneumovirus]	ABQ43390.1
phosphoprotein [Human metapneumovirus]	ABQ43386.1
phosphoprotein [Human metapneumovirus]	ABQ43391.1
phosphoprotein [Human metapneumovirus]	ACS16062.1
phosphoprotein [Human metapneumovirus]	AEZ68090.1
phosphoprotein [Human metapneumovirus]	AAK62967.1
phosphoprotein [Human metapneumovirus]	AEZ68093.1
phosphoprotein [Human metapneumovirus]	AEZ68088.1
phosphoprotein [Human metapneumovirus]	ABQ43392.1
phosphoprotein [Human metapneumovirus]	ABQ43393.1
phosphoprotein [Human metapneumovirus]	ABQ43384.1
phosphoprotein [Human metapneumovirus]	ABQ43394.1
phosphoprotein [Human metapneumovirus]	ABK96999.1
phosphoprotein [Human metapneumovirus]	AHV79489.1
phosphoprotein [Human metapneumovirus]	AGJ74235.1
phosphoprotein [Human metapneumovirus]	AAS22075.1
phosphoprotein [Human metapneumovirus]	AAS22115.1
phosphoprotein [Human metapneumovirus]	AIL17601.1
phosphoprotein [Human metapneumovirus]	ABK97000.1
phosphoprotein [Human metapneumovirus]	AHV79561.1
phosphoprotein [Human metapneumovirus]	AGT75040.1
phosphoprotein [Human metapneumovirus]	AAN52864.1
phosphoprotein [Human metapneumovirus]	ABK97001.1
phosphoprotein [Human metapneumovirus]	AGT74979.1
phosphoprotein [Human metapneumovirus]	AHV79955.1
phosphoprotein [Human metapneumovirus]	AGH27055.1
phosphoprotein [Human metapneumovirus]	AAV88361.1
phosphoprotein [Human metapneumovirus]	ABQ43397.1
phosphoprotein [Human metapneumovirus]	AGJ74173.1
P [Human metapneumovirus] [Human metapneumovirus]	AEK26904.1
phosphoprotein [Human metapneumovirus]	ACJ70104.1
phosphoprotein [Human metapneumovirus]	ABK97003.1
phosphoprotein [Human metapneumovirus]	AGT74955.1

235

TABLE 4-continued

hMPV NCBI Accession Numbers (Amino Acid Sequences)	
Virus	GenBank Accession
phosphoprotein [Human metapneumovirus]	AAN52856.1
phosphoprotein [Human metapneumovirus]	AAN52862.1
phosphoprotein [Human metapneumovirus]	AGJ74138.1
phosphoprotein [Human metapneumovirus]	AHV79613.1
phosphoprotein [Human metapneumovirus]	AGJ74060.1
phosphoprotein [Human metapneumovirus]	AAQ67684.1
phosphoprotein [Human metapneumovirus]	AEA02278.1
N [Human metapneumovirus] [Human metapneumovirus]	AEK26899.1
nucleoprotein [Human metapneumovirus]	ACS16061.1
nucleoprotein [Human metapneumovirus]	AAS88425.1
nucleoprotein [Human metapneumovirus]	YP_012605.1
nucleoprotein [Human metapneumovirus]	AHV79882.1
nucleoprotein [Human metapneumovirus]	AHV79774.1
nucleocapsid protein [Human metapneumovirus]	AAN52886.1
nucleoprotein [Human metapneumovirus]	AAS22082.1
nucleoprotein [Human metapneumovirus]	AHV79864.1
nucleoprotein [Human metapneumovirus]	AHV79828.1
nucleoprotein [Human metapneumovirus]	AGJ74084.1
nucleocapsid protein [Human metapneumovirus]	AAN52888.1
N [Human metapneumovirus] [Human metapneumovirus]	AIL23590.1
nucleoprotein [Human metapneumovirus]	AAK62966.1
nucleoprotein [Human metapneumovirus]	AHV79972.1
nucleoprotein [Human metapneumovirus]	AHV79470.1
nucleoprotein [Human metapneumovirus]	AHV79452.1
nucleoprotein [Human metapneumovirus]	AGJ74243.1
nucleoprotein [Human metapneumovirus]	AHV79533.1
nucleoprotein [Human metapneumovirus]	AGJ74181.1
nucleoprotein [Human metapneumovirus]	AHV79497.1
nucleoprotein [Human metapneumovirus]	AHV79702.1
nucleoprotein [Human metapneumovirus]	AHV79648.1
nucleoprotein [Human metapneumovirus]	AHV79435.1
putative nucleoprotein [Human metapneumovirus]	AGJ74260.1
nucleocapsid protein [Human metapneumovirus]	AAN52887.1
nucleoprotein [Human metapneumovirus]	AGU68386.1
nucleocapsid protein [Human metapneumovirus]	AAN52899.1
nucleoprotein [Human metapneumovirus]	AAR17673.1
nucleocapsid protein [Human metapneumovirus]	AAN52898.1
nucleoprotein [Human metapneumovirus]	AEA02277.1
nucleoprotein [Human metapneumovirus]	AHV79612.1
nucleoprotein [Human metapneumovirus]	AGU68416.1
nucleoprotein [Human metapneumovirus]	AGU68408.1
nucleoprotein [Human metapneumovirus]	AGU68370.1
nucleoprotein [Human metapneumovirus]	AAQ67683.1
nucleoprotein [Human metapneumovirus]	AGJ74137.1
nucleoprotein [Human metapneumovirus]	AGU68344.1
nucleocapsid protein [Human metapneumovirus]	ABK96997.1
nucleoprotein [Human metapneumovirus]	AGU68413.1
nucleocapsid protein [Human metapneumovirus]	AAN52891.1
nucleoprotein [Human metapneumovirus]	AGU68360.1
nucleoprotein [Human metapneumovirus]	AGU68353.1
nucleocapsid protein [Human metapneumovirus]	ABK96996.1
nucleoprotein [Human metapneumovirus]	AAR17666.1
N [Human metapneumovirus] [Human metapneumovirus]	AEK26903.1
nucleoprotein [Human metapneumovirus]	AGT75039.1

236

TABLE 4-continued

hMPV NCBI Accession Numbers (Amino Acid Sequences)	
Virus	GenBank Accession
nucleoprotein [Human metapneumovirus]	AGU68410.1
nucleoprotein [Human metapneumovirus]	AAS22074.1
nucleoprotein [Human metapneumovirus]	AHV79560.1
nucleoprotein [Human metapneumovirus]	AGT74978.1
nucleoprotein [Human metapneumovirus]	AGJ74128.1
nucleoprotein [Human metapneumovirus]	AAR17663.1
nucleoprotein [Human metapneumovirus]	AAR17662.1
nucleoprotein [Human metapneumovirus]	AAR17664.1
nucleoprotein [Human metapneumovirus]	AAR17657.1
nucleoprotein [Human metapneumovirus]	AAR17659.1
nucleoprotein [Human metapneumovirus]	AAR17661.1
nucleoprotein [Human metapneumovirus]	AGU68352.1
nucleoprotein [Human metapneumovirus]	AGU68373.1
nucleoprotein [Human metapneumovirus]	AGU68376.1
nucleoprotein [Human metapneumovirus]	AGU68342.1
nucleoprotein [Human metapneumovirus]	AGU68365.1
nucleoprotein [Human metapneumovirus]	AGU68363.1
nucleoprotein [Human metapneumovirus]	AGU68398.1
nucleoprotein [Human metapneumovirus]	AGU68348.1
nucleoprotein [Human metapneumovirus]	AGU68354.1
nucleoprotein [Human metapneumovirus]	AGU68391.1
nucleoprotein [Human metapneumovirus]	AGU68389.1
nucleoprotein [Human metapneumovirus]	AGU68399.1
nucleoprotein [Human metapneumovirus]	AGU68337.1
nucleoprotein [Human metapneumovirus]	AAR17660.1
nucleoprotein [Human metapneumovirus]	AAR17667.1
nucleoprotein [Human metapneumovirus]	AGU68402.1
nucleoprotein [Avian metapneumovirus type C]	CDN30025.1
nucleoprotein [Avian metapneumovirus]	AGZ87947.1
Nucleoprotein [Avian metapneumovirus type C]	CAL25113.1
nucleocapsid protein [Avian metapneumovirus]	ABO42286.1
nucleocapsid protein [Avian metapneumovirus]	AAK38430.1
nucleocapsid protein [Avian metapneumovirus]	AAK54155.1
nucleocapsid protein [Avian metapneumovirus]	AAK38426.1
nucleocapsid protein [Avian metapneumovirus]	AAK38425.1
nucleocapsid protein [Avian metapneumovirus]	AAK38424.1
nucleocapsid protein [Avian metapneumovirus]	AAF05909.1
nucleocapsid protein [Avian metapneumovirus]	AAK38435.1
nucleocapsid protein [Avian metapneumovirus]	AAK38428.1
nucleoprotein [Human metapneumovirus]	AAR17669.1
nucleocapsid protein [Avian metapneumovirus]	AAK38429.1
nucleocapsid protein [Avian metapneumovirus]	AAK38427.1
nucleocapsid protein [Avian metapneumovirus]	AAK38423.1
nucleocapsid protein [Avian metapneumovirus]	AAK38434.1
nucleoprotein [Human metapneumovirus]	AGU68338.1
nucleoprotein [Avian metapneumovirus]	YP_443837.1
nucleoprotein [Human metapneumovirus]	AGU68384.1
nucleocapsid protein [Avian metapneumovirus]	AAK38431.1
nucleoprotein [Human metapneumovirus]	AGU68405.1
nucleoprotein [Human metapneumovirus]	AGU68382.1
nucleoprotein [Human metapneumovirus]	AGU68395.1
nucleocapsid [Human metapneumovirus]	AAL35389.3
nucleoprotein [Human metapneumovirus]	AEZ68064.1

TABLE 5

Description	Sequence	SEQ ID NO:
PIV3 Nucleic Acid Sequences		
>gb KJ672601.1 : 4990-6609 Human parainfluenza virus 3 strain HPIV3/Homo sapiens/PER/ FLA4815/2008 [fusion glycoprotein F0]	ATGCCAATTTCAATACTGTTAATTATTACAACCATGATC ATGGCATCACACTGCCAAATAGACATCACAAACTACA GCATGTAGGTGATTGGTCAACAGTCCCAAAGGGATGA AGATATCACAAACTTCGAAACAAGATATCTAATCCTGA GTCTCATACCAAAATAGAAGATTCTAACTCTTGTGGTG ACCAACAGATCAAGCAATACAAGAGGTTATTGGATAGA CTGATCATTCCTTTATATGATGGACTAAGATTACAGAAG GATGTGATAGTGACTAATCAAGAATCCAATGAAACAC TGATCCAGAACAGAACGATTCTTTGGAGGGGTAATTGG AACTATTGCTCTAGGAGTAGCAACCTCAGCACAAATTAC AGCAGCAGTTGCTCTGGTTGAAGCCAAGCAGGCAAGAT	9

TABLE 5-continued

Description	Sequence	SEQ ID NO:
	CAGACATTGAAAACTCAAGGAAGCAATCAGGGACACA AATAAAGCAGTGCAGTCAGTTCAGAGCTCTGTAGGAAA TTTGATAGTAGCAATTAATCAGTCCAGGATTATGTCAA CAAAGAAATCGTGCCATCGATTGCGAGACTAGGTTGTG AAGCAGCAGGACTTCAGTTAGGGATTGCATTAACACAG CATTACTCAGAATTAACAAATATATTTGGTGATAACATA GGATCGTTTACAAGAAAAAGGAATAAAATTACAAGGTAT AGCATCATTATACCGTACAAATATCACAGAAATATTCAC AACATCAACAGTTGACAAATATGATATTTATGATCTATT ATTTACAGAATCAATAAAGGTGAGAGTTATAGATGTTGA TTTGAATGATTACTCAATAACCTTCCAGTCAGACTCCC TTTATTGACCAGACTGCTGAACACTCAAATCTACAAAGT AGATTCCATATCATACAATATCCAAAATAGAGAATGGTA TATCCCTCTTCCAGCCATATCATGACGAAAGGGCATT TCTAGTGGAGCAGATGTCAAAGAATGCATAGAAGCAT TCAGCAGTTATATATGCCCCTTCTGATCCAGGATTTGTACT AAACCATGAAATGGAGAGCTGTCTATCAGGAACCATAT CCCAATGTCCAAGAACCACAGTCACATCAGACATAGTTC CTAGGTATGCATTTGTCAATGGAGGAGTGGTTGCGAATT GTATAACAACACTACATGTACATGCAATGGTATCGGTAATA GAATCAACCAACCCTGATCAAGGAGTCAAAATTATA ACACATAAAGAATGTAATACAATAGGTATCAACGGAAT GCTATTCAACACAACAAAGAAGGAACTCTTGCAATCTA CACACCAGACGACATAACATTAACAATTTCTGTTGCAC TGATCCGATTGACATATCAATCGAGCTCAACAAGGCCAA ATCAGATCTTGAGGAATCAAAAGATGGATAAGAAGGT CAAATCAAAAGCTAGATTCTATTGGAAGTTGGCATCAAT CTAGCACTACAATCATAGTTATTTTGATAATGATGATTA TATTGTTTATAATTAATAACAATAATTACAATTGCAA TTAAGTATTACAGAATTCAAAAGAGAAATCGAGTGGAT CAAAATGATAAGCCGTATGTATTACAACAACAG	
gi 612507167 gb AHX22430.1 hemagglutinin- neuraminidase [Human parainfluenza virus 3]	ATGGAATACTGGAAGCACACCAACCACGGAAGGATGC TGGTAATGAGCTGGAGACATCCACAGCCACTCATGGCA ACAAGCTCACCACCAAGATAACATATATATTGTGGACG ATAACCCTGGTGTTATTAATCAATAGTCTTCATCATAGTG CTAACCTAATTCCATCAAAAGTGAAAGGCCCGCGAATC ATTGCTACAAGACATAAATAATGAGTTTATGGAAGTTAC AGAAAAGATCCAAGTGGCATCGGATAATACTAATGATC TAATACAGTCAGGAGTGAATACAAGGCTTCTTACAATTC AGAGTCATGTCCGAAATTATATACCAATATCATTGACAC AACAAATATCGGATCTTAGGAAATTCATTAGTGAAATTA CAATTAGAAATGATAATCAAGAAGTGCCACCACAAAGA ATAACACATGATGTGGGTATAAACTTTAAATCCAGAT GATTTCTGGAGATGCACGCTCTGGTCTTCCATCTTTGATG AAAACTCCAAAATAAGATTAATGCCGGGACCAGGATT ATTAGCTATGCCAAGCACTGTTGATGGCTGTGTGAGAAC CCCGTCCTTAGTGATAAATGATCTGATTTATGCTTACAC CTCAAATCTAATTACTCGAGTTGCCAGGATATAGGGAA ATCATATCAAGTATTACAGATAGGGATAATAACTGTAAA CTCAGACTTGGTACCTGACTTAAATCCTAGGATCTCTCA TACCTTCAACATAAATGACAATAGAAAGTCATGTTCTCT AGCACTCCTAAATACAGATGTATATCAACTGTGTTCAAC CCCAAAAGTTGATGAAAGATCAGATTATGCATCATCAG GCATAGAAGATATTGTACTTGATATTGTCAATTATGATG GCTCAATCTCGACAACAAGATTTAAGAATAATAATATAA GTTTGTATCAACCATATGCGGCATTATACCCATCTGTTG GACCAGGGATATACTACAAGGCAAAATAATTTCTC GGGTATGGAGGTCTTGAACATCCAATAATGAGAATGC AATCTGCAACACAACCTGGGTGTCTGGGAAAAACACAGA GAGACTGTAATCAAGCATCTCATAGTCCATGGTTTTCAG ATAGAAGGATGGTCAACTCTATAATTGTTGTTGACAAGG GCTTGAACCTCAGTTCCAAATTTGAAGGTATGGACGATAT CTATGAGACAAAATTAATGGGGTCAAGGAAGATTA CTTCTACTAGGTAACAAGATCTACATATACACAAGATCT ACAAGTTGGCACAGCAAGTTACAATTAGGAATAATTGA CATTACTGACTACAGTGATATAAGGATAAAATGGACAT GGCATAATGTGCTATCAAGACCAGGAAACAATGAATGT CCATGGGGACATTATGTCCGGATGGATGTATAACGGG AGTATATACCGATGCATATCCACTCAATCCCACAGGAAG CATTGTATCATCTGTATATTGGACTCACAAAATCGAG AGTCAACCCAGTCATAACTTACTCAACAGCAACCGAAA GGGTAAACGAGCTGGCTATCCGAAACAAACACTCTCA GCTGGGTACACAACAACAAGCTGCATTACACACTATAA CAAAGGGTATTGTTTTCATATAGTAGAAATAAATCATAA AAGCTTAAACACATTTCAACCATGTTGTTCAAAACAGA GATTCCAAAAGCTGCAGT	10

TABLE 5-continued

Description	Sequence	SEQ ID NO:
HPiV3_HN_Codon Optimized	ATGGAATACTGGAAGCACACCAACCACGGCAAGGACGC CGGCAACGAGCTGGAACACAGCACAGCCACACACGGCA ACAAGCTGACCAACAAGATCACCTACATCCTGTGGACC ATCACCTGGTGTCTGTAGCATCGTGTTCATCATCGTG CTGACCAATAGCATCAAGAGCGAGAAGGCCAGAGAGAG CCTGCTGCAGGACATCAACAACGAGTTCATGGAAGTGA CCGAGAAGATCCAGGTGGCCAGCGACAACACCAACGAC CTGATCCAGAGCGGCGTGAACACCCGGCTGTGACCATC CAGAGCCACGTGCAGAACTACATCCCATCAGCCTGACC CAGCAGATCAGCGACCTGCGGAAGTTCATCAGCGAGAT CACCATCCGGAACGACAACAGGAAGTGCCCCCAGAGA GAATCACCCACGACGTGGGCATCAAGCCCTGAACCCC GACGATTTCTGGCGGTGTACAAGCGGCCTGCCACGCTG ATGAAGACCCCCAAGATCCGGCTGATGCTTGGCCCTGG ACTGCTGGCCATGCTTACCACAGTGGATGGCTGTGTGCG GACCCCAAGCCTCGTGATCAACGATCTGATCTACGCCTA CACCAGCAACCTGATCACCCGGGGCTGCCAGGATATCG GCAAGAGCTACCAGGTGCTGCAGATCGGCATCATCACC GTGAATCCGACCTGGTGCCCGACCTGAACCTCGGATC AGCCACACCTTCAACATCAACGACAACAGAAAGAGCTG CAGCCTGGCTCTGCTGAACACCGACGTGTACCAGCTGTG CAGCACCCCAAGGTGGACGAGAGAAGCGACTACGCCA GCAGCGGCATCGAGGATATCGTGCTGGACATCGTGAAC TACGACGGCAGCATCAGCACCCCGGTTCAAGAACAA CAACATCAGCTTCGACCGACCTACGCCGCCCTGTACCC TTCTGTGGGCCCTGGCATCTACTACAAGGGCAAGATCAT CTTCTGGGCTACGGCGGCCTGGAACACCCCATCAACGA GAACGCCATCTGCAACACACCGGCTGCCCTGGCAAGA CCCAGAGAGACTGCAATCAGGCCAGCCACAGCCCTGG TTCAGCGACCGCAGAATGGTCAACTCTATCATCGTGGTG GACAAGGGCTGAACAGCGTGCCCAAGCTGAAAGTGTG GACAATCAGCATGCGCCAGAAGTACTGGGGCAGCGAGG GCAGACTTCTGTGCTGGGAAACAAGATCTACATCTACA CCGGTTCCACCGCTGGCACAGCAAACTGCAGCTGGGA ATCATCGACATCACCGACTACAGCGACATCCGGATCAA GTGGACCTGGCACAAACGTGCTGAGCAGACCCGGCAACA ATGAGTGCCCTTGGGGCCACAGCTGCCCGATGGATGTA TCACCGCGGTGTACACCGACGCCCTACCCCTGAATCCTA CCGGCTCCATCGTGTCCAGCGTGATCCTGGACAGCCAGA AAAGCAGAGTGAACCCCGTGATCACATACAGCACCGCC ACCGAGAGAGTGAACGAATGGCCATCAGAAACAAGAC CCTGAGCGCGGCTACACCACCAAGCTGCATCACAC ACTACAACAAGGGCTACTGCTTCCACATCGTGGAAATCA ACCACAAGTCCCTGAACACCTTCCAGCCCATGCTGTTCA AGACCGAGATCCCCAAGAGCTGCTCC	11
HPiV3_F_Codon Optimized	ATGCCCATCAGCATCCTGTGATCATCACCAATGATC ATGGCCAGCCACTGCCAGATCGACATCACAAGCTGCA GCACGTGGGCGTGCTCGTGAACAGCCCCAAGGGCATGA AGATCAGCCAGAACTTCGAGACACGCTACCTGATCCTGA GCCTGATCCCCAAGATCGAGGACAGCAACAGCTGCGGC GACCAGCAGATCAAGCAGTACAAGCGGCTGTGGACAG ACTGATCATCCCCCTGTACGACGGCCTGCGGCTGCAGAA AGACGTGATCGTGACCAACAGGAAGCAACGAGAACA CCGACCCCGGACCGAGAGATTCTTCGGCGGCGTGATCG GCACAATCGCCCTGGGAGTGGCCACAAGCGCCAGATT ACAGCCGCTGTGGCCCTGGTGAAGCCAGCAGGCCAG AAGCGACATCGAGAAGCTGAAAGAGGCCATCCGGGACA CCAACAAGGCCGTGCAGAGCTGCAGTCCAGCGTGGGC AATCTGATCGTGGCCATCAAGTCCGTGCAAGACTACGTG AACAAAGAAATCGTGCCCTCTATCGCCCGGCTGGGCTGT GAAGCTGCCGGACTGCAGCTGGGCATTGCCCTGACACA GCACTACAGCGAGCTGACCAACATCTTCGGCGACAACA TCGGCAGCCTGCAGGAAAAGGGCATTAGCTGCAGGGA ATCGCCAGCCTGTACCGCACCAACATCACCGAGATCTTC ACCACGACACCGTGGATAAGTACGACATCTACGACCT GCTGTTACACGAGCATCAAAGTGCAGTGTGACGCT GGACCTGAACGACTACAGCATCACCTGCAAGTGCAGC TGCCCTGTGACCGAGCTGCTGAACACCCAGATCTACA AGGTGGACAGCATCTCTACAACATCCAGAACCGCGAG TGGTACATCCCTCTGCCAGCCACATTATGACCAAGGGC GCCTTTCGGGCGGAGCCGACGTGAAAGAGTGCATCGA GGCTTTCAGCAGCTACATCTGCCCCAGCGACCTGGCTT CGTGTGAACCAAGAGATGGAAGCTGCCCTGAGCGGCA ACATCAGCCAGTGCCCGAGAACCACCGTGACCTCCGAC ATCGTGCCAGATACGCCCTTCTGTAATGGCGCGTGGTG	12

TABLE 5-continued

Description	Sequence	SEQ ID NO:
	GCCAACTGCATCACCACCCTGTACCTGCAACGGCATC GGCAACCGGATCAACCAGCCTCCCGATCAGGGCGTGAA GATTATCACCACAAAGAGTGTAAACCATTCGGCATCA ACGGCATGCTGTTCAATACCAACAAGAGGGCACCCTG GCCTTCTACACCCCGACGATATCACCTGAACAACTCC GTGGCTCTGGACCCCATCGACATCTCCATCGAGCTGAAC AAGGCCAAGAGCGACCTGGAAGAGTCCAAGAGTGGAT CCGGCCGAGCAACCAGAGCTGGACTCTATCGGCAGCT GGCACCAGAGCAGCACCATCATCTGATCCTGATTA TGATGATTATCCTGTTTCATCATCAACATTACCATCATCAC TATCGCCATTAAGTACTACCGGATCCAGAAACGGAACC GGGTGGACCAGAATGACAAGCCCTACGTGCTGACAAAC AAG	
PIV3 mRNA Sequences		
>gb KJ672601.1 : 4990-6609 Human parainfluenza virus 3 strain HPIV3/ <i>Homo sapiens</i> /PER/FLA4 815/2008 [fusion glycoprotein F0]	AUGCCAAUUUCAUACUGUUAUUUAUACAACCAUGA UCAUGGCAUCACACUGCCAAUAGACAUCACAAAACU ACAGCAUGUAGGUGUAUUGGUCAACAGUCCAAAGGG AUGAAGAUUACACAAACUUCGAAACAGAUUUCUAA UCCUGAGUCUCAUACCAAAUAGAAGAUUCUAACUC UUGUGGUGACCAACAGAUCAAGCAUACAAGAGGUUA UUGGAUAGACUGAUCAUUCUUUAUUGAUGGACUAA GAUUAACAGAGGAGUGAUGAGUCUAAUACAAGAAUC CAAUGAAAACACUGAUCCAGAACAGAACGAUUCUUU GGAGGGUAAUUGGAACUAUUGCUCUAGGAGUAGCAA CCUCAGCACAAAUACAGCAGCAGUUGCUCUGGUUGA AGCCAGCAGGCAAGAUACAGAUUGAAAAACUCAAG GAAGCAAUCAGGGACACAAUAAAGCAGUGCAGUCAG UUCAGAGCUCUGUAGGAAUUGAUUAGUAGCAAUUA AUCAGUCCAGGAUUAUGUCAACAAAGAAUUCUGGCCA UCGAUUGCGAGACUAGGUUGUGAAGCAGCAGGACUUC AGUUAAGGGAUUGCAUUAACACAGCAUUAUCAGAAU AACAAUUAUUAUUGGUGAUAACAUAGGAUCGUUACAA GAAAAAGGAUAAAUUAACAAGGUUAGCAUCAUUAU ACCGUACAAAUUACAGAAAUUAUACACACUACAAC AGUUGACAAAUUAGAUUAUUAUGAUCAUUAUUUACA GAAUCAUAAAGGUGAGAGUUAUAGAUUGUUAUUGA AUGAUUACUCAAAUACCCUCCAGUCAGACUCCUUU AUUGACCAGACUGCUGAACACUCAAAUUCUACAAGUA GAUUCCAUUAUCAUAAUUAUCCAAAUAGAGAAUGGU AUAUCCCUUCCAGCCAUUAUCAAGCAGAAAGGGGC AUUUCUAGGUGGAGCAGUUGCAAGAAUUGCAUAGAA GCAUUCAGCAGUUAUAUUGCCUUCUGAUCCAGGAU UUGUACUAAACCAUGAAUUGGAGAGCUGUUAUACAGG AAACAUUAUCCCAAUGUCCAGAACACAGUCACAUCA GACAUAGUCCUAGGUUAGCAUUAUGCAUUGGAGGAG UGGUUGCGAAUUGUAUAAACUACAUGUACAUUGCAA UGGUAUCGGUAAUAGAAUCAAACCAACCCUGAUCAA GGAGUCAAAAUUAUAAACAUAAAGAAUGUAUACAA UAGGUUAUACCGAAUUGCUAUUACAACACAAACAAAGA AGGAACUUCUGCAUUCUACACCCAGACGACAUACA UUAACAUAUUCUGUUGCAUCUUGAUCCGAUUGACAUU CAAUCGAGCUCAAAGGCCAAUUCAGAUUCUGAGGA AUCAAAAGAAUGGAUAAGAGGUCAAAUCAAAAGCUA GAUUCUAUUGGAAGUUGGCAUCAUUCAGCACUACAA UCAUAGUUAUUUGAUAAUGAUUAUUAUUGUUUAU AAUUAUUAUAAUAAUUAUUAUUAUUGCAUUAAGUAU UACAGAAUUCAAAAGAGAAUUCGAGUGGAUCAAUUG AUAAGCCGUAGUUAUUAACAACAAG	61
gi 612507167 gb AHX22430.1 hemagglutinin- neuraminidase [Human parainfluenza virus 3]	AUGGAUUAUCUGGAAGCACACCAACCACGAAAGGAUG CUGGUAAUAGAGCUGGAGACAUCCACAGCCACUCAUGG CAACAGCUCACCAACAAGAUAAUUAUUAUUGGG ACGAUAAACCCUGGUGUUAUUAUCAUAGUCUUCAUCA UAGUGCUAACUAAUCCAUCAAAAGUGAAAAGGCCCG CGAAUCAUUGCUACAAGACAUAAUUAUAGAUUUUUG GAAGUUAACAGAAAAGAUCCAAGUGGCAUCGGAAUUA CUAUAGAUCAUUAACAGUCAGGAGUGAAUACAAAGCU UCUUACAUAUUCAGAGUCAUGUCCAGAAUUAUUAACCA AUAUCAUUGACACAACAAUUAUCGGAUCUUAAGGAAU UCAUUAUGUAAAUACAUAUAGAAUAGAUUAUUAAGAA AGUGCCACCACAAAGAAUAAACAUGAUUGGGUUAUA AAACCUUUAUUAUCCAGAUUAUUCUGGAGUAGCACGU CUGGUUCUCCAUUCUUGAUGAAACUCCAAAAUAAG AUUAUUGCCGGACAGGAUUAUUAUGCUAUGCCAACG ACUGUUGAUGGCUGUGUCAGAACCCCGUCCUUAUGUA	62

TABLE 5-continued

Description	Sequence	SEQ ID NO:
	UAAAUGAUCUGAUUUUAUGCUUACACCUCAAAUCUAAU UACUCGAGGUUGCCAGGAUAUAGGGAUAUACUAAUCAA GUUUACAGAUAGGGAUAUAACUGUAAAACUAGACU UGGUACCUGACUUAAAUCUAGGAUCUCUUAUACCUU CAACAUAUAGACAAUAGAAUGUUAUUCUUAUAGCA CUCCUAAAUAACAGAUUAUAUACACUGUUAUACCCC CAAAAGUUGAUGAAAGAUCAUAUAGCAUACUAGG CAUAGAAAGAUUUGUAUUAUAGAUUUGCAUUAUAGAU GGCUCAAUCUGACAAACAAGAUUUAAGAAUAUAUA UAAUUUUUGAUCACCAUAUAGCGGCAUUAUACCAUC UGUUGGACAGGGAUAUAUACAAAGGCAAAUAUAUA UUUUCUGGUAUGGAGGUUUUAACAUCUCAAUAAUAG AGAAUGCAAUCUGCAACAACAUCUGGUGUCCUGGGAA AACACAGAGAGACUGUAUUAAGCAUCUUAUAGUCCA UGGUUUUCAGAUAGAAAGGUAUACUUAUAAUUG UUGUUGACAAAGGCUUUAACUAGUCCAAAUAUAGAA GGUUUGGACGAUAUUAUAGAGCAAAUUAUACUGGGG UCAGAAAGGAAGAUUACUUAUACUAGGUAAACAAGAU ACAUAUACACAAGAUUUAUAGUUGGCAAGCAAGUU ACAAUUAGGAUAUAUUAUAGAUUACUAGUAGAU AUAAGGAUAAAUGGACAUUGGCAUAAUGUGCUAUA GACCAGGAACAAUAGAUUCCAUUGGGCAUUAUAG UCCGGAUGGAUGUAUAACGGGUAUAUACCGAUGCA UAUCACUCUAAUCCACAGGAAGCAUUAUACUAGU UCAUAUUGGACUCAAUAAUAGAGUCAAUACCGAGU CAUAACUUAUCUAAACAGCAACCGAAGGUAACCGAG CUGGCUAUCCGAAACAAACACUUCAGCUGGGUACA CAACAACAAGCUGCAUUAACACUAUAACAAGGGUA UUGUUUUCAUAUAGUAGAAUAUAUCAAUAAAGCUUA AACACAUUUCAAACCAUGUUGUCAAACAGAGAUUUC CAAAAGCUGCAGU	
HPIV3_HN_Codon Optimized	AUGGAUUAUCUGGAAGCACACCAACCACGGCAAGGACG CCGGCAACGAGCUGGAAACACAGCAAGCCACACACGGC AACACAGCUGACCAACAAGAUCAUCCUAGUUGGA CCAUACCCUGGUGCUGCUGAGCAUCUGUUAUUAUAC GUGCUGACCAUAGCAUUAAGAGCGAGAAGGCCAGAG AGAGCCUGCUGCAGGCAUUAACAACGAGUUAUAGGA AGUGACCGAGAAGAUCCAGGUGGCGAGCAACACCC AACGACCUGAUCCAGAGCGCGUGAACACCCGGCUGCU GACCAUCCAGAGCCACGUGCAGAACUUAUCCCAUCA GCCUGACCCAGCAGAUCAAGCAGCUGCGGAAGUUAU AGCGAGAUCAACCAUCCGGAACGACAACAGGAAGUGC CCCCCAGAGAUAUACCCACGACGUGGGCAUUAAGCCC CUGAACCCCGCAGAUUUCUGGCGGUGAACAGCGGCC UGCCCAGCCUGAUGAAGACCCCAAGAUCCGGCUGAUG CCUGGCCUGGACUGCUGGCCAUGCCUACCAAGUGGA UGGCUGUGUGCGGACCCCAAGCUCUGUAUACCAUAC UGAUUAUAGCCUACACAGCAACCUGAUACCCGGGGC UGCCAGGAUAUCCGCAAGAGCUACCAAGGUGCUGAGA UCGGCAUUAUACCCUGAACUCCGACCUUGGUGCCCGAC CUGAACCCUGGUAUAGCACAACCUUAACAUAACGA CAACAGAAAGAGCUGCAGCCUGGCUUGCUGAACACC GACGUGUACCAAGCUGGACGACCCCAAGGUGGACG AGAGAAGCGAUACGCAGCAGCGCAUCCAGGAUAU CGUGCUGGACAUUAGAAUACGACGGCAGCAUACGC ACCACCCGGUUAAGAACAACAUAUAGCUUCGACCA GCCCUACGCCGCCUGUACCUUCUGUGGGCCUGGCA UCUACUACAAGGGCAAGAUUAUUCUUGGGCUACGG CGGCCUGGAACACCCCAUACAGGAACGCCAUUCGCA ACACCAACCGGCGGCCUGGCAAGACCCAGAGAGACUGC AAUCAGGCCAGCCACAGCCUUGGUUACGCGACCGCAG AAUGGUUAACUUAUUAUAGUGGUGGACAAAGGCCUG AACAGCGUGCCAAAGCUGAAAGUGUGGACAAUAGCA UGCGCCAGAACUACUGGGGACGAGGGCAGACUUCU GCUGCUGGGAACAAGAUUAUUAUACACCCGGUCC ACCAGCUGGCAAGCAAAUUGAGCUGGGAAUUAUUG ACAUACCCGACUACAGCGAUCCGGAUUAAGUGGACC UGGCAACAAGCUGGAGCAGACCCGGCAACAUAAGAU GCCUUGGGGCCACAGCUGCCCCGAUGGAUUAUACCC GGCGUGUACACCGACGCCUACCCCUGAUUAUACCCG CUCCAUCUGUCCAGCGUAUCCUGGACAGCCAGAAA AGCAGAGUGAACCCCGUAUUAUUAUAGCACCAGCCAC CGAGAGAGUAGAACGAAUUGGCAUAGAAACAGACC CUGAGCGCCGGUACACCAACCAAGCUGCAUACACA	63

TABLE 5-continued

Description	Sequence	SEQ ID NO:
HPV3_F_Codon Optimized mRNA sequence	CUACAACAAGGCUACUGCUUCCACAUCGUGGAAAUC AACCACAAGUCCUGAACACCUCCAGCCCAUGCUGUU CAAGACCGAGAUCCTCAAGAGCUGCUCC	64
	AUGCCCAUCAGCAUCUGCUGAUCAUACCAACAAUGAU CAUGGCCAGCCACUGCCAGAUCAUACCAAGCUGC AGCACGUGGCGUGCUCGUGAACAGCCCCAAGGGAU GAAGAUCAGCCAGAACUUCGAGACACGCUACCGAU CUGAGCCUGAUCCCCAAGAUCCGAGACAGCAACAGCU GCGGCCAGCAGCAGAUCAAGCAGUACAAGCGGCGUCU GGACAGACUGAUCAUCCCCUGUACGACGCGCUGCGGC UGCAGAAAGACGUGAUCCGAGCAACAGGAAAGCAA CGAGAACACCGACCCCGGACCGAGAGAUUCUUCGCGC GCGUGAUCCGACCAUCCGCGGAGUGGCCACAAAG CGCCAGAUUACAGCCGUGUGGCCUGUGGAGAGCCA AGCAGGCAGAGCGACAUCGAGAGCUGAAAGAGGC CAUCCGGGACACCAACAGGCGUGCAGAGCUGCAG UCCAGCGUGGCAUUCUGAUCCGUGCCAUCAAGUCCG UGCAGGACUACGUGAACAAAGAAUCCGUGCCUCU CGCCCGGCGUGGCGUGAGCUGCCGACUGCAGCUG GGCAUUGCCUGACACAGCACUACAGCGAGCUGACCAA CAUCUUCGGCGACAACAUCCGAGCUCGAGGAAAG GGCAUUAAGCUGCAGGGAUCCGCGCUGUACCGCA CCAACAUCACCGAGAUUCUACCCAGCAGCCGUGGAU AAGUACGACAUCAACGACUGCUGUACCGAGAGCA UCAAGUGCGCGUGAUCCGAGGACCUGAACGACUA CAGCAUACCCUGCAAGUGCGGCGUCCUGCUGACCA GACUGCUGAACCCAGAUCAACAGGUGGACAGCAU CUCCUACAACAUCAGAACCGGAGUGGUACAUCCUC UGCCAGCCACAUUAGCAAGGCGCCUUCUGGGC GGAGCCGACGUGAAAGAGUGCAUCGAGGCUUCAGCA GCUACAUCUGCCCGAGCGACCCUGGCUUCUGCUGAAC CACGAGAUUGAAAGCUGCUGAGCGGCAACUACGCC AGUGCCCGAGAACCCGUGACCUCCGACAUCCUGCCC AGAUACGCCUUCGUGAAUGGCGGCGUGGUGCCAAU GCAUACCCAGCCUGUACCGCAACCGCAUCGCGCAAC CGGAUCAACAGCCUCCGAGCAGGCGUGAAGAUUA UACCCACAAGAGUGUAACACCAUCGCGCAUACAGGC AUGCUGUUAUUAACCAACAAAGAGGCGACCCUGGCCU UCUACACCCCGAGCAUACCCUGAACCAUCCGUG GCUUGGACCCCAUCGACAUCCAUCCGAGCUGAACAA GGCCAGAGCGACCGUGAAGAGUCCAAAGAGUGGAUC CGGCGAGCAACAGAGCUGGACUUAUCGCGAGCU GGCACCAGAGCAGCACCAUCAUCGUGAUCCUGAUU AUGAUGAUUAUCCUGUUAUCAUACAACAUUACCAUCA UCACUAUCGCCAUUAAGUACUACCGGAUCCAGAAACG GAACCGGUGGACAGAAUGACAAGCCUACGUGCUG ACAAACAAG	

45

TABLE 6

PIV3 Amino Acid Sequences		SEQ ID NO:
Description	Sequence	
>gi 612507166 gb AHX22429.1 fusion glycoprotein F0 [Human parainfluenza virus 3]	MPISILLIITTMIMASHCQIDIT KLQHVGLVNSPKGMKISQNFET RYLILSLIPKIEDSNSCGDQQIK QYKRLDLRLIIPLYDGLRLQKDV IVTNQESNENTDPRTERFPGGVI GTIALGVATSAQITAVALVEAK QARSDIEKLKEAIRDTNKAVQSV QSSVGNLIVAIKSVQDYVNKEIV PSIARLGCEAAGLQGLIALTQHY SELTNIFGDNIGSLQEKIKLQG IASLYRTNITEIFTTSTVDKYDI YDLLFTESI KVRVIDVDLNDYSI TLQVRLPLLTRLNLTQIYKVDSI SYNIQNRWYIPLPSHIMTKGAF LGGADVKECIEAFSSYICPSDPG FVLNHEMESCLSGNISQCPRTTV TSDIVPRYAFVNGGVVANCITTT	13 50 55 60 65

TABLE 6-continued

PIV3 Amino Acid Sequences		SEQ ID NO:
Description	Sequence	
gi 612507167 gb AHX22430.1 hemagglutinin-neuraminidase [Human parainfluenza virus 3]	CTCNGIGNRINQPPDQGVKIITH KECNTIGINGMLFNTNKEGTLAF YTPDDITLNNVALDPIDISIEL NKAKSDLEESKEWIRRSNQKLDS IGSWHQSSTTIIIVILIMMIILFI INITIITIAIKYRIQKRNRVDQ NDKPYVLNTNK MEYWKHTNHGKDAGNELETSTAT HGNKLTNKITYILWTITLVLLSI VFIIVLTNSIKSEKARESLLQDI NNEFMVETKIQVASDNTNDLIQ SGVNTRLTLTIQSHVQNYIPISLT QQISDLRKFISEITIRNDNQEV PQRITHDVGIKPLNPDDFWRCTS GLPSLMKTPKIRLMPGGLLAMP TTVDGCVRTPSLVINDLIYATS	14

247

TABLE 6-continued

PIV3 Amino Acid Sequences		SEQ ID NO:
Description	Sequence	
	NLITRGCDIGKSYQVLQIGIIT	5
	VNSDLVPDLNPRISHTFNINDNR	
	KSCSLALLNTDQVQLCSTPKVDE	
	RSDYASSGIEDIVLDIVNYDGS I	
	STTRFKNNNISFDQPYAALYPSV	
	GPGIYYKGKIIIFLGYGGLLEHPIN	10
	ENAIcntTGCpGKTQRDCNQASH	
	SPWFSDRRMVNSIIVVDKGLNSV	
	PKLKVWTISMQRQNYWGSEGRLLL	

248

TABLE 6-continued

PIV3 Amino Acid Sequences		SEQ ID NO:
Description	Sequence	
	LGNKIYIYTRSTSWHSLQLGLII	
	DITDYSDIRIKWTWHNVLSRPGN	
	NECPWGHSCPDGCITGVYTDAYP	
	LNPTGSI VSSVILDSQKSRVNPV	
	ITYSTATERNVELAIRNKTLSAG	
	YTTTSCI THYNKGYCFHIVEINH	
	KSLNTFQPMFLKTEIPKSCS	

TABLE 7

PIV3 NCBI Accession Numbers (Nucleic Acid and Amino Acid Sequences)	
Description	GenBank Accession
Fusion glycoprotein F0 [Human parainfluenza virus 3] HPIV3/ <i>Homo sapiens</i> /PER/FLA4815/2008	KJ672601.1[:4990-6609 AHX22429 (Fusion protein)
hemagglutinin-neuraminidase [Human parainfluenza virus 3] HPIV3/ <i>Homo sapiens</i> /PER/FLA4815/2008	KJ672601.1[:6724-8442 AHX22430 (HN protein)
Recombinant PIV3/PIV1 virus fusion glycoprotein (F) and hemagglutinin (HN) genes, complete cds; and RNA dependent RNA polymerase (L) gene, partial cds.	AF016281 AAC23947 (hemagglutinin)
Recombinant PIV3/PIV1 virus fusion glycoprotein (F) and hemagglutinin (HN) genes, complete cds; and RNA dependent RNA polymerase (L) gene, partial cds.	AF016281 AAC23947 (fusion protein)
hemagglutinin-neuraminidase [Human parainfluenza virus 3]	BAO32044.1
hemagglutinin-neuraminidase [Human parainfluenza virus 3]	BAO32051.1
C protein [Human parainfluenza virus 3]	NP_599251.1
C protein [Human parainfluenza virus 3]	ABZ85670.1
C protein [Human parainfluenza virus 3]	AGT75164.1
C protein [Human parainfluenza virus 3]	AAB48686.1
C protein [Human parainfluenza virus 3]	AHX22115.1
C protein [Human parainfluenza virus 3]	AGW51066.1
C protein [Human parainfluenza virus 3]	AGW51162.1
C protein [Human parainfluenza virus 3]	AGT75252.1
C protein [Human parainfluenza virus 3]	AGT75188.1
C protein [Human parainfluenza virus 3]	AGW51218.1
C protein [Human parainfluenza virus 3]	AGW51074.1
C protein [Human parainfluenza virus 3]	AGT75323.1
C protein [Human parainfluenza virus 3]	AGT75307.1
C protein [Human parainfluenza virus 3]	AHX22131.1
C protein [Human parainfluenza virus 3]	AGW51243.1
C protein [Human parainfluenza virus 3]	AGT75180.1
C protein [Human parainfluenza virus 3]	AGT75212.1
C protein [Human parainfluenza virus 3]	AGW51186.1
C protein [Human parainfluenza virus 3]	AHX22075.1
C protein [Human parainfluenza virus 3]	AHX22163.1
C protein [Human parainfluenza virus 3]	AGT75196.1
C protein [Human parainfluenza virus 3]	AHX22491.1
C protein [Human parainfluenza virus 3]	AHX22139.1
C protein [Human parainfluenza virus 3]	AGW51138.1
C protein [Human parainfluenza virus 3]	AGW51114.1
C protein [Human parainfluenza virus 3]	AGT75220.1
C protein [Human parainfluenza virus 3]	AHX22251.1
RecName: Full = Protein C; AltName: Full = VP18 protein	P06165.1
C protein [Human parainfluenza virus 3]	AHX22187.1
C protein [Human parainfluenza virus 3]	AGT75228.1
C protein [Human parainfluenza virus 3]	AHX22179.1
C protein [Human parainfluenza virus 3]	AHX22427.1
C protein [Human parainfluenza virus 3]	AGW51210.1
nonstructural protein C [Human parainfluenza virus 3]	BAA00922.1
C protein [Human parainfluenza virus 3]	AHX22315.1
C protein [Human parainfluenza virus 3]	AGW51259.1
C protein [Human parainfluenza virus 3]	AHX22435.1
C protein [Human parainfluenza virus 3]	AHX22123.1

TABLE 7-continued

PIV3 NCBI Accession Numbers (Nucleic Acid and Amino Acid Sequences)	
Description	GenBank Accession
C protein [Human parainfluenza virus 3]	AHX22299.1
C protein [Human parainfluenza virus 3]	AGW51267.1
unnamed protein product [Human parainfluenza virus 3]	CAA28430.1
C protein [Human parainfluenza virus 3]	AGW51178.1
C protein [Human parainfluenza virus 3]	AHX22411.1
RecName: Full = Protein C	P06164.1
phosphoprotein [Human parainfluenza virus 3]	NP_067149.1
phosphoprotein [Human parainfluenza virus 3]	AAB48685.1
phosphoprotein [Human parainfluenza virus 3]	AHX22498.1
phosphoprotein [Human parainfluenza virus 3]	AHX22490.1
phosphoprotein [Human parainfluenza virus 3]	AGT75259.1
phosphoprotein [Human parainfluenza virus 3]	AGW51137.1
phosphoprotein [Human parainfluenza virus 3]	AGW51145.1
phosphoprotein [Human parainfluenza virus 3]	AGT75298.1
phosphoprotein [Human parainfluenza virus 3]	AGW51113.1
phosphoprotein [Human parainfluenza virus 3]	AGT75203.1
phosphoprotein [Human parainfluenza virus 3]	AGT75163.1
phosphoprotein [Human parainfluenza virus 3]	AHX22506.1
phosphoprotein [Human parainfluenza virus 3]	AGW51129.1
phosphoprotein [Human parainfluenza virus 3]	AHX22194.1
phosphoprotein [Human parainfluenza virus 3]	AGT75211.1
phosphoprotein [Human parainfluenza virus 3]	AHX22258.1
phosphoprotein [Human parainfluenza virus 3]	AGW51121.1
phosphoprotein [Human parainfluenza virus 3]	AGT75282.1
phosphoprotein [Human parainfluenza virus 3]	AHX22146.1
phosphoprotein [Human parainfluenza virus 3]	AHX22138.1
phosphoprotein [Human parainfluenza virus 3]	AHX22322.1
phosphoprotein [Human parainfluenza virus 3]	AHX22370.1
phosphoprotein [Human parainfluenza virus 3]	AHX22098.1
phosphoprotein [Human parainfluenza virus 3]	AHX22130.1
phosphoprotein [Human parainfluenza virus 3]	AHX22418.1
phosphoprotein [Human parainfluenza virus 3]	AHX22114.1
phosphoprotein [Human parainfluenza virus 3]	AHX22410.1
phosphoprotein [Human parainfluenza virus 3]	AGT75306.1
phosphoprotein [Human parainfluenza virus 3]	AHX22170.1
phosphoprotein [Human parainfluenza virus 3]	AHX22266.1
phosphoprotein [Human parainfluenza virus 3]	AHX22090.1
phosphoprotein [Human parainfluenza virus 3]	AGT75195.1
phosphoprotein [Human parainfluenza virus 3]	AHX22226.1
phosphoprotein [Human parainfluenza virus 3]	AHX22178.1
phosphoprotein [Human parainfluenza virus 3]	AHX22122.1
phosphoprotein [Human parainfluenza virus 3]	AHX22186.1
phosphoprotein [Human parainfluenza virus 3]	AHX22066.1
phosphoprotein [Human parainfluenza virus 3]	AHX22522.1
phosphoprotein [Human parainfluenza virus 3]	AGW51225.1
phosphoprotein [Human parainfluenza virus 3]	BAN29032.1
phosphoprotein [Human parainfluenza virus 3]	ABZ85669.1
phosphoprotein [Human parainfluenza virus 3]	AHX22426.1
phosphoprotein [Human parainfluenza virus 3]	AHX22058.1
phosphoprotein [Simian Agent 10]	ADR00400.1
phosphoprotein [Human parainfluenza virus 3]	AHX22250.1
phosphoprotein [Human parainfluenza virus 3]	AHX22434.1
phosphoprotein [Human parainfluenza virus 3]	AHX22298.1
phosphoprotein [Human parainfluenza virus 3]	AHX22442.1
phosphoprotein [Human parainfluenza virus 3]	AHX22074.1
phosphoprotein [Human parainfluenza virus 3]	AGW51153.1
phosphoprotein [Human parainfluenza virus 3]	AGW51241.1
phosphoprotein [Human parainfluenza virus 3]	AHX22210.1
phosphoprotein [Human parainfluenza virus 3]	AGW51105.1
phosphoprotein [Human parainfluenza virus 3]	AGT75251.1
phosphoprotein [Human parainfluenza virus 3]	AHX22362.1
phosphoprotein [Human parainfluenza virus 3]	AHX22474.1
phosphoprotein [Human parainfluenza virus 3]	AGW51217.1
phosphoprotein [Human parainfluenza virus 3]	ALG60038.1
phosphoprotein [Human parainfluenza virus 3]	AHX22378.1
phosphoprotein [Human parainfluenza virus 3]	AGW51057.1
phosphoprotein [Human parainfluenza virus 3]	AGT75187.1
phosphoprotein [Human parainfluenza virus 3]	AGW51233.1
phosphoprotein [Human parainfluenza virus 3]	AHX22482.1
phosphoprotein [Human parainfluenza virus 3]	AGW51161.1
phosphoprotein [Human parainfluenza virus 3]	AHX22306.1
phosphoprotein [Human parainfluenza virus 3]	AHX22162.1
phosphoprotein [Human parainfluenza virus 3]	ACJ70087.1
phosphoprotein [Human parainfluenza virus 3]	AHX22466.1

TABLE 7-continued

PIV3 NCBI Accession Numbers (Nucleic Acid and Amino Acid Sequences)	
Description	GenBank Accession
phosphoprotein [Human parainfluenza virus 3]	AHX22346.1
phosphoprotein [Human parainfluenza virus 3]	AGW51089.1
phosphoprotein [Human parainfluenza virus 3]	AGW51073.1
phosphoprotein [Human parainfluenza virus 3]	AGW51185.1
phosphoprotein [Human parainfluenza virus 3]	AGW51065.1
phosphoprotein [Human parainfluenza virus 3]	ABY47603.1
phosphoprotein [Human parainfluenza virus 3]	AGW51049.1
phosphoprotein [Human parainfluenza virus 3]	AHX22330.1
phosphoprotein [Human parainfluenza virus 3]	AGW51250.1
phosphoprotein [Human parainfluenza virus 3]	AGT75227.1
phosphoprotein [Human parainfluenza virus 3]	AGW51282.1
phosphoprotein [Human parainfluenza virus 3]	AGW51209.1
phosphoprotein [Human parainfluenza virus 3]	AGW51193.1
phosphoprotein [Human parainfluenza virus 3]	AGT75322.1
phosphoprotein [Human parainfluenza virus 3]	AGT75219.1
phosphoprotein [Human parainfluenza virus 3]	AGW51258.1
phosphoprotein [Human parainfluenza virus 3]	AGW51041.1
phosphoprotein [Human parainfluenza virus 3]	ACD99698.1
phosphoprotein [Human parainfluenza virus 3]	AGW51266.1
phosphoprotein [Human parainfluenza virus 3]	AGT75179.1
phosphoprotein [Human parainfluenza virus 3]	AHX22282.1
phosphoprotein [Human parainfluenza virus 3]	AGW51169.1
phosphoprotein [Human parainfluenza virus 3]	AGW51274.1
phosphoprotein [Human parainfluenza virus 3]	AGW51201.1
phosphoprotein [Human parainfluenza virus 3]	AGW51177.1
RecName: Full = Phosphoprotein; Short = Protein P	P06162.1
P protein [Human parainfluenza virus 3]	AAA66818.1
phosphoprotein [Human parainfluenza virus 3]	AAA46866.1
phosphoprotein [Human parainfluenza virus 3]	BAA00031.1
polymerase-associated nucleocapsid phosphoprotein (version 2) - parainfluenza virus type 3 [Human parainfluenza virus 3]	RRNZP5
phosphoprotein [Human parainfluenza virus 3]	AGT75171.1
phosphoprotein [Human parainfluenza virus 3]	BAA00921.1
D protein [Human parainfluenza virus 3]	NP_599250.1
D protein [Human parainfluenza virus 3]	AHX22377.1
D protein [Human parainfluenza virus 3]	AHX22121.1
D protein [Human parainfluenza virus 3]	AGT75297.1
D protein [Human parainfluenza virus 3]	AGW51136.1
D protein [Human parainfluenza virus 3]	AGW51242.1
D protein [Human parainfluenza virus 3]	AGW51112.1
D protein [Human parainfluenza virus 3]	AHX22497.1
D protein [Human parainfluenza virus 3]	AHX22145.1
D protein [Human parainfluenza virus 3]	AGT75202.1
D protein [Human parainfluenza virus 3]	AHX22385.1
D protein [Human parainfluenza virus 3]	AGW51216.1
D protein [Human parainfluenza virus 3]	AGT75281.1
D protein [Human parainfluenza virus 3]	AGT75194.1
D protein [Human parainfluenza virus 3]	AHX22521.1
D protein [Human parainfluenza virus 3]	AGW51120.1
D protein [Human parainfluenza virus 3]	AGT75313.1
D protein [Human parainfluenza virus 3]	AHX22249.1
D protein [Human parainfluenza virus 3]	AHX22097.1
D protein [Human parainfluenza virus 3]	AGW51144.1
D protein [Human parainfluenza virus 3]	AHX22089.1
D protein [Human parainfluenza virus 3]	AHX22225.1
D protein [Human parainfluenza virus 3]	AHX22137.1
D protein [Human parainfluenza virus 3]	AHX22065.1
D protein [Human parainfluenza virus 3]	AGW51224.1
D protein [Human parainfluenza virus 3]	AGT75210.1
D protein [Human parainfluenza virus 3]	AHX22393.1
D protein [Human parainfluenza virus 3]	AGT75258.1
D protein [Human parainfluenza virus 3]	AHX22345.1
D protein [Human parainfluenza virus 3]	AGT75250.1
D protein [Human parainfluenza virus 3]	AHX22113.1
D protein [Human parainfluenza virus 3]	AGW51232.1
D protein [Human parainfluenza virus 3]	AHX22057.1
D protein [Human parainfluenza virus 3]	AHX22209.1
D protein [Human parainfluenza virus 3]	AGW51056.1
D protein [Human parainfluenza virus 3]	AHX22161.1
D protein [Simian Agent 10]	ADR00402.1
D protein [Human parainfluenza virus 3]	AHX22361.1
D protein [Human parainfluenza virus 3]	AGW51281.1
D protein [Human parainfluenza virus 3]	AGW51184.1
D protein [Human parainfluenza virus 3]	AGW51160.1

TABLE 7-continued

PIV3 NCBI Accession Numbers (Nucleic Acid and Amino Acid Sequences)	
Description	GenBank Accession
D protein [Human parainfluenza virus 3]	AHX22465.1
D protein [Human parainfluenza virus 3]	AHX22329.1
D protein [Human parainfluenza virus 3]	AGW51064.1
D protein [Human parainfluenza virus 3]	AGW51040.1
D protein [Human parainfluenza virus 3]	AGT75226.1
D protein [Human parainfluenza virus 3]	AHX22425.1
D protein [Human parainfluenza virus 3]	AHX22305.1
D protein [Human parainfluenza virus 3]	AGW51249.1
D protein [Human parainfluenza virus 3]	AHX22481.1
D protein [Human parainfluenza virus 3]	AHX22281.1
D protein [Human parainfluenza virus 3]	AGW51048.1
D protein [Human parainfluenza virus 3]	AHX22297.1
D protein [Human parainfluenza virus 3]	AGW51088.1
D protein [Human parainfluenza virus 3]	AGT75305.1
D protein [Human parainfluenza virus 3]	AHX22185.1
D protein [Human parainfluenza virus 3]	AGW51104.1
D protein [Human parainfluenza virus 3]	AHX22081.1
D protein [Human parainfluenza virus 3]	AGW51192.1
D protein [Human parainfluenza virus 3]	AHX22489.1
D protein [Human parainfluenza virus 3]	AHX22441.1
D protein [Human parainfluenza virus 3]	AHX22409.1
D protein [Human parainfluenza virus 3]	AHX22369.1
D protein [Human parainfluenza virus 3]	AHX22321.1
D protein [Human parainfluenza virus 3]	AHX22073.1
D protein [Human parainfluenza virus 3]	AGW51152.1
D protein [Human parainfluenza virus 3]	AGW51072.1
D protein [Human parainfluenza virus 3]	AGT75321.1
D protein [Human parainfluenza virus 3]	AHX22257.1
D protein [Human parainfluenza virus 3]	AHX22129.1
D protein [Human parainfluenza virus 3]	AHX22417.1
D protein [Human parainfluenza virus 3]	AGT75218.1
D protein [Human parainfluenza virus 3]	AHX22265.1
D protein [Human parainfluenza virus 3]	AGT75178.1
D protein [Human parainfluenza virus 3]	AHX22433.1
D protein [Human parainfluenza virus 3]	AGW51273.1
D protein [Human parainfluenza virus 3]	AGW51208.1
D protein [Human parainfluenza virus 3]	AGT75170.1
D protein [Human parainfluenza virus 3]	AGT75162.1
D protein [Human parainfluenza virus 3]	AGW51257.1
D protein [Human parainfluenza virus 3]	AGW51200.1
D protein [Human parainfluenza virus 3]	AGW51176.1
D protein [Human parainfluenza virus 3]	AGT75186.1
D protein [Human parainfluenza virus 3]	AGW51265.1
D protein [Human parainfluenza virus 3]	AGW51168.1

TABLE 8

Signal Peptides	
Description	Sequence
HuIgG ₁ signal peptide	METPAQLLFLLLLWLPD TTG
IgE heavy chain epsilon -1 signal peptide	MDWTWILFLVAAATRVHS
Japanese encephalitis PRM signal sequence	MLGNSGQRVVFTILLLL VAPAYS
VSVg protein signal sequence	MKCLLYLAF LFIGNCA
Japanese encephalitis JEV signal sequence	MWLVS LAIVTACAGA

45

TABLE 9

hMPV/PIV Cotton Rat Challenge Study Design					
Group	n	Test Article	[conc]/μg	Route	Challenge
1	5	Placebo	n/a	IM	hMPV/A2
2	5	hMPV vaccine mRNA	30	IM	hMPV/A2
3	5	hMPV vaccine mRNA	15	IM	hMPV/A2
4	5	hMPV vaccine mRNA	10	IM	hMPV/A2
55	5	hMPV/PIV3 vaccine mRNA (15/15)	30	IM	hMPV/A2
6	5	FI-hMPV	n/a	IM	hMPV/A2
7	5	Placebo	n/a	IM	PIV3
8	5	PIV3 vaccine mRNA	30	IM	PIV3
60	9	PIV3 vaccine mRNA	15	IM	PIV3
10	5	PIV3 vaccine mRNA	10	IM	PIV3
11	5	hMPV/PIV3 vaccine mRNA (15/15)	30	IM	PIV3
12	5	FI-PIV3	n/a	IM	PIV3
65	60				

TABLE 10

Strain	Nucleic Acid Sequence	SEQ ID NO:
Betacoronavirus Nucleic Acid Sequence		
gb KJ156934.1 : 21405-25466 Middle East respiratory syndrome coronavirus isolate Riyadh_14_2013, spike protein (nucleotide)	ATGATACACTCAGTGTCTTCTACTGATGTTCTTGTAAACACC TACAGAAAAGTTACGTTGATGTAGGGCCAGATTCTGTAAAG TCTGCTTGTATTGAGGTTGATATACAACAGACCTTCTTTGA TAAACTTGGCCCTAGGCCAATTGATGTTTCTAAGGCTGAC GGTATTATATACCCCTCAAGGCCGTACATATTCTAACATAA CTATCACTTATCAAGGCTTTTTCCTATCAGGGAGACCAT GGTGATATGATGTTTACTCTGCAGGACATGCTACAGGCA CAACTCCACAAAAGTTGTTTGTAGCTAATCTTCTCAGGA CGTCAAACAGTTTGCTAATGGGTTTGTGCTCCGTATAGGA GCAGCTGCCAATTCCTAGGCACTGTTATTATTAGCCCATC TACCAGCGCTACTATACGAAAATTACCCTGCTTTTATGC TGGGTTCTTCAGTTGGTAATTTCTCAGATGGTAAATGGG CCGCTTCTTCAATCATACTCTAGTTCTTTTGCCGATGGAT GTGGCACTTTACTTAGAGCTTTTATTGTATTCTAGAGCCT CGCTCTGGAATCATGTCTGCTGGCAATTCCTATACTTC TTTTGCCACTTATCACACTCCTGCAACAGATTGTTCTGATG GCAATTACAATCGTAATGCCAGTCTGAACCTTTTAAAGGA GTATTTTAATTTACGTAACCTGCACCTTTATGTACACTTATA ACATTACCGAAGATGAGATTTTAGAGTGGTTTGGCATTAC ACAAACTGCTCAAGGTGTTCACTCTTCTCATCTCGGTATG TTGATTTGTACGGCGGCAATATGTTTCAATTTGCCACCTTG CCTGTTTATGATACTATTAAGTATTATCTATCATTCTCTCA CAGTATTGTTTCTATCCAAAGTGATAGAAAAGCTTGGGCT GCCTTCTACGTATATAAACTTCAACCGTTAACTTTCTGTT GGATTTTCTGTTGATGGTTATATACGAGAGCTATAGACT GTGGTTTAAATGATTTGTCAAACTCCACTGCTCATATGAA TCCTTCGATGTTGAATCTGGAGTTTATTAGTTTCGTCTTT CGAAGCAAAACCTTCTGGCTCAGTTGTGGAACAGGCTGAA GGTGTTGAATGTGATTTTTCACCTCTTCTGCTGGCACACC TCCTCAGGTTTATAATTTCAAGCGTTTGGTTTTTACCAATT GCAATTATAATCTTACCAAAATGCTTTCACTTTTTCTGTG AATGATTTTACTTGTAGTCAAATATCTCCAGCAGCAATTGC TAGCAACTGTTATTCTTCACTGATTTTGGATTATTTTTCAT ACCCACTTAGTATGAAATCCGATCTCAGTGTAGTTCTGCT GGTCCAATATCCAGTTTAAATATAAAGCAGTCTTTTCTAA TCCACATGTTTGATCTTAGCGACTGTTCTCATAACTTA CTACTATTACTAAGCCTCTTAAAGTACAGCTATATTAACAA GTGCTCTCGTCTTCTTCTGATGATCGTACTGAAGTACCTC AGTTAGTGAACGCTAATCAATACTCACCTGTGTATCCATT GTCCCATCCACTGTGTGGGAAGACGGTGATTATTATAGGA AACAACTATCTCCACTGGAAGGTGGTGGCTGGCTTGTGC TAGTGGCTCAACTGTTGCATGACTGAGCAATTACAGATG GGCTTTGGTATTACAGTTTCAATATGGTACAGACACCAATA GTGTTTGGCCCAAGCTTGAATTTGCTAATGACACAAAAT TGCCTCTCAATTAGGCAATTGCGTGGAAATATTCCTCTATG GTGTTTGGGGCGTGGTGTTTTTCAGAAATGACAGCTGTA GGTGTTTCGACAGCAGCGCTTTGTTTATGATGCGTACCAGA ATTTAGTTGGCTATTATTCTGATGATGGCACTACTACTGT CTGCGTGTCTGTGTAGTGTCTCTGTTCTGTCATCTATGA TAAAGAACTAAACCCACGCTACTCTATTGTTAGTGTGTT GCATGTGAACACATTTCTTCTACCATGTCTCAATACTCCCG TTCTACGCGATCAATGCTTAAACGGCGAGATTCTACATAT GGCCCCCTTCAGACACCTGTTGGTTGTGCTTAGGACTTGT TAATTCCTCTTTGTTCTGATAGAGGACTGCAAGTTGCCTCTCG GTCAATCTCTCTGTGCTCTTCTGACACACCTAGTACTCTC ACACCTCGCAGTGTGCGCTCTGTGCCAGGTGAAATGCGCT TGGCATCCATTGCTTTAATCATCCATTAGGTTGATCAA CTTAATAGTAGTTATTTTAAATTAAGTATACCCACTAATTT TTCCTTTGGTGTGACTCAGGAGTACATTCAGACAACCAATTC AGAAAGTTACTGTTGATTGTAACAGTACGTTTGCAATGG TTTCCAGAAGTGTGAGCAATTACTGCGCGAGTATGGCCAG TTTTGTTCCAAATAAACCAGGCTCTCCATGGTGCCAATTT ACGCCAGGATGATTCTGTACGTAATTTGTTTGGCAGCGTG AAAAGCTCTCAATCATCTCTTATCATACACAGGTTTGGAG GTGACTTTAATTTGACACTTCTAGAACCCTGTTCTATATCT ACTGGCAGTCGTAGTGCACGTAGTGCTATTGAGGATTTGC TATTTGACAAAGTCACTATAGCTGATCTGGTTATATGCA AGGTTACGATGATTGTATGCAGCAAGGTCAGCATCAGCT CGTGATCTTATTGTTGCTCAATATGTGGCTGGTTATAAGT ATTACCTCCTCTTATGGATGTTAATATGGAAGCCGCGTATA CTTCACTTTGCTTGGCAGCATAGCAGGTGTTGGCTGGACT GCTGGCTTATCCTCTTTGCTGCTATTCCATTGACACAGAG TATYTTTTATAGGTTAAACGGTGTGGCATTACTCAACAG GTTCTTTCAGAGAACCAAGCTTATTGCCAATAAGTTTA ATCAGGCTCTGGGAGCTATGCAACAGGCTTCACTACAAC	20

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	TAATGAAGCTTTTCGGAAGGTT CAGGATGCTGTGAACAAC AATGCACAGGCTCTATCCAAATTAGCTAGCGAGCTATCTA ATACTTTTGGTGCTATTTCCGCCTCTATTGGAGACATCATA CAACGTCCTTGATGTTCTCGAACAGGACGCCCAAATAGACA GACTTATTAATGGCCGTTTGACAACTAAATGCTTTTGTT GCACAGCAGCTTGTCGTTCCGAATCAGCTGCTCTTTCCGC TCAATTGGCTAAAGATAAAGTCAATGAGTGTGTCAAGGCA CAATCCAAAGCCTTCGGATTTCGGGTCAAGGCACACATA TAGTGTCCTTTGTTGTAATGCCCTAATGGCCTTTACTTT ATGCATGTTGGTTATTACCCTAGCAACCACATTGAGTTGT TTCTGCTTATGGTCTTTGCGATGCAGCTAACCTACTAATT GTATAGCCCTGTTAATGGCTACTTTATTAATAAATAATAAC ACTAGGATTGTTGATGAGTGGTCATATACTGGCTCGTCCTT CTATGCACCTGAGCCCATCACCTCTCTTAATACTAAGTATG TTGCACCACAGGTGACATACCAAAACATTTCTACTAACCT CCTCCTCCTCTTCTCGGCAATTCACCGGGATTGACTTCC AAGATGAGTTGGATGAGTTTTTCAAAATGTTAGCACCAG TATACCTAATTTTGGTTCTCTAACACAGATTAATACTACAT TACTCGATCTTACCCTACGAGATGTTGTCTCTTCAACAAGTT GTTAAAGCCCTTAATGAGTCTTACATAGACCTTAAAGAGC TTGGCAATTATACTTATTACAACAAATGGCCGTGGTACAT TTGGCTTGGTTTCATTGCTGGGCTTGTGCCTTAGCTCTAT GCGTCTTCTTACATGTGCTGCACCTGGTTGTGGCACAAAC TGTATGGGAAAACCTAAGTGTAATCGTTGTTGTATAGAT ACGAGGAATACGACCTCGAGCCGCATAAGGTTTCATGTTCA CTAA	
MERS S FL SPIKE 2cEMC/2012 (XBaI change(T to G)) (nucleotide)	ATGATACACTCAGTGTCTTCTACTGATGTTCTTGTTAACACC TACAGAAAGTTACGTTGATGTAGGGCCAGATTCTGTTAAG TCTGCTTGTAATTGAGGTTGATATACAACAGACTTTCTTTGA TAAAACTTGGCCTAGGCCAATTGATGTTTCTAAGGCTGAC GGTATTATATACCCCTCAAGGCCGTACATATTCTAACATAA CTATCACTTATCAAGGTCTTTTTCCCTATCAGGGAGACCAT GGTGATATGTATGTTACTCTGCAGGACATGCTACAGGCA CAACTCCACAAAAGTTGTTGTAGCTAATCTCTCAGGA CGTCAAAACAGTTTGTCTAATGGGTTTGTGCTCCGTATAGGA GCAGCTGCCAATTCCACTGGCACTGTTATTATTAGCCCATC TACCAGCGCTACTATACGAAAATTTACCCTGCTTTTATGC TGGGTTCTTCAGTTGGTAATTTCTCAGATGGTAAATGGG CCGCTTCTTCAATCATACTCTAGTTCTTTTGCCCGATGGAT GTGGCACTTTACTTAGAGCTTTTATTGTATTCTGGAGCCT CGCTCTGGAATCATTTGCTGCTGGCAATTCCTATACCTTC TTTTGCCACTTATCACACTCCTGCAACAGATTGTTCTGATG GCAATTACAATCGTAATGCCAGTCTGAACCTCTTTAAGGA GTATTTTAATTTACGTAACCTGCACCTTTATGTACACTTATA ACATTACCGAAGATGAGATTTTAGAGTGGTTTGGCATTAC ACAAACTGCTCAAGGTGTTCACTCTTCTCATCTCGGTATG TTGATTTGTACGGCGGCAATATGTTTCAATTTGCCACCTTG CCTGTTTATGATACTATTAAAGTATTATTCTATCATTCTCTCA CAGTATTCGTTCTATCCAAAGTGATAGAAAAGCTTGGGCT GCCTTCTACGTATATAAACTTCAACCGTTAACTTTCTGTT GGATTTTCTGTTGATGGTTATATACGACAGCTATAGACT GTGGTTTTAATGATTTGTCAAACTCCACTGCTCATATGAA TCCTTCGATGTTGAATCTGGAGTTTATTCAGTTTCGTCCTT CGAAGCAAAACCTTCTGGCTCAGTTGTGGAACAGGCTGAA GGTGTTGAATGTGATTTTTCACTCTTCTGTCTGGCACACC TCCTCAGGTTTATAATTTCAAGCGTTTGGTTTTTACCAATT GCAATTATAATCTTACCAAATTGCTTTCACTTTTTTCTGTG AATGATTTTACTTGTAGTCAAATATCTCCAGCAGCAATTGC TAGCAACTGTTATTCTTCACTGATTTTGGATTACTTTTCAT ACCCACTTAGTATGAAATCCGATCTCAGTGTAGTTCTGCT GGTCCAATATCCAGTTTAAATTATAAACAGTCCTTTTCTAA TCCCACATGTTTGATTTTAGCGACTGTTCCCTCATAACCTTA CTACTATTACTAAGCCTCTAAGTACAGCTATATTAAACAA GTGCTCTCGTCTTCTTCTGATGATCGTACTGAAGTACCTC AGTTAGTGAACGCTAATCAATACTCACCTGTGTATCCATT GTCCCATCCACTGTGTGGGAAGACGGTGATTATTATAGGA AACAACATATCTCCACTTGAAGGTGGTGGCTGGCTTGTGTC TAGTGGCTCAACTGTTGCCATGACTGAGCAATTACAGATG GGCTTTGGTATTACAGTTCAATATGGTACAGACACCAATA GTGTTTGCCCCAAGCTTGAATTTGCTAATGACACAAAAT TGCCTCTCAATTAGGCAATTGCGTGGAATATTCCTCTATG GTGTTTTCGGCCGTGGTGTTTTTCAGAAATGACACAGCTGTA GGTGTTTCGACAGCAGCGCTTTGTTTATGATGCGTACCAGA ATTTAGTTGGCTATTATTCTGATGATGGCAACTACTACTGT TTGCGTGTGTGTAGTGTCTCTGTTTCTGTGCATCTATGAT AAAGAAACTAAAACCCACGCTACTCTATTGTGTAGTGTG	21

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	CATGTGAACACATTCTTCTACCATGTCTCAATACTCCCGT TCTACGCGATCAATGCTTAAACGGCGAGATTCTACATATG GCCCCCTTCAGACACCTGTTGGTTGTCTCCTAGGACTTGTT AATTCTCTTTGTTTCGTAGAGGACTGCAAGTTGCCTCTTGG TCAATCTCTCTGTGCTCTTCTGACACACCTAGTACTCTCA CACCTCGCAGTGTGCGCTCTGTTCCAGGTGAAATGCGCTT GGCATCCATTGCTTTTAATCATCCTATTAGGTTGATCAAC TTAATAGTAGTTATTTAAATTAAGTATACCCACTAATTTT TCCTTTGGTGTGACTCAGGAGTACATTGAGACAACCATTC AGAAAGTTACTGTTGATTGTAACAGTACGTTTGCAATGG TTTCCAGAAGTGTGAGCAATTACTGCGCGAGTATGGCCAG TTTTGTTCAAAATAAACAGGCTCTCCATGGTGCCAATTT ACGCCAGGATGATTCTGTACGTAATTTGTTTGCAGCGTG AAAAGCTCTCAATCATCTCTATCATACAGGTTTGGAG GTGACTTTAATTTGACACTTCTGGAACCTGTTTCTATATCT ACTGGCAGTCGTAGTGCAGTAGTGCTATTGAGGATTTGC TATTTGACAAAGTCACTATAGCTGATCCTGGTTATATGCA AGGTTACGATGATTGCATGCAGCAAGGTCCAGCATCAGCT CGTGATCTTATTGTGCTCAATATGTGGCTGGTTACAAAGT ATTACCTCCTCTTATGGATGTTAATATGGAAGCCGCTATA CTTTATCTTTGCTTGGCAGCATAGCAGGTGTTGGCTGGACT GCTGGCTTATCCTCCTTTGCTGCTATTCCATTGACACAGAG TATCTTTTATAGGTTAAACGGTGTGGCATTACTCAACAGG TTCTTTTCAGAGAACCAAAGCTTATTGCCAATAAGTTTAA TCAGGCTCTGGGAGCTATGCAACAGGCTTCACTACAACT AATGAAGCTTTTCAGAAGGTTCAAGATGCTGTGAACAACA ATGCACAGGCTCTATCCAAATTAGCTAGCGAGCTATCTAA TACTTTTGGTGCTATTTCGCTCTATTGGAGACATCATAC AACGCTCTTGATGTTCTCGAACAGGACGCCAAATAGACAG ACTTATTAATGGCCGTTTGACAACTAAATGCTTTTGTG CACAGCAGCTTGTTCGTTCCGAATCAGTGCTCTTTCGCT CAATTGGCTAAAGATAAAGTCAATGAGTGTGCAAGGCAC AATCCAAGCGTCTCGGATTTTGCGGTCAAGGCACACATAT AGTGTCTTTGTTGTAATGCCCTAATGGCCTTTACTTCA TGCATGTTGGTTATTACCTAGCAACACATTGAGGTTGTT TCTGCTTATGGTCTTTGCGATGCAGCTAACCTACTAATTG TATAGCCCTGTAAATGGCTACTTTATTAATACTAATAACA CTAGGATTGTTGATGAGTGGTCATATACTGGCTCGTCTTC TATGCACCTGAGCCCATTAACCTCCCTTAATACTAAGTATGT TGCACCACAGGTGACATACCAAAACATTCTACTAACCTC CCTCCTCCTCTCTCGGCAATTCCACCGGGATTGACTTCCA AGATGAGTTGGATGAGTTTTCAAAAATGTTAGCACCAGT ATACCTAATTTTGGTTCCCTAACACAGATTAATACTACATT ACTCGATCTTACCTACGAGATGTTGTCTTCAACAAGTTG TTAAAGCCCTTAATGAGTCTTACATAGACCTTAAAGAGCT TGGCAATTATACTTATTACAACAAATGGCCGTGGTACATT TGGCTTGGTTTCATTGTGCTGGCTTGTGCCTTAGCTCTATG CGTCTTCTTCATACTGTGCTGCATGGTTGTGGCACAACCT GTATGGGAAAACCTAAGTGAATCGTTGTGTGATAGATA CGAGGAATACGACCTCGAGCCGCATAAGGTTTATGTTTAC TAA	
Novel_MERS_S2_ subunit_trimeric vaccine (nucleotide)	ATGATCCACTCCGTGTTCTCTCATGTTCTGTTGACCCC CACTGAGTCAGACTGCAAGCTCCCGCTGGGACAGTCCCTG TGTGCGCTGCCTGACACTCTAGCACTTGACCCACAGCTC CGTGCGGTGCGTGCCTGGCGAAATGCGGCTGGCCTCCATC GCCTTCAATCACCAATCCAAGTGATCAGCTGAATAGCT CGTATTTCAAGCTGTCCATCCCCACGAATTCTCGTTGCGG GTCACCCAGGAGTACATCCAGACCACAATTCAGAAGGTCA CCGTCGATTGCAAGCAATACGTGTGCAACGGCTTCCAGAA GTGCGAGCAGCTGCTGAGAGAAATACGGGCAGTTTTCGAGC AAGATCAACCAGGCGCTGCATGGAGCTAACTTGCGCCAGG ACGACTCCGTGCGCAACCTCTTTGCTCTGTGAAGTCATCC CAGTCTCTCCCAATCATCCGGGATTCGGAGGGGACTTCA ACCTGACCTCTCTGGAGCCGTGTGATCAGCACCGGTAG CAGATCGGCGCGCTCAGCCATTGAAGATCTTCTGTTTCGAC AAGGTCACCATCGCCGATCCGGGCTACATGCAGGGATACG ACGACTGTATGCAGCAGGACAGCCTCCGCGAGGGACCT CATCTGCGCGCAATACGTGGCCGGTACAAAGTGTGCCT CCTCTGATGGATGTGAACATGGAGGCCGCTTATACTTCGT CCTGTCTCGGCTCTATCGCCGGCGTGGGTGGACCGCCGG CCTGTCTCTCTCGCCGCTATCCCTTTGCAACATCCATT TCTACCGGCTCAACGGCGTGGGCATTACTCAACAAGTCT GTCCGGAACCAAGGTTGATCGCAACAAGTTCAATCA GGCCCTGGGGGCCATGCAGACTGGATTCACTACGACTAAC GAAGCGTTCCAGAAGGTCCAGGACGCTGTGAACAACAAC GCCCAGGCGCTCTAAAGCTGGCCTCCGAACCTAGCAACA	22

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	CCTTCGGAGCCATCAGCGCATCGATCGGTGACATAATTCA GCGGCTGGACGTGCTGGAGCAGGACGCCAGATCGACCG CCTCATCAACGGACGGCTGACCACCTTGAATGCCTTCGTG GCACAACAGCTGGTCCGGAGCGAATCAGCGGCACCTTCCG CCCAATCGCCAAGGACAAAGTCAACGAATGCGTGAAGG CCCAGTCCAAGAGGTCCGGTTTCTGCGGTCAAGGAACCCA TATTGTGTCCTTCGTGCGTGAACGCGCCCAACGGTCTGTACT TTATGCACGTCGGCTACTACCCGAGCAATCATATCGAAGT GGTGTCCGCCTACGGCCTGTGCGATGCGCTAACCCCACT AACTGTATTGCCCTTGTGAACGGATATTTTATTAAGACCA ACAACACCCGCATTGTGGACGAATGGTCATACCCGGTTC GTCTTCTACGCGCCCGAGCCCATCACTTCACTGAACACC AAATACGTGGCTCCGCAAGTGACCTACCAGAATCATCCA CCAATTTGCCGCGCCGCTGTCTCGGAAACAGCACCGGAAT TGATTTCCAAGATGAACCTGGACGAATTTCTCAAGAACGTG TCCACTTCCATTCCCAACTTCGGAAGCCTGACACAGATCA ACACCACCCCTTCTCGACCTGACCTACGAGATGCTGAGCCT TCAACAAGTGGTCAAGGCCCTGAACGAGAGCTACATCGAC CTGAAGGAGCTGGGCAACTATACCTACTACAACAAGTGGC CGGACAAGATTGAGGAGATTCTGTGAAAATCTACCACAT TGAAAACGAGATCGCCAGAATCAAGAAGCTTATCGGCGA AGCC	
MERS_S0_Full-length Spike protein (nucleotide, codon optimized)	ATGGAAACCCCTGCCAGCTGCTGTTCTGCTGCTGCTGTG GCTGCCTGATACCACCGGCAGCTATGTGGACGTGGGCCCC GATAGCGTGAAGTCCGCCTGTATCGAAGTGGACATCCAGC AGACCTTTTTCGACAAGACCTGGCCAGACCATCGACGT GTCCAAGGCCGACGGCATCATCTATCCACAAGGCCGGACC TACAGCAACATCACCATTACCTACCAAGGCCCTGTTCCCAT ATCAAGGCCGACCACGGCGATATGTACGTGACTCTGCGCG CCACGCCACCGGCACACACCCAGAACTGTTCTGTGGCC AACTACAGCCAGGACGTGAAGCAGTTCGCCAACGGCTTCG TCGTGGCGATTGGCGCCGCTGCCAATAGCACCGGCACAGT GATCATCAGCCCCAGCACCGGCCACCATCCGGAAGATC TACCCCGCCTTCATGCTGGGAGCTCCGTGGCAATTTCA GCGACGGCAAGATGGGCCGTTCTTCAACCACACCCCTGGT GCTGCTGCCCGATGGCTGTGGCACACTGCTGAGAGCCTTC TACTGCATCCTGGAACCCAGAAGCGGCACCACTGCCCTG CCGGCAATAGCTACACAGCTTCGCCACCTACCACACACC CGCCACCGATTGCTCCGACGGCAACTACAACCGGAACGCC AGCCTGAACAGCTTCAAAGAGTACTTCAACCTGCGGAAGT GCACCTTCATGTACACCTACAATATCACCAGGACGAGAT CCTGGAATGGTTCCGCATCACCAGACCGCCAGGGCGTG CACCTGTTTCAGCAGCAGATACGTGGACCTGTACGGCGGCA ACATGTTCCAGTTTGCCACCCCTGCCCGTGTACGACACCATC AAGTACTACAGCATCATCCCCACAGCATCCGGTCCATCC AGAGCGACAGAAAAGCCTGGGCCGCTTCTACGTGTACAA GCTGCAGCCCCTGACCTTCTGCTGGACTTCAGCTGGAC GGCTACATCAGACGGGCCATCGACTGCGGCTTCAACGACC TGAGCCAGCTGCACTGCTCTACGAGAGCTTCGACGTGGA AAGCGCGTGTACAGCGTGTCCAGCTTCGAGGCCAAGCCT AGCGGCAGCGTGGTGGAAACAGGCTGAGGGCGTGGAAATGC GACTTCAGCCCTCTGCTGAGCGGCACCCCTCCCCAGGTGT ACAACTTCAAGCGGCTGGTGTTCACCAACTGCAATTACAA CCTGACCAAGCTGCTGAGCCTGTTCTCGTGAACGACTTC ACCTGTAGCCAGATCAGCCCTGCCGCCATTGCCAGCAACT GCTACAGCAGCCTGATCTGGACTACTTCAGCTACCCCT GAGCATGAAGTCCGATCTGAGCGTGTCTCCGCGGAGCCC ATCAGCCAGTTCAACTACAAGCAGAGCTTCAGCAACCCTA CCTGCCTGATTCTGGCCACCGTGCCCCACAATCTGACCAC CATCACCAGCCCCGAAAGTACAGCTACATCAACAAGTGC AGCAGACTGCTGTCCGACGACCGGACCGAAGTGGCCAGC TCGTGAACGCCAACCAGTACAGCCCTGCGTGTCCATCGT GCCAGCACCGTGTGGGAGGACGGCGACTACTACAGAAA GCAGCTGAGCCCCCTGGAAGCGCGCGGATGGCTGGTGGCT TCTGGAAGCACAGTGGCCATGACCGAGCAGCTGCAGATG GGCTTTGGCATCACCGTGCAGTACGGCACCGACACCAACA GCGTGTGCCCAAGCTGGAATTCGCCAATGACACCAAGAT CGCCAGCCAGCTGGGAACTGCGTGGAAATCTCCCTGTAT GGCGTGTCCGACGGGGCGTGTCCAGAATTGCACAGCAG TGGGAGTGCGGCAGCAGAGATTCTGTACGATGCCTACCA GAACCTCGTGGGCTACTACAGCGACGACGGCAATTACTAC TGCCTGCGGGCTGTGTGTCGTGCCCGTGTCCGTGATCTA CGACAAAGAGACAAAGACCCACGCCACACTGTTCGGCTCC GTGGCCTGCGAGCACATCAGCTCCACCATGAGCCAGTACT CCCGCTCCACCCGGTCCATGCTGAAGCGGAGAGATAGCAC CTACGGCCCCCTGCAGACACCTGTGGGATGTGTGCTGGGC	23

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	CTCGTGAACAGCTCCCTGTTTGTGGAAGATTGCAAGCTGC CCCTGGGCCAGAGCCTGTGTGCCCTGCCAGATACCCCTAG CACCCTGACCCCTAGAAGCGTGCCTCTGTGCCCCGGCGAA ATGCGGCTGGCCTCTATCGCCTTCAATCACCCATCCAGGT GGACCAGCTGAACCTCCAGCTACTTCAAGCTGAGCATCCCC ACCAATTCAGCTTCGGCGTGACCCAGGAGTACATCCAGA CCACAATCCAGAAAGTGACCGTGGACTGCAAGCAGTACGT GTGCAACGGCTTTCAGAAAGTGCAGACAGCTGCTGCGCGAG TACGGCCAGTTCTGCAGCAAGATCAACCAGGCCCTGCACG GCGCCAACCTGAGACAGGATGACAGCGTGGGAACCTGTT CGCCAGCGTGAAAAGCAGCCAGTCCAGCCCCATCATCCCT GGCTTCGGCGGGCAGCTTTAACCTGACCCCTGCTGGAACCTG TGTCCATCAGCACCGGCTCCAGAAGCGCCAGATCCGCCAT CGAGGACCTGCTGTTCGACAAAGTGACCATTGCCGACCCC GGCTACATGCAGGGCTACGACGATTGCATGCAGCAGGGCC CAGCCAGCGCCAGGGATCTGATCTGTGCCAGTATGTGGC CGGCTACAAGGTGCTGCCCCCCTGATGGACGTGAACATG GAAGCCGCCTACACCTCCAGCCTGCTGGGCTCTATTGCTG GCGTGGGATGGACAGCGCGCTGTCTAGCTTTGCCGCCAT CCCTTTCGCCCAGAGCATCTTACCGGCTGAACGGCGTG GGCATCACACAACAGGTGCTGAGCGAGAACCAGAAGCTG ATCGCCAACAAGTTTAACCAAGCACTGGGCGCCATGCAGA CCGGCTTCAACCACCAACGAGGCTTCAGAAAGGTGCA GGACGCCGTGAACAACAACGCCAGGCTCTGAGCAAGCT GGCCCTCCGAGCTGAGCAATACCTTCGGCGCCATCAGCGCC TCCATCGGCGACATCATCCAGCGGCTGGACGTGCTGGAAC AGGACGCCCAGATCGACCGGCTGATCAACGGCAGACTGA CCACCCTGAACGCCCTTCGTGGCACAGCAGCTCGTGCGGAG CGAATCTGCCGCTCTGTCTGCTCAGCTGGCCAAGGACAAA GTGAACGAGTGGCTGAAGGCCAGTCCAAGCGGAGCGGC TTTTGTGGCCAGGGCACCCACATCGTGTCTTCGTCTGTGAA TGCCCCAACGGCTGTACTTTATGCACGTGGGCTATTACC CCAGCAACCACATCGAGGTGGTGTCCGCCTATGGCCTGTG CGACGCCGCCAATCCTACCACTGTATCGCCCCCGTGAA GGCTACTTCATCAAGACCAACAACCCGGATCGTGGACG AGTGGTCTTACACAGGCAGCAGCTTCTACGCCCCCGAGCC CATCACTCCCTGAACACCAATACGTGGCCCCCAAGTG ACATACCAGAACATCTCCACCAACCTGCCCCCTCCACTGC TGGGAAATTCACCGGCATCGACTTCCAGGACGAGCTGGA CGAGTTCTTCAAGAAGCTGTCCACCTCCATCCCAACTTCG GCAGCCTGACCCAGATCAACACCACTCTGCTGGACCTGAC CTACGAGATGCTGTCCCTGCAACAGGTGCTGAAAGCCCTG AACGAGAGCTACATCGACCTGAAAGAGCTGGGGAACCTAC ACCTACTACAACAAGTGGCCTTGGTACATTTGGCTGGGCT TTATCGCCGGCCTGGTGGCCCTGGCCCTGTGCGTGTCTTTC ATCCTGTGCTGACCGGCTGCGGCACCAATTGCATGGGCA AGCTGAAATGCAACCGGTGCTGCGACAGATACGAGGAAT ACGACCTGGAACCTCACAAAGTGCATGTGCAC	
Betacoronavirus mRNA Sequences		
gb KJ156934.1 : 21405-25466 Middle East respiratory syndrome coronavirus isolate Riyadh_14_2013, spike protein (nucleotide)	AUGAUAACACUCAGUGUUUUCUACUGAUGUUUCUUGUUAAC ACCUACAGAAAGUUAACGUUGAUGUAGGGCCAGAUUCUG UUAAGUCUGCUUGUAUUGAGGUUGAUUAACAACAGACC UUCUUUGAUAAAACUUGGCCUAGGCCAAUUGAUGUUUC UAAGGCUAGCGUAUUUAUUAACCCUACAGGCCGUACAU AUUCUAACAUAACUAUCUAUAAGGUCUUUUUCCCU AUCAGGGAGACCAUGGUGUAUGUAUGUUUAUCUCUGCA GGACAUGCUCAGGCACAAUCUACAAAAGUUGUUUGU AGCUAACUAUUCUACGAGCGUAAACAGUUUGCUAAUG GGUUUUGUCGUCGUUAUAGGAGCAGUCGCAAUUCCACUG GCACUGUUUAUUAUAGCCCAUCUACAGCGCUACUAUAC GAAAAUUUACCCUGCUUUUAUGCUGGGUUCUUCAGUU GGUAAUUUUCAGAUUGGUAAAAUGGCCGCUUCUUCAA UCAUAUCUAGUUUUUUGCCGAUGGAUGUGGCACUU UACUUAGAGCUUUUAUUGUAUUCUAGAGCCUCGCUUCU GGAAAUCAUUGCCUGCUGGCAAUCCUAUAUCUUCUU UGCCACUUAUCACAUCCUGCAACAGAUUGUUCUGAUGG CAAUUACAACUAGUAGCCAGUCUGAACUCUUUUAAAGG AGUAUUUUAAUUUACGUAACUGCACCUUUUAUGUACACU UAUAACAUAUACCAAGAUAGAUUUUAGAGUGGUUUUG CAUUAACAACUAGCUAAGGUGUACACCUUCUUCUACU UCGGUUUGUUAUUGUAUGCGCGCAAUUGUUUCAAU UUGCCACCUUGCCUGUUUAUGAUACUAUUAAGUAUUU UCUUAUUAUCCUACAGUAUUCGUUCUAUCCAAAGUGAU AGAAAGCUUGGCCUGCCUUUCACGUAUAUAAACUUA ACCGUUAACUUCCUGUUGGAUUUUUCUGUUGAUGGUU	65

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	AUAUACGCAGAGCUAUAAGACUGUGUUUUAAUGAUUUG UCACAACUCCACUGUCUAUAUGAAUCCUUCGAGUUGGAA UCUGGAGUUUAUUCAGUUUCGUCUUUCGAAGCAAAACC UUCUGGCUCAGUUGUGGAACAGGCUGAAGGUUUGAAU GUGAUUUUACCCUUCUUGUCUGGCACACCUCUCAGG UUUAUAAUUUCAAAGCGUUUGUUUUUACCAUUGCAAU UAUAUUCUUAACCAAUUGCUUUACAUUUUUUGUGAA UGAUUUUACUUGAGUCAAAUAUCUCCAGCAGCAAUUG CUAGCAACUGUUAUUCUACUGAUUUUGGAUUUUUU UCAUACCCACUAGUAUGAAUCCGAUCUCAGUGUUAG UUCUGCUGGUCCAAUAUCCAGUUUAUUUAACAGU CCUUUUCUAUACCAUUGAUUUUGAUUAGCGACUGUUU CUCAUAAACCUUACUUAUUACUAAAGCCUUAAGUACA GCUAUUAUAACAGGUCUCUGUCUUUUUGAUGAU CGUACUGAAGUACCUAGUUAGUAAACGCUAAUCAAUA CUCACCCUGUGUAUCCAUUGUCCCAUCCAGUGUGGGA AGACGGUGAUUUUAUAGGAAACAAUUCUCCACUUG AAGGUGGUGGCUGGCUUGUUGCUAGUGGCUCACUGUU GCCAUGACUGAGCAAUUACAGAUGGCUUUGGUUUUAC AGUUCAAUAUGGUACAGACCAAUAGUUUGCCCCA AGCUUGAAUUUGCUAAUGACACAAAAUUGCCUCUCAA UUAGGCAAUUGCGUGGAAUUAUCCUUAUGGUUUUC GGGCCGUGUGUUUUUAGAAUUGCACAGCUGUAGGUG UUCGACAGCAGCGCUUUGUUUAUGAUGCGUACCGAAU UUAGUUUGCUAUUUUUCUGAUGAUGGCAACUACUACUG UCUGCGUGCUUGUGUUAGUGUUCUUGUUUCUGUACUCU AUGAUAAAGAAACUAAACCCACGCUACUCUUAUUUGGU AGUGUUGCAUGUGAACACAUUUUCUACCAUGUCUCA AUACUCCCGUUCUACGCGAUCAAUGCUUAAACGCGGAGA UUCUACAUAUGGCCCCUUCAGACACCUUGUUUGGUUGU CCUAGGACUUUUUAUUCUUCUUUGUUCGUAAGGACU GCAAGUUGCCUUCUGGUCAAUUCUUGUGCUUUCUUG ACACACCUAGUACUCUACACCUCCAGUGUGCGCUCUG UGCCAGGUGAAUUGCGCUUGGCAUCCAUUGCUUUUAU CAUCCCAUUCAGGUUGAUCAAUUAUAGUAGUUUUU UAAAUUAAGUAUACCCACUAAUUUUUCCUUGGUGUGA CUCAGGAGUACAUUCAGACAACCAUUCAGAAAGUUACU GUUGAUUGUAAACAGUACGUUUGCAAUGGUUUCAGAA GUGUGAGCAAUUAUGCGCGAGUAUGGCCAGUUUUUGU CCAAAUAUAAACAGGCUCUCCAUUGGUGCCAAUUUACGCC AGGAUGAUUCUGUACGUAAUUGUUUGCGAGCGUGAAA AGCUCUCAAUCAUCCUUAUUAUACAGGUUUUGGAGGU GACUUUAUUUGACACUUCUAGAACCUGUUUCUAUAUC UACUGGCAGUCGUAGUGCACGUAGUGCUAUUGAGGAUU UGCUAUUUAGACAAAGUACUUAUAGCUGAUCCUGGUUU AUGCAAGGUUACGAUGAUUGUAUGCAGCAAGGUCCAGC AUCAGCUCUGUACUUAUUUGUGCUCAAUAGUGGCUG GUUAUAAAGUAUUAUCCUUCUUAUGGAUGUUAUAUG GAAGCCGCGUAUACUUCUUAUGCUUGGCAGCAUAGCA GGUGUUGGCUGGACUGCUGGCUUAUCCUUCUUGCUGCU AUUCCAUUUGCACAGAGUAUUAUUUAAGGUUAAACGG UGUUGGCAUUAUCUAAACAGGUUUCUUCAGAGAACCAAA AGCUUAUUGCCAAUAGUUUAUACAGGCUCUGGGAGCU AUGCAAAAGGCUCUACUACAACUAAUGAAGCUUUUCG GAAGGUUCAGGAUGCUGUGAACAAUAGCAAGGCUC UAUCCAAAUUAGCUAGCGAGCUAUAUUAUUCUUUGGU GCUAUUUCCGCCUCUAUUUGGAGACAUAUACAACGUCUU GAUGUUCUGAACAGGACGCCCAAUAGACAGACUUUAU UAAUGGCCGUUUGACAACACUAAUAGCUUUUGUUGCAC AGCAGCUUGUUCGUUCCGAAUACAGCUGCUUUUCCGCUC AAUUGGCUAAAGAUAAAGUCAUAGAGUGUGUCAAAGGCA CAAUCCAAAGCGUUCUGGAUUUUGCGGUCAGGCACACA AUAGUGUCCUUUGUUGUAAUAGCCCCUAAUGGCCUUUA CUUUUAGCAUGUUGGUUAUUACCUAGCAACACAUUG AGGUUGUUUCUGCUUAUGGUUUUGCGAUGCAGCUAAC CCUACUAAUUGUAUAGCCCCUGUUAAUGGCUACUUUAU UAAAAUUAUACACUAGGAUUGUUGAUGAGUGGUCAU AUACUGGCUCGUCCUUCUUAUGCACCUGAGCCCAUACCU CUCUUAUAUUAAGUAUGUUGCACCACAGGUGACAUACC AAAAUUAUUAUACUAAUCCUCCUCCUUCUUCUCGGCA AUUCCACCGGAUUGACUUCUAAAGUAGUUGGAUGAG UUUUUCAAUAAUGUAGCACCAGUAUACCUAAUUUUUG UUCUUAACACAGAUUAUAUACUACAUUACUUGAUCUUA CUACGAGAUUGUUCUUCUAAUAGUUUUAAAGCCC UUAUAGAGUCUUAUAGACCUUAAAGAGCUUGGCAU UAUACUUAUUAUCAAUAAUGGCCGUGGUACAUUUGGU UGGUUUCAUUGCUGGCUGUUGGCCUUAUCUUAUGCG	

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	UCUUCUUCUAUCUGUGCUGCAGUGGUGUGGCACAAACU GUAUGGGAAAACUUAAGUGUAUCGUGUGUGAUAAGA UACGAGGAUACGACCUCGAGCCGCAUAGGUUCAUGU UCACUAA	
MERS S FL SPIKE 2cEMC/2012 (XBaI change(U to G)) (nucleotide)	AUGAUACACUCAGUGUUUCUACUGAUGUUUUGUUAAC ACCUACAGAAAAGUUAACGUUGAUGUAGGGCCAGAUUCUG UUAAGUCUGCUUGUAUUGAGGUGUAUAUACAACAGACU UUCUUUGAUAAAACUUGGCCUAGGCCAAUUGAUGUUUC UAAGGUGACGGUAUUUAUACCCUACAAGGCCGUAUACU AUUCUAAACUAACUAUCACUUAUCAAGGUUUUUUUUUUU AUCAGGGAGACCAUGGUGUAUUGUUGUUUACUCUGCA GGACAUGCUACAGGCACAACUCCACAAAAGUUGUUUGU AGCUAAACUAUUCUCAGGACGUCAAAAGUUUGCUAUG GGUUUGUCGUCGUAUAGGAGCAGCUGCCAAUUCACUG GCACUGUUUAUUAUAGCCCAUCUACAGCGCUACUAUAC GAAAAAUUUAACCCUGCUUUUAUGCUGGGUUCUUCAGUU GGUAAUUUCUACAGUUGGUAUUGGGCCGCUUCUUCAA UCAUACUCUAGUUCUUUUUGCCGAGUGAUGGGCACUU UACUUAGAGCUUUUAUUGUAUUCUGGAGCCUCGCUUCU GGAAAUCAUUGUCCUGCGGCAAUUCCUAUACUUCUUU UGCCACUUAUCACACUCCUGCAACAGAUUUGUCUGAUGG CAAUUAACAUCGUAUUGCCAGUCUGAACUUCUUUAAGG AGUAUUUUAAUUUACGUAACUGCACCUUUUAUGUACACU UAUAACAUAUACCGAAGAGAGAUUUUAAGUGGUUUUGG CAUUAACAACAUCGUCUAAGGUGUUCACCUUCUUCUACU UCGGUAUGUGAUUUUGUACGGCGGCAUAUGUUUCAU UUGCCACCUUGCCUGUUUAUGAUACUAUUAGUAUUUAU UCUAUCAUUCUACAGUAUUCGUUCUAUCCAAAGUGAU AGAAAAGCUUGGGCUGCCUUCUACGUAUAUAAAACUUC ACCGUUAACUUCUUGGUUGGAUUUUUUGUUUGAUGGU AUAUACGACAGCUAUAAGACUGUGUUUUUAUGAUUUUG UCACAACUCCACUGCUCAUAUGAAUCCUUCGAGUGUAG UCUGGAGUUUAUUCAGUUUCGUCUUUCGAGCAAAACC UUCUGGCUCAUUGUGGAACAGGCUAAGGUGUUGAAU GUGAUUUUCCACCUUCUUGUGUGGCACACCUUCUCAGG UUUAUAAUUUACAAGCGUUUGGUUUUUAACAAUUGCAAU UAUAUUCUUAACAAUUGCUUCUUCUUCUUUCUGUGAA UGAUUUUACUUGUAGUCAAUAUUCUCCAGCAGCAAUUG CUAGCAACUGUUAUUCUUCACUGAUUUUGGAUUACUUU UCAUACCCACUUAUGUAUAAAUCGUAUCUAGUGUUAG UUCUGCUGGUCCAAUAUCCAGUUUAUUAUAAACAGU CCUUUUCUAAUCCCAUGUUUGAUUUUAGCGACUGUUC CUCAUAACCUUAUCUAUUAUUAAGCCUUCUUAAGUACA GCUAUAUUAACAAGUGCUUCUGUCUUUCUUGAUGAU CGUACUGAAGUACCUAGUUAUGUAACGCUAAUCAAUA CUCACCCUGUGUAUCCAUGUCCCAUCCAGUGUGGGGA AGACGGUGAUUAUUAUAGGAAACAACUAUUCUCCACUUG AAGGUGGUGGCUUGGCUUGUUGUAGUGGCUCACUGUU GCCAUGACUGAGCAUUAAGAUUGGCUUUGGUUAUAC AGUUCAAUUGGUACAGACACCAUAGUUGUUGCCCCA AGCUUGAAUUUGCUAAUGACACAAAAUUGCCUUCUCAA UUAAGGCAUUGCGUGGAAUAUUCUUCUUAUGGUGUUUC GGGCCUGUGUUUUUACGAAUUGCACAGCUGUAGGUG UUCGACAGCAGCGCUUUGUUUAUGAUGCGUACAGAAU UUAGUUGGCUAUAUUCUGAUGAUGGCAACUACUACUG UUUGCGUGCUUGUUGUAGUUGCCUGUUUCUGUACUUCU AUGAUAAAGAAACUAAAAACCACGCUACUUAUUGGU AGUGUUGCAUGUGAACACAUUUUCUUAACAUUGCUCA AUACUCCGCUUCACGCGAUCAAUGCUUAAACGGCGAGA UUUAACAUAUGGCCCCUUCAGACACCUUGUUGGUGUGU CCUAGGACUUGUUAUUCUUCUUUGUUGUAGGAGACU GCAAGUUGCCUUCUGGUAUUCUUCUGGCUUCUCCUG ACACACCUAGUACUUCACACCUUCGAGUGUGCGCUUCG UUCAGGUGAAUUGCGCUUGGCAUCCAUUGCUUUUAU CAUCCUAUUCAGGUUGAUCAACUUAUAGUAGUUAUUU UAAAUAAGUAUACCCACUAAUUUUUCUUGGUGUGA CUCAGGAGUACAUUCAGACAACCAUUCAGAAAGUUAU GUUGAUUGUAAACAGUACGUUGCAAUGGUUUCAGAA GUGUGAGCAAUUAUCUGCGCGAGUAUGGCCAGUUUGUU CCAAAUAUAAACAGGCUCUCCAUUGGUGCCAAUUACGCC AGGAUGAUUCUGUACGUAUUUGUUUGCGAGCGUGAAA AGCUUCUAAUACUUCUUAUUAUACAGGUUUUGGAGGU GACUUUAUUUAGACAUUCUGGAACCUUUUCUAUAUC UACUGGAGUUGUAGUGCACGUAGUGCUAUAUGAGGAU UGCUAUAUAGACAAAGUACUAUAGCUGAUCCUGGUUAU AUGCAAGGUUACGAUGAUUGCAUGCAGCAGGUCCAGC	66

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	AUCAGCUCGUGAUCUUUUUUGUGUCUAAUUGUGGCUG GUUACAAAGUAUUACCUCUCUUAUGGAUGUUAUAUG GAAGCCGCGUAUACUUCAUCUUUGCUUGGCAGCAUAGCA GGUGUUGGCUGGACUGCUGGCUUUACCUCUUUGCUGCU AUUCCAUUUGCACAGAGUAUCUUUUUAUGGUUAAACGG UGUUGGCAUUACUCAAACAGGUUCUUUCAGAGAACCAAA AGCUUAUUGCCAAUAAAGUUUAUACAGGCUCUGGGAGCU AUGCAAACAGGCUCUACUACAACUAAUGAAGCUUUUCA GAAGGUUCAGGAUGCUGUAACAACAAGCAGGCAGCUC UAUCCAAAUUAGCUAGCGAGCUAUCUAAUACUUUUGGU GCUAUUUCCGCCUCUAUUGGAGACAUACAAACGUCUU GAUGUUCUCGAACAGGACGCCAAAUAGACAGACUUUAU UAAUGGC CGUUUGACAACACUAAAUAGCUUUUUGUUCAC AGCAGCUUGUUCGUUCCGAUACAGCUGCUCUUUCCGCUC AAUUGGC UAAAGAUAAAGUCAUAGAGUGUGUCAAGGCA CAAUCCAAGCGUUCUGGAUUUUGCGGUCAAGGCACACAU AUAGUGUCUUUUGUUAUAAUGCCCCUAUUGGC UUUA CUUCAUGCAUGUUGGUUAUUACCCUAGCAACCAUAUGA GGUUGUUUCUGCUUAUGGUCUUUGCGAUGCAGCUAACCC CUACUAAUUGUAUAGCCCCUGUUAUUGGCUAUUUAUU AAAACUAAUAAACACUAGGAUUGUUGAUGAGUGGUCUAU UACUGGCUCGUCUUUUAUGCACCUGAGCCCAUUACCUCC CCUUAUACUAAGUAUGUUGCACCAGGUGACAUACCA AAACAUUUUCUAUAAACCUCCUCCUCCUUCUUCGGCAA UUCCACCGGGAUUGACUUCCAAGAUAGAUUGGAUGAGU UUUUCAAAAGUUAAGCACCAGUAUACCUAAUUUUGGU UCCCUAAACAGAUUAAUACUAUUAUCGUAUCUUACC UACGAGAUUGUUCUUAACAAGUUGUUAAGCCCU UAAUGAGUCUUACUAAGACCUAAAGAGCUUGGCAAUU AUACUUAUUAACAACAAUUGGCGUGGUACAUAUUGGCUU GGUUUCAUUGCUGGGCUUGUUGCCUUAAGCUCUAUGCGU CUUCUUAUACUGUGCUGCAGUGUUGGCGACAAACUG UAUGGGAUAAACUUAAGUGUAAUCGUGUUGUGUAUAGAU ACGAGGAUACGACCUCGAGCCGCAUAAAGGUUCAUGUUC ACUAA	
Novel_MERS_S2_subunit_trimeric_vaccine (nucleotide)	AUGAUCCACUCCGUGUUCUCCUCAUGUUCUGUUGACC CCCACUGAGUCAGACUGCAAGCUCUCCGUGGACAGUCC CUGUGUGCGCUGCCUGACACUCCUAGCACUCUGACCCCA CGCUCUGUGCGUCCGUGUCCUGGCGAAUUGCGGUGGCC UCCAUCCGCUUCAAUCCACCAUCCAAUGGGAUCAGCUG AAUAGCUCGUUUUCAAAGCUGUCCAUCCCAAGAACUUC UCGUUCGCGGUCACCCAGGAGUACAUCAGACCAUAUU CAGAAGGUCACCGUUGAUUGCAAGCAUACGUGUGCAAC GGCUUCCAGAAGUGCGAGCAGCUGCUGAGAGAAUACGG GCAGUUUUGCAGCAAGAUCAACAGGCGCUGCAUGGAGC UAAUUGCGCCAGGACGACUCCGUGCGCAACCUUUUGC CUCUGUAAGUCAUCCAGUCCUCCCAAUCAUCCCGGG AUUCGAGGGGACUUAACCUAGACCUCCUGGAGCCCGU GUCAUACAGCACCGUAGCAGAUCCGCGCGCUCAGCCAU UGAAGAUUCUUGUUCGACAAGGUCACCAUCGCCGAUCC GGGCUACAUGCAGGGAUACGACGACUGUAUGCAGCAGG GACCAGCUCUCCGCGAGGGACCUCAUCUGCGCGCAUACG UGGCCGGUACAAGUGCUGCCUCCUUGAUGGAUGUG AACAUUGGAGGCGCUUAUACUUCGUCCUUGCUCGGCUCU AUCGCCGCGUGGGGUGGACCGCCGGCUGUCCUCCUUC GCCGCUAUCCCUUUGCAUAAUCCAUUUUUAACCGGCUC AACGGCUGGGCAUUAUCUACAAGUCCUGUCGGAGAAC CAGAAGUUGAUCCGCAACAAAGUUCUUAUAGGCCUGGG GGCCAUGCAGACUGGAUUCACUACGACUAAACAGCGUU CCAGAAGGUCCAGGACGCUUGUAAACAACGCCACAGGC GCUUCUCAAAGCUGGCCUCCGAACUCAGCAACACCUUCGG AGCCAUCAGCGCAUCGAUCGGUGACAUAUUUAGCGGCU GGACGUGCUGGAGCAGGACGCCAGAUCCAGCGCCUUAU CAACGAGCAGGUGACCAACCUUGAUGCCUUCGUGGCACA ACAGCUGGUCGCGAGCGAAUCAGCGGCACUUUCCGCCCA ACUCGCCAAGGACAAGUCAAAGAUUGCGUAGGAGCCCA GUCCAAGAGGUCCGGUUUCUGCGGUCAAGGAACCCAUAU UGUGUCCUUCGUCGUAACGCGCCCAACGGUCUGUACUU UAUGCACGUCGGCUACUACCGAGCAUUAUUAUCGAAGU GGUGUCCGCCUACGGCUGUGCGAUGCCGCUAACCCAC UAAUGUAUUGCCCUUGAAGCGGAUUAUUUAUUAAGA CCAACAAACCCGCAUUGGAGCAUAGGUCAUACACCG GUUCGUCUUUACGCGCCGAGCCCAUCACUUCACUGA ACACCAAUACGUGGCUCGCAAGUAGACCUACCAAGACA UCUCCACCAAUUGCCGCGCGCUGCUCGGAAACAGCA CCGGAUUGAUUUCCAAGAUACUGGACGAUUCUUC	67

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	AAGAACGUGUCCACUCCAUUCCCAACUUCGGAAGCCUG ACACAGAUAACACCAACCCUUCUCGACCUAGCCUACGAG AUGCUGAGCCUUAACAAGUGGUCAGGCCUGAAGCAG AGCUACAUCGACCGUAGGAGCUGGGCAACUUAUACCUAC UACAACAAGUGGCCGACAAAGUUGAGGAGAUUCUGUC GAAAAUCUACCACAUUGAAAACGAGAUCCGCAAGAUCA AGAAGCUUAUCGGCGAAGCC	
MERS_S0_Full-length Spike protein (nucleotide, codon optimized)	AUGGAAACCCUGCCCAGCUGCUGUUCUGCUGCUGCUG UGGCUGCCUGAUACCACCGGCAGCUAUGUGGACGUGGGC CCCGAUAGCGUGAAGUCCGCCUGUAUCGAAGUGGACAU CAGCAGACCCUUUUUGGACAAAGACCCUGGCCAGACCCAU GACGUGUCCAGGCCGACGGCAUCAUCUAUCCACAAGGC CGGACCUACAGCAACAUACCAUUAUCCUACAGGCCUG UUCCCAUAUACAAGGCGACACGGCGAUUAUGACGUGUAC UCUGCCGGCCACGCCACCGGCACCAACCCAGAAACUG UUCGUGGCCAAUACAGCCAGGACGUGAAGCAGUUCGCC AACGGCUUCGUCGUGCGGAUUGGCCCGCUGCCAAUAGC ACCGGCACAGUGAUCAUCAGCCCGACACCCAGAGCCACC AUCCGGAAGAUUACCCCGCCUUCUUGCUGGGCAGCUC GUGGGCAAUUUACGACGGCAAGAUUGGCCCGGUUCUU CAACCACACCCUGGUGCUGCUGCCGAUGGCUGUGGCAC ACUGCUGAGAGCCUUCUACUGCAUCCUGGAACCCAGAAG CGGCAACACUGCCUGCCGGCAUAGCUACACAGCUU CGCCACCUACCAACACCCGCCACCGAUUGCUCGACGG CAACUACAACCGGAACGCCAGCCUGAACAGCUUCAAGA GUACUUAACCCUGCGGAACUGCACCUCUAGUACACCUA CAAUAUACCCGAGGACGAGAUCCUGGAUUGGUUCGGCA UACCCAGACCCGCCAGGGCGUGCAUCCUGUACAGCAGCA GAUACGUGGACCUUACGGCGGCAACUUGUCCAGUUU GCCACCCUGCCCGUGUACGACCAUACAGUACUACAGC AUCAUCCCCCAGCAUCCGUGUCCAUACAGCGACAGCA AAAGCCUGGGCCGCCUUCUACGUGUACAGCUGCAGCCC CUGACCUUCUGCUGGACUUCAGCGUGGACCGCUACAU AGACGGGCCAUAGCUGCGGCCUACAGCCUGAGCCAG CUGCAGCUGCUCUACGAGAGCUUCGACGUGGAAAGCGGC GUGUACAGCGUGUCCAGCUUCGAGGCCAAGCUAGCGGC AGCGUGGUGGAACAGGCUAGGGCGUGGAUUGCGACUU CAGCCCUUCUGCUGAGCGGCACCCUCCCCAGGUGUACAA CUUACAGCGCGUGGUGUUACCAACUGCAAUUAACAACCU GACCAAGCUGCUGAGCCUGUUCUCCGUGAACGACUUCAC CUGUAGCCAGAUACGCCUGCCGCCAUUGCCAGCAACUG CUACAGCAGCCUGAUCCUGGACUACUUCAGCUACCCCU GAGCAUGAAGUCCGAUUCGAGCGUGUCUCCGCCGGACC CAUCAGCCAGUUAACUACAAGCAGAGCUUCAGCAACCC UACCUGCCUGAUUUCGGCCACCGUGCCCAACAUCUGAC CACCAUCACCAAGCCUGAAGUACAGCUACUACAACA GUGCAGCAGACUGCUGUCGACGACCGGACCGAAGUGCC CCAGCUUGUAGACGCCAACAGUACAGCCCUUGCGUGUC CAUCGUGCCAGCACCGUGUGGAGGACGGCGACUACUA CAGAAAGCAGCUGAGCCCCUGGAAGCGGGCGAUGGCU GGUGGCUUCUGGAAGCACAGUGGCCAUGACCGAGCAGCU GCAGAUUGGGCUUUGGCAUACCGUGCAGUACGGCACCGA CACCAACAGCGUGUGCCCCAAGCUGGAUUCGCCAAUGA CACCAAGAUCCGACGACGUGGGAAACUGCGUGGAUA CUCCCUUGAUUGCGUGUCGAGCGGGCGUGUUCAGAA UUGCAGCAGUGGGAGUGCGGCAGCAGAGAUUCUGU ACGAUGCCUACCAAGAACCUUGGGCUACUACAGCGACG ACGGCAAUUAUACUGCCUGCGGGCUUGUGUCCGUGC CCGUGUCCUGAUUACGACAAAGAGACAAAGACCCACG CCACAUCUGUUCGGCUCCGUGGCCUGCGAGCAUACAGCU CCACCAGAGCCAGUACUCCCGUCCACCGGUCUCCAGC UGAAGCGGAGAGAUAGCAUCCUACGGCCCCUGCAGACAC CUGUGGGAUGUGUGCUGGGCCUCUGUAACAGCUCCUGU UUGUGGAAGAUUGCAAGCUGCCCCUGGGCAGAGCCUGU GUGCCUGCCAGAUACCCUAGCACCCUGACCCUAGAA GCGUGCGCUUCUGCCCGGCAAAUGCGGCUUGGCCUA UCGCCUUAUAUACCCAUCCAGGUGGACAGCUGAACU CCAGCUACUUAAGCUGAGCAUCCCAACCAUUCAGCU UCGGCGUGACCCAGGAGUACAUCCAGACCAACAUCAGA AAGUGACCGUGGACUGCAAGCAGUACGUGCAACGGC UUUCAGAAUGCGAACAGCUGCUGCGGAGUACGGCCAG UUCUGCAGCAAGAUCAACAGGCCUUGACGGCGCAAC CUGAGACAGGAUGACAGCUGCGGAACUGUUCGCCAGC GUGAAAGCAGCCAGUCCAGCCCCAUACUCCUGGCUUC GGCGGCGACUUAACUAGACCUUGCUGGAACUGUGUCC AUCAGCACCGGCUCCAGAGCGCCAGAUCCGCCAUCGAG	68

TABLE 10-continued

Strain	Nucleic Acid Sequence	SEQ ID NO:
	GACCGUGUUCGACAAAGUGACCAUUGCCGACCCCGGC UACAUGCAGGGCUACGACGAUUGCAUGCAGCAGGCCCCA GCCAGCGCCAGGAUCUGAUUCUGUGCCCAGUAUGUGGCC GGCUACAAGGUGCGUGCCCCCGAUGGACGUGAACAU GAAGCCGCCUACACCUCCAGCCUGUGGGUCUAUUGCU GGCGUGGGAUGGACAGCCGGCCUGUCUAGCUUUGCCGCC AUCCUUUCGCCAGAGCAUCUUCUACCGGCGAACCGGC GUGGGCAUCACACAAGGUGUGAGCGAGAACCAGAA GCUGAUCGCCAACAGUUUAACCGGCACUGGGCGCCAU GCAGACCGGCUUACACCAACCAACGAGGCCUUCAGAAA GGUGCAGGACGCGUGAACACAACGCCAGGCUUCUGAG CAAGCUGGCCUCCGAGCUGAGCAUACCUUGCGGCCAU CAGCGCCUCCAUCCGCGCAUCAUCCAGCGGCGUGAGCU GUGGGAACAGGACGCCAGAUAGCGGCGUACUACCGG CAGACUGACCAACCCUGAACGCCUUCUGGCGACAGCAGCU CGUGCGGAGCGAAUCUGCCGCUUCUGUCUAGCUGGC CAAGGACAAGUGAACGAGUGCGUGAAGGCCAGUCCA AGCGGAGCGGCUUUGUGGCCAGGCGACCCCAUCUGUGU CCUUCGUCGUGAAUGCCCCAACGGCCUGUACUUUAUGC ACGUGGGCUAUUACCCAGCAACCAUCAGGUGUGUGU CCGCCUAUGGCCUGUGCGACGCCGCCAAUCCUACCAACU GUAUCGCCCGGUGAACGGCUACUUAUCAAGACCAACA ACACCCGGAUCGUGGACGAGUGGUCUACACAGGCGACA GCUUCUACGCCCGGAGCCAUACCUCCUGAACACCA AAUACGUGGCCCGCAAGUGACAUAACAGAACAUCCUCCA CCAACCGGCCCUCCACUGCUGGGAAAUUCCACCGGCA UCGACUUCAGGACGAGCUGGACGAGUUCUUAAGAAGC UGUCCACCUCCAUCCCAACUUCGGCAGCUGACCCAGA UCAACACCAUCUGCUGGACCGUACUACGAGAUGCUGU CCUGCAACAGGUCUGAAGGCCUGAACGAGAGCUACA UCGACUGAAAGAGCUGGGGAACUACCCUACUACAACA AGUGGCCUUGGUACAUAUUGGCGGCUUAUCGCGGCC UGGUGGCCUGGCCUGUGCGUGUUCUUAUCCUGUGCU GCACCGGCGCGGCCAACAUUGCAUGGGCAAGCUGAAU GCAACCGGUGCGCGACAGAUACGAGGAUACGACCUGG AACCUCACAAAGUGCAUGUGCAC	

TABLE 11

Betacoronavirus Amino Acid Sequences		
Strain	Amino Acid Sequence	SEQ ID NO:
gb KJ156934.1 : 21405-25466 Middle East respiratory syndrome coronavirus isolate Riyadh_14_2013, spike protein (amino acid)	MIHSVFLMFLLTPTESYVDVGPDSVKSACIEVDIQTFDDKT WPRPIDVSKADGIIYPQGRYSNITITYQGLFPYQGDHDMY VYSAGHATGTTpQKLFVANYSDVKQFANGFVVRIGAAANS TGTVIIISPSTSATIRKIYPAPMLGSSVGNFSDGKMRFFNHTL VLLPDGCGTLRLAPYCIILEPRSGNHCPAGNSYTSFATYHTPA TDCSDGNYNRNASLNSFKEYFNLRNCTFMYTYNITTEDEILEW FGITQTAQGVHLFSSRYVDLYGGNMFQFATLPVYDTIKYYSII PHSIRSIQSDRKAWAAFVYKLQPLTLFLDPSVDGYIRRAIDC GFNDLSQLHCSYESFDVESGVYSVSSFEAKPSGSVVEQAEV ECDPSPLLSGTTPQVYNFKRLVFTNCNYNLTKLKLSLFSVNDFT CSQISPAAIASNCYSSLILDYFSYPLSMKSDLVSSAGPISQFN YKQSFNPTCLILATVPHNLTITIKPLKYSYINKCSRLLSDDR EVPQLVNANQYSPCVSIVPSTVWEDGDYRQKLSPLEGGGW LVASGSTVAMTEQLQMGFGITVQYGTDTNSVCPKLEFANDT KIASQLGNCVEYSLYGVSGRGVFNCTAVGVRQRFVYDA YQNLVGYSDDGNYCYLRACVSVPVSVIYDKETKTHATLFG SVACEHISSTMSQYSRSTRMLKRRDSTYGPLQTPVGCVLGL VNSSLFVEDCKLPLGQSLCALPDTPTSTLTPRSVRSVPGEMRLA SIARNHPIQVDQLNSSFYKLSIPTNFSGVVTQEIQTIIQKVT DCKQYVCNGFQKCEQLLREYGFQCSKINQALHGANLRQDD VRNLFASVKSSQSPPIIPGFGDFNLTLLEPVSTSTGSRARS EDLLFDKVTIADPGYMQYDDCMQGPASARDLICAQYVA GYKVLPLMDVNMEEAAYTSSLLGSIAGVGWTAGLSPPAIPF AQSFYRLNGVGITQQVLSNQKLIANKFNQALGAMQTGFTT TNEAPrKVQDAVNNAQALSKLASELSNTFGAISASIGDIIQR LDVLEQDAQIDRLINGRLTTLNAPVQQLVRSESAALSAQLA KDKVNECVKAQSKRSGFCQGTHIVSFVVPNGLYFMHV GYPSNHIEVVSAYGLCDAANPTNCIAPVNGYFIKTNNTTRIV DEWSTGSSFYAPEPITSLNTKYVAPQVTYQNIQNTNLPPLLG NSTGIDFQDELDEFKVNSTIPNFGSLTQINTLLDLTYEMLS	24

TABLE 11-continued

Betacoronavirus Amino Acid Sequences		
Strain	Amino Acid Sequence	SEQ ID NO:
	LQQVVKALNESYIDLKELGNYTYYNKWPWYIWLGFIAGLVA LALCVFFILCCTGCGTNCMGKCLKNRCDDRYEEDLEPHKV HVH	
MERS S FL SPIKE 2cEMC/2012 (XbaI change (T to G)) (amino acid)	MIHSVFLLMFLLTPTESYVDVGPDSVKSACIEVDIQQTFFDKT WPRPIDVSKADGIIYPQGRITYSNITITYQGLFPYQGDHGD VYSAGHATGTTTPQKLFVANYSQDVKQFANGFVVRIGAAANS TGTVIIISPSTSATIRKIYPAFMLGSSVGNFSDGKMGRFFNHTL VLLPDGCGTLLRAFYCILEPRSGNHCPAGNSYTSFATYHTPA TDCSDGNYNRNASLNSFKEYFNLRNCTFMYTYNITEDEILEW FGITQTAQGVHLFSSRYVDLYGGNMFQFATLPVYDTIKYYSII PHSIRSIQSDRKAWAAFVYKQLPLTFLLDVSDGYIRRAIDC GFNDLSQLHCSYESFDVESGVYSVSFEAKPSGSVVEQAEGV ECDPSPLLSGTTPQVYNFKRLVFTNCNINLTCLLSLFSVNDFT CSQISPAAIASNCYSSLILDYFSYPLSMKSDLSVSSAGPISQFN YKQSFSPNPTCLILATVPHNLTTITKPLKYSYINKCSRLSDDR EVPQLVNANQYSPCVSIVPSTVWEDGDYRKLSPLEGGGW LVASGSTVAMTEQLQMGFGITVQYGTDTNSVCPKLEFANDT KIASQLGNCVEYSYLVGVSGRGVFQNTAVGVRQQRFFVYDA YQNLVGYSDDGNYCLRACVSVPVSVIYDKETKTHATLFG SVACEHISSTMSQYSRSTRMLKRRDSTYGLQTPVGCVLGL VNSSLFVEDCKLPLGQSLCALPDTPTSTLTPRSVRSVPGEMRLA SIAFNHPIQVDQLNSSFYKLSIPTNFSFGVTQEYIQTIIQKVT DCKQYVCNGFQKCEQLLREYGFQCSKINQALHGANLRQDDS VRNLFASVKSSQSSPIIPGFGGDFNLTLLEPVSISTGSRARS EDLLFDKVTIADPGYMQGYDDCMQQGPASARDLCAQYVA GYKVLPLMDVMNEAAYTSSLLGSIAGVGWTAGLSSFAAIPF AQSIFYRLNGVGITQQVLSQKLIANKFNQALGAMQTGFTT TNEAFQKVQDAVNNAQALSCLASELSNTFGAISASIGDIIQR LDVLEQDAQIDRLINGRLTTLNFAVQAQQLVRSESAALSAQLA KDKVNECVKAQSKRSGFCGQGTHIVSFVVNAPNGLYFMHV GYPSNHIIEVVSAYGLCDAANPTNCIAPVNGYFIKTNNTRIV DEWYSYTGSSFYAPEPITSLNTKYVAPQVTYQNI STNLPPPLG NSTGIDFQDELDEFKKNVSTSI PNFGSLTQINTTLTLYEMLS LQQVVKALNESYIDLKELGNYTYYNKWPWYIWLGFIAGLVA LALCVFFILCCTGCGTNCMGKCLKNRCDDRYEEDLEPHKV HVH	25
Novel_MERS_S2_ subunit trimeric vaccine (amino acid)	MIHSVFLLMFLLTPTESDCKLPLGQSLCALPDTPTSTLTPRSVR SVPGEMRLASIAFNHPIQVDQLNSSFYKLSIPTNFSFGVTQEYI QTTIQKVTVDCKQYVCNGFQKCEQLLREYGFQCSKINQALH GANLRQDDSVRNLFASVKSSQSSPIIPGFGGDFNLTLLEPVSIS TGSRARSIAEDLLFDKVTIADPGYMQGYDDCMQQGPASAR DLICAQYVAGYKVLPLMDVMNEAAYTSSLLGSIAGVGWTA GLSSFAAIPFAQSIFYRLNGVGITQQVLSQKLIANKFNQAL GAMQTGFTTTNEAFQKVQDAVNNAQALSCLASELSNTFG AISASIGDIIQRDVLQDAQIDRLINGRLTTLNFAVQAQQLVR ESAALSAQLAKDKVNECVKAQSKRSGFCGQGTHIVSFVVNA PNGLYFMHVGYPSNHIIEVVSAYGLCDAANPTNCIAPVNGY FIKTNNTRIVDEWYSYTGSSFYAPEPITSLNTKYVAPQVTYQNI STNLPPPLLGSTGIDFQDELDEFKKNVSTSI PNFGSLTQINTTL LDLTYEMLSQVVKALNESYIDLKELGNYTYYNKWPDKIE EILSKIYHIENEIARIKKLIGEA	26
Isolate A1- Hasa_1_2013 (NCBI accession #AGN70962)	MIHSVFLLMFLLTPTESYVDVGPDSVKSACIEVDIQQTFFDKT WPRPIDVSKADGIIYPQGRITYSNITITYQGLFPYQGDHGD VYSAGHATGTTTPQKLFVANYSQDVKQFANGFVVRIGAAANS TGTVIIISPSTSATIRKIYPAFMLGSSVGNFSDGKMGRFFNHTL VLLPDGCGTLLRAFYCILEPRSGNHCPAGNSYTSFATYHTPA TDCSDGNYNRNASLNSFKEYFNLRNCTFMYTYNITEDEILEW FGITQTAQGVHLFSSRYVDLYGGNMFQFATLPVYDTIKYYSII PHSIRSIQSDRKAWAAFVYKQLPLTFLLDVSDGYIRRAIDC GFNDLSQLHCSYESFDVESGVYSVSFEAKPSGSVVEQAEGV ECDPSPLLSGTTPQVYNFKRLVFTNCNINLTCLLSLFSVNDFT CSQISPAAIASNCYSSLILDYFSYPLSMKSDLSVSSAGPISQFN YKQSFSPNPTCLILATVPHNLTTITKPLKYSYINKCSRLSDDR EVPQLVNANQYSPCVSIVPSTVWEDGDYRKLSPLEGGGW LVASGSTVAMTEQLQMGFGITVQYGTDTNSVCPKLEFANDT KIASQLGNCVEYSYLVGVSGRGVFQNTAVGVRQQRFFVYDA YQNLVGYSDDGNYCLRACVSVPVSVIYDKETKTHATLFG SVACEHISSTMSQYSRSTRMLKRRDSTYGLQTPVGCVLGL VNSSLFVEDCKLPLGQSLCALPDTPTSTLTPRSVRSVPGEMRLA SIAFNHPIQVDQLNSSFYKLSIPTNFSFGVTQEYIQTIIQKVT DCKQYVCNGFQKCEQLLREYGFQCSKINQALHGANLRQDDS VRNLFASVKSSQSSPIIPGFGGDFNLTLLEPVSISTGSRARS	27

TABLE 11-continued

Betacoronavirus Amino Acid Sequences		
Strain	Amino Acid Sequence	SEQ ID NO:
	EDLLFDKVTIADPGYMQGYDDCMQQGPASARDLI CAQYVA GYKVLPPPLMDVNMEAAAYTSSLLGSIAGVGWTAGLSSFAAIPF AQSI FYRLNGVGITQQVLS ENQKLIANKFNQALGAMQTGFTT TNEAFRKVD AVNNNAQALS KLASELSNTFGAISASIGDI IQR LDVLEQDAQIDRLINGRLTTLN AFVAQQLVRSESAALSAQLA KDKVNECVKAQSKRSGFCGQGTHIVSFVVNAPNGLYFMHV GYPSNHIEVVSAYGLCDAANPTNCIAPVNGYFIKTNNTTRIV DEWSYTGSSFYAPEPITSLNTKYVAPHVTYQNI STNLPPPLL G NSTGIDFQDELDEFFKNVSTSI PNFGSLTQINTTLLDLTYEMLS LQQVV KALNESYIDLKELGNYTYYNKWPWYIWLGFIAGLVA LALCVFFILCCTGCGTNCMGK LKCNRC DRYEEDYLEPHKV HVH	
Middle East respiratory syndrome coronavirus S protein UniProtKB- R9UQ53	MIHSVFLLMFLLTPTESYVDVGPDSVKSACIEVDIQQTFDDKT WPRPIDVSKADGIIYPQGRYSNITITYQGLFPYQGDHGD MY VYSAGHATGTT PQKLFVANYSDVKQFANGFVVRIGAAANS TGTVIIISPSTSATIRKIYPAFMLGSSVGNFSDGKMGRFFNHTL VLLPDGCGTLLRAFYCILEPRSGNHCPAGNSYTSFATYHTPA TDCSDGNYNRNASLNSFKEYFNLRNCTFMYTYNITEDEILEW FGITQTAQGVHLFSSRYVDLYGGNMQFATLPVYDTIKYYSII PHSIRSIQSDRKAWAAFVYVKLQPLTFLDFSDVGYIRRAIDC GFNDLSQLHCSYESFDESFGVSVSSFEAKPSGSVVEQAEV ECDFSPLLSGTPPQVYNFKRLVFTNCNYNLT KLLSLFSVNDFT CSQISPAAIASNCYSSLLIDYFSYPLSMKSDLSVSSAGPISQFN YKQSF SNPTCLILATVPHNLTITKPLKYSYINKCSRLSDDR T EVPQLVNANQYSPCVSIVPSTVWEDGDYRQKLSPLEGGGW LVASGSTVAMTEQLQMGFGITVQYGTDTNSVCPKLEFANDT KIASQLGNCVEYSLYGVSGRGVFNCTAVGVRQQRFFVYDA YQNLVGYSDDGNYCYLRACVSVPVSVIYDKETKTHATLFG SVACEHISSTMSQYSRSTRSMLKRRDSTYGPLQTPVGCVLGL VNSSLFVEDCKLPLGQSLCALPDTPTSLTPRSVRSVPGEMLRA SIAPNHP IQVDQLNSSYFKLSIPTNFSGVTQEQYIQTTIQKVTV DCKQYVCNGFQKCEQLLREYGFQCSKINQALHGANLRQDDS VRNLFASVKSSQSP IIPGFGGDFNLTLLEPVSI STGSR SARSAI EDLLFDKVTIADPGYMQGYDDCMQQGPASARDLI CAQYVA GYKVLPPPLMDVNMEAAAYTSSLLGSIAGVGWTAGLSSFAAIPF AQSI FYRLNGVGITQQVLS ENQKLIANKFNQALGAMQTGFTT TNEAFRKVD AVNNNAQALS KLASELSNTFGAISASIGDI IQR LDVLEQDAQIDRLINGRLTTLN AFVAQQLVRSESAALSAQLA KDKVNECVKAQSKRSGFCGQGTHIVSFVVNAPNGLYFMHV GYPSNHIEVVSAYGLCDAANPTNCIAPVNGYFIKTNNTTRIV DEWSYTGSSFYAPEPITSLNTKYVAPHVTYQNI STNLPPPLL G NSTGIDFQDELDEFFKNVSTSI PNFGSLTQINTTLLDLTYEMLS LQQVV KALNESYIDLKELGNYTYYNKWPWYIWLGFIAGLVA LALCVFFILCCTGCGTNCMGK LKCNRC DRYEEDYLEPHKV HVH	28
Human SARS coronavirus (SARS-CoV) (Severe acute respiratory syndrome coronavirus) Spike glycoprotein UniProtKB- P59594	MFIFLLFLTLTSGSDLDRCTTFDDVQAPNYTQHTSSMRGVY PDEIFRSDTLYLTDQLFLPFYSNVTGFHTINH TFGNPV I PKDG IYFAATEKSNVVRGWVFGSTMNKQSQV I IINNSTNVIRAC NFELCDNPFFAVSKPMGTQHTMIFDNFNCTFEYI SDAFSLD VSEKSGNPKHLREFVFNKNDGFLVYKGYQPIDVVRDLPSGF NTLKP I FKLPLGINITNFRAILTAFSPAQDIWGTSAAYFVGYL KPTTFMLKYDENGITIDAVDCSQNPLAELKCSVKSFEIDKGI YQTSNFRVVP SGDVVRFPNITNLCPFGEVFNATKFP SVYAW E RKKISNCVADYSVLNSTFFSTFKCYGVSATKLN DLCF SNVY ADSFVVKGDDVRQIAPGQTGVIADYN YKL PDDFMGCVLAW NTRNIDATSTGNYNKYRYLRHGLRPFERDISNVPFSPDGK PCTPPALNCYWPLNDYGFYTTTGIGYQPYRVVLSFELLNAP ATVCGPKLSTD LIKNQCVNFNFNGLTGTGVLTPSKRFQPFQ QFGRDVSDFDTSVRDPKTSEILDISP CSFGGVSVITPGTNASSE VAVLYQDVNCTDVSTAIHADQLTPAWRIYSTGNNV FQTQAG CLIGA EHVDTSYECDIPIGAGICASYHTVSLLRSTSQKSIVAYT MSLGADSSIAYSNNITAIPTNFSISITTEVMFVPSMAKTSVDCN MYICGDS TE CANLLQYGSFCTQLNRALSGIAAEQDRNTR EV FAQVKQMYKTPTLKYFGGFNFSQLPDPLKPTKR SFIEDLLFN KVTLADAGFMKQYGECLGDINARDL ICAQKFNGLTVLPPLL TDDMIAAYTAALVSGTATAGWTFGAGAALQIPFAMQ MAYR FNGIGVTQNVLYENQKQIANQFNKAISIQIESLT TTSTALGKL QDVVNQNAQALNTLVKQLSSNFGAISSVLNDILSR LDKVEAE VQIDRLITGR LQSLQTYVTQQLIRAAEIRASANLAATKMSEC VLGQSKRVDFCGKYHLSMFPQAAPHGVVFLHVTVVPSQER NFTTAPAI CHEGKAYFPREGVVFVNGTSWFTQRNFSPQII TT DNTFVSGNCDVVIGIINN TVYDPLQPELDSFKEELDKYFKNH TSPDVLGDISGINASVNIQKEIDRLNEVAKNLNESLIDLQE	29

TABLE 11-continued

Betacoronavirus Amino Acid Sequences		
Strain	Amino Acid Sequence	SEQ ID NO:
	LGKYEQYIKWPWYVWLGFIAGLIAIVMTILLCCMTSCCSCL KGACSCGSCCKFDEDDSEPVLKGVKLHYT	
Human coronavirus OC43 (HCoV-OC43) Spike glycoprotein UniProtKB-P36334	MFLILLISLPTAFAVIGDLKCTSDNINDKDTGPPPISTDTVDVT 30 NGLGTYVYVLDREVYLNNTTLFLNGYYPSTGSTRNMAKGSVL LSRLWFKPPFLSDFINGIFAKVKNTKVIKDRVMYSEFPITIGS TFVNTSYSVVVQPRITNSTQDGDNKLQGLLEVSVCCQYNMCE YPQTI CHPNLGNHRKELWHLDTGVVSCLYKRNFYDYNAD YLYFHFYQEGGTFYAYFTDTGVVTKFLFNVLGMALSHYYV MPLTCNSKLTLEYWVTPLTSRQYLLAFNQDGIIFNAEDCMSD FMSEIKCKTQSIAPPTGVYELNGYTVQPIADVYRRKPNLPNC NIEAWLNDKSVSPNLNWERKTFSCNCFNMSSLSMFIQADSFT CNNIDAAKIYGMCFSSITIDKFAIPNGRKVDLQGLNGLYLSQF NYRIDTTATSCQLYLNLPANVSVSFRPNPSTWNKRFGFIEDS VFKPRPAGVLTNHDVVYAQHCFAKPNFCPCKLNGSCVGS PGKNNGIGTCTPAGTNYLTCDNLCTPDPIITFTGTQKCPQTKSL VGIGEHCSGLAVKSDYCGGNSCTCRPQAFGLWSADSCLOGD KCNIFANFILHDVNSGLTCTSDLQKANTDIIILGVCVNYDLYGI LGQGI FVEVNATYYNSWQNLLYDSNGNLYGPRDYIINRTFMI RSCYSGRVSAAFHANSSEPALLFRNI KCNYVFNNSLTRQLQPI NYFDSYLGCVVNAYNSTAISVQTCDLTVGSGYCVDYSKNRR SRGAI TTGYRFTNPEPPTVNSVNDLSLEPVGGLYEIQIPSEPTIG NMVEFIQTSSPKVTIDCAAFVCGDYAACKSQLVEYGSFCDNI NAILTEVNELLDTTQLQVANSLMNGVTLSTKLKDGVNFNVD DINFSPLVGLGSECSKASSRAIEDLLFDKVKLSDVGFVEAY NNCTGGAEIRDLCVQSYKGIKVLPLLSLENQISGYTLAATSA SLFPWPWTAAGVPFYLNVQYRINGLGV TMDVLSQNQKLIAN AFNNALYAIQEGFDATNSALVKIQAVVNANA EALNNLLQQL SNRFGAISASLQEILSRDLAEAEAQIDRLINGRLTALNAYVS QQLSDSTLVKFSAAQAMEKVNCEKVSQSSRINFCGNGNHIIS LVQNAPYGLYFIHFSYVPTKYVTVARVSPGLCIAGDRGIAPKS GYFVNVTWMTYTGSGYYPPEPI TENNVVMSTCAVNYTK APYVMLNTSIPNLPDFKEELDQWFKNQTSVAPDLSLDYINVT FLDLQVEMNRLQEAIKVLNQSYINLKDIGTYEYVVKWPWYV WLLICLAGVAMLVLLFFICCTGCGTSCFKKCGGCCDDYTG YQELVIKTSHDD	
Human coronavirus HKU1 (isolate N5) (HCoV-HKU1) Spike glycoprotein UniProtKB-Q0ZME7	MFLIIFILPTTLAVIGDFNCTNSFINDYNKTIPIRISEDDVDVSLG31 LGTYVYVLRVYLNNTLLFTGYFPKSGANFRDLALKGSYILST LWYKPPFLSDFNNGIFSKVKNTKLYVNNTLYSEFSTIVIGSVF VNTSYTIVVQPHNGILEITACQYTMCEYPHTVCKSKGSIRNES WHIDSSEPLCLFKKNFTYNVSADWLYFHFYQERG VFYAYYA DVGMPTTFLFSLYLGITILSHYYVMPLTCAINSSNTDNETLEY WVTPLSRRQYLLNFDEHGVITNAVDCSSFLSEIQCTQSFAP NTGVYDLSGFTVKPVATVYRRIPNLPDCIDNWLNNVSVSP LNWERRIFSNCNFNLS TLLRLVHVSFSCNNLDKSKIFGSCFN SITVDKFAIPNRRRDDLQLGSSGFLQSSNYKIDISSSSCQLYYS LPLVNVTTNNFNPSSWNNRYGFGSFNLSYDVVYSDHCFSVN SDFCPCADPSVVNSCAKSKPPSAICPAGTKYRHCDLDTTLVY KNWCRCCLPDP ISTYSPNTCPQKVVVGIGEHCPGLGINEE KCGTQLNHSSCFCSFDAFLGWSFDS CINSNNRCNIFSNFIENGIN SGTTCSDNLLYSNTEISTGVCVNYDLYGITGQIFKEVSAAY YNNWQNLLYDSNGNII GFKDFLTNKTYTILPCYSGRVSAAFY QNSSSPALLYRNKCSYVLNNISFISQPFYFDSYLGCVLNAV LTSYSVSSCDLRMGSGFCIDYALPSSRRKRGISSPYRFVTFEP FNVSFVNDVSVETVGGLFEIQIPTNFTIAGHEEFIQTSSPKVTIDC SAFVCSNYAACHDLLSEYGTFCDNINSILNEVNDLLDITQLQV ANALMQGVTLSSNLTNLHSDVDNIDFKSLGGLGSGCQSSS RSLLEDLLFNKVKLSDVGFVEAYNNCTGGSEIRDLLCVQSFN GIKVLPPILSETQISGYTTAATVAAMFPWWSAAAGVPFSLNVQ YRINGLV TMDVLNKNQKLIANAFNKALLSIQNGFTATNSAL AKIQSVVNANAQALNSLLQQLFNKFGAISSSLQEILSRDLNLE AQVQIDRLINGRLTALNAYVSQQLSDITLIKAGASRAIEKVNE CVKQSQSPRINFCGNGNHLISLVQNAPYGLLFIHFSYKPTSPKT VLVSPGLCLSGDRGIAPKQGYFIKQNDSSWMTGSSYYPEPIS DKNVVFMNSCSVNFTKAPFIYLNNSIPNLSDFEAEELSLWFKN HTSIAPNLTFNSHINATFLDLYEEMNVIQESIKSLNSSFNLKEI GTYEMYVKWPWYIWLLIVLFII FLMLFFICCTGCGSACFSK CHNCCDEYGGHNDFVIKASHDD	
Novel_SARS_S2	MFIFLLFLTLSGSDLDRALSGIAAEQDRNTREVFQVQKQMY 32 KTPTLKYFGGFNFSQLPDPLKPTKRSFIEDLLFNKVTLADAG FMKQYGECLGDINARDLICAQKFNGLTVLPPLLTDDMIAAYT AALVSGTATAGWTFGAGAAQIPFAMQAMAYRFNGIGVTQN VLYENQKQIANQFNKAISQIQESLTTTSTALGKLQDVVNQNA	

TABLE 11-continued

Betacoronavirus Amino Acid Sequences		
Strain	Amino Acid Sequence	SEQ ID NO:
	QALNTLVKQLSSNFGAISSVLNDILSRLDKVEAEVQIDRLITG RLQSLQTYVTQQLIRAAEIRASANLAATKMSECVLGQSKRV DFCGKGYHLSFPQAAPHGVVFLHVTYVPSQERNFTTAPAIC HEGKAYFPREGVVFVFNQTSWFIQNRNFFSPQIITDNTFVSGN CDVVIGIINNNTVYDPLQPELDSFKEELDKYFNHTSPDVLG DISGINASVVNIQKEIDRLNEVAKNLNESLIDLQELGKYEYI KWPWYVWLGFIAGLIAIVMTILLCCMTSCCSCCLKGACSCGS CCKFDEDDSEPVKGVKLHYT	
Novel_MERS_S2	MIHSVFLLMFLLTPTESDCKLPLGQSLCALPDTPSTLTPRSVR SVPGEMRLASIAFNHPIQVDQLNNSYFKLSIPTNFSFGVTQEYI QTTIQKVTVDCKQYVCNGFQKCEQLLREYGFCSKINQALH GANLRQDDSVRNLFASVKSSQSSPIIPGFGGDFNLTLLEPVSIS TGSRARSASIEDLLFDKVTIADPGYMQGYDDCMQQGPASAR DLICAQYVAGYKVLPLMDVNMEAAYTSSLGSIAGVGWTA GLSSFAAIPFAQSI FYRLNGVGITQQVLSENQKLIANKFNQAL GAMQTGFTTTNEAPQKVQDAVNNAQALSKLASELSNTFG AISASIGDIIQRLDVLEQDAQIDRLINGRLTTLNAFVAQQLVRS ESAALSAQLAKDKVNECVKAQSKRSGFCGQGTHIVSFVVNA PNGLYFMHVGYPSNHIEVVSAYGLCDAANPTNCIAPVNGY FIKTNNTRIVDEWSTYTGSSFYAPEPITSLNTKYVAPQVTYQNI STNLPPPLLGNSTGIDFQDELDEFFKNVSTIPNFGSLTQINTTL LDLTYEMLSLQQVVKALNESYIDLKELGNITYYNKWP	33
Novel_Trimeric_SARS_S2	MFIFLLFLTSTSGSDLDRALSGIAAEQDRNTREVFQVKQMY KTPTLKYFGGFNFSQILPDPLKPTKRSFIEDLLFNKVTLADAG FMKQYGECLGDINARDLCAQKFNGLTVLPPLLDDMIAAYT AALVSGTATAGWTFGAGAAQIPFAMQAMAYRFGIGVTON VLYENQKQIANQFNKAISQIQESLTTTSTALGKLQDVVNQNA QALNTLVKQLSSNFGAISSVLNDILSRLDKVEAEVQIDRLITG RLQSLQTYVTQQLIRAAEIRASANLAATKMSECVLGQSKRV DFCGKGYHLSFPQAAPHGVVFLHVTYVPSQERNFTTAPAIC HEGKAYFPREGVVFVFNQTSWFIQNRNFFSPQIITDNTFVSGN CDVVIGIINNNTVYDPLQPELDSFKEELDKYFNHTSPDVLG DISGINASVVNIQKEIDRLNEVAKNLNESLIDLQELGKYEYI KWPWYVWLGFIAGLIAIVMTILLCCMTSCCSCCLKGACSCGS CCKFDEDDSEPVKGVKLHYT	34

TABLE 12

Full-length Spike Glycoprotein Amino Acid Sequences (<i>Homo sapiens</i> strains)				
GenBank Accession	Country	Collection Date	Release Date	Virus Name
AFY13307	United Kingdom	Sep. 11, 2012	Dec. 5, 2012	Betacoronavirus England 1, complete genome
AFS88936		Jun. 13, 2012	Sep. 27, 2012	Human betacoronavirus 2c EMC/2012, complete genome
AGG22542	United Kingdom	Sep. 19, 2012	Feb. 27, 2013	Human betacoronavirus 2c England-Qatar/2012, complete genome
AHY21469	Jordan	2012	May 4, 2014	Human betacoronavirus 2c Jordan-N3/2012 isolate MG167, complete genome
AGH58717	Jordan	April 2012	Mar. 25, 2013	Human betacoronavirus 2c Jordan-N3/2012, complete genome
AGV08444	Saudi Arabia	May 7, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Al-Hasa_12_2013, complete genome
AGV08546	Saudi Arabia	May 11, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Al-Hasa_15_2013, complete genome
AGV08535	Saudi Arabia	May 12, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Al-Hasa_16_2013, complete genome
AGV08558	Saudi Arabia	May 15, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Al-Hasa_17_2013, complete genome
AGV08573	Saudi Arabia	May 23, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Al-Hasa_18_2013, complete genome

TABLE 12-continued

Full-length Spike Glycoprotein Amino Acid Sequences (<i>Homo sapiens</i> strains)				
GenBank Accession	Country	Collection Date	Release Date	Virus Name
AGV08480	Saudi Arabia	May 23, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Al- Hasa_19_2013, complete genome
AGN70962	Saudi Arabia	May 9, 2013	Jun. 10, 2013	Middle East respiratory syndrome coronavirus isolate Al- Hasa_1_2013, complete genome
AGV08492	Saudi Arabia	May 30, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Al- Hasa_21_2013, complete genome
AHI48517	Saudi Arabia	May 2, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Al- Hasa_25_2013, complete genome
AGN70951	Saudi Arabia	Apr. 21, 2013	Jun. 10, 2013	Middle East respiratory syndrome coronavirus isolate Al- Hasa_2_2013, complete genome
AGN70973	Saudi Arabia	Apr. 22, 2013	Jun. 10, 2013	Middle East respiratory syndrome coronavirus isolate Al- Hasa_3_2013, complete genome
AGN70929	Saudi Arabia	May 1, 2013	Jun. 10, 2013	Middle East respiratory syndrome coronavirus isolate Al- Hasa_4_2013, complete genome
AGV08408	Saudi Arabia	Jun. 19, 2012	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Bisha_1_2012, complete genome
AGV08467	Saudi Arabia	May 13, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Buraidah_1_2013, complete genome
AID50418	United Kingdom	Feb. 10, 2013	Jun. 18, 2014	Middle East respiratory syndrome coronavirus isolate England/2/2013, complete genome
AJD81451	United Kingdom	Feb. 10, 2013	Jan. 18, 2015	Middle East respiratory syndrome coronavirus isolate England/3/2013, complete genome
AJD81440	United Kingdom	Feb. 13, 2013	Jan. 18, 2015	Middle East respiratory syndrome coronavirus isolate England/4/2013, complete genome
AHB33326	France	May 7, 2013	Dec. 7, 2013	Middle East respiratory syndrome coronavirus isolate FRA/UAE, complete genome
AIZ48760	USA	June 2014	Dec. 14, 2014	Middle East respiratory syndrome coronavirus isolate Florida/USA- 2_Saudi Arabia_2014, complete genome
AGV08455	Saudi Arabia	Jun. 4, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Hafr-Al- Batin_1_2013, complete genome
AHI48561	Saudi Arabia	Aug. 5, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Hafr-Al- Batin_2_2013, complete genome
AHI48539	Saudi Arabia	Aug. 28, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Hafr-Al- Batin_6_2013, complete genome
AIZ74417	France	Apr. 26, 2013	Mar. 10, 2015	Middle East respiratory syndrome coronavirus isolate Hu-France (UAE) - FRA1_1627- 2013_BAL_Sanger, complete genome
AIZ74433	France	May 7, 2013	Mar. 10, 2015	Middle East respiratory syndrome coronavirus isolate Hu-France - FRA2_130569-2013_IS-HTS, complete genome
AIZ74439	France	May 7, 2013	Mar. 10, 2015	Middle East respiratory syndrome coronavirus isolate Hu-France - FRA2_130569-2013_InSpU_Sanger, complete genome
AIZ74450	France	May 7, 2013	Mar. 10, 2015	Middle East respiratory syndrome coronavirus isolate Hu-France - FRA2_130569- 2013_Isolate_Sanger, complete genome
AKK52602	Saudi Arabia	Feb. 10, 2015	Jun. 8, 2015	Middle East respiratory syndrome coronavirus isolate Hu/Riyadh_KSA_2959_2015, complete genome

TABLE 12-continued

Full-length Spike Glycoprotein Amino Acid Sequences (<i>Homo sapiens</i> strains)				
GenBank Accession	Country	Collection Date	Release Date	Virus Name
AKK52612	Saudi Arabia	Mar. 1, 2015	Jun. 8, 2015	Middle East respiratory syndrome coronavirus isolate Hu/Riyadh_KSA_4050_2015, complete genome
AHN10812	Saudi Arabia	Nov. 6, 2013	Mar. 24, 2014	Middle East respiratory syndrome coronavirus isolate Jeddah_1_2013, complete genome
AID55071	Saudi Arabia	Apr. 21, 2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Jeddah_C10306/KSA/2014-04-20, complete genome
AID55066	Saudi Arabia	2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Jeddah_C7149/KSA/2014-04-05, complete genome
AID55067	Saudi Arabia	2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Jeddah_C7569/KSA/2014-04-03, complete genome
AID55068	Saudi Arabia	Apr. 7, 2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Jeddah_C7770/KSA/2014-04-07, complete genome
AID55069	Saudi Arabia	Apr. 12, 2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Jeddah_C8826/KSA/2014-04-12, complete genome
AID55070	Saudi Arabia	Apr. 14, 2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Jeddah_C9055/KSA/2014-04-14, complete genome
AHE78108	Saudi Arabia	Nov. 5, 2013	May 1, 2014	Middle East respiratory syndrome coronavirus isolate MERS-CoV- Jeddah-human-1, complete genome
AKL59401	South Korea	May 20, 2015	Jun. 9, 2015	Middle East respiratory syndrome coronavirus isolate MERS- CoV/KOR/KNH/002_05_2015, complete genome
ALD51904	Thailand	Jun. 17, 2015	Jul. 7, 2015	Middle East respiratory syndrome coronavirus isolate MERS- CoV/THA/CU/17_06_2015, complete genome
AID55072	Saudi Arabia	Apr. 15, 2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Makkah_C9355/KSA/Makkah/2014- 04-15, complete genome
AHC74088	Qatar	Oct. 13, 2013	Dec. 23, 2013	Middle East respiratory syndrome coronavirus isolate Qatar3, complete genome
AHC74098	Qatar	Oct. 17, 2013	Dec. 23, 2013	Middle East respiratory syndrome coronavirus isolate Qatar4, complete genome
AHI48572	Saudi Arabia	Aug. 15, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Riyadh_14_2013, complete genome
AGV08379	Saudi Arabia	Oct. 23, 2012	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Riyadh_1_2012, complete genome
AID55073	Saudi Arabia	Apr. 22, 2014	Nov. 12, 2014	Middle East respiratory syndrome coronavirus isolate Riyadh_2014KSA_683/KSA/2014, complete genome
AGV08584	Saudi Arabia	Oct. 30, 2012	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Riyadh_2_2012, complete genome
AGV08390	Saudi Arabia	Feb. 5, 2013	Sep. 17, 2013	Middle East respiratory syndrome coronavirus isolate Riyadh_3_2013, complete genome
AHI48605	Saudi Arabia	Mar. 1, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Riyadh_4_2013, complete genome
AHI48583	Saudi Arabia	Jul. 2, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Riyadh_5_2013, complete genome

TABLE 12-continued

Full-length Spike Glycoprotein Amino Acid Sequences (<i>Homo sapiens</i> strains)				
GenBank Accession	Country	Collection Date	Release Date	Virus Name
AHI48528	Saudi Arabia	Jul. 17, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Riyadh_9_2013, complete genome
AHI48594	Saudi Arabia	Jun. 12, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Taif_1_2013, complete genome
AHI48550	Saudi Arabia	Jun. 12, 2013	Feb. 6, 2014	Middle East respiratory syndrome coronavirus isolate Wadi-Ad- Dawasir_1_2013, complete genome
AIY60558	United Arab Emirates	Mar. 7, 2014	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi/Gayathi_UAE_2_2014, complete genome
AIY60538	United Arab Emirates	Apr. 10, 2014	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi_UAE_16_2014, complete genome
AIY60528	United Arab Emirates	Apr. 10, 2014	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi_UAE_18_2014, complete genome
AIY60588	United Arab Emirates	Apr. 13, 2014	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi_UAE_26_2014, complete genome
AIY60548	United Arab Emirates	Apr. 19, 2014	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi_UAE_30_2014, complete genome
AIY60568	United Arab Emirates	Apr. 17, 2014	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi_UAE_33_2014, complete genome
AIY60518	United Arab Emirates	Apr. 7, 2014	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi_UAE_8_2014, complete genome
AIY60578	United Arab Emirates	Nov. 15, 2013	Dec. 6, 2014	Middle East respiratory syndrome coronavirus strain Abu Dhabi_UAE_9_2013, complete genome
AKJ80137	China	May 27, 2015	Jun. 5, 2015	Middle East respiratory syndrome coronavirus strain ChinaGD01, complete genome
AHZ64057	USA	May 10, 2014	May 14, 2014	Middle East respiratory syndrome coronavirus strain Florida/USA- 2_Saudi Arabia_2014, complete genome
AKM76229	Oman	Oct. 28, 2013	Jun. 23, 2015	Middle East respiratory syndrome coronavirus strain Hu/Oman_2285_2013, complete genome
AKM76239	Oman	Dec. 28, 2013	Jun. 23, 2015	Middle East respiratory syndrome coronavirus strain Hu/Oman_2874_2013, complete genome
AKI29284	Saudi Arabia	Jan. 6, 2015	May 27, 2015	Middle East respiratory syndrome coronavirus strain Hu/Riyadh-KSA- 2049/2015, complete genome
AKI29265	Saudi Arabia	Jan. 21, 2015	May 27, 2015	Middle East respiratory syndrome coronavirus strain Hu/Riyadh-KSA- 2343/2015, complete genome
AKI29255	Saudi Arabia	Jan. 21, 2015	May 27, 2015	Middle East respiratory syndrome coronavirus strain Hu/Riyadh-KSA- 2345/2015, complete genome
AKI29275	Saudi Arabia	Jan. 26, 2015	May 27, 2015	Middle East respiratory syndrome coronavirus strain Hu/Riyadh-KSA- 2466/2015, complete genome
AKK52582	Saudi Arabia	Feb. 10, 2015	Jun. 8, 2015	Middle East respiratory syndrome coronavirus strain Hu/Riyadh_KSA_2959_2015, complete genome

TABLE 12-continued

Full-length Spike Glycoprotein Amino Acid Sequences (<i>Homo sapiens</i> strains)				
GenBank Accession	Country	Collection Date	Release Date	Virus Name
AKK52592	Saudi Arabia	Mar. 1, 2015	Jun. 8, 2015	Middle East respiratory syndrome coronavirus strain Hu/Riyadh_KSA_4050_2015, complete genome
AHZ58501	USA	Apr. 30, 2014	May 13, 2014	Middle East respiratory syndrome coronavirus strain Indiana/USA- 1_Saudi Arabia_2014, complete genome
AGN52936	United Arab Emirates	2013	Jun. 10, 2013	Middle East respiratory syndrome coronavirus, complete genome

TABLE 13

Description	Sequence	SEQ ID NO :
MeV Nucleic Acid Sequences		
GC_F_MEASLES_ B3.1 Sequence, NT (5' UTR, ORF, 3' UTR) Sequence Length: 1864	TCAAGCTTTTGGACCCTCGTACAGAAGCTAATACGACT CACTATAGGGAAATAAGAGAGAAAAAGAGTAAGAA GAAATATAAGAGCCACCATGGGTCTCAAGGTGAACGTC TCTGCCGTATTCATGGCAGTACTGTTAACTCTCCAAACA CCCGCCGGTCAAATTCATTGGGGCAATCTCTTAAGAT AGGGGTAGTAGGAATAGGAAGTGCAAGCTACAAAGTT ATGACTCGTTCAGCCATCAATCATAGTCATAAAATT AATGCCCAATATAACTCTCTCAATAATGCACGAGGG TAGAGATTGCAGAAATACAGGAGACTACTAAGAACAGTT TTGGAACCAATTAGGGATGCACCTAATGCAATGACCCA GAACATAAGGCCGGTTCAGAGCGTAGCTTCAAGTAGGA GACACAAGAGATTGCGGGAGTAGTCTTGGCAGGTGCG GCCCTAGGTGTTGCCACAGCTGCTCAGATAACAGCCGG CATTGCATCTACCGGTCCATGCTGAATCTCAGGCCAT CGACAATCTGAGAGCGAGCCTGGAACCTACTAATCAGG CAATTGAGGCAATCAGACAAGCAGGGCAGGAGATGAT ATTGGCTGTTAGGGGTGTTCAAGACTACATCAATAATG AGCTGATACCGTCTATGAACAGCTATCTTGTGATCTA ATCGGTACAGAAGCTCGGGCTCAAATGCTTAGATACTA TACAGAAATCTGTGTTATTTGGCCCCAGCCTACGGG ACCCATATCTGCGGAGATATCTATCCAGGCTTTGAGTT ATGCACCTGGAGGAGATCAATAAGGTGTTAGAAAAG CTCGGATACAGTGGAGGCGATTACTAGGCATCTTAGA GAGCAGAGGAATAAAGGCTCGGATAACTCAGTCGAC ACAGAGTCTACTTCATAGTCTCAGTATAGCCTATCCG ACGCTGTCCGAGATTAAAGGGGTGATTGCCACCGGCT AGAGGGGTCTCGTACAACATAGGCTCTCAAGAGTGGT ATACCACTGTGCCCAAGTATGTTGCAACCAAGGGTAC CTTATCTCGAATTTTGATGAGTCATCATGTACTTTCATG CCAGAGGGGACTGTGTGAGCCAAATGCCTGTACCC GATGAGTCTCTGCTCCAAGATGCCTCCGGGGTCCA CCAAGTCTGTGCTCGTACACTCGTATCCGGGTCTTTTG GGAACCGGTTCAATTTATCAAGGGAACCTAATAGCC AATTGTGCATCAATTCTTGTAGTGTACACAACAGGT ACGATTATTAATCAAGACCCGTGACAAGATCCTAACATA CATTGCTGCCGATCGCTGCCGGTAGTCGAGGTGAACG GCGTGACCATCCAAGTCCGGAGCAGGAGGTATCCAGA CGCTGTGTACTTGCACAGAATTGACCTCGGTCTCCCAT ATCATTTGAGAGGTTGGACGTAGGGACAAATCTGGGG AATGCAATTGCCAAATTGGAGGATGCCAAGGAATTGTT GGAATCATCGGACCAGATATTGAGAAGTATGAAAGGTT TATCGAGCACTAGCATAGTCTACATCCTGATTGCAGTG TGTCTTGGAGGGTTGATAGGGATCCCCACTTAAATATGT TGCTGCAGGGGGCGTTGTAACAAAAAGGGAGAACAG TTGGTATGTCAAGACCAGGCCATAAGCCTGACCTTACA GGAACATCAAAATCCTATGTAGATCGCTTTGATGATA ATAGGCTGGAGCCTCGGTGGCCAAGCTTCTTGCCCTT GGGCTCCCCCAGCCCTCTCCCTTCTGCAACCGT ACCCCGTGGTCTTTGAATAAAGTCTGAGTGGGCGGC	35

TABLE 13-continued

Description	Sequence	SEQ ID NO:
GC_F_MEASLES_ B3.1 ORF Sequence, NT	ATGGGTCTCAAGGTGAACGTCTCTGCCGTATTCATGGC AGTACTGTAACTCTCCAAACACCCGCCGGTCAAATTC ATTGGGGCAATCTCTCTAAGATAGGGGTAGTAGGAATA GGAGTGCAGCTACAAAGTTATGACTCGTTCAGCCA TCAATCATTAGTCATAAAATTAATGCCCAATATAACTCT CCTCAATAACTGCACGAGGGTAGAGATTGCAGAAATACA GGAGACTACTAAGAACAGTTTGGAAACCAATTAGGGAT GCACCTTAATGCAATGACCCAGAACATAAGGCCGGTTCA GAGCGTAGCTTCAAGTAGGAGACACAAGAGATTTGCG GGAGTAGTCCTGGCAGGTGCGGCCCTAGGTGTTGCCAC AGCTGCTCAGATAACAGCCGGCATTGCACCTTACCAGGT CCATGCTGAACCTCTCAGGCCATCGACAATCTGAGAGCG AGCCTGGAACTACTAATCAGGCAATTGAGGCAATCAG ACAAGCAGGGCAGGAGATGATATTGGCTGTTCCAGGGTG TCCAAGACTACATCAATAATGAGCTGATACCGTCTATG AACCAGCTATCTGTGATCTAATCGGTGAGAAGCTCGG GCTCAAATTGCTTAGATACTATACAGAAATCCTGTCATT ATTGGCCCCAGCCTACGGGACCCCATATCTGCGGAGA TATCTATCCAGGCTTTGAGTTATGCACTTGGAGGAGAT ATCAATAAGGTGTTAGAAAAGCTCGGATACAGTGGAG GCGATTTACTAGGCATCTTAGAGAGCAGAGGAATAAAG GCTCGGATAACTCAGTCGACACAGAGTCTACTTCAT AGTCTCTAGTATAGCCTATCCGACGCTGTCGAGATTA AGGGGGTGATTGTCCACCGGCTAGAGGGGGTCTCGTAC AACATAGGCTCTCAAGAGTGGTATACCACTGTGCCAA GTATGTTGCAACCCAGGGTACCTTATCTCGAATTTTGA TGAGTCATCATGTACTTTTCATGCCAGAGGGGACTGTGT GCAGCCAAAATGCCTTGTACCCGATGAGTCTCTGCTC CAAGAATGCCTCCGGGGTCCACCAAGTCTGTGCTCG TACACTCGTATCCGGGTCTTTTGGGAACCGGTTCAATTT ATCACAAGGGAACCTAATAGCCAATTGTGCATCAATTC TTTGTAAAGTGTACACAACAGGTACGATTATTAATCAA GACCTTGACAAGATCCTAACATACATTGCTGCCGATCG CTGCCCGGTAGTCGAGGTGAACGGCGTGACCATCCAAG TCGGGAGCAGGAGGTATCCAGACGCTGTGTACTTGCAC AGAATTGACCTCGGTCTCCCATATCATTGGAGAGGTT GGACGTAGGGACAAATCTGGGGAATGCAATTGCCAAA TTGGAGGATGCCAAGGAATTGTTGGAATCATCGGACCA GATATTGAGAAGTATGAAAGGTTTATCGAGCACTAGCA TAGTCTACATCCTGATTGCAGTGTGTCTTGGAGGGTTGA TAGGGATCCCCACTTTAATATGTTGCTGCAGGGGGCGT TGTAACAAAAGGGAGAACAAAGTTGGTATGTCAAGAC CAGGCCTAAAGCTGACCTTACAGGAACATCAAATCC TATGTAAGATCGCTTTGA	36
GC_F_MEASLES_ B3.1 mRNA Sequence (assumes T100 tail) mRNA Sequence Length: 1925	G*GGGAAATAAGAGAGAAAAGAAGAGTAAGAAGAAAT ATAAGAGCCACCATGGGTCTCAAGGTGAACGTCTCTGC CGTATTATGGCAGTACTGTTAACTCTCCAAACACCCG CCGGTCAAATTCATTGGGGCAATCTCTAAGATAGGG GTAGTAGGAATAGGAAGTGCAAGCTACAAAGTTATGA CTCGTTCAGCCATCAATCATATAGTCATAAAATTAATGC CCAATATAACTCTCCTCAATAACTGCACGAGGGTAGAG ATTGCAGAATACAGGAGACTACTAAGAACAGTTTGGGA ACCAATTAGGGATGCACTTAATGCAATGACCCAGAACA TAAGGCCGGTTCAGAGCGTAGCTTCAAGTAGGAGACAC AAGAGATTTGCGGGAGTAGTCTGCGCAGGTGCGGCCCT AGGTGTTGCCACAGCTGCTCAGATAACAGCCGGCATTG CACTTCACCGGTCCATGCTGAACCTCTCAGGCCATCGAC AATCTGAGAGCGAGCCTGGAACCTACTAATCAGGCAAT TGAGGCAATCAGACAAGCAGGGCAGGAGATGATATTG GCTGTTCAAGGTGTCCAAGACTACATCAATAATGAGCT GATACCGTCTATGAACAGCTATCTTGTGATCTAATCG GTCAGAAGCTCGGGCTCAAATTGCTTAGATACTATACA GAAATCCTGTGATTATTTGGCCCCAGCCTACGGGACCC CATATCTGCGGAGATATCTATCCAGGCTTTGAGTTATGC ACTTGGAGGAGATATCAATAAGGTGTTAGAAAAGCTCG GATACAGTGGAGGCGATTACTAGGCATCTTAGAGAGC AGAGGAATAAAGGCTCGGATAACTCACGTCGACACAG AGTCTTACTTCATAGTCTCAGTATAGCCTATCCGACGC TGTCCGAGATTAAAGGGGTGATTGTCCACCGGCTAGAG GGGGTCTCGTACAACATAGGCTCTCAAGAGTGGTATAC CACTGTGCCCCAAGTATGTTGCAACCCAGGGTACCTTA TCTCGAATTTTGATGAGTCATCATGTACTTTTCATGCCAG AGGGGACTGTGTGCAGCCAAAATGCCTTGTACCCGATG AGTCTCTGCTCCAAGATGCCTCCGGGGGTCCACCAA GTCCTGTGCTCGTACACTCGTATCCGGGTCTTTTGGGAA	37

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	CCGGTTTATTTATCACAGGGAACCTAATAGCCAATT GTGCATCAATTCTTTGTAAGTGTTCACACACAGGTACG ATTATTAAATCAAGACCCCTGACAAGATCCTAACATACAT TGCTGCCGATCGCTGCCCGGTAGTCGAGGTGAACGGCG TGACCATCCAAGTCGGGAGCAGGAGGTATCCAGACGCT GTGTACTTGCACAGAATTGACCTCGGTCTCCCATATCA TTGGAGAGGTTGGACGTAGGGACAAATCTGGGGAATG CAATTGCCAAATTGGAGGATGCCAAGGAATTGTTGGAA TCATCGGACCAGATATTGAGAAGTATGAAAGGTTTATC GAGCACTAGCATAGTCTACATCCTGATTGCAGTGTGTC TTGGAGGGTTGATAGGGATCCCCACTTTAATATGTTGCT GCAGGGGGCGTTGTAAACAAAAGGGAGAAACAAAGTTGG TATGTCAAGACCAGGCCTAAAGCCTGACCTTACAGGAA CATCAAAATCCTATGTAAGATCGCTTGTATGATAATAG GCTGGAGCCTCGGTGGCCAAGCTTCTTGCCCTTGGGC CTCCCCCAGCCCTCTCCCTTCTGCACCCGTACCC CCGTGGTCTTTGAATAAAGTCTGAGTGGCGCGCAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAATCTAG	
GC_F_MEASLES_ D8 Sequence, NT (5' UTR, ORF, 3' UTR) Sequence Length: 1864	TCAAGCTTTGGACCTCGTACAGAAGCTAATACGACT CACTATAGGGAAATAAGAGAGAAAAGAAGTAAGAA GAAATATAAGAGCCACCATGGGTCTCAAGGTGAACGTC TCTGTCATATTTCATGGCAGTACTGTTAACTCTTCAAACA CCCACCGGTCAATCCATGGGGCAATCTCTTAAGAT AGGGGTGGTAGGGTAGGAAGTGCAAGCTACAAAGTT ATGACTCGTTCAGCCATCAATCATTAGTCATAAAGTT AATGCCCAATATAACTCTCCTCAACAAATGCACGAGGG TAGGGATTGCAGAATACAGGAGACTACTGAGAACAGTT CTGGAACCAATTAGAGATGCCTTAATGCAATGACCCA GAATATAAGACCGGTTAGAGTGTAGCTTCAAGTAGGA GACACAAGAGATTTGCGGGAGTTGTCTGGCAGGTGCG GCCCTAGGCGTTGCCACAGCTGCTCAAATAACAGCCGG TATTGCACTTACCAGTCCATGCTGAACTCTCAAGCCAT CGACAATCTGAGAGCGAGCCTAGAACTACTAATCAGG CAATTGAGGCAATCAGACAAGCAGGGCAGGAGATGAT ATTGGCTGTTAGGGGTGTTCAAGACTACATCAATAATG AGCTGATACCGTCTATGAATCAACTATCTTGTGATTTAA TCGGCCAGAAGCTAGGGCTCAAATTGCTCAGATACTAT ACAGAAATCCTGTCTATTATTGGCCCCAGCTTACGGGA CCCCATATCTGCGGAGATATCTATCCAGGCTTTGAGCT ATGCGCTTGGAGGAGATATCAATAAGGTGTTGGAAAAG CTCGGATACAGTGGAGGTGATCTACTGGGCATCTTAGA GAGCAGAGGAATAAAGGCCCGGATAACTCACGTCGAC ACAGAGTCTACTTCAATTGTACTCAGTATAGCCTATCCG ACGCTATCCGAGATTAAAGGGGTGATTGTCCACCGGCT AGAGGGGGTCTCGTACAACATAGGCTCTCAAGAGTGGT ATACCACTGTGCCCAAGTATGTTGCAACCCAGGGTAC CTTATCTCGAATTTTGATGAGTCATATGCACCTTTCATG CCAGAGGGGACTGTGTGCAGCCAGAATGCCCTGTACCC GATGAGTCTCTGCTCCAAGAATGCTCCGGGGGTCCA CTAAGTCTGTGCTCGTACACTCGTATCCGGGTCTTTCG GGAACCGGTTCAATTTATCACAGGGGAACCTAATAGCC AATTGTGCATCAATCCTTTGCAAGTGTACACAACAGG AACAATCATTAAATCAAGACCTGACAAGATCCTAACAT ACATTGTGTCCGATCACTGCCCGGTGGTCGAGGTGAAT GCGGTGACCATCCAAGTCGGGAGCAGGAGGTATCCGG ACGCTGTGTAATTGCACAGGATTGACCTCGGTCTTCCC ATATCTTTGGAGAGGTTGGACGTAGGGACAAATCTGGG GAATGCAATTGCTAAGTTGGAGGATGCCAAGGAATTGT TGGAGTCATCGGACCAGATATTGAGGAGTATGAAAGGT TTATCGAGCACTAGTATAGTTTACATCTGATTGCAGTG TGTCTTGGAGGATTGATAGGGATCCCCGCTTAAATATGT TGCTGCAGGGGGCGTTGTAAACAAGAAAGGAGAACAAG TTGGTATGTCAAGACCAGGCCTAAAGCCTGATCTTACA GGAACATCAAAATCCTATGTAAGGTCACTCTGATGATA ATAGGCTGGAGCCTCGGTGGCCAAGCTTCTTGCCCTT GGGCTCCCCCAGCCCTCTCCCTTCTGCACCCGCT ACCCCGTGGTCTTTGAATAAAGTCTGAGTGGCGGC	38
GC_F_MEASLES_ D8 ORF Sequence, NT	ATGGGTCTCAAGGTGAACGTCTCTGTCATATTCATGGC AGTACTGTTAACTCTTCAAACACCCACCGTCAAATCC ATTGGGGCAATCTCTTAAGATAGGGGTGGTAGGGTA GGAAGTGCAAGCTACAAAGTTATGACTCGTTCACGCCA TCAATCATTAGTCATAAAGTTAATGCCCAATATAACTCT CCTCAACAATTGCACGAGGGTAGGGATTGCAGAATACA	39

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	GGAGACTACTGAGAACAGTTCTGGAACCAATTAGAGAT GCACTTAATGCAATGACCCAGAATATAAGACCGGTTCA GAGTGTAGCTTCAAGTAGGAGACACAAGAGATTTGCGG GAGTTGTCTTGGCAGGTGCGGCCCTAGGCGTTGCCACA GCTGCTCAATAACAGCCGGTATTGCAC TTCACAGTC CATGCTGAAC TCTCAAGCCATCGACAATCTGAGAGCGA GCC TAGAACTACTAATCAGGCAATTGAGGCAATCAGA CAAGCAGGGCAGGAGATGATATTGGCTGTT CAGGGTGT CCAAGACTACATCAATAATGAGCTGATACCGTCTATGA ATCAACTATCTTGTGATTAAATCGGCCAGAAGCTAGGG CTCAAATTGCTCAGATACTATACAGAAATCCTGT CATT ATTTGGCCCCAGCTTACGGGACCCCATATCTGCGGAGA TATCTATCCAGGCTTTGAGCTATGCGCTTGGAGGAGAT ATCAATAAGGTGTTGGAAAAGCTCGGATACAGTGGAG GTGATCTACTGGGCATCTTAGAGAGCAGAGGAATAAAG GCCCGGATAACTCAGTCGACACAGAGTCTTACTTCAT TGTA CTAGTATAGCCTATCCGACGCTATCCGAGATTA AGGGGGTGATTGTCCACCGGCTAGAGGGGGTCTCGTAC AACATAGGCTCTCAAGAGTGGTATACCCTGTGCCAA GTATGTTGCAACCC AAGGGTACCTTATCTCGAATTTGA TGAGTCATCATGCAC TTTCATGCCAGAGGGGACTGTGT GCAGCCAGAATGCCTTGTACCCGATGAGTCTCTGCTC CAAGAATGCCTCCGGGGTCCACTAAGTCTGTGCTCG TACACTCGTATCCGGGCTTTTCGGGAACCGGTT CATT ATCACAGGGGAACCTAATAGCCAATTGTGCATCAATCC TTTGCAAGTGTTACACAACAGGAACAATCATTAATCAA GACCCTGACAAGATCCTAACATACATTGCTGCCGATCA CTGCCCGGTGGTCGAGGTGAATGGCGTGACCATCCAAG TCGGGAGCAGGAGGTATCCGGACGCTGTGTACTTGCAC AGGATTGACCTCGGTCTCCCATATCTTTGGAGAGGTT GGACGTAGGGACAAATCTGGGGAATGCAATTGCTAAGT TGGAGGATGCCAAGGAATGTTGGAGTCATCGGACCAG ATATTGAGGAGTATGAAAGGTTTATCGAGCACTAGTAT AGTTTACATCTGATTGCAGTGTGTCTTGGAGGATTGAT AGGGATCCCCGCTTTAATATGTTGCTGCAGGGGGCGTT GTAACAAGAAGGGAGAACAGTTGGTATGTCAAGACC AGGCCTAAGCCTGATCTTACAGGAACATCAAAATCCT ATGTAAGGTCACCTCTGA	
GC_F_MEASLES_ D8 mRNA Sequence (assumes T100 tail) Sequence Length: 1925	G*GGGAAATAAGAGAGAAAAGAAGTAAGAAGAAAT ATAAGAGCCACCATGGGTCTCAAGGTGAACGCTCTCTGT CATATTCCATGGCAGTACTGTTAACTCTTCAAACACCCAC CGGTCAAATCCATTGGGGCAATCTCTCTAAGATAGGGG TGGTAGGGGTAGGAAGTGCAAGCTACAAAGTTATGACT CGTTCCAGCCATCAATCATTAGTCATAAAGTTAATGCC CAATATAACTCTCCTCAACAATTGCACGAGGGTAGGGA TTGCAGAAATACAGGAGACTACTGAGAACAGTTCTGGAA CCAATTAGAGATGCAC TTAATGCAATGACCCAGAATAT AAGACCGGTTCAGAGTGTAGCTTCAAGTAGGAGACACA AGAGATTTGCGGGAGTTGTCTGGCAGGTGCGGCCCTA GGCGTTGCCACAGCTGCTCAATAACAGCCGGTATTGC ACTTCACCACTCATGCTGAACCTCAAGCCATCGACA ATCTGAGAGCGAGCCTAGAACTACTAATCAGGCAATT GAGGCAATCAGACAAGCAGGGCAGGAGATGATATTGG CTGTT CAGGGTGTCCAAGACTACATCAATAATGAGCTG ATACCGTCTATGAATCAACTATCTTGTGATTAAATCGGC CAGAAGCTAGGCTCAAATTGCTCAGATACTATACAGA AATCCTGTCAATTATTGGCCCCAGCTTACGGGACCCCAT ATCTGCGGAGATATCTATCCAGGCTTTGAGCTATGCGC TTGGAGGAGATATCAATAAGGTGTTGGAAAAGCTCGGA TACAGTGGAGGTGATCTACTGGGCATCTTAGAGAGCAG AGGAATAAAGGCCCGGATAACTCAGTCGACACAGAG TCCTACTTCATTGTACTCAGTATAGCCTATCCGACGCTA TCCGAGATTAAGGGGGTGATTGTCCACCGGTAGAGGG GGTCTCGTACAACTAGGCTCTCAAGAGTGGTATACCA CTGTGCCCAAGTATGTTGCAACCC AAGGGTACCTTATC TCGAATTTTGATGAGTCATCATGCACTTTCATGCCAGAG GGGACTGTGTGCAGCCAGAATGCCTTGTACCCGATGAG TCCTCTGCTCCAAGAATGCCTCCGGGGTCCACTAAGT CCTGTGCTCGTACACTCGTATCCGGGTCTTTCGGGAACC GGTTCATTTTATCACAGGGGAACCTAATAGCCAATTGT GCATCAATCCTTTGCAAGTGTTACACAACAGGAACAAT CATTAATCAAGACCCCTGACAAGATCCTAACATACATTG CTGCCGATCACTGCCCGGTGGTCGAGGTGAATGGCGTG ACCATCCAAGTCGGGAGCAGGAGGTATCCGACGCTGT GTACTTGACAGGATTGACCTCGGTCTCCCATATCTTT GGAGAGGTTGGACGTAGGGACAAATCTGGGGAATGCA	40

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	ATTGCTAAGTTGGAGGATGCCAAGGAATTGTTGGAGTC ATCGGACCAGATATTGAGGAGTATGAAAGGTTATCGA GCACTAGTATAGTTTACATCCTGATTGCAGTGTGTCTTG GAGGATTGATAGGGATCCCCGCTTTAATATGTTGCTGC AGGGGGCGTTGTAAACAAGAAGGGAGAACAGTTGGTA TGTC AAGACCAGGCC TAAAGCCTGATCTTACAGGAACA TCAAAATCCTATGTAAGGTCACTCTGATGATAATAGGC TGGAGCCTCGGTGGCCAAGCTTCTTGGCCCTTGGGCCTC CCCCCAGCCCCCTCCCTTCCCTGCACCCGTACCCCCG TGGTCTTTGAATAAAGTCTGAGTGGGCGGCAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAATCTAG	
GC_H_MEASLES_ B3 Sequence, NT (5' UTR, ORF, 3' UTR) Sequence Length: 2065	TCAAGCTTTTGGACCTCGTACAGAAGCTAATACGACT CACTATAGGGAATAAGAGAGAAAAAGAGTAAGAA GAAATATAAGAGCCACCATGTCAACGCAACGAGACCG GATAAATGCCCTTCTACAAGATAACCCCTATCCCAAGG GAAGTAGGATAGTTTATTAACAGAGAACATCTTATGATT GACAGACCCTATGTTCTGCTGGCTGTTCTGTCTGCATG TTTCTGAGCTTGATCGGATTGCTGGCAATTGCAGGCATT AGACTTCATCGGGCAGCCATCTACACCGCGGAGATCCA TAAAAGCCTCAGTACCAATCTGGATGTGACTAACTCCA TCGAGCATCAGGTCAAGGACGTGCTGACACCACTCTTT AAAATCATCGGGGATGAAGTGGGCTGAGAACACCTC AGAGATTCACTGACCTAGTGAAATTCATCTCGGACAAG ATTAAATTCCTTAATCCGGATAGGGAGTACGACTTCAG AGATCTCACTTGGTGCATCAACCGCCAGAGAGGATCA AACTAGATTATGATCAATAGTGTGAGATGTGGCTGCT GAAGAGCTCATGAATGCATTGGTGAACCTCACTCTACT GGAGACCAGAACACCACTCAGTTCTTAGCTGTCTCAA AGGGAAACTGCTCAGGGCCACTACAATCAGAGGTCA ATTCTCAAACTGTGCTGCTCTTGTGGACTGTACTT AGGTCGAGGTTACAAATGTGTATCTATAGTCACTATGA CATCCCGGGAATGTATGGGGGAACCTACCTAGTTGAA AAGCCTAATCTGAACAGCAAGGGTCAGAGTTGTCACA ACTGAGCATGTACCGAGTGTGAAAGTAGGTGTGATCA GAAACCCGGGTTTGGGGCTCCGTTGTTCCATATGACA AACTATTTTGAACAACAGTCAGTAATGGTCTCGGCAA CTGTATGGTGGCTTTGGGGGAGCTCAAACCTCGAGCCC TTTGTCAAGGGGACGATTCTATCATAATTCCTATCAGG GATCAGGGAAAGGTGTGAGCTTCCAGCTCGTCAAGCTG GGTGTCTGGAATCCCCAACCGACATGCAATCCTGGGT CCCCTTATCAACGGATGATCCAGTGGTAGACAGGCTTT ACCTCTCATCTCACAGAGGTGTCATCGCTGACAAATCAA GCAAAATGGGCTGTCCGACAAACGAAACAGATGACA AGTTGCGAATGGAGACATGCTTCCAGCAGGCGTTAAA GGTAAATCCAAGCACTCTGCGAGAATCCCGAGTGGGT ACCATTGAAGGATAACAGGATTCCTTCATACGGGGTCC TGTCTGTTGATCTGAGCTGACGGTTGAGCTTAAATCA AAATTGCTTCGGGATTCCGGCCATTGATCACACACGGC TCAGGGATGGACCTATACAAATCCAACCTGCAACATGT GTATTGGCTGACTATTCCGCCAATGAGAAATCTAGCCT TAGGCGTAATCAACACATTGGAGTGATACCGAGATT AAGGTTAGTCCCAACCTCTTCACTGTCCCAATTAAGGA AGCAGGCGAAGACTGCCATGCCCCAACATACCTACCTG CGGAGGTGGACGGTGATGTCAAACCTCAGTTCCAACCTG GTGATTCTACCTGGTCAAGATCTCCAATATGTTTGGCA ACCTACGATACCTCCAGGGTTGAGCATGCTGTGGTTTA TTACGTTTACAGCCCAAGCCGCTCATTTTCTACTTTTA TCTTTTAGGTTGCCTATAAAGGGGTCCCAATCGAAC TACAAGTGAATGCTTCATATGGGATCAAAACTCTGG TGCCGTCACTTCTGTGTGCTTGCGGACTCAGAATCCGGT GGACTTATCACTCACTCTGGGATGGTGGCATGGGAGT CAGCTGCACAGCTACCCGGGAAGATGGAACCAATCGC AGATAATGATAATAGGCTGGAGCCTCGGTGGCCAAGCT TCTTGGCCCTTGGGCTCCCCAGCCCTCTCCCTT CCTGCACCCGTACCCCGTGGTCTTTGAATAAAGTCTG AGTGGGCGGC	41
GC_H_MEASLES_ B3 ORF Sequence, NT	ATGTCAACGCAACGAGACCGGATAAATGCCCTCTACAA AGATAAACCTTATCCCAAGGAAGTAGGATAGTTATTA ACAGAGAACATCTTATGATTGACAGACCTATGTTCTG CTGGCTGTTCTGTTCTGTCATGTTTCTGAGCTTGATCGGA TTGCTGGCAATTGCAGGCATTAGACTTCATCGGGCAGC CATCTACACCGGAGATCCATAAAGCCTCAGTACCA ATCTGGATGTGACTAACTCCATCGAGCATCAGGTCAAG	42

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	GACGTGCTGACACCACTCTTTAAATCATCGGGGATGA AGTGGGCTGAGAACCTCAGAGATTCAGTACCTAG TGAAATTCATCTCGGACAAGATTAAATTCCTTAATCCG GATAGGGAGTACGACTTCAGAGATCTCACTTGGTGCAT CAACCCGCGAGAGAGGATCAAACTAGATTATGATCAAT ACTGTGCAGATGTGGCTGCTGAAGAGCTCATGAATGCA TTGGTGAACCTCAACTCTACTGGAGACCAGAACCAACCAC TCAGTTCCTAGCTGTCTCAAAGGGAACTGCTCAGGGC CCACTACAATCAGAGGTCAATTCTCAAACATGTCGCTG TCCTTGTGGACTTGACTTAGGTTCGAGGTACAAATGTG TCATCTATAGTCACTATGACATCCAGGGAATGTATGG GGGAACTTACCTAGTTGAAAAGCCTAATCTGAACAGCA AAGGGTCAGAGTTGTCACAACTGAGCATGTACCGAGTG TTTGAAGTAGGTGTGATCAGAAACCCGGGTTGGGGGC TCCGGTGTTCATATGACAAACTATTTTGAGCAACCAG TCAGTAATGGTCTCGGCAACTGTATGGTGGCTTTGGGG GAGCTCAAACCTCGAGCCCTTTGTACGGGGACGATTC TATCATAATCCCTATCAGGGATCAGGGAAAGGTGTCA GCTTCCAGCTCGTCAAGCTGGGTGTCTGAAATCCCA ACCGACATGCATCCTGGGTCCCTTATCAACGGATGA TCCAGTGGTAGACAGGCTTTACCTCTCATCTCACAGAG GTGTCACTCGCTGACAATCAAGCAAAATGGGCTGTCCCG ACAACACGAACAGATGACAAGTTCGAATGGAGACAT GCTTCCAGCAGCGTGTAAAGTAAATCCAAAGCACTC TGCAGAAATCCGAGTGGGTACCATTAAGGATAACAG GATTCCTTCATACGGGGTCTGTCTGTTGATCTGAGTCT GACGGTTGAGCTTAAATCAAAATTCCTCGGGATTCTG GGCCATTGATCACACACGGCTCAGGGATGGACCTATAC AAATCCAACCTGCAACAATGTGTATTGGCTGACTATTCC GCCAATGAGAAATCTAGCCTTAGGCGTAATCAACACAT TGGAGTGGATACCGAGATTCAGGTAGTCCCAACCTC TTCATGTCCCAATTAAGGAAGCAGGCGAAGACTGCCA TGCCCCAACATACCTACCTGCGGAGGTGGACGGTGATG TCAAACCTCAGTTCCAACTGGTGATTCTACCTGGTCAA GATCTCCAATATGTTTGGCAACCTACGATACCTCCAG GGTTGAGCATGCTGTGGTTTATTACGTTTACAGCCCAA GCCGCTCATTTTCTTACTTTTATCCTTTTAGGTTGCCTAT AAAGGGGTCCCAATCGAACTACAAGTGAATGCTTCA CATGGGATCAAAAACCTCTGGTGCCTCACTTCTGTGTG CTTGCGGACTCAGAAATCCGGTGGACTTATCACTCACTCT GGGATGGTGGGCATGGAGTCAGTGCACAGCTACCCG GGAAGATGGAACCAATCGCAGATAA	
GC_H_MEASLES_ B3 mRNA Sequence (assumes T100 tail) Sequence Length: 2126	G*GGGAAATAAGAGAGAAAAGAGTAAGAAGAAAT ATAAGAGCCACCATGTCAACCGAACGAGACCGGATAA ATGCCTTCTACAAGATAACCTTATCCCAAGGGAAGT AGGATAGTTATTAAACAGAGAACATCTTATGATTGACAG ACCCTATGTTCTGCTGGCTGTTCTGTTTCGTATGTTTCT GAGCTTGATCGGATTGCTGGCAATTGACAGGCATTAGAC TTCATCGGGCAGCCATCTACCCGCGGAGATCCATAAA AGCCTCAGTACCAATCTGGATGTGACTAACTCCATCGA GCATCAGGTCAAGGACGTGCTGACACCCTCTTTAAAA TCATCGGGGATGAAGTGGGCTGAGAACACCTCAGAG ATTCACTGACCTAGTGAATTCATCTCGGACAAGATTA AATTCTTAATCCGATAGGGAGTACGACTTCAGAGAT CTCATTGGTGCATCAACCCGCCAGAGAGGATCAAACT AGATTATGATCAATACTGTGCAGATGTGGCTGCTGAAG AGCTCATGAATGCATTGGTGAACCTCAACTCTACTGGAG ACCAGAACAACTCACTCAGTTCCTAGCTGTCTCAAAGGG AAACCTGCTCAGGGCCCACTACAATCAGAGGTCAATTCT CAAACATGTCGCTGCTTGTGGACTTGTAATAGGTCT GAGGTTACAATGTGTATCTATAGTCACTATGACATCC CAGGGAATGTATGGGGGAACCTACCTAGTTGAAAAGCC TAATCTGAACAGCAAAGGTCAGAGTTGTCACAACTGA GCATGTACCGAGTGTGTAAGTAGGTGTGATCAGAAAC CCGGGTTTGGGGGCTCCGGTGTTCATATGACAAACTA TTTTGAGCAACCACTCAGTAATGGTCTCGGCAACTGTA TGGTGGCTTTGGGGGAGCTCAAACCTCGCAGCCCTTGT CACGGGGACGATTCTATCATAATTCCTATCAGGGATC AGGGAAAGGTGTGAGCTTCCAGCTCGTCAAGCTGGGTG TCTGGAATCCCAACCGACATGCAATCCTGGGTCCCC TTATCAACGGATGATCCAGTGGTAGACAGGCTTTACCT CTCATCTCACAGAGGTGTCATCGCTGACAAATCAAGCAA AATGGGCTGTCCCGACAACACGACAGATGACAAGTTG CGAATGGAGACATGCTTCAGCAGCGGTGTAAAGGTAA AATCCAAGCACTCTGCGAGAATCCGAGTGGGTACCAT TGAAGGATAACAGGATTCTTTCATACGGGGTCTGTCT	43

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	GTTGATCTGAGTCTGACGGTTGAGCTTAAATCAAAAT TGCTTCGGGATTTCGGGCCATTGATCACACACGGCTCAG GGATGGACCTATACAAATCCAACCTGCAACAATGTGTAT TGGCTGACTATTCCGCCAATGAGAAATCTAGCCTTAGG CGTAATCAACACATTGGAGTGGATACCGAGATTCAAGG TTAGTCCCAACCTCTTCACTGTCCCAATTAAGGAAGCA GGCGAAGACTGCCATGCCCAACATACCTACCTGCGGA GGTGGACGGTGATGTCAAACCTCAGTCCCACTGGTGA TTCTACCTGGTCAAGATCTCCAATATGTTTTGGCAACCT ACGATACCTCCAGGGTTGAGCATGCTGTGGTTTATTAC GTTTACAGCCCAGCCGCTCATTTTCTTACTTTTATCCT TTTAGGTTGCCATATAAAGGGGGTCCCAATCGAATACA AGTGAATGCTTCACATGGGATCAAAACCTCTGGTGCC GTCACCTTCTGTGTGCTTGGGACTCAGAATCCGGTGGA CTTATCACTCACTCTGGGATGGTGGGCATGGGAGTCAG CTGCACAGCTACCCGGGAAGATGGAACCAATCGCAGAT AATGATAATAGGCTGGAGCCTCGGTGGCCAAGCTTCTT GCCCCCTTGGGCTCCCCCAGCCCCCTCTCCCCCTTCCTG CACCCGTACCCCGTGGTCTTTGAATAAAGTCTGAGTG GGCGGCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAATC TAG	
GC_H_MEASLES_ D8 Sequence, NT (5' UTR, ORF, 3' UTR) Sequence Length: 2065	TCAAGCTTTTGGACCTCGTACAGAAGCTAATACGACT CACTATAGGGAATAAGAGAGAAAAGAAGTAAGAA GAAATATAAGAGCCACCATGTCAACCAACGAGACCG GATAAATGCCCTTCTACAAGACAACCCCATCCTAAGG GAAGTAGGATAGTTATTAACAGAGAACATCTTATGATT GATAGACCTTATGTTTTGCTGGCTGTTCTATTCTGTCATG TTTCTGAGCTTGATCGGGTTGCTAGCCATTGCAGGCATT AGACTTCATCGGGCAGCCATCTACCCGCAGAGATCCA TAAAAGCCTCAGCACCAATCTGGATGTAACCTAACTCAA TCGAGCATCAGGTTAAGGACGTGCTGACACCACTCTTC AAGATCATCGGTGATGAAGTGGCTTGAGGACACCTCA GAGATTCACTGACCTAGTGAAGTTCATCTCTGACAAGA TTAAATTCCTTAATCCGGACAGGGAATACGACTTCAGA GATCTCACTTGGTGTATCAACCCGCCAGAGAGAATCAA ATTGGATTATGATCAATACTGTGCAGATGTGGCTGCTG AAGAACTCATGAATGCATTGGTGAACCTCAACTCTACTG GAGACCAGGGCAACCAATCAGTTCCTAGCTGTCTCAA GGGAAACTGCTCAGGGCCCACTACAATCAGAGGCCAAT TCTCAAACATGTCGCTGTCCCTGTTGGACTTGATTTAA GTCGAGGTTACAATGTGTATCTATAGTCACTATGACA TCCCGGGAATGTACGGGGGAACCTACCTAGTGGAAAA GCCTAATCTGAGCAGCAAGGGTCAGAGTTGTCACAAC TGAGCATGCACCGAGTGTTTGAAGTAGGTGTTATCAGA AATCCGGGTTTGGGGCTCCGGTATTCATATGACAAA CTATCTTGAGCAACAGTCAGTAATGATTTCAAGCACT GCATGGTGGCTTTGGGGGAGCTCAAGTTCAGAGCCCTC TGTCACAGGGAAGATCTATCACAATCCCTATCAGGG ATCAGGGAAGGTGTGACGTTCCAGCTTGTCAAGCTAG GTGCTGGAAATCCCAACCGACATGCAATCCTGGGTC CCCCTATCAACGGATGATCCAGTGATAGACAGGCTTTA CCTCTCATCTCACAGAGCGTTATCGCTGACAATCAAG CAAAATGGGCTGTCCCGACAACCGACAGATGACAA GTTGCGAATGGAGACATGCTTCCAGCAGGCGTGAAGG GTAAATCCAAGCACTTTGCGGAATCCCGAGTGGACA CCATTGAAGGATAACAGGATTCCCTTCATACGGGGTCTT GTCTGTTGATCTGAGTCTGACAGTTGAGCTTAAATCA AAATTGTTTCAGGATTTCGGGCCATTGATCACACACGGT TCAGGGATGGACCTATACAAACTCAACCACAACAATAT GTATTGGCTGACTATCCCGCAATGAAGAACCTGGCCT TAGGTGTAATCAACACATTGGAGTGGATACCGAGATT AAGGTTAGTCCCAACCTCTTCACTGTTCCAATTAAGGA AGCAGGCGAGGACTGCCATGCCCAACATACCTACCTG CGGAGGTGGATGGTGTGATGTCAAACCTCAGTTCATCTG GTGATTCTACCTGGTCAAGATCTCCAATATGTTCTGGCA ACCTACGATACTTCAGAGTTGAACATGCTGTAGTTTAT TACGTTTACAGCCCAAGCCGCTCATTTTCTTACTTTTAT CCTTTTAGGTTGCCGTGAAGGGGGTCCCCATTGAATTA CAAGTGGAATGCTTCACATGGGACCAAACTCTGGTG CCGTCACTTCTGTGTGCTTGGGACTCAGAACTCTGGTGG ACATATCACTCACTCTGGGATGGTGGGATGGGAGTCA GCTGCACAGCCACTCGGAAGATGGAACCGCCGCAG	44

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	ATAGTGATAATAGGCTGGAGCCTCGGTGGCCAAGCTTC TTGCCCCCTGGGCCTCCCCCAGCCCCCTCCCCCTTCC TGCACCCGTACCCCGTGGTCTTTGAATAAAGTCTGAG TGGGCGGC	
GC_H_MEASLES_ D8 ORF Sequence, NT	ATGTCAACACACGAGACCGGATAAATGCCTTCTACAA AGACAACCCCATCCTAAGGGAAGTAGGATAGTTATTA ACAGAGAACATCTTATGATTGATAGACCTTATGTTTTGC TGGCTGTTCTATTTCGTATGTTTTCTGAGCTTGATCGGGT TGCTAGCCATTGCAGGCATTAGACTTCATCGGGCAGCC ATCTACACCGCAGAGATCCATAAAAGCCTCAGCACCAA TCTGGATGTAACCTCAATCGAGCATCAGGTTAAGG ACGTGCTGACACCACTCTCAAGATCATCGGTGATGAA GTGGGCTTGAGGACACCTCAGAGATTCAGTACCTAGT GAAGTTCATCTCTGACAAGATTAAATTCCTTAATCCGG ACAGGGAATACGACTTCAGAGATCTCACTTGGTGATC AACCCGCCAGAGAGAATCAAAATGGATTATGATCAATA CTGTGCAGATGTGGCTGCTGAAGAACTCATGAATGCAT TGGTGAACCTCAACTCTACTGGAGACCAGGCAACCAAT CAGTTCTAGCTGTCTCAAAGGGAACCTGCTCAGGGCC CACTACAATCAGAGGCCAATTCTCAAACATGTCGCTGT CCGTGTGGACTTGTTAATTAAAGTCGAGGTTACAATGTGT CATCTATAGTCACTATGACATCCAGGGAATGTACGGG GGAACCTTACCTAGTGGAAAAGCCTAATCTGAGCAGCAA AGGGTCAGAGTTGTCACACTGAGCATGCACCGAGTGT TTGAAGTAGGTGTTATCAGAAATCCGGGTTTGGGGCT CCGGTATTCCATATGACAACTATCTTGAGCAACCAAGT CAGTAATGATTTCAGCAACTGCATGGTGGCTTTGGGGG AGCTCAAGTTCGACGCCCTCTGTCAAGGGAAGATTCT ATCACAATTCCTATCAGGGATCAGGGAAGGTGTCAG CTTCCAGCTTGTCAAGCTAGGTGTCTGGAAATCCCAA CCGACATGCAATCCTGGGTCCCCCTATCAACGGATGAT CCAGTGATAGACAGGCTTTACCTCTCATCTCACAGAGG CGTTATCGCTGACAATCAAGCAAAATGGGCTGTCCCGA CAACACGGACAGATGACAAGTTGCGAATGGAGACATG CTTCCAGCAGGCGTGAAGGGTAAATCCAAGCACTTT GCGAGAATCCCGAGTGGACACCAATGAAGGATAACAG GATTCTTTCATACGGGGTCTTGTCTGTTGATCTGAGTCT GACAGTTGAGCTTAAATCAAAATGTTTCAGGATTCTG GGCCATTGATCACACAGGTTCAAGGATGGACCTATAC AAATCCAACCAACAATATGATTGGCTGACTATCCC GCCAATGAAGAACCTGGCCTTAGGTGTAATCAACACAT TGGAGTGGATACCGAGATTCAAGGTTAGTCCCAACCTC TTCCTGTTCCAATTAAGGAAGCAGGCAGGACTGCCA TGCCCCAACATACCTACCTGCGGAGGTGGATGGTGATG TCAAACTCAGTTCCAATCTGGTGATTCTACCTGGTCAAG ATCTCCAATATGTTCTGGCAACCTACGATACTTCAGA GTTGAACATGCTGTAGTTTATTACGTTTACAGCCCAAGC CGCTCATTTTCTTACTTTTATCTTTTAGGTTGCCTGTAA GGGGGGTCCCCATTGAATTACAAGTGAATGCTTCACA TGGGACCAAAAACCTCTGGTGCCGTCACCTTCTGTGTGCTT GCGGACTCAGAATCTGGTGGACATATCACTCACTCTGG GATGGTGGGCATGGGAGTCAGCTGCACAGCCACTCGGG AAGATGGAACCAAGCCGAGATAG	45
GC_H_MEASLES_ D8 mRNA Sequence (assumes T100 tail) Sequence Length: 2126	G*GGGAAATAAGAGAGAAAAGAAGAGTAAGAAGAAAT ATAAGAGCCACCATGTCAACACACGAGACCGGATAA ATGCCTTCTACAAAGACAACCCCATCCTAAGGGAAGT AGGATAGTTATTAAACAGAGAACATCTTATGATTGATAG ACCTTATGTTTTGCTGGCTGTTCTATTTCGTATGTTTCTG AGCTTGATCGGGTTGCTAGCCATTGCAGGCATTAGACT TCATCGGGCAGCCATCTACACCGCAGAGATCCATAAAA GCCTCAGCACCAATCTGGATGTAACCTCAATCGAG CATCAGGTTAAGGACGTGCTGACACCACTCTCAAGAT CATCGGTGATGAAGTGGGCTTGAGGACACCTCAGAGAT TCACTGACCTAGTGAAGTTCATCTCTGACAAGATTAAA TTCCTTAATCCGACAGGGAATACGACTTCAGAGATCT CACTTGGTGATCAACCCGCCAGAGAGAATCAAAATGG ATTATGATCAATACGTGACAGATGTGGCTGCTGAAGAA CTCATGAATGCATTGGTGAACTCAACTCTACTGGAGAC CAGGGCAACCAATCAGTTCTAGCTGTCTCAAAGGGAA ACTGCTCAGGGCCCACTACAATCAGAGGCCAATTCTCA AACATGTCGCTGTCCCTGTTGGACTTGTTAAGTCGA GGTTACAATGTGTCATCTATAGTCACTATGACATCCCA GGGAAATGTACGGGGGAACCTACCTAGTGGAAAAGCCT AATCTGAGCAGCAAGGGTCAGAGTTGTCACAACTGAG CATGCACCGAGTGTGTAAGTAGGTGTTATCAGAAATC	46

[illegible]

GC_F_MEASLES_	UCAAGCUUUUGGACCCTTCGUACAGAAAGCUAAUACGAC	69
B3.1	UCACUAUAGGGGAAAUAGAGAGAAAGAAAGAGAGUAAG	
Sequence, NT (5'	AAGAAAUAUAGAGCCACCAUGGGGUCUACAGGUGAA	
UTR, ORF, 3'	CGUCUCUGCCCGUAUUAUAGGCGAGUACUGUUAACUCUC	
UTR)	CAAAACACCCCGCGGUCAAUAUACUCUCCUGGGCAAUUCU	
Sequence Length:	CUAAGAUAGGGGUGAGUAGGAAUAGGAAGUGCAAGCU	
1864	ACAAAGUUAUGACUCGUUCCAGCCAUCAAUUAUAGU	
	CAUAAAUAUUAUGCCCAAUAUAACUCUCCUCAAUAAC	
	UGCACGAGGGUAGAGAUGCGAGAAUACAGGAGACUA	
	CUAAGAACAAGUUUUGGAACCAAUUAGGGAUGCACUU	
	AAUGCAAUGACCCAGAAACAUAAAGGCCGCUACAGCG	
	UAGCUUCAAGUAGGAGACACAAGAGAUUUGCGGGAG	
	UAGUCCUGGCAGGUGCGGCCCUAGGUGUUGCCACAGC	
	UGCUCAGAUAAACAGCCGGCAUUGCACUACCCGGUCC	
	AUGCUGAAACUCUCAGGCCAUCCAGCAUUCUGAGAGCGA	
	GCCUGGAAACUACUAAUACAGGCAAAUUGAGGCAAUACG	
	ACAAGCAGGGCAGGAGAUUAUUGGCGUUCAGGG	
	UGUCCAAAGACUAUAUCAAUAUAGAGCGUAUACCGUCU	
	AUGAACCCAGCUAUCUUGUGAGUACUAAUCGGUCAGAAGC	
	UCGGGCUCAAUUGCUUAGAUACUAACAGAAUCCU	
	GUCAUUAUUUGGCCCCAGCCUACGGGACCCCAUAUCU	
	CGGGAGUAUCUAUCCAGGCUUUAGGUGUAUGCAUUC	
	GGAGGGAUAUACAAUAAGGUGUUAGAAAAGCUCGGA	
	UACAGUGGAGGCGAUUUUACUAGGCAUCUUAGAGAGC	
	AGAGGAUAAAGGCUUCGGAUAACUCACGUCGCACAG	
	AGUCCUACUUCUAUAGUCCUCAGUAUAGCCUAUCCGAC	
	CGUGUCCGAGAUUAAAGGGGGUUAUUGUCCACCGGCUA	
	GAGGGGUGUCUGUACACAUAGGCUUCUCAAAGAGUGG	
	UAUACCAUCUGGCCCAAGUAUGUUGCAACCCAAAGGU	
	ACCUAUCUGCAAUUUUGAGUAGUACUAUGUACUU	
	UCAUGCCAGAGGGGACUGUGUGCAGCCAAAAGUCCUU	
	GUACCCUGAGUGUCCUCUGCUCUCCAAGAUUGCCUCCG	
	GGGUCCACCAAGUCCUGUGCGUGUACUCUGUAUCCG	
	GGUCUUUUGGGAACCGGUUCAUUUAUACAAGGGA	
	ACCUAUAGCCAAUUGUGCAUAUUCUUUGUAAGU	
	GUUACACAACAGGUAAGUAUUUAUUAUACAAGACCCUGA	
	CAAGAUCUAACAUAUUGCUGCCGAUCGCGGCCG	
	GUAGUCGAGGUGAACAAGCGUGACAUUACAGUCGGGA	
	GCAGGGAGUAUCAGACGCGUGUGUAUUGCAAGAAU	

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	UGACCUCGGUCCUCCCAUAUCAUUGGAGAGGUUGGAC GUAGGGACAAAUUCUGGGGAAUGCAAUUGCCAAAUUG GAGGAUGCCAAGGAUUUGUUGGAAUCAUCGGACCAG AUAUUGAGAAGUAUGAAAGGUUUUUCGAGCACUAGC AUAUCUACAUCUUGAUUUGCAGUGUGUCUUGGAGGG UUGAUAGGGAUCCCAUUUAAUUGUUGCUGCAGG GGGCGUUGUAAACAAAAGGAGAAACAAGUUGGUAUG UCAAGACCAGGCCUAAAGCCUAGCCUUACAGGAACAU CAAAUCCUAUGUAAGAUCGUUUUGAUUAUAAUAGG CUGGAGCCUCGGUGGCCAAGCUUCUUGCCCUUGGGC CUCUCCCCAGCCCCUCCUCCUUCUGCACCCGUACC CCCGUGGUCUUUGAAUAAAGUCUGAGUGGGCGGC	
GC_F_MEASLES_ B3.1 ORF Sequence, NT	AUGGGUCUCAAGGUGAACGUCUCUGCCGUAUUC AUGG CAGUACUGUUAACUCUCCAAACACCCGCCGUCAAAU UCAUUGGGGCAUUCUCUAAGAUAAGGGUAGUAGG AAUAGGAAGUGCAAGCUACAAGUUUAUGACUCGUUC CAGCCAUCAAUCAUUGUCAUAAAAUUAUUGCCCAAU AUAACUCUCCUCAAUAACUGCACGAGGGUAGAGAUG CAGAAUACAGGAGACUACUAAGAACAGUUUUGGAAC CAAUUAAGGGAUGCAUUAAUGCAAUGACCAGAACAU AAGGCCGGUUCAGAGCGUAGCUUCAAGUAGGAGACAC AAGAGAUUUGCGGAGUAGUCCUGGCAGGUGCGGCC UAGGUGUUGCCACAGCUGCUCAGAUAAAGCCGGCAU UGCACUUCACCGGUCCAUGCUGAACUCACAGGCCAUC GACAAUCUGAGAGCGAGCCUGGAAACUACUAAUCAGG CAAUUGAGGCAUACAGACAGCAGGGCAGGAGAUAGA UAUUGGCGUUCAGGGUGUCCAAGACUACAUCAAUA AUGAGCUGAUACCGUCUAUGAACAGCUAUCUUGUGA UCUAUUCGGUCAGAAGCUCGGGCUCAAUUGCUUAGA UACUAUACAGAAAUCCUGUCAUUAUUUGGCCAGCC UACGGGACCCCAUAUCUGCGGAGAUUAUUAUCCAGGC UUUGAGUUAUGCACUUGGAGGAGAUUAUCAAUAGGU GUUAGAAAGCUCGGAUACAGUGGAGGCGAUUUACU AGGCAUCUUAGAGAGCAGAGGAUUAAGGCUCGGAU AACUCACGUCGACACAGAGUCCUACUUAUAGUCCUC AGUAUAGCCUAUCCGACGCUUCCGAGAUUAAGGGG UGAUUGUCCACCGGCUAGAGGGGUCUCGUACAACAU AGGCUCUCAAGAGUGGUUAACACUGUGCCCAAGUAU GUUGCAACCAAGGUACCUUAUCUCGAAUUUUGAUG AGUCAUCUAGUACUUUUAUGCCAGAGGGGACUGUGU GCAGCCAAAUGCCUUGUACCCGAUGAGUCCUCUGCU CCAAGAAUGCCUCCGGGGUCCACCAAGUCCUGUGCU CGUACACUCGUAUCCGGGUCUUUUGGGAAACGGUUA UUUUAUCAAGGGGAACCUAAUAGCCAAUUGUGCAUC AAUUCUUUGUAAGUGUUAACAACAAGGUACGAUUAU UAAUCAAGACCUCGACAAGAUCCUAACAUACUUGCU GCCGAUCGUCGCCCCGUAGUCGAGGUGAACGGCGUGA CCAUCCAGUCGGGAGCAGGAGGUUAUCAGACGCGU GUACUUGCACAGAAUUGACCUCGGUCCUCCAUUAUA UUGGAGAGGUUGGACGUAGGACAAAUUGGGGAU GCAAUUGCCAAAUUGGAGGAUGCCAAGGAUUUGUUG GAAUCAUCGGACCAGAUUAUGAGAUAUGAAAGGU UUAUCGAGCACUAGCAUAGUCUACUCCUGAUUGCAG UGUGUCUUGGAGGGUUGAUAGGGAUCCCAUUAUA UAUGUUGCUGCAGGGGCGUUGUAACAAAAGGGAG AACAAUUGGUAUGUCAAGACCAGGCCUAAAGCCUGA CCUUAACAGGAACAUAUAAUCCUAUGUAAGAUCGCUU UGA	70
GC_F_MEASLES_ B3.1 mRNA Sequence (assumes T100 tail) mRNA Sequence Length: 1925	G*GGGAAUAAGAGAGAAAAGAAGAGUAAGAAGAAA UUAUAGAGCCACCAUGGGUCUCAAGGUGAACGUCUCU GCCGUUAUUGGAGUACUUAACUCUCCAAACAC CCGCCGUCAAAUUAUUGGGGCAUUCUCUAAGAU AGGGGUGAUGAGAAUAGGAAGUGCAAGCUACAAGU UAUGACUCGUUCCAGCCAUCAAUUAUGUCAUAAA UUAUAGCCCAAUAUAACUCUCCUCAAUAACUGCACGA GGGUAGAGAUUGCAGAAUACAGGAGACUACUAAAGAA CAGUUUUGGAACCAAUUAAGGAUGCACUUAUUGCAA UGACCCAGAACAUAAAGCCGGUUCAGAGCGUAGCUUC AAGUAGGAGACACAAGAGAUUUGCGGGAGUAGUCCU GGCAGGUGCGGCCUAGGUGUUGCCACAGCUGCUCAG AUAACAGCCGGCAUUGCACUUCACCGGUCCAUUGCUA ACUCUCAGGCCAUCGACAAUCUGAGAGCGAGCCUGGA AACUACUAAUACAGGCAUUGAGGCAUUCAGACAAGCA GGGCAGGAGAUGAUUAUUGGCUUUCAGGGUGUCCAA GACUACAUCAAUAUAGAGCUGAUACCGUCUAUGAACC	71

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	AGCUAUCUUGGAUCUAAUCGGUCAGAAGCUCGGGCU CAAAUUGCUUAGAUACUUAACAGAAUCCUGUCAUU AUUUGGCCCCAGCCUACGGGACCCCAUUAUCUGCGGAG AUAUCUAUCCAGGCUUUGAGUUUAUGCAUUGGAGGA GAUAUCAUAAGGUGUUAAGAAAGCUCGGAUACAGU GGAGGCGAUUUACUAGGCAUCUAGAGAGCAGAGGA AUAAGGCUCGGAUAAUCACGUCGACACAGAGUCCU ACUUCUAUAGUCCUCAGUAUAGCCUAUCCGACGUCGUC CGAGAUUAAGGGGUGAUUUGCCACCGGCUAGAGGG GGUCUCGUACAACAUAAGGCUCUCAAGAGUGGUAUACC ACUGUGCCCAAGUAUGUUGCAACCAAGGGUACCUUA UCUCGAAUUUUGAGAGUCAUCAUGUAUUUUAUGCC AGAGGGGACUGUGGAGCAGCAAAUUGCCUUGUACCCG AUGAGUCCUCUGCUCCAAGAAUGCCUCCGGGGUCCA CCAAGUCCUGUGCUCGUACACUCGUUACCGGGUCCUU UGGGAACCGGUUCAUUUAUACAAGGAACCUAAU AGCCAAUUGUGCAUCAAUUCUUUGUAAGUGUUAAC AACAGGUACGAUUAUUAUACAGACCCUGACAGAUC CUAACAUACAUAUGCUGCCGAUCGUCGCCGGUAGUCG AGGUGAACGGCGUGACCAUCCAGUCGGGAGCAGGAG GUUCCAGACGCGUGUACUUGCACAGAAUUGACCTUC GGUCCUCCCAUAUCAUUGGAGAGGUUGGACGUAGGG ACAAUUCUGGGGAUUGCAUUGCCAAUUGGAGGAU GCCAAGGAUUUGUUGGAUUAUCGACAGAUUUG AGAAGUAUGAAAGGUUAUCGAGCAUAGCAUAGUC UACAUCCUGAUUGCAGUGUGCUUGGAGGGUUGAU GGGAUCCCCACUUUAUUGUUGCUGCAGGGGGCGUU GUAAACAAAAGGGAGAAAGUUGGUUAUGUCAAGAC CAGGCCUAAAGCCUGACCUUACAGGAACAUAUAAUUC CUAUGUAAGAUCGCUUUGAUGAUAAUAGGCGUGAGC CUCGGUGGCCAAGCUUUCUUGCCCUGGGCCUCCCC CAGCCCCUCCUCCUUCUGCACCUGUACCCCCGUGG UCUUUGAAUAAAGUCUGAGUGGGCGGCAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAUUCUAG	
GC_F_MEASLES_ D8 Sequence, NT (5' UTR, ORF, 3' UTR) Sequence Length: 1864	UCAAGCUUUUGGACCCUGUACAGAAGCUAAUACGAC UCACUAUAGGGAAUAAGAGAGAAAGAAGAGUAAG AAGAAUAUAAGAGCCACCAUGGGUCUACAAGGUGAA CGUCUCUGUCAUAUUAUGGCAUACUGUUAACUCUU CAAACACCCACCGGUCAAUCCAUUGGGGCAUUCUU CUAAGAUAGGGGUGUAGGGGUAGGAAGUGCAAGCU ACAAAGUUAUGACUCGUUCCAGCCAUCAUUAUAGU CAUAAAGUUAUUGCCCAAUAUAUCUCCUACAACAAU UGCACGAGGGUAGGGAUUGCAGAAUACAGGAGACUA CUGAGAACAGUUCUGGAACCAUUAAGAUAGCACUU AAUGCAAUGACCCAGAAUAUAGACCGGUUCAGAGU GUAGCUUCAAGUAGGAGACACAAGAGAUUUGCGGGA GUUGUCCUGGCAGGUGCGGCCUAGGCGUUGCCACAG CUGUCUCAAUAACAGCCGGUUAUGCACUUCACAGUC CAUGCUGAACUCUCAAGCCAUCCGACAAUCUGAGAGCG AGCCUAGAAACUACUAUACAGGCAAUUGAGGCAAUCA GACAAGCAGGGCAGGAGAUUAUUGGCGUUCAGG GUGUCCAAAGACUACAUCAAUAAUAGCUGAUACCGUC UAUGAAUCAAUUAUCUUGUUAUUAUUGCGCCAGAA GCUAGGGCUCAAUUGCUCAGAUACUAUACAGAAUUC CUGUCAUUAUUGGCCCCAGCUUACGGGACCCCAUUA CUGCGGAGAUUAUUAUCCAGGCUUUGAGCUAUGCGCU UGGAGGAGAUUAUUAAGGUGUUGGAAAGCUCGG AUAACUGGAGGUGAUUACUGGGCAUUAUAGAGAG CAGAGGAUUAAGGCCCGGAUAAUCACGUCGACACA GAGUCCUACUUAUUGUACUCAGUAUAGCCUAUCCGA CGCUAUCGAGAUUAAGGGGGUGAUUUGCCACCGGCU AGAGGGGUCUCGUACAACAUAAGGCUUCUACAGAGUG GUUAUACACUGGCCAAGUAUGUUGCAACCAAGGG UACCUUAUCUGAAUUUUGAUGAGUACAUAUGCACUU UCAUGCCAGAGGGGACUGUGGAGCCAGAAUUGCCUU GUACCCGAGUAGUCUCUGUCUCCAGAAUUGCCUCCGG GGGUCCAUAAGUCCUGUGCUCGUACAUCGUUAUCCG GGUCUUUCGGGAACCGGUUCAUUUAUACAAGGGGA ACCUAAUAGCCAAUUGUGCAUCAAUCCUUUGCAAGUG UUAACAACAGGAACAUAUUAUUAAGACCCUGAC AAGAUCUAAACAUAUUAUGCUGCCGAUACUUGCCGG UGGUCGAGGUGAAUGGCGUGACCAUCCAGUCGGGA GCAGGAGGUUCCGGACGCGUGUACUUGCACAGGAU UGACCUUGGUCUCCCAUUAUUAUUGGAGAGGUUGGAC	72

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	GUAGGGACAAAUCUGGGGAAUGCAAUUGCUAAGUUG GAGGAUGCCAGGAUUGUUGGAGUCAUCGGACCAG AUAUUGAGGAGUAUGAAAGGUUUAUCGAGCACUAGU AUAGUUUACAUCUGAUUGCAGUGUGUCUUGGAGGA UUGAUAGGGAUCCCCGCUUUAUUGUUGCUGCAGG GGGCGUUGUAACAAGAGGGAGAACAGUUGGUUUG UCAAGACCAGGCCUAAAGCCUGAUCUUACAGGAACAU CAAAUCCUAUGUAAGGUCACUCUGAUGAUAAUAGG CUGGAGCCUCGGUGGCCAAGCUUCUUGCCCCUUGGGC CUCCCCCAGCCCCUCUCCCCUCCUGCACCCGUACC CCCGUGGUCUUUGAAUAAAGUCUGAGUGGGCGGC	
GC_F_MEASLES_ D8 ORF Sequence, NT	AUGGGUCUCAAGGUGAACGUCUCUGUCAUAUUAUG GCAGUACUGUUAACUCUCAAACACCCACCGGUCAAA UCCAUUGGGGCAAUCUCUCUAAAGUAGGGGUGGUAG GGGUAGGAAGUGCAAGCUACAAGUUUAUGACUCGUU CCAGCCAUCAAUCAUUGUCUAAAGUUAAUGCCCAA UAUAAUCUCUCCAACAUAUUGCAGGAGGUAGGGAUU GCAGAAUACAGGAGACUACUGAGAACAGUUCUGGAA CCAAUUAGAGAUAGCUAAUAGCAAUGACCCAGAAUA UAAGACCGGUUCAGAGUGUAGCUUCAAUAGGAGAC ACAAGAGAUUUGCGGGAGUUGUCUUGGCAGGUGCGG CCCUAGGCGUUGCCACAGCUGCUCAAUAACAGCCGG UAUUGCACUUCACCAGUCCAUGCUGAACUCUCAAGCC AUCGACAAUCUGAGAGCGAGCCUAGAAACUACUAAUC AGGCAAUUGAGGCAUUCAGACAAGCAGGGCAGGAGA UGAUAUUGGCUUGUUCAGGGUGUCCAAGACUACAUC AUAAUGAGCUGAUACCGUCUAGAAUCAAUAUCUU GUGAUUUAAUCGGCCAGAAGCUAGGGCUCAAUUGC UCAGAUACUAUACAGAAUCCUGUCAUUAUUUGGCC CAGCUUACGGGACCCCAUAUCUGCGGAGAUUUAUC CAGGCUUUGAGCUAUGCGCUUGGAGGAGAUUCAAU AAGGUGUUGGAAAAGCUCGGAUACAGUGGAGGUGAU CUACUGGGCAUCUUAGAGAGCAGAGGAUAAAGGCC GGAUAACUCACGUCGACACAGAGUCCUUAUUAUUGU ACUCAGUAUAGCCUUAUCCGACGCUAUCGAGAUUAAG GGGGUGAUUUGUCCACCGGCUAGAGGGGUGUCUGUACA ACAUAGGCUCUCAAGAGUGGUUAUACCUUGGCCCAA GUAUGUUGCAACCAAGGGUACCUUAUCUGAAUUUU GAUGAGUCAUCAUGCACUUUAUGCCAGAGGGGACUG UGUGCAGCCAGAAUGCCUUGUACCGAUGAGUCCUCU GCUCCAAGAAUGCCUCCGGGGUCCACUAAGUCCUGU GCUUGUACACUCGUAUCCGGGUCUUUCGGGAACCGGU UCAUUUUUAUCACAGGGGAACUAAUAGCCAAUUGUC AUCAAUCCUUUGCAAGUGUUAACAACAGGAACAUC AUUAUACAAGACCCUGACAAGAUCCUAAUAUACAUUG CUGCCGAUCACUGCCCGGUGGUCGAGGUGAAUGGCGU GACCAUCCAAGUCGGGAGCAGGAGUAUCCGGACGCU GUGUACUUGCACAGGAUUGACCUUGGUCUCCCAUAU CUUUGGAGAGGUUGGACGUAAGGACAAUUCUGGGGA AUGCAAUUGCUAAGUUGGAGGAUGCCAAGGAUUUGU UGGAGUCAUCGGACCAGAUUAUUGAGGAGUAUGAAAG GUUUUUCGAGCACUAGUAUAGUUACAUCUGAUUG CAGUGUGUCUUGGAGGAUUGAUAGGUAUCCCGCUU UAAUAUGUUGCUGCAGGGGGCGUUGUAACAAGAGG GAGAAACAAGUUGUAUGUCAAGACCAGGCCUAAAGCC UGAUCUUAACAGGAACAUCAAAUCCUAUGUAAGGUC ACUCUGA	73
GC_F_MEASLES_ D8 mRNA Sequence (assumes T100 tail) Sequence Length: 1925	G*GGGAAUUAAGAGAGAAAAGAAGUAAGAAGAAA UUAAGAGCCACCAUGGGUCUCAAGGUAAACGUCUCU GUCAUAUUCUAGGCAUACUGUUAACUCUCAAACAC CCACCGGUCAAAUCUUAUGGGGAAUCUCUUAAGAU AGGGUGUGUAGGGUAGGAAGUGCAAGCUACAAGU UAUGACUCGUUCAGCCAUCAUUAUAGUCAUAAAG UUAUUGCCAAUAUAACUCUCCUCAACAUAUGCACGA GGGUAGGGAUUGCAGAAUACAGGAGACUACUGAGAA CAGUUCUGGAACCAAUAAGAGUAGCACUUAUUGCAA UGACCCAGAAUAUAAGACCGGUUCAGAGUGUAGCUUC AAGUAGGAGACACAAGAGAUUUGCGGAGUUGUCCU GGCAGGUGCGGCCUAGGCGUUGCCACAGCUGUCUCAA AUAACAGCCGGUAUUGCACUUCACCAGUCCAUUGCUA ACUCUCAAGCCAUUCGACAAUCUGAGAGCGAGCCUAGA AACUACUAAUCAGGCAUUGAGGCAUUCAGACAAGCA GGGCAGGAGAUUAUUGGUGUUCAGGGUGUCCAA GACUACAUAUAUAGAGCUGAUACCGUCUAGAAUUC AACUAUCUUGUAUUUAUUGGCCAGAGCUGAGGGC	74

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	UCAAAUUGCUCAGAUACUAUACAGAAAUCUGUCAUU AUUUGGCCCCAGCUUACGGGACCCCAUAUUCUGCGGAG AUUAUCUAUCCAGGCUUUGAGCUAUGCGCUUGGAGGA GAUAUCAAAUAGGUGUUGGAAAAGCUCGGAUACAGU GGAGGUGAUUCUUGGGCAUCUUAGAGAGCAGAGGA AUAAAGGCCCGGAUAAUCACGUCGACACAGAGUCCU ACUUCAUUGUACUCAGUAUAGCCUAUCCGACGCUAUC CGAGAUUAAAGGGGUGAUUGUCCACCGGCUAGAGGG GGUCUCGUACAACAAGGCUUCUAGAGUGGUUAUAC ACUGUGCCCAAGUAUGUUGCAACCCAAGGGUACCUUA UCUCGAAUUUUGAUGAGUCAUCAUGCAUUCUUGCC AGAGGGGACUGUGUGCAGCCAGAAUGCCUUGUACCCG AUGAGUCCUCUGUCUCCAAGAAUGCCUCCGGGGGUCCA CUAAGUCCUGUGUCGUACACUCGUUACCGGGUCCUU CGGGAACCGGUUCAUUUUAUCACAGGGGAACCUAAUA GCCAAUUGUGCAUCAAAUCUUUGCAAGUGUACACAA CAGGAACAUAUUAUUAUACAGACCUGACAAGAUCCU AACAUACAUAUGCUGCCGAUCACUGCCCUGGUGUCGAG GUGAAUGGCGUGACCAUCUAGUCGGGAGCAGGAGG UAUCCGGACGUCUGUAUUGCACAAGGAUUGACCUCG GUCCUCCCAUAUCUUUGGAGAGGUUGGACGUAGGGAC AAAUCUGGGGAUGCAAUUGCUAAGUUGGAGGAUGC CAAGGAAUUGUUGGAGUCAUCGACCAAGUAUUGAG GAGUAUGAAAGGUUAUUCGAGCAUAGUAUAGUUUA CAUCCUGAUUGCAGUGUGUCUUGGAGGAUUGAUAGG GAUCCCCGCUUUAUAUUGUUGCUGAGGGGCGUUGU AACAGAAGGGAGAACAAGUUGGUUUGUCAAGACCA GGCCUAAAGCCUGAUCUUAACAGGAACAUCAAAUCU AUGUAAGGUCACUCUGAUGAUAAUAGGCUAGGACCU CGGUGGCCAAGCUUCUUGCCCCUUGGGCCUCCCCCA GCCCUCUCCCCUUCUGCACCCGUAUCCCGUGGUC UUUUAAUAAAGUCUGAGUGGGCGCAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAUCUAG	
GC_H_MEASLES_ B3 Sequence, NT (5' UTR, ORF, 3' UTR) Sequence Length: 2065	UCAAGCUUUUGGACCCUCGUACAGAAGCUAAUACGAC UCACUAUAGGGAAAUAAGAGAGAAAAGAGAGUAAG AAGAAUAUAAGAGCCACCAUGUCACCGCAACGAGAC CGGAUAAUAGCCUUUAUCAAAGAUAAACCUUAUCCCA AGGGAAGUAGGAUAGUUAUUAACAGAGAAACUUAU UGAUUGACAGACCCUAUGUUCUGCUGGCUUUCUGUU CGUCAUUGUUUCUGAGCUUGAUCGGAUUGCUGGCAU UGCAGGCAUUAAGACUUAUCGGGCAAGCAUACACC GCGGAGAUCCAUAAGCCUCAGUACCAAUUCUGGAUG UGACUAAUCCAUUCGAGCAUCAGGUCAAGGACGUGCU GACACCAUCUUAUAAAUCAUCGGGGAUGAAGUGGGC CUGAGAACCUCAGAGAUUCACUGACCUAGUGAAAU UCAUCUCGGACAAGAUUAAAUUCCUAAUCCGGAUAG GGAGUACGACUUCAGAGAUUCACUUGGUGCAUCAA CCGCCAGAGAGAUCAAUAAGAUUAUGAUCAAUACU GUGCAGAUUGGCGUGCUGAAGAGCUAUGAAUGCAU UGGUGAACUCAACUCUACUGGAGACCAGAACAAACAC UCAGUUCUAGCUGUCUCAAAGGGAACUGCUCAGGG CCCACUACAAUCAGAGGUCAAUUCUAAACAUUGUCG UGUCCUUGUUGGACUUGUACUUAAGGUCGAGGUACA AUGUGUCAUCUAUAGUCACUAUGACAUCCAGGGAAU GUAUGGGGGAACCUACCUAGUUGAAAAGCCUAAUCU GAACAGCAAAGGGUCAGAGUUGUCAACUAGGCAU GUACCGAGUGUUUGAAGUAGGUGUGAUCAGAAACCC GGGUUUGGGGCUCCGUGUUCUUAUAGACAAACUA UUUUGAGCAACAGUCAGUAUUGGUCUGGCAACUGU AUGGUGGCUUUGGGGAGCUCAAACUCGACGCCUUU GUCACGGGACGAUUCUAUUAUUAUCCUUAUCAGGG AUCAGGGAAGGUGUCAGCUUCCAGCUCGUCAGCUG GGUGUCUGGAAAUCCCAACGACAUCAUCCUGGG UCCCUUAUACACGGAUGAUCCAGUGGUAAGACAGGCU UUAUCUUCUACUUCACAGAGGUGUCAUCGUGACAAU CAAGCAAAUUGGGCUGUCCGACAACGACAGAGAUG ACAAGUUGCGAAUUGGAGACAUGCUUCCAGCAGGCGUG UAAAGGUAAAAUCCAGCACUCUGCGAGAAUCCGAG UGGGUACCAUUGAAGGAUAACAGGAUUCUUAUAC GGGGUCCUGUCUGUUGAUCUGAGUCUGACGGUUGAG CUUAAAAUCAAAUUGCUUCCGGGAUUCGGGCCAUUG AUCACACACGGCUCAGGGAUGGACCUAUAACAAUCCA ACUGCAACAAGUGUAUUGGCUAGCUAUCUCCGCCAAU GAGAAUUCUAGCCUUAAGCGUAUAUACAACAUUGGA	75

TABLE 13-continued

Description	Sequence	SEQ ID NO :
	GUGGAUACCGAGAUUCAAGGUUAGUCCCAACCUUUC ACUGUCCCAUUAAGGAAGCAGGCGAAGACUGCCAUG CCCAACAUACCUACCUACCGGAGGUGGACGGUGAUGU CAAACUCAGUUCACACCGUGGAUUCUACCGUGCAA GAUCUCCAAUUGUUUGGCAACCUACGAUACCUCCA GGGUUGAGCAUGCUGUGUUUAUACGUUUACAGCC CAAGCCGCUCAUUUUUCUACUUUUUACCUUUUAGGU GCCUAUAAGGGGUGCCAAUCGAACUACAAGUGGAA UGCUUACAUGGGAUCAAACCUUGGUGCCGUCACU UCUGUGUGCUUGCGGACUCAGAAUCGUGGACUUU CACUCACUCUGGGAUGGUGGGCAUGGGAGUCAGCUGC ACAGCUACCCGGGAAGAUGGAACCAUCGAGAUAAU GAUAAUAGGCUUGGAGCCUGCGUGGCAAGCUUUCUGC CCCUUGGGCCUCCCCCAGCCCCUCCUCCUUCUGC ACCCGUACCCCGUGGUCUUUGAAUAAAGUCUGAGUG GCGCGC	
GC_H_MEASLES_ B3 ORF Sequence, NT	AUGUCACCGCAACGAGACCGGAUAAAUGCCUUCUACA AAGAUAACCCUUAUCCCAAGGGAAGUAGGAUAGUUA UUAACAGAGAACAUUUUAGUUGACAGACCCUAUG UUCUGCUGGCUGUUUCUGUUCGUCAUGUUUCUGAGCUU GAUCGGAUUGCUGGCAUUGCAGGCAUAGACUUCA UCGGGCAGCCAUCUACACCGCGGAGAUCCAUAAGC CUCAGUACCAUUCUGGAUGUGACUAAUCUACGAGC AUCAGGUCAAGGACUGUGUACACACUUAUAAAAU CAUCGGGAUGAAGUGGGCCUGAGAACCUCAGAGA UUCACUGACCUAGUGAAAUCAUCUGGACAAGAUUA AAUUCUUAUCCGGAUAGGGAGUACGACUUCAGAG AUCUCACUUGGUGCAUACCCCGCAGAGAGGAUCAA ACUAGAUUAUGAUCAAUACUGUGCAGAUUGGCGUC UGAAGAGCUCUAGAAUGCAUUGGUGAACUACACUCU ACUGGAGACCAGAACACACUACAGUUCUAGCUGUC UCAAAGGGAACUGUCAGGGCCACUACAACAGAG GUCAAUUCUCAAACAUUGCGUGUCCUUGUUGGACUU GUACUUAGGUCGAGGUUACAAGUGUCUACUUAUGU CACUAGACAUCCAGGGAUUGUUGGGGAACCUAC CUAGUUGAAAAGCCUAAUCUGAACAGCAAGGGGUA GAGUUGUCACAACUGAGCAUGUACCGAGUGUUUGAA GUAGGUGUGAUCAGAAACCCGGUUUGGGGGUCUG GUGUCCAUUAGACAAACUUAUUUGAGCAACAGUCA GUAAUGGUCUGGCAACUGUUGGUGGUUUUGGGG AGCUCAAACUCGAGCCUUUGUCACGGGACGAUUC UAUCUAUAUCCUUAUCAGGGAUCAGGGAAGGUGU CAGCUUCAGCUCGUCAGCUGGGUGUCUGGAAUCC CCAACCGACAUGCAUCCUGGUUCCCUUAUACACGG AUGAUCCAGUGGUAGACAGGCUUUAACUUCUACUCA CAGAGGUGUCAUCGUGACAAUACAGCAAAUUGGCU GUCCCGACAACACGAACAGAUAGAUUGCGAAUGG AGACAUGCUUCAGCAGGCGUGUAAAGGUAAAUCCA AGCACUCUGCGAGAAUCCGAGUGGGUACCAUUGAAG GAUAACAGGAUUCUUAUACGGGUCUGUCUGUUG AUCUGAGUCUGACGGUUGAGCUUAAAUCAAAUUG CUUCGGGAUUCGGGCCAUUGAUCACACCGGCUAGG GAUGGACCUAUAACAAUCCACUGCAACAAUGUGUAU UGGCUGACUUAUCCGCCAAUGAGAAUUCUAGCCUAG GCGUAAUACAACAUUGGAGUGGAUACCGAGAUUCA AGGUUAGUCCCAACCUUUCACUGUCCCAUUAAGGA AGCAGGCGAAGACUGCCAUUGCCCAACUACCUACCU GCGGAGGUGGACGGUGAUGUCAAUCAGUUCACAC UGGUGAUUCUACUGGUCAGAUUCCAAUUGUUU UGGCAACCUACGAUACUCCAGGUGUAGCAUGCUGU GGUUUAUUAACGUUACAGCCAGCGCUCAUUUUCU UACUUUUUACCUUUUAGGUUGCCUUAUAAAGGGGUC CCAUUCGAACUACAAGUGGAUUGCUUACAUGGGAU AAAAAUCUGGUGCGUCACUUCUGUGUGCUGCGGA CUCAGAAUCCGGUGGACUUAUCACUACUUGGGAUG GUGGGCAUGGGAGUCAGCUGCACAGCUACCCGGGAAG AUGGAACCAUUCGAGUAA	76

TABLE 13-continued

Description	Sequence	SEQ ID NO:
GC_H_MEASLES_ B3 mRNA Sequence (assumes T100 Tail) Sequence Length: 2126	G*GGGAAUAAGAGAGAAAAGAGAGUAAGAAGAAA UAUAAGAGCCACCAUGUCACCGCAACGAGACCGGAUA AAUGCCUUCUACAAAGAUAAACCUUAUCCCAAGGGAA GUAGGAUAGUUAUUAACAGAGAACAUUUUAUGAUUG ACAGACCCUAUGUUUCUGCUGGCUGUUCUGUUCGUCAU GUUUCUGAGCUUGAUCGGAUUGCUGGCAAUUGCAGG CAUUAAGACUUCACUCGGGCAGCCAUUCACCCGCGGAG AUCCAUAAAAGCCUCAGUACCAAUUGGAUGUGACUA ACUCCAUUCGAGCAUCAGGUCAAGGACGUGCUGACACC ACUCUUUAAAUAUCUGGGGAUGAAGUGGGCUGAG AACACCUCAGAGAUUCACUGACCUGAGAAAUUCAUC UCGGACAAGAUUAAAUUCUUAUCCGGAUAGGGAG UACGACUUCAGAGAUCACUUGGUGCAUACACCCGC CAGAGAGGAUCAAACUAGAUUAUGAUCAAUACUGUG CAGAUGUGGCUGCUGAAGAGCUCAUGAAUGCAUUGG UGAACUCAACUCUACUGGAGACCAGAACACCAUCA GUUCCUAGCUGUCUCAAAGGAAACUGUCAGGGCCC ACUACAAUCAGAGGUCAAUUCUCAAACAUUGCGUGU CCUUGUUGGACUUGUAUUAGGUCGAGGUUACAAUG UGUCAUCUAUAGUCACUAUGACAUCACAGGGAUGUA UGGGGGAACCUACCUAGUUUAAAAGCCUAAUUCGAAC AGCAAAGGGUCAGAGUUGUCACAACUGAGCAUGUACC GAGUGUUUGAAGUAGGUGUAUCAGAAACCCGGGU UGGGGGCUCCGGUGUUCCAUAUGACAACUAUUUG AGCAACCAAGUCAGUAAUGGUCUCGGCAACUGUAUGGU GGCUUUGGGGAGCUCAAACUCGCAAGCCUUUGUCAC GGGGACGAUUCUAUCAAUAAUCCUAUACAGGAUACAG GGAAAGGUGUCAGCUUCAGCUCGUAAGCUGGGUGU CUGGAAAUCCCCAACCGACAUGCAAUCUGGGUCCCC UUAUCAACGGAUGAUCAGUGGUAGACAGGCUUUAAC UCUCUUCUACAGAGGUGUACUCGUGACAUAACAGC AAAAUGGGCUGUCCCGACAACAGAAAGAUAGACAAG UUGCGAAUGGAGACAUUCUCCAGCAGGCGUGUAAA GGUAAAUAUCCAGCACUCUGCGAGAAUCCCGAGUGGG UACCAUUGAAGGAUAAACAGGAUUCUUAUACGGGG UCCUGUCUGUUUGAUUCUGAGUCUGACGGUUGAGCUUA AAAUCAAAAUGUCUUCGGGAUUCGGGCCAUUGAUACAC ACACGGCUCAGGGAUGGACCUAUACAAUCCAAUCUGC AACAAUUGUAUUGGUGUACUUAUCCGCCAAUGAGA AAUCUAGCCUUAAGGCGUAUAACACAUAUGGAGUGG AUACCGAGAUUAAGGUUAGUCCCAACCUUUCACUG UCCCAAUUAAGGAAGCAGGCGAAGACUGCCAUGCCCC AACAUACCUACUGCGGAGGUGGACGGUGAUGUCAA CUCAGUCCAAACUGGUGAUUCUACCGGUGCAAGAUUC UCCAAUAGUUUUGGCAACCUACGAUACCUCCAGGGU UGAGCAUGCUGUGGUUAUUAUCGUUAUACGCCAAGC CGCUCAUUUUCUUAUUUUAUCCUUUAGGUUGCCUA UAAAGGGGGUCCCAAUCGAACTUACAAGUGGAUUGCU UCACAUUGGGAUCAAACUUCUGGUGCCGUCAUUCUG UGUGCUUGCGGACUCAGAAUCCGGUGGACUUAUCACU CACUCUGGGAUGGUGGGAUGGAGUACGUCGACAG CUACCCGGGAAGAUUGAACCAAUCGAGAUAAUGAUA AUAGGCGUGAGCCUCGUGGGCAAGCUUCUUGCCCCU UGGGCCUCCCCCAGCCCCUCCUCCCCUUCUGCACCC GUACCCCUGGUGCUUUGAAUAAAGUCUGAGUGGGCG GCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAUCUAG	77
GC_H_MEASLES_ D8 Sequence, NT (5' UTR, ORF, 3' UTR) Sequence Length: 2065	UCAAGCUUUUGGACCCUCGUACAGAAGCUAAUACGAC UCACUAUAGGGGAAUAAGAGAGAAAAGAGAGUAAG AAGAAAUAUAGAGCCACCAUGUCACCACAACGAGAC CGGAUAAUUGCCUUUAUCAAAGACAACCCCAUCCUA AGGGAAGUAGGAUAGUUAUUAACAGAGAACAUUUUA UGAUUGAUAGACCUUAUGUUUUGCUGGCUGUUCUAU UCGUCAUGUUUCUGAGCUUGAUCGGGUUGCUAGCCAU UGCAGGCAUUAAGACUUCACUGGGCAGCCAUUACACC GCAGAGAUCCAUAAGGCCUCAGCACCAAUUCUGGAUG UAACUAAUCUAAUUCGAGCAUCAGGUUAAGGACGUGCU GACACCACUCUUAAGAUCAUCGUGAUGAAGUGGGC UUGAGGACACCUAGAGAUUCACUGACCUGUGAAGU UCAUCUCUGACAAGAUUAAAUUCCUAAUCCGGACAG GGAAUACGACUUCAGAGAUUCACUUGGUGUAUCAAC CCGCCAGAGAGAAUCAAUUGGAUUAUGAUCAAUAC UGUGCAGAUUGGCUGUGAAGAACUCAUGAAUGCA UUGGUGAACUCACUCUACUGGAGACCAGGCAACCA	78

TABLE 13-continued

Description	Sequence	SEQ ID NO:
	AUCAGUCCUAGCUGUCUCAAAGGGAAACUGCUCAGG GCCACUACAACAGAGGCCAAUUCUCAAACUAGUCG CUGUCCUGUUGGACUUGUAUUUAAGUCGAGGUUAC AAUGUGUCAUCUAUAGUCACUAUGACAUCACAGGGAA UGUACGGGGAAACUUAACUAGUGGAAAAGCCUAAUC UGAGCAGCAAAGGGUCAGAGUUGUCACAACUGAGCA UGCACCGAGUGUUUGAAGUAGGUGUUAUCAGAAAUC CGGUGUUUGGGGUCGCGUAUUCUAUAGACAACUA UCUUGAGCAACCGAGUCAGUAUUAUUCAGCAACUGC AUGGUGGCUUUGGGGAGCUCAGUUCGAGCCUCU GUCACAGGGAAGAUUCUACAACUCCUACAGGG AUCAGGGAAGGUGUCAGCUUCCAGCUUGUCAAGCUA GGUGUCUGGAAAUCCCAACCGACAUGCAUCCUGGG UCCCCUUAACCGAUGAUCCAGUGAUGACAGGCU UUACCUCAUCUACAGAGGCGUUAUCGUGACAAU CAAGCAAAUUGGGCUGUCGACAAACGACAGAU ACAAGUUGCGAAUGGAGACAUGCUUCCAGCAGGCGUG UAAGGGUAAAAUCCAGCACUUUGCGAGAAUCCGAG UGGACACCAUUGAAGGAUAACAGGAUCCUUAUACG GGGUCUUGUCUGUUAUCUGAGUCUGACAGUUGAGC UUAAAAUCAAUUUGUUUCAGGAUUCGGGCCAUUGA UCACACCGGUUCAGGGAUGGACCUAUACAUAUCCAA CCACAACAAUAGUAUUGGCGUACUACCCGCCAAUG AAGAACCUGGCCUUAGGUGUAUUAACAUUGGAG UGGAUACCGAGAUUUAAGGUUAUCCCAACCUUUA CUGUUCCAAUUAAGGAAGCAGGCGAGGACUGCCAUGC CCCAACAUAACCUACUGCGGAGGUGGAUGGUGAUGUC AAACUCAGUUCCAAUCUGGUGAUUUAUCCUGGUCAG AUCUCCAAUAGUUCUGGCAACCUACGAUACUUCAG AGUUGAACAUUGCUGUAGUUUAUUAACGUUACAGCCC AAGCCGCUCAUUUUUAUACUUUUUAUCCUUUAGGUUG CCUGUAAGGGGGUCCCAUUGAAUUAAGUUGGAA UGCUCACAUGGGACCAAAACUUGGUGCCGUCACU UCUGUGGCUUGCGGACUCAGAAUCUGGUGGACUA UCACUCACUCUGGGUUGGUGGCAUGGAGUCAGCUG CACAGCCACUCGGGAAGAUAGAACAGCCGCGAGAUAG UGAUAAUAGGCUUGGAGCCUUGGUGGCAAGCUUCUG CCCCUUGGGCCUCCCCCAGCCCCUCCUCCUCCUCCUG CACCCGUACCCCGUGGUCUUUGAAUAAAGUCUGAGU GGGCGGC	
GC_H_MEASLES_ D8 ORF Sequence, NT	AUGUCACCACAACGAGACCGGAUAAUAGCCUUCUACA AAGACAACCCCAUCCUAAAGGGAAGUAGGAUAGUUU UAAACAGAGAACAUUUUAUGAUUGAUAGACCUUAUGU UUUGCUGGCGUUCUAUUCGUAUGUUUCUGAGCUU GAUCGGGUUGCUAGCCAUUGCAGGCAUUAAGACUUAU CGGGCAGCCAUCUACACCGCAGAGAUCCAUAAGGCC UCAGCACCAUUCUGGAUGUAACUAACUAUUCGAGCA UCAGGUUAAGGACGUGCUGACACCAUCUUAAGAU AUCGGUGAUGAAGUGGCUUGAGGACACCUAGAGA UUCACUGACCUAGUGAAGUUAUCUUCUGACAAGAUUA AAUUCUUAUACCGGACAGGGAUACGACUUCAGAGA UCUCACUUGGUGUAUACCCGCCAGAGAGAAUCAA UUGGAUUAUGAUCAAUACUGGCAUGUGGCGUCU GAAGAACUCAUGAAUGCAUUGGUGAACUACUAC UGGAGACCAGGGCAACCAUAGUUCUAGCUGUCUC AAAGGGAACUGUCAGGGGCCACUACAACAGAGGC CAAUUCUCAAACAUUGCGUGUCCUGUUGGACUUGU AUUUAAGUCGAGGUUAUUAUGUGUCAUAUAGUCA CUAUGACAUCCAGGGAUUAACGGGGGAACUUAACCU AGUGGAAAGGCCUAAUUCUGAGCAGCAAGGGUACAGA GUUGUCACAACUGAGCAUGCACCGAGUGUUUGAAGU AGGUGUUUAUGAAAUCCGGGUUUGGGGCUCCGCU AUUCCAUAGACAACUAUUCUAGGCAACAGUCAGU AAUGAUUUCAGCAACUGCAUGGUGGCUUUGGGGAG CUCAAGUUCGAGCCUUCUGUCACAGGGAAGAUUCUA UCACAAUCCCUAUCAGGGAUACGGGAAGGUGUCAG CUUCCAGCUUGUCAAGCUAGGUGUCUGGAAUCCCA ACCGACAUGCAAUUCUGGGUCCCCUAUACCGGAGU AUCCAGUGAUAGACAGGCUUUAUCCUUAUCUACAG AGGCGUUAUCGUGACAUAACAGCAAAUUGGCGUC CCGACAACACGGACAGAUACAAGUUGCGAAUGGAGA CAUGCUUCCAGCAGGCGUGUAAGGGUAAAAUCCAGC ACUUGCGAGAAUCCCGAGUGGACACCAUUGAAGGAU AACAGGAUUCUUAUACGGGGUUCUUGUCUGUAGUC UGAGUCUGACAGUUGAGCUUAAAAUCAAUUGUUU CAGGAUUCGGGCCAUUGAUACACACGCUUCAGGGAU	79

Description	Sequence	SEQ ID NO:
GC_H_MEASLES_ D8 mRNA Sequence (assumes T100 tail) Sequence Length: 2126	GGACCUAUACAAAUCCAAACCACAACAAUUGUAUUGG CUGACUAUCCCGCCAAUGAAGAACCUUGGCCUUAAGGUG UAAUCAACACAUUGGAGUGGAUACCAGAUUCAAAGG UUAGUCCCAACCUCUUCACUGUUCCAAUUAAGGAAGC AGGCGAGGACUGCCAUAGCCCAACAUACCUACCGUGCG GAGGUGGAUGGUGAUGUCAAAACUCAGUUCCAAUCUG GUGAUUUCACUGGUCAGGAUUCCAAUUAUGUUCUGG CAACCUCAGUAUCUUCACAGAUUGAACAUUGCUGUAGU UUAUUAAGCUUUAACAGCCCAAGCCGCUCAUUUUUCUAC UUUUUACUUUUUAGGUUGGUCUGUAAGGGGGUCCCCA UUGAAUUACAAGUGGAAUGCUUCACAUUGGGACCAAA AACUCUGGUGCCGUCACUUCUGUGUGCUUGCGGACUC AGAAUCUGGUGGACAUUAUACUACUCUGGGAUGGU GGGCAUGGGAGUCAGCUGCACAGCCACUCGGGAAGAU GGAACCAAGCCGACAGUAG	80

TABLE 14

MeV Amino Acid Sequences		
Description	Sequence	SEQ ID NO:
GC_F_MEASLES_B3.1 ORF Sequence, AA	MGLKVNVS AVFMAVLLTLQTPAGQIHWGNLSKIGVV GIGSASYKVMTRSSHQSLVIKLMPNITLLNNCTRVEIA EYRLLRRTVLEPIRDALNMTQNI RPPVQSVASSRRHK RFAGVVLGAGALGVATAAQITAGIALHRSMLNSQAID NLRASLETTNQAI EAIRQAGQEMILAVQGVQDYINNE LIPSMNQLSCDLIGQKLGLKLLRYYTEILSLFGPSLRDP ISAEISIQALS YALGGDINKVLEKLGYSGGDLLGILES R GIKARITHVDTESYFIVLSIAYPTLSEIKGVIVHRLGVS YNIGSQEWYTTVPKYVATQGYLISNFDESSCTFMPEG TVCSQNALYPMSPLLQEC LGSTKSCARTLVSGSFGN RFILSQGNLIANCASILCKCYTTGTIINQDPDKILTYIAA DRCPVVEVNGVTIQVGSRRYPDAVYLHRIDLGPPI SLE RLDVG TNLGNIAKLEDAKELLESDQILRSMKGLSST SIVYILIAVCLGGLIGIPTLICCCGRGCNKKGEQVGMSR PGLKPDLTGTSKSYVRSL*	47
GC_F_MEASLES_D8 ORF Sequence, AA	MGLKVNVS VIFMAVLLTLQTPGQIHWGNLSKIGVVG VGSASYKVMTRSSHQSLVIKLMPNITLLNNCTRVGIAE YRLLRRTVLEPIRDALNMTQNI RPPVQSVASSRRHKR FAGVVLGAGALGVATAAQITAGIALHQSMNLNSQAIDN LRASLETTNQAI EAIRQAGQEMI LAVQGVQDYINNELI PSMNQLSCDLIGQKLGLKLLRYYTEILSLFGPSLRDPIS AEISIQALS YALGGDINKVLEKLGYSGGDLLGILES RGI KARITHVDTESYFIVLSIAYPTLSEIKGVIVHRLGVS Y NIGSQEWYTTVPKYVATQGYLISNFDESSCTFMPEGT VCSQNALYPMSPLLQEC LGSTKSCARTLVSGSFGNR FILSQGNLIANCASILCKCYTTGTIINQDPDKILTYIAAD HCPVVEVNGVTIQVGSRRYPDAVYLHRIDLGPPI SLE LDVG TNLGNIAKLEDAKELLESDQILRSMKGLSSTS IVYILIAVCLGGLIGIPALICCCGRGCNKKGEQVGMSRP GLKPDLTGTSKSYVRSL*	48
GC_H_MEASLES_B3 ORF Sequence, AA	MSPQRDRINAFYKDNPHYKGS RIVINREHLMIDRPYVL LAVLFVMPLSLIGLLAIAGIRLHRAAIYTA EIHKSLSTN LDVTNSIEHQVDVLTPLFKI IGDEVGLRTPQRFTDLV KFISDKIKFLNPDREYDFRDLTWCINPPERIKLDYDQY CADVAAEELMNALVNSTLLETRTTQFLAVSKGNCS GPTTIRGQFSNMSLSLDDL YLGRGYNVSSIVTMTSQG MYGGTYLVEKPNLNSKGS ELSQLSMYRVFEVGVIRNP GLGAPVFHMTNYEQPVSNGLNCMV ALGELKLAAL CHGDDSIIPYQSGSGKVSFQLVKLG VVKSPTDMQSW VPLSTDDPVDRLYLSSH RGV IADNQAKWAVPTTRT DDKLRMETCFQQACKGKI QALCENPEWVPLKDNRIPS YGVLSVDLSLTVELKIKIASGFGPLITHGSGMDLYKSN CNNVYWLTI PPMRNALGVINTLEWIPRFKVS PNLFV PIKEAGEDCHAPT YLPAEVDGDV KLSNLVILPGQDL QYVLATYDTSRVEHAVVYVYSPSRFSYFYFPFRLPIK GVPIELQVECTWDQKLWCRHFCVLADSESGGLITHS GMVGMGVSC TATREDGTNR*	49
GC_H_MEASLES_D8 ORF Sequence, AA	MSPQRDRINAFYKDNPHPKGS RIVINREHLMIDRPYVL LAVLFVMPLSLIGLLAIAGIRLHRAAIYTA EIHKSLSTN LDVTNSIEHQVDVLTPLFKI IGDEVGLRTPQRFTDLV KFISDKIKFLNPDREYDFRDLTWCINPPERIKLDYDQY CADVAAEELMNALVNSTLLETRATNQFLAVSKGNCS GPTTIRGQFSNMSLSLDDL YLSRGYNVSSIVTMTSQGM YGGTYLVEKPNLSSKGS ELSQLSMHRVFEVGVIRNPG LGAPVFHMTNYLEQPVSNDFSNCMVALGELKFAALC HREDSITIPYQSGSGKVSFQLVKLG VVKSPTDMQSW VPLSTDDPVIDRLYLSSH RGV IADNQAKWAVPTTRTD DKLRMETCFQQACKGKI QALCENPEWVPLKDNRIPSY GVLSVDLSLTVELKIKIVSGFGPLITHGSGMDLYKSNH NNMYWLTI PPMKNLALGVINTLEWIPRFKVS PNLFV PIKEAGEDCHAPT YLPAEVDGDV KLSNLVILPGQDL QYVLATYDTSRVEHAVVYVYSPSRFSYFYFPFRLPV RGVPIELQVECTWDQKLWCRHFCVLADSESGGHITH SGMVGMGVSC TATREDGTSRR*	50

TABLE 15

MeV NCBI Accession Numbers (Amino Acid Sequences)		
Type	Virus Name	GenBank Accession
hemagglutinin	hemagglutinin [Measles virus strain Moraten]	AAF85673.1
hemagglutinin	hemagglutinin [Measles virus strain Rubeovax]	AAF85689.1
hemagglutinin	hemagglutinin [Measles virus]	AAF89824.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAA91369.1
hemagglutinin	hemagglutinin [Measles virus]	BAJ23068.1
hemagglutinin	hemagglutinin protein [Measles virus]	BAB39848.1
hemagglutinin	hemagglutinin [Measles virus]	AAA50551.1
hemagglutinin	RecName: Full = Hemagglutinin glycoprotein	P08362.1
hemagglutinin	hemagglutinin [Measles virus]	AAB63802.1
hemagglutinin	hemagglutinin [Measles virus]	AAA56650.1
hemagglutinin	hemagglutinin [Measles virus]	AAA56642.1
hemagglutinin	hemagglutinin [Measles virus]	AAA74936.1
hemagglutinin	hemagglutinin protein [Measles virus]	BAH56665.1
hemagglutinin	hemagglutinin [Measles virus]	ACC86105.1
hemagglutinin	hemagglutinin [Measles virus strain Edmonston-Zagreb]	AAF85697.1
hemagglutinin	hemagglutinin [Measles virus]	AAR89413.1
hemagglutinin	hemagglutinin [Measles virus]	AAA56653.1
hemagglutinin	RecName: Full = Hemagglutinin glycoprotein	P35971.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94916.1
hemagglutinin	hemagglutinin [Measles virus]	AAC03036.1
hemagglutinin	hemagglutinin [Measles virus]	AAF85681.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94927.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94925.1
hemagglutinin	hemagglutinin protein [Measles virus]	BAB39835.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94931.1
hemagglutinin	hemagglutinin [Measles virus genotype A]	AFO84712.1
hemagglutinin	hemagglutinin [Measles virus]	AAA56639.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94926.1
hemagglutinin	hemagglutinin protein [Measles virus]	BAB39836.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94929.1
hemagglutinin	RecName: Full = Hemagglutinin glycoprotein	P06830.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94928.1
hemagglutinin	hemagglutinin protein [Measles virus]	BAB39837.1
hemagglutinin	hemagglutinin [Measles virus]	AAA74935.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43780.1
hemagglutinin	hemagglutinin [Measles virus]	BAA09952.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43815.1
hemagglutinin	hemagglutinin [Measles virus]	AAF28390.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94923.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43785.1
hemagglutinin	hemagglutinin [Measles virus]	ABD34001.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43782.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43781.1
hemagglutinin	hemagglutinin [Measles virus]	BAH22353.1
hemagglutinin	hemagglutinin [Measles virus]	AAC35878.2
hemagglutinin	hemagglutinin protein [Measles virus]	AAL86996.1
hemagglutinin	hemagglutinin [Measles virus]	CAA76066.2
hemagglutinin	hemagglutinin [Measles virus]	AAA46428.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43803.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94918.1
hemagglutinin	hemagglutinin [Measles virus]	AAF72162.1
hemagglutinin	hemagglutinin [Measles virus]	AAM70154.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43776.1
hemagglutinin	hemagglutinin [Measles virus genotype D4]	ACT78395.1
hemagglutinin	hemagglutinin [Measles virus genotype D7]	AAL02030.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43789.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43774.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94920.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94922.1
hemagglutinin	hemagglutinin [Measles virus]	ABB59491.1
hemagglutinin	hemagglutinin protein [Measles virus]	BAB39843.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43804.1
hemagglutinin	hemagglutinin [Measles virus]	AAX52048.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94930.1
hemagglutinin	hemagglutinin [Measles virus]	AAA74526.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43814.1
hemagglutinin	hemagglutinin [Measles virus]	ABB59493.1
hemagglutinin	hemagglutinin [Measles virus genotype D4]	AAL02019.1
hemagglutinin	Hemagglutinin [Measles virus]	CAB94919.1
hemagglutinin	hemagglutinin protein [Measles virus]	AAL86997.1
hemagglutinin	hemagglutinin [Measles virus genotype C2]	AAL02017.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43769.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43808.1
hemagglutinin	hemagglutinin [Measles virus]	BAO97032.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43805.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43777.1

TABLE 15-continued

MeV NCBI Accession Numbers (Amino Acid Sequences)		
Type	Virus Name	GenBank Accession
hemagglutinin	hemagglutinin [Measles virus]	AAL67793.1
hemagglutinin	hemagglutinin [Measles virus]	AAF89816.1
hemagglutinin	hemagglutinin [Measles virus genotype D4]	AAL02020.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43786.1
hemagglutinin	hemagglutinin protein [Measles virus strain MVi/New Jersey.USA/45.05]	AEP40452.1
hemagglutinin	hemagglutinin [Measles virus]	AAA74531.1
hemagglutinin	hemagglutinin [Measles virus]	AAB63800.1
hemagglutinin	hemagglutinin [Measles virus]	AAO21711.1
hemagglutinin	hemagglutinin [Measles virus genotype D8]	ALE27189.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43810.1
hemagglutinin	hemagglutinin [Measles virus]	AAF89817.1
hemagglutinin	hemagglutinin [Measles virus genotype D6]	AAL02022.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43800.1
hemagglutinin	hemagglutinin protein [Measles virus genotype B3]	AGA17219.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43770.1
hemagglutinin	hemagglutinin protein [Measles virus strain MVi/Texas.USA/4.07]	AEP40444.1
hemagglutinin	hemagglutinin [Measles virus]	AAX52047.1
hemagglutinin	hemagglutinin [Measles virus]	AAB63794.1
hemagglutinin	hemagglutinin [Measles virus]	AAB63796.1
hemagglutinin	hemagglutinin [Measles virus]	AAA74528.1
hemagglutinin	hemagglutinin [Measles virus]	AAB63774.1
hemagglutinin	hemagglutinin [Measles virus]	AAB63795.1
hemagglutinin	hemagglutinin [Measles virus]	AAA74519.1
hemagglutinin	hemagglutinin protein [Measles virus]	CAB43778.1
fusion protein	fusion protein [Measles virus strain Moraten]	AAF85672.1
fusion protein	fusion protein [Measles virus]	AAA56645.1
fusion protein	fusion protein [Measles virus strain Rubeovax]	AAF85688.1
fusion protein	fusion protein [Measles virus]	AAF85680.1
fusion protein	fusion protein [Measles virus]	AEE30359.1
fusion protein	fusion protein [Measles virus]	BAA09957.1
fusion protein	fusion protein [Measles virus]	AAV84957.1
fusion protein	fusion protein [Measles virus MeV-eGFP_Edm-tag]	AII16636.1
fusion protein	fusion protein [Measles virus]	ABY58018.1
fusion protein	fusion protein [Measles virus]	BAA19838.1
fusion protein	fusion protein [Measles virus]	AAA56641.1
fusion protein	F protein [Measles virus]	ABK40529.1
fusion protein	fusion protein [Measles virus]	AAA56652.1
fusion protein	fusion protein [Measles virus]	ABY58017.1
fusion protein	fusion protein [Measles virus]	ABB71645.1
fusion protein	fusion protein [Measles virus]	NP_056922.1
fusion protein	fusion protein [Measles virus strain AIK-C]	AAF85664.1
fusion protein	fusion protein [Measles virus]	BAB60865.1
fusion protein	fusion protein [Measles virus]	BAA09950.1
fusion protein	fusion protein [Measles virus strain MVi/New York.USA/26.09/3]	AEP40403.1
fusion protein	fusion protein [Measles virus]	AAA74934.1
fusion protein	fusion protein [Measles virus]	CAB38075.1
fusion protein	fusion protein [Measles virus strain MVi/Texas.USA/4.07]	AEP40443.1
fusion protein	fusion protein [Measles virus]	AAF02695.1
fusion protein	fusion protein [Measles virus]	AAF02696.1
fusion protein	fusion protein [Measles virus]	AAT99301.1
fusion protein	fusion protein [Measles virus]	ABB71661.1
fusion protein	fusion protein [Measles virus]	BAK08874.1
fusion protein	fusion protein [Measles virus]	AAF02697.1
fusion protein	fusion protein [Measles virus genotype D4]	AFY12704.1
fusion protein	fusion protein [Measles virus strain MVi/California.USA/16.03]	AEP40467.1
fusion protein	fusion protein [Measles virus genotype D8]	AHN07989.1
fusion protein	fusion protein [Measles virus]	AAA46421.1
fusion protein	fusion protein [Measles virus]	AAA56638.1
fusion protein	fusion protein [Measles virus strain MVi/Virginia.USA/15.09]	AEP40419.1
fusion protein	fusion protein [Measles virus genotype D8]	ALE27200.1
fusion protein	fusion protein [Measles virus genotype D8]	AFY12695.1
fusion protein	fusion protein [Measles virus genotype D8]	ALE27248.1
fusion protein	fusion protein [Measles virus genotype D8]	ALE27224.1
fusion protein	fusion protein [Measles virus]	AAT99300.1
fusion protein	fusion protein [Measles virus]	BAH96592.1
fusion protein	fusion protein [Measles virus strain MVi/California.USA/8.04]	AEP40459.1
fusion protein	fusion protein [Measles virus genotype D8]	AIG94081.1
fusion protein	fusion protein [Measles virus]	BAA09951.1
fusion protein	fusion protein [Measles virus genotype D8]	ALE27194.1

TABLE 15-continued

MeV NCBI Accession Numbers (Amino Acid Sequences)		
Type	Virus Name	GenBank Accession
fusion protein	fusion protein [Measles virus]	BAA33871.1
fusion protein	fusion protein [Measles virus strain MVi/Washington.USA/18.08/1]	AEP40427.1
fusion protein	fusion protein [Measles virus]	ABY21182.1
fusion protein	fusion protein [Measles virus genotype D8]	ALE27284.1
fusion protein	fusion protein [Measles virus]	ACA09725.1
fusion protein	fusion protein [Measles virus genotype D8]	ALE27314.1
fusion protein	fusion protein [Measles virus genotype G3]	AFY12712.1
fusion protein	fusion protein [Measles virus genotype D8]	ALE27368.1
fusion protein	RecName: Full = Fusion glycoprotein F0; Contains: RecName: Full = Fusion glycoprotein F2; Contains: RecName: Full = Fusion glycoprotein F1; Flags: Precursor	P35973.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53713.1
	unnamed protein product [Measles virus]	CAA34588.1
fusion protein	fusion protein [Measles virus]	CAA76888.1
fusion protein	fusion protein [Measles virus genotype B3.1]	AIY55563.1
fusion protein	fusion protein [Measles virus]	ADO17330.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53703.1
fusion protein	fusion protein [Measles virus genotype B3]	AGA17208.1
fusion protein	fusion protein [Measles virus]	AAL29688.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53706.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53701.1
fusion protein	fusion protein [Measles virus genotype B3]	ALE27092.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53714.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53694.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53668.1
fusion protein	fusion protein [Measles virus]	ACC86094.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53670.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53707.1
fusion protein	fusion protein [Measles virus genotype B3]	AGA17216.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53671.1
fusion protein	fusion protein [Measles virus strain MVi/New Jersey.USA/45.05]	AEP40451.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53684.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53688.1
fusion protein	fusion protein [Measles virus genotype B3]	AGA17214.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53683.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53667.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53686.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53685.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53681.1
	unnamed protein product [Measles virus]	CAA34589.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53678.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53710.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53669.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53664.1
fusion protein	fusion protein [Measles virus]	AAA50547.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53679.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53709.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53672.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53697.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53689.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53676.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53675.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53663.1
fusion protein	fusion protein [Measles virus]	BAA19841.1
fusion protein	fusion protein [Measles virus]	AAF02701.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53680.1
fusion protein	fusion protein [Measles virus genotype H1]	AIG53674.1
C protein	C protein [Measles virus strain Moraten]	AAF85670.1
C protein	RecName: Full = Protein C	P03424.1
C protein	C protein [Measles virus]	ACN54404.1
C protein	C protein [Measles virus]	ACN54412.1
C protein	RecName: Full = Protein C	P35977.1
C protein	C protein [Measles virus]	AAF85678.1
C protein	C protein [Measles virus]	ABD33998.1
C protein	unnamed protein product [Measles virus]	CAA34586.1
C protein	C protein [Measles virus]	BAJ51786.1
C protein	C protein [Measles virus]	BAA33869.1
C protein	virulence factor [Measles virus]	ABO69700.1
C protein	C protein [Measles virus]	NP_056920.1
C protein	C protein [Measles virus]	ADO17333.1
C protein	C protein [Measles virus]	ACC86082.1
C protein	C protein [Measles virus]	BAA33875.1
C protein	C protein [Measles virus]	ABY21189.1
C protein	C protein [Measles virus]	BAE98296.1

TABLE 15-continued

MeV NCBI Accession Numbers (Amino Acid Sequences)		
Type	Virus Name	GenBank Accession
C protein	C protein [Measles virus]	ADU17782.1
C protein	C protein [Measles virus strain MVi/Virginia.USA/15.09]	AEP40417.1
C protein	C protein [Measles virus]	ADU17814.1
C protein	C protein [Measles virus]	ADU17798.1
C protein	C protein [Measles virus genotype D4]	AFY12700.1
C protein	C protein [Measles virus]	ADU17784.1
C protein	C protein [Measles virus strain MVi/California.USA/16.03]	AEP40465.1
C protein	C protein [Measles virus]	ABB71643.1
C protein	C protein [Measles virus]	AEI91027.1
C protein	C protein [Measles virus]	ADU17874.1
C protein	C protein [Measles virus]	ADU17903.1
C protein	C protein [Measles virus]	CAA34579.1
C protein	C protein [Measles virus]	ADU17790.1
C protein	C protein [Measles virus]	ADU17800.1
C protein	C protein [Measles virus]	ABB71667.1
C protein	unnamed protein product [Measles virus]	CAA34572.1
C protein	C protein [Measles virus strain MVi/Arizona.USA/11.08/2]	AEP40433.1
C protein	C protein [Measles virus]	ADU17830.1
C protein	C protein [Measles virus]	ADU17947.1
C protein	C protein [Measles virus]	ADU17818.1
C protein	C protein [Measles virus strain MVi/New Jersey.USA/45.05]	AEP40449.1
C protein	C protein [Measles virus strain MVi/Texas.USA/4.07]	AEP40441.1
C protein	C protein [Measles virus]	ADU17864.1
C protein	C protein [Measles virus]	ADU17838.1
C protein	C protein [Measles virus]	ADU17881.1
C protein	C protein [Measles virus strain MVi/Washington.USA/18.08/1]	AEP40425.1
C protein	C protein [Measles virus]	ADU17927.1
C protein	C protein [Measles virus]	ADU17953.1
C protein	C protein [Measles virus]	ADU17889.1
C protein	C protein [Measles virus]	ADU17963.1
C protein	C protein [Measles virus]	ADU17893.1
C protein	C protein [Measles virus]	ADU17820.1
C protein	C protein [Measles virus]	ABB71651.1
C protein	C protein [Measles virus]	ADU17786.1
C protein	C protein [Measles virus]	ADU17862.1
C protein	C protein [Measles virus]	ADU17923.1
C protein	C protein [Measles virus]	ADU17959.1
C protein	C protein [Measles virus]	ADU17951.1
C protein	C protein [Measles virus]	ADU17916.1
C protein	C protein [Measles virus]	ADU17957.1
C protein	C protein [Measles virus]	ADU17925.1
C protein	C protein [Measles virus]	ADU17901.1
C protein	C protein [Measles virus]	ADU17887.1
C protein	C protein [Measles virus]	ADU17832.1
C protein	C protein [Measles virus]	ADU17891.1
C protein	C protein [Measles virus]	ADU17961.1
C protein	C protein [Measles virus]	ADU17872.1
C protein	C protein [Measles virus]	ADU17929.1
C protein	C protein [Measles virus]	ADU17908.1
C protein	C protein [Measles virus]	ADU17910.1
C protein	C protein [Measles virus]	ADU17921.1
C protein	C protein [Measles virus]	ADU17824.1
C protein	C protein [Measles virus strain MVi/Pennsylvania.USA/20.09]	AEP40473.1
C protein	C protein [Measles virus]	ADU17828.1
C protein	C protein [Measles virus]	ADU17812.1
C protein	C protein [Measles virus genotype D8]	AFY12692.1
C protein	nonstructural C protein [Measles virus]	ABA59559.1
C protein	RecName: Full = Protein C	Q00794.1
C protein	nonstructural C protein [Measles virus]	ADO17934.1
C protein	nonstructural C protein [Measles virus]	ACJ66773.1
C protein	C protein [Measles virus genotype G3]	AFY12708.1
C protein	RecName: Full = Protein C	P26035.1
C protein	C protein [Measles virus]	BAA84128.1
nucleoprotein	RecName: Full = Nucleoprotein; AltName: Full = Nucleocapsid protein; Short = NP; Short = Protein N	Q77M43.1
nucleoprotein	nucleocapsid protein [Measles virus strain Rubeovax]	AAF85683.1
nucleoprotein	RecName: Full = Nucleoprotein; AltName: Full = Nucleocapsid protein; Short = NP; Short = Protein N	Q89933.1
nucleoprotein	nucleocapsid protein [Measles virus strain AIK-C]	AAF85659.1
nucleoprotein	nucleoprotein [Measles virus]	ABI54102.1
nucleoprotein	nucleoprotein [Measles virus]	AAA56643.1

TABLE 15-continued

MeV NCBI Accession Numbers (Amino Acid Sequences)		
Type	Virus Name	GenBank Accession
nucleoprotein	nucleoprotein [Measles virus]	AAC03050.1
nucleoprotein	nucleoprotein [Measles virus]	AAA18990.1
nucleoprotein	nucleoprotein [Measles virus]	AAA56640.1
nucleoprotein	RecName: Full = Nucleoprotein; AltName: Full = Nucleocapsid protein; Short = NP; Short = Protein N	P35972.1
nucleoprotein	RecName: Full = Nucleoprotein; AltName: Full = Nucleocapsid protein; Short = NP; Short = Protein N	P10050.1
nucleoprotein	N protein [Measles virus]	BAB60956.1
nucleoprotein	RecName: Full = Nucleoprotein; AltName: Full = Nucleocapsid protein; Short = NP; Short = Protein N	B1AAA7.1
nucleoprotein	nucleoprotein [Measles virus]	AAA18991.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46894.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46871.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46872.1
nucleoprotein	nucleoprotein [Measles virus]	ABU49606.1
nucleoprotein	nucleocapsid protein [Measles virus]	AAA75494.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46883.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46892.1
nucleoprotein	unnamed protein product [Measles virus]	CAA34584.1
nucleoprotein	nucleoprotein [Measles virus]	AAA18997.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46863.1
nucleoprotein	nucleoprotein [Measles virus]	AEF30352.1
nucleoprotein	nucleoprotein [Measles virus]	ABI54103.1
nucleoprotein	nucleocapsid protein [Measles virus]	AAA46433.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46902.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46873.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46906.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74547.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74537.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46862.1
nucleoprotein	nucleocapsid protein [Measles virus]	BAA09961.1
nucleoprotein	nucleoprotein [Measles virus]	AAO15875.1
nucleoprotein	nucleoprotein [Measles virus]	AAO15871.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46882.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60124.1
nucleoprotein	nucleoprotein [Measles virus]	ABI54104.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46869.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46880.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74541.1
nucleoprotein	nucleocapsid protein [Measles virus strain MVi/New Jersey.USA/45.05]	AEP40446.1
nucleoprotein	nucleoprotein [Measles virus]	ABI54110.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46903.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46899.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46901.1
nucleoprotein	nucleocapsid protein [Measles virus]	ABB71640.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60113.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60114.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60116.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46895.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60121.1
nucleoprotein	nucleoprotein [Measles virus]	ABI54111.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46889.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46898.1
nucleoprotein	nucleoprotein [Measles virus genotype B3]	ALE27083.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60118.1
nucleoprotein	nucleocapsid protein [Measles virus]	CAA34570.1
nucleoprotein	nucleoprotein [Measles virus]	AAC29443.1
nucleoprotein	nucleocapsid protein [Measles virus strain MVi/Washington.USA/18.08/1]	AEP40422.1
nucleoprotein	nucleoprotein [Measles virus]	AAO15872.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46874.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74550.1
nucleoprotein	nucleocapsid protein [Measles virus]	ABB71648.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46900.1
nucleoprotein	nucleoprotein [Measles virus]	BAH22440.1
nucleoprotein	nucleocapsid protein [Measles virus]	AAA46432.1
nucleoprotein	nucleocapsid protein [Measles virus]	BAA33867.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74539.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60115.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60123.1
nucleoprotein	nucleocapsid protein [Measles virus]	ABB71664.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60125.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74546.1
nucleoprotein	nucleoprotein [Measles virus]	CAB46886.1
nucleoprotein	nucleoprotein [Measles virus]	BAH22350.1

TABLE 15-continued

MeV NCBI Accession Numbers (Amino Acid Sequences)		
Type	Virus Name	GenBank Accession
nucleoprotein	nucleoprotein [Measles virus]	CAB46867.1
nucleoprotein	nucleocapsid protein [Measles virus]	BAA09954.1
nucleoprotein	nucleoprotein [Measles virus]	AAO15873.1
nucleoprotein	nucleocapsid protein [Measles virus]	AEP95735.1
nucleoprotein	nucleoprotein [Measles virus]	AAL37726.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74549.1
nucleoprotein	RecName: Full = Nucleoprotein; AltName: Full = Nucleocapsid protein; Short = NP; Short = Protein N	P26030.1
nucleoprotein	nucleoprotein [Measles virus ETH55/99]	AAK07777.1
nucleoprotein	nucleoprotein [Measles virus genotype B3]	AGA17238.1
nucleoprotein	nucleoprotein [Measles virus]	AEF30351.1
nucleoprotein	nucleoprotein [Measles virus genotype B3]	AGA17242.1
nucleoprotein	nucleoprotein [Measles virus ETH54/98]	AAK07776.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74548.1
nucleoprotein	nucleoprotein [Measles virus]	AAA19221.1
nucleoprotein	nucleoprotein [Measles virus]	AAC03039.1
nucleoprotein	nucleoprotein [Measles virus]	AAA19223.1
nucleoprotein	nucleoprotein [Measles virus genotype B3]	AGA17241.1
nucleoprotein	nucleoprotein [Measles virus]	CAB60122.1
nucleoprotein	nucleoprotein [Measles virus]	CAC34599.1
nucleoprotein	nucleoprotein [Measles virus]	AAC03042.1
nucleoprotein	nucleoprotein [Measles virus]	CAC34604.1
nucleoprotein	nucleoprotein [Measles virus]	AAA74544.1
nucleoprotein	nucleocapsid protein [Measles virus]	NP_056918.1
V Protein	RecName: Full = Non-structural protein V	Q9IC37.1
V Protein	RecName: Full = Non-structural protein V	Q9EMA9.1
V Protein	V protein [Measles virus]	ACN54411.1
V Protein	V protein [Measles virus]	ACN54403.1
V Protein	V protein [Measles virus]	AEP95742.1
V Protein	V protein [Measles virus strain MVi/Virginia.USA/15.09]	AEP40416.1
V Protein	V protein [Measles virus]	ADU17801.1
V Protein	V protein [Measles virus]	ADU17849.1
V Protein	V protein [Measles virus]	ABB71642.1
V Protein	V protein [Measles virus genotype D8]	AFY12693.1
V Protein	V protein [Measles virus]	YP_003873249.2
V Protein	V protein [Measles virus strain MVi/Arizona.USA/11.08/2]	AEP40432.1
V Protein	RecName: Full = Non-structural protein V	P26036.1
V Protein	V protein [Measles virus strain MVi/California.USA/16.03]	AEP40464.1
V Protein	V protein [Measles virus strain MVi/California.USA/8.04]	AEP40456.1
V Protein	V protein [Measles virus]	ABY21188.1
V Protein	V protein [Measles virus strain MVi/Washington.USA/18.08/1]	AEP40424.1
V Protein	V protein [Measles virus]	BAH96581.1
V Protein	V protein [Measles virus]	ABB71666.1
V Protein	RecName: Full = Non-structural protein V	P60168.1
V Protein	V protein [Measles virus]	BAH96589.1
V Protein	V protein [Measles virus]	ADU17954.1
V Protein	V protein [Measles virus strain MVi/New York.USA/26.09/3]	AEP40400.1
V Protein	V protein [Measles virus]	ABY21196.1
V Protein	virulence factor [Measles virus]	ABO69701.1
V Protein	V protein [Measles virus]	ABB71650.1
V Protein	V protein [Measles virus]	ACC86086.1
V Protein	V protein [Measles virus genotype D4]	AFY12702.1
V Protein	V protein [Measles virus strain MVi/New Jersey.USA/45.05]	AEP40448.1
V Protein	V protein [Measles virus]	BAE98295.1
V Protein	V protein [Measles virus]	ACC86083.1
V Protein	V protein [Measles virus]	ACU57139.1
V Protein	V protein [Measles virus]	ADO17334.1
V Protein	V protein [Measles virus]	ADU17930.1
V Protein	V protein [Measles virus genotype G3]	AFY12710.1
V Protein	V protein [Measles virus strain MVi/Pennsylvania.USA/20.09]	AEP40472.1
V Protein	phosphoprotein [Measles virus]	ADU17839.1
V Protein	V protein [Measles virus]	ADU17894.1
V Protein	V protein [Measles virus]	ACN50010.1
V Protein	V protein [Measles virus]	ADU17892.1
V Protein	unnamed protein product [Measles virus]	CAA34585.1
V Protein	V protein [Measles virus]	ABD33997.1

TABLE 16

Name	Sequence	SEQ ID NO:
Flagellin Nucleic Acid Sequences		
NT (5' UTR, ORF, 3' UTR)	TCAAGCTTTTGGACCCTCGTACAGAAGCTAATACGACTCACTAT AGGGAAATAAGAGAGAAAAGAAGAGTAAGAAGAAATATAAG AGCCACCATGGCACAAGTCATTAATACAAACAGCCTGTGCGTG TTGACCCAGAATAACCTGAACAAATCCCAGTCCGCACTGGGCA CTGCTATCGAGCGTTTGTCTTCCGGTCTGCGTATCAACAGCGCG AAAGACGATGCGGCAGGACAGGCGATTGCTAACCGTTTACC CGAACATCAAAGGTCTGACTCAGGCTTCCCGTAACGCTAACGA CGGTATCTCCATTGCGCAGACCCTGAAGGCGCGCTGAACGAA ATCAACAACAACCTGCAGCGTGTGCGTGAACCTGGCGGTTCA CTGCGAATGGTACTAACTCCCAGTCTGACCTCGACTCCATCCAG GCTGAAATCACCCAGCGCCTGAACGAAATCGACCGTGATCCG GCCAGACTCAGTTCACCGCGCTGAAGTCTTGGCGCAGGACAA CACCCGTGACCATCCAGGTTGGTGCCAAACGCGTGAACATATC GATATTGATTAAAGAAATCAGCTCTAAACACTGGGACTTGG ATAAGCTTAATGTCCAGATGCCTACACCCGAAAGAACTGC TGTAACCGTTGATAAAACTACCTATAAAATGGTACAGATCCT ATTACAGCCAGAGCAATACTGATATCCAACTGCAATTGGCG GTGGTGCAACGGGGTTACTGGGGCTGATATCAAAATTTAAAGA TGGTCAATACTATTTAGATGTTAAAGGCGGTGCTTCTGCTGGTG TTTATAAGCCCTTATGATGAACTACAAGAAAGTAAATAT TGATACGACTGATAAACTCCGTTGGCACTGCGGAAGCTACA GCTATTCGGGGAACGGCCACTATAACCCACAACCAATTTGCTG AAGTAACAAAAGAGGGTGTGATACGACCACAGTTGCGGCTCA ACTTGCTGAGCAGGGGTTACTGGCGCCGATAAGGACATACT AGCCTTGTAAGCACTATCGTTTGGAGATAAAACCGTAAGGTTA TTGATGGTGGCTATGCAAGTGAAGTGGGCGACGATTTCTATGC CGCTACATATGATGAGAAACAGGTGCAATTACTGCTAAACCC ACTACTTATACAGATGGTACTGGCGTGTCTCAAACCTGGAGCTGT GAAATTTGGTGGCGCAATGGTAAATCTGAAGTTGTTACTGCT ACCGATGGTAAGACTTACTTAGCAAGCGACTTGACAAACATA ACTTCAGAACAGGCGGTGAGCTTAAAGAGGTTAATACAGATAA GACTGAAAACCCACTGCAGAAATGATGCTGCCTTGGCACAG GTTGATACACTTCGTTCTGACCTGGGTGCGGTTGAGAACCGTT CAACTCCGCTATCACCAACCTGGGCAATACCGTAAATAACCTG TCTTCTGCCCGTAGCCGTATCGAAGATTCCGACTACGCAACCGA AGTCTCCAACATGTCTCGCGCGCAGATTCTGAGCAGGCGCGT ACCTCCGTTCTGGCGCAGGCGAACCAGGTTCCGCAAAACGTCC TCTCTTACTGCGTTGATAATAGGCTGGAGCCTCGGTGGCCATG CTTCTTGGCCCTTGGGCTCCCCCAGCCCTCCTCCCTTCTCTG CACCCGTACCCCGTGGTCTTTGAATAAAGTCTGAGTGGGCGG C	51
ORF Sequence, NT	ATGGCACAAGTCATTAATACAAACAGCCTGTGCTGTTGACCC AGAATAACCTGAACAAATCCCAGTCCGCACTGGGCACTGCTAT CGAGCGTTTGTCTTCCGGTCTGCGTATCAACAGCGCGAAAGAC GATGCGGCAGGACAGGCGATTGCTAACCCTTTTACCGGAACA TCAAAGGTCTGACTCAGGCTTCCCGTAACGCTAACGACGCTAT CTCCATTGCGCAGACCCTGAAGGCGCGCTGAACGAAATCAAC AACAACTGCAGCGTGTGCGTGAACCTGGCGGTTCACTGTCGA ATGGTACTAACTCCAGTCTGACCTCGACTCCATCCAGGCTGAA ATCACCCAGCGCTGAACGAAATCGACCGTGTATCCGGCCAGA CTCAGTTCAACGGCGTGAAAGTCCCTGGCGCAGGACAAACCCCT GACCATCCAGGTTGGTGCCACGACGGTGAACCTATCGATATT GATTTAAAGAAATCAGCTCTAAACACTGGGACTTGATAAGC TTAATGTCCAAGATGCCTACACCCGAAAGAACTGCTGTAAAC CGTTGATAAACTACCTATAAAATGGTACAGATCCTATTACA GCCAGAGCAATACTGATATCCAACTGCAATTGGCGGTGGTG CAACGGGGTTACTGGGGCTGATATCAAAATTTAAAGATGGTCA ATACTATTTAGATGTTAAAGGCGGTGCTTCTGCTGGTGTATT AAGCCACTTATGATGAACTACAAGAAAGTTAATATTGATAC GACTGATAAACTCCGTTGGCAACTGCGGAAGCTACAGCTATT CGGGGAACGGCCACTATAACCCACAACCAATTTGCTGAAGTAA CAAAAGAGGGTGTGATACGACCACAGTTGCGGCTCAACTTGC TGCAGCAGGGTTACTGGCGCCGATAAGGACAATACTAGCCCTT GTAAACTATCGTTTGGAGATAAAACCGTAAGGTTATTGATG GTGGCTATGCAAGTGAAGTGGGCGACGATTCTATGCGCTAC ATATGATGAGAAAACAGGTGCAATTACTGCTAAAACCACTACT TATACAGATGGTACTGGCGTTGCTCAAACCTGGAGCTGTGAAAT TTGGTGGCGCAATGGTAAATCTGAAGTTGTTACTGCTACCGAT GGTAAGACTTACTTAGCAAGCGACCTTGACAAACATAACTTCA GAACAGGCGGTGAGCTTAAAGAGGTTAATACAGATAAGACTG AAAACCACTGCAGAAATTTGATGCTGCCTTGGCACAGGTTGA TACACTTCGTTCTGACCTGGGTGCGGTTGAGAACCGTTTCAACT CCGCTATCACCACCTGGGCAATACCGTAAATAACCTGTCTTCT GCCCCGTAGCCGTATCGAAGATTCCGACTACGCAACCGAAGTCT	52

TABLE 16-continued

Name	Sequence	SEQ ID NO:
	CCAACATGTCCTCGCGCGCAGATTCTGCAGCAGGCCGGTACCTC CGTTCTGGCGCAGGCCGAACAGGTTCCGCAAAACGTCCTCTCTT TACTGCGT	
mRNA Sequence (assumes T100 tail)	G*GGGAAUAAGAGAGAAAAGAAGAGUAAGAAGAAUAUAA GAGCCACCAUGGCACAAGUCAUUAUACAAACAGCCUGUCGC UGUUGACCCAGAAUAACUGAACAAAUCCAGUCCGCACUGG GCACUGCUAUCGAGCGUUUGUCUCCGGUCUGCGUAUCAACA GCGCGAAAGACGAUGCGGCAGGACAGGC GAUUGCUAACCGU UUACCGGAACAUCAAAGGUCUGACUCAGGCUUCCGUAACG CUAACGACGGUAUCUCCAUUGCGCAGACCACUGAAGGCGCGC UGAACGAAAUCAACAACACCGUCAGCGUGUGCGUGAACUGG CGGUUCAGUCUGCGAAUGGUACUAACUCCAGUCUGACCUUG ACUCCAUCCAGGCUGAAUACCCAGCGCCUGAACGAAUUCG ACCGUGUAUCCGGCCAGACUCAGUUCACCGGCGUGAAAGUCC UGGCGCAGGACAACACCCUGACCAUCCAGGUUGGUGCCAACG ACGGUGAAACUAUCGAUAUUGAUUUAAGAAGAAUACGCUU AAAACACUGGGACUUAAGCUUAAUGUCCAAAGUCCUAC ACCCGAAAGAAACUGUCUGUAACCGUUGAUAAAACUACCUAU AAAAAUGGUACAGAUCCUAUUAACAGCCCAGAGCAAUACUGAU AUCCAAACUGCAAUUGGCGGUGGUGCAACGGGGGUUACUGG GGCUGAUUAUCAAUUUAAGAUGGUCAAUACUAUUUAGAUG UUAAAGGCGGUGCUUCUGCUGGUGUUUAUAAAGCCACUUAU GAUGAAACUACAAAGAAAGUUAUUAUUGAUACGACUGAUAA AACUCCGUUGGCAACUGCGGAAGCUACAGCUAUUCGGGGAAC GGCCACUAUAACCCACAACCAAUUGCUAAGUAACAAAGA GGGUGUUGAUACGACCACAGUUGCGGCUCAACUUGCUGCAGC AGGGGUUACUGGCGCCGAUAAGGACAAUACUAGCCUUGUAA AACUAUCGUUUGAGGAUAAAAACGGUAAGGUUAUUGAUGGU GGCUAUGCAGUAAAAUGGGCGACGAUUUCUUGCCGCUACA UAUGAUGAGAAACAGGUGCAAUUACUGCUAAAACCAUAC UUAUAUCAGAUUGUACUGGCGUUGCUCAAACUGGAGCUGUGA AAUUUGGUGGCGCAAUUGGUAAAUUGAAGUUGUACUGCU ACCGAUGGUAAAGACUUAUUAAGCAAGCGACCUUGACAACA AACUUCAGAACAGGCGGUGAGCUUAAAGAGGUUAUACAGA UAAGACUGAAACCCACUGCAGAAAAUUGAUGCUGCCUUGGC ACAGGUUGAUACAUUCGUUCUGACCUUGGUGCGGUUCAGAA CCGUUUAACUCGCUUAUCACCAACUGGGCAAUACCGUAAA UAACUGUCUUCUGCCCGUAGCCGUUACGAAGAUUCCGACUA CGCAACCGAAGUCUCCACAUGUCUCCGCGCAGAUUCUGCA GCAGGCGGUACCUUCGUUCUGGCGCAGGCGAACAGGUUCC GCAAAACGUCUCUUCUUAUCUGCGUUGAUAAUAGGCUUGAGC CUCCGUGGCAUGCUUCUUGCCCCUUGGGCCUCCCCCAGCC CCUCCUCCCCUUCUGACCCCGUACCCCGUGGUCUUUGAAU AAAGUCUGAGUGGCGGCAAAAAAAAAAAAAAAAAAAAAA AA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAUCUAG	53
Flagellin mRNA Sequences		
NT (5' UTR, ORF, 3' UTR)	UCAAGCUUUGGACCCUCGUACAGAACGUAUACGACUCACU AUAGGGAAUAAGAGAGAAAAAGAGUAAGAAAGAAUAUA AGAGCCACCAUGGCACAAGUCAUUAUACAAACAGCCUGUCG CUGUUGACCCAGAAUAACUGAACAAAUCCAGUCCGCACUG GGCACUGCUAUUCGAGCGUUUGUCUUCGGUCUGCGUAUCAAC AGCGCGAAAGACGAUGCGGCAGGACAGGCGAUUGCUAACCGU UUUACCGCGAACAUCAAAGGUUGACUCAGGCUUCCCGUAAC GCUAACGACGGUAUCUCCAUUGCGCAGACCACUGAAGGCGCG CUGAACGAAAUCAACAACACUGCAGCGUGUGCGUGAACUG GCGGUUCAGUCUGCGAAUGGUACUAACUCCAGUCUGACCUC GACUCCAUCCAGGCUGAAAUACCCAGCGCUGAACGAAAUUC GACCGUGUAUCCGGCCAGACUCAGUUAACGGCGUGAAAGUC CUGGCGCAGGACAACACCCUGACCAUCCAGGUUGGUGCCAAC GACGGUGAAACUAUCGAUAUUGAUUUAAAGAAAUACAGTUC UAAAAACACUGGACUUGAUAAAGCUAAUGUCCAAAGAUCCU ACACCCCGAAAGAAACUGUCUGUAACCGUUGAUAAAACUACCU AUAAAAAUGGUACAGAUCCUAUUAACAGCCAGAGCAAUACUG AUAUCCAAACUGCAAUUGGCGGUGUGCAACGGGGGUUACU GGGGCUGAUUAUCAAUUUAAGAUGGUCAAUACUUAUAGA UGUUAAAGGCGGUGCUUCUGCUGGUGUUUAUAAAGCCACUU AUGAUGAAACUACAAAGAAAGUUAUUAUUGAUACGACUGAU AAAAUCUCCGUUGGCAACUGCGGAAGCUACAGCUAUUCGGGA ACGGCCACUAUAACCCACAACCAAUUGUGAAGUAACAAAA GAGGGUGUUGAUACGACCACAGUUGCGGCUCAACUUGCUGCA GCAGGGGUUACUGGCGCCGAUAAGGACAAUACUAGCCUUGUA AAACUAUCGUUUGAGGAUAAAAACGGUAAGGUUAUUGAUGG UGGCUAUGCAGUAAAAUGGGCGACGAUUUCUUGCCGCUAC	81

TABLE 16-continued

Name	Sequence	SEQ ID NO :
	AUAUGAUGAGAAAACAGGUGCAAUUAUCUGCUAAAACCACUA CUUAUACAGAUUGGUACUGGCGUUGUCUAAACUGGAGCUGUG AAAUUUGUGGCGCAAAUUGGUAAAUCUGAAGUUGUUAUCUGC UACCGAUGGUAAAGACUUACUUAGCAAGCGACCUUGACAAACA UAACUUCAGAAACAGGCGUGAGCUUAAAGAGGUUAAUACAG AUAAGACUGAAAACCCACUGCAGAAAUAUGAUGCUGCCUUGG CACAGGUUGAUACACUUCGUUCUGACCUGGGUGCGGUUCAGA ACCGUUUCAAUCGCUAUCACCAACCUUGGGCAAUACCGUAA AUAACCUUGUCUUCUGCCGUAGCCGUUAUGAAGAUUCCGACU ACGCAACCGAAGUCUCCAACAUUGUCUGCGCGCAGAUUCUGC AGCAGGCCGGUACCUCCGUUCUGGCGCAGGCGAACAGGUUC CGCAAAACGUCUUCUUACUGCGUUGAUAAUAGGCGUGGAG CCUCGGUGGCCAUGCUUCUUGCCCCUUGGGCCUCCCCCAGC CCCUCCUCCCCUUCUGCACCCGUACCCCGUGGUCUUUGAA UAAAGUCUGAGUGGGCGGC	
ORF Sequence, NT	AUGGCACAAGUCAUUAAUACAACAGCCUGUCGCGUUGACCC CAGAAUAACCGUAACAAAUCCAGUCCGACUGGGCACUGCU AUCGAGCGUUUGUCUUCGCGUCUGCGUAUCAACAGCGCGAAA GACGAUGCGGCGAGGACAGGCGAUUGCUAACCGUUUACCGCG AACAUCAAAGGUCUGACUCAGGCUUCCCGUAACGCUAACGAC GGUAUCUCCAUGCGCAGACCACUGAAGGCGCGCUGAACGAA AUCACAAACAACCGUCAGCGUGUGCGUGAACUGGCGGUUCAG UCUGCGAAUGGUACUAAUCUCCAGUCUGACCUAGCUCCAUUC CAGGCGUAAAUAACCCAGCGCCUGAACGAAAUCGACCGUGUA UCCGGCCAGACUCAGUUCAAACGGCGUGAAAGUCCUGGCGCAG GACAAACCCUGACCAUCCAGGUUGGUGCCAACGACGGUGAA ACUAUCGAUAUUGAUUUAAAAGAAUACGCUUAAAACACU GGGACUUGAUAAAGCUUAAUGUCCAAGAUGCCUACACCCCGAA AGAAACUGCUGUAACCGUUGAUAAAACUACCUAUAUAAAUG GUACAGAUCCUAUUAACAGCCCAGAGCAAUACUGAUUCCAAA CUGCAAUUGGCGGUGUGCAAACGGGGGUUACUGGGGUGAU AUCAAAUUUAAAGAUGGUCAAUACUUAUUAAGUUGUAAAGG CGGUGCUUCUGCUGGUGUUUAUAAAGCCACUUAUGAUGAAA CUACAAAGAAAGUUAUUAUUGAUACGACUGAUAAAACUCCG UUGGCAACUGCGGAAGCUACAGCUAUUCGGGGAACGGCCACU AUAACCCCAACAACCAAUUGCUGAAGUAAACAAAGAGGGUGU UGAUACGACCAAGUUGCGGCUAACUUGCUGCAGCAGGGGU UACUGGCGCCGAUAGGACAAUACUAGCCUUGUAAAACUUAUC GUUUGAGGAUAAAACGGUAAGGUUAUUGAUGGUGGCUAUG CAGUGAAAUGGGCGACGAUUCUAGUCCGCUACAUUGAU GAGAAAACAGGUGCAAUACUGCUAAAACCAUACUUAUACA GAUGGUACUGGCGUUGCUCAAACUGGAGCUUGUAUUAUUGG UGGCGCAAUUGGUAAAUCUGAAGUUGUACUGCUACCGAUG GUAAGACUUAUCUAGCAAGCGACCUUGACAAAACUAUACUUA GAACAGGCGGUAGCUUAAAAGAGGUUAUACAGAUAAAGACU GAAAACCCACUGCAGAAAUAUGAUGCUGCCUUGGCACAGGUU GAUACACUUCGUUCUGACCUUGGUGCGGUUCAGAACCGUUC AACUCCGCUAUCACCAACUGGCGAAUACCGUAAAUAACUG UCUUUGCCCGUAGCGUAUCGAAGAUCGACUACGCAACC GAAGUCUCCAAGUCUCUGCGCGCAGAUUCUGCAGCAGGCC GGUACCUCCGUUCUGGCGCAGGCGAACAGGUUCCGCAAAAC GUCCUCUUCUUACUGCGU	82
mRNA Sequence (assumes T100 tail)	G*GGGAAUAAAGAGAGAAAAGAAGAGUAAGAAGAAAUUAA GAGCCACCAUGGCACAAAGUCAUUAAUACAACAGCCUGUCGC UGUUGACCCAGAAUAACCGUAACAAUCCAGUCCGCACUGG GCACUGCUAUCGAGCGUUUGUCUUCGCGUCUGCGUAUCAACA GCGCGAAAGACGAUGCGGCGAGGCGGGAUUGCUAACCGUU UUACCGCGAACAUCAAAGGUCUGACUCAGGCUUCCCGUAACG CUAACGACGGUAUCUCCAUUGCGCAGACACUGAAGGCGCGC UGAACGAAAUCAAACAACUCCAGCAGUGUGCGUGAACUGG CGGUUCAGUCUGCGAAUGGUACUAAUCUCCAGUCUGACCUUG ACUCCAUUCAGGCGUAAAUAACCCAGCGCCUGAACGAAUUCG ACCGUGUAUCCGGCCAGACUAGUUAACGCGUGAAGAUCC UGGCGCAGGACAAACCCUGACCAUCCAGGUUGGUGCCAACG ACGGUGAAACUAUCGAUAUUGAUUAAAAGAAAUACGCUUCU AAAACACUGGGACUUGAUAAAGCUUAAUGUCCAAGAUGCCUAC ACCCCGAAAGAAACUGCUGUAACCGUUGAUAAAACUACCUAU AAAAUUGGUACAGAUCCUAUUAACAGCCAGAGCAAUACUGAU AUCCAAACUGCAAUUGGCGGUGGUGCAAACGGGGGUUACUGG GGCUGAUUAUCAAUUUAAAGAUGGUCAAUUAUUAGAUG UUAAAGGCGGUUCUUCUGCUGGUGUUUAUAAAGCCACUUAU GAUGAAACUACAAGAAAGUUAUUAUUGAUACGACUGAUAA AACUCCGUUGGCAACUGCGGAAGCUACAGCUAUUUCGGGAAC GGCCACUAUAAACCAACAACCAAUUGCUGAAGUAAACAAGA GGGUGUUGAUACGACCACAGUUGCGGCUAACUUGCUGCAGC	83

TABLE 16-continued

Name	Sequence	SEQ ID NO :
	AGGGGUUACUGGCGCCGAUAAGGACAAUACUAGCCUUGUAA AACUAUCGUUUGAGGAUAAAAACGGUAAGGUUAUUGAUGGU GGCUAUGCAGUGAAAAUGGGCGACGAUUUCUUGCCGCUACA UAUGAUGAGAAAAACAGGUGCAAUACUGCUAAAAACACUAC UUAUACAGAUUGGUACUGGCGUUGCUCAAACUGGAGCUGUGA AAUUUGGUGGCGCAAUGGUAAAUCUGAAGUUGUACUGCU ACCGAUGGUAAAGACUUAUAGCAAGCGACCUUGACAAACA AACUUCAGAACAGGCGGUGAGCUUAAAGAGGUUAAUACAGA UAAGACUGAAAAACCCACUGCAGAAAAUUGAUGCUGCCUUGGC ACAGGUUGAUACACUUCGUUCUGACCUGGUGCGGUUCAGAA CCGUUUAACUCCGCUAUCACCAACUGGGCAUACCGUAAA UAACCUGUCUUCUGCCCGUAGCCGUUUCGAAGAUUCCGACUA CGCAACCGAAGUCUCCAAUGUUCGCGCGCAGAUUCUGCA GCAGGCCGUAUCCUUCGUUCUGGCGCAGGCGAACAGGUUCC GCAAAACGUCUCUUCUUCUACUGCGUUGAUAAUAGGCUUGAGC CUCGUGGCGCAUGCUUCUUGCCCUUGGGCCUCCCCCAGCC CCUCCUCCCCUUCUGCACCCGUACCCGUGGUCUUUGAAU AAAGUCUGAGUGGGCGGCAAAAAAAAAAAAAAAAAAAAAA AA AAAUCUAG	

TABLE 17

Name	Sequence	SEQ ID NO :
Flagellin Amino Acid Sequences		
ORF Sequence, AA	MAQVINTNSLSLLTQNNLNKSQSALGTAIERLSSGLRINSKDDAA GQAIANRFTANIKGLTQASRNANDGISIAQTTEGALNEINNNLQRV RELAVQSANGTNSQSDLDLSIQAEITQRLNEIDRVSGQTQFNGVKVL AQDNTLTIQVGANDGETIDIDLKEISSKTLGLDKLNVQDAYTPKET AVTVDKTTYKNGTDPITAQSNTDIQTAIGGGATGVTGADIKFKDG QYYLDVKGGASAGVYKATYDETTKKVNIDTTDKTPLATAEATAI RGTATI THNQIAEVTKEGVDTTTAAQLAAAGVTGADKDNLSLV KLSFEDKNGKVIDGGYAVKMGDDFYAATYDEKGTGAI TAKTTTTYT DGTGVAQTGAVKFGGANGSEVVTATDGKTYLASDLDKHNFRT GGELKEVNTDKTENPLQKIDAALAQVDTLRSDLGAVQNRFNSAIT NLGNTVNNLSARSRIEDSDYATEVSNMSRAQILQQAGTSVLAQA NQVPQNVLSLLR	54
Flagellin- GS linker- circum- sporozoite protein (CSP)	MAQVINTNSLSLLTQNNLNKSQSALGTAIERLSSGLRINSKDDAA GQAIANRFTANIKGLTQASRNANDGISIAQTTEGALNEINNNLQRV RELAVQSANGTNSQSDLDLSIQAEITQRLNEIDRVSGQTQFNGVKVL AQDNTLTIQVGANDGETIDIDLKQINSQTLGLDLTLNVQKQYKVS TAATVTGYADTTIALDNSTFKASATGLGGTDQKIDGDLKFDDTTG KYYAKVTVTGGTGKDGYYEVSVDKTNGEVLTAGGATSPLTGGLP ATATEDVKNVQVANADLTEAKAALTAAGVTGTASVVKMSYTDN NGKTIDGGLAVKVGDDYYSATQNKDGSISINTTKYTADDGTSKTA LNKLGADGKTEVVSIGGKTYAASKAEGHNFKAQPDLAEEAATT TENPLQKIDAALAQVDTLRSDLGAVQNRFNSAITNLGNTVNNLTS ARSRIEDSDYATEVSNMSRAQILQQAGTSVLAQANQVPQNVLSLL RGGGSGGGGSMAPDPNPNPNPNPNPNPNPNPNPNPNPNPNPN <u>NPANPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPN</u> <u>ANPNPNPNKNQNGQGHNMPNDPNRNVDENANANNVKNNN</u> <u>NEEPSDKHIEQYLKKIKNSISTEWSPCSVTCGNGIQVRIKPGSANKP</u> <u>KDELVDYENDIEKKICKMEKCSSVFNVVNS</u>	55
Flagellin- RPVT linker- circum- sporozoite protein (CSP)	MMAPDPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPN ANPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNPNKN QNGQGHNMPNDPNRNVDENANANNVKNNNNEEPSDKHIEQY LKKIKNSISTEWSPCSVTCGNGIQVRIKPGSANKPKDELVDYENDIEK KICKMEKCSSVFNVVNSRPVTMAQVINTNSLSLLTQNNLNKSQSA LGTAIERLSSGLRINSKDDAAGQAIANRFTANIKGLTQASRNAND GISIAQTTEGALNEINNNLQRVRELAVQSANGTNSQSDLDLSIQAEIT QRLNEIDRVSGQTQFNGVKVLAQDNTLTIQVGANDGETIDIDLKQI NSQTLGLDLTLNVQKQYKVSDDTAATVTGYADTTIALDNSTFKASAT GLGGTDQKIDGDLKFDDTTGKYYAKVTVTGGTGKDGYYEVSVD KTNGEVLTAGGATSPLTGGLPATATEDVKNVQVANADLTEAKAA LTAAGVTGTASVVKMSYTDNNGKTIDGGLAVKVGDDYYSATQNK DGSISINTTKYTADDGTSKALNKLGGADGKTEVVSIGGKTYAA SKAEGHNFKAQPDLAEEAATTENPLQKIDAALAQVDTLRSDLG AVQNRFNSAITNLGNTVNNLTSARSRIEDSDYATEVSNMSRAQILQ QAGTSVLAQANQVPQNVLSLLR	56

TABLE 18

Human Metapneumovirus Mutant Amino Acid Sequences		
Strain	Sequence	SEQ ID NO:
HMPV_SC_ DSCAV1_ 4MMV	MSWKVVIIFSLITPQHGLKESYLEESCSTITEGYLSVLRTGWYTNVFTLEVGD DVENLTCSDGPSLIKTELDTKSALRELKTVSADQLAREEQIENPGSGSFVLG AIALGVAAAAAVTAGVAICTIRLESEVTAINNALKKTNEAVSTLGNGVRV LAFVRELKDFVSKNLTRALNKNKCDIDDLKMAVSFSQFNRRFLNVVRQFS DNAGITPAISLDLMTDAELARAVPNMPTSAGQIKMLLENRAMVRRKGFIL CGVYGSSVIYMQLPFIGVIDTPCWIVKAAPSCSEKKGNYACLLREDQGWY Q ¹ NAGSTVYYPNEKDCETR ² GDHVFCDTAAGINVAEQSKECNINISTTNYPC KVSTGRHPISMVALSPLGALVACYKGVSCSIGSNRVGIKQLNKGCSYITNQ DADTVTIDNTVYQLSKVEGEQHVIKGRPVSSSFDPIKFPEDQFQVALDQVFENI NIENSQALVDQSNRILSSAEKGNTGFIIVIIILIAVLGSSMILVSIFIIKKTKKPTG APPELSGVTNNGFIPHN	85
HMPV_SC_ DSTRIC_ 4MMV	MSWKVVIIFSLITPQHGLKESYLEESCSTITEGYLSVLRTGWYTNVFTLEVGD DVENLTCSDGPSLIKTELDTKSALRELKTVSADQLAREEQIENPGSGSFVLG AIALGVAAAAAVTAGVAICTIRLESEVTAINNALKKTNEAVSTLGNGVRV LATAVRELKDFVSKNLTRAINKNKCDIDDLKMAVSFSQFNRRFLNVVRQFS DNAGITPAISLDLMTDAELARAVPNMPTSAGQIKMLLENRAMVRRKGFIL CGVYGSSVIYMQLPFIGVIDTPCWIVKAAPSCSEKKGNYACLLREDQGWY Q ¹ NAGSTVYYPNEKDCETR ² GDHVFCDTAAGINVAEQSKECNINISTTNYPC KVSTGRHPISMVALSPLGALVACYKGVSCSIGSNRVGIKQLNKGCSYITNQ DADTVTIDNTVYQLSKVEGEQHVIKGRPVSSSFDPIKFPEDQFQVALDQVFENI NIENSQALVDQSNRILSSAEKGNTGFIIVIIILIAVLGSSMILVSIFIIKKTKKPTG APPELSGVTNNGFIPHN	86
HMPV_SC_ DM_Krarup_ T74LD185P	MSWKVVIIFSLITPQHGLKESYLEESCSTITEGYLSVLRTGWYTNVFTLEVGD DVENLTCSDGPSLIKTELDTKSALRELKTVSADQLAREEQIENPGSGSFVLG AIALGVAAAAAVTAGVAIAKTIRLESEVTAINNALKKTNEAVSTLGNGVRV LATAVRELKDFVSKNLTRAINKNKCDIPDLKMAVSFSQFNRRFLNVVRQFS DNAGITPAISLDLMTDAELARAVPNMPTSAGQIKMLLENRAMVRRKGFIL GVYGS ¹ SVIYMQLPFIGVIDTPCWIVKAAPSCSEKKGNYACLLREDQGWYC Q ² NAGSTVYYPNEKDCETR ³ GDHVFCDTAAGINVAEQSKECNINISTTNYPC VSTGRHPISMVALSPLGALVACYKGVSCSIGSNRVGIKQLNKGCSYITNQD ADTVTIDNTVYQLSKVEGEQHVIKGRPVSSSFDPIKFPEDQFQVALDQVFENI ENSQALVDQSNRILSSAEKGNTGFIIVIIILIAVLGSSMILVSIFIIKKTKKPTGA PPELSGVTNNGFIPHN	87
HMPV_SC_ TM_Krarup_ T74LD185PD454N	MSWKVVIIFSLITPQHGLKESYLEESCSTITEGYLSVLRTGWYTNVFTLEVGD DVENLTCSDGPSLIKTELDTKSALRELKTVSADQLAREEQIENPGSGSFVLG AIALGVAAAAAVTAGVAIAKTIRLESEVTAINNALKKTNEAVSTLGNGVRV LATAVRELKDFVSKNLTRAINKNKCDIPDLKMAVSFSQFNRRFLNVVRQFS DNAGITPAISLDLMTDAELARAVPNMPTSAGQIKMLLENRAMVRRKGFIL GVYGS ¹ SVIYMQLPFIGVIDTPCWIVKAAPSCSEKKGNYACLLREDQGWYC Q ² NAGSTVYYPNEKDCETR ³ GDHVFCDTAAGINVAEQSKECNINISTTNYPC VSTGRHPISMVALSPLGALVACYKGVSCSIGSNRVGIKQLNKGCSYITNQD ADTVTIDNTVYQLSKVEGEQHVIKGRPVSSSFDPIKFPEDQFQVALDQVFENI ENSQALVDQSNRILSSAEKGNTGFIIVIIILIAVLGSSMILVSIFIIKKTKKPTGA PPELSGVTNNGFIPHN	88
HMPV_SC_ 4M_Krarup_ T74LS170LD185P	MSWKVVIIFSLITPQHGLKESYLEESCSTITEGYLSVLRTGWYTNVFTLEVGD DVENLTCSDGPSLIKTELDTKSALRELKTVSADQLAREEQIENPGSGSFVLG AIALGVAAAAAVTAGVAIAKTIRLESEVTAINNALKKTNEAVSTLGNGVRV LATAVRELKDFVLKNLTRAINKNKCDIPDLKMAVSFSQFNRRFLNVVRQFS DNAGITPAISLDLMTDAELARAVPNMPTSAGQIKMLLENRAMVRRKGFIL GVYGS ¹ SVIYMQLPFIGVIDTPCWIVKAAPSCSEKKGNYACLLREDQGWYC Q ² NAGSTVYYPNEKDCETR ³ GDHVFCDTAAGINVAEQSKECNINISTTNYPC VSTGRHPISMVALSPLGALVACYKGVSCSIGSNRVGIKQLNKGCSYITNQD ADTVTIDNTVYQLSKVEGEQHVIKGRPVSSSFDPIKFPEDQFQVALDQVFENI ENSQALVDQSNRILSSAEKGNTGFIIVIIILIAVLGSSMILVSIFIIKKTKKPTGA PPELSGVTNNGFIPHN	89
HMPV_SC_ 5M_Krarup_ T74LS170LD185PD454N	MSWKVVIIFSLITPQHGLKESYLEESCSTITEGYLSVLRTGWYTNVFTLEVGD DVENLTCSDGPSLIKTELDTKSALRELKTVSADQLAREEQIENPGSGSFVLG AIALGVAAAAAVTAGVAIAKTIRLESEVTAINNALKKTNEAVSTLGNGVRV LATAVRELKDFVLKNLTRAINKNKCDIPDLKMAVSFSQFNRRFLNVVRQFS DNAGITPAISLDLMTDAELARAVPNMPTSAGQIKMLLENRAMVRRKGFIL GVYGS ¹ SVIYMQLPFIGVIDTPCWIVKAAPSCSEKKGNYACLLREDQGWYC Q ² NAGSTVYYPNEKDCETR ³ GDHVFCDTAAGINVAEQSKECNINISTTNYPC VSTGRHPISMVALSPLGALVACYKGVSCSIGSNRVGIKQLNKGCSYITNQD ADTVTIDNTVYQLSKVEGEQHVIKGRPVSSSFDPIKFPEDQFQVALDQVFENI ENSQALVDQSNRILSSAEKGNTGFIIVIIILIAVLGSSMILVSIFIIKKTKKPTGA PPELSGVTNNGFIPHN	90

SEQUENCE LISTING

Sequence total quantity: 147

SEQ ID NO: 1 moltype = DNA length = 1620
 FEATURE Location/Qualifiers
 source 1..1620
 mol_type = genomic DNA
 organism = Human metapneumovirus

SEQUENCE: 1

atgagctgga	agggtggtgat	tatcttcagc	ctgctgatta	cacctcaaca	cggcctgaag	60
gagagctacc	tgggaagagag	ctgctccacc	atcaccgagg	gctacctgag	cgtgctgcgg	120
accggctggt	acaccaacgt	gttcacccctg	gagggtggcg	acgtgggagaa	cctgacctgc	180
agcgacggcc	ctagcctgat	caagaccgag	ctggacctga	ccaagagcgc	tctgagagag	240
ctgaagaccg	tgtccgccga	ccagctggcc	agagaggaaac	agatcgagaa	ccctcggcag	300
agcagattcg	tgtctggcgcg	catcgctctg	ggagtcgccg	ctgccgctgc	agtgacagct	360
ggagtggcca	tgtctaagac	catcagactg	gaaagcgagg	tgacagccat	caacaatgcc	420
ctgaagaaga	ccaacgaggg	cgtgagcacc	ctgggcaatg	gagtgagagt	gctggccaca	480
gccgtgctgg	agctgaagga	cttcgtgagc	aagaacctga	ccagagccat	caacaagaac	540
aagtgcgaca	tcgatgacct	gaagatggcc	gtgagcttct	cccagttcaa	cagacggttc	600
ctgaacgtgg	tgagacagtt	ctccgacaac	gctggaatca	cacctgccat	tagcctggac	660
ctgatgaccg	acgcgagct	ggctagagcc	gtgcccaaca	tgcccaccag	cgttgccag	720
atcaagctga	tgtctggagaa	cagagccatg	gtgctggagaa	agggtctcgg	catcctgatt	780
ggggtgtatg	gaagctccgt	gatctacatg	gtgcagctgc	ccatcttcgg	cgtgatcgac	840
acaccctgct	ggatcgtgaa	ggccgctcct	agctgctccg	agaagaaagg	aaactatgcc	900
tgtctgctga	gagaggacca	gggctggtac	tgccagaacg	ccggaagcac	agtgactat	960
cccaacgaga	aggactgcga	gaccagaggg	gaccacgtgt	tctgcgacac	cgtgcccga	1020
atcaacgtgg	ccgagcagag	caaggagtgc	aacatcaaca	tcagcacaac	caactacccc	1080
tgcaagtgga	gcaccggacg	gcaccccatc	agcatgggtg	ctctgagccc	tctgggcgct	1140
ctggtggcct	gctataaggg	cgtgtcctgt	agcatcgcca	gcaatcgggt	gggcatcatc	1200
aagcagctga	acaagggatg	ctcctacatc	accaaccagg	acgcccagac	cgtgaccatc	1260
gacaacaccg	gtgaccaggt	gagcaagggt	gagggcgagc	agcacgtgat	caagggcaga	1320
cccgtgagct	ccagcttcga	ccccatcaag	ttccctgagg	accagttcaa	cgtggccctg	1380
gaccaggtgt	ttgagaacat	cgagaacagc	caggccctgg	tggaccagag	caacagaatc	1440
ctgtccagcg	ctgagaagcg	caacaccggc	ttcatcattg	tgatcattct	gatcgccgtg	1500
ctgggcagct	ccatgatcct	ggtgagcatc	ttcatcatta	tcaagaagac	caagaacccc	1560
accggagccc	ctcctgagct	gagcggcgctg	accaacaatg	gttctattcc	ccacaactga	1620

SEQ ID NO: 2 moltype = DNA length = 1620
 FEATURE Location/Qualifiers
 source 1..1620
 mol_type = genomic DNA
 organism = Human metapneumovirus

SEQUENCE: 2

atgtcttggg	aagtgatgat	catcatttcg	ttactcataa	caccccagca	cgggctaagg	60
gagagttatt	tgggaagaatc	atgtagtact	ataactgagg	gataccctcag	tgttttaaga	120
acaggctggt	acaactaatgt	cttcacatta	gaagttgggt	atgttgaaaa	tcttacatgt	180
actgatggac	ctagcttaaat	caaaaacagaa	cttgatctaa	caaaaagtgc	ttaaggggaa	240
ctcaaaacag	tctctgctga	tcagttggcg	agagaggagc	aaattgaaaa	tcccagacaa	300
tcaagatttg	tcttaggtgc	gatagctctc	ggagttgcta	cagcagcagc	agtcacagca	360
ggcattgcaa	tagccaaaac	cataaggcct	gagagtggag	tgaatgcaat	taaagggtgt	420
ctcaaaacaa	ctaataaagc	agtatccaca	ttaggggaatg	gtgtgctggg	cctagccact	480
gcagtggagag	agctaaaaaga	atttgtgagc	aaaaacctga	ctagtgcaat	caacagggaac	540
aaatgtgaca	ttgctgatct	gaagatggct	gtcagcttca	gtcaattcaa	cagaagattt	600
ctaaatgttg	tgccggcagtt	ttcagacaat	gcaggggataa	caccagcaat	atcattggac	660
ctgatgactg	atgctgagtt	ggccagagct	gtatcataca	tgccaacatc	tgcaaggcag	720
ataaaaactga	tgttggagaa	ccgcgcaatg	gtaaggagaa	aaggatttgg	aatcctgata	780
ggggtctacg	gaagctctgt	gatttcatatg	gttcaattgc	cgatcttttg	tgtcatagat	840
acaccttggt	ggatccatcaa	ggcagctccc	tcttgctcag	aaaaaaacgg	gaattatgct	900
tgccctctaa	gagaggatca	agggtgggtat	tgtaaaaatg	caggatctac	tgtttactac	960
ccaaatgaaa	aagactgcga	aacaagaggt	gatcatgttt	tttgtgacac	agcagcaggg	1020
atcaatgttg	ctgagcaatc	aagagaatgc	aacatcaaca	tatctactac	caactaccca	1080
tgcaaaagtc	gcacagggaag	acaccctata	agcatgggtg	cactatcacc	tctcgggtgt	1140
ttggtggcct	gctataaagg	ggtaagctgc	tcgattggca	gcaattgggt	tggaatcatc	1200
aaacaattac	ccaaaggctg	ctcatacata	accaaccagg	atgcagacac	tgtaacaatt	1260
gacaataccg	tgtatcaact	aagcaaaagt	gaaggtgaac	agcatgtaac	aaaagggaga	1320
ccagtttcaa	gcagttttga	tccaatcaag	tttcctgagg	atcagttcaa	tgttgcgctt	1380
gatcaagctc	tcgaaacagt	caggcactag	tggaccagtc	aaacaaaatt	1440	
ctaaacagtg	cagaaaaagg	aaacactggt	ttcattatcg	tagtaatttt	ggttgctggt	1500
cttggtctaa	ccatgatttc	agtgagcatc	atcatcataa	tcaagaaaaa	aaggaagccc	1560
acaggagcac	ctccagagct	gaatggtgtc	accaacggcg	gtttcatacc	acatagttag	1620

SEQ ID NO: 3 moltype = DNA length = 1620
 FEATURE Location/Qualifiers
 source 1..1620
 mol_type = genomic DNA
 organism = Human metapneumovirus

SEQUENCE: 3

atgtcttggg	aagtgatgat	tatcatttcg	ttactcataa	cacctcagca	tggactaaaa	60
gaaagtattt	tagaagaatc	atgtagtact	ataactgaag	gatatctcag	tgttttaaga	120
acaggttggt	acaccaatgt	ctttacatta	gaagttgggt	atgttgaaaa	tcttacatgt	180

-continued

```

actgatggac ctagcttaat caaaacagaa cttgacctaa ccaaaagtgc ttaagagaa 240
ctcaaaacag tttctgtgta tcagtttagc agagaagaac aaattgaaaa tcccagacaa 300
tcaaggtttg tcctaggtgc aatagctctt ggagttgcca cagcagcagc agtcacagca 360
ggcattgcaa tagccaaaac tataaggett gagagtgaag tgaatgcaat caaaggtgct 420
ctcaaaacaa ccaatgagggc agtatcaaca ctaggaaatg gagtgcgggt cctagccact 480
gcagtaagag agctgaaaga atttgtgagc aaaaacctga ctagtgccgat caacaagaac 540
aagtggtgaca ttgctgattt gaagatggct gtcagcttca gtcagttcaa cagaagattc 600
ctaaatgttg tgcggcagtt ttcagacaat gcagggataa caccagcaat atcattggac 660
ctgatgaatg atgctgagct ggcacagagc gtatcatata tgccaacatc tgcaggacag 720
ataaaactaa tggtagagaa ccgtgcaatg gtgaggagaa aaggatttgg aatcttgata 780
ggggtctacg gaagctctgt gatttacatg gtccagctgc cgatctttgg tgcataaaat 840
acaccttggt ggataatcaa ggcagctccc tcttgttcag aaaaagatgg aaattatgct 900
tgctctctaa gagaggatca aggggtggtat tgtaaaaatg caggatccac tgtttactac 960
ccaaatgaaa aagactgcga aacaagaggt gatcatgttt tttgtgacac agcagcaggg 1020
atcaatgttg ctgagcaatc aagagaatgc aacatcaaca tatctaccac caactaccca 1080
tgcaaaagtc gcacaggaag acacctatc agcatggttg cactatcacc tctcggtgct 1140
ttggtagctt gctacaaaag ggtagctgc tcgactggca gtaatcaggt tggaaataatc 1200
aaacaactac ctaaggctg ctcatacata actaaccagg acgcagacac tgaacaaatt 1260
gacaaacactg tgtatcaact agcaaaagtt gaggggtgaac agcatgtaat aaaagggaga 1320
ccagtttcaa gcagttttga tccaatcagg tttcctgagg atcagttcaa tgttgcgctt 1380
gatcaagtct ttgaaagcat tgaaaacagt caagcactag tggaccagtc aaacaaaatt 1440
ctgaacagtg cagaaaaagg aaacactgggt ttcattattg taataatttt gattgctggt 1500
cttgggttaa ccatgatttc agtgagcatc atcatcataa tcaaaaaaac aaggaagccc 1560
acagggggac ctcggagctt gaatggtggt accaacggcg gtttcatacc gcatagttag 1620

```

```

SEQ ID NO: 4      moltype = DNA length = 1725
FEATURE          Location/Qualifiers
source           1..1725
                 mol_type = genomic DNA
                 note = Human respiratory syncytial virus
                 organism = unidentified

```

```

SEQUENCE: 4
atggagttgc caatcctcaa aacaaatgca attaccacaa tccttgctgc agtcacactc 60
tgtttcgctt ccagtcacaaa catcactgaa gaattttatc aatcaaatgc cagtgcagtt 120
agcaaaagct atcttagtgc tctaagaact ggttggtata ctagtgttat aactatagaa 180
ttaagtaata tcaagggaaa taagtgtaat ggaacagatg ctaaggtaaa attgataaaa 240
caagaattag aaaaatataa aatgctgta acagaattgc agttgctcat gcaaaagcaca 300
ccagcagcca acaatcgagc cagaagagaa ctaccaaggt ttatgaatta tacactcaat 360
aatacccaaa ataccaatgt aacattaagc aagaaaagga aaagaagatt tcttggtctt 420
ttggttaggtg ttgtagctgc aatcgccagt ggcattgctg tatctaaggt cctgcacctc 480
gaaggggaaag tgaacaaaat caaaagtgtc ctactatcca caaacaaggg tgtagtcagc 540
ttatcaaatg gagtttagtg cttaaccagc aaagtgttag acctcaaaaa ctatatagat 600
aaacagttgt tacctattgt gaacaagcaa agctgcagca tatcaaacat tgaactgtg 660
atagagttcc aacaaaagaa caacagacta ctagagatta ccagggaatt tagtgttaat 720
gcaggtgtaa ctacacctgt aagcacttat atgttaacta atagtgaatt attatcatta 780
atcaatgata tgcctataac aatgatcag aaaaagttaa tgtccaacaa tgttcaata 840
gttagacagc aaagttaact tatcatgtcc ataataaagg aggaagtctt agcatatgta 900
gtacaattac cactatatgg ttgaatagat acacctgtt ggaaactgca cacatcccct 960
ctatgtacaa ccaacacaaa ggaagggtcc aacatctgct taacaagaac cgacagagga 1020
tggtattgtg acaatgcagg atcagtatct ttcttcccac aagctgaaac atgtaaaagt 1080
caatcgaatc gggatttttg tgacacaatg aacagtttaa cattaccaag tgaagtaaat 1140
ctctgcaaca ttgacattat caacccaaa tatgattgca aaattatgac ttcaaaaaa 1200
gatgtaagca gctccgttat cacatctcta ggagccattg tgtcatgcta tggcaaaact 1260
aaatgtacag catccaataa aaatcggtgg atcataaaga cattttctaa cgggtgtgat 1320
tatgtatcaa ataagggggg ggatactgtg tctgtaggtg atacattata ttatgtaaat 1380
aagcaagaag gcaaaagtct ctatgtaaaa ggtgaaccaa taataaattt ctatgaccca 1440
tagtggttcc cctctgatga atttgatgca tcaatatctc aagtcaatga gaagattaac 1500
cagagcctag catttattct taaatccgat gaattattac ataagttaa tgctggtaaa 1560
tccaccacaa atatcatgat aactactata attatagtga ttatagtaat attgttatca 1620
ttaattgcag ttggactgct cctatactgc aaggccagaa gcacaccagt cacactaagt 1680
aaggatcaac tgagtgggat aaataatatt gcatttagta actga 1725

```

```

SEQ ID NO: 5      moltype = AA length = 539
FEATURE          Location/Qualifiers
source           1..539
                 mol_type = protein
                 note = Human metapneumovirus isolate
                 organism = unidentified

```

```

SEQUENCE: 5
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPRQ SRFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VFSQFNRFR LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLNRRAM VRRKGFGILI GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEDQFNVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELSGV TNNGFIPHN 539

```

-continued

SEQ ID NO: 6 moltype = AA length = 539
 FEATURE Location/Qualifiers
 source 1..539
 mol_type = protein
 organism = Human metapneumovirus

SEQUENCE: 6

MSWKVMIIIS	LLITPQHGLK	ESYLEESCST	ITEGYLSVLR	TGWYTNVFTL	EVGDVENLTC	60
TDGPSLIKTE	LDLTKSALRE	LKTVSADQLA	REEQIENPRQ	SRFVLGAIAL	GVATAAAVTA	120
GIAIAKTIRL	ESEVNAIKGA	LKQTNEAVST	LGNGVRVLAT	AVRELKEFVS	KNLTSAINRN	180
KCDIADLKMA	VSFSQFNRRF	LNVRVQFSDN	AGITPAISLD	LMTDAELARA	VSYMPTSAGQ	240
IKMLLENRAM	VRKGFGLILI	GVYGS SVIYM	VQLPIFGVID	TPCWIIKAAP	SCSEKNGNYA	300
CLLREDQGWY	CKNAGSTVYY	PNEKDCETRG	DHVFCDTAAG	INVAEQSREC	NINISTTNYP	360
CKVSTGRHPI	SMVALSPLGA	LVACYKGVSC	SIGSNWVGII	KQLPKGCSYI	TNQDADTVTI	420
DNTVYQLSKV	EGEQHVIGKR	PVSSSFDPPIK	FPEDQFNVAL	DQVFESIENS	QALVDQSNKI	480
LNSAEKGNTG	FIIIVILVAV	LGLTMISVSI	IIIIKKTRKP	TGAPPELNGV	TNGGFIPHS	539

SEQ ID NO: 7 moltype = AA length = 539
 FEATURE Location/Qualifiers
 source 1..539
 mol_type = protein
 organism = Human metapneumovirus

SEQUENCE: 7

MSWKVMIIIS	LLITPQHGLK	ESYLEESCST	ITEGYLSVLR	TGWYTNVFTL	EVGDVENLTC	60
TDGPSLIKTE	LDLTKSALRE	LKTVSADQLA	REEQIENPRQ	SRFVLGAIAL	GVATAAAVTA	120
GIAIAKTIRL	ESEVNAIKGA	LKTTNEAVST	LGNGVRVLAT	AVRELKEFVS	KNLTSAINKN	180
KCDIADLKMA	VSFSQFNRRF	LNVRVQFSDN	AGITPAISLD	LMNDAELARA	VSYMPTSAGQ	240
IKMLLENRAM	VRKGFGLILI	GVYGS SVIYM	VQLPIFGVIN	TPCWIIKAAP	SCSEKNGNYA	300
CLLREDQGWY	CKNAGSTVYY	PNEKDCETRG	DHVFCDTAAG	INVAEQSREC	NINISTTNYP	360
CKVSTGRHPI	SMVALSPLGA	LVACYKGVSC	STGSNQVGII	KQLPKGCSYI	TNQDADTVTI	420
DNTVYQLSKV	EGEQHVIGKR	PVSSSFDPPIR	FPEDQFNVAL	DQVFESIENS	QALVDQSNKI	480
LNSAEKGNTG	FIIIVILIAV	LGLTMISVSI	IIIIKKTRKP	TGAPPELNGV	TNGGFIPHS	539

SEQ ID NO: 8 moltype = AA length = 574
 FEATURE Location/Qualifiers
 source 1..574
 mol_type = protein
 note = Human respiratory syncytial virus
 organism = unidentified

SEQUENCE: 8

MELPILKTNA	ITITLAAVTL	CFASSQNITE	EFYQSTCSAV	SKGYLSALRT	GWYTSVITIE	60
LSNIKENKCN	GTDKVKLIK	QELDKYKNAV	TELQLLMQST	PAANNRRARRE	LPRFMNYTLN	120
NTKNTNVTLS	KKRKRRLGFI	LLGVGSAIAS	GIAVSKVLHL	EGEVNKKISA	LLSTNKAVVS	180
LSNGVSVLTS	KVLDLKNYID	KQLLPIVKNQ	SCSISNIETV	IEFQQKNNRL	LEITREFSVN	240
AGVTPPVSTY	MLTNSSELLSL	INDMPITNDQ	KKLMSNNVQI	VRQQSYSIMS	IIKEEVLAYV	300
VQLPLYGVID	TPCWKLHTSP	LCTTNTKEGS	NICLTRTDRG	WYCDNAGSVS	FFPQAETCKV	360
QSNRVFCDTM	NSLTLPSSEVN	LCNIDIFNPK	YDCKIMTSKT	DVSSSVITSL	GAIVSCYGKT	420
KCTASNKNRG	IIKTFSNGCD	YVSNKGVDTV	SVGNTLYYVN	KQEGKSLYVK	GEPIINFYDP	480
LVFPSDEPDA	SISQVNEKIN	QSLAFIRKSD	ELLHNVNAGK	STTNIMITTI	IIVIIIVILLS	540
LIAVGLLLYC	KARSTPVTLT	KDQLSGINNI	AFSN			574

SEQ ID NO: 9 moltype = DNA length = 1617
 FEATURE Location/Qualifiers
 source 1..1617
 mol_type = genomic DNA
 note = Human parainfluenza virus 3
 organism = unidentified

SEQUENCE: 9

atgccaat	ttcaactactgt	aattattaca	accatgatca	tggcatcaca	ctgccaaata	60
gacatcaca	aactacagca	tgtagggtgta	ttgggtcaaca	gtcccaaagg	gatgaagata	120
tcacaaaact	tcgaacaacg	atatctaata	ctgagttctca	taccaaaaat	agaagattct	180
aactcttg	gtgaccaaca	gatcaagcaa	tacaagaggt	tattggatag	actgatcatt	240
cctttatatg	atggactaag	attacagaag	gatgtgatag	tgactaatca	agaatccaat	300
gaaaacactg	atcccgagaac	agaacgattc	tttgaggagg	taattggaac	tattgtctcta	360
ggagtagcaa	cctcagcaca	aattacagca	gcagttgctc	tggttgaagc	caagcaggca	420
agatcagaca	ttgaaaaact	caaggaagca	atcaggggaca	caaataaagc	agtcagtcga	480
gttcagagct	ctgtaggaaa	tttgatagta	gcaattaaat	cagtcocagga	ttatgtcaac	540
aaagaaatcg	tgccatcgat	tgcgagacta	gggtgtggaag	cagcaggact	tcagttagg	600
attgcattaa	cacagcatta	ctcagaatta	acaaatatat	ttggtgataa	cataggatcg	660
ttacaagaaa	aaggaataaa	attacaaggt	atagcatcat	tataccgtac	aaatatcaca	720
gaaatattca	caacatcaac	agttgacaaa	tatgatattt	atgatctatt	atttacagaa	780
tcaataaagg	tgagagttat	agatgttgat	ttgaatgatt	actcaataac	cctccaagtc	840
agactccctt	tattgaccag	actgctgaac	actcaaatct	acaaagtaga	ttccatatca	900
tacaatatcc	aaaatagaga	atgggtatatc	cctcttccca	gccatatcat	gacgaaaggg	960
gcatttctag	gtggagcaga	tgctcaagaa	tgcatagaag	cattcagcag	ttatatatgc	1020
ccttctgata	caggatttgt	actaaacct	gaaatggaga	gctgtctatc	aggaaacata	1080
tcccaatgtc	caagaaccac	agtcacatca	gacatagttc	ctaggtagtc	atttgtcaat	1140
ggaggagtgg	ttgcgaattg	tataacaact	acatgtacat	gcaatgggat	cggtaataga	1200
atcaaccaac	cactgatca	aggagtcaaa	attataacac	ataaagaatg	taataacaata	1260
ggtatcaacg	gaatgctatt	caacacaaac	aaagaaggaa	ctcttgcat	ctacacacca	1320

-continued

gacgacataa	cattaataca	tctgtgtgca	cttgatccga	ttgacatatt	aatcgagctc	1380
aacaaggcca	aatcagatct	tgagggaatca	aaagaatgga	taagaagggtc	aaatcaaaag	1440
ctagatttcta	ttggaagtgtg	gcacatcatct	agcactacaa	tcatagttat	tttgataaatg	1500
atgattatat	tgtttataat	taataataca	ataattacaa	ttgcaattaa	gtattacaga	1560
attcaaaaga	gaaatcgagt	ggatcaaaat	gataagccgt	atgtattaac	aaacaag	1617

SEQ ID NO: 10 moltype = DNA length = 1716
 FEATURE Location/Qualifiers
 source 1..1716
 mol_type = genomic DNA
 note = Human parainfluenza virus 3
 organism = unidentified

SEQUENCE: 10

atggaatact	ggaagcacac	caaccacgga	aaggatgctg	gtaatgagct	ggagacatcc	60
acagccactc	atggcaacaa	gctcaccac	aagataacat	atatattgtg	gacgataaacc	120
ctggtgttat	tatcaatagt	cttcacata	gtgctaacta	attccatcaa	aagtgaagaa	180
gcccgccaat	catgtgtaca	agacataaat	aatgagttaa	tggaagttac	agaaaagatc	240
caagtggcat	cggataatca	taatgatcta	atacagtcag	gagtgaatac	aagggtctct	300
acaattcaga	gtcatgtcca	gaattatata	ccaatatcat	tgacacaaca	aatatcggtat	360
cttaggaaat	tcattagtga	aattacaatt	agaaatgata	atcaagaagt	gccaccacaa	420
agaataaac	atgatgtggg	tataaaacct	ttaaatccag	atgatttctg	gagatgcacg	480
tctggtcttc	catctttgat	gaaaactcca	aaaataagat	taatgccggg	accaggatta	540
ttagctatgc	caacgactgt	tgatggctgt	gtcagaaccc	cgtccttagt	gataaaatgat	600
ctgatttatg	cttacacctc	aaatctaatt	actcgaggtt	gccaggatat	agggaaatca	660
tatcaagtat	tacagatagg	gataataact	gtaaactcag	acttgggtacc	tgacttaaat	720
cctaggatct	ctcacacctt	caacataaat	gacaatagaa	agtcattgtt	tctagcactc	780
ctaaatacag	atgtatatca	actgtgttca	accccaaaag	ttgatgaaag	atcagattat	840
gcatacatcg	gcatagaaga	tattgtactt	gatattgtca	attatgatgg	ctcaatctcg	900
acaacaagat	ttaagaataa	taataataag	tttgatcaac	catatgcggc	attataacca	960
tctgttggac	cagggatata	ctcaaaaggc	aaaataatat	ttctcgggta	tgagggtctt	1020
gaacatccaa	taaatgagaa	tcacatctgc	aacacaaact	gggtgcctgg	gaaaacacag	1080
agagactgta	atcaagcttc	tcatagtcca	tggttttccag	atagaaggat	gggtcaactct	1140
ataattgttg	ttgacaaggg	cttgaactca	gttccaaaaa	tgaaggatat	gacgatattct	1200
atgagacaaa	attactgggg	gtcagaagga	agattacttc	tactaggtaa	caagatctac	1260
atatacacaa	gatctacaag	ttggcacagc	aagttacaat	taggaataat	tgacattact	1320
gactacagtg	atataaggat	aaaatggaca	tggcataatg	tgctatcaag	accaggaaac	1380
aatgaatgtc	catggggaca	ttcatgtccg	gatggatgta	taacgggagt	atataccgat	1440
gcataatccac	tcaatcccac	aggaagcatt	gtatcatctg	tcataattgga	ctcacaaaaa	1500
tcgagagtgca	accagctcat	aacttactca	acagcaaccg	aaagggtaaa	cgagctggct	1560
atccgaaaca	aaacactctc	agctgggtac	acaacaacaa	gctgcattac	acactataac	1620
aaagggtatt	gttttcatat	agtagaaata	aatcataaaa	gcttaaacac	atttcaaccc	1680
atgttgttca	aaacagagat	tccaaaaagc	tgacgt			1716

SEQ ID NO: 11 moltype = DNA length = 1716
 FEATURE Location/Qualifiers
 misc_feature 1..1716
 note = Synthetic Polynucleotide
 source 1..1716
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 11

atggaatact	ggaagcacac	caaccacggc	aaggacgccc	gcaacgagct	ggaaaccagc	60
acagccacac	acggcaacaa	gctgaccaac	aagatcacct	acatcctgtg	gaccatcacc	120
ctggtgtctg	tgagcatcgt	gttcacatc	gtgctgacca	atagcatcaa	gagcgagaag	180
gccagagaga	gctgtgtgca	ggacatcaac	aacgagtcca	tggaagtgc	cgagaagatc	240
caggtggcca	gcgacaacac	caacgacctg	atccagagcg	gcgtgaacac	ccggctgtctg	300
accatccaga	gccacgtgca	gaactacatc	cccacagccc	tgacccagca	gatcagcgac	360
ctgagggaagt	tcatacgcga	gatcaccatc	cggaacgaca	accagggaagt	gccccccag	420
agaatcaccc	acgacgtggg	catcaagccc	ctgaacccc	acgatttctg	gcggtgtaca	480
agcggcctgc	ccagcctgat	gaagaccccc	aagatccggc	tgatgcctgg	ccctggactg	540
atggccatgc	ctaccacagt	ggatggctgt	gtgcgggacc	ccagcctcgt	gatcaacgat	600
ctgatctacg	cctacaccag	caacctgac	acccggggct	gccaggatat	cggcaagagc	660
taccaggtgc	tgcatagctg	catcatcacc	gtgaactccg	acctgggtgc	cgacctgaac	720
cctcggatca	gccacacctt	caacatcaac	gacaacagaa	agagctgcag	cctggctctg	780
ctgaacaccc	agcgtgacca	gctgtgcagc	acccccaaag	tggacgagag	aagcgactac	840
gccagcagcg	gcacagagga	tatcgtgctg	gacatcgtga	actacgacgg	cagcatcagc	900
accacccggg	tcaagaacaa	caacatcagc	ttcgaccagc	cctaagccgc	cctgtaccct	960
tctgtgggcc	ctggcatcta	ctacaagggc	aagatcatct	tcctgggcta	cgccggcctg	1020
gaacacccca	tcaacgagaa	cgccatctgc	aacaccaccg	gctgccctgg	caagacccag	1080
agagactgca	atcaggccag	ccacagcccc	tggttcagcg	accgcagaat	gggtcaactct	1140
atcatcgtgg	tggacaaggg	cctgaacagc	gtgcccgaag	tgaaagtgtg	gacaatcagc	1200
atgcgccaga	actactgggg	cagcgagggc	agacttctgc	tgctgggaaa	caagatctac	1260
atctacaccc	gggtccaccg	ctggcacagc	aaactgcagc	tgggaaatcat	cgacatcacc	1320
gactacagcg	acatccggat	caagtggacc	tggcacacag	tgctgagcag	accggccaac	1380
aatgagtgc	cttggggcca	cagctgcccc	gatggatgta	tcaccggcgt	gtacacggac	1440
gcctaccccc	tgaatcctac	cggctccatc	gtgtccagcg	tgatcctgga	cagccagaaa	1500
agcagagtga	accccgatg	cacatacagc	accgccaccg	agagagtga	cgaactggcc	1560
atcagaaaca	agaccctgag	cgcggctac	accaccacaa	gctgcacac	acactacaac	1620
aagggtact	gcttccacat	cgtggaatc	aaccacaagt	ccctgaacac	cttccagccc	1680

-continued

atgctgttca agaccgagat ccccaagagc tgetcc 1716

SEQ ID NO: 12 moltype = DNA length = 1617
 FEATURE Location/Qualifiers
 misc_feature 1..1617
 note = Synthetic Polynucleotide
 source 1..1617
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 12
 atgcccata gcatcctgct gatcatcacc acaatgatca tggccagcca ctgccagatc 60
 gacatcacca agctgcagca cgtgggcgtg ctggtgaaca gcccgaagg catgaagatc 120
 agccagaact tcgagacacg ctacctgac ctgagcctga tcccgaagat cgaggacagc 180
 aacagctcgc gcgaccagca gatcaagcag tacaagcggc tgcgtggacag actgatcattc 240
 cccctgtacg acgacctgcg gctgcagaaa gacgtgatcg tgaccaacca ggaaagcaac 300
 gagaacaccg acccccgacg cgagagattc ttccggcggc tgatcggcac aatcgccctg 360
 ggagtggcca caagcgccca gattacagcc gctgtggccc tgggtggaag caagcaggcc 420
 agaagcgaca tcgagaagct gaaagaggcc atccgggaca ccaacaaggc cgtgcagagc 480
 gtgcagtcca gcgtgggcaa tctgatcgtg gccatcaagt ccgtgcagga ctacgtgaac 540
 aaagaaatgc tgccctctat cgcccggtg ggctgtgaag ctgcccgaact gcagctgggc 600
 attgccctga cacagacta cagcgagctg accaaccatt ctggcgacaa catcggcagc 660
 ctgcagggaaa agggcattaa gctgcaggga atcgccagcc tgtaccgcac caacatcacc 720
 gagatcttca ccaccagcac cgtggataag tacgacattc acgacctgct gttcaccgag 780
 agcatcaaaag tgcgctgtag cgacgtggac ctgaacgact acagcatcac cctgcaagt 840
 cggtgcccc tgctgaccag actgctgaac accagatct acaagggtga cagcatctcc 900
 tacaacatcc agaaccgaga gtggtacatc cctctgccca gccacattat gaccaagggc 960
 gcttctcttg gcggagccga cgtgaaagag tgcctcagag ccttcagcag ctacatctgc 1020
 ccacgcgacc ctggtctcgt gctgaaccac gagatggaaa gctgctcag cggaacatc 1080
 agccagtacc ccagaaccac cgtgacctcc gacatcgtgc ccagatacgc cttcgtgaat 1140
 ggccgctggt tggccaactg catcaccacc acctgtacct gcaacggcat cggcaaccgg 1200
 atcaaccagc gtcctcgatc gggcgtgaag attatcacc acaaagagt taacaccatc 1260
 ggcatcaacg ctatctgtgt caataccaac aaagagggca ccttggcctt ctacaccccc 1320
 gacgatatca ccctgaacaa ctccgtgggt ctggacccca tcgacatctc catcgagctg 1380
 aacaaggcca agagcgagct ggaagagtcc aaagagtgga tccggcggag caaccagaag 1440
 ctggactcta tcggcagctg gcaccagagc agcaccacca tcctcgtgat cctgattatg 1500
 atgattatcc tgttcacat caacattacc atcatcacta tcgccattaa gtactaccgg 1560
 atccagaaac ggaaccgggt ggaccagaat gacaagccct acgtgctgac aaacaag 1617

SEQ ID NO: 13 moltype = AA length = 539
 FEATURE Location/Qualifiers
 source 1..539
 mol_type = protein
 note = Human parainfluenza virus 3
 organism = unidentified

SEQUENCE: 13
 MPISILLIIT TMIMASHCQI DITKLQHVGV LVNSPKGMKI SQNFETRYLI LSLIPKIEDS 60
 NSCGDQQIKQ YKRLDLRLII PLYDGLRLQK DVIVTNQESN ENTDPRTFR FGGVIGTIAL 120
 GVATSAQITA AVALVEAKQA RSDIEKLKEA IRDTNKAVQS VQSSVGNLIV AIKSVQDYVN 180
 KEIVPSIARL GCEAAGLQLG IALTQHYSEL TNIFGDNIGS LQEKGIKLQG IASLYRTNIT 240
 EIFTTSTVDK YDIYDLLEFE SIKVRVIDVD LNDYSITLQV RLPPLTRLLN TQIYKVDNIS 300
 YNIQNEWYI PLPSHIMTKG AFLGGADVKE CIEAFSSYIC PSDPGFVLNH EMESCLSGNI 360
 SQCPRTTSTS DIVPRYAFVN GGVVANCITT TCTCNGIGNR INQPPDQGVK IITHKECNTI 420
 GINGMLFNTN KEGTLAFYTP DDITLNNNSVA LDPIDISIEL NKAASDLEES KEWIRRSNQK 480
 LDSIGSWHQS STTIIVILIM MITLFIINIT IITIAIKYYR IQKRNRVDQN DKPYVLTNK 539

SEQ ID NO: 14 moltype = AA length = 572
 FEATURE Location/Qualifiers
 source 1..572
 mol_type = protein
 note = Human parainfluenza virus 3
 organism = unidentified

SEQUENCE: 14
 MEYWKHTNHG KDAGNELETS TATHGNKLTN KITYILWTIT LVLLSIVFII VLTNSIKSEK 60
 ARESLLQDIN NEFMEVTEKI QVSDNTNDL IQSGVNTRLR TIQSHVQNYI PISLTQQISD 120
 LRKFISEITI RNDNQEVPPQ RITHDVGIKP LNPDDFWRCT SGLPSLMKTP KIRLMPGPGL 180
 LAMPITVDGC VRTPSLVIND LIYAYTSNLI TRGCQDIGKS YQVLQIGIIT VNSDLVPDLN 240
 PRISHTFNIN RDKSCSLAL LNTDVYQLCS TPKVDSRSDY ASSGIEDIVL DIVNYDGSIS 300
 TTRFKNNIS FDQPYAALYP SVGPPIYYKG KIIFLGYGGL EHPINENAIC NTTGCPGKTQ 360
 RDCNQASHSP WFSRRMVNS IIVVDKGLNS VPKLKVWTIS MRQNYWGSEG RLLLLGNKIY 420
 IYTRSTSWHS KLQLGIIDIT DYSDIRIKWT WHNVLSRPGN NECPWGHSCP DGCITGVYTD 480
 AYPLNPTGSI VSSVILDSQK SRVNPVITYS TATERVNELA IRNKTLSAGY TTTSCITHYN 540
 KGYCFHIVEI NHKSLNTPQP MLFKTEIPKS CS 572

SEQ ID NO: 15 moltype = AA length = 20
 FEATURE Location/Qualifiers
 REGION 1..20
 note = Synthetic Polypeptide
 source 1..20
 mol_type = protein

-continued

```

                                organism = synthetic construct
SEQUENCE: 15
METPAQLLFL LLLWLDPDTG                                20

SEQ ID NO: 16      moltype = AA  length = 18
FEATURE           Location/Qualifiers
REGION           1..18
                  note = Synthetic Polypeptide
source          1..18
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 16
MDWTWILFLV AAATRVHS                                18

SEQ ID NO: 17      moltype = AA  length = 24
FEATURE           Location/Qualifiers
REGION           1..24
                  note = Synthetic Polypeptide
source          1..24
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 17
MLGSNSGQQRV VFTILLLLVA PAYS                        24

SEQ ID NO: 18      moltype = AA  length = 17
FEATURE           Location/Qualifiers
REGION           1..17
                  note = Synthetic Polypeptide
source          1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 18
MKCLLYLAFL FIGVNCA                                17

SEQ ID NO: 19      moltype = AA  length = 15
FEATURE           Location/Qualifiers
REGION           1..15
                  note = Synthetic Polypeptide
source          1..15
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 19
MWLVSLAIVT ACAGA                                  15

SEQ ID NO: 20      moltype = DNA  length = 4062
FEATURE           Location/Qualifiers
source          1..4062
                  mol_type = genomic DNA
                  note = Middle East respiratory syndrome coronavirus
                  organism = unidentified

SEQUENCE: 20
atgatacact cagtggttct actgatgttc ttgttaaac ctacagaaag ttacgttgat 60
gtagggccag attctgtaa gtctgcttgt attgaggttg atatacaaca gacctctttt 120
gataaaactt ggcctaggcc aattgatgtt tctaaggctg acggtattat ataccctcaa 180
ggcgttacat attctaacat aactatcact tatcaaggtc tttttcccta tcagggagac 240
catggtgata tgtatgttta ctctgcagga catgctacag gcacaactcc acaaaagttg 300
ttttagctca actattctca ggacgtcaaa cagtttgcta atgggtttgt cgtccgtata 360
ggagcagctg ccaattccac tggcactggt attattagcc catctaccag cgctactata 420
cgaaaaattt accctgcttt ttgctgggt tcttcagttg gtaatttctc agatggtaaa 480
atgggcccgt tcttcaatca tactctagtt ctttggcccg atggatgtgg cactttactt 540
agagcttttt attgtattct agagcctcgc tctggaaatc attgtcctgc tggcaattcc 600
tatacttctt ttgccactta tcacactcct gcaacagatt gttctgatgg caattacaat 660
cgtaatgcca gtctgaactc ttttaaggag tattttaatt tacgtaactg cacccttatg 720
tacacttata acattaccga agatgagatt ttagagtggg ttggcattac acaaaactgc 780
caaggtgttc accctctctc atctcgggat gttgatttgt acggcggcaa tatgtttcaa 840
tttgccacct tgccgtgtta tgatactatt aagtattatt ctatcattcc tcacagtatt 900
cgttctatcc aaagtgatag aaaagcttgg gctgccttct acgtatataa acttcaaccg 960
ttaactttcc tgttggattt ttctgttgat ggttatatac gcagagctat agactgtggt 1020
tttaatgatt tgtcacaaact ccactgctca tatgaatcct tcatgttgga atctggaggt 1080
tattcagttt cgtctttcga agcaaaacct tctggctcag ttgtggaaca ggctgaaggt 1140
gttgaatgtg atttttcacc tcttctgtct ggcacacctc ctcagggtta taatttcaag 1200
cgtttggttt ttaccaattg caattataat cttaccaaat tgctttcact tttttctgtg 1260
aatgatttta cttgtagtca aatattctca gcagcaattg cttagcaactg ttattcttca 1320
ctgattttgg attatttttc ataccactt agtatgaaat ccatctcag tgtagttctt 1380
gctggtccaa tatccagtt taattataaa cagtcctttt ctaatccac atgtttgatc 1440
ttagcgactg ttctcataa ccttactact attactaagc ctcttaagta cagctatatt 1500
aacaagtgtc ctctgttctt ttctgatgat cgtactgaag tacctcagtt agtgaacgct 1560
aatcaatact caccctgtgt atccattgtc ccatccactg tgtgggaaga cggtgattat 1620
tataggaaac aactatctcc acttgaaggt ggtggtctgc ttgttgctag tggtcact 1680

```

-continued

```

gttgccatga ctgagcaatt acagatgggc ttggttatta cagttcaata tggtagacag 1740
accaatagtg tttgocccaa gcttgaattt gctaatgaca caaaaattgc ctctcaatta 1800
ggcaattgctg tggaaatattc cctctatggt gtttcgggccc gtggtgtttt tcagaattgc 1860
acagctgtag gtgttcgaca gcagcgcttt gtttatgatg cgtaccagaa tttagttggc 1920
tattattctg atgatggcaa ctactactgt ctgctgtgctt gtgttagtgt tctgttttct 1980
gtcatctatg ataagaacac taaaaccacac gctactctat ttggtagtgt tgcattgtga 2040
cacatttctt ctaccatgtc tcaatactcc cgttctacgc gatcaatgct taaacggcga 2100
gattctacat atggocccct tcagacacct gttggttgtg tctaggagat tgttaattcc 2160
tctttgttctg tagaggactg caagttgcct ctcggtcaat ctctctgtgc tcttctctgac 2220
acacctagta ctctcacacc tcgcagtggtg cgctctgtgc caggtgaaat gcgcttggca 2280
tccattgctt ttaatcatcc cattcagggt gatcaactta atagtagtta ttttaatta 2340
agtataccca ctaatttttc ctttggtgtg actcaggagt acattcagac aaccattcag 2400
aaagttactg ttgattgtaa acagtaacgtt tgcaatggtt tccagaagtg tgagcaatta 2460
ctgcgcgagt atggocccct ttgttccaaa ataaaccagg ctctccatgg tgccaattta 2520
cgccaggatg attctgtacg taatttggtt gcgagcgtga aaagctctca atcatctcct 2580
atcataccag gttttggagg tgactttaat ttgacacttc tagaacctgt ttctatatct 2640
actggcagtc tagtgctgac tagtgctatt gaggtttgc tatttgacaa agtcaactata 2700
gctgatcctg gttatatgca aggttacgat gattgtatgc agcaaggtcc agcatcagct 2760
cgtgatctta ttgtgtctca atatgtggct ggttataaag tattacctcc tcttatggat 2820
gttaatatgg aagcgcgcta tacttcatct ttgcttgcca gcatagcagg tgttggtctg 2880
actgctggct tatcctcctt tgctgtctatt ccatttgcac agagtatytt ttataggtta 2940
aacggtgttg gcattactca acaggttctt tcagagaacc aaaagcttat tgccaataag 3000
tttaaatcagg ctctggggagc tatgcaaaaca ggcttccact caactaatga agcttttctg 3060
aaggttcagg atgctgtgaa caacaatgca caggctctat ccaaattagc tagcgagcta 3120
tctaatactt ttggtgctat ttccgcctct attggagaca tcatacaacg tcttgatgtt 3180
ctcgaaacagg acgcccacaa agacagactt attaatggcc gtttgacaac actaaatgct 3240
tttgttgca acagcgttgt tcgttccgaa tcagctgtct ttccgctca attggcctaa 3300
gataaagtca atagtggtgt caaggcacaa tccaagcgtt ctggattttg cgttcaaggc 3360
acacatatag tgcctcttgt tgtaaatgcc cctaagtgcc ttactttat gcattgttgt 3420
tattacccta gcaaccacat tgaggttgtt tctgtctatg gtctttgcga tgcagctaac 3480
cctactaatt gtatagcccc tgtaaatggc tactttatta aaactaatga cactaggatt 3540
gttgatgagt ggatcatatac tggctcgctc ttctatgcac ctgagcccat cactctctct 3600
aatactaaat atgttgcaac acaggtgaca taccaaaaca ttctactata cctccctcct 3660
cctcttctcg gcaattccac cgggattgac ttccaagatg agttggatga gtttttcaaa 3720
aatgttagca ccagtatacc taatttttgt tctctaacac agattaatac tacattactc 3780
gatcttaact acgagatggt gtctcttcaa caagttgtta aagcccttaa tgagtcttac 3840
atagacctta aagagcttgg caattatact tattacaaca aatggcctg gtacatttgg 3900
cttggtttca ttgctgggct tgttgcccta gctctatgct tcttctctat actgtgctgc 3960
actggttgtg gcacaaaactg tatgggaaaa cttaagtgtat atcgttgttg tgatagatac 4020
gaggaatacag acctcgagcc gcataaggtt catgttctact aa 4062

```

```

SEQ ID NO: 21      moltype = DNA length = 4062
FEATURE
misc_feature       1..4062
                    note = Synthetic Polynucleotide
source             1..4062
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 21
atgatacact cagtggttct actgatgttc ttgttaaacac ctacagaaaag ttacgttgat 60
gtagggccag attctgttaa gtctgcttgt attgaggttg atatacaaca gactttcttt 120
gataaaactt ggcctagggc aattgatgtt tctaaggctg acggtattat ataccctcaa 180
ggcgtacat attctaacat aactatacct tatcaaggctc tttttcccta tcagggagac 240
catggtgata tgtatgttta ctctgcagga catgctacag gcacaactcc aaaaagttg 300
tttgtagctca actattctca ggaagctcaaa cagtttgcta atgggtttgt cgtccgtata 360
ggagcagctg ccaattccac tggcactggt attattagcc catctaccag cgctactata 420
cgaaaaattt accctgcttt tatgctgggt tcttcagttg gtaatttctc agatggtaaa 480
atgggcgctc tcttccatca tactctagtt cttttgcccg atggatgtgg cactttactt 540
agagcttttt attgtattct ggagcctcgc tctggaatcc attgtcctgc tggcaattcc 600
tatacttctt ttgccactta tcacactcct gcaacagatt gttctgatgg caattacaat 660
cgtaatgcca gtctgaactc ttttaaggag tattttaatt tacgtaactg cacttttatg 720
tacacttata acattaccga agatgagatt ttagagtggg ttggcattac acaaactgct 780
caaggtgttc accctctctc atctcgggat gttgatttgt acggcggaac tatgtttcaa 840
tttgccacct tgctgtttta tgatactatt aagattattt ctatcattcc tcacagtatt 900
cgttctatcc aaagtgatg aaaagcttgg gctgccttct acgtatataa acttcaaccg 960
ttaactttcc tgttggattt ttctgttgat ggttatatac gcagagctat agactgtggt 1020
tttaatgatt tgtcacaact ccactgctca tatgaatcct tcgatgttga atctggagtt 1080
tattcagttt cgtctttcga agcaaaaact ctggctcag ttgtggaaca ggctgaaggt 1140
gttgaatgtg atttttcaac tcttctgtct ggcacacctc ctacaggtta taatttcaag 1200
cgtttggttt ttaccaattg caattataat cttaccaaat tgctttcact tttttctgtg 1260
aatgatttta cttgtagtca aatatctcca gcagcaattg ctagcaactg ttattcttca 1320
ctgatttttg attacttttc ataccactt agtatgaaat ccgatctcag tgttagttct 1380
gctggtccaa tatccaggtt taattataaa cagtcctttt ctaatcccac atgtttgatt 1440
ttagcgactg ttctctataa ccttactact attactaagc ctcttaagta cagctatatt 1500
aacaagtgtc ctgctcttct ttctgatgat cgtactgaag tacctcagtt agtgaacgct 1560
aatcaatact caccctgtgt atccattgtc ccatccactg tgtgggaaga cggtgattat 1620
tataggaaac aactatctcc acttgaaggt ggtggctggc ttgttgctag tggctcaact 1680
gttgccatga ctgagcaatt acagatgggc ttggttatta cagttcaata tggtagacag 1740
accaatagtg tttgocccaa gcttgaattt gctaatgaca caaaaattgc ctctcaatta 1800

```

-continued

```

ggcaattgcy tggaaatattc cctctatggt gtttcgggccc gtggtgtttt tcagaattgc 1860
acagctgtag gtgttcgaca gcagcgcttt gtttatgatg cgtaccagaa tttagtggc 1920
tattattctg atgatggcaa ctactactgt ttgcgtgctt gtgttagtgt tccgtgttct 1980
gtcatctatg ataaagaac taaaaccac gctactctat ttggtagtgt tgcagtga 2040
cacatttctt ctaccatgtc tcaatactcc cgttctacgc gatcaatgct taaacggcga 2100
gattctacat atggccccc tccagacact gttggttgtg tccaggact tgttaattcc 2160
tctttgttcg tagaggactg caagttgcct ctggtcaat ctctctgtgc tcttctgac 2220
acacctagta ctctcacacc tcgcagtgtg cgtctgttcc caggtgaaat gcgcttgga 2280
tccattgctt ttaatcatcc tattcagggt gatcaactta atagtagtta ttttaaatta 2340
agtataccca ctaatttttc ctttgggtgt actcaggagt acattcagac aaccattcag 2400
aaagttactg ttgattgtaa acagtacgtt tgcaatgggt tccagaagtg tgagcaatta 2460
ctgcgcgagt atggccagtt ttgttccaaa ataaaccagg ctctccatgg tgccaattta 2520
cgccaggatg attctgtacg taatttgttt gcgagcgtga aaagctctca atcatctcct 2580
atcataccag tttttggagg tgactttaat ttgacacttc tggaaacctg tctctatct 2640
actggcagtc gtagtgacg tagtgctatt gaggatttgc tatttgaca agtcactata 2700
gctgatccga gttatatgca aggttacgat gattgcacgc agcaaggctc agcatcagct 2760
cgtgatctta tttgtgctca atatgtggct ggttacaaag tattacctcc tcttatggat 2820
gttaatatgg aagccgcgta tacttcatct ttgcttgga ccatagcagg tgttggtcgg 2880
actgctggct tatctctctt tgctgctatt ccatttgcac agagtatctt ttataggtta 2940
aacggtgttg gcattactca acaggttctt tcagagaacc aaaagcttat tgccaataag 3000
ttaatcagg ctctgggagc tatgcaaaac ggcttcacta caactaatga agcttttcag 3060
aaggttcagg atgctgtgaa caacaatgca caggctctat ccaattagc tagcgagcta 3120
tctaatactt ttggtgctat ttccgcctct attggagaca tcatacaacg tcttgatgtt 3180
ctcgaaacagg acgcccacaa agacagactt attaatggcc gtttgacaac actaaatgct 3240
ttgttgcac agcagcttgt tcgttccgaa tcagctgctc ttccgcctca attggctaaa 3300
gataaagtca atgagtgtgt caaggcacaa tccaagcgtt ctggattttg cggccaaggc 3360
acacatatag tgccttctgt tgtaaatgcc cctaattggc tttactctat gcatgttgtt 3420
tattacccta gcaaccacat tgaggttgtt tctgcttatg gtctttgcga tgcagctaac 3480
cctactaatt gctatgcccc tggttaatggc tactttatta aaactaataa cactaggatt 3540
gttgatgagt ggtcatatac tggctcgtcc ttctatgcac ctgagcccat tactccctt 3600
aatactaagt atgttgcacc acaggtgaca taccaaaaca ttctactaa cctccctcct 3660
cctcttctcg gcaatttccac cgggattgac ttccaagatg agttggatga gtttttcaaa 3720
aatgttagca ccagtatacc taatttttgt tccctaacac agattaatac tacattactc 3780
gatcttacct acagagtgtt gtctcttcaa caagttgtta aagcccttaa tgagtcttac 3840
atagacctta aagagcttgg caattatact tattacaaca aatggccgtg gtacatttgg 3900
cttggtttca ttgctgggct tgttgcccta gctctatgcg tcttctctat actgtgctgc 3960
actggttgtg gcacaaactg tatgggaaaa cttaagtgtg atcgttgttg tgatagatac 4020
gaggaatacgc acctcgagcc gcataagggt catgttcaat aa 4062

```

```

SEQ ID NO: 22      moltype = DNA length = 1845
FEATURE            Location/Qualifiers
misc_feature       1..1845
                    note = Synthetic Polynucleotide
source             1..1845
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 22
atgataccact ccgtgttctt cctcatgttc ctgttgaccc ccaactgagtc agactgcaag 60
tcccgcgtgg gacagtcctt gtgtgcgctg cctgacactc ctgacactct gacccacgc 120
tccgtgcggg cggtgctctg cgaaatgcgg ctggcctcca tcgcttcaa tcaccaatc 180
caagtggatc agctgaatag ctctgatttc aagctgtcca tcccacgaa cttctcgctc 240
ggggtcacc caggatcatc ccagaccaca attcagaagg tcacgctcga ttgcaagcaa 300
tacgtgtgca ccggttccca gaagtgcgag cagctgctga gagaatacgg gcagttttgc 360
agcaagatca accaggcgct gcatggagct aacttgcgcc aggcagactc cgtgcgcaac 420
ctctttgcct ctgtgaagtc atcccagtc tcccacatca tccgggatt cggaggggac 480
ttcaacctga cctcctgga gcccggtgct atcagcaccg gtagcagatc ggcgcgctca 540
gccattgaag atcttctgtt cgacaaggtc accatcgccg atccgggcta catgcaggga 600
tacgacgact gtatgcagca gggaccagcc tccgcgaggg acctcatctg cgcgcaatac 660
gtggccgggt acaaaagtgt gctcctctg atggatgtga acatggaggc cgttatact 720
tcgtccctgc tcggctctat cgcgcgctg ggggtggacc cgggcctgtc ctccctgcc 780
gctatccctt ttgcacaate cattttctac cggctcaacg gcgtgggcat tactcaaca 840
gtcctgtcgg agaaccagaa gttgatcgca aacaagtcca atcaggccct gggggccatg 900
cagactggat tcaactacgac taacgaagcg ttccagaagg tccaggacgc tgtgaacaac 960
aacgcccagg cgctctcaaa gctggcctcc gaactcagca acaccttcgg agccatcagc 1020
gcatcgatcg gtgacataat tcagcggctg gacgtgctgg agcaggacgc ccagatcgac 1080
cgctcatca acggacggct gaccaccttg aatgctctg tggcacaaca gctggctcgg 1140
agcgaatcag ggcacatttc cgcccaactc gccaaaggaca aagtcaacga atgcgtgaag 1200
gccagtcaca agaggctcgg tttctgcggt caaggaaacc atattgtgtc cttcgtcgtg 1260
aacgcgcccc acggtctgtg ctttatgcac gtccgctact acccgagcaa tcatatcgaa 1320
gtggtgtccg cctacggcct gtgcgatgcc gctaacccca ctaactgtat tgcccctgtg 1380
aacggatat tttatgaag caacaacacc cgcatttgtg acgaatggtc atacaccgt 1440
tcgtccttct acgcgcccga gcccatcact tcaactgaac ccaaatcgtt ggctccgcaa 1500
gtgacctacc agaactctc caccaatttg ccgcgcgcg tgctcggaaa cagcaccgga 1560
attgatttcc aagatgaact ggacgaatcc ttcaagaacg tgtccacttc cattcccaac 1620
ttcggaagcc tgacacagat caacaccacc cttctcgacc tgacctacga gatgctgagc 1680
cttcaacaag tggtaaggc cctgaacgag agctacatcg acctgaagg gctgggcaac 1740
tatacctact acaacaagt gccggacaag attgaggaga ttctgtcgaa aatctaccac 1800
attgaaaacg agatcgccag aatcaagaag cttatcgccg aagcc 1845

```

-continued

```

SEQ ID NO: 23      moltype = DNA length = 4071
FEATURE           Location/Qualifiers
misc_feature       1..4071
                   note = Synthetic Polynucleotide
source            1..4071
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 23
atggaaccct ctgcccagct gctgttctctg ctgctgctgt ggctgctga taccaccggc 60
agctatgtgg acgtggggccc cgatagcgtg aagtcgcct gtatcgaagt ggacatccag 120
cagacctttt tcgacaagac ctggcccaga cccatcgacg tgtccaaggc cgacggcatc 180
atctatccac aaggccggac ctacagcaac atcaccatta cctaccaggg cctgttccca 240
tatcaaggcg accacggcga tatgtacgtg tactctgccc gccacggcac cggcaccaca 300
ccccagaaac tgttcgtggc caactacagc caggacgtga agcagttcgc caacggcttc 360
gtcgtgcgga ttggcgccgc tgccaatagc accggcacag tgatccatcag cccacgaccc 420
agcggccacca tccggaagat ctaccccgcc ttcctgtctg gcagctccgt gggcaatttc 480
agcgacggca agatggggcg gttcttcaac cacacctggg tctgtctgcc cgatggctgt 540
ggcacactgc tgagagcctt ctactgcctc ctggaaccaca gaagcggcaa ccactgccct 600
gcccggcaata gctacaccag cttcgccacc taccacacac ccgccaccga ttgtccgac 660
ggcaactaca accggaacgc cagcctgaac agcttcaaa agtacttcaa cctgcgggaa 720
tgacacttca tgtacaccct caatatcacc gaggacgaga tccctggaatg gttcggcatc 780
acccagaccg cccaggggcg gcaactgttc agcagcagat acgtggacct gtaaggcggc 840
aacatgttcc agtttgccac cctgcccgtg tacgacacca tcaagtacta cagcatcctc 900
ccccacagca tccgggtccat ccagagcgac agaaaagcct gggccgcctt ctacgtgtac 960
aagctgcagc cctgtgacct cctgtgggac ttcagcgtgg accgctacat cagacggggc 1020
atcgactgcg gcttcaacga cctgagccag ctgcaactgct cctacgagag cttcgacgtg 1080
gaaagcggcg tgtacagcgt gtcacgcttc gaggccaagc ctacggcgag cgtgtgtgga 1140
caggttgagg gcgtggaatg gcaactcagc cctctgtgta gcggcaccgc tcccagggtg 1200
tacaacttca agcgggtggt gttcaccaac tgcgaattaca acctgaccaa gctgtctgagc 1260
ctgttctcgc tgaacgactt caactgtagc cagatcagcc ctgcccgcct tgcacgaac 1320
tgctacagca cctgatcctt ggaactcttc agctacccc tgagcatgaa gtccgatctg 1380
agcgtgtcct gcgcgggacc catcagccag ttcaactaca agcagagctt cagcaaccct 1440
acctgcctga ttctggccac cgtgcccacc aatctgacca ccatcaccaa gcccctgaag 1500
tacagctaca tcaacaagtg cagcagactg ctgtccgacg accggaccga agtgcccacg 1560
ctcgtgaacg ccaaccagta cagcccctgc gtgtccatcg tgcccagcac cgtgtgggag 1620
gacggcgact actacagaaa gcagctgagc cccctggaag gcggcggatg gctgtgtggt 1680
tctggaagca ctactggccat gacgagcag ctgcagatgg gctttggcat caccgtgcag 1740
tacggcaccg acaccaacag cgtgtgcccc aagctggaat tcgccaatga caccagatc 1800
gccagccagc tgggaaactg cgtggaatac tccctgtatg gcgtgtccgg acggggcggtg 1860
ttccagaatt gcacagcagt gggagtgccg cagcagagat tctgtacga tgctaccag 1920
aacctcgtgg gctactacag cgacgacggc aattactact gcctgggggc ctgtgtgttc 1980
gtgcccgtgt ccgtgatcta cgacaaagag acaaagacc cgcaccactt gttcggctcc 2040
gtggcctgcg agcacatcag ctccaccatg agccagtact cccgtccacc ccggtccatg 2100
ctgaagcgga gagatagcac ctacggcccc ctgcagacac ctgtgggatg tgtgtgtggc 2160
ctcgtgaaca gtcctcgtgt tctggaagat tgcgaagctg cccctgggca gacccgtgtg 2220
gccctgcccag ataccctag caccctgacc cctagaagcg tgcgtctgtg gccggcgaa 2280
atgcggctgg cctctatcgc cttcaatcac cccatccagg tggaccagct gaactccagc 2340
tacttcaagc tgagcatccc caccacactt agcttcggcg tgacccagga gtacatccag 2400
accacaatcc agaaagtgcg cgtggactgc aagcagtagt tgtgcaacgg ctttcagaag 2460
tgccaacagc tgtgcgcgga gtacggcccag ttctgcagca agatcaacca ggccctgcac 2520
ggcgccaacc ttgagacagga tgacagcgtg cggaaacctg tcgccagcgt gaaaagcagc 2580
cagtcacagc ccatcctccc tggtctcggc ggcgacttta acctgacctt gctggaacct 2640
gtgtccatca gcaccggctc cagaagcgcc agatccgcca tcgaggacct gctgttcgac 2700
aaagtgacca ttgccgaccc cggctacatg cagggtctacg acgattgcat gcagcagggc 2760
ccagccagcg ccagggatct ggtctgtgcc cagtatgtgg ccggtacaaa ggtgctgccc 2820
cccctgatgg acgtgaacat ggaagccgcc tacacctcca gcctgctggg ctctatttgt 2880
ggcgtgggat ggacagccgg cctgtctagc tttgcgcgca tccctttcgc ccagagcatc 2940
ttctaccggc tgaacggcgt gggcatcaca caacagggtg tgagcgagaa ccagaagctg 3000
atcgccaaca agtttaacca ggcactgggc gccatgcaga ccggcttcac caccaccaac 3060
gaggccttca gaaagggtga ggacgcccgtg aacaacaacg cccaggctct gagcaagctg 3120
gcctccgagc tgagcaatca cttcgccgac atcagcgccct ccatcgcgca catcatccag 3180
cggctggagc tgttggaaca ggacgcccag atcgaccggc tgatcaacgg cagactgacc 3240
acctgaaacg ccttcgtggc acagcagctc gtgcggagcg aatctgcgcg tctgtctgtc 3300
cagctggcca aggacaaaagt gaacagatgc gtgaaggccc agtccaagcg gagcgcttt 3360
tgtggccagg gcacccacat cgtgtccttc gtctgtaatg cccccaacgg cctgtacttt 3420
atgcacgtgg gctattaccc cagcaaccac atcgaggtgg tgcgcgcta tggcctgtgc 3480
gacgcccgca atcctaccaa ctgtatcgcc cccgtgaacg gctacttcat caagaccaac 3540
aacacccgga tctgtggacga gtgtgtctac gcttctacgc ccccgagccc 3600
atcacctccc tgaacaccaa atacgtggcc ccccaagtga cataccagaa catctccacc 3660
aacctgcccc ctccactgct gggaaattcc accggcatcg acttccagga cgagctggac 3720
gagttcttca agaacgtgtc cactccatc cccaacttcg gcagcctgac ccagatcaac 3780
accactctgc tggacctgac ctacgagatg ctgtccctgc aacaggtcgt gaaagccctg 3840
aacgagagct acatcgacct gaaagagctg ggggaactaca cctactacaa caagtggcct 3900
tggtagattt ggctgggctt tatcgccggc ctgggtggcc tggccctgtg cgtgttcttc 3960
atcctgtgct gcacccgctg cggcaccaat tgcattgggca agctgaaatg caaccggctg 4020
tgcgacagat acgaggaata cgacctggaa cctcacaag tgcatgtgca c 4071

```

```

SEQ ID NO: 24      moltype = AA length = 1353
FEATURE           Location/Qualifiers

```

-continued

source 1..1353
 mol_type = protein
 note = Middle East respiratory syndrome coronavirus
 organism = unidentified

SEQUENCE: 24

```

MIHVSFLLMF LLTPTESYVD VGPDSVKSAC IEVDIQQTFF DKTWPRPIDV SKADGIIYPQ 60
GRITYSNITIT YQGLFPYQGD HGDYVYSAG HATGTPQKL FVANYSQDVK QFANGFVVRI 120
GAAANSTGTV IISPSTSATI RKIYPAPMLG SSVGNFSDGK MGRFFNHTLV LLPDGC GTLL 180
RAFYCILEPR SGNHCPAGNS YTSFATYHTP ATDCSDGNYN RNASLNSFKE YFNLRNCTFM 240
YTYNITEDEI LEWFGITQTA QGVHLFSSRY VDLYGGMFQ FATLPVYDTI KYYSIIPHSI 300
RSIQSDRKAW AAFVYVKLQP LTFLLDFSVD GYIRRAIDCG FNDLSQLHCS YESFDVESGV 360
YSVSSFEAKP SGSVVEQAEQ VECDFSPLLS GTTPQVYNFK RLVFTNCNIN LTKLLSLFSV 420
NDFTCSQISP AAIASNCYSS LILDYFSYPL SMKSDLVSS AGPISQFNYK QSFNPTCLI 480
LATVPHNLTT ITKPLKYSYI NKCSRLLSDD RTEVPQLVNA NQYSPCVSIV PSTVWEDGDY 540
YRKQLSPLEG GGLVVASGST VAMTEQLQMG FGITVQYGTD TNSVCPKLEF ANDTKIASQL 600
GNCVEYSLYG VSGRGVFQNC TAVGVRQORF VYDAYQNLVG YSDDGNYC LRACVSVFVS 660
VIYDKETKTH ATLFSGVACE HISSTMSQYS RSTRMLKRR DSTYGPLQTP VGCVLGLVNS 720
SLFVEDCKLP LGQSLCALPD TPSTLTFRSV RSVPGEMRLA SIAFNHPIQV DQLNSSFYKL 780
SIPTNFSFGV TQEIYQTTIQ KVTVDCKQYV CNGFQKCEQL LREYQGFCSK INQALHGANL 840
RQDDSVRNLF ASVKSSQSSP IIPGFGGDFN LTLLEPVSIS TGSRSARSAI EDLLFDKVTI 900
ADPGYMQGYD DCMQGPASA RDLICAYVA GYKVLPLPMD VNMEAAYTSS LLGSIAGVGW 960
TAGLSSFAAI AAIASNCYSS LILDYFSYPL NGVGITQQVL SENQKLIANK FNQALGAMQT GFTTTNEAFR 1020
KVQDAVNNA QALSKLASEL SNTFGAISAS IGDIIQRLDV LEQDAQIDRL INGRLLTLNA 1080
FVAQQLVRSE SAALSAQLAK DKVNECVKAQ SKRSGFCGQG THIVSFVNA PNGLYFMHVG 1140
YYPNHNIEVY SAYGLCDAAN PTNCIAPVNG YFIKTNNTRI VDEWSTYTGSS FYAPEPITSL 1200
NTKYVAPQVT YQNISTNLPP PLLGNSTGID FQDELDEFK NVSTSIPIFG SLTQINTTLL 1260
DLTYEMLSLQ QVVKALNESY IDLKELGNYT YYNKWPYIW LGFIAGLVAL ALCVFFILCC 1320
TGCGTNCMGK LKCNRCDDRY EYDLEPHKV HVH 1353

```

SEQ ID NO: 25 moltype = AA length = 1353
 FEATURE Location/Qualifiers
 REGION 1..1353
 note = Synthetic Polypeptide
 source 1..1353
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 25

```

MIHVSFLLMF LLTPTESYVD VGPDSVKSAC IEVDIQQTFF DKTWPRPIDV SKADGIIYPQ 60
GRITYSNITIT YQGLFPYQGD HGDYVYSAG HATGTPQKL FVANYSQDVK QFANGFVVRI 120
GAAANSTGTV IISPSTSATI RKIYPAPMLG SSVGNFSDGK MGRFFNHTLV LLPDGC GTLL 180
RAFYCILEPR SGNHCPAGNS YTSFATYHTP ATDCSDGNYN RNASLNSFKE YFNLRNCTFM 240
YTYNITEDEI LEWFGITQTA QGVHLFSSRY VDLYGGMFQ FATLPVYDTI KYYSIIPHSI 300
RSIQSDRKAW AAFVYVKLQP LTFLLDFSVD GYIRRAIDCG FNDLSQLHCS YESFDVESGV 360
YSVSSFEAKP SGSVVEQAEQ VECDFSPLLS GTTPQVYNFK RLVFTNCNIN LTKLLSLFSV 420
NDFTCSQISP AAIASNCYSS LILDYFSYPL SMKSDLVSS AGPISQFNYK QSFNPTCLI 480
LATVPHNLTT ITKPLKYSYI NKCSRLLSDD RTEVPQLVNA NQYSPCVSIV PSTVWEDGDY 540
YRKQLSPLEG GGLVVASGST VAMTEQLQMG FGITVQYGTD TNSVCPKLEF ANDTKIASQL 600
GNCVEYSLYG VSGRGVFQNC TAVGVRQORF VYDAYQNLVG YSDDGNYC LRACVSVFVS 660
VIYDKETKTH ATLFSGVACE HISSTMSQYS RSTRMLKRR DSTYGPLQTP VGCVLGLVNS 720
SLFVEDCKLP LGQSLCALPD TPSTLTFRSV RSVPGEMRLA SIAFNHPIQV DQLNSSFYKL 780
SIPTNFSFGV TQEIYQTTIQ KVTVDCKQYV CNGFQKCEQL LREYQGFCSK INQALHGANL 840
RQDDSVRNLF ASVKSSQSSP IIPGFGGDFN LTLLEPVSIS TGSRSARSAI EDLLFDKVTI 900
ADPGYMQGYD DCMQGPASA RDLICAYVA GYKVLPLPMD VNMEAAYTSS LLGSIAGVGW 960
TAGLSSFAAI PFAQSIYFRL NGVGITQQVL SENQKLIANK FNQALGAMQT GFTTTNEAFR 1020
KVQDAVNNA QALSKLASEL SNTFGAISAS IGDIIQRLDV LEQDAQIDRL INGRLLTLNA 1080
FVAQQLVRSE SAALSAQLAK DKVNECVKAQ SKRSGFCGQG THIVSFVNA PNGLYFMHVG 1140
YYPNHNIEVY SAYGLCDAAN PTNCIAPVNG YFIKTNNTRI VDEWSTYTGSS FYAPEPITSL 1200
NTKYVAPQVT YQNISTNLPP PLLGNSTGID FQDELDEFK NVSTSIPIFG SLTQINTTLL 1260
DLTYEMLSLQ QVVKALNESY IDLKELGNYT YYNKWPYIW LGFIAGLVAL ALCVFFILCC 1320
TGCGTNCMGK LKCNRCDDRY EYDLEPHKV HVH 1353

```

SEQ ID NO: 26 moltype = AA length = 615
 FEATURE Location/Qualifiers
 REGION 1..615
 note = Synthetic Polypeptide
 source 1..615
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 26

```

MIHVSFLLMF LLTPTESDCK LPLGQSLCAL PDTPSTLTFR SVRSVPGEML LASIAFNHPI 60
QVDQLNSSFY KLSIPTNFSF GVTQEIYQTT IQKVTVDCKQ YVCNGFQKCE QLLREYGFQFC 120
SKINQALHGA NLRQDDSVRN LFASVKSSQS SPIIPGFGGD FNLTLLEPVS ISTGSRARS 180
ATIEDLLFDKVTIADPGYMQG YDDCMQGPAS SARDLICAY VAGYKVLPLP MDVNMEAAYT 240
SSLGSIAGV QALSKLASEL AIPFAQSIYF RLNGVGITQQ VLSNQKLIANK NFQNALGAM 300
QTGFTTTNEA FQKVQDAVN NAQALSKLAS ELSNTFGAIS ASIGDIIQRL DVLEQDAQID 360
RLINGRLTTL NAFVAQQLVR SESAALSAQL AKDKVNECVK AQSKRSGFCG QGTHIVSFV 420
NAPNGLYFMH VGYYPNHNIE VVSAYGLCDA ANPTNCIAPV NGYFIKTNNT RIVDEWSTY 480
SSFYAPEPIT SLNTKYVAPQ VTYQNISTNL PPPLGNSTG IDFQDELDEF FKNVSTSIPI 540
FGSLTQINTT LLDLYEMLSL LQQVVKALNE SYIDLKELGN YTYYNKWPDK IEEILSKIYH 600

```

-continued

IENEIARIKK LIGEA 615
 SEQ ID NO: 27 moltype = AA length = 1353
 FEATURE Location/Qualifiers
 source 1..1353
 mol_type = protein
 note = Middle East respiratory syndrome coronavirus
 organism = unidentified
 SEQUENCE: 27
 MIHVSFLLMF LLTPTESYVD VGPDSVKSAC IEVDIQQTFF DKTWPRPIDV SKADGIIYPQ 60
 GRTYSNITIT YQGLFPYQGD HGDYMYVSAG HATGTPQKL FVANYSQDVK QFANGFVVRI 120
 GAAANSTGTV IISPSTSATI RKIYPAMFLG SSVGNFSDGK MGRFFNHTLV LLPDGCCTLL 180
 RAFYCILEPR SGNHCPAGNS YTSFATYHTP ATDCSDGNYN RNASLNSFKE YFNLRNCTFM 240
 YTYNITEDEI LEWFGITQTA QGVHLFSSRY VDLYGGMNFQ FATLPVYDTI KYYSIIPHSI 300
 RSIQSDRKAW AAFVYKLPQ LTFLLDFSVD GYIRRAIDCG FNDLSQLHCS YESFDVESGV 360
 YSVSSFEAKP SGSVVEQAEQ VECDFSPLLS GTPPQVYNFK RLVFTNCNIN LTKLLSLFSV 420
 NDFTCQSISP AAIASNCYSS LILDYFSYPL SMKSDLSVSS AGPISQFNK QSFNPTCLI 480
 LATVPHNLTT ITKPLKYSYI NKCSRLSDD RTEVPQLVNA NQYSPCVSIV PSTVWEDGDY 540
 YRKQLSPLRG GGLVASGST VAMTEQLQMG FGITVQYGTD TNSVCPKLEF ANDTKIASQL 600
 GNCVEYSLYG VSGRGVQNC TAVGVRQQRV VYDAYQNLVG YSDDGNYYC LRACVSVVPS 660
 VIYDKETKTH ATLPGSVACE HISSTMSQYS RSTRSMLKRR DSTYGPLQTP VGCVLGLVNS 720
 SLFVEDCKLP LGQSLCALPD TPSTLTPRSV RSVPGEMRLA SIAFNHPIQV DQLNSSYFKL 780
 SIPTNFSFGV TQEIYQTTIQ KVTVDCKQYV CNGFQKCEQL LREYGFQCSK INQALHGANL 840
 RQDDSVRNLF ASVKSQSSP IIPGFGGDFN LTLLEPVSIS TGSRSARSAI EDLLFDKVTI 900
 ADPGYMQGYD DCMQGPASA RDLICAYVA GYKVLPLMD VNMEAAYTSS LLGSIAGVGW 960
 TAGLSSFAAI PFAQSIFYRL NGVGITQQVL SENQKLIANK FNQALGAMQT GFTTTNEAFR 1020
 KVQDAVNNA QALSKLASEL SNTFGAISAS IGDIIQLRDV LEQDAQIDRL INGRLTTLNA 1080
 FVAQQLVRSE SAALSAQLAK DKVNECVKAQ SKRSGFCGQG THIVSFVNA PNGLYPMHVG 1140
 YYPNSHIEVV SAYGLCDAAN PTNCIAPVNG YFIKTNNTRI VDEWSTYGSF FYAPEPITSL 1200
 NTKYVAPHVT YQNISTNLPP PLLGNSTGID FQDELDEFK NVSTSIPIFG SLTQINTTLL 1260
 DLTYEMLSLQ QVVKALNESY IDLKELGNYT YYNKWPWYIW LGFIAGLVAL ALCVFILCC 1320
 TCGTNCMGK LKCNRCDDRY EBYDLEPHKV HVH 1353
 SEQ ID NO: 28 moltype = AA length = 1353
 FEATURE Location/Qualifiers
 source 1..1353
 mol_type = protein
 note = Middle East respiratory syndrome coronavirus
 organism = unidentified
 SEQUENCE: 28
 MIHVSFLLMF LLTPTESYVD VGPDSVKSAC IEVDIQQTFF DKTWPRPIDV SKADGIIYPQ 60
 GRTYSNITIT YQGLFPYQGD HGDYMYVSAG HATGTPQKL FVANYSQDVK QFANGFVVRI 120
 GAAANSTGTV IISPSTSATI RKIYPAMFLG SSVGNFSDGK MGRFFNHTLV LLPDGCCTLL 180
 RAFYCILEPR SGNHCPAGNS YTSFATYHTP ATDCSDGNYN RNASLNSFKE YFNLRNCTFM 240
 YTYNITEDEI LEWFGITQTA QGVHLFSSRY VDLYGGMNFQ FATLPVYDTI KYYSIIPHSI 300
 RSIQSDRKAW AAFVYKLPQ LTFLLDFSVD GYIRRAIDCG FNDLSQLHCS YESFDVESGV 360
 YSVSSFEAKP SGSVVEQAEQ VECDFSPLLS GTPPQVYNFK RLVFTNCNIN LTKLLSLFSV 420
 NDFTCQSISP AAIASNCYSS LILDYFSYPL SMKSDLSVSS AGPISQFNK QSFNPTCLI 480
 LATVPHNLTT ITKPLKYSYI NKCSRLSDD RTEVPQLVNA NQYSPCVSIV PSTVWEDGDY 540
 YRKQLSPLRG GGLVASGST VAMTEQLQMG FGITVQYGTD TNSVCPKLEF ANDTKIASQL 600
 GNCVEYSLYG VSGRGVQNC TAVGVRQQRV VYDAYQNLVG YSDDGNYYC LRACVSVVPS 660
 VIYDKETKTH ATLPGSVACE HISSTMSQYS RSTRSMLKRR DSTYGPLQTP VGCVLGLVNS 720
 SLFVEDCKLP LGQSLCALPD TPSTLTPRSV RSVPGEMRLA SIAFNHPIQV DQLNSSYFKL 780
 SIPTNFSFGV TQEIYQTTIQ KVTVDCKQYV CNGFQKCEQL LREYGFQCSK INQALHGANL 840
 RQDDSVRNLF ASVKSQSSP IIPGFGGDFN LTLLEPVSIS TGSRSARSAI EDLLFDKVTI 900
 ADPGYMQGYD DCMQGPASA RDLICAYVA GYKVLPLMD VNMEAAYTSS LLGSIAGVGW 960
 TAGLSSFAAI PFAQSIFYRL NGVGITQQVL SENQKLIANK FNQALGAMQT GFTTTNEAFR 1020
 KVQDAVNNA QALSKLASEL SNTFGAISAS IGDIIQLRDV LEQDAQIDRL INGRLTTLNA 1080
 FVAQQLVRSE SAALSAQLAK DKVNECVKAQ SKRSGFCGQG THIVSFVNA PNGLYPMHVG 1140
 YYPNSHIEVV SAYGLCDAAN PTNCIAPVNG YFIKTNNTRI VDEWSTYGSF FYAPEPITSL 1200
 NTKYVAPHVT YQNISTNLPP PLLGNSTGID FQDELDEFK NVSTSIPIFG SLTQINTTLL 1260
 DLTYEMLSLQ QVVKALNESY IDLKELGNYT YYNKWPWYIW LGFIAGLVAL ALCVFILCC 1320
 TCGTNCMGK LKCNRCDDRY EBYDLEPHKV HVH 1353
 SEQ ID NO: 29 moltype = AA length = 1255
 FEATURE Location/Qualifiers
 source 1..1255
 mol_type = protein
 note = Human SARS coronavirus
 organism = unidentified
 SEQUENCE: 29
 MFIFLLFLTL TSGSDLRDT TFDVQAPNY TQHTSSMRGV YYPDEIFRSD TLYLTQDLFL 60
 PFYSNVTGFH TINHTFGNPV IPFKDGIYFA ATEKSNVVRG WVFSTTMNKK SQSVIIINNS 120
 TNVIRACNF ELCNPPFAV SPKMGQTQHT MIFDNAFNCT FEYISDAFSL DVSEKSGNFK 180
 HLREFVFKNK DGFLYVYKGY QPIDVVRDLP SGFNTLKPIF KLPLGINITN FRAILTAFSP 240
 AQDIWGTSA AYPFGYKPT TFMKYDENG TITDAVDCSQ NPLAELKCSV KSFEIDKGIY 300
 QTSNFRVVPV GDVVRFPNIT NLCPFGEVFN ATKFPVSVYAW ERKKISNCVA DYSLVYNSTF 360
 FSTFKCYGVV ATKLNDLCFS NVYADSFFVK GDDVRQIAPG QTGVIAADNY KLPPDFMGCV 420
 LAWNTRNIDA TSTGNYNYKY RYLHKGKLRP FERDISNVFP SPDGKPCPTP ALNICYWPLND 480

-continued

```

YGFYTTTGIG YQPYRVVLS FELLNAPATV CGPKLSTDLI KNQCVNPNFN GLTGTGVLTP 540
SSKRFPQFPQ FGRDVSDFTD SVRDPKTSEI LDISPCSPGG VSVITPGTNA SSEVAVLYQD 600
VNCTDVSTAI HADQLTPAWR IYSTGNNVFO TQAGCLIGAE HVDTSYECDI PIGAGICASY 660
HTVSLLRSTS QKSIVAYTMS LGADSSIAYS NNTIAIPTNF SISITTEVMP VMAKTSVDC 720
NMYICGDSTE CANLLQLQYS FCTQLNRALS GIAAEQDRNT REVFAQVKQM YKTPTLKYFG 780
GFNFSQLLPD PLKPTKRSFI EDLLFNKVTI ADAGFMKQYG ECLGGINARD LICAQKFNGL 840
TVLPPLLTDD MIAAYTAALV SGTATAGWTF GAGAALQIPF AMQMAYRFNG IGVTONVLYE 900
NQKQIANQFN KAISQIQESL TTTSTALGKL QDVVNQNAQA LNTLVKQLSS NFGAISSVLN 960
DILSRLDKVE AEVQIDRLIT GRLQSLQTYV TQQLIRAAEI RASANLAATK MSECVLGQSK 1020
RVDFCGKGXV LMSFPQAAPH GVVFLHVTYV PSQERNFTTA PAICHEGKAY FPREGVVFVN 1080
GTSWFITQRN FFSQIITTD NTFVSGNCDV VIGIINNTVY DPLQPELDSF KEELDKYFKN 1140
HTSPDVLDGD ISGINASVVN IQKEIDRLNE VAKNLNESLI DLQELGKYEQ YIKWPWYVWL 1200
GFIAGLIAIV MVTILLCCMT SCCSCLKGAC SCGSCCKFDE DSEFVLKGV KLHYT 1255

```

```

SEQ ID NO: 30      moltype = AA length = 1353
FEATURE
source            Location/Qualifiers
                  1..1353
                  mol_type = protein
                  note = Human coronavirus
                  organism = unidentified

```

```

SEQUENCE: 30
MFLILLISLP TAFAVIGDLK CTSNDINDKD TGPPPISTDT VDVNTGLGTY YVLDRVYLNT 60
TLFLNGYYPT SGSTYRNMAL KGSVLLSRLW FKPPFLSDFI NGIFAKVKNT KVIKDRVMYS 120
EFPATITGST FVNTSYSVVV QPRTINSTQD GDNKLQGLLE VSVCCQYNMCE YPQTICHPNL 180
GNHRKELWHL DTGVVSCLYK RNFTYDVNAD YLYPHFYQEG GTFYAYFTDT GVVTKFLFNV 240
YLGMAISHYY VMPLTCSNKL TLEYWVTPLT SRQYLLAFNQ DGIIFNAEDC MSDFMSEIKC 300
KTQSIAPPTG YVELNGYTVQ PIADVYRRKP NLPNCNIEAW LNDKSVSPSL NWERKTFSNC 360
NFMSSSLMSF IQADSFTCNN IDAAKIYGM C FSSITIDKFA IPNGRKVDLQ LGNLGYLQSF 420
NYRIDTTATS CQLYYNLPA NVSVSRRFNS TWNKRFGFIE DSVFKPRPAG VLTNHDVVYA 480
QHCFAKAPNF CPCKLNGSCV GSGPGKNGI GTCPAGTNYL TCDNLCTPDP ITFTGTYKCP 540
QTKSLVGIGE HCSGLAVKSD YCGGNSCTCR PQAFLGWSAD SCLQGDKN CNI FANFILHDVN 600
SGLTCTSDLQ KANTDIILGV CVNYDLYGIL GQGIFVEVNA TYYNSWQNL L YDSNGNLYGF 660
RDYIINRTFM IRSCYSGRVS AAFHANSSEP ALLFRNICKN YVFNNSLTRQ LQPINYFDSY 720
LGCVVNAYNS TAISVQTCDL TVGSGYCVDY SKNRRSRGAI TTGYRFTNFE PFTVNSVND 780
LEPVGGLEYE QIPSEFTIGN MVEFIQTSSP KVTIDCAAFV CGDYAACKSQ LVEYGSFCDN 840
INAILTEVNE LLDTTQQLVA NSLMNGVTL S TKLDGVNFN VDDINFSPVL GCLGSECSKA 900
SSRSAIEDLL FDKVKLSDVG FVEAYNNCTG GAEIRDLCV QSYKGKVL P LLSNQISG 960
YTLAATSASL PPPWTAAGV PFYLVQYRI NGLGVMTDVL SQNQKLIANA FNNALYAIQE 1020
GPDATNSALV KIQAVVNANA EALNNLLQQL SNRFGAISAS LQELSLRLDA LEAEAQIDRL 1080
INGRLTALNA LKVFSAQAM LVKFSAAQAM EKVNECVKSG SSRINFCGNG NHIISLVQNA 1140
PYGLYFIHFS YVPTKYVTAR VSPGLCIAGD RGIAPKSGYF VNVNNTWMTY GSGYYPPEPI 1200
TENNVMVMT CAVNYTKAPY VMLNTSIPNL PDFKEELDQW FKNQTSVAPD LSLDYINVTF 1260
LDLQVEMNRL QEAIKVLNQS YINLKDIGTY EYVVKWPWYV WLLICLAGVA MLVLLFFICC 1320
CTGCGTSCPK KCGGCCDDYT GYQELVIKTS HDD 1353

```

```

SEQ ID NO: 31      moltype = AA length = 1351
FEATURE
source            Location/Qualifiers
                  1..1351
                  mol_type = protein
                  note = Human coronavirus
                  organism = unidentified

```

```

SEQUENCE: 31
MFLIIFILPT TLAVIDGFNC TNSFINDYK TIPRISEDV V DVSGLGLGTY VLNRVYLNTT 60
LLFTGYFFPKS GANFRDLALK GSIYLSLWY KPPFLSDFN GIFSKVKNTK LYVNNTLYSE 120
FSTIVIGSVF VNTSYTIVVQ PHNGILEITA CQYTMCEYPH TVCKSKGSIR NESWHIDSSE 180
PLCLFKKNFT YNVSADWLYF HFYQERGVFY AYYADVGMPT TFLFSLYLGT ILSHYVVMPL 240
TCNAISSNTD NETLEYWVTP LSRRQYLLNF DEHGVITNAV DCSSSFLSEI QCKTQSFAPN 300
TGVYDLSGFT VKPVATVYRR IPNLPCDDID NWLNNSVSPS PLNWERRIFS NCFNLSLTL 360
RLVHVDSPFC NNLDKSKIFG SCFNSITVDK FAIPNRRRDD LQLGSSGFLQ SSNYKIDISS 420
SSCQLYSLP LVNVTINNFN PSSWNRRYGF GSFNLSSYDV VYSDHCFSVN SDFPCPADPS 480
VVNSCAKSKP PSAICPAGTK YRHCDLDTTL YVKNWCRCSC LPDPITYSP NTCPOKKVVV 540
GIGEHCPGLG INEEKCGTQL NHSSCFCSPD AFLGWSFDCS ISNNRCNIFS NFIFNGINS 600
TTCSDNLLYS NTEISTGVCV NYDLYGITGQ GIFKEVSAAY YNNWQNLLYD SNGNIIGFKD 660
FLTNTYITIL PCYSGRVSAA FYQNSSPAL LYRNLCYSYV LNNISFISQP FYFDSYLGCV 720
LNAVNLTSYS VSSCDLRMG S GFCIDYALPS SRRKRRGISS PYRFVTFEPF NVSFVND 780
TVGGLFEIQI PTNFTIAGHE EFTQTSSPKV TIDCSAFVCS NYAACHDLLS EYGTFCNDIN 840
SILNEVNDLL DTQLQVANA LMQGVTLSSN LNTNLHSDVD NIDPKSLGCG LGSQCSSSR 900
SLEDLLFNK VKLSDVGFVE AYNNTGGSE IRDLLCVQSF NGIKVLPPI L SETQISGYTT 960
AATVAAMFPP WSAAGVPFS LNVQYRINGL GVTMDVLNKN QKLIANAENK ALLSIQNGFT 1020
ATNSALAKIQ SVVNANAQAL NSLLQQLFNK FGAISSSLQE ILSRLDNLEA QVQIDRLING 1080
RLTALNAYVS QQLSDITLIK AGASRAIEKV NECVKSQSPR INFCGNGNHI LSLVQNPAYG 1140
LLFIHFSYKP TSFKTVLVSP GLCLSGDRGI APKQGYFIKQ NDSWMFTGSS YYPEPISDK 1200
NVVFMNSCSV NFKTAPPIYL NNSIPNLSDF EABELSLWFKN HTSIAPNLTF NSHINATFLD 1260
LYYEMNVIQE SIKSLNSSFI NLKEIGTYEM YVKWPWYIWL LIVILFIIFL MILFFICCT 1320
GCGSACFSKC HNCCDEYGGH NDFVIKASHD D 1351

```

```

SEQ ID NO: 32      moltype = AA length = 526
FEATURE
REGION            Location/Qualifiers
                  1..526

```


-continued

```

note = Synthetic Polypeptide
source      1..526
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 32
MFIFLLFLTL TSGSDLDRAL SGIAAEQDRN TREVFAQVKQ MYKTPTLKYF GGFNFSQILP 60
DPLKPTKRSF IEDLLFNKVT LADAGFMKQY GECLGDINAR DLICAQKFNG LTVLPPLTLD 120
DMIAAYTAAL VSGTATAGWT FGAGAAQIP FAMQMAYRFN GIGVTQNVLY ENQKQIANQF 180
NKAIISQIQES LTTTSTALGK LQDVVNQNAQ ALNTLVKQLS SNFGAISSVL NDILSRLDKV 240
EAEVQIDRLI TGRQLSLQTY VTQQLIRAAE IRASANLAAT KMSECVLGQS KRVDPCGKGY 300
HLMSPFQAAP HGVVFLHVTY VPSQERNFTT APAICHEGKA YFPREGVVFV NGTSWFITQR 360
NFFSPQIITT DNTFVSGNCD VVIGIINNTV YDPLQPELDS FKEELDKYFK NHTSPDVDLG 420
DISGINASVV NIQKEIDRLN EVAKNLNESL IDLQELGKYE QYIKWPWYVW LGFIAGLIAI 480
VMVTILLCCM TSCCSCCKGA CSCGSCCKFD EDDSEPVKLG VKLHYT 526

SEQ ID NO: 33      moltype = AA length = 588
FEATURE           Location/Qualifiers
REGION            1..588
note = Synthetic Polypeptide
source            1..588
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 33
MIHSVFLLMF LLTPTESDCK LPLGQSLCAL PDTPSTLTPR SVRSVPGEMR LASIAFNHPI 60
QVDQLNSSYF KLSIPTNRSF GVTQEIYIQT IQKVTVDCKQ YVCNGFQKCE QLLREYGQFC 120
SKINQALHGA NLRQDDSVRN LFAVSKSSQS SPIIPGFGGD FNLTLLEPVS ISTGSRARS 180
AIEDLLFDKV TIADPGYMQG YDDCMQGGPA SARDLICAQY VAGYKVLPLP MDVNMEAAYT 240
SSLLGSIAGV GWTAGLSSFA AIPFAQSIFY RLNGVGITQQ VLSENQKLIK NKFNQALGAM 300
QTGFTTTNEA FQKVQDAVNN NAQALSCLAS ELSNTFGAIS ASIGDIIQRL DVLEQDAQID 360
RLINGRLTTL NAFVAQQLVR SESAAQSAQL AKDKVNECVK AQSKRSGFCG QGTHIVSFVV 420
NAPNGLYFMH VGYYPNSHIE VVSAYGLCDA ANPTNCIAPV NGYPIKTNNT RIVDEWSYTG 480
SSFYAPEPIT SLNTKYVAPQ VTYQNISTNL PPPLLGNSGT IDPQDELDEF FKNVSTSIPN 540
FGSLTQINTT LLDLTYEMLS LQQVVKALNE SYIDLKELGN YTYYNKWP 588

SEQ ID NO: 34      moltype = AA length = 526
FEATURE           Location/Qualifiers
REGION            1..526
note = Synthetic Polypeptide
source            1..526
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 34
MFIFLLFLTL TSGSDLDRAL SGIAAEQDRN TREVFAQVKQ MYKTPTLKYF GGFNFSQILP 60
DPLKPTKRSF IEDLLFNKVT LADAGFMKQY GECLGDINAR DLICAQKFNG LTVLPPLTLD 120
DMIAAYTAAL VSGTATAGWT FGAGAAQIP FAMQMAYRFN GIGVTQNVLY ENQKQIANQF 180
NKAIISQIQES LTTTSTALGK LQDVVNQNAQ ALNTLVKQLS SNFGAISSVL NDILSRLDKV 240
EAEVQIDRLI TGRQLSLQTY VTQQLIRAAE IRASANLAAT KMSECVLGQS KRVDPCGKGY 300
HLMSPFQAAP HGVVFLHVTY VPSQERNFTT APAICHEGKA YFPREGVVFV NGTSWFITQR 360
NFFSPQIITT DNTFVSGNCD VVIGIINNTV YDPLQPELDS FKEELDKYFK NHTSPDVDLG 420
DISGINASVV NIQKEIDRLN EVAKNLNESL IDLQELGKYE QYIKWPWYVW LGFIAGLIAI 480
VMVTILLCCM TSCCSCCKGA CSCGSCCKFD EDDSEPVKLG VKLHYT 526

SEQ ID NO: 35      moltype = DNA length = 1864
FEATURE           Location/Qualifiers
misc_feature      1..1864
note = Synthetic Polynucleotide
source            1..1864
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 35
tcaagctttt ggaccctcgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatgggtct caagggtgaac gtctctgccg 120
tattcatggc agtactgtta actctccaaa caccgccggg tcaaatcatc tggggcaatc 180
tctctaagat aggggtagta ggaataggaa gtgcaagcta caaagttatg actcggtcca 240
gccatcaatc attagtcata aaattaatgc ccaatataac tctcctcaat aactgcacga 300
gggtagagat tgcagaatac aggagactac taagaacagt tttggaacca attagggatg 360
cacttaatgc atagaccagg aacataaggc cgggttcagag cgtagcttca agtaggagac 420
acaagagatt tgcgggagat gctctggcag gtgcggccct aggtgttgcc acagctgctc 480
agataacagc cggcattgca cttcaccggt ccatgctgaa ctctcaggcc atcgacaatc 540
tgagagcgag cctggaacat actaatcagg caattgaggg aatcagacaa gcagggcagg 600
agatgatatt ggtgttcag ggtgtccaag actacatcaa taatgagctg ataccgtcta 660
tgaaccagct atcttgtgat ctaatcggtc agaagctcgg gctcaaattg cttagatact 720
atacagaaat cctgtcatta ttggcccca gcctacggga ccccatatct gcggagatat 780
ctatccaggc tttgagttat gcaactggag gagatatcaa taaggtgtta gaaaagctcg 840
gatacagtggt aggcgattta ctaggcatct tagagagcag aggaataaag gctcgggataa 900
ctcacgtcga cacagagtc tacttcatag tctcagtat agcctatccg acgctgtccg 960
agattaaggg ggtgattgtc caccggctag aggggggtct gtacaacata ggctctcaag 1020
agtggataac cactgtgccc aagtatgttg caaccaagg gtacattatc tcgaattttg 1080
atgagtcatc atgtactttc atgccagagg ggaactgtgtg cagccaaaat gccttgtacc 1140

```

-continued

```

cgatgagtc tctgtccaa gaatgcctcc ggggggtccac caagtctgt gctcgtacac 1200
tcgtatccgg gtcttttggg aaccgggttca ttttatcaca agggaaacct atagccaatt 1260
gtgcatcaat tctttgttaag tgttacacaa caggtacgat tattaatcaa gacctgaca 1320
agatcctaac atacattgct gccgatcgct gcccggtagt cgaggtgaac ggcgtgacca 1380
tccaagtcgg gagcaggagg tatccagacg ctgtgtactt gcacagaatt gacctcggtc 1440
ctcccatatc attggagagg ttggacgtag ggacaaatct ggggaatgca attgccaaat 1500
tggaggatgc caaggaattg ttggaatcat cggaccagat attgagaagt atgaaagggt 1560
tatcgagcac tagcatagtc tacatcctga ttgcagtgtg tcttggaggg ttgataggga 1620
tccccacttt aatatgttgc tgcagggggc gttgtaacaa aaagggagaa caagtgtgta 1680
tgtcaagacc aggcctaagg cctgacctta caggaacatc aaaatcctat gtaagatcgc 1740
tttgatgata ataggctgga gctcgggtg ccaagcttct tgcctcttgg gctcccccc 1800
agccctcctc ccccttctct caccctgacc cccgtggtct ttgaataaag tctgagtggg 1860
cggc 1864

```

```

SEQ ID NO: 36      moltype = DNA length = 1653
FEATURE
misc_feature      1..1653
                  note = Synthetic Polynucleotide
source            1..1653
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 36
atgggtctca aggtgaacgt ctctgccgta ttcattggcag tactgttaac tctccaaaca 60
cccgccggtc aaattcattg gggcaatctc tctaagatag gggtagtagg aataggaaagt 120
gcaagctaca aggttatgac tcgttccagc catcaatcat tagtcaataa attaagccc 180
aatataactc tctccaataa ctgcacgagg gtagagattg cagaatacac gagactacta 240
agaacagttt tggaaaccaat tagggatgca cttaatgcaa tgacccagaa cataaaggccg 300
gttcagagcg tagcttcaag taggagacac aagagatttg cgggagtagt cctggcagggt 360
gcgcccttag gtgttgccac agctgctcag ataacagccg gcattgcact tcaccgggtc 420
atgctgaact ctcaggccat cgacaatctg agagcggagc ttgaaactac taatcaggca 480
attgagggca tcagacaagc agggcaggag atgatattgg ctgttcaggg tgtccaagac 540
tacatcaata atgagctgat accgtctatg aaccagctat cttgtgatct aatcgggtcag 600
aagctcgggc tcaaatgtct tagatactat acagaaatcc tgtcattatt tggccccagc 660
ctacgggacc ccatatctgc ggagatatct atccaggctt tgagttatgc acttggagga 720
gatatcaata aggtgttaga aaagctcgga tacagtggag gcgatttact aggcattctta 780
gagagcagag gaataaaggc tcggataaact cacgtcgaca cagagtccta ctcatagtc 840
ctcagtatag cctatccgag gctgtccgag attaaggggg tgattgtcca ccggctagag 900
ggggctctcg acaacatagg ctctcaagag ttggtatacca ctgtgcccaa gtatgttgca 960
acccaagggt accttatctc gaattttgat gactcatcat gtactttcat gccagagggg 1020
actgtgtgca gccaaaatgc cttgtacccg atgagtcctc tgctccaaga atgcctccg 1080
gggtccacca agcctctgtg ctgtacactc gtatccgggt cttttgggaa ccgggttcatt 1140
ttatcacaag ggaacctaat agccaattgt gcatcaattc tttgtaaagt ttacacaaca 1200
ggtacgatta ttaatacaga cctgacaaag atcctaacat acattgctgc cgatcgctgc 1260
ccggtagtcg aggtgaacgg cgtgaccatc caagtcggga gcaggaggta tccagacgct 1320
gtgtacttgc acagaattga cctcggtcct cccatcatcat tggagagggt ggacgtaggg 1380
acaaatctgg ggaatgcaat tgccaaattg gaggatgcc aagaaattgt ggaatcatcg 1440
gaccagatat ttgagaagtat gaaaggttta tcgagcacta gcatagtcta catcctgatt 1500
gcagtggtgc ttggagggtt gatagggatc cccactttaa tatgtgtctg cagggggcg 1560
tgtaacaaaa agggagaaca agttggtatg tcaagaccag gcctaaagcc tgaccttaca 1620
ggaacatcaa aatcctatgt aagatcgctt tga 1653

```

```

SEQ ID NO: 37      moltype = DNA length = 1925
FEATURE
misc_feature      1..1925
                  note = Synthetic Polynucleotide
source            1..1925
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 37
ggggaaataa gagagaaaag aagagtaaga agaaatataa gagccaccat ggggtctcaag 60
gtgaacgtct ctgccgtatt catggcagta ctgttaactc tccaaacacc cgccgggtcaa 120
attcattggg gcaatctctc taagataggg gtagtaggaa taggaagtgc aagctacaaa 180
gttatgactc gttccagcca tcaatcatta gtcataaaat taatgcccaa tataactctc 240
ctcaataact gtcacgagggt agagattgca gaatacagga gactactaag aacagttttg 300
gaaccaatta gggatgcact taatgcaatg acccagaaca taaggccgggt tcagagcgta 360
gcttcaagta ggagacacaa gagatttgct ggagtagtcc tggcagggtc gcccttaggt 420
gttggccacg ctgctcagat aacagccggc attgcacttc accgggtccat gctgaactct 480
caggccatcg acaatctgag agcgagcctg gaaactacta atcaggcaat tgaggcaatc 540
agacaagcag ggcaggagat gatattggct gttcagggtg tccaagacta catcaataat 600
gagctgtatac cgtctatgaa ccagctatct tgtgatctaa tcggtcagaa gctcggggtc 660
aaattgctta gatactatac agaaatcctg tcattatttg gccccagcct acgggacccc 720
atatctcgcg agatatctat ccaggctttg agttatgcac ttggaggaga tatcaataag 780
gtgttagaaa agctcggata cagtggaggg gatttactag gcatcctaga gagcagagga 840
ataaaggctc ggataactca ctcgacaca gagtccactc tcatagtcct cagtatagcc 900
tatccgacgc tgtccgagat taaggggggt attgtccacc ggctagaggg ggtctcgtac 960
aacataggct ctcaagagtg gtataccact gtgcccaagt atgttgcaac ccaagggtag 1020
cttatctcga attttgatga gtcacatgt actttcatgc cagaggggac tgtgtgcagc 1080
caaaatgcct tgtaccgat gagtccctc ctccaagaat gcctccgggg gtccaccaag 1140
tcctgtgctc gtacactcgt atccgggtct tttgggaacc ggttcatttt atcacaaggg 1200

```

-continued

```

aacctaataag ccaattgtgc atcaattctt tgtaagtgtt acacaacagg tacgattatt 1260
aatcaagacc ctgacaagat cctaacatac attgctgccc atcgctgccc ggtagtcgag 1320
gtgaacggcg tgaccatcca agtcgggagc aggaggtatc cagacgctgt gtacttgcac 1380
agaattgacc tcggtctctc catatcattg gagaggttgg acgtaggggac aaatctgggg 1440
aatgcaattg ccaaatgtga ggatgccaag gaattgttgg aatcatcgga ccagatattg 1500
agaagtatga aaggtttatc gagcactagc atagtctaca tcctgattgc agtgtgtctt 1560
ggagggttga tagggatccc cactttaata tgttgctgca gggggcggtt taacaaaaag 1620
ggagaacaag ttggtatgtc aagaccaggc ctaaagcctg accttacagg aacatcaaaa 1680
tcctatgtaa gatcgctttg atgataatag gctggagcct cggtggccaa gcttcttgcc 1740
ccttgggcct cccccagcc cctctccccc ttctcgcacc cgtacccccg tggctcttga 1800
ataaagtctg agtgggcggc aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1860
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1920
tctag 1925

```

```

SEQ ID NO: 38      moltype = DNA length = 1864
FEATURE           Location/Qualifiers
misc_feature       1..1864
                   note = Synthetic Polynucleotide
source            1..1864
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 38
tcaagctttt ggacctctgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatgggtct caagggtgaac gtctctgtca 120
tattcatggc agtactgtta actcttcaaa caccaccggg tcaaatccat tggggcaatc 180
tctctaagat aggggttgga ggggtaggaa gtgcaagcta caaagttatg actcgttcca 240
gccatcaatc attagtcata aagttaatgc ccaatataac tctcctcaac aattgcacga 300
gggtagggat tgcagaactc aggagactac tgagaacagt tctggaacca attagagatg 360
cacttaatgc aatgaccagg aatataagac cgggttcagag tgtagcttca agtaggagac 420
acaagagatt tgcgggagtt gtcctggcag gtgcggccct aggcgttgcc acagctgctc 480
aaataacagc cggttattgca cttcaccagt ccatgctgaa ctctcaagcc atcgacaatc 540
tgagagcgag cctagaaact actaatcagg caattgaggc aatcagacaa gcagggcagg 600
agatgatatt ggctgttcag ggtgtccaag actacatcaa taatgagctg ataccgtcta 660
tgaatcaact atcttgtgat ttaatcggcc agaagctagg gctcaaatg ctcagatact 720
atacagaaat cctgtcatta tttggcccca gcttacggga ccccatatct gcggagatat 780
ctatccaggc tttgagctat gcgcttgagg gagatataca taagggtgtg gaaaagctcg 840
gatacagttg aggtgatcta ctgggcatct tagagagcag aggaataaag gcccgataa 900
ctcacgtcga cacagagtcc tacttcattg tactcagtat agcctatccg acgctatccg 960
agattaaggg ggtgattgtc caccggctag aggggggtct gtacaacata ggctctcaag 1020
agtgtatata cactgtgccc aagtatgttg caaccgaagg gtacctatc tcgaattttg 1080
atgagtcata atgcatttcc atgccagagg ggactgtgtg cagccagaat gccttgtacc 1140
cgatgagtc tctgtcccaa gaatgcctcc ggggggtccac taagtctctg gctcgtacac 1200
tcgtatccgg gtccttcggg aaccggttca ttttatcaca ggggaaccta atagccaatt 1260
gtgcatcaat cctttgcaag tgttacacaa caggaacaat cattaatcaa gaccctgaca 1320
agatcctaac atacattgct gccgatcact gcccggtggt cgagggtgaat gccgtgacca 1380
tccaagtcgg gagcaggagg tatccggacg ctgtgtactt gcacaggatt gacctcggtc 1440
ctcccatatc tttggagagg ttggacgtag ggacaaatct ggggaatgca attgctaagt 1500
tgaggagatg caaggaattg ttggagtcac cggaccagat attgaggagt atgaaaggtt 1560
tatcgagcac tagtatagtt tacatcctga ttgcagtgtg tcttggagga ttgataggga 1620
tccccgcttt aatatgttgc tgcagggggc gttgtaacaa gaaggggagaa caagtgtgta 1680
tgtcaagacc aggcctaagg cctgatctta caggaacatc aaaatcctat gtaaggtcac 1740
tctgatgata ataggttgga gccctgggtg ccaagcttct tgccccctgg gcctccccc 1800
agccccctct ccccttctct gacccgtacc cccgtggtct ttgaataaag tctgagtggt 1860
cggc 1864

```

```

SEQ ID NO: 39      moltype = DNA length = 1653
FEATURE           Location/Qualifiers
misc_feature       1..1653
                   note = Synthetic Polynucleotide
source            1..1653
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 39
atgggtctca aggtgaacgt ctctgtcata ttcattggcag tactgttaac tcttcaaa 60
cccaccggtc aaatccattg gggcaatctc tctaagatag ggggtgtagg ggtaggaa 120
gcaagctaca aagttatgac tcgttccagc catcaatcat tagtcaataa gttaatgccc 180
aatataactc tctccaacaa ttgcacgagg gtagggattg cagaatacag gagactactg 240
agaacagttc tggaaaccaat tagagatgca cttaatgcaa tgaccagaa tataagaccg 300
gttcagagtg tagcttcaag taggagacac aagagatttg cgggagttgt cctggcagg 360
ggggccctag gcgttgccac agctgtctca ataacagccg gtattgcact tcaccagtcc 420
atgctgaact ctcaagccat cgacaatctg agagcgagcc tagaaactac taatcaggca 480
attgaggcaa tcagacaagg agggcaggag atgatatttg ctgttcaggg tgtccaagac 540
tatatcaata atagctgat accgtctatg aatcaactat cttgtgattt aatcgccagg 600
aagctagggc tcaaatgtct cagatactat acagaaatcc tgtcattatt tggccccagc 660
ttacggggacc ccatatctgc ggagatatct atccaggctt tgagctatgc gcttgaggga 720
gatatacata aggtgttgga aaagctcgga tacagtggag gtgatctact gggcatctta 780
gagagcagag gaataaaggc ccggataact cacgtcgaca cagagtccta cttcattgta 840
ctcagtatag cctatccgac gctatccgag attaaggggg tgattgtcca ccggctagag 900
ggggtctctg acaacatagg ctctcaagag tggatatacca ctgtgccccg gtatgttgca 960

```

-continued

```

acccaagggt accttatctc gaattttgat gagtcacat gcactttcat gccagagggg 1020
actgtgtgca gccagaatgc cttgtaccgc atgagtcctc tgctccaaga atgcctccgg 1080
gggtccacta agtcctgtgc tcgtacactc gtatccgggt ctttcgggaa ccggttcatt 1140
ttatcacagg ggaacctaat agccaattgt gcatcaatcc tttgcaagtg ttacacaaca 1200
ggaacaatca ttaatcaaga ccttgacaag atcctaacat acattgctgc cgatcactgc 1260
ccggtgggtcg aggtgaatgg cgtgaccatc caagtcggga gcaggaggta tccggacgct 1320
gtgtacttgc acaggattga cctcggtcct cccatatctt tggagagggt ggacgtaggg 1380
acaaatctgg ggaatgcaat tgctaagttg gaggatgcca aggaattggt ggagtcactc 1440
gaccagatat tgaggagtat gaaaggttta tcgagcacta gtatagttta catcctgatt 1500
gcagtgtgtc ttggaggatt gatagggatc cccgctttaa tatgttgctg cagggggcgt 1560
tgtaacaaga agggagaaca agttggtatg tcaagaccag gcctaaagcc tgatcttaca 1620
ggaacatcaa aatcctatgt aaggtcactc tga                                     1653

```

```

SEQ ID NO: 40      moltype = DNA length = 1925
FEATURE            Location/Qualifiers
misc_feature       1..1925
                    note = Synthetic Polynucleotide
source             1..1925
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 40
ggggaataaa gagagaaaag aagagtaaga agaaatataa gagccaccat gggtctcaag 60
gtgaacgtct ctgtcatatt catggcagta ctgttaactc ttcaaacacc caccggtcaa 120
atccattggg gcaatctctc taagataggg gtggtagggg taggaagtgc aagctacaaa 180
gttatgactc gttccagcca tcaatcatta gtcataaagt taatgcccaa tataactctc 240
ctcaacaatt gcacgagggt agggattgca gaatacagga gactactgag aacagttctg 300
gaaccaatta gagatgcact taatgcaatg acccagaata taagaccggg tcagagtgtg 360
gcttcaagta ggagacacaa gagatttgcg ggagttgtcc tggcagggtc ggccctaggc 420
gttgccacag ctgctcaaat aacagccggg attgcacttc accagtcact gctgaactct 480
caagccatcg acaatctgag agcgagccta gaaactacta atcaggcaat tgaggcaatc 540
agacaagcag ggcaggagat gatattggct gttcagggtg tccaagacta catcaataat 600
gagctgatac cgtctatgaa tcaactatct tgtgatttaa tcggccagaa gctaggggctc 660
aaattgtctc gatactatac agaatacctg tcattatttg gcccagctt acggggacccc 720
atatctgcgg agatatctat ccaggctttg agctatgcgc ttggaggaga tatcaataag 780
gtgttgaaaa agctcggata cagtggaggt gatctactgg gcatcttaga gagcagagga 840
ataaaggccc ggataactca cgtcgacaca gactcctact tcattgtact cagtatagcc 900
tatccgagcg tatccgagat taagggggtg attgtccacc ggctagaggg ggtctcgtac 960
aacataggct ctcaagatg gtataccact gtgcccaggt atgttgcaac ccaaggggtac 1020
cttatctcga attttgatga gtcacatgac acttctcatg cagaggggac tgtgtgcagc 1080
cagaatgcct tgtaccgat gagtctctctg ctccaagaat gcctccgggg gtccactaag 1140
tcctgtgtct gtacactcgt atccgggtct ttcgggaacc gggtcatttt atcacagggg 1200
aacctaatag ccaattgtgc atcaatcctt tgcaagtgtt acacaacagg aacaatcatt 1260
aatcaagacc ctgacaagat cctaacatac attgctgcgc atactgccc ggtggtcgag 1320
gtgaatggcg tgaccatcca agtcggggagc aggaggatgc cggacgctgt gtacttgac 1380
aggattgacc tcggtcctcc catatctttg gagaggttgg acgtagggac aaatctgggg 1440
aatgcaattg ctaagtgtga ggaagccaaag gaattgttgg agtcacgga ccagatattg 1500
aggagtgatg aaggtttatc gagcactagt atagtttaca tcctgattgc agtgtgtctt 1560
ggaggattga ttgggatccc cgctttaata tgttgcgtca gggggcgttg taacaagaag 1620
ggagaacaag ttggtatgtc aagaccaggc ctaaagcctg atcttacagg aacatcaaaa 1680
tcctatgtaa ggtcactctg atgataatag gctggagcct cgggtggccaa gcttcttgcc 1740
ccttgggctc cccccagcc cctcctcccc ttctgcgacc cgtacccccc ttgtctttga 1800
ataaagtctg agtgggcggc aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1860
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1920
tctag                                             1925

```

```

SEQ ID NO: 41      moltype = DNA length = 2065
FEATURE            Location/Qualifiers
misc_feature       1..2065
                    note = Synthetic Polynucleotide
source             1..2065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 41
tcaagccttt ggaccctcgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatgtcacc gcaacgagac cggataaatg 120
ccttctacaa agataacccct tatcccaagg gaagtaggat agttattaac agagaacatc 180
ttatgattga cagaccctat gttctgctgg ctgttctgtt cgtcatgttt ctgagcttga 240
tcggattgct ggcaattgca ggcattagac ttcactgggc agccatctac accgcggaga 300
tcataaaaag cctcagttacc aatctggatg tgactaaact catcgagcat cagggtcaagg 360
acgtgctgac accactcttt aaaatcatcg gggatgaagt gggcctgaga acacctcaga 420
gattcactga cctagtgaaa ttcatctcgg acaagattaa attccttaat ccggataggg 480
agtacgactt cagagatctc acttggtgca tcaaccgccc agagaggatc aaactagatt 540
atgatcaata ctgtgcagat gtgggtgctg aagagctcat gaatgcattg gtgaactcaa 600
ctctactgga gaccagaaca accactcagt tcctagctgt ctcaaggga aactgctcag 660
ggcccaactc aaccagaggt caattctcaa acatgtcgtg gtccttgttg gacttgtagt 720
taggtcaggg ttacaattgtg tcactatag tcactatgac atcccaggga atgtatgggg 780
gaacctacct agttgaaaag cctaactctga acagcaaaag gtcagagtgt tcacaactga 840
gcatgtaccg agtgtttgaa gtagggtgta tcagaaaccc gggtttgggg gctccgggtg 900
tccatagtag aaactatttt gagcaaccag tcagtaatgg tctcggcaac tgtatggtgg 960

```

-continued

```

ctttggggga gctcaaaactc gcagcccttt gtcacgggga cgattctatc ataattccct 1020
atcaggggatc agggaaaggt gtcagcttcc agctcgtcaa gctgggtgtc tggaaatccc 1080
caaccgacat gcaatcctgg gtcccttat caacggatga tccagtggta gacaggcttt 1140
acctctcatc tcacagaggt gtcacgtctg acaatcaagc aaaatgggtc gtcccgacaa 1200
cacgaacaga tgacaagtgt cgaatggaga catgcttcca gcaggcgtgt aaaggtaaaa 1260
tccaagcact ctgcgagaat cccgagtggt taccattgaa ggataacagg attccttcat 1320
acgggggtcct gtctgttgat ctgagctctga cgggtgagct taaaatcaaa attgcttcgg 1380
gattcgggcc attgatcaca cagggctcag ggatggacct atacaaatcc aactgcaaca 1440
atgtgtattg gctgactatt ccgccaatga gaaatctagc cttaggcgta atcaacacat 1500
tggagtggat accgagattc aaggttagtc ccaacctctt cactgtccca attaaggaag 1560
caggcgaaga ctgccaatgcc ccaacatacc tacctgcgga ggtggacggt gatgtcaaac 1620
tcagttccaa cctggtgatt ctacctggtc aagatctcca atatgttttg gcaacctacg 1680
atacctccag ggttgagcat gctgtggttt attacgttta cagcccaagc cgctcatttt 1740
cttactttta tctcttttag ttgcctataa aggggggtccc aatcgaaact caagtggaa 1800
gcttcacatg ggatcaaaaa ctctggtgcc gtcacttctg tgtgcttgcg gactcagaat 1860
ccggtggact tatcactcac tctgggatgg tgggcattgg agtcagctgc acagctaccc 1920
gggaagatgg aaccaactgc agataatgat aataggctgg agcctcggtg gccaaagcttc 1980
ttgccccttg ggctccccc cagcccctcc tccccttctc gcacccgtac ccccggtggtc 2040
ttgaataaaa gtctgagtg gcggc 2065

```

```

SEQ ID NO: 42      moltype = DNA length = 1854
FEATURE           Location/Qualifiers
misc_feature       1..1854
                   note = Synthetic Polynucleotide
source            1..1854
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 42
atgtcacgcg aacgagaccg gataaatgcc ttctacaaag ataaccctta tcccaaggga 60
agtaggatag ttattaacag agaacatctt atgattgaca gacctatgt tctgctggct 120
gttctgttgc tcatgtttct gagcttgatc ggattgctgg caattgcagg cattagactt 180
catcgggcag cctctacac cgcgggagatc cataaaagcc tcagtaccaa tctggatgtg 240
actaactcca tcgagcatca ggtcaaggac gtgctgacac cactctttaa aatcatcggtg 300
gatgaagtgg gcctgagaac acctcagaga ttcactgacc tagtgaatt catctcggac 360
aagattaaat tccctaattc ggataggag tagcacttca gagatctcac ttggtgcac 420
aaccgcgccg agaggatcaa actagattat gatcaatact gtgcagatgt ggctgctgaa 480
gagctcatga atgcattggg gaactcaact ctactggaga ccagaacaac cactcagttc 540
ctagctgtct caaagggaata ctgctcaggg cccactacaa tcagagggtca attctcaaac 600
atgtcgctgt cctgttgga cttgtactta ggtcgaggtt acaatgtgtc atctatagtc 660
actatgacat cccagggaat gtaggggga acctacctag ttgaaaagcc taatctgaac 720
agcaaaaggt cagagtgtgc acaactgagc atgtaccgag tgtttgaagt aggtgtgatc 780
agaaacccgg gtttgggggc tccggtgttc catatgacaa actatttga gcaaccagtc 840
agtaaatggtc tcggcaactg tatggtggct ttgggggagc tcaaaactgc agccctttgt 900
cacggggacg attctatcat aattccctat cagggatcag ggaaagggtg cagcttccag 960
ctcgtcaagc tgggtgtctg gaaatcccca accgacatgc aatcctgggt ccccttatca 1020
acggatgac cagtggtaga caggctttac ctctcatctc acagaggtgt catcgctgac 1080
aatcaagcaa aatgggctgt ccgcacaaca cgaacagatg acaagttgcg aatggagaca 1140
tgcttcacag aggcgtgtta aggtaaaatc caagcactct gcgagaatcc cgagtgggta 1200
ccattgaagg ataacaggat tcttctatac ggggtcctgt ctgttgatct gactctgacg 1260
gttgagctta aaatcaaaat tgcttcggga ttcggggcat tgatcacaca cggctcaggg 1320
atggacctat acaaatccaa ctgcaacaat gtgtattggc tgactattcc gccaatgaga 1380
aatctagcct tagcgtaat caacacattg gagtggatgc cgagattcaa ggttagtccc 1440
aacctcttca ctgtcccaat taaggaagca ggcgaagact gccatgcccc aacataccta 1500
cctcgggagg tggacggtga tgtcaaaact agttccaacc tgggtattct acctgggtcaa 1560
gatctccaat atgttttggc aacctacgat acctccaggg ttgagcatgc tgtggtttat 1620
tacgtttaca gcccaagccg ctcatcttct tactttttat cttttagggt gcctataaag 1680
gggggtccaa tcgaactaca agtggaatgc ttcacatggg atcaaaaact ctggtgcccgt 1740
cacttctgtg tgcttcggga ctcagaatcc ggtggactta tcaactactc tgggtggtg 1800
ggcatgggag tcagctgcac agctaccccg gaagatggaa ccaatcgacg ataa 1854

```

```

SEQ ID NO: 43      moltype = DNA length = 2126
FEATURE           Location/Qualifiers
misc_feature       1..2126
                   note = Synthetic Polynucleotide
source            1..2126
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 43
ggggaaataa gagagaaaag aagagtaaga agaaatataa gagccaccat gtcaccgcaa 60
cgagaccgga taaatgcctt ctacaaagat aacccttacc ccaaggggaag taggatagtt 120
attaacagag aacatcttat gattgacaga cctatgttct tgcgggctgt tctgttcgtc 180
atgtttctga gcttgatcgg attgctggca attgcaggca ttagacttca tcgggcagcc 240
atctacaccg ccgagatcca taaaagcctc agtaccatcc tggatgtgac taactccatc 300
gagcatcagg tcaaggacgt gctgacacca ctcttataaa tcatcgggga tgaagtgggc 360
ctgagaacac ctcagagatt cactgacctc gtgaaattca tctcggacaa gattaaattc 420
cttaatccgg ataggagta cgacttcaga gatctcactt ggtgcaccaa cccgcagag 480
aggatcaaac tagattatga tcaatactgt gcagatgtgg ctgctgaaga gctcatgaat 540
gcattgggtg actcaactct actggagacc agaacaacca ctcagttcct agctgtctca 600
aagggaact gctcagggcc cactacaatc agaggtcaat tctcaaacat gtcgctgtcc 660

```

-continued

```

ttgttggtact tgtacttagg tctaggttac aatgtgtcat ctatagtcac tatgacatcc 720
caggggaatgt atgggggaac ctacctagtt gaaaagccta atctgaacag caaaggggtca 780
gagttgtcac aactgagcat gtaccgagtg tttgaagtag gtgtgatcag aaaccgggt 840
ttgggggctc cggtgttcca tatgacaaac tattttgagc aaccagtcag taatggcttc 900
ggcaactgta tgggtggcttt gggggagctc aaactcgagc ccctttgtca cggggacgat 960
tctatcataa ttccctatca gggatcaggg aaaggtgtca gcttccagct cgtcaagctg 1020
ggtgtctgga aatccccaac cgacatgcaa tcctgggtcc ccttatcaac ggatgatcca 1080
gtggtagaca ggctttacct ctcatctcac agaggtgtca tgcctgacaa tcaagcaaaa 1140
tgggtgtgct cgacaacacg aacagatgac aagttgagaa tggagacatg cttccagcag 1200
gcgtgtaaa gtaaaatcca agcactctgc gagaatccc agtgggtacc attgaaggat 1260
aacaggattc cttcatacgg ggtcctgtct gttgatctga gtcgacgggt tgagcttaaa 1320
atcaaaattg cttcgggatt cgggccattg atcacacacg gtcagggat ggacctatac 1380
aaatccaact gcaacaatgt gtattggctg actattccgc caatgagaaa tctagcctta 1440
ggcgtaact acacatttga gtggataccg agattcaagg ttagtcccaa cctcttcaac 1500
gtcccaatta aggaagcagg cgaagactgc catgccccaa catacctacc tgcggaggtg 1560
gacgggtgat tcaaaactcag ttccaacctg gtgattctac ctggtcaaga tctccaatat 1620
gttttggcaa cctacgaggt tcaccgggtt gagcatgctg tggtttatta cgtttacagc 1680
ccaagccgct cattttctta cttttatcct tttaggttgc ctataaaggg ggtcccaatc 1740
gaactacaag tggaaatgct cacatgggat caaaaactct ggtgccgtca cttctgtgtg 1800
cttgccgact cagaatccgg tggacttatc actcactctg ggatgggtgg catgggagtc 1860
agctgcacag ctacccggga agatggaacc aatcgagat aatgataata ggctggagcc 1920
tcggtggcca agcttcttgc cccttgggccc tccccccagc ccctcctccc cttctgcac 1980
cgtacccttc gtgtcttctt aataaagtct gagtgggcgg caaaaaaaaa aaaaaaaaaa 2040
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 2100
aaaaaaaaaa aaaaaaaaaa atctag 2126

```

```

SEQ ID NO: 44      moltype = DNA length = 2065
FEATURE           Location/Qualifiers
misc_feature       1..2065
                   note = Synthetic Polynucleotide
source            1..2065
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 44
tcaagctttt ggacctctgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatgtcacc acaacgagac cggataaatg 120
ccttctacaa agacaacccc catcctaagg gaagttagat agttattaac agagaacatc 180
ttatgattga tagaccttat gttttgctgg ctgttctatt cgtcatgttt ctgagcttga 240
tcgggtgtct agccatttga ggcattagac ttcacggggc agccatctac accgcagaga 300
tcataaaagc cctcagcacc aatctggatg taactaactc aatcgagcat caggttaagg 360
acgtgctgac accactcttc aagatcatcg gtgatgaagt gggcttgagc acacctcaga 420
gattcactga cctagtgaag ttcatctctg acaagattaa attccttaac cgggacaggg 480
aatacgaact cagagatctc acttgggtga tcaacccgcc agagagaatc aaattggatt 540
atgatcaata ctgtgcagat gtggctgctg aagaactcat gaatgcattg gtgaactcaa 600
ctctactgga gaccagggca accaatcagt tcctagctgt ctcaaaaggga aactgctcag 660
ggccactcac aatcagaggc caattctcaa acatgtcgtc gtccctgttg gacttgtatt 720
taagtcgagg ttacaattgt tcacttatag tcaactatgac atcccaggga atgtacgggg 780
gaacttacct agtggaaaag cctaacttga gcagcaaaagg gtcagagattg tcacaactga 840
gcattgcacc aggttttgaa gtaggtgtta tcagaaatcc gggtttgggg gctccgggat 900
tccatatgac aaactatctt gagcaaccag tcagtaatat tttcagcaac tgcattgggtg 960
ctttggggga gctcaagttc gcagccctct gtcacaggga agattctatc acaattccct 1020
atcagggatc agggaaagggt actgcttccc agcttgtcaa gctaggtgtc tggaaatccc 1080
caaccgacat gcaatcctgg gtcccccctat caacggatga tccagtgaata gacaggcttt 1140
acctctcatc tcacagaggc gttatcgctg acaatcaaac aaaatgggct gtcccagaaa 1200
cacggacaga tgacaagtgt cgaatggaga catgcttcca gcaggcgtgt aagggtaaaa 1260
tccaagcact ttgcgagaat cccgagtggg caccattgaa ggataacagg attccttcat 1320
acgggggtct gtctgttgat ctgagcttga cagttgagct taaaatcaaa attgtttcag 1380
gattcggggc attgatcaca cagggttcag ggatggacct atacaatatc aaccacaaca 1440
atatgtattg gctgactatc ccgccaatga agaactctggc cttaggtgta atcaacacat 1500
tggagtggat accgagattc aagggttagt ccaacctctt cactgttcca attaagggaag 1560
caggcgagga ctgccaatgccc caacataacc tacctgcgga ggtggatggt gatgtcaaac 1620
tcagttccaa tctgggtgatt ctacctggtc aagatctcca atatgttctg gcaacctacg 1680
atacttccag agttgaacat gctgtagttt attacgttta cagcccaagg cgtctatctt 1740
cttactttta tccttttagg ttgctgttaa ggggggtccc cattgaatta caagtggaa 1800
gcttcacatg ggacaaaaaa ctctgggtgc gtcacttctg tgtgcttgcg gactcagaat 1860
ctggtggaca tatcactcac tctgggatgg tgggcatggg agtcagctgc acagccactc 1920
gggaagatgg aaccagccgc agatagtgat aataggctgg agcctcgggt gccaaagcttc 1980
ttgccccttg ggctcctccc cagcccctcc tccccttctc gcacccgtac ccccggtgtc 2040
tttgaataaa gtctgagttg gcggc 2065

```

```

SEQ ID NO: 45      moltype = DNA length = 1854
FEATURE           Location/Qualifiers
misc_feature       1..1854
                   note = Synthetic Polynucleotide
source            1..1854
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 45
atgtcaccac aacgagaccg gataaatgcc ttctacaaag acaaccccca tcctaaggga 60

```

-continued

```

agtaggatat ttattaacag agaacatctt atgattgata gaccttatgt tttgctggct 120
gttctattcg tcatgtttct gagcttgatc ggggttgctag ccattgcagg cattagactt 180
catcgggcag ccatctacac cgcagagatc cataaaagcc tcagcaccaa tctggatgta 240
actaactcaa tcgagcatca ggttaaggac gtgctgacac cactcttcaa gatcatcggt 300
gatgaagtgg gcttgaggac acctcagaga ttcactgacc tagtgaagt catctctgac 360
aagattaaat tccttaatcc ggacagggaa tacgacttca gagatctcac ttggtgtatc 420
aaccgcgcag agagaatcaa attggattat gatcaatact gtgcagatgt ggctgctgaa 480
gaactcatga atgcattggt gaactcaact ctactggaga ccaggggcaac caatcagttc 540
ctagctgtct caaagggaaa ctgctcaggg cccactacaa tcagaggcca attctcaaac 600
atgtcgctgt cccgtgtgga ctgtatttta agtcgaggtt acaatgtgtc atctatagtc 660
actatgacat cccagggaat gtacggggga acttacctag tggaaaagcc taatctgagc 720
agcaaaagggt cagagtgtgc acaactgagc atgcaccgag tgtttgaagt aggtgttatc 780
agaaatccgg gtttggggggc tcgggtattc catatgacaa actatcttga gcaaccagtc 840
agtaatgatt tcagcaactg cctgggtggc ttggggggagc tcaagttcgc agccctctgt 900
cacagggaag attctatcac aatccctat cagggatcag ggaaagggtg cagcttccag 960
cttgtcaagc taggtgtctg gaaatcccca accgacatgc aatcctgggt cccctatca 1020
acggatgatc cagctgtaga caggctttac ctctcatctc acagaggcgt tatcgctgac 1080
aatcaagcaa aatgggtgtg ccgcacaaca cggacagatg acaagttgcg aatggagaca 1140
tgcttccagc aggcgtgtaa gggtaaaatc caagcacttt gcgagaatcc cgagtggaca 1200
ccattgaagg ataacaggat tccttcatac ggggtcttgt ctggtgatct gagtctgaca 1260
gttgagctta aaatcaaaat tgtttcagga ttccggccat tgatccacac cggttcaggg 1320
atggacatct acaaatccaa ccacaacaat atgtattggc tgactatccc gccaatgaag 1380
aacctggcct taggtgtaat caacacattg gagtgggata cgagattcaa ggttagtccc 1440
aacctcttca ctgttccaat taagggaagc ggcgaggact gccatgcccc aacataccta 1500
cctgcggagg tggatgggtg atctcaaatc agttccaatc tgggtgattct acctgggtcaa 1560
gatctccaat atgttctggc aacctacgat acttccagag ttgaacatgc tgtagtttat 1620
tacgtttaca gcccaagccg ctcatcttct tacttttacc cttttaggtt gcctgtaagg 1680
ggggtcccca ttgaattaca agtggaatgc ttcacatggg accaaaaact ctggtgccgt 1740
cactctgtg tgcttgccga ctcagaatct ggtggacata tcaactcactc tgggtaggtg 1800
ggcatgggag tcagctgcac agccactcgg gaagatggaa ccagcccgag atag 1854

```

```

SEQ ID NO: 46      moltype = DNA length = 2126
FEATURE            Location/Qualifiers
misc_feature        1..2126
                    note = Synthetic Polynucleotide
source              1..2126
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 46
ggggaaataa gagagaaaag aagagtaaga agaaatataa gagccaccat gtcaccacaa 60
cgagaccgga taaatgcctt ctacaaagac aacccccatc ctaagggaag taggatagtt 120
attaacagag aacatcttat gattgataga ccttatgttt tgctggctgt tctatctgtc 180
atgtttctga ccttgatcgg gttgctagcc attgcaggca ttagacttca tcgggcagcc 240
atctacaccg cagtagtcca taaaagcctc agcaccaatc tggatgtaac taactcaatc 300
gagcatcagg ttaaggacgt gctgacacca ctctcaaga tcacgtgtga tgaagtgggc 360
ttgaggacac ctacagagat cactgacctg gtgaagtcca tctctgacaa gattaaattc 420
cttaatccgg acaggggaata cgacttcaga gatctcactt ggtgtatcaa cccgccagag 480
agaatcaaat tggattatga tcaatactgt gcagatgtgg ctgctgaaga actcatgtaat 540
gcattggtga actcaactct actggagacc agggcaacca atcagttcct agctgttcca 600
aagggaatct gctcaggggc cactacaatc agaggccaat tctcaaacat gtcgctgttc 660
ctgttggaat tgtatttaag tcgaggttac aatgtgtcat ctatagtcac tatgacatcc 720
cagggaatgt acgggggaac ttacctagtg gaaaagccta atctgagcag caaagggtca 780
gagttgtcac aactgagcat gcaccgagtg tttgaagtag gtgttatcag aaatccgggt 840
ttgggggctc cggtattcca tatgacaaac tatcttgagc aaccagtcag taatgatttc 900
agcaactgca ttgtggcctt gggggagctc aagttcgag cctctgttca cagggaagat 960
tctatcacaa ttcctcatca gggatcaggg aaaggtgtca gcttcagctt tgtcaagcta 1020
ggtgtctgga aatccccaac cgacatgcaa tcctgggtcc ccctatcaac ggatgatcca 1080
gtgatagaca ggctttacct ctcatctcac agaggcggtta tcgctgacaa tcaagcaaaa 1140
tgggctgtcc cgacaacacg gacagatgac aagttgcgaa tggagacatg cttccagcag 1200
gcgtgtaagg gtaaaatcca agcactttgc gagaatcccg agtggacacc attgaaggat 1260
aacagggatc cttcatagcg ggtctgtctg gttgatctga gtcgacagtg tagctttaa 1320
atcaaaattg tttcaggatt cgggccattg atcacacacg gttcagggat ggacctatac 1380
aaatccaacc acaacaatat gtattggctg actatcccgc caatgaagaa cctggcctta 1440
gggttaatca acacattgga gtggataccg agattcaagg ttagtcccaa cctcttcaact 1500
gttccaatta aggaagcagg cgaggactgc catgccccaa catacctacc tgcggaggtg 1560
gatggtgatg tcaaaactcag ttccaatctg gtgattctac ctggtcaaga tctccaatat 1620
gttctggcaa cctacgatac ttccagagtt gaacatgctg tagtttatta cgtttacagc 1680
ccaagccgct cattttctta cttttatcct tttagggtgc ctgtaagggg ggtccccatt 1740
gaattacaag tggaatgctt cacatgggac caaaaactct ggtgccgtca cttctgtgtg 1800
cttgcggact cagaactctg tggacatata actcactctg ggatgggtgg catggggagt 1860
agctgcacag ccaatcgagg agatggaacc agccgcagat agtgataata ggctggagcc 1920
tcggtggcca agcttcttgc ccttggggcc tccccccagc cctctctccc cttctctgac 1980
ccgtaccccc gtggtctttg aataaagtct gagtggggcg caaaaaaaa aaaaaaaa 2040
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 2100
aaaaaaaaaa aaaaaaaaaa atctag 2126

```

```

SEQ ID NO: 47      moltype = AA length = 550
FEATURE            Location/Qualifiers
REGION              1..550

```

-continued

```

note = Synthetic Polypeptide
source      1..550
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 47
MGLKVNVS AV FMAVLLTLQT PAGQIHWGNL SKIGVVGIGS ASYKVMTRSS HQSLVIKLMF 60
NITLLNNCTR VEIAEYRRL RTVLEPIRDA LNAMTQNIRP VQSVASSRRH KRFAGVVLAG 120
AALGVATAAQ ITAGIALHRS MLNSQAIDNL RASLETTNQA IEAIRQAGQE MILAVQGVQD 180
YINNELIPSM NQLSCDLIGQ KLGLKLLRY Y TEILSLFGPS LRDPISAEIS IQALSVALGG 240
DINKVLEKLG YSGGDLGIL ESRGIKARIT HVDTESYFIV LSIAYPTLSE IKGVIVHRLE 300
GVSYNIGSQE WYTTVPKYVA TQGYLISNFD ESSCTFMPEG TVCSQNALYP MSPLLQECLR 360
GSTKSCARTL VSGSFGNRFI LSQGNLIANC ASILCKCYTT GTIINQDPDK ILTYIAADRC 420
PVVEVNGVTI QVGSRRYPDA VYLHRIDLGP PISLERLDVG TNLGNIAIKL EDAKELLESL 480
DQILRSMKGL SSTSIYILI AVCLGGLIGI PTLICCCRGR CNKKGEQVGM SRPGLKPDLT 540
GTSKSYVRS L 550

SEQ ID NO: 48      moltype = AA length = 550
FEATURE           Location/Qualifiers
REGION           1..550
                 note = Synthetic Polypeptide
source           1..550
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 48
MGLKVNVS VI FMAVLLTLQT PTGQIHWGNL SKIGVVGIGS ASYKVMTRSS HQSLVIKLMF 60
NITLLNNCTR VGIAEYRRL RTVLEPIRDA LNAMTQNIRP VQSVASSRRH KRFAGVVLAG 120
AALGVATAAQ ITAGIALHRS MLNSQAIDNL RASLETTNQA IEAIRQAGQE MILAVQGVQD 180
YINNELIPSM NQLSCDLIGQ KLGLKLLRY Y TEILSLFGPS LRDPISAEIS IQALSVALGG 240
DINKVLEKLG YSGGDLGIL ESRGIKARIT HVDTESYFIV LSIAYPTLSE IKGVIVHRLE 300
GVSYNIGSQE WYTTVPKYVA TQGYLISNFD ESSCTFMPEG TVCSQNALYP MSPLLQECLR 360
GSTKSCARTL VSGSFGNRFI LSQGNLIANC ASILCKCYTT GTIINQDPDK ILTYIAADHC 420
PVVEVNGVTI QVGSRRYPDA VYLHRIDLGP PISLERLDVG TNLGNIAIKL EDAKELLESL 480
DQILRSMKGL SSTSIYILI AVCLGGLIGI PALICCCRGR CNKKGEQVGM SRPGLKPDLT 540
GTSKSYVRS L 550

SEQ ID NO: 49      moltype = AA length = 617
FEATURE           Location/Qualifiers
REGION           1..617
                 note = Synthetic Polypeptide
source           1..617
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 49
MSPQRDRINA FYKDNYPYK SRIVINREHL MIDRPYVLLA VLFVFMFLSLI GLLAIAGIRL 60
HRAAIYTAEI HKSLSTNL DV TNSIEHQVKD VLTPLFKIIG DEVGLRTPQR FTDLVKFISD 120
KIKFLNPDR E YDFRDLTWC I NPPERIKLDY DQYCADVAE ELMNALVNST LLETRTTTQF 180
LAVSKGNC SG PTTIRGQFS N MSLSLLDLYL GRGYNVSSIV TMTSQMGYGG TYLVEKPNLN 240
SKGSELSQL S MYRVFEVGI RNPGLGAPVF HMTNYFEQPV SNGLGNCMVA LGELKLAALC 300
HGDDSI IIPY QSGKGVSFQ LVKLGWVKSP TDMQSWVPLS TDDPVVDRLY LSSHRGVIAD 360
NQAKWAVPT T RTDDKL RMET CFQQACKGKI QALCENPEWV PLKDNRI PSY GVLSVDLSLT 420
VELKIKIAS G FGPLITHGSG MDLYKSNCNN VYWLTI PPMR NLALGVINTL EWIPRFK VSP 480
NLFTVPIKEA GEDCHAPT YL PAEVDGDVKL SSNLVILPGQ DLQYVLATYD TSRVEHAVVY 540
YVYSPSRSF S YFYPRFLPI K GVPIELQVEC FTWDQKLWCR HFCVLADSES GGLITHSGMV 600
GMGVSTATR EDG TNRR 617

SEQ ID NO: 50      moltype = AA length = 617
FEATURE           Location/Qualifiers
REGION           1..617
                 note = Synthetic Polypeptide
source           1..617
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 50
MSPQRDRINA FYKDNPHPKG SRIVINREHL MIDRPYVLLA VLFVFMFLSLI GLLAIAGIRL 60
HRAAIYTAEI HKSLSTNL DV TNSIEHQVKD VLTPLFKIIG DEVGLRTPQR FTDLVKFISD 120
KIKFLNPDR E YDFRDLTWC I NPPERIKLDY DQYCADVAE ELMNALVNST LLETRATNQF 180
LAVSKGNC SG PTTIRGQFS N MSLSLLDLYL SRGYNVSSIV TMTSQMGYGG TYLVEKPNLS 240
SKGSELSQL S MHRVFEVGI RNPGLGAPVF HMTNYLEQPV SNDFSNCMVA LGELKFAALC 300
HREDSI IIPY QSGKGVSFQ LVKLGWVKSP TDMQSWVPLS TDDPVIDRLY LSSHRGVIAD 360
NQAKWAVPT T RTDDKL RMET CFQQACKGKI QALCENPEWT PLKDNRI PSY GVLSVDLSLT 420
VELKIKIVSG FGPLITHGSG MDLYKSNHNN MYWLTI PPMK NLALGVINTL EWIPRFK VSP 480
NLFTVPIKEA GEDCHAPT YL PAEVDGDVKL SSNLVILPGQ DLQYVLATYD TSRVEHAVVY 540
YVYSPSRSF S YFYPRFLPVR GVPIELQVEC FTWDQKLWCR HFCVLADSES GGHITHSGMV 600
GMGVSTATR EDG TSRR 617

SEQ ID NO: 51      moltype = DNA length = 1729
FEATURE           Location/Qualifiers
misc_feature     1..1729
                 note = Synthetic Polynucleotide

```


-continued

```

source          1..1729
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 51
tcaagctttt ggacctcgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaat ataagagcca ccatggcaca agtcattaat acaaacagcc 120
tgtcgtgtgt gaccagaat aacctgaaca aatcccgatc cgcactgggc actgctatcg 180
agcgtttgtc ttccggtctg cgtatcaaca gcgcgaaaaga cgatgcggca ggacagggca 240
tgtctaaccg ttttaccgcg aacatcaaag gtctgactca ggcttcccg t aacgctaaccg 300
acggtatctc cattgcgag accactgaag gcgcgctgaa cgaaatcaac aacaacctgc 360
agcgtgtgcg tgaactggcg gttcagtcgt cgaatggtag taactcccag tctgacctcg 420
actccatcca ggctgaaatc acccagcgcc tgaacgaaat cgaccgtgta tccggccaga 480
ctcagttcaa cggcggtgaaa gtccgtggcg aggacaacac cctgaccatc caggttggtg 540
ccaacgacgg tgaactatc gatattgatt taaaagaaat cagctctaaa acactgggac 600
tgtataagct taatgtccaa gatgcctaca ccccgaaaaga aactgctgta accgttgata 660
aaactaccta taaaaatggt acagatccta ttacagccca gagcaatact gatatccaaa 720
ctgcaattgg cgggtgggtga acgggggtta tctggggctga tatcaaat t aaagatgggtc 780
aatactattt agatgttaaa ggcggtgctt ctgctgggtg ttataaagcc acttatgatg 840
aaactacaaa gaaagttaat attgatacga ctgataaaac tccgttggca actgcggaag 900
ctacagctat tcgggggaacg gccactataa cccacaacca aattgctgaa gtaacaaaag 960
aggggtgtga tacgaccaca gttgcggctc aactgtgctg agcaggggtt actggcgccg 1020
ataaggacaa tactagcctt gtaaaactat cgtttgagga taaaaacggt aaggttattg 1080
atgggtgcta tgcagtgaat atgggcgacg atttctatgc cgctacatat gatgagaaaa 1140
caggtgcaat tactgttaaa accactactt atacagatgg tactggcggt gctcaaacgt 1200
gagctgtgaa atttggtggc gcaaatggta aatctgaagt tgttactgct accgatggta 1260
agacttactt agcaagcgac ctgacaaaaa ataacttcag aacaggcggt gagcttaaaag 1320
agggttaaac agataagact gaaaacccac tgcagaaaaa tgatgctgcc ttggcacagg 1380
ttgatacact tcgttctgac ctgggtgcgg ttcagaacgg tttcaactcc gctatcacca 1440
acctgggcaa taccgtaaat aacctgtctt ctgcccgtag ccgtatcgaa gattccgact 1500
acgcaaccga agtctccaac atgtctcgcg cgcagattct gcagcaggcc ggtacctccg 1560
ttctggcgca ggcgaacatg gttccgcaaa acgtcctctc tttactgcgt tgataatagg 1620
ctggagcttc ggtggccatg ctctctgccc ccttgggccc cccccagccc ctctctccc 1680
tctcgacccc gtacccccgt ggtctttgaa taaagtctga gtgggcggc 1729

SEQ ID NO: 52      moltype = DNA length = 1518
FEATURE           Location/Qualifiers
misc_feature       1..1518
                  note = Synthetic Polynucleotide
source            1..1518
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 52
atggcacaaag tcattaatac aaacagcctg tcgctgttga cccagaataa cctgaacaaa 60
tcccagtcgg cactgggcac tgctatcgag cgtttgtctt ccggtctgag tatcaacagc 120
gcgaaagacg atgcggcagg acaggcgatt gctaaccggt ttaccgcgaa catcaaaagt 180
ctgactcagg cttcccgtaa gcctaacgac ggtatctcca ttgcgcagac cactgaaggc 240
gcgctgaacg aaatcaacaa caacctgcag cgtgtgcgtg aactggcggt tcagtctgag 300
aatggtacta actcccagtc tgacctcgac tccatccagg ctgaaatcac ccagcgccgtg 360
aacgaaatcg accgtgtatc cggccagact cagttcaacg gcgtgaaaag cctggcgagc 420
gacaaacccc tgaccatcca ggttgggtgc aacgacgggt aaactatcga tattgatcta 480
aaagaaatca gctctaaaac actgggactt gataagctta atgtccaaga tgcctacacc 540
ccgaaagaaa ctgctgtaac cgttgataaa actacctata aaaatggtag agatcctatt 600
acagcccaga gcaatactga tatccaaact gcaattggcg gtggtgcaac ggggggttact 660
ggggctgata tcaaatatta agatgtgcaa tactatttag atgttaaaag cgggtgctct 720
gctggtgttt ataaagccac ttatgatgaa actacaaaaga aagttaatat tgatacgact 780
gataaaaact cgttggcaac tgcggaagct acagctattc ggggaacggc cactataaac 840
cacaacccaa ttgctgaagt aacaaaagag ggtgttgata cgaccacagt tgcggctcaa 900
cttgcctcag caggggttat tggcgccgat aaggacaata ctagcctgtg aaaactatcg 960
tttgaggata aaaaacggtta ggttattgat ggtggctatg cagtgaatat gggcgacgat 1020
ttctatgccg ctacatatga tgagaaaaca ggtgcaatta ctgctaaaac cactacttat 1080
acagatggta ctggcggttg tcaaaactgga cgtgtgaaat ttggtggcgc aaatggtaaa 1140
tctgaagttg ttactgttac cgatggtaag acttacttag caagcgacct tgacaaacat 1200
aacttcagaa caggcggtga gcttaaaag gtttaatacag ataagactga aaaccactg 1260
cagaaaattg atgctgcctt ggcacaggtt gatacacttc gttctgacct ggggtcggtt 1320
cagaaccggt tcaactccgc tatcccaac ctgggcaata ccgtaaaata cctgtcttct 1380
gcccgtagcc gtatcgaaag ttccgactac gcaaccggaag tctccaacat gtctcgcgcg 1440
cagattctgc agcaggccgg tacctccgtt ctggcgcgag cgaaccaggt tccgcaaaa 1500
gtcctctctt tactgcgt

```

```

SEQ ID NO: 53      moltype = RNA length = 1790
FEATURE           Location/Qualifiers
misc_feature       1..1790
                  note = Synthetic Polynucleotide
source            1..1790
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 53
ggggaaataa gagagaaaag aagagtaaga agaaatataa gagccaccat ggcaacaagtc 60
attaatacaa acagcctgtc gctgttgacc cagaataacc tgaacaaatc ccagtcgca 120

```

-continued

```

ctgggcactg ctatcgagcg tttgtcttcc ggtctgcgta tcaacagcgc gaaagacgat 180
gcggcaggac aggcgattgc taaccgtttt accgcgaaca tcaaaggtct gactcaggct 240
tcccgtaacg ctaacgacgg tatctccatt gcgcagacca ctgaaggcgc gctgaacgaa 300
atcaacaaca acctgcagcg tgtgcgtgaa ctggcggttc agtctgcgaa tggactaac 360
tcccagtcctg acctcgactc catccaggct gaaatcacc agcgctgaa cgaaatcgac 420
cgtgtatccg gccagactca gttcaacggc gtgaaagtc tggcgcgagg caacaccctg 480
accatccagg ttggtgccaa cgacggtgaa actatcgata ttgatttaa agaatcagc 540
tctaaaacac tgggacttga taagcttaat gtccaagatg cctacacccc gaaagaaact 600
gctgtaaccg ttgataaaac tacctataaa aatggtacag atcctattac agcccagagc 660
aatactgata tccaaactgc aattggcggg ggtgcaacgg gggttactgg ggctgatatc 720
aaatttaaag atggtcaata ctatttagat gttaaaggcg gtgcttctgc tgggtgttat 780
aaagccactt atgatgaac tacaaagaaa gttaatatgt atacgactga taaaactccg 840
ttggcaactg cggaagctac agctattcgg ggaacggcca ctataaccca caaccaaatt 900
gctgaagtaa caaaagaggg tgttgatagc accacagttg cggtcaact tgctgcagca 960
ggggttactg gcgcgataa ggacaatact agccttgtaa aactatcggt tgaggataaa 1020
aacggtaagg ttattgatgg tggctatgca gtgaaaatgg gcgacgattt ctatgccgct 1080
acatatgatg agaaaacagg tgcaattact gctaaaacca ctacttatac agatgggtact 1140
ggcgttgctc aaactgagc tgtgaaatgt ggtggcgcaa atggtaaatc tgaagtgtt 1200
actgtaccg atggtgaagc ttacttagca agcgacctg acaaacataa cttcagaaca 1260
ggcgtgagc ttaaagaggt taatacagat aagactgaaa acccactgca gaaaattgat 1320
gctgccttgg cacaggttga tacacttcgt tctgacctgg gtgcggttca gaaccgttct 1380
aaactccgta tcaccaactc gggcaatacc gtaataaacc tgtcttctgc ccgtagccgt 1440
atcgaagatt ccgactacgc aaccgaagtc tccaacatgt ctgcgcgca gattctgcag 1500
caggccggtg cctccgttct ggcgcaggcg aaccaggttc cgcaaaacgt cctctcttta 1560
ctgcgttgat aatagcggtg agcctcggtg gccatgcttc ttgccccctg ggccctcccc 1620
cagccctcc tccccttct gcaccgtac ccccgtggtc tttgaataaa gtctgagtg 1680
gcggcaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1740
aaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaatctag 1790

```

```

SEQ ID NO: 54      moltype = AA length = 506
FEATURE           Location/Qualifiers
REGION            1..506
                  note = Synthetic Polypeptide
source            1..506
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 54
MAQVINTNSL SLLTQNNLNK SQSALGTAIE RLSSGLRINS AKDDAAGQAI ANRFTANIKG 60
LTQASRNAND GISIAQTTEG ALNEINNNLQ RVRELAVQSA NGTNSQSDLD SIQAEITQRL 120
NEIDRVSGQT QFNGVKVLAQ DNTLTIQVGA NDGETIDIDL KEISSKTLGL DKLNVQDAYT 180
PKETAVTVDK TTYKNGTDPI TAQSNNDIQT AIGGGATGVT GADIKFKDGQ YYLDVKGGAS 240
AGVYKATYDE TTKVNIIDTT DKTPLATAEA TAIRGTATIT HNQIAEVTKE GVDTTTVAAQ 300
LAAAGVTGAD KDNLSLVKLS FEDKNGKVID GGYAVKMGDD FYAATYDEKT GAITAKTTTY 360
TDGTGVAQTG AVKFGGANGK SEVVTTATDGK TYLASDLDKH NFRTGGELKE VNTDKTENPL 420
QKIDALAQVQ DTLRSDLGAV QNRFNSAITN LGNTVNNLSS ARSRIEDSDY ATEVSNMSRA 480
QILQQAGTSV LAQANQVPQN VLSLLR 506

```

```

SEQ ID NO: 55      moltype = AA length = 698
FEATURE           Location/Qualifiers
REGION            1..698
                  note = Synthetic Polypeptide
source            1..698
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 55
MAQVINTNSL SLLTQNNLNK SQSALGTAIE RLSSGLRINS AKDDAAGQAI ANRFTANIKG 60
LTQASRNAND GISIAQTTEG ALNEINNNLQ RVRELAVQSA NSTNSQSDLD SIQAEITQRL 120
NEIDRVSGQT QFNGVKVLAQ DNTLTIQVGA NDGETIDIDL KQINSQTLGL DTLNVQKQYK 180
VSDTAATVTG YADTTIALDN STFKAATGL GGTDQKIDGD LKFDDTTGKY YAKVTVTGGT 240
GKDGYYEVSV DKTNGEVTLA GGATSPLTGG LPATATEDVK NVQVANADLT EAKAALTAAG 300
VTGTASVVKM SYTDNNGKTI DGLAVKVG DYYSATQNKD GSISINTTKY TADDGTSKTA 360
LNKLGADGK TEVVSIGGKT YAASKAEGHN FKAQPDLEA AATTTENPLQ KIDAALAQVD 420
TLRSDLGAVQ NRPNSAITNL GNTVNNLTSR RSRIEDSDYA TEVSNMSRAQ ILQQAGTSVL 480
AQANQVPQNV LSLLRGGGGS GGGGMMAPD PNANPNANPN ANPNANPNAN PNANPNANPN 540
ANPNANPNAN PNANPNANPN ANPNANPNAN PNANPNANPN ANPNANPNKN NQNGGQGHNM 600
PNDPNRNVDN NANANNAVKN NNNEEPSDKH IEQYLKKIKN SISTEWSPCS VTCNGIQVR 660
IKPGSANKPK DELDYENDIE KKICKMEKCS SVFNVVNS 698

```

```

SEQ ID NO: 56      moltype = AA length = 692
FEATURE           Location/Qualifiers
REGION            1..692
                  note = Synthetic Polypeptide
source            1..692
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 56
MMAPDPNANP NANPNANPNA NPNANPNANP NANPNANPNA NPNANPNANP NANPNANPNA 60
NPNANPNANP NANPNANPNA NPNKNQNG QGHNMNDPN RNVDENANAN NAVKNNNNEE 120
PSDKHIEQYL KKIKNSISTE WSPCSVTCGN GIQVRIKPGS ANKPKDELDDY ENDIEKKICK 180

```

-continued

```

MEKCSSVFNV VNSRPVTMAQ VINTNSLSLL TQNNLNKSQS ALGTAIERLS SGLRINSKAD 240
DAAGQAIAMR FTANIKGLTQ ASRNANDGIS IAQTTEGALN EINNQLQRVR ELAVQSANST 300
NSQSDLDLSIQ AEITQRLENEI DRVSGQTQFN GVKVLAQDNT LTIQVGANDG ETIDIDLKQI 360
NSQTLGLDLTL NVQQKYKVSD TAATVTGYAD TTIALDNSTF KASATGLGGT DQKIDGDLKF 420
DDTTGKYYAK VTVTGGTGKD GYYEVSVDKT NGEVTLAGGA TSPLTGGLPA TATEDVKNVQ 480
VANADLTEAK AALTAAGVTG TASVVKMSYT DNNKGTIDGG LAVKVGDDYY SATQNKDGS I 540
SINTTKYTAD DGTSTALNK LGGADGKTEV VSIGGKTYAA SKAEGHNFKA QPDLAEAAAT 600
TTENPLQKID AALAQVDTLR SDLGAVQNRN NSAITNLGNT VNNLTSARSR IEDSDYATEV 660
SNMSRAQLIQ QAGTSVLAQA NQVPQNVLSL LR 692

```

```

SEQ ID NO: 57      moltype = RNA length = 1620
FEATURE           Location/Qualifiers
source            1..1620
                  mol_type = genomic RNA
                  organism = Human metapneumovirus

```

```

SEQUENCE: 57
atgagctgga aggtggtgat tatcttcagc ctgctgatta cacctcaaca cggcctgaag 60
gagagctacc tggagagagag ctgctccacc atcaccgagg gctacctgag cgtgctgcgg 120
accggtggtt acaccaacgt gtccaccctg gaggtgggag acgtggagaa cctgacctgc 180
agcgacggcc ctgacctgat caagaccgag ctggacctga ccaagagcgc tctgagagag 240
ctgaagaccg tgtccgccga ccagctggcc agagaggaa acatcgagaa ccctcggcag 300
agcagattcg tgctggggcg catcgctctg ggagtcggcg ctgcccgtgc agtgacagct 360
ggagtgggca ttgctaagac catcagactg gaaagcgagg tgacagccat caacaatgcc 420
ctgaagaaga ccaacgaggg cgtgagcacc ctgggcaatg gagtggaggt gctggccaca 480
gccgtgcccc agctgaagga ctctcgtagc aagaacctga ccagagccat caacaagaac 540
aagtgcgaca tcgatgacct gaagatggcc gtgagcttct cccagttcaa cagacgggtc 600
ctgaacgttg tgagacagtt ctccgacaac gctggaatca cacctgccat tagcctggac 660
ctgatgaccg acgcgcagct ggctagagcc gtgcccaca tgcccaccag cgtcgccag 720
atcaagctga tgctggagaa cagagccatg gtgcccagaa agggcttcgg catcctgatt 780
ggggtgatg gaagctccgt gatctacatg gtgcagctgc ccatcttcgg cgtgatcgac 840
acacctgctt ggtcgtgaa ggccgctcct agctgctccg agaagaaagg aaactatgcc 900
tgtctgctga gagagggatg gggctgggtt tgccagaacg ccggaagcac agtgtactat 960
cccaacgaga aggactcgga gaccagaggg gaccacgtgt tctgcgacac cgtgcccaga 1020
atcaacgttg ccagcagagc aaggagtggt aacatcaaca tcagcacaac caactacccc 1080
tgcaaggtga gcaccggagc gcaccccatc agcatgggtg ctctgagccc tctggggcgt 1140
ctggtggcct gctataaggg cgtgtcctgt agcatcgcca gcaatcggtt gggcatcatc 1200
aagcagctga acaagggtat ctctacatc accaaccagg acgcgcagac cgtgaccatc 1260
gacaacaccg tgtaccagct gagcaaggtg gagggcgagc agcagctgat caagggcaga 1320
cccgtgagct ccagcttcga ccccatcaag ttccctgagg accagttcaa cgtggccctg 1380
gaccaggtgt ttgagaacat caagaacagg caggccctgg tggaccagag caacagaatc 1440
ctgtccagcg ctgagaaggg caaacccggc ttcacatttg tgatcattct gatcgccgtg 1500
ctgggcagct ccattgatct ggtgagcatc ttcacatta tcaagaagac caagaacccc 1560
accggagccc ctctgagct gagcggcggt accaacaatg gcttcattcc ccacaactga 1620

```

```

SEQ ID NO: 58      moltype = RNA length = 1620
FEATURE           Location/Qualifiers
source            1..1620
                  mol_type = genomic RNA
                  organism = Human metapneumovirus

```

```

SEQUENCE: 58
atgtcttggg aagtgatgat catcatttcg ttactcataa caccgccaga cgggctaaga 60
gagagtgatt tggagaatgc atgtagtact ataactgagg gatacctcag tgttttaaga 120
acaggctggtt acactaatgt ctccacatta gaagttggtg atgttgaaaa tcttacatgt 180
actgatggac ctagtctaat caaaacagaa cttgatctaa caaaaagtgc ttttaaggaa 240
ctcaaaacag tctctgctga tcagttggcg agagaggagc aaattgaaaa tcccagacaa 300
tcaagatttg tcttaggtgc gatagctctc ggagttgcta cagcagcagc agtcacagca 360
ggcatttgcaa tagccaaaac cataaggcct gagagtgagg tgaatgcaat taaggtgct 420
ctcaaaacaa ctaatgaagc agtatccaca ttagggaatg gtgtgcgggt cctagccact 480
gcagtgagag agctaaaaga atttgtgagc aaaaacctga ctagtgcatt caacaggaa 540
aaatgtgaca ttgtgatctt gaagatgggt gtcagcttca gtcaattcaa cagaagattt 600
ctaaatgttg tgcggcagct ttacagcaat gcagggataa caccagcaat atcattggac 660
ctgatgactg atgctgagtt ggccagagct gtatcataca tgccaacatc tgcaggggcag 720
ataaaactga tgttgagaa cgcgcgaatg gtaaggagaa aaggatttgg aatcctgata 780
ggggtctacg gaagctctgt gatttacatg gttcaattgc cgtcttttgg tgcatagat 840
acaccttggt ggatcatcaa ggcagctccc tcttgctcag aaaaaaacgg gaattatgct 900
tgccctcctaa gagaggatca aggggtggtt tgtaaaaatg caggatctac tgtttactac 960
ccaaatgaaa aagactgcga aacaagaggt gatcatgttt tttgtgacac agcagcaggg 1020
atcaatgttg ctgagcaatc aagagaatgc atcatcata accaaccagg atgcagacac tgtaacaatt 1080
tgcaaatgca gcacaggaa acaccctata agcatgggtg cactatcacc tctcggtgct 1140
ttggtggcct gctataaagg ggtaagctgc tcgattggca gcaattgggt tggaaatcac 1200
aaacaattac ccaaggctg ctcatcata accaaccagg atgcagacac tgtaacaatt 1260
gacaataaccg tgtatcaact aagcaaatgt gaaggtgaac agcatgtaat aaaagggaga 1320
ccagtttcaa gcagttttga tccaatcaag ttccctgagg atcagttcaa tgttgcgctt 1380
gatcaagctc tcgaagcagt tgagaacagt caggcactag tggaccagtc aaacaaaatt 1440
ctaaacagtg cagaaaaagg aaacactggt ttcattatcg tagtaatttt ggttgctggt 1500
cttggtctaa ccattgattt agtgagcatc atcatcataa tcaagaaaac aaggaagccc 1560
acaggagcac ctccagagct gaatggtgtc accaaccggc gtttcatacc acatagttag 1620

```

```

SEQ ID NO: 59      moltype = RNA length = 1620

```

-continued

```

FEATURE                               Location/Qualifiers
source                               1..1620
                                     mol_type = genomic RNA
                                     organism = Human metapneumovirus

SEQUENCE: 59
atgtcttgga aagtgatgat tatcatttcg ttactcataa cacctcagca tggactaaaa 60
gaaagtattt tagaagaatc atgtagtact ataactgaag gatatctcag tgttttaaga 120
acaggttggt acaccaatgt ctttacatta gaagttgggt atgttgaaaa tcttacatgt 180
actgatggac ctactctaat caaaacagaa cttgacctaa ccaaaagtgc ttttaagagaa 240
ctcaaaacag tttctgctga tcagttagcg agagaagaac aaattgaaaa tcccagacaa 300
tcaagggttg tcctaggtgc aatagctctt ggagttgcca cagcagcagc agtcacagca 360
ggcattgcaa tagccaaaac tataaggcctt gagagtgaag tgaatgcaat caaagggtgct 420
ctcaaaacaa ccaatgaggg agtatcaaca ctaggaaatg gagtgcgggt cctagccact 480
gcagtaagag agctgaaaga atttctgagc aaaaacctga ctagtgcgat caacaagaac 540
aagtgtgaca ttgctgattt gaagatggct gtcagcttca gtcagttcaa cagaagattc 600
ctaaatgttg tgcggcagtt ttacagacaat gcagggataa caccagcaat atcattggac 660
ctgatgaaag atgctgagct ggcagagctt gtatcataca tgccaacatc tgcaggacag 720
ataaaactaa tggtagagaa ccgtgcaatg gtgaggagaa aaggatttgg aatcttgata 780
ggggtctacg gaagctctgt gatttacatg gtccagctgc cgatctttgg tgtcataaat 840
acacctgtgt ggataaactc ggcagctccc tcttgttcag aaaaagatgg aaattatgct 900
tgccctctaa gagaggatca aggggtggat tgtaaaaatg caggatccac tgtttactac 960
ccaaatgaaa aagactgcga aacaagaggt gatcatgttt tttgtgacac agcagcaggg 1020
atcaatgttg ctgagcaatc aagagaatgc aacatcaaca tatctaccac caactaccca 1080
tgcaaaagta gcacaggaag acaccctatc agcatgggtg cactatcacc tctcggtgct 1140
ttggtagctt gctacaaaagg ggttagctgc tgcactggca gtaatcaggt tggaaataatc 1200
aaacaactac ctaaaagctg ctcatacata actaaccagg acgcagacac tgtaacaatt 1260
gacaacactg tgtatcaact aagcaaaagt gaggggtgaac agcatgtaat aaaagggaga 1320
ccagtttcaa gcagttttga tccaatcagg ttctctgagg atcagttcaa tgttgcgctt 1380
gatcaagctc ttgaaagcgt tgaaaaacag caagcactag tggaccagtc aaacaaaatt 1440
ctgaacagtg cagaaaaagg aaacactggt ttcattattg taataatttt gatttgctgt 1500
cttgggttaa ccatgatttc agtgagcatc atcatcataa tcaaaaaaac aaggaagccc 1560
acaggggcac ctccggagct gaatgggtgt accaacggcg gtttcatacc gcatagttag 1620

```

```

SEQ ID NO: 60                      moltype = RNA length = 1725
FEATURE                             Location/Qualifiers
source                             1..1725
                                   mol_type = genomic RNA
                                   note = Human respiratory syncytial virus
                                   organism = unidentified

SEQUENCE: 60
atggagttgc caatocctcaa aacaaatgca attaccacaa tccttgctgc agtcacactc 60
tgtttcgctt ccagtcacaaa catcactgaa gaattttatc aatcaacatg cagtgcagtt 120
agcaaaaggct atcttagtgc tctaagaact ggttggtata ctagtgttat aactatagaa 180
ttaagtaata tcaaggaaaaa taagtgtaat ggaacagatg ctaaggtaaa attgataaaa 240
caagaattag ataaatataa aaatgtctgt acagaattgc agttgtcat gcaaaagcaca 300
ccagcagcca acaatcgagc cagaagagaa ctaccaaggt ttatgaatta tacactcaat 360
aatccaaaaa ataccaatgt aacattaagc aagaaaaagg aaagaagatt tcttggtctt 420
ttgttaggtg ttgtagctgc aatcgccagt ggcattgtct tatctaaggt cctgcacctc 480
gaaggggaaag tgaacaaaaa caaaaagtgt ctactatcca caaacaaggc tgtagtcagc 540
ttatcaaatg gagttagttg cttaaccagc aaagtgttag acctcaaaaa ctatatagat 600
aaacagttgt tacctattgt gaacaagcaa agctgcagca tatcaaacat tgaacctgtg 660
atagagttcc aacaaaagaa caacagacta ctagagatta ccagggaatt tagtgttaat 720
gcaggtgtaa ctacacctgt aagcacttat atgttaacta atagtgaatt attatcatta 780
atcaatgata tgccataaac aaatgatcag aaaaagttaa tgtccacaaa tgttcaataa 840
gttagacagc aaagttgctc tatcatgtcc ataataaagg aggaagtctt agcatatgta 900
gtacaattac cactatatgg tgtaatagat acacctgtgt ggaaactgca cacatcccct 960
ctatgtacaa ccaacacaaa ggaagggtcc aacatctgct taacaagaac cgacagagga 1020
tggtattgtg acaatgcagg atcagtatct ttctccacac aagctgaaac atgtaagatt 1080
caatcgaaac gggatttttg tgacacaatg aacagtttaa cattaccaag tgaagtaaat 1140
ctctgcaaca ttgacatatt caaccccaaa tatgattgca aaattatgac ttcaaaaaa 1200
gatgtaagca gctccgttat cacatctcta ggagccattg tgtcatgcta tggcaaaaact 1260
aaatgtacag catccaataa aaatcgtggg atcataaaga cattttctaa cgggtgtgat 1320
tatgtatcaa ataagggggg ggatactgtg tctgtaggta atacattata ttatgtaaat 1380
aagcaagaag gcaaaagtct ctatgtaaaa ggtgaaccac taataaattt ctatgacca 1440
ttagtgttcc cctctgatga atttgatgca tcaatatctc aagtcaatga gaagattaac 1500
cagagcctag catttattcg taaatccgat gaattattac ataagttaa tgctgtgtaa 1560
tccaccacaa atatcatgat aactactata attatagtga ttatagtaat attgttatca 1620
ttaattgcag ttgactgct cctatactgc aaggccagaa gcacaccagt cacactaagt 1680
aaggatcaac tgagtgggtat aaataatatt gcatttagta actga 1725

```

```

SEQ ID NO: 61                      moltype = RNA length = 1617
FEATURE                             Location/Qualifiers
source                             1..1617
                                   mol_type = genomic RNA
                                   note = Human parainfluenza virus
                                   organism = unidentified

SEQUENCE: 61
atgccaaatt caatactggt aattattaca accatgatca tggcatcaca ctgccaaata 60
gacatcacia aactacagca tgtagggtga ttggtcaaca gtcccaagg gatgaagata 120

```

-continued

```

tcacaaaact tcgaacaacg atatctaate ctgagttcta taccaaaaat agaagattct 180
aactcttctg gtgaaccaac gatcaagcaa tacaagaggt tattggatag actgatcatt 240
cctttatatg atggactaac attacagaag gatgtgatag tgactaatca agaatccaat 300
gaaaacactg atcccagaac agaacgattc tttggagggg taattggaac tattgctcta 360
ggagtagcaa cctcagcaca aattacagca gcagttgctc tggttgaagc caagcaggca 420
agatcagaca ttgaaaaact caagggaagca atcagggaca caaataaagc agtgcagtca 480
gttcagagct ctgtaggaaa tttgatagta gcaattaaat cagtccagga ttatgtcaac 540
aaagaaatcg tgccatcgat tgcgagacta ggtttggaag cagcaggact tcagttaggg 600
attgcattaa cacagcatta ctcaagaatta acaaatatat ttggtgataa cataggatcg 660
ttacaagaaa aaggaataaa attacaaggt atagcatcat tataccgtac aaatatacaca 720
gaaatattca caacatcaac agttgacaaa tatgatattt atgatctatt attacagaa 780
tcaataaagg tgagagttat agatgttgat ttgaatgatt actcaataac cctccaagtc 840
agactccctt tattgaccag actgctgaac actcaaatct acaaagtaga ttccatatca 900
tacaatatcc aaaatagaga atggatatc cctcttccca gccatatcat gacgaaaggg 960
gcatttctag gtggagcaga tgtcaaaagaa tgcatagaag cattcagcag ttatatatgc 1020
ccttctgac caggatttgc actaaacct gaaatggaga gctgtctatc aggaaacata 1080
tcccgaatgc caagaaccac agtcacatca gacatagttc ctaggatgc atttgtcaat 1140
ggaggagtggt ttgcgaattg tataacaact acatgtacat gcaatgggat cggtaataga 1200
attcaaccaac cacctgatca aggagtcaaa attataacac ataaagaatg taatacaata 1260
ggtatcaacg caatgctatt caacacaaa caaggaaggaa ctcttgcatc ctacacacca 1320
gacgacataa cattaaacaa tttctgttga cttgatccga ttgacatata aatcgagctc 1380
aacaaggcca aatcagatct tgaggaatca aaagaatgga taagaagggt aatcaaaaag 1440
ctagattcta ttggaagttg gcataaatct agcactacaa tcatagttat ttgataatg 1500
atgattatat tgtttataat taatataaca ataattacaa ttgcaattaa gtattacaga 1560
attcaaaaaga gaaatcgagt ggatcaaaat gataagccgt atgtattaac aaacaag 1617

```

```

SEQ ID NO: 62      moltype = RNA length = 1716
FEATURE           Location/Qualifiers
source            1..1716
                  mol_type = genomic RNA
                  note = Human parainfluenza virus 3
                  organism = unidentified

```

```

SEQUENCE: 62
atggaatact ggaagcacac caaccacgga aaggatgctg gtaatgagct ggagacatcc 60
acagccactc atggcaacaa gctcaccac aagataacat atatatgtg gacgataacc 120
ctgggtgtat tatcaatagt cttcatcata gtgctaacta attccatcaa aagtgaagg 180
gccgcgaat cattgtcata agacataaat aatgagttaa tggaagtac agaaaagatc 240
caagtggcat cggataatac taatgatcta atacagtcag gagtgaatac aaggcttctt 300
acaattcaga gtcattgtcc gaattatata ccaatatcat tgacacacaa aatatcggt 360
cttaggaaat tcattagtga aattacaatt agaaatgata atcaagaagt gccaccacaa 420
agaataacac atgatgtggg tataaaacct ttaaatccag atgatttctg gagatgcacg 480
ttctgtcttc catcttgat gaaaactcca aaaataagat taatgccggg accaggatta 540
ttagctatgc caacgactgt tgatggctgt gtcagaaccc cgtccttagt gataaatgat 600
ctgatttatg cttacacctc aaatctaatt actcgaggtt gccaggatat agggaaatca 660
tatcaagtat tacagatagg gataataact gtaaacctag acttggtacc tgacttaaat 720
cctaggatct ctcatacctt caacataaat gacaatagaa agtcatgttc tctagcactc 780
ctaaatacac atgtatatca actgtgttca accccaaaag ttgatgaaag atcagattat 840
gcatcatcag gcatagaaga tattgtactt gatattgtca attatgatg ctcaatctcg 900
acaacaagat ttaagaataa taatataagt ttgatcaac catatgcggc attataccca 960
ttctgtggac caggatata ctacaaaggg aaaataatat ttctcgggta tggaggtctt 1020
gaacatccaa taatagagaa tccaactctg aacacaaact ggtgtctcgg gaaaacacag 1080
agagactgta atcaagcatc tcatagtcca tgggtttcag atagaaggat ggccaactct 1140
ataaattgtt ttgacaaggg cttgaactca gttccaaaat tgaaggatg gacgatattc 1200
atgagacaaa attactgggg gtcagaagga agattacttc tactaggtaa caagatctac 1260
atatacacia gatcacaag ttggcacagc aagttacaat taggaataat tgacattact 1320
gactacagtg atataaggat aaaaatggga tggcataatg tgctatcaag accaggaaac 1380
aatgaatgtc catggggaca ttcattgtcc gatggatgta taacgggagt atataccgat 1440
gcataccac tcaatccac aggaagcatt gtatcatctg tcatattgga ctacacaaaa 1500
tcgagagtca acccagtcac aacttactca acagcaaccg aaagggtaaa cgagctggct 1560
atccgaaaca aaacactctc agctgggtac acaacaacaa gctgcattac aactataac 1620
aaagggtatt gttttcatat agtagaaata aatcataaaa gcttaaacac atttcaaccc 1680
atgtgtgtta aaacagagat tccaaaaagc tgcagt 1716

```

```

SEQ ID NO: 63      moltype = RNA length = 1716
FEATURE           Location/Qualifiers
misc_feature       1..1716
                  note = Synthetic Polynucleotide
source            1..1716
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 63
atggaatact ggaagcacac caaccacggc aaggacgccg gcaacgagct ggaaccacgc 60
acagccacac acggcaacaa gctgaccaac aagatcacct acatcctgtg gaccatcacc 120
ctgggtgtgc tgactcactg gttcatcatc gtgctgacca atagcatcaa gagcgagaag 180
gccagagaga gactgctgca ggacatcaac aacgagttca tggaagtgc cgagaagatc 240
cagggtggca gcaacaacac caacgacctg atccagagcg gcgtgaacac ccggtgctgt 300
accatccaga gccacgtgca gaactacatc cccatcagcc tgaccagca gatcagcgac 360
ctgcgggaag tccatcagca gatcaccatc cggaacgaca accaggaagt gccccccag 420
agaatcacc acgagctggg catcaagccc ctgaaccccg acgatttctg gcggtgtaca 480

```

-continued

```

agcggcctgc ccagcctgat gaagaccccc aagatccggc tgatgcctgg cctggactg 540
ctggccatgc ctaccacagt ggaatggctgt gtgcgggaccc ccagcctcgt gatcaacgat 600
ctgatctacg cctacaccag caacctgatc acccggggct gccaggatat cggcaagagc 660
taccaggtgc tgcagatcgg catcatcacc gtgaactccg acctgggtgc cgacctgaac 720
cctcggatca gccacacctt caacatcaac gacaacagaa agagctgcag cctggctctg 780
ctgaacaccg acgtgtacca gctgtgcagc acccccaagg tggacgagag aagcgactac 840
gccagcagcg gcacagagga tatcgtgctg gacatcgtga actacgacgg cagcatcagc 900
accacccggg tcaagaacaa caacatcagc ttcgaccagc cctacgcgcg cctgtaccct 960
tctgtggggc ctggcatcta ctacaagggc aagatcatct tcctgggcta cggcggcctg 1020
gaacacccca tcaacgagaa ccacatctgc aacaccaccg gctgccttgg caagaccag 1080
agagactgca atcaggccag ccacagcccc tgggtcagcg accgcagaat ggtcaactct 1140
atcatcgtgg tggacaaggg cctgaacagc gtgcccaagc tgaagtgctg gacaatcagc 1200
atgcgccaga actactgggg cagcgagggc agacttctgc tgcctgggaa caagatctac 1260
atctacaccc ggtccacagc ctggcacagc aaactgcagc tgggaatcat cgacatcacc 1320
gactacagcg acatcccgat caagtggacc tggcacaacg tgcctgagcag acccgccaac 1380
aatgagtgcg cttggggcca cagctgcccc gatggatgta tcacggcgct gtacaccgac 1440
gcctaccccc tgaatcctac cggctccatc gtgtccagcg tgatcctgga cagccagaaa 1500
agcagagtga acccctgat cacatacagc accgccaccg agagagtga cgaactggcc 1560
atcagaaaca agaccctgag gcgcggctac accaccacaa gctgcacac acactaac 1620
aagggtact gcttcacat cgtggaaatc aaccacaagt cctgaacac cttccagccc 1680
atgctgttca agaccgagat ccccaagagc tgctcc 1716

```

```

SEQ ID NO: 64      moltype = RNA length = 1617
FEATURE            Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 64
atgcccatca gcatcctgct gatcatcacc acaatgatca tggccagcca ctgccagatc 60
gacatcacca agctgcagca cgtgggcgtg ctctgaaca gccccaggc catgaagatc 120
agccagaact tcgagacagc ctacctgatc ctgagcctga tccccaaagt caggacagc 180
aacagctcgc gcgaccagca gatcaagcag tacaagcggc tgcaggacag actgacatc 240
ccccctgtacg acggcctgcg gtgcagaaa gacgtgatcg tgaccaacca ggaagcaac 300
gagaacaccg acccccgagc cgagagattc ttcggcgcg tgatcgccac aatcgccctg 360
ggagtggcca caagcgccca gattacagcc gctgtggccc tgggtggaagc caagcaggcc 420
agaagcgaca tcgagaagct gaaagaggcc atccgggaca ccaacaaggc cgtgcagagc 480
gtgcagtcca gcgtgggcaa tctgatcgtg gccatcaagt ccgtgcagga ctacgtgaac 540
aaagaaatcg tgccctctat gcgccggctg ggtgtgaaag ctgcggact gcagctgggc 600
attgccctga cacagacta cgcgagctg accaacatct tcggcgacaa catcggcagc 660
ctgcaggaaa agggcattaa gctgcaggga atcgccagcc tgtaccgcac caacatcacc 720
gagatcttca ccagaaccac cgtggataag tacgacatct acgacctgct gttcacgcag 780
agcatcaaa gtcgcgtgat cgcagtgga cgtgaacgact acagcatcac cctgcaagtg 840
cggctgcccc tgcagaccag actgctgaac acccagatct acaaggtgga cagcatctcc 900
tacaacatcc agaaccgca gtggtacatc cctctgcccc gccacattat gaccaagggc 960
gcctttctcg gcggagccga cgtgaaagag tgcacgagg ccttcagcag ctacatctgc 1020
cccagcgacc ctggtctcgt gctgaaccac gagatggaaa gctgcctgag cggcaacatc 1080
agccagtgc ccagaaccac cgtgacctcc gacatcgtgc ccagatacgc cttcgtgaat 1140
ggcggcgtgg tggccaactg catcaccacc acctgtacct gcaacggcat cggcaaccgg 1200
atcaaccagc ctcccgatca gggcgtgaag attatcaccc acaagagtg taacaccatc 1260
ggcatcaacg gcctgaacta caataccaac aaagaggcca cctggcctt ctacaccccc 1320
gcgatatca cctgaacaa ctccgtggct ctggacccca tcgacatctc catcgagctg 1380
aacaaggcca agcgcagcct ggaagagtcc aaagagtgga tcggcgagg caaccagaa 1440
ctggactcta tcggcagctg gcaccagagc agcaccacca tcacgtgat cctgattatg 1500
atgattatcc tgttcacat caacattacc atcatcacta tcgccattaa gtactaccg 1560
atccagaaac ggaaccgggt ggaccagaat gacaagccct acgtgctgac aaacaag 1617

```

```

SEQ ID NO: 65      moltype = RNA length = 4062
FEATURE            Location/Qualifiers
source              1..4062
                    mol_type = genomic RNA
                    note = Middle East respiratory syndrome coronavirus
                    organism = unidentified

```

```

SEQUENCE: 65
atgatacact cagtgtttct actgatgttc ttgttaaac ctacagaaa ttacgttgat 60
gtaggggcag attctgttaa gtctgcttgt attgaggttg atatacaaa gacctctttt 120
gataaaactt ggcctaggcc aattgatgtt tctaaggctg acggtattat ataccctcaa 180
ggccgtacat attctaact aactatcact tatcaaggtc tttttcccta tcagggagac 240
catggtgata tgtatgttta ctctgcagga catgctacag gcacaactcc aaaaaagtg 300
ttttagctca actattctca ggacgtcaaa cagtttgcta atgggttgt cgtccgtata 360
ggagcagctg ccaattccac tggcactgtt attattagcc catctaccag cgctactata 420
cgaaaaattt accctgtttt tatgctgggt tcttcagttg gtaattctc agatggtaaa 480
atggggcgct tctcaatca tactctagtt cttttgccc atggatgtgg cactttaact 540
agagcttttt attgtattct agagcctcgc tctggaatc attgtcctgc tggcaattcc 600
tatacttctt ttgccactta tcacactcct gcaacagatt gttctgatgg caattacaat 660
cgtaatgcca gtctgaactc tttaaggag tattttaatt tacgttaact cacctttatg 720
tacacttata acattaccga agatgagatt ttagagtggt ttggcattac aaaaactgct 780
caaggtgttc acctcttctc atctcggat gttgatttgt acggcgga tatgtttcaa 840

```

-continued

```

tttgccacct tgcctgttta tgatactatt aagtattatt ctatcattcc tcacagtatt 900
cgttctatcc aaagtgatag aaaagcttgg gctgccttct acgtatataa acttcaaccg 960
ttaactttcc tgttggattt ttctgttgat gggtatatac gcagagctat agactgtggt 1020
tttaatgatt tgtcacaaact ccaactgctca tatgaatcct tcgatgttga atctggagtt 1080
tattcagttt cgtcttttoga agcaaaacct tctggctcag ttgtggaaca ggctgaaggt 1140
gttgaatgtg attttttacc tcttctgtct ggcacacctc ctcaggttta taatttcaag 1200
cgtttggttt ttaccaattg caattataat cttaccaaact tgccttcact ttttctgtg 1260
aatgatttta cttgtatgca aatatctcca gcagcaattg ctagcaactg ttattcttca 1320
ctgatttttg attatttttc ataccacttt agtatgaaat ccgatctcag tgttagttct 1380
gctggtccaa tatccagttt taattataaa cagtcctttt ctaatcccac atgtttgatc 1440
ttagcgactg ttctctataa ccttactact attactaagc ctcttaagta cagctatatt 1500
aacaagtgtc ctctgtctct ttctgatgat cgtactgaag tacctcagtt agtgaacgct 1560
aatcaatact caccctgtgt atccattgtc ccattccactg tgtgggaaga cggtgattat 1620
tataggaaac accctatctcc acttgaaggt ggtggctggc ttgttgctag tggctcaact 1680
gttgccatga ctgagcaatt acagatgggc tttggtatta cagtccaata tggtagagac 1740
accaatagtg tttgcccmaa gcttgaattt gctaatgaca caaaaattgc ctctcaatta 1800
ggcaattgct tggtaatttc cctctatggt gtttcgggccc gttggtttt tcagaattgc 1860
acagctgtag gtgttcgaca gcagcgtctt gtttatgatg cgtaccagaa tttagttggc 1920
tattattctg atgatggcaa ctactactgt ctgctgtctt gtgttagtgt tcctgtttct 1980
gtcatctatg ataaagaaac taaaaccccac gctactctat ttggtagtgt tgcagtgtga 2040
cacatttctt ctaccatgtc tcaatactcc cgttctacgc gatcaatgct taaacggcga 2100
gattctacat atggcccctt tcagacacct gttggttggt tcctaggact tgttaattcc 2160
tctttgttgc tagaggactg caagttgcct ctctctgtgc tcttctgtac 2220
acacctagta ctctcacacc tcgcagttgt cgtctgtgct caggtgaaat gcgcttgcca 2280
tcatttgctt ttaatcatcc cattcaggtt gatcaactta atagtgtta ttttaatta 2340
agtataccca ctaatttttc ctttgggtgt actcaggagt acattcagac aaccattcag 2400
aaagttaact ttgattgtaa acagtagctt tgcgaatggt tccagaagtg tagcaatta 2460
ctgcgcgagt atggccagtt ttgttccaaa ataaaccagg ctctccatgg tgccaattta 2520
cgccaggatg attctgtacg taatttgttt gcgagcgtga aaagctctca atcatctct 2580
atcataccag gttttggagg tgactttaat ttgacacttc tagaacctgt ttctatatct 2640
actggcagtc gtagtgcaag tagtgctatt gaggatttgc tatttgacaa agtcaactata 2700
gctgatcctg gttatatgca aggttacgat gattgtatgc agcaaggtcc agcatcagct 2760
cgtgatctta ttgttgctca atatgtggct gggtataaag tattacctcc tcttatggat 2820
gttaatatgg aagccgcgtt tacttcatct ttgcttgcca gcatagcagg tgttggtgg 2880
actgctggtt tatctctctt tgctgctatt ccatttgcac agagtatyt ttatagggtta 2940
aacggtgttg gcattactca acaggttctt tcagagaacc aaaagcttat tgccaataag 3000
tttaatcagg ctctgggagc tatgcaaaaca ggcttcaact caactaatga agcttttcgg 3060
aagggttcagg atgctgtgaa caacaatgca caggctctat ccaaattagc tagcgagcta 3120
tctaataact ttggtgctat ttccgcctct attggagaca tcatacaacg tcttgatgtt 3180
ctcgaaacagg agccccaact agacagactt attaatggcc gtttgacaac actaaatgct 3240
tttgttgcaac agcagcttgt tcgttccgaa tcagctgctc tttccgctca attggctaaa 3300
gataaagtca atgagtggtt caaggcacaa tccaagcgtt ctggattttg cggccaaggc 3360
acacatatag tgcctgttgt tctaattgcc tttactttat gcattgttgt 3420
tattacccta gcaaccacat tgaggtttgt tctgcttatg gtcttttgcg tcagctaac 3480
cctactaatt gtatagcccc tgttaattgg tactttatta aaactaataa cactaggatt 3540
gttgatgagt ggtcatatac ttgctcgtcc ttctatgcac ctgagcccat cactctctt 3600
aatactaagt atgttgcaac acaggtgaca taccaaaaca tttctactaa cctccctct 3660
cctcttctcg gcaattccac cgggattgac ttccaagatg agttggatga gtttttcaaa 3720
aatgttagca ccagttatcc taatttttgt tctctaaccac agattaatac tacattactc 3780
gatcttacct acgagatggt gtctcttcaa caagttgtta aagcccttaa tgagtcttac 3840
atagacctta aagagcttgg caattatact tattacaaca aatggccgtg gtacatttgg 3900
cttggtttca ttgctgggct gtttgccctta gctctatgct tcttctcat actgtgctgc 3960
actggttgtg gcacaaaactg tatgggaaaa cttaagtgtta atcgttgttg tgatagatac 4020
gaggaatacg acctcgagcc gcataaggtt catgttccact aa 4062

```

```

SEQ ID NO: 66      moltype = RNA length = 4062
FEATURE            Location/Qualifiers
misc_feature        1..4062
                    note = Synthetic Polynucleotide
source              1..4062
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 66
atgatacact cagtggtttct actgatgttc ttgttaaacac ctacagaaaag ttacgttgat 60
gtaggggccag attctgttaa gtctgcttgt attgaggttg atatacaaca gactttcttt 120
gataaaactt ggccataggcc aattgatgtt tctaaggctg accgtattat ataccctcaa 180
ggccgtacat attctaact aactatcact tatcaaggctc ttttcccta tcaggagac 240
catggtgata tgtatgttta ctctgcagga catgctacag gcacaactcc acaaaagttg 300
ttttagctca actattctca ggacgtcaaa cagtttgcta atgggttgtt cgtccgtata 360
ggagcagctg ccaattccac tggcactgtt attattagcc catctaccag cgtactata 420
cgaaaaattt accctgcttt tatgctgggt tcttcagttg gtaatttctc agatggtaaa 480
atgggcccgt tcttcaatca tactctagtt cttttgcccg atggatgttg cactttactt 540
agagcttttt attgtattct ggagcctcgc tctggaaatc attgtctgc tggcaattcc 600
tatacttctt ttgcccacta tcacactcct gcaacagatt gttctgatgg caattacaat 660
cgtaatgcca gtctgaactc ttttaaggag tattttaatt tacgtaactg cacttttatg 720
tacacttata acattaccga agatgagatt ttagagtgtt ttggcaattc acaaaactgct 780
caaggtgttc acctctcttc atctcggtat gttgatttgt acggcggcaa tatgtttcaa 840
ttgcccactt tgcctgttta tgatactatt aagtattatt ctatcattcc tcacagtatt 900
cgttctatcc aaagtgatag aaaagcttgg gctgccttct acgtatataa acttcaaccg 960

```

-continued

ttaacttttc	tgttggtatt	tctgttgat	ggttatatac	gcagagctat	agactgtggt	1020
ttaaatgatt	tgtcacaaact	ccactgctca	tatgaatcct	tcgatgttga	atctggaggt	1080
tattcagttt	cgtctttcga	agcaaaacct	tctggctcag	ttgtggaaca	ggctgaaggt	1140
gttgaatgtg	atttttcacc	tcttctgtct	ggcacacctc	ctcagggtta	taatttcaag	1200
cgtttgggtt	ttaccaattg	caattataat	cttaccaaat	tgttttcact	ttttctgtg	1260
aatgatttta	ctttagtgca	aatatctcca	gcagcaattg	ctagcaactg	ttattcttca	1320
ctgattttgg	attacttttc	atacccactt	agtatgaaat	ccgatctcag	tgttagttct	1380
gctgggtccaa	tatcccagtt	taattataaa	cagtcctttt	ctaataccac	atgtttgatt	1440
ttagcgactg	ttcctcataa	ccttactact	attactaagc	ctcttaagta	cagctatatt	1500
aacaagtgtc	ctcgtctttc	ttctgatgat	cgtactgaag	tacctcagtt	agtgaacgct	1560
aatcaatact	caccctgtgt	atccattgtc	ccatccactg	tgtgggaaga	cggtgattat	1620
tataggaaac	aactatctcc	acttgaaggt	ggtggctggc	ttgttgctag	tggctcaact	1680
gttgccatga	ctgagcaatt	acagatgggc	tttggtatta	cagttcaata	tggtaacagc	1740
accaatagtg	tttgcccmaa	gcttgaattt	gctaatgaca	caaaaattgc	ctctcaatta	1800
ggcaattgctg	tgggaatttc	cctctatggt	gtttcggggc	gtggtgtttt	tcagaattgc	1860
acagctgtag	gtgttgacga	gcagcgcttt	gtttatgatg	cgtaccagaa	tttagttggc	1920
tattattctg	atgatgtcaa	ctactactgt	ttgcgtgctt	gtgttagtgt	tcctgtttct	1980
gtcatctatg	ataaagaac	taaaaccac	gctactctat	ttggtagtgt	tgcagtgtga	2040
cacatttctt	ctaccatgtc	tcaatactcc	cgttctacgc	gatcaatgct	taaaacggcg	2100
gattctacat	atggcccctt	tcagacacct	gttggtttgtg	tcctaggact	tgttaatttc	2160
tctttgttgc	tagaggactg	caagttgcct	cttggtcaat	ctctctgtgc	tcttccctgac	2220
acacctaagt	ctctcacacc	tcgcagtggt	cgtctgttgc	caggtgaaat	gcgcttggca	2280
tccattgtct	ttaatcatcc	tattcaggtt	gatcaactta	atagtagtta	ttttaaatga	2340
agtataccga	ctaatttttc	ccttggtgtg	actcaggagt	acattcagac	aaccattcag	2400
aaagttaactg	ttgattgtaa	acagtagctt	tgcgaatggt	tcagaagtgt	tgagcaatta	2460
ctgcgcgagt	atggccagtt	ttgttccaaa	ataaaccagg	ctctccatgg	tgccaattta	2520
cgccagtagt	attctgtacg	taatttgttt	gcgagcgtga	aaagctctca	atcatctcct	2580
atcataccag	gttttggagg	tgaacttaat	ttgacacttc	tggaaactgt	ttctatatct	2640
actggcagtc	gtagtgcacg	tagtcttatt	gaggatttgc	tatttgacaa	agtcactata	2700
gctgatccctg	gttatatgca	aggttacgat	gattgcctgc	agcaaggctc	agcatcagct	2760
cgtgatctta	tttgtgctca	atatgtggct	ggttacaaag	tattacctcc	tcttatggat	2820
gttaatatgg	aagccgcgta	tacttcatct	ttgcttgcca	gcatagcagg	tgttggtctg	2880
actgctggct	tatcctcctt	tgtctgtatt	ccatttgcac	agagtatctt	ttatagggtta	2940
aaagggtgttg	gcattactca	acaggttcct	tcagagaacc	aaaagcttat	tgccaataag	3000
tttaatacagg	ctctggggagc	tatgcaaaaca	ggcttccacta	caactaatga	agcttttcag	3060
aagggttcagg	atgctgtgaa	caacaatgca	caggctctat	ccaaattagc	tagcgagcta	3120
tctaataactt	ttggtgctat	ttccgcctct	attggagaca	tcatacaacg	tcttgatggt	3180
ctcgaaacagg	acgcccacaa	agacagactt	attaatggcc	gtttgacaac	actaaatgct	3240
tttgttgca	agcagcttgt	tcgttccgaa	tcagctgctc	tttccgctca	attggcctaaa	3300
gataaagtca	atagtgctgt	caaggcacaa	tccaagcgtt	ctggattttg	cggtaacaggc	3360
acacatatag	tgtcttttgt	tgtaaatgcc	cctaattggcc	tttacttcac	gcattgttgg	3420
tattacccta	gcaaccacat	tgaggttgtt	tctgttatg	gtctttgcga	tcagactaac	3480
cctactaatt	gtatagcccc	gttaaatggc	tactttatta	aaactaataa	cactaggatt	3540
gttgatgagt	ggctcatatac	tggctcgctc	ttctatgcac	ctgagcccat	tacctccctt	3600
aatactaagt	atgttgcacc	acaggtgaca	tacaaaaaca	tttctactaa	cctccctcct	3660
cctcttctcg	gcaattccac	cgggattgac	ttccaagatg	agttggatga	gtttttcaaa	3720
aatgttagca	ccagtatacc	taatttttgt	tccttaacac	agattaatac	tacattactc	3780
gatcttacct	acagatgtgt	gtctcttcaa	caagttgtta	aagcccttaa	tgagtcttac	3840
atagacctta	aagagcttgg	caattatact	tattacaaca	aatggccgtg	gtacatttgg	3900
cttggtttca	ttgtgtgggt	tgttgcccta	gctctatgct	tcttcttcac	actgtgctgc	3960
actggttgtg	gcacaaaactg	tatgggaaaa	cttaagtgtg	atcgttgttg	tgatagatac	4020
gaggaatac	acctcgagcc	gcataaggtt	catgttctact	aa		4062

SEQ ID NO: 67 moltype = RNA length = 1845
 FEATURE Location/Qualifiers
 misc_feature 1..1845
 note = Synthetic Polynucleotide
 source 1..1845
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 67

atgateccact	ccgtgttctct	cctcatgttc	ctgttgaccc	ccactgagtc	agactgcaag	60
ctcccgcctgg	gacagtcctc	gtgtgcgctg	cctgacactc	ctagcactct	gaccccacgc	120
tcctgtcgggt	cgggtgcctgg	cgaaatgcgg	ctggcctcca	tcgcttcaaa	tcaccaatc	180
caagtggatc	agctgaatag	ctcgtatttc	aagctgtcca	tcctccacga	cttctcgttc	240
ggggtcaccc	aggagtacat	ccagaccaca	attcagaagg	tcaccgtcga	ttgcaagcaa	300
tacgtgtgca	acggcttcca	gaagtgcgag	cagctgctga	gagaatacgg	gcagttttgc	360
agcaagatca	accaggcgct	gcattggagct	aacttgcgcc	aggacgactc	cgtgcgcaac	420
ctctttgcct	ctgtgaagtc	atcccagctc	tccccaatca	tcccgggatt	cggaggggac	480
ttcaacctga	ccctcctgga	gcccggtgtg	atcagcacgc	gtagcagatc	ggcgcgctca	540
gccattgaag	atcttctgtt	cgacaaggtc	accatcgccg	atccgggcta	catgcaggga	600
tacgacgact	gtatgcagca	gggaccagcc	tccgcgaggg	acctcatctg	cgcgcaatac	660
gtggccgggtg	acaaagtgtc	gcctcctctg	atggatgtga	acatggaggc	cgttataact	720
tcgtccctgc	tcggctctat	gcgcggcggtg	gggtggaccg	cggcctgttc	ctccttcgcc	780
gctatccctt	ttgcacaatc	cattttctac	cggctcaaac	gcgtgggcat	tactcaacaa	840
gtcctgtcgg	agaaccagaa	gttgatcgca	aacaagtcca	atcaggccct	gggggccaatg	900
cagactggat	tactacagac	taacgaagcg	ttccagaagg	tccaggacgc	tgtgaacaac	960
aacgcccagg	cgtctccaaa	gctggcctcc	gaactcagca	acaccttcgc	agccatcagc	1020
gcacgcatcg	gtgacataat	tcagcggctg	gacgtgctgtg	agcaggacgc	ccagatcgac	1080

-continued

cgccctcatca	acggacggct	gaccaccttg	aatgccttcg	tggcacaaca	gctgggtccgg	1140
agcgaaatcag	cggcactttc	cgcccaactc	gccaaggaca	aagtcacaga	atgcgtgaag	1200
gcccagtcaca	agagggtccg	tttctgcggt	caaggaaacc	atattgtgtc	cttcgtcgtg	1260
aacgcgcccc	acgggtctgta	ctttatgcac	gtcgggtact	acccagagcaa	tcatatcgaa	1320
gtgggtgtccg	cctaoggcct	gtgcgatgcc	gctaacccca	ctaactgtat	tgccctgtg	1380
aacggatatt	ttattaagac	caacaacacc	cgcattgttg	acgaatggtc	atacaccggt	1440
tcgtccttct	acgcgcgcga	gcccatact	tcactgaaca	ccaaatacgt	ggctccgcaa	1500
gtgacctacc	agaacatctc	caccaatttg	ccgcgcgcgc	tgctcggaaa	cagcacccga	1560
attgatattcc	aagatgaact	ggacgaattc	ttcaagaacg	tgctccattc	cattcccaac	1620
ttcggaagcc	tgacacagat	caacaccacc	cttctcgacc	tgacctacga	gatgtcgagc	1680
cttcaacaag	tggtcaaggc	cctgaacgag	agctacatcg	acctgaagga	gctgggcaac	1740
tatacctact	acaacaagtg	gccggacaag	attgaggaga	ttctgtcgaa	aatctaccac	1800
attgaaaacg	agatcgccag	aatcaagaag	cttatcggcg	aagcc		1845

```

SEQ ID NO: 68      moltype = RNA length = 4071
FEATURE            Location/Qualifiers
misc_feature       1..4071
                    note = Synthetic Polynucleotide
source             1..4071
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 68
atggaaaccc ctgccagct gctgttctct ctgctgtgtt ggctgctga taccaccggc 60
agctatgttg acgtggggcc ctagatcgct aagtcgcct gtatcgaagt ggacatccag 120
cagacctttt tcgacaagac ctggcccaga cccatcgacg tgtccaaggc cgacggcattc 180
atctatccac aaggccggac ctacagcaac atcaccatta cctaccaggg cctgttccca 240
tatcaaggcg accacggcga tatgtacgtg tactctgccc gccacggcac cggcaccaca 300
ccccagaaac tgttcgtggc caactacagc caggacgtga agcagttcgc caacggcttc 360
gtcgtgcgga ttggcgctgc tgccaatagc accggcacag tgatcatcag cccacgcacc 420
agcgcaccaca tccggaagat ctaccccgcc ttcattgctg gcagctccgt gggcaatttc 480
agcgacggca agatggggcg gttcttcaac cacaccctgg tgcgtgtgcc cgatggctgt 540
ggcacactgc tgagagcctt ctactgcatt ctggaaacca gaagcggcaa ccactgccct 600
gcccggcaata gctacaccag ctccggccacc taccacacac ccgcccacga ttgctccgac 660
ggcaactaca accggaacgc cagcctgaac agcttcaaa agtacttcaa cctgcggaac 720
tgcaccttca tgtacacctt caatatcacc gaggacgaga tccctggaat gttcggcatc 780
acccagaccg cccagggcgt gcacctgttc agcagcagat acgtggacct gtacggcggc 840
aacatgttcc agtttgccac cctgcccgtg tacgacacca tcaagtacta cagcatcatc 900
ccccacagca tccgggtccat ccagagcgac agaaaagcct gggccgcctt ctacgtgtac 960
aagctgcagc cctcgacctt cctgctggac ttcagcgtgg acggctacat cagacggggc 1020
atcgactcgc gtttcaacga cctgagccag ctgcaactgc cctacgagag cttcgactgt 1080
gaaagcggcg tgtacagcgt gtccagcttc gaggccaagc ctacggcgag cgtgggtgaa 1140
caggctgagc gctggaatg cgaactcagc cctctgtgta ggcggacccc tccccagggt 1200
tacaacttca agcggctggt gttcaccac tgcaattaca acctgacca gctgctgagc 1260
ctgttctccg tgaacgactt cacctgtagc cagatcagcc ctgcgcccat tgccagcaac 1320
tgctacagca cctcgatcct ggactacttc agctacccc tgagcatgaa gtccgatctg 1380
agcgtgtcct ccgcccggac ctacggccag ttcaactaca agcagagctt cagcaaccct 1440
acctgctga tttcggccac cgtgcccacc aatctgacca ccatcaccaa gcccctgaag 1500
tacagctaca tcaacaagtg cagcagactg ctgtccgacg accggaccga agtgcccag 1560
ctcgtgaacg ccaaccagta cagcccctgc gtgtccatcg tgcccagcac cgtgtgggag 1620
gacggcgact actacagaaa gcagctgagc cccctggaag ggcggcgatg gctgggtggc 1680
cttggaaagc cagtggccat gacggagcag ctgcagatgg gctttggcat caccgtgcag 1740
tacggcaacg acaccaacag cgtgtgcccc aagctggaat tcgccaatga caccaagatc 1800
gccagccagc tgggaaactg cgtggaatac tccctgtatg gcgtgtccgg acggggcgtg 1860
ttccagaatt gcaacagagt gggagtgcgg cagcagagat tcgtgtacga tgcctaccag 1920
aaacctcgtg gctatcacag gcacgacggc aattactact gcctgcgggc ctgtgtgtcc 1980
gtgcccgtgt ccgtgatcta cgacaagag acaaagacc acgcccacat gttcggctcc 2040
gtggcctcgc agcacatcag ctcccacctg agccagta ctccgctccac ccgggtccatg 2100
ctgaagcgga gagatgcac ctacggcccc ctgcagacac ctgtgggatg tgtgtgggc 2160
ctcgtgaaca gctccctgtt tgtggaagat tgcaagctgc cctggggcca gagcctgtgt 2220
gccctgccag atacccttag caccctgacc cctagaagcg tgcgctctgt gcccgcgaa 2280
atgcggctgg cctctatcgc cctcaatcac tggaccagct gaactccagc 2340
tacttcaagc tgagcatccc caccacttc agcttcggcg tgacccagga gtacatccag 2400
accacaatcc agaagtgac cgtggactgc aagcagtagc tgtgcaacgg ctttcagaag 2460
tgccaacagc tgcctgcgca gtacggccag ttctgcagca agatcaacca gcccctgcac 2520
ggcgccaacc tgagacagga tgacagcgtg cggaaacctg tcgcccagct gaaaagcagc 2580
cagtcacgac ccacatccc tggcttcggc ggcgacttta acctgacctt gctggaacct 2640
gtgtccatca gcaccggctc cagaagcgcc agatccgcca tcgaggacct gctgttcgac 2700
aaagtgaaca ttgcgacccc cggctacatg cagggctacg acgattgcat gcagcgggc 2760
ccagccagcg ccagggatct gatctgtgcc cagtatgtgg ccggtacaaa ggtgctgccc 2820
cccctgatgg acgtgaacat ggaagccgcc tacacctcca gcctgtctgg ctcctattgt 2880
ggcgtgggat tggacggcg cctgtctagc tttgcggcca tcccttcgc ccagagcatc 2940
ttctaccggc tgaacggcgt gggcatcaca caacaggtgc tgagcgagaa ccagaagctg 3000
atcgccaaca agtttaacca ggcactgggc gccatgcaga ccggtctcac caccaccaac 3060
gaggccttca gaaaggtgca ggaagccgtg aacaacaacg cccaggctct gagcaagctg 3120
gcccctcagc tgagcaaatc cttcggcgcc atcagcgctt ccatcgcgga catcatccag 3180
cggctggacg tgcctggaac gaacggccag atcgaccggc tgatcaacgg cagactgacc 3240
acctgaacg ccttcgtggc acagcagctc gtgcggagcg aatctgcgcg tctgtctgct 3300
cagctggcca aggacaagtg gaacagtgct gtgaaggccc agtccaagcg gagcggctt 3360
tgtggccagg gcacccacat cgtgtccttc gtctgtgaatg cccccaacgg cctgtacttt 3420

```

-continued

```

atgcacgtgg gctattaccc cagcaaccac atcgaggtgg tgtccgccta tggcctgtgc 3480
gacgcccgcga atcctaccaa ctgtatcgcc cccgtgaacg gctacttcat caagaccaac 3540
aacaccccga tctgtggacga gtggtcctac acaggcagca gcttctacgc ccccgagccc 3600
atcacctccc tgaacaccaa atacgtggcc ccccaagtga cataccagaa catctccacc 3660
aacctgcccc ctccactgct gggaaattcc accggcatcg acttccagga cgagctggac 3720
gagttcttca agaacgtgtc cactccatc cccaacttcg gcagcctgac ccagatcaac 3780
accactctgc tggacctgac ctacgagatg ctgtccctgc aacaggctcg gaaagccctg 3840
aacgagagct acatcgacct gaaagagctg gggaaactaca cctactacaa caagtggcct 3900
tggtagacatt ggctgggctt tatcgccggc ctggtggccc tggccctgtg cgtgttcttc 3960
atcctgtgct gcaccggctg cggaccaat tgcattgggca agctgaaatg caaccggctg 4020
tcgcacagat acgaggaata cgcactggaa cctcacaag tgcattgtgca c 4071

```

```

SEQ ID NO: 69      moltype = RNA length = 1864
FEATURE           Location/Qualifiers
misc_feature      1..1864
                  note = Synthetic Polynucleotide
source            1..1864
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 69
tcaagctttt ggaccctcgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatgggtct caaggtgaac gtctctgccg 120
tattcatggc agtactgtta actctccaaa caccgcccgg tcaaatccat tggggcaatc 180
tctctaagat aggggtagta ggaataggaa gtgcaagcta caaagttatg actcgttcca 240
gccatcaatc attagtctata aaattaatgc ccaatataac tctctcaat aactgcacga 300
gggtagagat tgcagaatac aggagactac taagaacagt tttggaacca attagggatg 360
cacttaatgc aatgaccagg aacataaggc cgggttcagag cgtagcttca agtaggagac 420
acaagagatt tgcgggagta gtctggcag gtgcggccct aggtgttgcc acagctgtct 480
agataaacgc cggcattgca ctccaccggt ccatgctgaa ctctcaggcc atcgacaatc 540
tgagagcgag cctggaaact actaatcagg caattgaggc aatcagacaa gcagggcagg 600
agatgatatt ggctgttcag ggtgtccaag actacatcaa taatgagctg ataccgtcta 660
tgaaccagct atcttgtgat ctaatcggtc agaagctcgg gctcaaatgg cttagatact 720
atacagaaat cctgtcatta ttggcccca gccacggga ccccatatct gcggagatat 780
ctatccaggc tttgagttat gcacttgag gagatatcaa taaggtgtta gaaaagctcg 840
gatacagtgagg ggcgatttta ctaggcatct tagagagcag aggaataaag gctcggataa 900
ctcagctcga cacagagctc tacttcatag tctcagtat agcctatccg acgctgtccg 960
agattaaggg ggtgattgtc caccggctag agggggtctc gtacaacata ggtctcaag 1020
agtggatata cactgtgccc aagtatgttg caaccacaagg gtaccttatc tcgaattttg 1080
atgagtcata atgtactttc atgccagagg ggaactgtgt cagccaaaat gccttgtacc 1140
cgatgagtc tctgctccaa gaatgcctcc gggggtccac caagtctctg gctcgtacac 1200
tcgtatccgg gtcttttggg aaccggttca ttttatcaca agggaacctc atagccaatt 1260
gtgcatcaat tctttgttaag tgttacacaa caggtagcat tattaatcaa gacctgaca 1320
agatcctaac atacattgct gccgatcgct gcccggtagt cgaggtgaac ggctgacca 1380
tccaagtcgg gagcaggagg tatccagacg ctgtgtactt gcacagaatt gacctcggtc 1440
ctcccatatc attggagagg ttggacgtag ggacaaatct ggggaatgca attgccaaat 1500
tggaggatgc caaggaattg ttggaatcat cggaccagat attgagaagt atgaaagggt 1560
tatcgagcac tagcatagtc tacatcctga ttgcagtgtg tcttggaggg ttgataggga 1620
tcccactttt aatatgttgc tgcagggggc gttgtacaac aaaggggaga caagttggta 1680
tgtcaagacc aggcctaaag cctgaccctta caggaaacatc aaaatcctat gtaagatcgc 1740
tttgatgata ataggctgga gccctcggtg ccaagcttct tgccccctgg gctcccccc 1800
agccctctct ccccttctcg caccctgacc cccgtggtct ttgaataaag tctgagtggg 1860
cggc 1864

```

```

SEQ ID NO: 70      moltype = RNA length = 1653
FEATURE           Location/Qualifiers
misc_feature      1..1653
                  note = Synthetic Polynucleotide
source            1..1653
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 70
atgggtctca aggtgaacgt ctctgccgta ttcattggcag tactgttaac tctccaaaca 60
cccgcgggtc aaattcattg gggcaatctc tctaagatag gggtagtagg aataggaaat 120
gcaagctaca aagttatgac tctgtccagc catcaatcat tagtcaataa attaatgcc 180
aatataactc tctcaataa ctgcacgagg gtagagattg cagaatacag gagactacta 240
agaacagttt tggaaaccaat tagggatgca cttaatgcaa tgacccagaa cataaggccg 300
gttcagagcg tagcttcaag taggagacac aagagatttg cgggagtagt cctggcagggt 360
gcggccctag gtgttgcac agctgctcag ataacagccg gcattgcact tcaccggtcc 420
atgtggaact ctacggccat cgacaatctg agagcgagcc tggaaactac taatcaggca 480
attgaggcaa tcagacaagc agggcaggag atgatatttg ctgttcaggg tgtccaagac 540
tacatcaata atgagctgat accgtctatg aaccagctat cttgtgatct aatcggctcag 600
aagctcgggc tcaaatgtct tagatactat acagaaatcc tgtcattatt tggccccagc 660
ctacgggacc ccatatctgc ggagatatct atccaggctt tgagtattgc acttggagga 720
gatataataa aggtgttaga aaagctcgga tacagtggag gcgatttact aggcattctta 780
gagagcagag gaataaaggc tcggataact cagctcgaca cagagtccta cttcatagtc 840
ctcagtatag cctatccgac gctgtccgag attaaggggg tgattgtcca ccggctagag 900
ggggtctcgt acaacatagg ctctcaagag ttgtatacca ctgtgcccaa gtatgttgca 960
acccaagggt acccttatctc gaattttgat gactcatcat gtacttccat gccaggggg 1020
actgtgtgca gccaaaatgc cttgtaccgg atgagtcctc tgctccaaga atgcctccgg 1080

```

-continued

```

gggtccacca agtctctgtgc tctgacactc gtatccgggt cttttgggaa ccggttcatt 1140
ttatcacaa ggaacctaat agccaattgt gcatcaattc tttgtaagtg ttacacaaca 1200
ggtacgatta ttaatcaaga ccttgacaag atcctaacat acattgtctg cgtacgtctg 1260
ccggtagtcg aggtgaacgg cgtgaccatc caagtcggga gcaggaggtg tccagacgct 1320
gtgtacttgc acagaattga cctcggtcct cccatatcat tggagaggtt ggacgtaggg 1380
acaaatctgg ggaatgcaat tgccaaattg gaggatgcca aggaattgtt ggaatcatcg 1440
gaccagatat tgagaagtat gaaaggttta tcgagcacta gcatagtcta catcctgatt 1500
gcagtgtgtc ttggagggtt gatagggatc cccactttaa tatgttctg cagggggcg 1560
tgtaacaaaa agggagaaca agttggtatg tcaagaccag gcctaaagcc tgaccttaca 1620
ggaacatcaa aatcctatgt aagatcgctt tga 1653

```

```

SEQ ID NO: 71      moltype = RNA length = 1925
FEATURE
misc_feature       1..1925
                    note = Synthetic Polynucleotide
source             1..1925
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 71
ggggaataa gagagaaaag aagagtaaga agaaatataa gagccaccat ggggtctcaag 60
gtgaacgtct ctgccgtatt catggcagta ctgttaactc tccaaacacc cgccgggtcaa 120
attcattggg gcaatctctc taagataggg gtagtaggaa taggaagtgc aagctacaaa 180
gttatgactc gtccagccca tcaatcatta gtcataaaat taatgcccaa tataactctc 240
ctcaataact gcacgagggg agagattgca gaatacagga gactactaag aacagttttg 300
gaaccaatta gggatgcact taatgcaatg acccagaaca taaggccggg tcagagcgta 360
gcttcaagta ggagacacaa gagatttgcg ggagtagtcc tggcaggtgc ggccttaggt 420
gttgcacacg ctgctcagat aacagccggc attgcactc accggtccat gctgaactct 480
caggccatcg acaatctgag agcgagcctg gaaactacta atcaggcaat tgaggcaatc 540
agacaagcag gcagcgagat gatattggct gttcagggtg tccaaagcta catcaataat 600
gagctgatag cgtctatgaa ccagctatct tgtgatctaa tcggtcagaa gctcgggctc 660
aaattgctta gatactatc agaaatcctg tcattatttg gcccagcct acgggacccc 720
atatctgcgg agatatctat ccaggctttg agttatgcac ttggaggaga tatcaataag 780
gtgtagaaaa agctcggata cagtggaggc gatttactag gcatcttaga gagcagagga 840
ataaaggctc ggataactca ctcgacaca gagtccactc tcatagtcct cagtatagcc 900
tatccgacgc tgtccgagat taaggggggtg attgtccacc ggctagaggg ggtctcgtac 960
aacataggct ctcaagagtg gtataccact gtgcccaagt atgttgcaac ccaaggggtac 1020
cttatctcga attttgatga gtcattcatg actttcatgc cagaggggac tgtgtgcagc 1080
caaaatgcct tgtaccgatg gagtctctcg ctccaagaat gcctccgggg gtccaccaag 1140
tctgtgtctc tgcaactcgt atccgggtct tttgggaacc ggttcatttt atcacaaggg 1200
aacctaatag ccaattgtgc atcaattctt tgaagtgtt acacaacagg tacgattatt 1260
aatcaagacc ctgacaagat cctaacatac attgtgtcgg atcgctgccg ggtagtcgag 1320
gtgaacggcg tgaccatcca agtcgggagc aggaggtatc cagacgtgtg gtacttgcac 1380
agaattgacc tcggtctctc catatcattg gagaggttgg acgtagggac aaactctggg 1440
aatgcaattg ccaaatggga ggtatgccaag gaattgttgg aatcatcgga ccagatattg 1500
agaagtatga aaggtttatc gagcactagc atagtctaca tctgtattgc agtgtgtctt 1560
ggagggttga ttgggatccc cactttaata tgttgcgtgc gggggcggtt taacaaaaag 1620
ggagaacaag ttggtatgtc aagaccaggc cttaaagcctg accttacagg aacatcaaaa 1680
tctatgttaa gctcgctttg atgataatag gctggagcct cggtggccaa gcttcttgcc 1740
ccttgggcct ccccccctcc tctctgcacc cgtacccccg tggctcttga 1800
ataaagtctg agtgggctgc aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1860
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1920
tctag 1925

```

```

SEQ ID NO: 72      moltype = RNA length = 1864
FEATURE
misc_feature       1..1864
                    note = Synthetic Polynucleotide
source             1..1864
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 72
tcaagctttt ggaccctcgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatgggtct caagggtgaac gtctctgtca 120
tattcatggc agtactgtta actcttcaaa caccacccgg tcaaatccat tggggcaatc 180
tctctaagat aggggtggta ggggtaggaa gtgcaagcta caaagttatg actcgttcca 240
gccaataatc attagtcata aagttaatgc ccaatataac tctcctcaac aattgtcacga 300
gggtagggat tgcagaatac aggagactac tgagaacagt tctggaacca attagagatg 360
cacttaatgc aatgaccagc aatataagac cggttcagag ttagcttca agtaggagac 420
acaagagatt tgcgggagtt gtctcggcag gtgcggccct aggcgttgcc acagctgctc 480
aaataacagc cggtatttgc cttcaccagt ccatgctgaa ctctcaagcc atcgacaatc 540
tgagagcgag cctagaaact actaatcagg caattgaggc aatcagacaa gcagggcagg 600
agatgatatt ggctgttcag ggtgtccaag actacatcaa taatgagctg ataccgtcta 660
tgaatcaact atctgttgat ttaatcgccc agaagctagg gctcaaatg ctcatgact 720
atacagaaat cctgtcatta ttggcccca gcttacggga cccatatct gcggagatat 780
ctatccaggc tttgagctat gcgcttgagg gagatatcaa taagggtgtg gaaaagctcg 840
gatacagtg aggtgatcta ctgggcatct tagagagcag aggaataaag gcccgataa 900
ctcacgtcga cacagagtcc tacttcattg tactcagtat agcctatccg acgctatccg 960
agattaaggg ggtgatgtgc caccgctag agggggctc gtacaacata ggctctcaag 1020
agtgttatc cactgtgccc aagtatgtt caacccaagg gtacctatc tcgaattttg 1080

```

-continued

atgagtcac	atgcactttc	atgccagagg	ggactgtgtg	cagccagaat	gccttgtacc	1140
cgatgagtc	tctgtcctca	gaatgcctcc	gggggtccac	taagtccctgt	gctcgtacac	1200
tcgtatccgg	gtctttccgg	aaccggttca	ttttatcaca	ggggaacct	atagccaatt	1260
gtgcatcaat	cctttgcaag	tgttacacaa	caggaacaat	cattaatcaa	gacctgaca	1320
agatcctaac	atacattgct	gccgatcact	gcccgggtgt	cgagggtgaat	ggcgtgacca	1380
tccaagtcgg	gagcaggagg	tatccggacg	ctgtgtactt	gcacaggatt	gacctcggtc	1440
ctcccatatc	tttggagagg	ttggacgtag	ggacaaatct	ggggaatgca	attgctaagt	1500
tggaggatgc	caagggaattg	ttggagtcac	cggaccagat	attgaggagt	atgaaagggt	1560
tatcgagcac	tagtatagtt	tacatcctga	ttgcagtgtg	tcttggagg	ttgataggga	1620
tccccgcttt	aatatgttgc	tgcagggggc	gttgaacaa	gaaggagaa	caagtgtgta	1680
tgtcaagacc	aggcctaagg	cctgatctta	caggaacatc	aaaatcctat	gtaaggtcac	1740
tctgatgata	ataggctgga	gcctcgggtg	ccaagcttct	tgccccttgg	gcctcccccc	1800
agccccctct	ccccttctctg	caccctgacc	cccgtgtgtc	ttgaataaag	tctgagtggtg	1860
cggc						1864

SEQ ID NO: 73 moltype = RNA length = 1653
 FEATURE Location/Qualifiers
 misc_feature 1..1653
 note = Synthetic Polynucleotide
 source 1..1653
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 73

atgggtctca	agggtgaacgt	ctctgtcata	ttcatggcag	tactgttaac	tcttcaaaca	60
cccaccggtc	aaatcctatg	gggcaatctc	tctaagatag	gggtggtagg	ggtaggaagt	120
gcaagctaca	aagttatgac	tcgttccagc	catcaatcat	tagtcaataa	gttaatgcc	180
aataataact	tcctcaacaa	ttgcacgagg	gtagggtattg	cagaatacac	gagactactg	240
agaacagttc	tggaaaccaat	tagagatgca	cttaatgcaa	tgaccagaa	tataagaccg	300
gttcagagtg	tagtttcaag	taggagacac	aagagatttg	cgggagttgt	cctggcaggt	360
gcggccctag	gcgttgccac	agctgtctca	ataacagccg	gtattgcact	tcaccagtcc	420
atgtggaact	ctcaagccat	cgacaatctg	agagcgagcc	tagaaactac	taatcaggca	480
attgaggcaa	tcagacaagc	agggcaggag	atgatatggg	ctgttcaggg	tgtccaagac	540
tacatcaata	atgagctgat	accgtctatg	aatcaactat	cttgtgattt	aatcggccag	600
aagctagggc	tcaaatgtct	cgataactat	acagaaatcc	tgtcattatt	tggccccagc	660
ttacgggacc	ccatatctgc	ggagatatct	atccaggctt	tgagctatgc	gcttggagg	720
gatatcaata	agggtgttga	aaagctcgga	tacagtggag	gtgatctact	gggcatctta	780
gagagcagag	gaataaaggc	ccggataact	cacgtcgaca	cagagtccta	cttcattgta	840
ctcagtatag	ccatcccgac	gctatccgag	atlaaggggg	tgattgtcca	ccggctagag	900
ggggtctctg	acaacatagg	ctctcaagag	tggatatacca	ctgtgcccaa	gtatgttgca	960
acccaagggt	accctatctc	gaattttgat	gagtcacat	gcactttcat	gccagagggg	1020
actgtgtgca	gccagaatgc	cttgtaccgg	atgagtcctc	tgctccaaga	atgcctccgg	1080
gggtccacta	agtcctgtgc	tcgtacactc	gtatccgggt	ctttcgggaa	ccgggttcatt	1140
ttatcacagg	ggaacaattg	agccaattgt	gcataaatcc	tttgcaagtg	ttacacaaca	1200
ggaacaatca	ttaatcaaga	ccctgacaag	atcctaact	acattgtctg	cgatcactgc	1260
ccggtggctg	agggtgaatg	cgtgaccatc	caagtccgga	gcaggaggta	tcgggacgct	1320
gtgtacttgc	acaggattga	cctcggtcct	cccatatctt	tggagaggtt	ggacgttagg	1380
acaaatctgg	ggaatgcaat	tgctaaagtg	gaggatgcca	aggaattggt	ggagtcactg	1440
gaccagatat	tgaggaggtat	gaaagggtta	tcgagcacta	gtatagttta	catcctgatt	1500
gcagtggtgc	ttggaggagt	gatagggtac	cccgttttaa	tatgttctgc	cagggggcgt	1560
tgtaacaaga	agggagaaca	agttgggtat	tcaagaccag	gcctaaagcc	tgatcttaca	1620
ggaacatcaa	aatcctatgt	aagggtcactc	tga			1653

SEQ ID NO: 74 moltype = RNA length = 1925
 FEATURE Location/Qualifiers
 misc_feature 1..1925
 note = Synthetic Polynucleotide
 source 1..1925
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 74

ggggaaataa	gagagaaaag	aagagtaaga	agaaatataa	gagccaccat	gggtctcaag	60
gtgaacgtct	ctgtcatatt	catggcagta	ctgttaactc	ttcaaacacc	caccggtcaa	120
atccattggg	gcaatctctc	taagataggg	gtggtagggg	taggaagtgc	aagctacaaa	180
gttatgactc	gttccagcca	tcaatcatta	gtcataaagt	taatgcccaa	tataactctc	240
ctcaacaatt	gcacgagggt	agggattgca	gaatacagga	gactactgag	aacagttctg	300
gaaccaatta	gagatgcact	taatgcaatg	accagaata	taagaccggt	tcagagtgt	360
gcttcaagta	ggagacacaa	gagatttgct	ggagttgtcc	tggcagggtc	ggccctaggc	420
gttgccacag	ctgtctcaat	aacagccggt	attgcacttc	accagtcact	gctgaactct	480
caagccatcg	acaatctgag	agcagaccta	gaaactacta	atcaggcaat	tgaggcaatc	540
agacaagcag	ggcaggagat	gatattggct	gttcagggtg	tccaagacta	catcaataat	600
gagctgatac	cgtctatgaa	tcaactatct	tgtgatttaa	tcggccagaa	gctagggtct	660
aaattgctca	gatactatac	agaaatcctg	tcattatttg	gcccagcctt	acgggacccc	720
atatctcgcg	agatatctat	ccaggctttg	agctatgcgc	ttggaggaga	tatcaataag	780
gtgttgga	agctcggata	cagtggaggt	gatctactgg	gcattctaga	gagcagagga	840
ataaaggccc	ggataactca	cgtcgacaca	gagtcctact	tcattgtact	cagtatagcc	900
tatccgacgc	tatccgagat	taaggggggtg	attgtccacc	ggctagaggg	ggtctcgtac	960
aacataggct	ctcaagagtg	gtataccact	gtgcccaagt	atgttgcaac	ccaaggggtac	1020
cttatctcga	attttgatga	gtcatcatgc	acttctatgc	cagaggggac	tgtgtgcagc	1080
cagaatgcct	tgtacccgat	gagtcctctg	ctccaagaat	gcctccgggg	gtccactaag	1140

-continued

tccctgtgctc	gtacactcgt	atccgggtct	tccgggaacc	gggttcatttt	atcacagggg	1200
aacctaatag	ccaattgtgc	atcaatcctt	tgcaagtgtt	acacaacagg	aacaatcatt	1260
aatcaagacc	ctgacaagat	cctaacatac	attgctgccc	atcactgccc	ggtggtcgag	1320
gtgaatggcg	tgaccatcca	agtcgggagc	aggaggtatc	cggacgctgt	gtacttgac	1380
aggattgacc	tcggtcctcc	catatctttg	gagagggttg	acgtagggac	aaatctgggg	1440
aatgcaattg	ctaagtggga	ggaatgccaag	gaattgttgg	agtcacgga	ccagatattg	1500
aggagtatga	aaggtttatc	gagcactagt	atagtttaca	tcctgattgc	agtgtgtctt	1560
ggaggattga	tagggatccc	cgtttaata	tgttgctgca	ggggcggttg	taacaagaag	1620
ggagaacaag	ttggtatgtc	aagaccaggc	ctaaagcctg	atcttacagg	aacatcaaaa	1680
tcctatgtaa	ggctactctg	atgataatag	gctggagcct	cgggtggccaa	gcttcttgcc	1740
ccttgggcct	ccccccagcc	ctctctcccc	ttcctgcacc	cgtacccccg	tggtctttga	1800
ataaagtctg	agtggcgcg	aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	1860
aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	aaaaaaaaaa	1920
tctag						1925

SEQ ID NO: 75 moltype = RNA length = 2065
 FEATURE Location/Qualifiers
 misc_feature 1..2065
 note = Synthetic Polynucleotide
 source 1..2065
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 75

tcaagctttt	ggaccctcgt	acagaagcta	atacagactca	ctatagggaa	ataagagaga	60
aaagaagagt	aagaagaagt	ataagagcca	ccatgtcacc	gcaacgagac	cggataaatg	120
ccttctacaa	agataaacct	tatcccaagg	gaagtaggat	agttattaac	agagaacatc	180
ttatgattga	cagaccctat	gttctgctgg	ctgttctgtt	cgtcatgttt	ctgagcttga	240
tcggattgct	ggcaattgca	ggcattagac	ttcatcgggc	agccatctac	accgcggaga	300
tcataaaaag	ccctcagacc	aatctggatg	tgactaaactc	catcgagcat	cagggtcaagg	360
acgtgctgac	accactcttt	aaaatcatcg	gggatgaagt	gggcctgaga	acacctcaga	420
gattcactga	ctcagtgaac	ttcatctcgg	acaagattaa	attccttaat	ccgगतaggg	480
agtagcactt	cagagatctc	acttgggtga	tcaacccgcc	agagaggatc	aaactagatt	540
atgatcaata	ctgtgcagat	gtggctgctg	aagagctcat	gaatgcattg	gtgaactcaa	600
ctctactgga	gaccagaaca	accactcagt	tcctagctgt	ctcaaaggga	aactgctcag	660
ggcccaactac	aatacagaggt	caattctcaa	acatgtcgct	gtccttgttg	gacttgtact	720
taggtcgagg	ttacaatgtg	tcactatag	tcactatgac	atcccaggga	atgtatgggg	780
gaacctacct	agttgaaaag	cctaactctga	acagcaaaag	gtcagagtgt	tcacaactga	840
gcatgtaccg	agtgtttgaa	gtagggtgtga	tcagaaaacc	gggtttgggg	gctccgggtg	900
tcctatgac	aaactatttt	gagcaaccag	tcagtaatgg	tctcggaac	tgtatgggtg	960
ccttggggga	gctcaaaactc	gcagcccttt	gtcacgggga	cgattctatc	ataattccct	1020
atcagggatc	agggaaaagt	gtcagcttcc	agctcgtaaa	gctgggtgtc	tggaatccc	1080
caaccgacat	gcaatcctgg	gtccccttat	caacggatga	tcagtggtga	gacaggcttt	1140
acctctcatc	tccagaggtg	gtcatcgctg	acaatcaagc	aaaatgggct	gtcccgaaca	1200
cacgaacaga	tgacaagtgt	gcaatggaga	catgcttcca	gcaggcggtg	aaaggtaaaa	1260
tccaagcact	ctgcgagaat	cccgagtggtg	taccattgaa	ggataacagg	attccttcat	1320
acggggctct	gtctgttgat	ctagctctga	cgggttagct	taaaatcaaa	attgcttcgg	1380
gattcggggc	attgatcaca	cacggctcag	ggatggacct	atacaaatcc	aactgcaaca	1440
atgtgtattg	gctgactatt	cgcaccaatga	gaaatctagc	cttaggcgtga	atcaacacat	1500
tggagtggat	accgagattc	aagggttagtc	ccaacctctt	cactgtccca	attaagggaag	1560
caggcgaaaga	ctgccatgcc	ccaacatacc	tacctcgga	gggtggacggt	gatgtcaaac	1620
tcagttccaa	cctgggtgatt	ctacctgggtc	aagatctcca	atatgttttg	gcaacctacg	1680
atactccag	ggctgtgatt	gtctgtggtt	attacgttta	cagcccaagc	cgctcatttt	1740
cttactttta	tccttttagg	ttgcctataa	aggggggtccc	aatcgaaacta	caagtgggaat	1800
gcttcacatg	ggatcaaaaa	ctctgtgtgc	gtcacttctg	tgtgcttgcg	gactcagaat	1860
ccggtggact	tatcatctac	ctctgggatg	tgggcatggg	agtcagctgc	acagctaccc	1920
gggaagatgg	aaccaatcgc	agataatgat	aataggctgg	agcctcggtg	gccaagcttc	1980
ttgccccttg	ggcctcccc	cagccctctc	tccccttctt	gcacccgtac	ccccgtgggt	2040
tttgaataaa	gtctgagtgg	gcggc				2065

SEQ ID NO: 76 moltype = RNA length = 1854
 FEATURE Location/Qualifiers
 misc_feature 1..1854
 note = Synthetic Polynucleotide
 source 1..1854
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 76

atgtcacccg	aacgagaccg	gataaatgcc	ttctacaaag	ataaccctta	tcccaaggga	60
agtaggatag	ttattaacag	agaacatctt	atgattgaca	gacctatgt	tctgctggct	120
gttctgttcg	tcattgtttc	gagcttgatc	ggattgtcgg	caattgcagg	cattagactt	180
catcggggcag	ccactctacac	cgcggagatc	cataaaaagc	tcagtaccaa	tctggatgtg	240
actaactcca	tcgagcatca	ggtcaaggac	gtgctgacac	cactctttaa	aatcatcggtg	300
gatgaagtgg	gcctgagaac	acctcagaga	ttcactgacc	tagtgaaatt	catctcggac	360
aagattaaat	tccttaatcc	ggataggag	tacgacttca	gagatctcac	ttgggtgcac	420
aaacccgccag	agaggatcaa	actagattat	gatcaatact	gtgcagatgt	ggctgctgaa	480
gagctcatga	atgcattggt	gaactcaact	ctactggaga	ccagaacaac	cactcagttc	540
ctagctgtct	caaaggga	ctgctcagg	cccactacaa	tcagagggtca	attctcaaac	600
atgtcgctgt	cctgtgtgga	ctgtgactta	ggtcgaggtt	acaatgtgtc	atctatagtc	660
actatgacat	cccagggaat	gtatggggga	acctacctag	ttgaaaagcc	taatctgaac	720

-continued

```

agcaaagggt cagagttgtc acaactgagc atgtaccgag tgtttgaagt aggtgtgatc 780
agaaaacccgg gtttgggggc tccggtgttc catatgacaa actattttga gcaaccagtc 840
agtaatgttc tcggcaactg tatggtggct ttgggggagc tcaaaactcg agccctttgt 900
cacggggacg attctatcat aattccctat cagggatcag ggaagggtgt cagcttccag 960
ctcgtaaacg tgggtgtctg gaaatcccca accgacatgc aatcctgggt ccccttatca 1020
acggatgato cagtggtaga caggctttac ctctcatctc acagagggtg catcgctgac 1080
aatcaagcaa aatgggctgt cccgacaaca cgaacagatg acaagttgct aatggagaca 1140
tgcttccagc aggcgtgtta aggtaaaatc caagcactct gcgagaatcc cagtggtgta 1200
ccattgaagg ataacaggat tccttcatac ggggtcctgt ctgttgatct gagtctgacg 1260
gttgagctta aatcaaaaat tgcttcggga ttccggccat tgatcacaca cggctcaggg 1320
atggacctat acaaatccaa ctgcaacaat gtgtattggc tgactattcc gccaatgaga 1380
aatctagcct taggcgtaat caacacattg gagtggatgc cgagattcaa ggttagtccc 1440
aacctcttca ctgtcccact taaggaagca ggccgaagact gccatgcccc aacataccta 1500
cctgcggagg ttgcgggtga tgtcaaaact agttccaaac tgggtgattc acctggtcaa 1560
gatctccaat atgttttggc aaactacgat acctccaggg ttgagcatgc tgtggtttat 1620
tagctttaca gcccaagcgc ctcatcttct tacttttctc cttttaggtt gcctataaag 1680
ggggtcccaa tcgaactaca agtggaatgc ttccatggg atcaaaaaac ctggtgccgt 1740
cacttctgtg tgcttgcgga ctcagaatcc ggtggactta tcaactcactc tgggtgggtg 1800
ggcatgggag tcagctgcac agctaccccg gaagatggaa ccaatcgagc ataa 1854

```

```

SEQ ID NO: 77      moltype = RNA length = 2126
FEATURE           Location/Qualifiers
misc_feature       1..2126
                   note = Synthetic Polynucleotide
source            1..2126
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 77
ggggaataa gagagaaaag aagagtaaga agaaatataa gagccaccat gtcaccgcaa 60
cgagaccgga taaatgcctt ctacaagat aacccttacc ccaagggag taggatagtt 120
attaacagag aacatcttat gattgacaga ccctatgttc tgctggctgt tctgttctgc 180
atgtttctga cgttgatcgg attgctggca attgcaggca ttagacttca tccggcagcc 240
atctacaccg cggagatcca taaaagcctc agtaccaaac tggatgtgac taactccatc 300
gagcatcagg tcaaggacgt gctgacacca ctctttaaaa tcatcgggga tgaagtgggc 360
ctgagaacac ctacagatgt cactgaccta gtgaaattca tctcgacaaa gattaaattc 420
cttaatccgg atagggagta cgacttcaga gatctcactt ggtgcaccaa cccgccagag 480
aggatcaaac tagattatga tcaatactgt gcagatgtgg ctgctgaaga gctcatgaat 540
gcattgtgtg actcaactct actggagacc agaacaacca ctacgttctc agctgtctca 600
aagggaactc gctcagggcc ctacataatc agaggtcaat tctcaaacat gtcgctgtcc 660
ttgttggaat gtacttagg tcaggtttac aatgtgtcat ctatagtcac tatgacatcc 720
cagggaatgt tggggggaac ctaccatagt gaaaagccta atctgaacag caaaggggtc 780
gagttgtcac aactgagcat gtaccgagtg tttgaagtag gtgtgatcag aaaccggggt 840
ttgggggctc cgggtttcca tatgacaaac tattttgagc aaccagtcag taatggtctc 900
ggcaactgta tgggtgcttt gggggagctc aaactcgag ccctttgtca cggggacgat 960
tctatcataa ttccctatca gggatcaggg aaaggtgtca gcttcacgct cgtcaagctg 1020
ggtgtctgga aatcccaaac gcacatgcac tcctgggtcc ccttatcaac ggatgatcca 1080
tggttagaca ggctttacct ctcatctcac agaggtgtca tcgctgacaa tcaagcaaaa 1140
tggtgtgtcc cgacaacacg aacagatgac aagttgcgaa tggagacatg cttccagcag 1200
gcgtgtaaag gtaaaatcca agcactctgc gagaatcccg agtgggtacc attgaaggat 1260
aacaggatcc ctacatacgg ggtcctgtct gttgatctga gtctgacggt tgagcttaaa 1320
atcaaaattg cttcggtagt cgggccattg atcacacacg gctcagggat ggacctatac 1380
aaatccaact gcaacaatgt gtattggctg actattccgc caatgagaaa tctagcctta 1440
ggcgtaatca acacattgga gtggataccg agattcaagg ttagtcccaa cctcttctac 1500
gtcccaatta tgaagcagcg cgaagactgc catgccccaa catacctacc tgcggaggtg 1560
gacggtgatg tcaaacctcag ttccaaacct gtgattctac ctggtcaaga tctccaatat 1620
gttttggcaa cctacgatac ctccaggggt gagcatgctg ttggtttatta cgtttacagc 1680
ccaagccgct cattttctta cttttacct tttagggtgc ctataaaggg ggtcccaatc 1740
gaactacaag tggaaatcgc cacatgggat caaaaactct ggtgccgtca cttctgtgtg 1800
cttgccgact cagaatccgg tggacttacc actcactctg ggtggtggg catggggagt 1860
agctgcacag ctaccgggga agatggaacc aatcgagat aatgataata ggctggagcc 1920
tcggtggcca agcttcttgc cctctgggcc tccccccagc cctcctccc cttcctgcac 1980
cgtaccccc gtggtctttg aataaaagtct gagtgggcgg caaaaaaaaa aaaaaaaaaa 2040
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 2100
aaaaaaaaaa aaaaaaaaaa atctag 2126

```

```

SEQ ID NO: 78      moltype = RNA length = 2065
FEATURE           Location/Qualifiers
misc_feature       1..2065
                   note = Synthetic Polynucleotide
source            1..2065
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 78
tcaagctttt ggaccctcgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatgtcacc acaacgagac cggataaatg 120
ccttctacaa agacaacccc catcctaagg gaagtaggat agttattaac agagaacatc 180
ttatgattga tagaccttat gttttgctgg ctgttctatt cgtcatgttt ctgagcttga 240
tcgggttgct agccattgca gccattagac ttcatcgggc agccatctac accgcagaga 300
tccataaaag cctcagcacc aatctggatg taactaactc aatcgagcat caggttaagg 360

```

-continued

```

acgtgctgac accactcttc aagatcatcg gtgatgaagt gggcttgagg acacctcaga 420
gattcactga cctagtgaag ttcatctctg acaagattaa attccttaat ccggacaggg 480
aatacgcactt cagagatctc acttggtgta tcaacccgcc agagagaaac aaattggatt 540
atgatcaata ctgtgcagat gtggctgctg aagaactcat gaatgcattg gtgaactcaa 600
ctctactgga gaccagggca accaatcagt tcctagctgt ctcaaaggga aactgctcag 660
ggcccactac aatcagaggc caattctcaa acatgtcgtc gtccctgttg gacttgtatt 720
taagtcgagg ttacaatgtg tcacttatag tcactatgac atcccaggga atgtacgggg 780
gaacttacct agtggaagg cctaacttga gcagcaagg gtcagagttg tcacaactga 840
gcattgcacc agtggttgaa gtagggtgta tcagaaatcc ggggttgagg gctccgggat 900
tccatatgac aaactatctt gagcaaccag tcagtaatga ttccagcaac tgcattgggtg 960
ctttggggga gctcaagttc gcagccctct gtcacaggga agattctatc acaattccct 1020
atcagggatc agggaaagggt gtacgcttcc agcttggtcaa gctaggtgtc tggaaatccc 1080
caaccgacat gcaatcctgg gtcccccctat caacggatga tccagtgtga gacaggcttt 1140
acctctctac tcacagaggc gttatcgctg acaatcaagc aaaatgggct gtcccacaa 1200
cacggacaga tgacaagtgt cgaatggaga catgcttcca gcaggcgtgt aagggtaaaa 1260
tccaagcact ttgcgagaat cccgagtgtg caccattgaa ggataacagg attccttcat 1320
acgggggtctt gtcgtgtgat cagtgctgag cagttgagct taaaaatcaa attgtttcag 1380
gattcgggcc attgatcaca cagggttcag ggtaggacct atacaaatcc aaccacaaca 1440
atatgtattg gctgactatc gcgccaatga agaacctggc cttaggtgtg atcaacacat 1500
tggagtggat accgagattc aagggttagtc ccaacctctt cactgttcca attaagggaag 1560
caggcgaggga ctgcatgccc ccaacatacc tacctgcgga ggtggatggt gatgtcaaac 1620
tcagttccaa tctggtgatt ctacctgggtc aagatctcca atatgttctg gcaacctacg 1680
atacttccag agttgaacat gctgtagttt attacgttta cagcccaagc cgctcatttt 1740
cttactttta tcccttttagg ttgctgttaa ggggggtccc cattgaatta caagtgggat 1800
gcttcacatg ggacccaaaaa ctctggtgcc gtcacttctg tgtgcttgcg gactcagaat 1860
ctgggtggaca tatcactcac tctgggatgg tgggcattgg agtcagctgc acagccactc 1920
gggaagatgg aaccagccgc agatagtgat aataggctgg agcctcgggt gccaaagcttc 1980
ttgcccttg ggctccccc cagcccctcc tccccttctc gcaccggtac ccccggtgtc 2040
ttgaataaaa gtctgagtgg ggggc 2065

```

```

SEQ ID NO: 79      moltype = RNA length = 1854
FEATURE           Location/Qualifiers
misc_feature       1..1854
                   note = Synthetic Polynucleotide
source            1..1854
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 79
atgtcaccac aacagagaccg gataaatgcc ttctacaaag acaaccccca tcctaaggga 60
agtaggatag ttattaacag agaacatctt atgattgata gaccttatgt ttgtctggct 120
gtttctattc tcattgtttc agccttgatc ggggtgctag ccattgcagg cattagactt 180
catcggggcag ccatctacac cgcagagatc cataaaagcc tcagcaccaa tctggatgta 240
actaaactca tgcagcagatc ggttaaggac gtgctgacac cactcttcaa gatcatcggt 300
gatgaagtgg gcttgaggac acctcagaga ttcactgacc tagtgaagtt catctctgac 360
aagattaaat tccttaatcc ggacagggaa tacgacttca gagatctcac ttggtgtatc 420
aaaccgcca agagaatcaa attggattat gatcaatact gtgcagatgt ggctgctgaa 480
gaactcctga atgcattggt gaactcaact ctactggaga ccagggcaac caatcagttc 540
tagctgtctc caaagggaaa ctgctcaggg ccactacaaa tcagaggcca attctcaaac 600
atgtcgtctg cctctgttga cttgtattta agtcgaggtt acaatgtgtc atctatagtc 660
actatgacat cccagggaat gtaacgggga acctacctag tggaaaagcc taatctgagc 720
agcaaaaggt cagagttgtc acaactgagc atgcaccgag tgtttgaagt aggtgttatc 780
agaaatccgg gtttgggggc tccggatttc catatgacaa actatcttga gcaaccagtc 840
agtaatgatt tcagcaactc catgggtggt ttgggggagc tcaagttcgc agccctctgt 900
cacaggggaag attctatcac aattccctat cagggatcag ggaaaggtgt cagcttcocag 960
cttgtcaagc tagtgtctcg gaaatcccca accgacatgc aatcctgggt cccctatcca 1020
acgggatgac cagtgtatga caggctttac ctctcatctc acagaggcgt tatcgtgac 1080
aatcaagcaa aatgggctgt ccgcacaaca cggacagatg acaagttgag aatggagaca 1140
tgcttccagc aggcgtgtta gggtaaaatc caagcacttt gcgagaatcc cgagtggaca 1200
ccattgaagg ataacaggat tccttcatac ggggtcttgt ctgttgatct gactctgaca 1260
gttagcttta aaatcaaaat tgtttcagga ttcgggcatc tgatcacaca cggttcaggg 1320
atggacctat acaaatccaa ccacaacaat atgtattggc tgactatccc gccaatgaag 1380
aacctggcct taggtgtaat caacacattg gagtggatgc cgagattcaa ggtagtccc 1440
aacctcttca ctgttccaat taagggaagc ggcgaggact gccatgcccc aacataccta 1500
cctgcggagg tggatggtga tgtcaaaact agttccaatc tgggtattct acctggtcaa 1560
gatctccaat atgttctggc aacctacgat acttccagag ttgaacatgc tgtagtttat 1620
tacgtttaca gcccaagccg ctcatcttct tacttttctc cttttagggt gcctgtaagg 1680
ggggtcccca ttgaattaca agtggaatgc ttcacatggg accaaaaact ctggtgccgt 1740
cactctgtgt tgcttgctga ctcagaatct ggtggacata tcaactcactc tgggatgggtg 1800
ggcatgggag tcagctgcac agccactcgg gaagatggaa ccagccgcag atag 1854

```

```

SEQ ID NO: 80      moltype = RNA length = 2126
FEATURE           Location/Qualifiers
misc_feature       1..2126
                   note = Synthetic Polynucleotide
source            1..2126
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 80
ggggaataaa gagagaaaag aagagtaaga agaaataaa gagccaccat gtcaccacaa 60

```

-continued

```

cgagaccgga taaatgcctt ctacaaagac aacccccatc ctaaggggaag taggatagtt 120
attaacagag aacatcttat gattgtaga ccttatgttt tgctggctgt tctattcgtc 180
atgtttctga gcttgatcgg gttgctagcc attgcaggca ttagacttca tcgggcagcc 240
atctacaccg cagagatcca taaaagcctc agcaccatc tggatgtaac taactcaatc 300
gagcatcagg ttaaggacgt gctgacacca ctcttcaaga tcctcgtgga tgaagtgggc 360
ttgaggacac ctcagagatt cactgaccta gtgaagtcca tctctgacaa gattaaattc 420
cttaatccgg acaggggaata cgacttcaga gatctcactt ggtgatcaa cccgccagag 480
agaatcaaat tggattatga tcaatactgt gcagatgtgg ctgctgaaga actcatgaat 540
gcattggtga actcaactct actggagacc agggcaacca atcagttcct agctgtctca 600
aagggaaact gctcagggcc cactacaatc agaggccaat tctcaaacat gtcgctgtcc 660
ctgttggaat tgtatttaag tcgaggttac aatgtgtcat ctatagtcac tatgacatcc 720
caggggaatgt acgggggaac ttacctagtgt gaaaagccta atctgagcag caaagggtca 780
gagttgtcac actgagcagt gcaccgagtg ttgaagtag gtgtatcag aaatccgggt 840
ttggggggctc cggatttcca tatgacaaac tatcttgagc aaccagtcag taatgatttc 900
agcaactgca tgggtggcttt gggggagctc aagttcgag ccctctgtca caggggaagat 960
tctatcacia ttccctatca gggatcaggg aaaggtgtca gcttcagct tgtcaagcta 1020
gggtgtctga aatccccaac cgacatgcaa tcttgggtcc cctatcaac ggatgatcca 1080
gtgatagaca ggctttacct ctcatctcac agaggcggtta tcgctgacaa tcaagcaaaa 1140
tgggctgtcc cgacaacacg gacagatgac aagttgcgaa tggagacatg ctccagcag 1200
gcgtgtaagg gtaaaatcca agcactttgc agaatcccag agtggacacc attgaaggat 1260
aacaggattc cttcatcagg ggtctgtctt gttgatctga gtctgacagt tagcttaaaa 1320
atcaaaattg tttcaggatt cgggcatgtg atcacacacg gttcagggat ggacctatac 1380
aaatccaacc acaacaatat gtattggctg actatcccg caatgaagaa cctggcctta 1440
gggttaatca acacattgga gtggatccg agattcaagg ttagtcccaa cctcttcaat 1500
gttccaatta tgggaagcagg caggactgc catgccccaa catacctacc tgcggaggtg 1560
gatggtgatg tcaaaactcag ttccaatctg gtgattctac ctggtaaga tctccaatat 1620
gttctggcaa cctacgatac ttccagagtt gaacatgctg tagtttatta cgtttacagc 1680
ccaaagccgt cattttctta cttttatcct tttaggttgc ctgtaagggg ggtcccat 1740
gaattacaag tggatgtctt cacatgggac caaaaaactc ggtgcccgtc cttctgtgtg 1800
cttgcggact cagaatctgg tggacatata actcactctg ggatgggtgg catgggagtc 1860
agctgcacag ccactcggga accatggaac agccgcagat agtgataata ggctggagcc 1920
tcggtggcca cacttcttgc ccttggggcc tccccccagc cctcctccc cttcctgcac 1980
ccgtaccccc gtgtctcttg aataaagtct gagtggggcg caaaaaaaaa aaaaaaaaaa 2040
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 2100
aaaaaaaaaa aaaaaaaaaa atctag 2126

```

```

SEQ ID NO: 81      moltype = RNA length = 1729
FEATURE           Location/Qualifiers
misc_feature       1..1729
                   note = Synthetic Polynucleotide
source            1..1729
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 81
tcaagctttt ggaccctcgt acagaagcta atacgactca ctatagggaa ataagagaga 60
aaagaagagt aagaagaaat ataagagcca ccatggcaca agtcattaat acaaacagcc 120
tgctcgtgtt gaccagaat aacctgaaca aatcccagtc cgcactgggc actgctatcg 180
agcgtttgtc ttccggctcg cgtatcaaca gcgcgaaaga cgatgcggca ggacaggcga 240
ttgctaaccg ttttacccg ccaatcaaag gtctgactca ggcttcccg aacgctaacc 300
acggtatctc cattgcgcag accactgaag gcgcgctgaa cgaaatcaac aacaacctgc 360
agcgtgtgag tgaactggcg gttcagttct cgaatggtag taactccag tctgacctcg 420
actccatcca ggctgaaatc acccagcgcc tgaacgaaat cgaccgtgta tccggccaga 480
ctcagttcaa cggcgtgaaa gtctcggcgc aggacaacac cctgaccatc cagggttggtg 540
ccaacgacgg tgaactatc gatattgatt taaaagaaat cagctctaaa acactgggac 600
ttgataagct taatgtccaa gatgcctaca cccgaaaga aactgctgta accgttgata 660
aaactaccta taaaaatggt acagatccta ttacagccca gagcaatact gatataccaa 720
ctgcaattgg cggtgggtgca acgggggtta ctggggctga tatcaaat 780
aatactattt agatgttaaa ggcggtgctt ctgctggtgt ttataaagcc acttatgatg 840
aaactacaaa gaaagttaat attgatacga ctgataaaac tccgttggca actgcggaag 900
ctacagctat tcgggggaac gccactataa cccacaacca aattgctgaa gtaacaaaag 960
aggggtgtga tacgaccaca gttgcggctc aacttgcctg agcaggggtt actggcgccg 1020
ataaggacaa tactagcctt gtaaaactat cgtttgagga taaaaacggt aagggtattg 1080
atggtggcta tgcagtgaat atggcgacg atttctatgc cgctacatat gatgagaaaa 1140
caggtgcaat tactgtcaaa accactactt atacagatgg tactggcggt gctcaactg 1200
gagctgtgaa atttgggtgc gcaaatggta aatctgaagt tgttactgct accgatggta 1260
agacttactt agcaagcgac ctgacaaaac ataacttcag aacaggcggt gagcttaaa 1320
aggtaatac agataagact gaaaacccac tgcagaaaaa tgatgctgcc ttggcacagg 1380
ttgatacact tcgttctgac ctgggtgcgg ttcagaaacc ttcaactcc gctatcacca 1440
acctgggcaa tacgtgaaat aacctgtctt ctgcccgtag ccgtatcgaa gattccgaat 1500
acgcaaccga agtctccaac atgtctcgcg cgcagattct gcagcaggcc ggtacctccg 1560
ttctggcgca ggcgaaccag gttccgcaaa acgtcctctc ttactgctg tgataatagg 1620
ctggagcctc ggtggccatg cttcttgcgc cttgggcctc ccccgagccc ctcctcccct 1680
tctgcacccc gtacccccgt ggtctttgaa taaagtctga gtggcgccg 1729

```

```

SEQ ID NO: 82      moltype = RNA length = 1518
FEATURE           Location/Qualifiers
misc_feature       1..1518
                   note = Synthetic Polynucleotide
source            1..1518

```


-continued

```

mol_type = other RNA
organism = synthetic construct

SEQUENCE: 82
atggcacaag tcattaatac aaacagcctg tcgctgttga cccagaataa cctgaacaaa 60
tcccagtcgc cactggggcac tgctatcgag cgtttgtctt ccggtctgcg tatcaacagc 120
gcgaagacgc atgcgggcagg acaggcgatt gctaaccgtt ttaccgcgaa catcaaaagt 180
ctgactcagg cttcccgtaa cgtaacgac ggtatctcca ttgcgcagac cactgaaggc 240
gcgctgaacg aaatcaacaa caacctgcag cgtgtgcgtg aactggcggt tcagtctgcg 300
aatggtacta actcccagtc tgacctcgac tccatccagg ctgaaatcac ccagcgctcg 360
aacgaaatcg accgtgtatc ggcccgagct cagttcaacg gcgtgaaagt cctggcgcgag 420
gacaaacacc tgacctcca ggttggtgcc aacgacggtg aaactatcga tattgattta 480
aaagaaatca gctctaaaac actgggactt gataagctta atgtccaaga tgccacacc 540
ccgaaagaaa ctgctgtaac cgttgataaa actacctata aaaatggtac agatcctatt 600
acagcccaga gcaatactga tatccaaact gcaattggcg gtggtgcaac ggggggttact 660
ggggctgata tcaaatttaa agatggtcaa tactatttag atgttaaagg cgggtgcttct 720
gctggtgttt ataaagccac ttatgatgaa actacaaaga aagttaatat tgatacgact 780
gataaaaact cgttggcaac tgcggaagct acagctattc ggggaacggc cactataacc 840
cacacccaaa ttgctgaagt aacaaaagag ggtgttgata cgaccacagt tgcggctcaa 900
cttgctgcag caggggttac tggcgccgat aaggacaata ctagccttgt aaaactatcg 960
tttgaggata aaaacggtta ggttattgat cagtgtatag cagtgaaaat gggcgacgat 1020
ttctatgcgc ctacatatga tgagaaaaca ggtgcaatta ctgtctaaac cactacttat 1080
acagatggtg ctggcggtgc tcaaaactgga gctgtgaaat ttggtggcgc aaatggtaaa 1140
tctgaagttg ttaactgctac cgaatggtta acttacttag caagcgacct tgacaaacat 1200
aacttcagaa caggcggtga gcttaaagag gttaatacag ataagactga aaacccactg 1260
cagaaaattg atgctgcctt ggcacaggtt gatacacctc gttctgaact ggggtcggtt 1320
cagaaccgtt tcaactccgc tatcaccacac ctgggcaata ccgtataaaa cctgtcttct 1380
gcccgtagcc gtatcgaaga ttccgactac gcaaccggaag tctccaacat gtctcgcgcg 1440
cagattctgc agcaggcccg tacctccgtt ctggcgcgagg cgaaccaggt tccgcaaaaac 1500
gtccctctct tactcgct

```

```

SEQ ID NO: 83      moltype = RNA length = 1790
FEATURE           Location/Qualifiers
misc_feature       1..1790
                   note = Synthetic Polynucleotide
source             1..1790
                   mol_type = other RNA
                   organism = synthetic construct

```

```

SEQUENCE: 83
ggggaataaa gagagaaaag aagagtaaga agaaatataa gagccaccat ggcacaagtc 60
attaatacaa acagcctgtc gctgttgacc cagaataacc tgaacaaatc ccagtcogca 120
ctgggcactg ctatcgagcg ttgtctcttc ggtctgcgta tcaacagcgc gaaagacgat 180
cgggcaggac aggcgattgc taaccgtttt accgcgaaca tcaaaaggtct gactcaggct 240
tcccgtaacg ctaacgcagg tatctccatt cgcgagacca ctgaaggcgc gctgaacgaa 300
atcaacaaca acctgcagcg tgtgcgtgaa ctggcggttc agtctgcgaa tggtaactaac 360
tcccagtcct acctcgactc catccaggct gaaatcaccc agcgccctgaa cgaatcgcac 420
cgtgtatccg gccagactca gttcaacggc gtgaaagtcc tggcgcgagg caacaccctg 480
accatccagg ttggtgccaa gcacgggtgaa actatcgata ttgatttaaa agaaatcagc 540
tctaaaacac tgggacttga taagcttaat gtccaagatg cctacacccc gaaagaaact 600
gctgtaaccg ttgataaaac tacctataaa aatgggtacag atcctattac agcccagagc 660
aatactgata tccaaactgc aattggcggt ggtgcaacgg ggggttactg ggctgatatc 720
aaatttaag atggtcaata ctatttagat gttaaaggcg gtgcttctgc tgggtgttat 780
aaagccactt atgatgaaac tacaaagaaa gttaatatgt atacgactga taaaactccg 840
ttggcaactg cggaaagtac agctattcgg ggaacggcca ctataaccca caaccaaatt 900
gctgaagtaa caaaagaggg ttgtgatacg accacagttg cggctcaact tgcgtcagca 960
gggggttact ggcgcgataa ggacaatact agccttgtaa aactatcgtt tgaggataaa 1020
aacggttaag ttattgatgg tggctatgca gtgaaaaatg gcgacgattt ctatgccgct 1080
acatatgatg agaaaacagg tgcaattact gctaaaacca ctacttatac agatggtact 1140
ggcgcttctc aaactggagc tgtgaaattt ggtggcgcaa atggtaaatc tgaagttggt 1200
actgctaccg atggtgaagc ttacttagca agcgaccttg acaaacataa cttcagaaca 1260
ggcggtgagc ttaagagggt taatacagat aagactgaaa acccactgca gaaaattgat 1320
gctgccttgg cacaggttga tacacttcgt tctgacctgg gtgcgggtca gaaccgtttc 1380
aactccgcta tcaccaacct gggcaatacc gtaataaacc tgtcttctgc ccgtagccgt 1440
atcgaagatt ccgactacgc aaccgaagtc tccaacatgt ctgcgcgcga gattctgcag 1500
caggccggtg cctccgttct ggcgcaggcg aaccagggtc cgcaaaacgt cctctcttta 1560
ctgcgttgat aataggctgg agcctcggtg gccatgcttc ttgccccttg ggccctcccc 1620
cagccctccc tccccttctc gcacccgtac ccccggtgct tttgaataaa gtctgagtg 1680
cgggcaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1740
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaatctag

```

```

SEQ ID NO: 84      moltype = AA length = 13
FEATURE           Location/Qualifiers
source             1..13
                   mol_type = protein
                   note = Salmonella typhimurium
                   organism = unidentified

```

```

SEQUENCE: 84
LQVRRELAVQ SAN

```

13

```

SEQ ID NO: 85      moltype = AA length = 539

```

-continued

```

FEATURE          Location/Qualifiers
REGION           1..539
                 note = Synthetic Polypeptide
source           1..539
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 85
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAICKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAF AVRELKDFVS KNLTRALNKN 180
KCDIDDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLIC GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEQDFNVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 86      moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION           1..539
                 note = Synthetic Polypeptide
source           1..539
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 86
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAICKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAF AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLIC GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEHQWHVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 87      moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION           1..539
                 note = Synthetic Polypeptide
source           1..539
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 87
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLLKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAF AVRELKDFVS KNLTRAINKN 180
KCDIPDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEQDFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 88      moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION           1..539
                 note = Synthetic Polypeptide
source           1..539
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 88
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLLKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAF AVRELKDFVS KNLTRAINKN 180
KCDIPDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEQDFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 89      moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION           1..539
                 note = Synthetic Polypeptide
source           1..539
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 89

```

-continued

```

MSWKVVIIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLLKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVL KNLTRAINKN 180
KCDIPDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGILI GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVPCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELGSV TNGGFIPHN 539

```

```

SEQ ID NO: 90      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 90
MSWKVVIIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLLKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVL KNLTRAINKN 180
KCDIPDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGILI GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVPCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELGSV TNGGFIPHN 539

```

```

SEQ ID NO: 91      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 91
MSWKVVIIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL PVGDVENLTC 60
SDGPSLIKTE LDLLKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGILI GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVPCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELGSV TNGGFIPHN 539

```

```

SEQ ID NO: 92      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 92
MSWKVVIIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL PVGDVENLTC 60
SDGPSLIKTE LDLLKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGILI GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVPCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK FPEQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP TGAPPELGSV TNGGFIPHN 539

```

```

SEQ ID NO: 93      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 93
MSWKVVIIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLLKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGILI GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETRG DHVPCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420

```

-continued

DNTVYQLSKV	EGEQHVIGR	PVSSSFDPK	FPEDQFQVAL	DQVFENIENS	QALVDQSNRI	480
LSSAEKGNTG	FIIIVILIAV	LGSSMILVSI	FIIIKTKPK	TGAPPELGSV	TNNGFIPHN	539

```

SEQ ID NO: 94      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 94
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDLENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKPK  TGAPPELGSV TNNGFIPHN 539

```

```

SEQ ID NO: 95      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 95
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVL KNLTRAINKN 180
KCDIDDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKPK  TGAPPELGSV TNNGFIPHN 539

```

```

SEQ ID NO: 96      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 96
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLWRAINKN 180
KCDIDDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKPK  TGAPPELGSV TNNGFIPHN 539

```

```

SEQ ID NO: 97      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 97
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDLENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVL KNLWRAINKN 180
KCDIDDLKMA VSFSQFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKLMLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQAGSTVYY PNEKDCETRG DHVFCDTAAG INVAEQSKC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKPK  TGAPPELGSV TNNGFIPHN 539

```

```

SEQ ID NO: 98      moltype = AA  length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide

```

-continued

```

source          1..539
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 98
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL PVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETR  DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR  PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP  TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 99      moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 99
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIPDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETR  DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR  PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP  TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 100     moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 100
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCPIDDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETR  DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR  PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP  TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 101     moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 101
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180
KCDIDDLKMA VSFQSFNRRF LNVVRQFSDN AGITPAISLD LMTDAELARA VPNMPTSAGQ 240
IKMLLENRAM VRRKGFGLI  GVGSSVIYM VQLPIFGVID TPCWIVKAAP SCSEKKGNYA 300
CLLREDQGWY CQNAGSTVYY PNEKDCETR  DHVFCDTAAG INVAEQSKEC NINISTTNYP 360
CKVSTGRHPI SMVALSPLGA LVACYKGVSC SIGSNRVGII KQLNKGCSYI TNQDADTVTI 420
DNTVYQLSKV EGEQHVIGR  PVSSSFDPK  FPEDQFQVAL DQVFENIENS QALVDQSNRI 480
LSSAEKGNTG FIIIVILIAV LGSSMILVSI FIIIKTKKP  TGAPPELSCGV TNNGFIPHN 539

SEQ ID NO: 102     moltype = AA length = 539
FEATURE           Location/Qualifiers
REGION            1..539
                  note = Synthetic Polypeptide
source            1..539
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 102
MSWKVVIIFS LLITPQHGLK ESYLEESCST ITEGYLSVLR TGWYTNVFTL EVGDVENLTC 60
SDGPSLIKTE LDLTKSALRE LKTVSADQLA REEQIENPGS GSFVLGAIAL GVAAAAAVTA 120
GVAIAKTIRL ESEVTAINNA LKKTNEAVST LGNGVRVLAT AVRELKDFVS KNLTRAINKN 180

```

-continued

KCDIDDLKMA	VSFSQFNRRF	LNVRQFSDN	AGITPAISLD	LMTDAELARA	VPNMPTSAGQ	240
IKLMLENRAM	VRKGFGLI	GVYSSVIYM	VQLPIFGVID	TPCWIVKAAP	SCSEKKGNYA	300
CLLREDQGWY	CQNAGSTVYY	PNEKDCETRG	DHVPCDTAAG	INVAEQSKEC	NINISTTNYP	360
CKVSTGRHPI	SMVALSPLGA	LVACYKGVSC	SIGSNRVGII	KQLNKGCSYI	TNQDADTVTI	420
DNTVYQLSKV	EGEQHVIGKR	PVSSSFPIK	FPEDQFQVAL	DQVFENIENS	QALVDQSNRI	480
LSSAEKGNTG	FIIIVILIAV	LGSSMILVSI	FIIIKTKKP	TGAPPELGSV	TNNGFIPHN	539

SEQ ID NO: 103 moltype = AA length = 539
 FEATURE Location/Qualifiers
 REGION 1..539
 note = Synthetic Polypeptide
 source 1..539
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 103

MSWKVVIIIFS	LLITPQHGLK	ESYLEESCST	ITEGYLSVLR	TGWYTNVFTL	EVGDVENLTC	60
SDGPSLIKTE	LDLTKSALRE	LKTVSADQLA	REEQIENPGS	GSFVLGAIAL	GVAATAVTA	120
GVAIAKTIRL	ESEVTAINNA	LKKTNEAVST	LGNGVRVLAT	AVRELKDFVS	KNLTRAINKN	180
KCDIDDLKMA	VSFSQFNRRF	LNVRQFSDN	AGITPAISLD	LMTDAELARA	VPNMPTSAGQ	240
IKLMLENRAM	VRKGFGLI	GVYSSVIYM	VQLPIFGVID	TPCWIVKAAP	SCSEKKGNYA	300
CLLREDQGWY	CQNAGSTVYY	PNEKDCETRG	DHVPCDTAAG	INVAEQSKEC	NINISTTNYP	360
CKVSTGRHPI	SMVALSPLGA	LVACYKGVSC	SIGSNRVGII	KQLNKGCSYI	TNQDADTVTI	420
DNTVYQLSKV	EGEQHVIGKR	PVSSSFPIK	FPEDQFQVAL	DQVFENIENS	QALVDQSNRI	480
LSSAEKGNTG	FIIIVILIAV	LGSSMILVSI	FIIIKTKKP	TGAPPELGSV	TNNGFIPHN	539

SEQ ID NO: 104 moltype = AA length = 539
 FEATURE Location/Qualifiers
 REGION 1..539
 note = Synthetic Polypeptide
 source 1..539
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 104

MSWKVVIIIFS	LLITPQHGLK	ESYLEESCST	ITEGYLSVLR	TGWYTNVFTL	EVGDVENLTC	60
SDGPSLIKTE	LDLTKSALRE	LKTVSADQLA	REEQIENPGS	GSFVLGAIAL	GVAATAVTA	120
GVAIAKTIRL	ESEVTAINNA	LKKTNEAVST	LGNGVRVLAT	AVRELKDFVS	KNLTRAINKN	180
KCDIDDLKMA	VSFSQFNRRF	LNVRQFSDN	AGITPAISLD	LMTDAELARA	VPNMPTSAGQ	240
IKLMLENRAM	VRKGFGLI	GVYSSVIYM	VQLPIFGVID	TPCWIVKAAP	SCSEKKGNYA	300
CLLREDQGWY	CQNAGSTVYY	PNEKDCETRG	DHVPCDTAAG	INVAEQSKEC	NINISTTNYP	360
CKVSTGRHPI	SMVALSPLGA	LVACYKGVSC	SIGSNRVGII	KQLNKGCSYI	TNQDADTVTI	420
DNTVYQLSKV	EGEQHVIGKR	PVSSSFPIK	FPEDQFQVAL	DQVFENIENS	QALVDQSNRI	480
LSSAEKGNTG	FIIIVILIAV	LGSSMILVSI	FIIIKTKKP	TGAPPELGSV	TNNGFIPHN	539

SEQ ID NO: 105 moltype = AA length = 539
 FEATURE Location/Qualifiers
 REGION 1..539
 note = Synthetic Polypeptide
 source 1..539
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 105

MSWKVVIIIFS	LLITPQHGLK	ESYLEESCST	ITEGYLSVLR	TGWYTNVFTL	EVGDVENLTC	60
SDGPSLIKTE	LDLTKSALRE	LKTVSADQLA	REEQIENPGS	GSFVLGAIAL	GVAATAVTA	120
GVAIAKTIRL	ESEVTAINNA	LKKTNEAVST	LGNGVRVLAT	AVRELKDFVS	KNLTRAINKN	180
KCDIDDLKMA	VSFSQFNRRF	LNVRQFSDN	AGITPAISLD	LMTDAELARA	VPNMPTSAGQ	240
IKLMLENRAM	VRKGFGLI	GVYSSVIYM	VQLPIFGVID	TPCWIVKAAP	SCSEKKGNYA	300
CLLREDQGWY	CQNAGSTVYY	PNEKDCETRG	DHVPCDTAAG	INVAEQSKEC	NINISTTNYP	360
CKVSTGRHPI	SMVALSPLGA	LVACYKGVSC	SIGSNRVGII	KQLNKGCSYI	TNQDADTVTI	420
DNTVYQLSKV	EGEQHVIGKR	PVSSSFPIK	FPEDQFQVAL	DQVFENIENS	QALVDQSNRI	480
LSSAEKGNTG	FIIIVILIAV	LGSSMILVSI	FIIIKTKKP	TGAPPELGSV	TNNGFIPHN	539

SEQ ID NO: 106 moltype = DNA length = 1617
 FEATURE Location/Qualifiers
 misc_feature 1..1617
 note = Synthetic Polynucleotide
 source 1..1617
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 106

atgagctgga	agggtgcat	catcttcagc	ctgctgatca	cacctcagca	cggcctgaaa	60
gagagctacc	tggaagagtc	ctgcagcacc	atcacagagg	gctacctgtc	tgtgctgaga	120
accggctggt	acaccaacgt	gttcacactg	gaagtgggag	acgtcgagaa	tctgacatgc	180
tctgatggcc	ctagccctgat	caagaccgag	ctggatctga	ccaagagcgc	cctgagagaa	240
ctcaagaccg	tgtctggcga	ctagctggcc	agagaggaa	agatcgagaa	tcctggcagc	300
ggcagctttg	tgtctgggagc	cattgtctct	ggagtggctg	ctgctgcagc	tggtacagca	360
ggcgtggcca	tctgcaagac	catcagactg	gaaagcgaa	tgaccgcat	caacaacgcc	420
ctgaagaaga	caaacgaggc	cgctcagaca	ctcggaatg	gcgttagagt	gctggccttt	480
gccgtgctgc	agctgaagga	cttcgtgtcc	aagaacctga	cacgggacct	gaacaagaac	540
aagtgcgaca	tcgacgacct	gaagatggcc	gtgtctctta	gccagttcaa	ccggcggttt	600

-continued

```

ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgcgggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgtgt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgctgtctga gagaggacca aggtcgggat tgtcagaacg ccggcagcac cgtgtactac 960
cttaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctggggacc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa agcagctgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcaa cgtggccctg 1380
gaccaggtgt tcgagaacat ccgaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccattgacat ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 107      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 107
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tctgcaagac catcagactg gaaagcgaa tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tcgggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgcgggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgtgt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgctgtctga gagaggacca aggtcgggat tgtcagaacg ccggcagcac cgtgtactac 960
cttaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctggggacc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa agcagctgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagc accagtggca tgtggccctg 1380
gaccaggtgt tcgagaacat ccgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccattgacat ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 108      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 108
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaa tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcctgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tcgggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgcgggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgctgtctga gagaggacca aggtcgggat tgtcagaacg ccggcagcac cgtgtactac 960
cttaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020

```

-continued

```

atcaatgtgg cccagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcaccctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcggca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtct cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaagggt gaaggcgaa acacagtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagt accaacaatg gttcatccc tcacaac 1617

```

```

SEQ ID NO: 109      moltype = DNA length = 1617
FEATURE
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 109
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa acatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccggcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcgggtt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgcctgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctctgctg gagaggacca aggtgggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgcgctgga 1020
atcaatgtgg ccagcagagc caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcaccctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcggca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtct cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaagggt gaaggcgaa acacagtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgaga accagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagt accaacaatg gttcatccc tcacaac 1617

```

```

SEQ ID NO: 110      moltype = DNA length = 1617
FEATURE
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 110
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa acatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccggcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgctt aagaacctga caccggccat taacaagaac 540
aagtgcgaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcgggtt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgcctgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctctgctg gagaggacca aggtgggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgcgctgga 1020
atcaatgtgg ccagcagagc caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcaccctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcggca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtct cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaagggt gaaggcgaa acacagtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440

```


-continued

```

ctgtctagcg ccgagaaggg aaacaccggc ttcacatcg tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 111      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                     note = Synthetic Polynucleotide
source               1..1617
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 111
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg gaagtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctgactgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga cttcgtgctt aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgccggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgag aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaagg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgcccagac cgtgaccatc 1260
gacaacaccg tgatcagct gagcaaggtg gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgaga accagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcg tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 112      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                     note = Synthetic Polynucleotide
source               1..1617
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 112
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg cctgtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctgactgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgccggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgag aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaagg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgcccagac cgtgaccatc 1260
gacaacaccg tgatcagct gagcaaggtg gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcg tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 113      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617

```

-continued

```

note = Synthetic Polynucleotide
source      1..1617
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 113
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtctggt acaccaacgt gttcacactg cctgtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgctgcccga tcaagtggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctcgccaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga cttcgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccatcac tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtctggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggaactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatgggtg ctctgtctcc tctggggacc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgcccatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa acgacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttcctgaga accagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccattgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accgcgctc ctccagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

SEQ ID NO: 114      moltype = DNA   length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                    note = Synthetic Polynucleotide
source               1..1617
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 114
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtctggt acaccaacgt gttcacactg gaagtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgctgcccga tcaagtggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctcgccaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga cttcgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccatcac tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtctggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggaactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatgggtg ctctgtctcc tctggggacc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgcccatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa acgacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttcctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccattgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accgcgctc ctccagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

SEQ ID NO: 115      moltype = DNA   length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                    note = Synthetic Polynucleotide
source               1..1617
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 115
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120

```

-continued

```

accggtggtg acaccaacgt gttcacactg gaagtgggag acctcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgetgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgcggccag 720
atcaagctga tgcctgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgccctgctg gagaggacca aggcctggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgca gacaagaggg gaccacgtgt tctgtgatac cgcgctgga 1020
atcaatgtgg ccagcagagc caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcacctatt tctatgggtg ctctgtctcc tctggggagc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgcccagac cgtgaccatc 1260
gacaacaccc tgatcacgtg gagcaagggt gaaggcgaa acgacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacct cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgg tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 116      moltype = DNA length = 1617
FEATURE
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 116
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggaaagatc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtggtg acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgetgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgcggccag 720
atcaagctga tgcctgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgccctgctg gagaggacca aggcctggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgca gacaagaggg gaccacgtgt tctgtgatac cgcgctgga 1020
atcaatgtgg ccagcagagc caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcacctatt tctatgggtg ctctgtctcc tctggggagc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgcccagac cgtgaccatc 1260
gacaacaccc tgatcacgtg gagcaagggt gaaggcgaa acgacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacct cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgg tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 117      moltype = DNA length = 1617
FEATURE
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 117
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggaaagatc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtggtg acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgetgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctgt ggccggccat taacaagaac 540

```

-continued

```

aagtgcgaca tgcagcagct gaagatggcc gtgtccttta gccagttcaa ccggcgggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccctacac tgcgggccag 720
atcaagctga tgcctgagaa tagagccatg gtccgacgga aaggtctcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtgaa ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcacc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa acgacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctcttg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 118      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                     note = Synthetic Polynucleotide
source               1..1617
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 118
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagatc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg aaagtgggag acctcgagaa tctgacatgc 180
tctgatggcc ctgactgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgcga tcaagtgccc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgcctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaa tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgctt aagaacctgt ggcgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcgggtt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccctacac tgcgggccag 720
atcaagctga tgcctgagaa tagagccatg gtccgacgga aaggtctcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtgaa ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcacc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa acgacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctcttg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 119      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                     note = Synthetic Polynucleotide
source               1..1617
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 119
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagatc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg cctgtgggag acctcgagaa tctgacatgc 180
tctgatggcc ctgactgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgcga tcaagtgccc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgcctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaa tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcgggtt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccctacac tgcgggccag 720
atcaagctga tgcctgagaa tagagccatg gtccgacgga aaggtctcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtgaa ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcggtat tgtcagaacg ccggcagcac cgtgtactac 960

```

-continued

```

cctaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccgcgag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctggtggcctt gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcate 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgatcacgt gagcaaggtg gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccattgacct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accgcgctc ctcagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 120      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 120
atgagctgga aggtgggtcat catcttcagc ctgctgatca caccctcagca cggcctgaaa 60
gagagctacc tggaaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtggtt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctggggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaa tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cctcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga cttcgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaa gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgacctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctctgctg gagaggacca aggtctggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccgcgag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctggtggcctt gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcate 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgatcacgt gagcaaggtg gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccattgacct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accgcgctc ctcagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 121      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 121
atgagctgga aggtgggtcat catcttcagc ctgctgatca caccctcagca cggcctgaaa 60
gagagctacc tggaaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtggtt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctggggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaa tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cctcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga cttcgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcccta tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaa gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgacctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctctgctg gagaggacca aggtctggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccgcgag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctggtggcctt gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcate 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgatcacgt gagcaaggtg gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctgagg atcagttcca ggtggccctg 1380

```

-continued

```

gaccaggtgt tgcagaacat cgagaattcc caggctcttg tggaccagtc caacagaate 1440
ctgtctagcg ccgagaaggg aaacacccgc ttcacatcgc tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagt accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 122      moltype = DNA length = 1617
FEATURE
misc_feature        Location/Qualifiers
                     1..1617
                     note = Synthetic Polynucleotide
source              1..1617
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 122
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgctctt ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg cctagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctcggcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga ctctgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa cggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgcctcgaaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acacctgctc ggattgtgaa ggcgcctcct agctgtagcg agaagaagg caattacgcc 900
tgctctgtga gagaggacca aggtcggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgca gacaagaggg gaccacgtgt tctgtgatac cgcgcgtgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacccgcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaagggt gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcca cctatcaag ttccttgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctcttg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagt accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 123      moltype = DNA length = 1617
FEATURE
misc_feature        Location/Qualifiers
                     1..1617
                     note = Synthetic Polynucleotide
source              1..1617
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 123
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgctctt ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctcggcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga ctctgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa cggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgcctcgaaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acacctgctc ggattgtgaa ggcgcctcct agctgtagcg agaagaagg caattacgcc 900
tgctctgtga gagaggacca aggtcggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgca gacaagaggg gaccacgtgt tctgtgatac cgcgcgtgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacccgcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaagggt gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttccc acctatcaag ttccttgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctcttg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagt accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 124      moltype = DNA length = 1617
FEATURE
Location/Qualifiers

```

-continued

```

misc_feature      1..1617
                  note = Synthetic Polynucleotide
source            1..1617
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 124
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggagagatc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtctgtt acaccaacgt gtccacactg gaagtgggagc acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgcgga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga ctcgtgtccc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tcgccgccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtaa ggcgcctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacaa aggtcggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgcgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaacggcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttcctcagg accagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgtacct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

SEQ ID NO: 125      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                    note = Synthetic Polynucleotide
source               1..1617
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 125
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggagagatc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtctgtt acaccaacgt gtccacactg gaagtgggagc acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgcgga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga ctcgtgtccc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tcgccgccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtaa ggcgcctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacaa aggtcggtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgcgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaacggcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttcctcagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgtacct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

SEQ ID NO: 126      moltype = DNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                    note = Synthetic Polynucleotide
source               1..1617
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 126
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60

```

-continued

```

gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg gaagtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tctgcaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagtggaa ccggcgggtt 600
ctgaacgtcg tgcggcagtt tagcgacaa cccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccctacac tgcgggccag 720
atcaagctga tgcctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgtat 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgggtat tgtcagaacg ccggcagcac cgtgtactac 960
cttaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgtgga 1020
atcaatgtgg ccgagcagtg caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatgggtg ctctgtctcc tctggggacc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaaggctg cagctacatc accaaccagg acgcccatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa agcagctgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag tccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 127      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 127
atgagctgga aggtgggtcat catcttcagc ctgctgatca cactcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg gaagtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tctgcaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccttt 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga caccggccat gaacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcgggtt 600
ctgaacgtcg tgcggcagtt tagcgacaa cccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccctacac tgcgggccag 720
atcaagctga tgcctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgtgt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgggtat tgtcagaacg ccggcagcac cgtgtactac 960
cttaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgtgga 1020
atcaatgtgg ccgagcagtg caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatgggtg ctctgtctcc tctggggacc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaaggctg cagctacatc accaaccagg acgcccatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa agcagctgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag tccctgagg atcagttcaa cgtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 128      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 128
atgagctgga aggtgggtcat catcttcagc ctgctgatca cactcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg gaagtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tctgcaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480

```


-continued

```

gccgtgcgcg agctgaagga ctcgtgtgcc aagaacctga cacggggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgtgt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgccgtgctga gagaggacca aggctgggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgca gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaagggtg ccacggcgag gcacctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgatcagct gagcaagggt gaaggcgaa acacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagc accagtggca tgtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcctcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 129      moltype = RNA length = 1617
FEATURE
misc_feature        1..1617
                     note = Synthetic Polynucleotide
source              1..1617
                     mol_type = other RNA
                     organism = synthetic construct

```

```

SEQUENCE: 129
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggaaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg aaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa agatcgagaa tcttgccagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tgcctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctccggaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctga cacggggccat taacaagaac 540
aagtgcgaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggcgctcct agctgtagcg agaagaaggg caattacgcc 900
tgccgtgctga gagaggacca aggctgggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgca gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaagggtg ccacggcgag gcacctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgatcagct gagcaagggt gaaggcgaa acacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagc atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcctcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 130      moltype = RNA length = 1617
FEATURE
misc_feature        1..1617
                     note = Synthetic Polynucleotide
source              1..1617
                     mol_type = other RNA
                     organism = synthetic construct

```

```

SEQUENCE: 130
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tggaaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg aaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa agatcgagaa tcttgccagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tgcctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctccggaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctga cacggggccat taacaagaac 540
aagtgcgaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtga ggcgctcct agctgtagcg agaagaaggg caattacgcc 900

```

-continued

tgcctgctga	gagaggacca	aggetggtat	tgtcagaacg	cggcgagcac	cgtgtactac	960
cctaacgaga	aggactgcga	gacaagaggc	gaccacgtgt	tctgtgatac	cgccgctgga	1020
atcaatgttg	ccgagcagag	caaagagtgc	aacatcaaca	tcagcaccac	caactatccc	1080
tgcaaggtgt	ccaccggcag	gcaccctatt	tctatggttg	ctctgtctcc	tctgggagcc	1140
ctgggtggctt	gttataaggg	cgtgtcctgt	agcatcgcca	gcaacagagt	gggcatcatc	1200
aagcagctga	acaagggctg	cagctacatc	accaaccagg	acgccgatac	cgtgaccatc	1260
gacaacaccg	tgtatcagct	gagcaaggtg	gaaggcgaa	agcacgtgat	caagggcaga	1320
cctgtgtcca	gcagcttcga	ccctatcaag	ttccctgaga	accagttcca	ggtggccctg	1380
gaccaggtgt	tcgagaacat	cgagaattcc	caggctcttg	tggaaccagtc	caacagaatc	1440
ctgtctagcg	ccgagaaggg	aaacaccggc	ttcatcatcg	tgatcatcct	gatcgccgtg	1500
ctgggcagct	ccatgatcct	ggtgtccatc	ttcatcatta	tcaagaagac	caagaagccc	1560
accggcgctc	ctccagaact	gagcggagtg	accaacaatg	gcttcatccc	tcacaac	1617

SEQ ID NO: 131 moltype = RNA length = 1617
 FEATURE Location/Qualifiers
 misc_feature 1..1617
 note = Synthetic Polynucleotide
 source 1..1617
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 131

atgagctgga	agggtggtcat	catcttcagc	ctgctgatca	cacctcagca	cgccctgaaa	60
gagagctacc	tggaagagtc	ctgcagcacc	atcacagagg	gctacctgtc	tgtgctgaga	120
accggctggt	acaccaacgt	gttcacactg	gaagtgggcg	acgtcgagaa	tctgacatgc	180
tctgatggcc	ctagcctgat	caagaccgag	ctggatctgc	tcaagagcgc	cctgagagaa	240
ctcaagaccg	tgtctgcga	tcagctggcc	agagaggaac	agatcgagaa	tcctggcagc	300
ggcagctttg	tgtctgggagc	cattgtctctt	ggagtggctg	ctgctgcagc	tgttacagca	360
ggcgtggcca	tcgctaagac	catcagactg	gaaagcgaa	tgaccgccat	caacaacgcc	420
ctgaagaaga	caaacgaggg	cgtcagcaca	ctcggcaatg	gcgttagagt	gctggccaca	480
gccgtgcgcg	agctgaagga	cttcgtgctt	aagaacctga	cacgggccat	taacaagaac	540
aagtcgcgaca	tcctcgacct	gaagatggcc	gtgtccttta	gccagttcaa	ccggcggttt	600
ctgaacgtcg	tgcggcagtt	tagcgacaac	gccggaatca	caccagccat	cagcctggac	660
ctgatgacag	atgctgagct	ggctagagcc	gtgcctaaca	tgctacatc	tgccggccag	720
atcaagctga	tgctcgagaa	tagagccatg	gtccgacgga	aaggcttcgg	cattctgatt	780
ggcgtgtacg	gcagcagcgt	gatctatatg	gtgcagctgc	ctatcttcgg	cgtgatcgac	840
acaccctgct	ggatttgtgaa	ggccgctcct	agctgtagcg	agaagaaggg	caattacgcc	900
tgcctgctga	gagaggacca	aggctggtat	tgtcagaacg	cggcgagcac	cgtgtactac	960
cctaacgaga	aggactgcga	gacaagaggc	gaccacgtgt	tctgtgatac	cgccgctgga	1020
atcaatgttg	ccgagcagag	caaagagtgc	aacatcaaca	tcagcaccac	caactatccc	1080
tgcaaggtgt	ccaccggcag	gcaccctatt	tctatggttg	ctctgtctcc	tctgggagcc	1140
ctgggtggctt	gttataaggg	cgtgtcctgt	agcatcgcca	gcaacagagt	gggcatcatc	1200
aagcagctga	acaagggctg	cagctacatc	accaaccagg	acgccgatac	cgtgaccatc	1260
gacaacaccg	tgtatcagct	gagcaaggtg	gaaggcgaa	agcacgtgat	caagggcaga	1320
cctgtgtcca	gcagcttcga	ccctatcaag	ttccctgagg	atcagttcca	ggtggccctg	1380
gaccaggtgt	tcgagaacat	cgagaattcc	caggctcttg	tggaaccagtc	caacagaatc	1440
ctgtctagcg	ccgagaaggg	aaacaccggc	ttcatcatcg	tgatcatcct	gatcgccgtg	1500
ctgggcagct	ccatgatcct	ggtgtccatc	ttcatcatta	tcaagaagac	caagaagccc	1560
accggcgctc	ctccagaact	gagcggagtg	accaacaatg	gcttcatccc	tcacaac	1617

SEQ ID NO: 132 moltype = RNA length = 1617
 FEATURE Location/Qualifiers
 misc_feature 1..1617
 note = Synthetic Polynucleotide
 source 1..1617
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 132

atgagctgga	agggtggtcat	catcttcagc	ctgctgatca	cacctcagca	cgccctgaaa	60
gagagctacc	tggaagagtc	ctgcagcacc	atcacagagg	gctacctgtc	tgtgctgaga	120
accggctggt	acaccaacgt	gttcacactg	gaagtgggcg	acgtcgagaa	tctgacatgc	180
tctgatggcc	ctagcctgat	caagaccgag	ctggatctgc	tcaagagcgc	cctgagagaa	240
ctcaagaccg	tgtctgcga	tcagctggcc	agagaggaac	agatcgagaa	tcctggcagc	300
ggcagctttg	tgtctgggagc	cattgtctctt	ggagtggctg	ctgctgcagc	tgttacagca	360
ggcgtggcca	tcgctaagac	catcagactg	gaaagcgaa	tgaccgccat	caacaacgcc	420
ctgaagaaga	caaacgaggg	cgtcagcaca	ctcggcaatg	gcgttagagt	gctggccaca	480
gccgtgcgcg	agctgaagga	cttcgtgctt	aagaacctga	cacgggccat	taacaagaac	540
aagtcgcgaca	tcctcgacct	gaagatggcc	gtgtccttta	gccagttcaa	ccggcggttt	600
ctgaacgtcg	tgcggcagtt	tagcgacaac	gccggaatca	caccagccat	cagcctggac	660
ctgatgacag	atgctgagct	ggctagagcc	gtgcctaaca	tgctacatc	tgccggccag	720
atcaagctga	tgctcgagaa	tagagccatg	gtccgacgga	aaggcttcgg	cattctgatt	780
ggcgtgtacg	gcagcagcgt	gatctatatg	gtgcagctgc	ctatcttcgg	cgtgatcgac	840
acaccctgct	ggatttgtgaa	ggccgctcct	agctgtagcg	agaagaaggg	caattacgcc	900
tgcctgctga	gagaggacca	aggctggtat	tgtcagaacg	cggcgagcac	cgtgtactac	960
cctaacgaga	aggactgcga	gacaagaggc	gaccacgtgt	tctgtgatac	cgccgctgga	1020
atcaatgttg	ccgagcagag	caaagagtgc	aacatcaaca	tcagcaccac	caactatccc	1080
tgcaaggtgt	ccaccggcag	gcaccctatt	tctatggttg	ctctgtctcc	tctgggagcc	1140
ctgggtggctt	gttataaggg	cgtgtcctgt	agcatcgcca	gcaacagagt	gggcatcatc	1200
aagcagctga	acaagggctg	cagctacatc	accaaccagg	acgccgatac	cgtgaccatc	1260
gacaacaccg	tgtatcagct	gagcaaggtg	gaaggcgaa	agcacgtgat	caagggcaga	1320

-continued

```

cctgtgtcca gcagcttcga cctcatcaag ttccttgaga accagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 133      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 133
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg cctgtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgagggc cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgacctacac tgcgggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtgaa ggcgctcctc agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggaactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagtc caaagagtcg aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttcctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 134      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 134
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gttcacactg cctgtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgagggc cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgag agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgacctacac tgcgggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtgaa ggcgctcctc agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggaactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagtc caaagagtcg aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccttgaga accagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 135      moltype = RNA length = 1617

```

-continued

```

FEATURE                               Location/Qualifiers
misc_feature                           1..1617
                                         note = Synthetic Polynucleotide
source                                1..1617
                                         mol_type = other RNA
                                         organism = synthetic construct

SEQUENCE: 135
atgagctgga aggtggctcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa agatcgagaa tcttggcagc 300
ggcagctttg tgcctgggagc cattgctctt ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctccggaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga ctctgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa cggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgcagc atgctgagct ggctagagcc gtgcctaaca tgcctacatc tgcggccag 720
atcaagctga tgcctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacc gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acacctgctc ggatttgtaa ggcgctcctt agctgtagcg agaagaaggc caattacgcc 900
tgctgtctga gagaggacca aggcctggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccgagc gcaacctatt tctatgggtg ctctgtcttc tctgggagcc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgg tgatcctcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 136                       moltype = RNA length = 1617
FEATURE                               Location/Qualifiers
misc_feature                           1..1617
                                         note = Synthetic Polynucleotide
source                                1..1617
                                         mol_type = other RNA
                                         organism = synthetic construct

SEQUENCE: 136
atgagctgga aggtggctcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggctggt acaccaacgt gtccacactg gaagtgggag acctcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa agatcgagaa tcttggcagc 300
ggcagctttg tgcctgggagc cattgctctt ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggc cgtcagcaca ctccggaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga ctctgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa cggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgcagc atgctgagct ggctagagcc gtgcctaaca tgcctacatc tgcggccag 720
atcaagctga tgcctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacc gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acacctgctc ggatttgtaa ggcgctcctt agctgtagcg agaagaaggc caattacgcc 900
tgctgtctga gagaggacca aggcctggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactgcga gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccgagc gcaacctatt tctatgggtg ctctgtcttc tctgggagcc 1140
ctggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgg tgatcctcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 137                       moltype = RNA length = 1617
FEATURE                               Location/Qualifiers
misc_feature                           1..1617
                                         note = Synthetic Polynucleotide
source                                1..1617
                                         mol_type = other RNA
                                         organism = synthetic construct

SEQUENCE: 137

```

-continued

```

atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtctgtt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgtctgggagc cattgtctctt ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgcggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggtctcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtgaa ggcgctcctt agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcacc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgcatacagt gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctcttg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 138      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                     note = Synthetic Polynucleotide
source               1..1617
                     mol_type = other RNA
                     organism = synthetic construct

```

```

SEQUENCE: 138
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtctgtt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgtctgggagc cattgtctctt ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctgt ggcgggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgcggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggtctcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggatttgtgaa ggcgctcctt agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgtat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcacctatt tctatggtgg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcacc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgcatacagt gagcaaggtg gaaggcgaac agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctcttg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcactct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcacccc tcacaac 1617

```

```

SEQ ID NO: 139      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature         1..1617
                     note = Synthetic Polynucleotide
source               1..1617
                     mol_type = other RNA
                     organism = synthetic construct

```

```

SEQUENCE: 139
atgagctgga aggtgggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtctgtt acaccaacgt gtccacactg gaagtgggag acctcgagaa tctgacatgc 180
tctgatggcc ctgacctgat caagaccgag ctggatctgc tcaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcttggcagc 300
ggcagctttg tgtctgggagc cattgtctctt ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420

```

-continued

```

ctgaagaaga caaacgaggg cgtcagcaca ctccggcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgctt aagaacctgt ggccggccat taacaagaac 540
aagtgcgcaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaa cccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cttaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggtg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttcctgagg atcagttcca ggtggccctg 1380
gaccaggtgt ccgagaattc caggtcttgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccattgatct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 140      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 140
atgagctgga aggtgggtcat catcttcagc ctgctgatca caccacgca cggcctgaaa 60
gagagctacc tggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accgctggtt acaccaacgt gttcacactg cctgtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccggcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgcaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cttaacgaga aggactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatggttg ctctgtctcc tctgggagcc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgga gcaacagagt gggcatcatc 1200
aagcagctga acaagggtg cagctacatc accaaccagg acgccgatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaa agcacgtgat caagggcaga 1320
cctgtgtcca tcagcttcga ccctatcaag ttcctgagg atcagttcca ggtggccctg 1380
gaccaggtgt ccgagaacat cgagaattcc caggtcttgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccattgatct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctcagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 141      moltype = RNA length = 1617
FEATURE             Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 141
atgagctgga aggtgggtcat catcttcagc ctgctgatca caccacgca cggcctgaaa 60
gagagctacc tggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accgctggtt acaccaacgt gttcacactg gaagtgggcg acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccggcaatg gcgttagagt gctggccaca 480
gccgtgcgcg agctgaagga cttcgtgtcc aagaacctga cacgggccat taacaagaac 540
aagtgcgcaca tccctgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840

```

-continued

```

acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgccctgctga gagaggacca aggcctggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgga gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcacctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcggca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaagggt gaaggcgaa acacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagt accaacaatg gttcatccc tcacaac 1617

```

```

SEQ ID NO: 142      moltype = RNA length = 1617
FEATURE
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 142
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtgggt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa acatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaa tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccggaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga cacgggcatc taacaagaac 540
aagtgcctta tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgccggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggttagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgccctgctga gagaggacca aggcctggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgga gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcacctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcggca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaagggt gaaggcgaa acacgtgat caagggcaga 1320
cctgtgtcca gcagcttcga cctatcaag ttccctgagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgc tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagt accaacaatg gttcatccc tcacaac 1617

```

```

SEQ ID NO: 143      moltype = RNA length = 1617
FEATURE
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 143
atgagctgga aggtggtcat catcttcagc ctgctgatca cacctcagca cggcctgaaa 60
gagagctacc tgggaagagtc ctgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtgggt acaccaacgt gtccacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgtctgccga tcagctggcc agagaggaa acatcgagaa tcttggcagc 300
ggcagctttg tgctgggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg cctagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccggaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga cacgggcatc taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgccggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggttagagcc gtgcctaaca tgctacatc tgccggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgct ggattgtgaa ggcgctcct agctgtagcg agaagaagg caattacgcc 900
tgccctgctga gagaggacca aggcctggat tgtcagaacg ccggcagcac cgtgtactac 960
cctaacgaga aggactcgga gacaagaggc gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgttg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccacggcgag gcacctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctgggtggctt gttataaggg cgtgtcctgt agcatcggca gcaacagagt gggcatcatc 1200
aagcagctga acaagggtcg cagctacatc accaaccagg acgcccgtac cgtgaccatc 1260

```

-continued

gacaacaccg	tgtatcagct	gagcaaggtg	gaaggcgaac	agcacgtgat	caagggcaga	1320
cctgtgtcca	gcagcttcga	ccctatcaag	ttccctgagg	atcagttcca	ggtggccctg	1380
gaccaggtgt	tcgagaacat	cgagaattcc	caggctcttg	tggaaccagtc	caacagaatc	1440
ctgtctagcg	ccgagaaggg	aaacaccggc	ttcatcatcg	tgatcatcct	gatcgccgtg	1500
ctgggcagct	ccatgatcct	ggtgtccatc	ttcatcatta	tcaagaagac	caagaagccc	1560
accggcgctc	ctccagaact	gagcggagtg	accaacaatg	gcttcatccc	tcacaac	1617

SEQ ID NO: 144 moltype = RNA length = 1617
 FEATURE Location/Qualifiers
 misc_feature 1..1617
 note = Synthetic Polynucleotide
 source 1..1617
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 144

atgagctgga	aggtgggtcat	catcttcagc	ctgctgatca	cacctcagca	cggcctgaaa	60
gagagctacc	tggaagagtc	ctgcagcacc	atcacagagg	gctacctgtc	tgtgctgaga	120
accggctggt	acaccaacgt	gttcacactg	gaagtgggcg	acgtcgagaa	tctgacatgc	180
tctgatggcc	ctagcctgat	caagaccgag	ctggatctga	ccaagagcgc	cctgagagaa	240
ctcaagaccg	tgtctgccga	tcagctggcc	agagaggaac	agatcgagaa	tcctggcagc	300
ggcagctttg	tgtctgggagc	cattgtctctt	ggagtggctg	ctgctgcagc	tgttacagca	360
ggcgtggcca	tcgctaagac	catcagactg	gaaagcgaag	tgaccgccat	caacaacgcc	420
ctgaagaaga	caaacgagtc	cgtcagcaca	ctcggcaatg	gcgttagagt	gctggccaca	480
gccgtgctcg	agctgaagga	cttcgtgtcc	aagaacctga	cacgggccat	taacaagaac	540
aaagtgcgaca	tcgacgacct	gaagatggcc	gtgtccttta	gccagttcaa	ccggcgggtt	600
ctgaacgtcg	tgccgcagtt	tagcgacaac	gccggaatca	caccagccat	cagcctggac	660
ctgatgacag	atgctgagct	ggctagagcc	gtgcctaaca	tgctacatc	tgccggccag	720
atcaagctga	tgctcgagaa	tagagccatg	gtccgacgga	aaggcttcgg	cattctgatt	780
ggcgtgtacg	gcagcagcgt	gatctatatg	gtgcagctgc	ctatcttcgg	cgtgatcgac	840
acaccctgct	ggatttgtgaa	ggcgcctcct	agctgtagcg	agaagaaggg	caattacgcc	900
tgctctgtga	gagaggacca	aggctgggtat	tgtcagaacg	ccggcagcac	cgtgtactac	960
cctaaccgaga	aggactgcga	gaccaagagg	gaccacgtgt	tctgtgatac	cgccgctgga	1020
atcaatgtgg	ccgagcagag	caaagagtg	aacatcaaca	tcagcaccac	caactatccc	1080
tgcaaggtgt	ccaccggcag	gcaccctatt	tctatggtgg	ctctgtctcc	tctgggagcc	1140
ctggtggctt	gttataaggg	cgtgtcctgt	agcatcgcca	gcaacagagt	gggcatac	1200
aagcagctga	acaagggctg	cagctacatc	accaaccagg	acgcccagac	cgtgaccatc	1260
gacaacaccg	tgatcagct	gagcaaggtg	gaaggcgaac	agcacgtgat	caagggcaga	1320
cctgtgtcca	gcagcttccc	acctatcaag	ttccctgagg	atcagttcca	ggtggccctg	1380
gaccaggtgt	tcgagaacat	cgagaattcc	caggctcttg	tggaaccagtc	caacagaatc	1440
ctgtctagcg	ccgagaaggg	aaacaccggc	ttcatcatcg	tgatcatcct	gatcgccgtg	1500
ctgggcagct	ccatgatcct	ggtgtccatc	ttcatcatta	tcaagaagac	caagaagccc	1560
accggcgctc	ctccagaact	gagcggagtg	accaacaatg	gcttcatccc	tcacaac	1617

SEQ ID NO: 145 moltype = RNA length = 1617
 FEATURE Location/Qualifiers
 misc_feature 1..1617
 note = Synthetic Polynucleotide
 source 1..1617
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 145

atgagctgga	aggtgggtcat	catcttcagc	ctgctgatca	cacctcagca	cggcctgaaa	60
gagagctacc	tggaagagtc	ctgcagcacc	atcacagagg	gctacctgtc	tgtgctgaga	120
accggctggt	acaccaacgt	gttcacactg	gaagtgggcg	acgtcgagaa	tctgacatgc	180
tctgatggcc	ctagcctgat	caagaccgag	ctggatctga	ccaagagcgc	cctgagagaa	240
ctcaagaccg	tgtctgccga	tcagctggcc	agagaggaac	agatcgagaa	tcctggcagc	300
ggcagctttg	tgtctgggagc	cattgtctctt	ggagtggctg	ctgctgcagc	tgttacagca	360
ggcgtggcca	tcgctaagac	catcagactg	gaaagcgaag	tgaccgccat	caacaacgcc	420
ctgaagaaga	caaacgagtc	cgtcagcaca	ctcggcaatg	gcgttagagt	gctggccaca	480
gccgtgctcg	agctgaagga	cttcgtgtcc	aagaacctga	cacgggccat	taacaagaac	540
aaagtgcgaca	tcgacgacct	gaagatggcc	gtgtccttta	gccagttcaa	ccggcgggtt	600
ctgaacgtcg	tgccgcagtt	tagcgacaac	gccggaatca	caccagccat	cagcctggac	660
ctgatgacag	atgctgagct	ggctagagcc	gtgcctaaca	tgctacatc	tgccggccag	720
atcaagctga	tgctcgagaa	tagagccatg	gtccgacgga	aaggcttcgg	cattctgatt	780
ggcgtgtacg	gcagcagcgt	gatctatatg	gtgcagctgc	ctatcttcgg	cgtgatcgac	840
acaccctgct	ggatttgtgaa	ggcgcctcct	agctgtagcg	agaagaaggg	caattacgcc	900
tgctctgtga	gagaggacca	aggctgggtat	tgtcagaacg	ccggcagcac	cgtgtactac	960
cctaaccgaga	aggactgcga	gaccaagagg	gaccacgtgt	tctgtgatac	cgccgctgga	1020
atcaatgtgg	ccgagcagag	caaagagtg	aacatcaaca	tcagcaccac	caactatccc	1080
tgcaaggtgt	ccaccggcag	gcaccctatt	tctatggtgg	ctctgtctcc	tctgggagcc	1140
ctggtggctt	gttataaggg	cgtgtcctgt	agcatcgcca	gcaacagagt	gggcatac	1200
aagcagctga	acaagggctg	cagctacatc	accaaccagg	acgcccagac	cgtgaccatc	1260
gacaacaccg	tgatcagct	gagcaaggtg	gaaggcgaac	agcacgtgat	caagggcaga	1320
cctgtgtcca	gcagcttcca	ccctatcaag	ttccctgaga	accagttcca	ggtggccctg	1380
gaccaggtgt	tcgagaacat	cgagaattcc	caggctcttg	tggaaccagtc	caacagaatc	1440
ctgtctagcg	ccgagaaggg	aaacaccggc	ttcatcatcg	tgatcatcct	gatcgccgtg	1500
ctgggcagct	ccatgatcct	ggtgtccatc	ttcatcatta	tcaagaagac	caagaagccc	1560
accggcgctc	ctccagaact	gagcggagtg	accaacaatg	gcttcatccc	tcacaac	1617

-continued

```

SEQ ID NO: 146      moltype = RNA length = 1617
FEATURE            Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 146
atgagctgga aggtgggtcat catcttcagc ctgctgatca cactcagca cggcctgaaa 60
gagagctacc tggaagagtc gtgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtgggt acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgctgcccga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgtggggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttcaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccatcac tgcgggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgtc ggattgtgaa ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgaga aggaactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgcccatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaaac agcactgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctcagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

```

SEQ ID NO: 147      moltype = RNA length = 1617
FEATURE            Location/Qualifiers
misc_feature        1..1617
                    note = Synthetic Polynucleotide
source              1..1617
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 147
atgagctgga aggtgggtcat catcttcagc ctgctgatca cactcagca cggcctgaaa 60
gagagctacc tggaagagtc gtgcagcacc atcacagagg gctacctgtc tgtgctgaga 120
accggtgggt acaccaacgt gttcacactg gaagtgggag acgtcgagaa tctgacatgc 180
tctgatggcc ctagcctgat caagaccgag ctggatctga ccaagagcgc cctgagagaa 240
ctcaagaccg tgctgcccga tcagctggcc agagaggaac agatcgagaa tcctggcagc 300
ggcagctttg tgtggggagc cattgtctct ggagtggctg ctgctgcagc tgttacagca 360
ggcgtggcca tcgctaagac catcagactg gaaagcgaag tgaccgccat caacaacgcc 420
ctgaagaaga caaacgaggg cgtcagcaca ctccgcaatg gcgttagagt gctggccaca 480
gccgtgcccg agctgaagga ctctgtgtcc aagaacctga caccggccat taacaagaac 540
aagtgcgaca tcgacgacct gaagatggcc gtgtccttta gccagttgaa ccggcggttt 600
ctgaacgtcg tgcggcagtt tagcgacaac gccggaatca caccagccat cagcctggac 660
ctgatgacag atgctgagct ggctagagcc gtgcctaaca tgccatcac tgcgggccag 720
atcaagctga tgctcgagaa tagagccatg gtccgacgga aaggcttcgg cattctgatt 780
ggcgtgtacg gcagcagcgt gatctatatg gtgcagctgc ctatcttcgg cgtgatcgac 840
acaccctgtc ggattgtgaa ggccgctcct agctgtagcg agaagaaggg caattacgcc 900
tgctgtctga gagaggacca aggtcgtgat tgtcagaacg ccggcagcac cgtgtactac 960
cctaaccgaga aggaactgcga gacaagaggg gaccacgtgt tctgtgatac cgccgctgga 1020
atcaatgtgg ccgagcagag caaagagtgc aacatcaaca tcagcaccac caactatccc 1080
tgcaaggtgt ccaccggcag gcaccctatt tctatgggtg ctctgtctcc tctgggagcc 1140
ctggtggcct gttataaggg cgtgtcctgt agcatcgcca gcaacagagt gggcatcatc 1200
aagcagctga acaagggctg cagctacatc accaaccagg acgcccatac cgtgaccatc 1260
gacaacaccg tgtatcagct gagcaaggtg gaaggcgaaac agcactgat caagggcaga 1320
cctgtgtcca gcagcttcga ccctatcaag ttccctcagg atcagttcca ggtggccctg 1380
gaccaggtgt tcgagaacat cgagaattcc caggctctgg tggaccagtc caacagaatc 1440
ctgtctagcg ccgagaaggg aaacaccggc ttcacatcgt tgatcatcct gatcgccgtg 1500
ctgggcagct ccatgatcct ggtgtccatc ttcacatta tcaagaagac caagaagccc 1560
accggcgctc ctccagaact gagcggagtg accaacaatg gcttcatccc tcacaac 1617

```

What is claimed is:

1. A method of vaccinating a human subject against a betacoronavirus, the method comprising administering to the human subject a dose of a vaccine comprising a mes-

65

5-25 mol % neutral lipid, 25-55 mol % cholesterol, and 0.5-15 mol % PEG-modified lipid, wherein the mRNA comprises an open reading frame encoding a full-length betacoronavirus spike protein, wherein the open reading frame comprises nucleosides consisting of N1-methylpseudouridine, adenosine, guanosine, and cytidine, wherein

469

the dose is administered intramuscularly and comprises from 20-50 µg of the mRNA, and wherein the vaccine has an efficacy against betacoronavirus infection of at least 90%, wherein betacoronavirus infection is determined using polymerase chain reaction (PCR).

2. The method of claim 1, wherein the dose comprises 20 µg of the mRNA.

3. The method of claim 1, wherein the dose comprises 25 µg of the mRNA.

4. The method of claim 1, wherein the dose comprises 30 µg of the mRNA.

5. The method of claim 1, wherein the dose comprises 50 µg of the mRNA.

6. A method of vaccinating a human subject against a betacoronavirus, the method comprising administering to the human subject a dose of a vaccine comprising a messenger ribonucleic acid (mRNA) formulated in a lipid nanoparticle comprising 20-60 mol % ionizable cationic lipid, 5-25 mol % neutral lipid, 25-55 mol % cholesterol, and 0.5-15 mol % PEG-modified lipid, wherein the mRNA comprises an open reading frame encoding a full-length betacoronavirus spike protein, wherein the open reading frame comprises nucleosides consisting of N1-methylpseudouridine, adenosine, guanosine, and cytidine, wherein

470

the dose is administered intramuscularly and comprises from 10-15 µg of the mRNA, and wherein the vaccine has an efficacy against betacoronavirus infection of at least 90%, wherein betacoronavirus infection is determined using polymerase chain reaction (PCR).

7. A method of vaccinating a human subject against a betacoronavirus, the method comprising administering to the human subject a dose of a vaccine comprising a messenger ribonucleic acid (mRNA) formulated in a lipid nanoparticle comprising 20-60 mol % ionizable cationic lipid, 5-25 mol % neutral lipid, 25-55 mol % cholesterol, and 0.5-15 mol % PEG-modified lipid, wherein the mRNA comprises an open reading frame encoding a full-length betacoronavirus spike protein, wherein the open reading frame comprises nucleosides consisting of N1-methylpseudouridine, adenosine, guanosine, and cytidine, wherein the dose is administered intramuscularly and comprises from 10-100 µg of the mRNA, and wherein the vaccine has an efficacy against betacoronavirus infection of at least 90%, wherein betacoronavirus infection is determined using polymerase chain reaction (PCR).

8. The method of claim 6, wherein the dose comprises 10 µg of the mRNA.

* * * * *